EDUCORN'S

MATHEMATICS

PRACTICE PROBLEMS

MATHEMATICAL CONCEPTS FOR HIGHER CLASS

VOLUME-II

MATHEMATICS IS TRUE FRIEND....

Matrices and Determinants

Topic 1

Algebra of Matrices, Types of Matrices

Q 1. The number of 3 x3 non-singular matrices,

with four entries as 1 and all other entries as 0, is

[AJEEE]

(a) less than 4

(b) 5

(c) 6

(d) at least 7

Q 2. Let A be a 2×2 matrix with non-zero entries and let $A^2 = I$, where I is 2×2 identity, matrix.

Define Tr(A) = sum of diagonal elements of A and |A| = determinant of matrix A.

[AIEEE]

Statement I Tr(A) = **0**

Statement II |A| = 1.

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

StatementI

(c) Statement I is true, Statement II is false (d)

Statement I is false', Statement II is true

Q 3. If the matrices $A=\begin{bmatrix}2 & 1 & 3\\ 4 & 1 & 0\end{bmatrix}$ and $B=\begin{bmatrix}1 & -1\\ 0 & 2\\ 5 & 0\end{bmatrix}$, then AB will be

[WBJEE]

(a) $\begin{bmatrix}17 & 0\\ 4 & -2\end{bmatrix}$

(b) $\begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix}$

(c) $\begin{bmatrix}17 & 4\\ 0 & -2\end{bmatrix}$

(d) $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

Q 4. If $\begin{bmatrix} x-y-z \\ -y+z \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 3 \end{bmatrix}$, then the values of x, y and z

are respectively

[Kerala CEE]

(a) 5, 2, 2

(b) 1,−2,3

(c) 0,−3,.3

(d) 11,8, 3

(e) 4,1,3

Q 5. If $A = [a_{ij}]_{2 \times 2}$, where $a_{ij} = i + j$, then A is equal to

[BITSAT]

(a) $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

Q 6. If A and B are square matrices of the same order such that $(A+B)(A-B) = A^2 - B^2$, then $(ABA^{-1})^2$ is equal to

[KCET]

(a) B^2

(b) I

(c) A^2B^2

(d) A^2

Q 7. If $1, \omega, \omega^2$ are the cube roots of unity and if

$$\begin{bmatrix} 1+\omega & 2\omega \\ -2\omega & -b \end{bmatrix} + \begin{bmatrix} a & -\omega \\ 3\omega & 2 \end{bmatrix} = \begin{bmatrix} 0 & \omega \\ \omega & 1 \end{bmatrix},$$ then $a^2 + b^2$ is equal to

[Kerala CEE]

(a) $1 + \omega^2$

(b) $\omega^2 - 1$

(c) $1 + \omega$

(d) $(1 + \omega)^2$

(e) ω^2

Q 8. Let A and B be two symmetric matrices of same order. Then, the matrix AB − BA is

[EAMCET]

(a) a symmetric matrix

(b) a skew−symmetric matrix

(c) a null matrix

(d) the identity matrix

Q 9. If A is a square matrix, then

[WB JEE]

(a) $A + A^T$ is symmetric

(b) AA^T is skew–symmetric

(c) $A^T + A$ is skew–symmetric

(d) $A^T A$ is skew–symmetric

Q 10. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is such that $|A| = 0$ and $A^2 - (a + d) A + kI = 0$, then k is equal to

[BCECE]

(a) b + c

(b) a + d

(c) ab + cd

(d) zero

Q 11. If $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$ and A^2 is the identity matrix,

then x is equal to

[MPPET]

(a) -1

(b) 0

(c) 1

(d) 2

Q 12. If ω is a complex cube root of unity and $A = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, then A^{50} is

[RPET]

(a) $\omega^2 A$

(b) ωA

(c) A

(d) 0

Q 13. If A and B are square matrices of order 3 x 3, then which of the following is true ?

[RPET]

(a) AB $=0 \Rightarrow$ A = O or B = O

(b) det$(2AB) = 8$ det (A) det(B)

(c) $A^2 - B^2 = (A + B)(A - B)$

(d) det(A + B) = det(A) + det(B)

Q 14. If X and Y are 2×2 matrices such that $2X + 3Y = O$ and $X + 2Y = I$, where O and I denote the 2×2 zero matrix and the 2×2 identity matrix, then X is equal to

[J&K CET]

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$

(d) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

Q 15. Let A be a 2×2 matrix with real entries.

Let I be the 2×2 identity matrix. Denote by Tr (A), the sum of diagonal entries of A. Assume that $A^2 = I$.

Statement I If $A \neq I$ and $A \neq -I$, then det (A) = -1.

Statement II If $A \neq I$ and $A \neq -I$, then Tr $A \neq 0$

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

StatementI

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 16. If $A=\begin{bmatrix}1 & -2\\ 4 & 5\end{bmatrix}$ and $f(t) = t^2 - 3t + 7$, then

$f(A)+\begin{bmatrix}3 & 6\\ -12 & -9\end{bmatrix}$ is equal to

[BITSAT]

(a) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

(b) $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

(c) $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$

(d) $\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}$

Q 17. The characteristic roots of the matrix

$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$ are

[KCET]

(a) 1,3,6

(b) 1,2,4

(c) 4,5, 6

(d) 2,4,6

Q 18. If $A = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix}$, then A^4 is equal to

[Kerala CEE]

(a) 27 A

(b) 81 A

(c) 243 A

(d) 729 A

(e) 3A

Q 19. If $[1 \times 1]\begin{bmatrix}1 & 2 & 3\\ 0 & 5 & 1\\ 0 & 3 & 2\end{bmatrix}\begin{bmatrix}x\\ 1\\ -2\end{bmatrix}=0$, then the value of

x is

[OJEE]

(a) 0

(b) $\frac{2}{3}$

(c) $\frac{5}{4}$

(d) $-\frac{4}{5}$

Q 20. $\begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix}^4 = I,$ then

[VITEEE]

(a) a = 1 = 2b

(b) a = b

(c) $a = b^2$

(d) ab = 1

Q 21. If $2A+3B=\begin{bmatrix}2 & -1 & 4\\ 3 & 2 & 5\end{bmatrix}$ and $A+2B=\begin{bmatrix}5 & 0 & 3\\ 1 & 6 & 2\end{bmatrix}$, then

B is

[JamiaMillia Islamia]

(a) $\begin{bmatrix}8 & -1 & 2\\ -1 & 10 & -1\end{bmatrix}$

(b) $\begin{bmatrix}8 & 1 & 2\\ -1 & 10 & -1\end{bmatrix}$

(c) $\begin{bmatrix} 8 & 1 & -2 \\ -1 & 10 & -1 \end{bmatrix}$

(d) $\begin{bmatrix} 8 & 1 & 2 \\ 1 & 10 & 1 \end{bmatrix}$

Q 22. If O(A) = 2×3, O(B) = 3×2, and O(C) = 3×3, which one of the following is not defined ?

[Jamia Millia Islamia]

(a) CB + A'

(b) BAC

(c) C(A+B')'

(d) C(A+B')

Q 23. If $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, then which of the following statement is not correct?

[Jamia Millia Islamia]

(a) A is orthogonal matrix

(b) A' is orthogonal matrix

(c) Determinant A = 1

(d) A is not invertible

Q 24. If $E(\theta)=\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$ and θ and ϕ differ by an odd multiple of $\frac{\pi}{2}$, then $E(\theta)E(\phi)$ is a

[Manipal]

(a) unit matrix

(b) null matrix

(c) diagonal matrix

(d) None of these

Q 25. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, then $A^2 - 4A$ is equal to

[MPPET]

(a) $2I_3$

(b) $3I_3$

(c) $4I_3$

(d) $5I_3$

Q 26. If A is square matrix, A', is its transpose, then $\frac{1}{2}(A - A')$ is

[RPET]

(a) a symmetric matrix

(b) a skew−symmetric matrix

(c) a unit matrix

(d) an elementary matrix

Q 27. If $f(x) = x^2 - 5x$, $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, then $f(A)$ is equal to

[J&K CET]

(a) $\begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & -7 \\ -7 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 7 \\ 7 & 0 \end{bmatrix}$

Q 28. If $f(x) = x^2 + 4x - 5$ and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$, then

$f(A)$ is equal to

[Guj.CET]

(a) $\begin{bmatrix} 0 & -4 \\ 8 & 8 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 8 & 4 \\ 8 & 0 \end{bmatrix}$

Q 29. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and I is the unit matrix of order 2, then A^2 equals

[BITSAT]

(a) $4A - 3I$

(b) $3A - 4I$

(c) A − I

(d) A + I

Q 30. If $A = \begin{bmatrix} 1 & -5 & 7 \\ 0 & 7 & 9 \\ 11 & 8 & 9 \end{bmatrix}$, then trace of matrix A is

[AMU]

(a) 17

(b) 25

(c) 3

(d) 12

Q 31. If $A=\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$. The only correct

statement about the matrix y A is

[Jamia Millia Islamia]

(a) A is a zero matrix

(b) A = (-1)I, where I is a unit matrix

(c) A^{-1} does not exist

(d) A^2 = I

Q 32. If $A=\begin{bmatrix} \cos^2\alpha & \cos\alpha\sin\alpha \\ \cos\alpha\sin\alpha & \sin^2\alpha \end{bmatrix}$ and

$B=\begin{bmatrix} \cos^2\beta & \cos\beta\sin\beta \\ \cos\beta\sin\beta & \sin^2\beta \end{bmatrix}$ are two matrices such that the product AB is null matrix, then $\alpha - \beta$ is

[MHTCET]

(a) 0

(b) multiple of π

(c) an odd multiple of $\pi/2$

(d) None of the above

Q 33. If A is a symmetric matrix and $n \in N$, then A^n is

[Guj. CET]

(a) symmetric matrix

(b) a diagonal matrix

(c) skew−symmetric matrix

(d) None of the above

Q 34. If A and B are square matrices of size n x n such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will be always true ?

[AIEEE]

(a) AB = BA

(b) either of A or B is a zero matrix

(c) either of A or B is an identity matrix (d) A=B

Q 35. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, a, b $\in$ N. Then

[AIEEE]

(a) there exist more than one but finite number of B's such that AB =BA

(b) there exists exactly one B such that AB = BA

(c) there exists infinitely many B's such that AB =BA

(d) there cannot exist any B such that AB = BA

Q 36. If $A=\begin{bmatrix}3 & 5\\ 2 & 0\end{bmatrix}$ and $B=\begin{bmatrix}1 & 17\\ 0 & -10\end{bmatrix}$, then | AB| is equal to

[BITSAT]

(a) 80

(b) 100

(c) −110

(d) 92

Q 37. If $A=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$, then A^{100} is equal to

[UP SEE]

(a) $2^{100}A$

(b)2^{99} A

(c) 100A

(d) 299A

Q 38. If $A=\begin{bmatrix} 1 & -3 \\ 2 & k \end{bmatrix}$ and $A^2 - 4A + 10I = A$, then k is equal to

[KCET]

(a) 0

(b) -4

(c) 4 and not 1

(d) 1 or 4

Q 39. If A is an invertible matrix of order n, then the determinant of adj (A) is equal to

[EAMCET]

(a) $|A|^n$

(b) $|A|^{n+1}$

(c) $|A|^{n-1}$

(d) $|A|^{n+2}$

Q 40. If a square matrix A is such that $AA^T = I = A^T A$, then $|A|$ is equal to

[BCECE]

(a) 0

(b) ± 1

(c) ± 2

(d) None of these

Q 41. If $A(\alpha)=\begin{bmatrix}\cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha\end{bmatrix}$, then the matrix $A^2(\alpha)$ is

[MHT CET]

(a) $A(2\alpha)$

(b) $A(\alpha)$

(c) $A(3\alpha)$

(d) $A(4\alpha)$

Q 42. Let A, B and C be n × n matrices. Which one of the following is a correct statements?

[MPPET]

(a) If AB = AC, then B = C

(b) If $A^3 + 2A^2 + 3A + 5I = O$; then A is invertible

(c) If $A^2=O$, then A = O

(d) None of the above

Q 43. The matrix $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$ is

[MPPET]

(a) unitary

(b) orthogonal

(c) nilpotent

(d) involutory

Q 44. If U = [2 − 3 4], X = [0 2 3], $V = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ and

$Y = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$, then UV + XY

[RPET]

(a) [20]

(b) 20

(c) [−20]

(d) −20

Q 45. If A is a non−singular matrix such that A^3 = A + I, then the inverse of B = $A^6 - A^5$ is

[J&E CET]

(a) A

(b) A^{-1}

(c) $-A$

(d) $-A^{-1}$

Q 46. If A is a matrix such that $A^2 = A + I$,

where I is the unit matrix, then A^5 is equal to

[J&KCET]

(a) $5A + I$

(b) $5A + 2I$

(c) $5A + 3I$

(d) $5A + 4I$

Q 47. If $P = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$,

then $P^TQ\ P$ is

[IITJEE]

(a) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2005 \\ 2005 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ 2005 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q 48. If A is a skew symmetric matrix of order n and C is a column matrix of order n × 1, then C^T AC is

[AMU]

(a) an identity matrix of order n

(b) an identity matrix of order 1

(c) a zero matrix of order 1

(d) None of the above

Q 49. If $m[-3\ 4] + n[4\ -3] = [10\ -\ 11]$, then $3m + 7n$ is equal to

[EAMCET]

(a) 3

(b) 5

(c) 10

(d) 1

Q 50. If. $A+I=\begin{bmatrix}3 & -2\\ 4 & 1\end{bmatrix}$, then (A + I) . (A − I) is

equal to

[MHT CET]

(a) $\begin{bmatrix}-5 & -4\\ 8 & -9\end{bmatrix}$

(b) $\begin{bmatrix}-5 & 4\\ -8 & 9\end{bmatrix}$

(c) $\begin{bmatrix}5 & 4\\ 8 & 9\end{bmatrix}$

(d) $\begin{bmatrix}-5 & -4\\ -8 & -9\end{bmatrix}$

Q 51. If $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & a \\ 4 & b \end{bmatrix}$ and $(A + B)^2 = A^2 + B^2$. Then, a and b are respectively

(a) $1, -1$

(b) $2, -3$

(c) $-1, 1$

(d) $3, -2$

[MHT CET]

Q 52. If $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$, then A is equal to

[DCE]

(a) idempotent

(b) involuntary

(c) nilpotent

(d) scalar

Q 53. What must be the matrix X if

$2X + \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 7 & 2 \end{bmatrix}$?

[KCET]

(a) $\begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -3 \\ 2 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 6 \\ 4 & -2 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & -6 \\ 4 & -2 \end{bmatrix}$

Q 54. Let A and B are two square matrices such that AB = A and BA = B, then A^2 equals to

[Kerala CEE]

(a) B

(b) A

(c) I

(d) O

Q 55. If $P=\begin{bmatrix}1 & 2 & 3\\ 2 & 3 & 4\\ 3 & 4 & 5\end{bmatrix}\begin{bmatrix}-1 & -2\\ -2 & 0\\ 0 & -4\end{bmatrix}\begin{bmatrix}-4 & -5 & -6\\ 0 & 0 & 1\end{bmatrix}$, then P_{22} is

equal to

[OJEE]

(a) 40

(b) −40

(c) −20

(d) 20

Q 56. $A = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$, then A^2 is equal to

[MHTCET]

(a) null matrix

(b) unit matrix

(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Q 57. If A = [x y z], $B = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ and $C = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Then,

ABC=O, if

[MHTCET]

(a) $[ax^2 + by^2 + cz^2 + 2gxy + 2jyz + 2czx] = O$

(b) $[ax^2 + cy^2 + bz^2 + xy + yz + zx] = O$

(c) $[ax^2 + by^2 + cz^2 + 2hxy + 2by + 2cz] = O$

(d) $[ax^2+by^2+cz^2+2gzx+2hxy+2fyz] = O$

Q 58. $A=\begin{bmatrix} 0 & 3 & 3 \\ -3 & 0 & -4 \\ -3 & 4 & 0 \end{bmatrix}$ and $B=\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then B'(AB) is

[MHT CET]

(a) null matrix

(b) singular matrix

(c) unit matrix

(d) symmetric matrix

Q 59. If $A=\begin{bmatrix}6 & 8 & 5\\ 4 & 2 & 3\\ 9 & 7 & 1\end{bmatrix}$ is the sum of a symmetric matrix B and skew−symmetric matrix C, then B is

[MPPET]

(a) $\begin{bmatrix}6 & 6 & 7\\ 6 & 2 & 5\\ 7 & 5 & 1\end{bmatrix}$

(b) $\begin{bmatrix}0 & 2 & -2\\ -2 & 5 & -2\\ 2 & 2 & 0\end{bmatrix}$

(c) $\begin{bmatrix}6 & 6 & 7\\ -6 & 2 & -5\\ -7 & 5 & 1\end{bmatrix}$

(d) $\begin{bmatrix} 0 & 6 & -2 \\ 2 & 0 & -2 \\ -2 & -2 & 0 \end{bmatrix}$

Q 60. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$, then value of α for which $A^2 = B$ is

[IITJEE]

(a) 1

(b) -1

(c) 4

(d) No real values

Q 61. If $A=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$, I is the unit matrix of order 2 and a, b are arbitrary constants, then $(aI + bA)^2$ is equal to

[DCE]

(a) $a^2I - abA$

(b) $a^2I + 2abA$

(c) $a^2I + b^2A$

(d) None of the above

Q 62. $\cos\theta\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}+\sin\theta\begin{bmatrix}\sin\theta & -\cos\theta\\ \cos\theta & \sin\theta\end{bmatrix}$ is equal to

[AMU]

(a) $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

(b) $\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$

(c) $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$

(d) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

Q 63. If A and B are 3×3 matrices such that

AB = B and BA = A, then

[OJEE]

(a) A^2 =A and $B^2 \neq B$

(b) $A^2 \neq A$ and B^2 =B

(c) A^2 =A and B^2 =B

(d) $A^2 \neq$ A and $B^2 \neq$ B

Q 64. Matrix A is such that $A^2 = 2A - I$, where I

is the identity matrix, then for $n \geq 2$, A^n is equal to

[BCECE]

(a) $nA - (n-1)I$

(b) $nA - I$

(c) $2^{n-1}A - (n-1)I$

(d) $2^{n-1}A - I$

Q 65. If $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ 4 & 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$, then AB is equal to

[MPPET]

(a) $\begin{bmatrix} 5 & 1 & -3 \\ 3 & 2 & 6 \\ 14 & 5 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 11 & 4 & 3 \\ 1 & 2 & 3 \\ 0 & 3 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 8 & 4 \\ 2 & 9 & 6 \\ 0 & 2 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 & 2 \\ 5 & 4 & 3 \\ 1 & 8 & 2 \end{bmatrix}$

Q 66. If A and B are two matrices such that

both A+B = AB are defined, then

[J& K CET]

(a) A and B are of same order

(b) A is of order $m \times m$ and B is of order $n \times n$

(c) both A and B are of same order $n \times n$

(d) A is of order $m \times n$ and B is of order $n \times m$

Topic 2 Properties and Solution of Determinants

2010

Q 1. Passage Let p be an odd prime number and T be the following set of 2×2 matrices

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix}; a, b, c \in \{0, 1, 2, ..., p-1\} \right\}$$

[IIT JEE]

(i) The number of A in T_p such that A is either symmetric or skew–symmetric or both, and det (A) is divisible by p is

(a) $(p-1)^2$

(b) $2(p-1)$

(c) $(p-1)^2+1$

(d) $2p-1$

(ii) The number of A in T_p such that the trace of A is not divisible by p but det (A) is divisible by p is

[**Note:** The trace of a matrix is the sum of its diagonal entries.]

(a) $(p-1)(p^2-p+1)$

(b) $p^3 - (p - 1)^2$

(c) $(p - 1)^2$

(d) $(p - 1)(p^2 - 2)$

(iii) The number of A in T_p such that det (A) is not divisible by p, is .

(a) $2p^2$

(b) $p^3 - 5p$

(c) $p^3 - 3p$

(d) $p^3 - p^2$

Q 2. ω is an imaginary cube root of unity and

$\begin{vmatrix} x+\omega^2 & \omega & 1 \\ \omega & \omega^2 & 1+x \\ 1 & x+\omega & \omega^2 \end{vmatrix} = 0$, then one of the values of x is

[WBJEE]

(a) 1

(b) 0

(c) -1

(d) 2

Q 3. The value of the determinant $\begin{vmatrix} 15! & 16! & 17! \\ 16! & 17! & 18! \\ 17! & 18! & 19! \end{vmatrix}$ is

equal to

(a) $15! + 16!$

(b) $2(15!)\,(16!)\,(17!)$

(c) $15! + 16! + 17!$

(d) $16! + 17!$

(e) $2\,(15! + 16!)$

Q 4. If $\begin{vmatrix} x^2+x & 3x-1 & -x+3 \\ 2x+1 & 2+x^2 & x^3-3 \\ x-3 & x^2+4 & 3x \end{vmatrix} = a_0 + a_1x + a_2x^2 + \ldots$

$+a_7x^7$, the value of a_0 is

[Kerala CEE]

(a) 25

(b) 24

(c) 23

(d) 22

(e) 21

Q 5. Let a, b, c be such that $(b + c) \neq 0$ and

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^{n}c \end{vmatrix} = 0$$

Then the value of n is

[AIEEE]

(a) zero

(b) any even integer

(c) any odd integer

(d) any integer

Q 6. In the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$, the number of real solutions of the equations $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ is

[UP SEE]

(a) 0

(b) 2

(c) 1

(d) 3

Q 7. If $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$ and x, y, z are all distinct,

then xyz is equal to

[UPSEE]

(a) -1

(b) 1

(c) 0

(d) 3

Q 8. If $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix} = 0$, then a, b, c are

[KCET]

(a) in GP

(b) in HP

(c) equal

(d) in AP

Q 9. The value of $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$ is equal to

[KCET]

(a) 0

(b) 1

(c) xy^2

(d) log xyz

Q 10. If α, β, γ are the cube roots of unity, then the value of the determinant $\begin{vmatrix} e^{\alpha} & e^{2\alpha} & (e^{3\alpha}-1) \\ e^{\beta} & e^{2\beta} & (e^{3\beta}-1) \\ e^{\gamma} & e^{2\gamma} & (e^{3\gamma}-1) \end{vmatrix}$ is equal to

[Kerala CEE]

(a) -2

(b) -1

(c) 0

(d) 1

(e) 2

Q 11. If B is a non−singular matrix and A is a square matrix such that $B^{-1}AB$ exists, then det $(B^{-1}AB)$ is equal to

[Kerala CEE]

(a) det (A^{-1})

(b) det (B^{-1})

(c) det(B)

(d) det(A)

(e) $\det(AB^{-1})$

Q 12. If $A = \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x & x & 1 \end{vmatrix}$ and $I = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$, then $A^3 - 4A^2 +$ $3A + I$ is equal to

[Kerala CEE]

(a) $3I$

(b) I

(c) $-I$

(d) $-2I$

(e) $2I$

Q 13. If $A=\begin{bmatrix}1 & 2\\ 3 & 5\end{bmatrix}$, then the value of the determinant $|A-5A|$ is

[Kerala CEE]

(a) -6

(b) -5

(c) -4

(d) 4

(e) 6

Q 14. If $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = k \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, then k is equal to

[OJEE]

(a) 4

(b) 3

(c) 2

(d) 1

Q 15. If the value of the determinant $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix}$ is positive, then

[Jamia Millia Islamia]

(a) abc > 1

(b) abc> −8

(c) abc < −8

(d) $abc > -2$

Q 16. If $A = \begin{bmatrix} 3 & 2 & 4 \\ 1 & 2 & 1 \\ 3 & 2 & 6 \end{bmatrix}$ and A_{ij} are the cofactors of a_{ij}, then $a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$ is equal to

[MHTCET]

(a) 8

(b) 6

(c) 4

(d) 0

Q 17. If $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = k \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$, then the value of k is

[MPPET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 18. If $f(x)=\begin{vmatrix} 1+a & 1+ax & 1+ax^2 \\ 1+b & 1+bx & 1+bx^2 \\ 1+c & 1+cx & 1+cx^2 \end{vmatrix}$, where a, b, c are non−zero constants, then value of $f(10)$ is

[RPET]

(a) $10(b - a)(c - a)$

(b) $100(b - a)(c - b)(a - c)$

(c) 100 abc

(d) 0

Q 19. The value of $\begin{vmatrix} \log_5 729 & \log_3 5 \\ \log_5 27 & \log_9 25 \end{vmatrix} \begin{vmatrix} \log_3 5 & \log_{27} 5 \\ \log_5 9 & \log_5 9 \end{vmatrix}$ is

equal to

[J&KCET]

(a) 1

(b) 6

(c) $\log_5 9$

(d) $\log_3 5 - \log_5 81$

Q 20. The matrix $\begin{bmatrix} \lambda & 1 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & \lambda \end{bmatrix}$ is non−singular

[J&KCET]

(a) for all real values of λ

(b) only when $\lambda = \pm \frac{1}{\sqrt{2}}$

(c) only when $\lambda \neq 0$

(d) only when $\lambda = 0$

Q 21. The sum of the products of the elements of any row of a determinant A with the cofactors of the corresponding elements is equal to

[DCE]

(a) 1

(b) 0

(c) |A|

(d) $\frac{1}{2}$|A|

Q 22. $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$ is equal to

[BITSAT]

(a) 0

(b) $a + b + c$

(c) $(a + b + c)^2$

(d) $(a + b + c)^3$

Q 23. If $a \neq b \neq c$, the value of x which satisfies the equation $\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0,$ is

[UPSEE]

(a) x = 0

(b) x = a

(c) x = b

(d) x = c

Q 24. If matrix $\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ \lambda & -3 & 0 \end{bmatrix}$ is singular, then λ is equal to

[UPSEE]

(a) -2

(b) -1

(c) 1

(d) 2

Q 25. If the matrix M_r is given by

[Kerala CEE]

$M_r = \begin{bmatrix} r & r-1 \\ r-1 & r \end{bmatrix}, r = 1, 2, 3, \ldots\ldots,$ then the value of det (M_1) + det (M_2) + …+det (M) is

(a)

(b)

(c) $()^2$

(d) $()^2$

(e)

Q 26. If $f(\alpha)=\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix}$, then $f(\sqrt[3]{3})$ is equal to

[Kerala CEE]

(a) 1

(b) -4

(c) 4

(d) 2

(e) -2

Q 27. If $f(\theta)=\begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$. Then, for all θ.

[OJEE]

(a) $f(\theta) = 0$

(b) $f(\theta) = 1$

(c) $f(\theta) = -1$

(d) None of these

Q 28. If $f(x) = \begin{vmatrix} 1 & 2(x-1) & 3(x-1)(x-2) \\ x-1 & (x-1)(x-2) & (x-1)(x-2)(x-3) \\ x & x(x-1) & x(x-1)(x-2) \end{vmatrix}$. Then,

the value of $f(49)$ is

[OJEE]

(a) $49x$

(b) $-49x$

(c) 0

(d) 1

Q 29. The values of x for which the given matrix

$\begin{bmatrix} -x & x & 2 \\ 2 & x & -x \\ x & -2 & -x \end{bmatrix}$ will be non−singular, are

[WB JEE]

(a) $-2 \leq x \leq 2$

(b) for all x other than 2 and -2

(c) $x \geq 2$ (d) $x \leq -2$

Q 30. $\begin{vmatrix} \log e & \log e^2 & \log e^3 \\ \log e^2 & \log e^3 & \log e^4 \\ \log e^3 & \log e^4 & \log e^5 \end{vmatrix}$ is equal to

[BCECE]

(a) 0

(b) 1

(c) 4 log e

(d) 5 1og e

Q 31. If x, y, z are different from zero and

$\Delta = \begin{vmatrix} a & b-y & c-z \\ a-x & b & c-z \\ a-x & b-y & c \end{vmatrix} = 0$, then the value of the expression

$\frac{a}{x}+\frac{b}{y}+\frac{c}{z}$ is

[VITEEE]

(a) 0

(b) -1

(c) 1

(d) 2

Q 32. The value of $\begin{vmatrix} x+y & y+z & z+x \\ x & y & z \\ x-y & y-z & z-x \end{vmatrix}$ is equal to

[Jamia Millia Islamia]

(a) $2(x + y+z)^2$

(b) $2(x + y+z)^3$

(c) $(x + y + z)^2$

(d) 0

Q 33. The arbitrary constant on which the value of the determinant $\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \cos(p-d)a & \cos pa & \cos(p-d)a \\ \sin(p-d)a & \sin pa & \sin(p-d)a \end{vmatrix}$ does not depend, is

[Manipal]

(a) α

(b) p

(c) d

(d) a

Q 34. If one root of determinant $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$, is -9, then the other two roots are

[MPPET]

(a) $2, 7$

(b) $2, -7$

(c) $-2, 7$

(d) $-2, -7$

Q 35. The determinant $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$ is equal to zero for all values of α, if

[MP PET]

(a) a, b, c are in AP

(b) a, b, c are in GP

(c) a, b, c are in HP

(d) None of these

Q 36. A is a square matrix of order 4 and I is a unit matrix, then it is true that

[RPET]

(a) $\det(2A) = 2\det(A)$

(b) $\det(2A) = 16\det(A)$

(c) $\det(-A) = -\det(A)$

(d) $\det(A+ I) = \det(A) + I$

Q 37. If $x \neq 0$, $\begin{vmatrix} x+1 & 2x+1 & 3x+1 \\ 2x & 4x+3 & 6x+3 \\ 4x+4 & 6x+4 & 8x+4 \end{vmatrix} = 0$, then $x + 1$ is equal to

[J&KCET]

(a) x

(b) 0

(c) $2x$

(d) $3x$

Q 38. The coefficient of x in $f(x)=\begin{vmatrix} x & 1+\sin x & \cos x \\ 1 & \log(1+x) & 2 \\ x^2 & 1+x^2 & 0 \end{vmatrix}$

, $-1 < x \leq 1$, is

[Guj.CET]

(a) 1

(b) -2

(c) -1

(d) 0

Q 39. If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0$, $y \neq 0$, then D is

[AIEEE]

(a) divisible by neither x nor y

(b) divisible by both x and y

(c) divisible by x but not y

(d) divisible by y but not x

Q 40. If $c = 2\cos\theta$, then the value of the determinant $\Delta = \begin{vmatrix} c & 1 & 0 \\ 1 & c & 1 \\ 6 & 1 & c \end{vmatrix}$ is

[DCE]

(a) $\frac{\sin 4\theta}{\sin \theta}$

(b) $\frac{2\sin^2 2\theta}{\sin \theta}$

(c) $4 \cos^2 \theta (2 \cos \theta - 1)$

(d) None of these

Q 41. The Value of $\begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix}$ is

[KCET]

(a) x(x−p)(x−q)

(b) $(x-p)(x-q)(x+p+q)$

(c) $(p-q)(x-q)(x-p)$

(d) $pq(x-p)(x-q)$

Q 42. $\begin{vmatrix} \alpha & -\beta & 0 \\ 0 & \alpha & \beta \\ \beta & 0 & \alpha \end{vmatrix} = 0$, then

[Kerala CEE]

(a) $\frac{\alpha}{\beta}$ is one of the cube roots of unity

(b) α is one of the cube roots of unity

(c) β is one of the cube roots of unity

(d) $\alpha\beta$ is one of the cube roots of unity

(e) None of the above

Q 43. If $\begin{vmatrix} x & 3 & 6 \\ 3 & 6 & x \\ 6 & x & 3 \end{vmatrix} = \begin{vmatrix} 2 & x & 7 \\ x & 7 & 2 \\ 7 & 2 & x \end{vmatrix} = \begin{vmatrix} 4 & 5 & x \\ 5 & x & 4 \\ x & 4 & 5 \end{vmatrix} = 0$, then x is

equal to

[Kerala CEE]

(a) 9

(b) -9

(c) 0

(d) -1

(e) 1

Q 44. The value of the determinant

$$\begin{vmatrix} \cos\alpha & -\sin\alpha & 1 \\ \sin\alpha & \cos\alpha & 1 \\ \cos(\alpha+\beta) & -\sin(\alpha+\beta) & 1 \end{vmatrix}$$ is

[AMU]

(a) independent of α

(b) independent of β

(c) independent of α and β

(d) None of these

Q 45. If $\begin{bmatrix} 1 & 2 & x \\ 4 & -1 & 7 \\ 2 & 4 & -6 \end{bmatrix}$ is a singular matrix, then x is equal to

[EAMCET]

(a) 0

(b) 1

(c) -3

(d) 3

Q 46. $\begin{vmatrix} a-b+c & -a-b+c & 1 \\ a+b+2c & -a+b+2c & 2 \\ 3c & 3c & 3 \end{vmatrix}$ is

[OJEE]

(a) $6ab$

(b) ab

(c) $12\ ab$

(d) $2\ ab$

Q 47. Let [x] represent the greatest integer less than or equal to x, then the value of the determinant $\begin{vmatrix} [e] & [\pi] & [\pi^2-6] \\ [\pi] & [\pi^2-6] & [e] \\ [\pi^2-6] & [e] & [\pi] \end{vmatrix}$ is

[OJEE]

(a) -8

(b) 8

(c) 10

(d) None of these

Q 48. If $D_r = \begin{vmatrix} r & 1 & \frac{n(n+1)}{2} \\ 2r-1 & 4 & n^2 \\ 2^{r-1} & 5 & 2^n - 1 \end{vmatrix}$, then the value of $\sum_{r=0}^{n} D_r$ is

[BCECE]

(a) 0

(b) 1

(c) $\frac{n(n+1)(2n+1)}{6}$

(d) None of these

Q 49. Let a, b, c, be positive and not all equal, the value of the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is

[BCECE]

(a) positive

(b) negative

(c) zero

(d) None of these

Q 50. If $x = -5$ is a root of $\begin{vmatrix} 2x+1 & 4 & 8 \\ 2 & 2x & 2 \\ 7 & 6 & 2x \end{vmatrix} = 0$, then

the other roots are

[VITEEE]

(a) 3,3.5

(b) 1,3.5

(c) 1, 7

(d) 2, 7

Q 51. If $a^2 + b^2 + c^2 = -2$ and

$$f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & (1+b^2x) & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & (1+c^2x) \end{vmatrix}$$, then f(x) is a polynomial

of degree

[MPPET]

(a) 2

(b) 3

(c) 0

(d) 1

Q 52. If $\begin{vmatrix} 1 & 1 & 0 \\ 2 & 0 & 3 \\ 5 & -6 & x \end{vmatrix} = 29$, then x is

[RPET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 53. If $\begin{vmatrix} x & y & z \\ -x & y & z \\ x & -y & z \end{vmatrix} = kxyz$, then k is equal to

[J&KCET]

(a) 1

(b) 3

(c) 4

(d) 2

Q 54. If a, b, c are non−zero real numbers, then

$\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix}$ vanishes, when

[Guj.CET]

(a) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$

(b) $\frac{1}{a}-\frac{1}{b}-\frac{1}{c}=0$

(c) $\frac{1}{b}+\frac{1}{c}-\frac{1}{a}=0$

(d) $\frac{1}{b}-\frac{1}{c}-\frac{1}{a}=0$

Q 55. If $\begin{vmatrix} -12 & 0 & \lambda \\ 0 & 2 & -1 \\ 2 & 1 & 15 \end{vmatrix} = -360$, then the value of A, is

[DCE]

(a) -1

(b) -2

(c) -3

(d) 4

Q 56. The solutions of the equation $\begin{vmatrix} x & 2 & -1 \\ 2 & 5 & x \\ -1 & 2 & x \end{vmatrix} = 0$,

are

[BITSAT]

(a) $3,-1$

(b) $-3,1$

(c) $3, 1$

(d) $-3,-1$

Q 57. The value of $\begin{vmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{vmatrix} = 0$, when m

is equal to

[KCET, Kerala CEE]

(a) 6

(b) 5

(c) 4

(d) 1

(e) 2

Q 58. The value of the determinant,

$$\begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & 5 & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$$ is

[Kerala CEE]

(a) $5(\sqrt{6}-5)$

(b) $5\sqrt{3}(\sqrt{6}-5)$

(c) $\sqrt{5}(\sqrt{6}-\sqrt{3})$

(d) $\sqrt{2}(\sqrt{7}-\sqrt{5})$

(e) $3(\sqrt{5}-\sqrt{2})$

Q 59. The value of $\begin{vmatrix} \cos(x-a) & \cos(x+a) & \cos x \\ \sin(x+a) & \sin(x-a) & \sin x \\ \cos a \tan x & \cos a \cot x & \cos ec\, 2x \end{vmatrix}$ is

equal

[Kerala CEE]

(a) 1

(b) sin a cos a

(c) 0

(d) sin x cos x

(e) $\operatorname{cosec} 2x$

Q 60. The value of $\begin{vmatrix} 441 & 442 & 443 \\ 445 & 446 & 447 \\ 449 & 450 & 451 \end{vmatrix}$ is

[AMU]

(a) $441 \times 446 \times 4510$

(b) 0

(c) -1

(d) 1

Q 61. If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 40 & 3 & i \end{vmatrix} = x + iy$, then

[OJEE]

(a) x = 3, y = 1

(b) x = l, y = 3

(c) x = 0, y = 3

(d) x = 0,y = 0

Q 62. l, m, n are the pth, qth and rth terms of an GP and all positive, then $\begin{vmatrix} \log l & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix}$ equals

[OJEE]

(a) 3

(b) 2

(c) 1

(d) zero

Q 63. If $a_1, a_2, \ldots, a_n, \ldots\ldots\ldots$ are in GP and $a_i > 0$ for each i, then the determinant

$$\Delta = \begin{vmatrix} \log a_n & \log a_{n+2} & \log a_{n+4} \\ \log a_{n+6} & \log a_{n+8} & \log a_{n+10} \\ \log a_{n+12} & \log a_{n+14} & \log a_{n+16} \end{vmatrix}$$ is equal to

[BCECE]

(a) 0

(b) 1

(c) 2

(d) n

Q 64. If $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix} = (y - z)(z - x)(x - y)$

$\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$, then n is equal to

[BCECE]

(a) 2

(b) -2

(c) -1

(d) 1

Q 65. The roots of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$

[RPET]

(a) $-1,-2$

(b) $-1,2$

(c) $1,-2$

(d) 1,2

Q 66. $\begin{vmatrix} 1 & 2 & 3 \\ 1^3 & 2^3 & 3^3 \\ 1^5 & 2^5 & 3^5 \end{vmatrix}$ is equal to

[J&KCET]

(a) 1!213

(b) 1!3!5!

(c) 6!

(d) 9!

Q 67. If is ω a complex cube root of unity, then

$$\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$$ is equal to

[Guj.CET]

(a) -1

(b) 1

(c) 0

(d) ω

Q 68. If $a_1, a_2, \ldots, a_n, \ldots$, are in GP, then the determinant is equal to

[AIEEE]

$$\Delta = \begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+5} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$$

(a) 2

(b) 4

(c) 0

(d) 1

Q 69. If $f(x) = \begin{vmatrix} 1 & x & (x+1) \\ 2x & x(x-1) & x(x+1) \\ 3x(x-1) & x(x-1)(x-2) & x(x-1)(x+1) \end{vmatrix}$ then

$f(11)$ equals

[DCE]

(a) 0

(b) 11

(c) -11

(d) 1

Q 70. $\begin{vmatrix} 1+\sin^2\theta & \sin^2\theta & \sin^2\theta \\ \cos^2\theta & 1+\cos^2\theta & \cos^2\theta \\ 4\sin 4\theta & 4\sin 4\theta & 1+4\sin 4\theta \end{vmatrix} = 0$, then $\sin 4\theta$ equals to

[BITSAT]

(a) $1/2$

(b) 1

(c) $-1/2$

(d) -1

Q 71. The value of θ lying between $\theta = 0$ and $\frac{\pi}{2}$ and satisfying the equation $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix}$ is

[UP SEE]

(a) $\frac{7\pi}{24}$

(b) $\frac{5\pi}{24}$

(c) $\frac{11\pi}{2}$

(d) $\frac{\pi}{24}$

Q 72. If $A=\begin{bmatrix}3 & 5\\ 2 & 0\end{bmatrix}$ and $B=\begin{bmatrix}1 & 17\\ 0 & -10\end{bmatrix}$, then | AB| is equal to

[KCET]

(a) 80

(b) 100

(c) −110

(d) 92

Q 73. If $f(x) = \begin{vmatrix} x-3 & 2x^2-18 & 3x^3-81 \\ x-5 & 2x^2-50 & 4x^3-500 \\ 1 & 2 & 3 \end{vmatrix}$, then $f(1)$.

$f(3) + f(3) \cdot f(5) + f(5) \cdot f(1)$ is equal to

(a) $f(1)$

(b) $f(3)$

(c) $f(l)+ f(3)$

(d) $f(l) + f(5)$

(e) $f(l)+ f(3)+ f(5)$

[Kerala CEE]

Q 74. If $A_i = \begin{bmatrix} a^i & b^i \\ b^i & a^i \end{bmatrix}$ and if $|a| < 1$, $|b| < 1$, then

$\sum_{i=1}^{\infty} \det(A_i)$ is equal to

[Kerala CEE]

(a) $\frac{a^2}{(1-a)^2} - \frac{b^2}{(1-b)^2}$

(b) $\frac{a^2 - b^2}{(1-a^2)(1-b^2)}$

(c) $\frac{a^2}{(1-a)^2} + \frac{b^2}{(1-b)^2}$

(d) $\frac{a^2}{(1+a)^2} - \frac{b^2}{(1-b)^2}$

(e) $\frac{a}{1+a} - \frac{b}{1-b}$

Q 75. The value of $\begin{vmatrix} 1 & 1 & 1 \\ (2^x+2^{-x})^2 & (3^x+3^{-x})^2 & (5^x+5^{-x})^2 \\ (2^x-2^{-x})^2 & (3^x-3^{-x})^2 & (5^x-5^{-x})^2 \end{vmatrix}$ is

[AMU]

(a) 0

(b) 30^x

(c) 30^{-x}

(d) 1

Q 76. Let the determinant of a 3 x 3 matrix A be 6, then B is a matrix defined by B = $5A^2$. Then, determinant of B is

[OJEE]

(a) 180

(b) 100

(c) 80

(d) None of these

Q 77. The value of the determinant $\begin{vmatrix} x & a & b+c \\ x & b & c+a \\ x & c & a+b \end{vmatrix}$ =

0, if

[BCECE]

(a) x = a

(b) x = b

(c) x = c

(d) x has any value

Q 78. If $(\omega \neq 1)$ is a cubic root of unity, then

$$\begin{vmatrix} 1 & 1+i+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -1+\omega-i & -1 \end{vmatrix}$$ equals

[Jamia Millia Islamia]

(a) zero

(b) 1

(c) i

(d) ω

Q 79. If a = $1 + 2 + 4 + \ldots$ to n terms, b = $1 + 3 + 9 + \ldots$ to n terms and c = $1 + 5 + 25 + \ldots$ to n terms, then $\begin{vmatrix} a & 2b & 4c \\ 2 & 2 & 2 \\ 2^n & 3^n & 5^n \end{vmatrix}$ equals

[MPPET]

(a) $(30)^n$

(b) $(10)^n$

(c) 0

(d) $2^n + 3^n + 5^n$

Q 80. The integer represented by the

determinant $\begin{vmatrix} 215 & 342 & 511 \\ 6 & 7 & 8 \\ 36 & 49 & 54 \end{vmatrix}$ is exactly divisible by

[RPET]

(a) 146

(b) 21

(c) 20

(d) 335

Q 81. The matrix $\begin{bmatrix} 5 & 10 & 3 \\ -2 & -4 & 6 \\ -1 & -2 & b \end{bmatrix}$ is a singular matrix, if b is equal to

[RPET]

(a) -3

(b) 3

(c) 0

(d) for any value of b

Q 82. The determinant $\begin{vmatrix} 4+x^2 & -6 & -2 \\ -6 & 9+x^2 & 3 \\ -2 & 3 & 1+x^2 \end{vmatrix}$ is not divisible by

[J&KCET]

(a) x

(b) x^3

(c) $14 + x^2$

(d) x^5

Q 83. The minors of -4 and 9 and the cofactors of -4 and 9 in matrix $\begin{bmatrix} -1 & -2 & 3 \\ -4 & -5 & -6 \\ -7 & 8 & 9 \end{bmatrix}$ are respectively

[J&KCET]

(a) $42, 3, -42, 3$

(b) $-42, -3, 42, -3$

(c) $42, 3, -42, -3$

(d) $42, 3, 42, 3$

Q 84. In ΔABC if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$, then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to

[Guj. CET]

(a) $\frac{4}{9}$

(b) $\frac{9}{4}$

(c) $3\sqrt{3}$

(d) 1

Q 85. If $A=\begin{bmatrix}\alpha & 2\\ 2 & \alpha\end{bmatrix}$ and $|A^3| = 125$, then the value of α is

(a) ± 1

(b) ± 2

(c) ± 3

(d) ± 5

[IIT JEE]

Q 86. The value of the determinant

$$\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$$ is equal to

[DCE]

(a) 6xyz

(b) xyz

(c) 4xyz

(d) xy + yz + zx

Q 87. The value of $\begin{vmatrix} 1 & 1 & 1 \\ bc & ca & ab \\ b+c & c+a & a+b \end{vmatrix}$ is

[KCET]

(a) 1

(b) 0

(c) (a − b)(b − c)(c − a)

(d) (a + b)(b + c)(c + a)

Q 88. If a, b, c, d, e and f are in GP, then the value of $\begin{vmatrix} a^2 & d^2 & x \\ b^2 & e^2 & y \\ c^2 & f^2 & z \end{vmatrix}$

[Kerala CEE]

(a) depends on x and y

(b) depends on x and z

(c) depends on y and z

(d) depends on x, y and z

(e) independent of x, y and z

Q 89. If a square matrix A is such that $AA^T = I = A^T A$, then $|A|$ is equal to

[AMU]

(a) 0

(b) ± 1

(c) ± 2

(d) None of these

Q 90. Match the following elements of $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & 2 \\ 3 & -4 & 6 \end{bmatrix}$ with

their cofactors and choose the correct answer.

[EAMCET]

	Element		**Cofactor**

A.	−1	1.	−2
B.	1	2.	32
C.	3	3.	4
D.	6	4.	6
		5.	−6

A B C D

(a) 2 4 1 3

(b) 2 4 3 1

(c) 4 2 1 3

(d) 4 1 2 3

Q 91. The value of $\begin{vmatrix} 1990 & 1991 & 1992 \\ 1991 & 1992 & 1993 \\ 1992 & 1993 & 1994 \end{vmatrix}$ is

[EAMCET]

(a) 1992

(b) 1993

(c) 1994

(d) 0

Q 92. The value of the following determinant is

[BCECE]

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$$

(a) $(a-b)(b-c)(c-a)(a+b+c)$

(b) $abc(a+b)(b+c)(c+a)$

(c) $(a-b)(b-c)(c-a)$

(d) None of the above

Q 93. The value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ is

equal to

[Jamia Millia Islamia]

(a) $3 - x + y$

(b) $(1 - x)(1 + y)$

(c) xy

(d) $-xy$

Q 94. If is ω imaginary root of unity, then the value of $\begin{vmatrix} b & b\omega^2 & a\omega \\ b\omega & c & b\omega^2 \\ c\omega & a\omega & c \end{vmatrix}$ is

[RPET]

(a) $a^3 + b^3 + c^3$

(b) $a^2b - b^2c$

(c) 0

(d) $a^3 + b^3 + c^3 - 3abc$

Q 95. If are $1, \omega, \omega^2$ the cube roots of unity,

then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^n & \omega^{2n} & 1 \\ \omega^{2n} & 1 & \omega^n \end{vmatrix}$ is equal to

[AIEEE]

(a) 0

(b) 1

(c) ω

(d) ω^2

Q 96. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and vectors $(1, a, a^2)$,(l, b, b^2) and $(1, c, c^2)$ are non–coplanar, then the product abc equals

[AIEEE]

(a) 2

(b) -1

(c) 1

(d) 0

Q 97. If $\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$, then x is equal to

[DCE, RPET]

(a) $0, 2a$

(b) c,$2a$

(c) $0,3a$

(d) None of these

Q 98. The value of the determinant

$\begin{vmatrix} 1 & \cos(\alpha-\beta) & \cos\alpha \\ \cos(\alpha-\beta) & 1 & \cos\beta \\ \cos\alpha & \cos\beta & 1 \end{vmatrix}$ is

[UP SEE]

(a) 0

(b) 1

(c) $\alpha^2 - \beta^2$

(d) $\alpha^2 + \beta^2$

Q 99. $\begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b & ab & a+b \end{vmatrix}$ is equal to

[KCET]

(a) $\frac{1}{abc}(ab+bc+ca)$

(b) ab + bc + ca

(c) 0

(d) a + b + c

Q $100.$ If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ p+a & q+b & 2c \\ a & b & r \end{vmatrix} = 0,$

then $\frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}$ is equal to

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 101. The value of the determinant $\begin{vmatrix} 10! & 11! & 12! \\ 11! & 12! & 13! \\ 12! & 13! & 14! \end{vmatrix}$ is

[OJEE]

(a) $2(10!\ 11!)$

(b) $2(10!\ 13!)$

(c) $2(10!\ 11!\ 12!)$

(d) $2(11!\ 12!\ 13!)$

Q 102. If A is a 3×3 non-singular matrix, then

det $(A^{-1}$ adj A) is equal to

[J&KCET]

(a) det A

(b) 1

(c) $(\det A)^2$

(d) $(\det A)^{-1}$

Topic 3 Transpose, Adjoint and Inverse of Matrices

2010

Q 1. If then $A=\begin{bmatrix}1 & 2\\ -4 & -1\end{bmatrix}$, A^{-1} is

[WBJEE]

(a) $\frac{1}{7}\begin{bmatrix}-1 & -2\\ 4 & 1\end{bmatrix}$

(b) $\frac{1}{7}\begin{bmatrix}1 & 2\\ -4 & -1\end{bmatrix}$

(c) $\frac{1}{7}\begin{bmatrix}1 & 2\\ 4 & 1\end{bmatrix}$

(d) Does not exist

Q 2. If $A = \begin{bmatrix} x & -2 \\ 3 & 7 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} \frac{7}{34} & \frac{1}{17} \\ \frac{-3}{34} & \frac{2}{17} \end{bmatrix}$, then the value of x is

[Kerala CEE]

(a) 2

(b) 3

(c) −4

(d) 4

(e) -2

Q 3. If A is a non-singular matrix of order 3, then adj (adj A) is equal to

[BITSAT]

(a) A

(b) A^{-1}

(c) $\frac{1}{|A|}A$

(d) | A | A

(e) $\frac{1}{|A|}A^{-1}$

Q 4. Which one of the following is true always for any two non−singular matrices A and B of same order ?

[VITEEE]

(a) AB = BA

(b) $(AB)^t = A^tB^t$

(c) $(A+B)(A-B) = A^2-B^2$

(d) $(AB)^{-1} = B^{-1}A^{-1}$

(e) $AB = -BA$

Q 5. Let A be 2×2 matrix.

Statement I adj (adj **A**) = **A**

Statement II | adj A | = A

[AIEEE]

(a) Statement I is true, Statement II is true

Statement II is a correct explanation for Statement

I.

(b) Statement I is true, Statement II is true

Statement II is not a correct explanation for

Statement I.

(c) Statement I is true, Statement II is false.

(d) Statement I is false, Statement II is true.

Q 6. If $A=\begin{bmatrix}2 & 1 & 0\\ 0 & 2 & 1\\ 1 & 0 & 2\end{bmatrix}$, then | adj A | is equal to

[KCET]

(a) 0

(b) 9

(c) $\frac{1}{9}$

(d) 81

Q 7. If $\begin{bmatrix}1 & -1 & x\\ 1 & x & 1\\ x & -1 & 1\end{bmatrix}$ has no inverse, then the real value of x is

[EAMCET]

(a) 2

(b) 3

(c) 0

(d) 1

Q 8. If A and B are square matrices of the same order and AB $=3$I, then A^{-1} is equal to

[WBJEE]

(a) $3B$

(b) $\frac{1}{3}B$

(c) $3B^{-1}$

(d) $\frac{1}{3}B^{-1}$

Q 9. If A^T, B^T are transpose matrices of the square matrices A, B respectively, then $(AB)^T$ is equal to

[MPPET]

(a) A^TB^T

(b) AB^T

(c) BA^T

(d) B^TA^T

Q 10. The matrix $\begin{bmatrix} \lambda & -1 & 4 \\ -3 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix}$ is invertible, if

[MPPET]

(a) $\lambda \neq -17$

(b) $\lambda \neq -18$

(c) $\lambda \neq -19$

(d) $\lambda \neq -20$

Q 11. For $0 < \theta < \pi$, if $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ then

[J&K CET]

(a) $A^T = A$

(b) $A^T = -A$

(c) $A^2 = I$

(d) $A^T = A^{-1}$

Q 12. Let A be a square matrix all of whose entries are integers. Then, which one of the following is true ?

[AIEEE]

(a) If det(A) = ± 1, then A^{-1} need not exist

(b) If det(A) = ± 1, then A^{-1} exists but all its entries are not necessarily integers

(c) If det (A) $\neq \pm 1$, then A^{-1} exists and all its entries are non−integers

(d) If det (A) = ± 1, then A^{-1} exists and all its entries are integers

Q 13. If $A = \begin{bmatrix} 2x & 0 \\ x & x \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$, then x equals

[UP SEE]

(a) 2

(b) $-\frac{1}{2}$

(c) 1

(d) $\frac{1}{2}$

Q 14. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then A^{-1} is equal to

[KCET]

(a) $-\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

(b) $\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 4 \\ 1 & 3 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$

Q 15. The inverse of the matrix $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ is

[AMU, EAMCET]

(a) $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 3 & 1 \\ 4 & 3 & 8 \\ 3 & 4 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 3 & 4 \\ 3 & 4 & 3 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

Q 16. The inverse matrix of $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ is

[MHT CET]

(a) $\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}$

(b) $\begin{bmatrix} \frac{1}{2} & -4 & \frac{5}{2} \\ 1 & -6 & 3 \\ 1 & 2 & -1 \end{bmatrix}$

(c) $\frac{1}{2}\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 2 & 3 \end{bmatrix}$

(d) $\frac{1}{2}\begin{bmatrix} 1 & -1 & -1 \\ -8 & 6 & -2 \\ 5 & -3 & 1 \end{bmatrix}$

Q 17. The adjoint of the matrix $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ is

[J&K CET]

(a) $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

(b) $\begin{bmatrix} \sin\theta & \cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

(c) $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

(d) $\begin{bmatrix} -\sin\theta & \cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

Q 18. Let $A = \begin{bmatrix} 1 & 2 \\ -5 & 1 \end{bmatrix}$ and $A^{-1} = xA + yI$, then the value of x and y are

[DCE]

(a) $x = \frac{-1}{11}, y = \frac{2}{11}$

(b) $x = \frac{-1}{11}, y = \frac{-2}{11}$

(c) $x = \frac{1}{11}, y = \frac{2}{11}$

(d) $x = \frac{1}{11}, y = \frac{-2}{11}$

Q 19. If $A=\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{bmatrix}$, $10\ B=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & \alpha\\ 1 & -2 & 3\end{bmatrix}$ and B is the inverse of A, then the value of a is

[KCET, Jamia Millia Islamia]

(a) 2

(b) 0

(c) 5

(d) 4

Q 20. The matrix A satisfying the equation

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$ is

[AMU]

(a) $\begin{bmatrix} 1 & 4 \\ -1 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -4 \\ 1 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix}$

(d) None of these

Q 21. If $A=\begin{bmatrix} 1 & x \\ x^2 & 4y \end{bmatrix}$ and $B=\begin{bmatrix} -3 & 1 \\ 1 & 0 \end{bmatrix}$, adj $A+B=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then values of x and **y** are

[BCECE]

(a) 1,1

(b) ± 1, 1

(c) 1,0

(d) None of these

Q 22. If $A(\theta) = \begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}$ and AB = I, then

$(\sec^2\theta)B$ is equal to

[VITEEE]

(a) $A(\theta)$

(b) $A\left(\frac{\theta}{2}\right)$

(c) $A(-\theta)$

(d) $A\left(-\frac{\theta}{2}\right)$

Q 23. If $A^2 - A + I = O$, then the inverse of A is

[MP PET]

(a) I − A

(b) A – I

(c) A

(d) A + I

Q 24. If $A=\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$, then A^{-1} is equal to

[RPET]

(a) 2A

(b) A

(c) −A

(d) I

Q 25. $A=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$, then A^{-1} is

[J&KCET]

(a) A

(b) –A

(c) adj(A)

(d)−adj(A)

Q 26. If $\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}\begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then

[Guj. CET]

(a) a = 1, b = 1

(b) a = sin 2θ, b = cos 2θ

(c) a = cos 2θ, b = sin 2θ

(d) None of the above

Q 27. The inverse of the matrix $\begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$ is

[BITSAT, Guj. CET]

(a) $\frac{1}{11}\begin{bmatrix} 1 & 2 \\ -3 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 \\ -3 & 5 \end{bmatrix}$

(c) $\frac{1}{13}\begin{bmatrix} -2 & 5 \\ 1 & 3 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$

Q 28. Suppose A is a matrix of order 3 and B = | A | A^{-1}. If |A| = −5, then |B| is equal to

[Kerala CEE]

(a) 1

(b) -5

(c) -1

(d) 25

(e) -125

Q 29. If $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then A is equal to

[Kerala CEE]

(a) $-\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

(e) $\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$

Q 30. Inverse of the matrix $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ is

[AMU]

(a) $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$

(b) $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$

(c) $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$

(d) $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$

Q 31. If $A = \begin{bmatrix} a & b & 0 \\ -b & a & 0 \\ 0 & 0 & 1 \end{bmatrix}$, where $a^2 + b^2 = 1$, then adj

(A) is equal to

[J&KCET]

(a) A^{-1}

(b) A^T

(c) A

(d) –A

(Here, A^T is the transpose of A)

Q 32. The inverse of $\begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}$ is

[J&KCET]

(a) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 1/2 & 0 & 2 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/2 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 0 & 1/2 \\ 0 & 1/2 & 0 \\ 1/2 & 0 & 0 \end{bmatrix}$

Q 33. If $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{bmatrix}$ and B = (adj A), and C = 5A, then $\frac{|\text{adj } B|}{|C|}$ is equal to

[Kerala CEE]

(a) 5

(b) 25

(c) −1

(d) 1

(e) 125

Q 34. adj $\begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & -2 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & a & -2 \\ 1 & 1 & 0 \\ -2 & -2 & b \end{bmatrix}$, then [a b] is equal to

[EAMCET]

(a) [−4 1]

(b) $[-4\ -1]$

(c) $[4\ 1]$

(d) $[4\ -1]$

Q 35. If ω is a root of unity and $A=\begin{bmatrix}1 & 1 & 1\\ 1 & \omega & \omega^2\\ 1 & \omega^2 & \omega\end{bmatrix}$,

then A^{-1} is equal to

[BCECE]

(a) $\begin{bmatrix}1 & \omega & \omega^2\\ \omega^2 & 1 & \omega\\ \omega & \omega^2 & 1\end{bmatrix}$

(b) $\frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \\ 1 & 1 & 1 \end{bmatrix}$

(d) $\frac{1}{2}\begin{bmatrix} 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \\ 1 & 1 & 1 \end{bmatrix}$

Q 36. For non-singular square matrices A, B and C of the same order, $(AB^{-1}C)^{-1}$ is equal to

[MPPET]

(a) $A^{-1}BC^{-1}$

(b) $C^{-1}B^{-1}A^{-1}$

(c) CBA^{-1}

(d) $C^{-1}BA^{-1}$

Q 37. If $f(\theta)=\begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$ then $\{f(\theta)^{-1}\}$ is equal to

[J&K CET]

(a) $f(-\theta)$

(b) $f(\theta)^{-1}$

(c) $f(2\theta)$

(d) None of these

Q 38. Inverse of the matrix $A=\begin{bmatrix}1 & -2\\ 3 & 4\end{bmatrix}$ is

[Guj.CET]

(a) $\frac{1}{10}\begin{bmatrix}1 & -2\\ 3 & 4\end{bmatrix}$

(b) $\frac{1}{10}\begin{bmatrix}4 & 2\\ -3 & 1\end{bmatrix}$

(c) $\begin{bmatrix}4 & 2\\ -3 & 1\end{bmatrix}$

(d) $\frac{1}{10}\begin{bmatrix}4 & -2\\ -3 & 1\end{bmatrix}$

Q 39. If A is a singular matrix, then A adj (A) is a

[DCE]

(a) scalar matrix

(b) zero matrix

(c) identity matrix

(d) orthogonal matrix

Q 40. Let for any matrix M, M^{-1} exist which of the following is not true?

[OJEE]

(a) $|M^{-1}| = |M|^{-1}$

(b) $(M^2)^{-1} = (M^{-1})^2$

(c) $(M^T)^{-1} = (M^{-1})^T$

(d) $(M^{-1})^{-1} = M$

Q 41. If $A = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$, then $(A^{-1})^3$ is equal to

[Jamia Millia Islamia]

(a) $\frac{1}{27}\begin{bmatrix} 1 & -26 \\ 0 & 27 \end{bmatrix}$

(b) $\frac{1}{27}\begin{bmatrix} -1 & 26 \\ 0 & 27 \end{bmatrix}$

(c) $\frac{1}{27}\begin{bmatrix} 1 & -26 \\ 0 & -27 \end{bmatrix}$

(d) $\frac{1}{27}\begin{bmatrix} -1 & -26 \\ 0 & -27 \end{bmatrix}$

Q 42. If $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, then A^{-1} is

[MP PET]

(a) $-A$

(b) A

(c) 1

(d) None of these

Q 43. If $A=\begin{bmatrix}4 & 2\\ 3 & 4\end{bmatrix}$, |adj A| is equal to

[UPSEE]

(a) 6

(b) 16

(c) 10

(d) None of these

Q 44. If $\begin{bmatrix} 3 & 1 \\ 4 & 1 \end{bmatrix} X = \begin{bmatrix} 5 & -1 \\ 2 & 3 \end{bmatrix}$ then X is equal to

[AMU]

(a) $\begin{bmatrix} -3 & 4 \\ 14 & -13 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & -4 \\ -14 & 13 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 4 \\ 14 & 13 \end{bmatrix}$

(d) $\begin{bmatrix} -3 & 4 \\ -14 & 13 \end{bmatrix}$

Q 45. The adjoint matrix of $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ is

[MPPET]

(a) $\begin{bmatrix} 4 & 8 & 3 \\ 2 & 1 & 6 \\ 0 & 2 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$

(c) $\begin{bmatrix} 11 & 9 & 3 \\ 1 & 2 & 8 \\ 6 & 9 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & -2 & 1 \\ -1 & 3 & 3 \\ -2 & 3 & -3 \end{bmatrix}$

Q 46. If k is a scalar and I is a unit matrix of order 3, then adj (kI) is equal to

[RPET]

(a) k^3I

(b) k^2I

(c) $-k^3I$

(d) $-k^2I$

Topic 4 System of Linear Equations, Rank of Matrices

2010

Q 1. The number of 3×3 matrices A whose entries are either 0 or 1 and for which the system $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has exactly two distinct solutions, is

[IIT JEE]

(a) 0

(b) $2^9 - 1$

(c) 168

(d) 2

Q 2. Consider the system of linear equations

$x_1 + 2x_2 + x_3 = 3$ $2x_1 + 3x_2 + x_3 = 3$ $3x_1 + 5x_2 + 2x_3$

$= 1$

The system has

[AIEEE]

(a) Infinite number of solutions

(b) Exactly 3 solutions

(c) A unique solution

(d) No solution

Q 3. The system of equations, $x + y + z = 6$ x $+ 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$ has no solution, if .

[DCE]

(a) $\lambda = 3, \mu = 10$

(b) $\lambda \neq 3, \mu = 10$

(c) $\lambda \neq 3, \mu \neq 10$

(d) $\lambda = 3, \mu \neq 10$

Q 4. Consider the system of equations in x, y, z as

$x \sin 3\theta - y + z = 0$

$x \cos 2\theta + 4y + 3z = 0$

and $2x + 7y + 7z = 0$

If this system has a non-trivial solution, then for integer n, values of θ are given by

(a) $\pi\left(n+\frac{(-1)^n}{3}\right)$

(b) $\pi\left(n+\frac{(-1)^n}{4}\right)$

(c) $\pi\left(n+\frac{(-1)^n}{6}\right)$

(d) $\frac{n\pi}{2}$

Q 5. If the three linear equations x + 4ay + az = 0, x + 3by + bz = 0, x+2cy + cz = 0 have a non−

trivial solution, where a $\neq 0$, b $\neq 0$, c $\neq 0$, then ab + bc is equal to

[Kerala CEE]

(a) 2ac

(b) – ac

(c) ac

(d) -2ac

(e) a

Q 6. If the system of equations ax + ay − z = 0, bx− y + bz = 0 and −x + cy + cz = 0 has a non−trivial solution, then the value of $\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}$ is

[AMU]

(a) 0

(b) 1

(c) 2

(d) 3

Q 7. If B is an invertible matrix and A is a

matrix, then

[AMU]

(a) rank (BA) = rank (A)

(b) rank (BA) $\geq$ rank (B)

(c) rank (BA) > rank (A)

(d) rank (BA) > rank (B)

Q 8. The real value of & for which the system of equations $2kx - 2y + 3z = 0$, $x + ky + 2z = 0$, $2x + kz = 0$, has non−trivial solution is

[OJEE]

(a) 2

(b) -2

(c) 3

(d) -3

Q 9. If the system of homogeneous equations

$2x - y + z - 0$, $x - 2y + z = 0$, $\lambda x - y + 2z = 0$ has

infinitely many solutions, then

[J&K CET]

(a) $\lambda = 5$

(b) $\lambda = -5$

(c) $\lambda \neq \pm 5$

(d) None of these

Q 10. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that x = cy + bz, y = az + cx, and z = bx + ay have non−zero solution Then, $a^2 + b^2 + c^2 + 2abc$ is equal to

[AIEEE]

(a) 1

(b) 2

(c) −1

(d) 0

Q 11. The system of equations $x + y + z = 8$, $x - y + 2z - 6$, $3x + 5y - 7z = 14$ has **[OJEE]**

(a) no solution

(b) unique solution

(c) infinitely many solution

(d) None of the above

Q 12. The system of equations $x + y + z = 0$, $2x + 3y + z = 0$ and $x + 2y = 0$ has

[VITEEE]

(a) a unique solution; x =0, y = 0, z = 0

(b) infinite solutions

(c) no solution

(d) finite number of non−zero solutions

Q 13. The solution of (x, y, z) the equation

$$\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$ is(x, y, z)

[MHT CET]

(a) $(1,1,1)$

(b) $(0,-1,2)$

(c) $(-1,2,2)$

(d) $(-1,0,2)$

Q 14. If $\begin{bmatrix} 2 & -1 & 3 \\ 1 & 3 & -1 \\ 3 & 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \\ 10 \end{bmatrix}$, then $\begin{bmatrix} x \\ y \\ y \end{bmatrix}$ is equal to

(a) $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

(d) $\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$

[MP PET]

Q 15. The number of solutions of the system of equations

$$x - y + z = 2$$

$$2x + y - z = 5$$

$4x + y + z = 10$ is

[J&KCET]

(a) ∞

(b) 1

(c) 2

(d) 0

Q 16. Let a, b, c are positive real numbers. The following system of equations $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$,

$\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{z^2}{c^2}=1,-\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$, in x, y and z has

[Guj. CET]

(a) infinite solutions

(b) unique solution

(c) no solution

(d) finite number of solutions

Q 17. The number of non−trivial solutions of the system $x - y + z = 0$, $x + 2y - z = 0$, $2x + y + 3z = 0$ is

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 18. The system of equations $2x + y - 5 = 0$, $x - 2y + 1 = 0$, $2x - 14y - a = 0$, is consistent.

Then, a is equal to

[OJEE]

(a) 1

(b) 2

(c) 5

(d) None of these

Q 19. The simultaneous equations $Kx+2y-z=1$, $(K-1)y-2z=2$ and $(K+2)z=3$ have only one solution when

[VITEEE]

(a) $K=-2$

(b) $K=-1$

(c) $K=0$

(d) $K=1$

Q 20. If the rank of the matrix $\begin{bmatrix} -1 & 2 & 5 \\ 2 & -4 & a-4 \\ 1 & -2 & a+1 \end{bmatrix}$ is 1,

then the value of a is

[VITEEE]

(a) -1

(b) 2

(c) -6

(d) 4

Q 21. For the equations $x + 2y + 3z = 1$, $2x + y + 3z = 2$ and $5x + 5y + 9z = 4$

[MHTCET]

(a) there is only one solution

(b) there exists infinitely many solutions

(c) there is no solution

(d) None of the above

Q 22. The system of equations

[J&KCET]

$3x - y + 4z = 3$ $x + 2y - 3z = -2$ $6x + 5y + \lambda z = -$

3 has at least one solution, if

(a) $\lambda = -5$

(b) $\lambda = 5$

(c) $\lambda = 3$

(d) $\lambda = -13$

Q 23. If $x\begin{bmatrix}-3\\4\end{bmatrix} + y\begin{bmatrix}4\\3\end{bmatrix} = \begin{bmatrix}10\\-5\end{bmatrix}$, then

[J&KCET]

(a) $x = -2, y = 1$

(b) $x = -9, y = 10$

(c) $x = 22, y = 1$

(d) $x = 2, y = -1$

Q 24. The values of λ and μ for which the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$ have infinite number of solutions, are

[Guj. CET]

(a) $\lambda = 3, \mu = 10$

(b) $\lambda = 3, \mu = \neq 10$

(c) $\lambda \neq 3, \mu = 10$

(d) $\lambda \neq 3, \mu = \neq 10$

Q 25. If $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & -2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}$, then $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is equal to

[UP SEE]

(a) $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$

(c) $\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

Q 26. The values of a for which the system of equations $x + y + z = 0$, $x + ay + az = 0$, $x - ay + z = 0$, possesses non−zero solutions, are given by

[BCECE]

(a) $1, 2$

(b) $1, -1$

(c) $1, 0$

(d) None of these

Q 27. The value of λ such that $x + 3y + \lambda z = 0$, $2x + 4y - z = 0$, $x + 5y - 2z = 0$ has a non-trivial solution is

[J&KCET]

(a) -1

(b) 0

(c) 1

(d) 2

Q 28. The system of equations

$ax + y + z = \alpha - 1$

$x + \alpha y + z = \alpha - 1$

$x + y + \alpha z = \alpha - 1$

has no solution, if a is

[AIEEE]

(a) 1

(b) not -2

(c) either -2 or 1

(d) -2

Q 29. Let the homogeneous system of linear equations $px + y + z = 0$, $x + qy + z = 0$, and $x + y + r z = 0$, where $p, q, r \neq 1$, have a non-zero solution, then the value of $\frac{1}{1-p}+\frac{1}{1-q}+\frac{1}{1-r}$ is

[BITSAT, OJEE]

(a) -1

(b) 0

(c) 2

(d) 1

Q 30. The rank of the matrix $\begin{bmatrix} 4 & 1 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 6 & 0 & 2 & 0 \end{bmatrix}$ is

[AMU]

(a) 4

(b) 3

(c) 2

(d) 1

Q 31. The values of a for which the system of equations $ax + y + z = 0$, $x - ay + z = 0$, $x + y + z = 0$ possesses non−zero solution, are given by

[AMU]

(a) $1,2$

(b) $1,-1$

(c) 0

(d) None of these

Q 32. If a system of the equations

$(\alpha + 1)^3 x + (\alpha + 2)^3 y - (\alpha + 3)^3 = 0,$

$(\alpha + 1)x + (\alpha + 2)y - (\alpha + 3) = 0$, and $x + y - 1 = 0$ is consistent. What is the value of α ?

[OJEE]

(a) 1

(b) 0

(c) -3

(d) -2

Q 33. For what value of λ, the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = 0$ is consistent?

[MPPET]

(a) 1

(b) 2

(c) -1

(d) 3

Q 34. Given $2x - y + 2z = 2$, $x - 2y + 2z = -4$, $x + y + \lambda z = 4$ then the value of λ such that the given system of equation has no solution, is

[IIT JEE]

(a) 3

(b) 1

(c) 0

(d) -3

Q 35. For what value of k the following system of linear equations will have infinite solutions

$x - y + z = 3$, $2x + y - z = 2$ and $-3x - 2ky + 6z = 3$

[DCE]

(a) $k \neq 2$

(b) $k = 0$

(c) $k = 3$

(d) $k \in [2, 3]$

Q 36. If $A = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 8 & 10 \\ -6 & -12 & -15 \end{bmatrix}$, the rank of A is equal to

[UP SEE]

(a) 0

(b) 1

(c) 2

(d) 3

Q 37. The rank of $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$ is

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 38. $x_1 + 2x_2 + 3x_3 = 2x_1 + 3x_2 + x_3 = 3x_1 + x_2 + 2x_3 = 0$. This system of equation has

[OJEE]

(a) infinite solution

(b) no solution

(c) unique solution

(d) None of the above

Q 39. If a_1, a_2, a_3, a_4, a_5, a_6 are in AP with common difference $d \neq 0$, then the system of equations $a_1x + a_2y = a_3, a_4x + a_5y = a_6$ has

[J&KCET]

(a) infinite number of .solutions

(b) unique solution

(c) no solution

(d) Cannot say any thing

Q 40. The rank of the matrix $\begin{bmatrix} 4 & 2 & (1-x) \\ 5 & k & 1 \\ 6 & 3 & (1+x) \end{bmatrix}$ is 2, then

[J&K CET]

(a) $k = \frac{5}{2}, x = \frac{1}{5}$

(b) $k = \frac{5}{2}, x \neq \frac{1}{5}$

(c) $k = \frac{1}{5}, x = \frac{5}{2}$

(d) None of these

Q 41. If the system of equations x + ay = 0, az + y = 0 and ax + z = 0 has infinite solutions, then the value of a is

[IIT JEE]

(a) -1

(b) 1

(c) 0

(d) No real values

Q 42. If the system of linear equations x + 2ay + az = 0, x + $3$by + bz = 0 and x + 4cy + cz = 0 has a non−zero solution, then a, b, c

[AIEEE]

(a) are in AP

(b) are in GP

(c) are in HP

(d) satisfy $a+2b+3c=0$

Q 43. $x+ky-z=0$, $3x-ky-z=0$ and $x-3y+z=0$ has non-zero solution for k is equal to

[AMU]

(a) -1

(b) 0

(c) 1

(d) 2

Q 44. If $\begin{bmatrix} x+y+z \\ x+y \\ y+z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$, then the value of (x, y, z) is

[AMU]

(a) $(4,3,2)$

(b) $(3,2,4)$

(c) $(2,3,4)$

(d) None of the above

Q 45. The rank of the matrix $\begin{bmatrix} -1 & 2 & 5 \\ 2 & -4 & a-4 \\ 1 & -2 & a+1 \end{bmatrix}$ is

[AMU]

(a) 1 if a = 6

(b) 2 if a = 1

(c) 3 if a =2

(d) 4 if a = −6

Q 46. The number of solutions of the system of equations

$2x + y - z = 1$, $x - 3y + 2z = 1$, $x + 4y - 3z = 5$ is

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 47. If the points (x_1, y_1),(x_2, y_2) and (x_3, y_3) are collinear, then the rank of the matrix $\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$ will always be less than

[OJEE]

(a) 2

(b) 3

(c) 1

(d) None of these

Q 48. The system of equations $x + 3y + 2z = 0$, $3x + y + z = 0$ and $2x - 2y - z = 0$ **[J&KCET]**

(a) possesses a trivial solution only

(b) possesses a non−zero unique solution

(c) does not have a common non−zero solution

(d) has infinitely many solutions

Limits, Continuity and Differentiability

Useful Results & Formulae

1. Indeterminate forms are

$$\frac{0}{0}, \infty - \infty, \frac{\infty}{\infty}, 0 \times \infty, 1^{\infty}, 0^{0}, \infty^{0} .$$

2. (i) $\lim_{x \to 0} \frac{\sin x}{x} = \lim_{x \to 0} \cos x = \lim_{x \to 0} \frac{\tan x}{x} = 1$

(ii) $\lim_{x \to \infty} \frac{\sin x}{x} = \lim_{x \to \infty} \frac{\cos x}{x} = 0$

(iii) $\lim_{x \to a} \frac{\sin(x-a)}{x-a} = \lim_{x \to a} \frac{\tan(x-a)}{x-a} = 1$

(iv) $\lim_{x\to 0}\frac{e^{x\lambda}-1}{x}=\lambda(\lambda\neq 0)$

(v) $\lim_{x\to 0}\frac{a^{x}-1}{x}=\log_{e} a$

(vi) $\lim_{x\to 0}\frac{x^{n}-a^{n}}{x-a}=na^{n-1}$

(vii) $\lim_{x\to 0}\frac{\sin^{-1} x}{x}=\lim_{x\to 0}\frac{\tan^{-1} x}{x}=1$

(viii) If $\lim_{x\to a} f(x)=1$ and $\lim_{x\to a} g(x)=\infty$, then

$$\lim_{x\to a}[f(x)]^{g(x)}=e^{\lim_{x\to a}(f(x)-1)g(x)}$$

(ix) $\lim_{x\to 0}(1+\lambda x)^{1/x}=e^{\lambda}=\lim_{x\to\infty}\left(1+\frac{\lambda}{x}\right)^{x}$

(x) $\lim_{x\to 0}\frac{\log(1+x)}{x}=1$

(xi) $\lim_{n\to\infty}(1+x^n)^{\frac{1}{n}} = 0$

3. (i) $e^x = 1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+.....$

(ii) $e^{-x} = 1-\frac{x}{1!}+\frac{x^2}{2!}-\frac{x^3}{3!}+.....$

(iii) $a^x = 1+x\frac{\log_e a}{1!}+\frac{x^2}{2!}(\log_e a)^2+.....$

(iv) $\sin x = x-\frac{x^3}{3!}+\frac{x^5}{5!}-.....$

(v) $\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-.....$

(vi) $\tan x = x+\frac{x^3}{3}+\frac{2x^5}{15}+.....$

(vii) $\sin^{-1} x = x+\frac{1^2}{3!}x^3+\frac{1^2.3^2}{5!}x^5+.....$

(viii) $\cos^{-1} x = x - \frac{x^3}{6} + ...$

(ix) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - .....$

(x) $\frac{x^n - a^n}{x - a} = x^{n-1} + x^{n-2}a + x^{n-3}a^2 + ... + a^{n-1}$

4. If at x = a, left hand limit = right hand limit = value of function, then function is continuous at x = a.

5. If at $x = x_0$, if $Lf'(x_0) = Rf'(x_0)$ ie,

$\lim_{h\to 0}\frac{f(x_0-h)-f(x_0)}{(x_0-h)-(x_0)}=\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{(x_0+h)-(x_0)}$, then function is

differentiable at $x = x_0$

6. If f, g are continuous (differentiable) at point a,

then f ± g, fg are continuous (differentiable) at a

and if $g(a) \neq 0$, then $\frac{f}{g}$ is also continous

(differentiable) at a.

7. If a function is differentiable at a point, then it is

continuous also at that point but converse need not

be true.

8. If there is any break in a graph, then function is neither continuous nor differentiable at that point.

9. If there is any sharp corner in a graph, then it is not differentiable at that corner.

10. Newton–Leibnitz's formula

If $I(x) = \int_{\phi(x)}^{\psi(x)} f(t)dt,$ then

$$\frac{d}{dx}(I(x)) = f(\psi(x)).\left\{\frac{d}{dx}\psi(x)\right\} - f(\phi(x))\left\{\frac{d}{dx}\phi(x)\right\}$$

11. Sandwich theorem,

If $h(x) \leq f(x) \leq g(x)$

and $\lim_{x \to a} h(x) = L$ and $\lim_{x \to a} g(x) = L$,

then $\lim_{x \to a} f(x) = L$

Topic – 1 Limits of Algebraic, Exponential and Logarithmic Functions

Q 1. The value of $\lim_{x \to 0} \int_0^x \frac{t \log(1+t)}{t^4 + 4} dt$ is

[IIT JEE]

(a) 0

(b) $\frac{1}{12}$

(c) $\frac{1}{24}$

(d) $\frac{1}{64}$

Q 2. Let $f : R \to R$ be a positive increasing function with $\lim_{x\to\infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x\to\infty} \frac{f(2x)}{f(x)} = 1$ is equal to

[AIEEE]

(a) 1

(b) $\frac{2}{3}$

(c) $\frac{3}{2}$

(d) 3

Q 3. If $f(5) = 7$ and $f'(5) = 7$, then $\lim_{x \to 5} \frac{xf(5) - 5f(x)}{x - 5}$ is given by

[WB JEE]

(a) 35

(b) -35

(c) 28

(d) -28

Q 4. The value of $\lim_{x\to 0}\left(\frac{1+5x^2}{1+3x^2}\right)^{1/x^2}$ is

[WB JEE]

(a) e^2

(b) e

(c) $\frac{1}{e}$

(d) $\frac{1}{e^2}$

Q 5. $\lim_{x\to\infty}\left(\frac{x^3}{3x^2-4}-\frac{x^2}{3x+2}\right)$ is equal to

[Kerala CEE]

(a) $-\frac{1}{4}$

(b) $-\frac{1}{2}$

(c) 0

(d) $\frac{2}{9}$

(e) $-\frac{6}{5}$

Q 6. $\lim_{x\to 0}\left(\frac{x}{\sqrt{1+x}-\sqrt{1-x}}\right)$ is equal to

[Kerala CEE]

(a) 0

(b) 1

(c) 2

(d) -1

(e) -2

Q 7. $\lim_{x \to 2} = \frac{\sqrt{1+\sqrt{2+x}} - \sqrt{3}}{x-2}$ is equal to

[KCET]

(a) $\frac{1}{8\sqrt{3}}$

(b) $\frac{1}{\sqrt{3}}$

(c) $8\sqrt{3}$

(d) $\sqrt{3}$

Q 8. $\lim_{n\to\infty} \frac{3.2^{n+1} - 4.5^{n+1}}{5.2^n + 7.5^n}$ is equal to

[KCET]

(a) $\frac{3}{5}$

(b) $-\frac{4}{7}$

(c) $-\frac{20}{7}$

(d) 0

Q 9. Let f : R $\rightarrow$ R be a differentiable function such that f(3) = 3, f'(3) = $\frac{1}{2}$. Then, the value of $\lim_{x\to 3}\frac{\int_3^{f(x)} 2t^3}{x-3}dt$ is

[AMU]

(a) 25

(b) 26

(c) 27

(d) None of these

Q 10. Let $f(x)=\frac{\sqrt{x+3}}{x+1}$, then the value of $\lim_{x\to -3-0} f(x)$ is

[WB JEE]

(a) 0

(b) does not exist

(c) $\frac{1}{2}$

(d) $-\frac{1}{2}$

Q 11. $\lim_{x\to\infty} \frac{\sqrt{x^2+1}-\sqrt[3]{x^3+1}}{\sqrt[4]{x^4+1}-\sqrt[5]{x^4+1}}$ equals

[Jamia Millia Islamia]

(a) 1

(b) 0

(c) -1

(d) None of these

Q 12. Given $f(x) = \frac{ax+b}{x+1}, \lim_{x\to\infty} f(x) = 1$ and $\lim_{x\to 0} f(x) = 2$,

then $f(-2)$ is

[MHT CET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 13. $\lim_{x\to 1} (\log ex)^{1/\log x}$ is equal to

[MHT CET]

(a) e^{-1}

(b) e

(c) e^{2}

(d) 0

Q 14. If $\lim_{x\to 0}\frac{\log(x+a)-\log a}{x}+k\lim_{x\to e}\frac{\log x-1}{x-e}=1$, then the value of k is .

[MPPET]

(a) $1-\frac{1}{a}$

(b) $e(1 - a)$

(c) $e\left(1-\frac{1}{a}\right)$

(d) $e(1 + a)$

Q 15. $\lim_{x\to 0}\frac{(1+x)^8-1}{(1+x)^2-1}$ is equal to

[MPPET]

(a) 8

(b) 6

(c) 4

(d) 2

Q 16. If f(l) = 2 and f'(l) = 1, then value of

$\lim_{x \to 1} \frac{2x - f(x)}{x - 1}$ is

[RPET]

(a) -1

(b) 0

(c) 1

(d) 2

Q 17. If $0 < a < b$, then $\lim_{n\to\infty} \frac{a^n + b^n}{a^n - b^n}$

[J&K CET]

(a) equals 0

(b) equals -1

(c) equals 1

(d) does not exist

Q 18. $\lim_{n\to\infty} \frac{1^2 + 2^2 + 3^2 + ... + n^2}{n^3}$ is equal to

[Guj.CET]

(a) $\frac{1}{2}$

(b) $\frac{2}{3}$

(c) $\frac{1}{3}$

(d) $\frac{1}{6}$

Q 19. $\lim_{n\to\infty} r^n = 0$, then r is equal to

[Guj. CET]

(a) $\frac{4}{5}$

(b) $\frac{5}{4}$

(c) 2

(d) 1

Q 20. $\lim_{x \to -3} \frac{3x^2 + ax + a - 7}{x^2 + 2x - 3}$ exists, then a is equal to

[Guj.CET]

(a) 10

(b) 15

(c) −15

(d) -10

Q 21. If f be a function such that f(9) = 9 and f'

(9) = 3, then $\lim_{x \to 9} \frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3}$ is equal to

[UP SEE]

(a) 9

(b) 3

(c) 1

(d) None of these

Q 22. $\lim_{x \to -\infty} \frac{2x-1}{\sqrt{x^2+2x+1}}$ is equal to

[Kerala CEE]

(a) 2

(b) -2

(c) 1

(d) -1

(e) 0

Q 23. If $f(1) = 1$, $f'(1) = 2$, then $\lim_{x \to 1} \frac{\sqrt{f(x)} - 1}{\sqrt{x} - 1}$ is

[Kerala CEE]

(a) 2

(b) 4

(c) 1

(d) ½

(e) 3

Q 24. If $f : R \to R$ is defined by $f(x) = [x - 3] + |x-4|$ for $x \in R$, then $\lim_{x \to 3^-} f(x)$ is equal to

[EAMCET]

(a) -2

(b) -1

(c) 0

(d) 1

Q 25. $\lim_{x \to 0}(-1)^{[x]}$, where [•] denotes the greatest integer function is equal to

[OJEE]

(a) 0

(b) 1

(c) -1

(d) does not exist

Q 26. The value of $\lim_{x \to 2} \frac{5}{\sqrt{2} - \sqrt{x}}$ is

[WB JEE]

(a) $10\sqrt{2}$

(b) $+\infty$

(c) $-\infty$

(d) does not exist

Q 27. If $0 < p < q$, then $\lim_{n\to\infty}(q^n + p^n)^{1/n}$ is equal to

[BCECE]

(a) e

(b) p

(c) q

(d) 0

Q 28. If g(x) is a polynomial satisfying g(x)g(y)

= g(x) + g(y) + g(xy) − 2 for all real x and y and

g(2) = 5,then $\lim_{x\to 3}$ g(x)is

[VITEEE]

(a) 9

(b) 10

(c) 25

(d) 20

Q 29. The value of $\lim_{x \to 0} \frac{5^x - 5^{-x}}{2x}$ is

[Jamia Millia Islamia]

(a) log 5

(b) 0

(c) 1

(d) 2 log 5

Q 30. $\lim_{x\to\infty}\left(\frac{x+2}{x+1}\right)^{x+3}$ is

[MPPET]

(a) 1

(b) e

(c) e^2

(d) e^3

Q 31. $\lim_{x\to 1}\frac{x^m-1}{x^n-1}$ is equal to

[J&KCET]

(a) $\frac{n}{m}$

(b) $\frac{m}{n}$

(c) $\frac{2m}{n}$

(d) $\frac{2n}{m}$

Q 32. $\lim_{x \to 0} \frac{e^{5x} - e^{4x}}{x}$ is equal to

[J&KCET]

(a) 1

(b) 2

(c) 4

(d) 5

Q 33. If $\lim_{x\to 1}\frac{ax^2+bx+c}{(x-1)^2}=2$, then (a, b, c) is

[Guj. CET]

(a) $(2,-4,2)$

(b) $(2,4,2)$

(c) $(2,4,-2)$

(d) $(2,-4,-2)$

Q 34. $\lim_{x\to\frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2 - \frac{\pi^2}{16}}$ equals

[IIT JEE]

(a) $\frac{8}{\pi} f(2)$

(b) $\frac{2}{\pi} f(2)$

(c) $\frac{2}{\pi} f\left(\frac{1}{2}\right)$

(d) $4f(2)$

Q 35. If f(x) is differentiable function and $f'(0)$ = a, then $\lim_{x\to 0}\frac{2f(x)-3f(2x)+f(4x)}{x^2}$ is equal to

[DCE]

(a) 3a

(b) 2a

(c) 5a

(d) 4a

Q 36. If $\lim_{x\to\infty}\left[\frac{x^3+1}{x^2+1}-(ax+b)\right]=2$, then

[BITSAT]

(a) a = 1 and b = 1

(b) a = 1 and b = −1

(c) a= 1 and b = −2

(d) a = 1 and b = 2

Q 37. The value of $\lim_{x\to\infty}\left(\frac{x^2+bx+4}{x^2+ax+5}\right)$ is

[UP SEE]

(a) $\frac{b}{a}$

(b) 0

(c) 1

(d) $\frac{4}{5}$

Q 38. $\lim_{x\to\infty} \frac{(2x-3)(3x-4)}{(4x-5)(5x-6)}$ is equal to

[UP SEE]

(a) $\frac{1}{10}$

(b) 0

(c) $\frac{1}{5}$

(d) $\frac{3}{10}$

Q 39. $\lim_{x \to 1} \frac{e^{-x} - e^{-1}}{x - 1}$ is equal to

[OJEE]

(a) $\frac{1}{e}$

(b) $-\frac{1}{e}$

(c) 1

(d) None of these

Q 40. The value of $\lim_{x \to \infty} \sqrt{a^2x^2 + ax + 1} - \sqrt{a^2x^2 + 1}$ is

[WB JEE]

(a) $\frac{1}{2}$

(b) 1

(c) 2

(d) None of these

Q 41. $\lim_{x\to\infty}\frac{\int_0^{2x} xe^{x^2}dx}{e^{4x^2}}$ equals

[WBJEE]

(a) 0

(b) ∞

(c) 2

(d) $\frac{1}{2}$

Q 42. $\lim_{x\to 0}\frac{a^x - b^x}{e^x - 1}$ –is equal to

[WBJEE]

(a) $\log_e\left(\frac{a}{b}\right)$

(b) $\log_e\left(\frac{b}{a}\right)$

(c) $\log_e (ab)$

(d) $\log_e (a + b)$

Q 43. $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n} e^{r/n}$ is

[Jamia Millia Islamia]

(a) e

(b) $e - 1$

(c) $1 - e$

(d) $e + 1$

Q 44. If $\lim_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x}=e^2$, then the values of a and b are

[Jamia Millia Islamia]

(a) $a \in R, b \in R$

(b) $a = 1, b \in R$

(c) $a \in R, b = 2$

(d) $a = 1, b = 2$

Q 45. The value of $\lim_{x\to\infty}\left(\frac{x^2-2x+1}{x^2-4x+2}\right)^x$ is

[MHTCET]

(a) e^2

(b) e^{-2}

(c) e^6

(d) None of these

Q 46. Let f : R $\rightarrow$ R be a differentiable function having f(2) = 6, f'(2)= $\left(\frac{1}{48}\right)$. Then, $\lim_{x \to 2} \frac{\int_6^{f(x)} 4t^3 dt}{x-2}$ is equals

[MP PET]

(a) 18

(b) 12

(c) 36

(d) 24

Q 47. $\lim_{n\to\infty} \frac{2^{-n}(n^2+5n+6)}{(n+4)(n+5)}$ is equal to

[J&KCET]

(a) 0

(b) 1

(c) ∞

(d) $-\infty$

Q 48. $\lim_{x\to\infty}\left(1-\frac{4}{x-1}\right)^{3x-1}$ is equal to

[AMU]

(a) e^{12}

(b) e^{-12}

(c) e^4

(d) e^3

Q 49. $\lim_{x\to\infty}[\sqrt{x^2+2x-1}-x]$ is equal to

[EAMCET]

(a) ∞

(b) $\frac{1}{2}$

(c) 4

(d) 1

Q 50. $\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$ is equal to

[OJEE]

(a) e^4

(b) e^2

(c) e^3

(d) e

Q 51. Let $f(2) = 4$ and $f'(2) = 4$. Then,

$\lim_{x\to 2}\frac{xf(2)-2f(x)}{x-2}$ is given by

[OJEE]

(a) 2

(b) −2

(c) −4

(d) 3

Q 52. The value of $\lim_{x\to 0}\frac{e^x+\log(1+x)-(1-x)^{-2}}{x^2}$ is equal to

[WB JEE, Jamia Millia Islamia]

(a) 0

(b) -3

(c) -1

(d) infinity

Q 53. $\lim_{x \to 0} \frac{a^x + a^{-x} - 2}{x^2}$ is equal to

[MHTCET]

(a) $(\log a)^2$

(b) log a

(c) 0

(d) None of these

Q 54. Let $f(x)=\begin{cases}1, \text{ when } x \text{ is rational}\\ 0, \text{ when } x \text{ is irrational}\end{cases}$, then $\lim_{x\to 0} f(x)$ is

[MPPET]

(a) 0

(b) 1

(c) $\frac{1}{2}$

(d) None of these

Q 55. It is given that f'(a) exists, then

$\lim_{x \to a} \frac{xf(a) - af(x)}{x - a}$ is equal to

[RPET]

(a) f(a) − af'(a)

(b) f'(a)

(c) −f'(a)

(d) f(a) + af'(a)

Q 56. $\lim_{x \to 1} \frac{2x^2 + x - 3}{3x^3 - 3x^2 + 2x - 2}$ is equal to

[J&K CET]

(a) 1

(b) 2

(c) -1

(d) -2

Q 57. The value of $\lim_{x \to 0} \frac{e^{ax} - e^{bx}}{x}$ is equal to

[DCE]

(a) a + b

(b) a – b

(c) e^{ab}

(d) 1

Q 58. The value of the constant α and β such that $\lim_{x\to\infty}\left(\frac{x^2+1}{x+1}-\alpha x-\beta\right)=0$ are respectively

[BITSAT, OJEE]

(a) (1,1)

(b) $(-1,1)$

(c) $(1,-1)$

(d) $(0,1)$

Q 59. If $f : R \to R$ is defined by

$$f(x)=\begin{cases}\dfrac{x-2}{x^2-3x+2}, & \text{if } x \in R-\{1,2\} \\ 2, & \text{if } x=1 \\ 1, & \text{if } x=2\end{cases}$$

then $\lim\limits_{x\to 2}\dfrac{f(x)-f(2)}{x-2}$ is equal to

[EAMCET]

(a) 0

(b) -1

(c) 1

(d) $-\frac{1}{2}$

Q 60. If $\lim_{n\to\infty}\frac{1-(10)^n}{1+(10)^{n+1}}=\frac{-\alpha}{10}$, then the value of α is

[OJEE]

(a) 0

(b) -1

(c) 1

(d) 2

Q 61. For $x \in R$ $\lim_{x\to\infty}\left(\frac{x-3}{x+2}\right)^x$ is equal to

[Jamia Millia Islamia]

(a) e

(b) e^{-1}

(c) e^{-5}

(d) e^{5}

Q 62. $\lim_{x \to 0}\left[\frac{\sqrt{a+x}-\sqrt{a-x}}{4x}\right]$ is equal to

[MHT CET]

(a) $\frac{1}{2\sqrt{a}}$

(b) $\frac{1}{4\sqrt{a}}$

(c) $\frac{1}{3\sqrt{a}}$

(d) $\frac{2}{\sqrt{a}}$

Q 63. If $\lim_{x \to a}\frac{a^x - x^a}{x^x - a^a} = -1$, then a equals to

[MPPET]

(a) 1

(b) 0

(c) e

(d) (1/e)

Q 64. $\lim_{x \to a} \frac{\log(x-a)}{\log(e^x - e^a)}$ is equal to

[RPET]

(a) 0

(b) 1

(c) a

(d) does not exist

Q 65. If f(x) is the least integer not less than x and g(x) is the greatest integer not greater than x, then

$\lim_{x \to e+\pi}$ {f(x) + g(x)} is equal to

[J&K CET]

(a) 9

(b) 13

(c) 1

(d) None of these

Q 66. The value of $\lim_{n\to\infty} \frac{x^n}{x^n+1}$, where $x < -1$ is

[J&KCET]

(a) $\frac{1}{2}$

(b) $-\frac{1}{2}$

(c) 1

(d) None of these

Q 67. If f(a) = 2, f'(a) = 1, g(a) = 3, g'(a) = −1,

then $\lim_{x \to a} \frac{f(a)g(x) - f(x)g(a)}{x - a}$ is equal to

[J&K CET]

(a) 6

(b) 1

(c) −1

(d) −5

Q 68. $\lim_{x \to 0}(1 - ax)^{1/x}$ is equal to

[J&KCET]

(a) e^{-a}

(b) e

(c) e^{a}

(d) 1

Q 69. If f(x) is differentiable and strictly increasing function, then the value of $\lim_{x\to 0}\frac{f(x^2)-f(x)}{f(x)-f(0)}$ is

[IITJEE]

(a) 1

(b) 0

(c) -1

(d) 2

Q 70. Let f : R $\rightarrow$ R be a differentiable function and f(1) = 4.

Then, the value of $\lim_{x \to 1} \frac{\int_4^{f(x)} 2t}{x-1} dt$, if f'$(1)$ = 2 is

[Kerala CEE]

(a) 16

(b) 8

(c) 4

(d) 2

(e) None of these

Q 71. $\lim_{x \to 0}\left[\frac{2^x - 1}{\sqrt{1+x} - 1}\right]$ is equal to

[Kerala CEE]

(a) $\log_e 2$

(b) $\log_e \sqrt{2}$

(c) $\log_e 4$

(d) 2

(e) $\frac{1}{2}$

Q 72. If $f(x)=\begin{cases} x^2-3, & 2<x<3 \\ 2x+5, & 3<x<4 \end{cases}$, the equation whose roots are $\lim\limits_{x\to 3^-} f(x)$ and $\lim\limits_{x\to 3^+} f(x)$, is

[OJEE]

(a) $x^2 - 7x + 3 = 0$

(b) $x^2 - 20x + 66 = 0$

(c) $x^2 - 17x + 66 = 0$

(d) $x^2 - 18x + 60 = 0$

Q 73. The value of $\lim_{x \to \infty} \frac{\sqrt{1+x^4} - (1+x^2)}{x^2}$ is equal to

[BCECE]

(a) 0

(b) −1

(c) 2

(d) None of these

Q 74. $\lim_{x \to 0} \frac{e^{1/x}}{e^{1/x+1}}$ is equal to

[BCECE]

(a) 0

(b) 1

(c) does not exist

(d) None of these

Q 75. $\lim_{x \to 0} \left[\frac{3^x + 3^{-x} - 2}{x^2} \right]$ is equal to

[MHT CET]

(a) $2 \log 3$

(b) $-(\log 3)^2$

(c) $-2 \log 3$

(d) $(\log 3)^2$

Q 76. $\lim_{x \to \infty} \left(\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x} \right)$ is equal to

[J&K CET]

(a) $1/2$

(b) 0

(c) 1

(d) None of these

Q 77. The value of $\lim_{x\to\infty}\left(\frac{3x-4}{3x+2}\right)^{\frac{x+1}{3}}$ is equal to

[J&KCET]

(a) $e^{-1/3}$

(b) $e^{-2/3}$

(c) e^{-1}

(d) e^{-2}

Q 78. $\lim_{h\to 0}\frac{f(2h+2+h^2)-f(2)}{f(h-h^2+1)-f(1)}$, given that $f'(2) = 6$ and $f'(1) = 4$

[IIT JEE]

(a) does not exist

(b) is equal to $-3/2$

(c) is equal to $3/2$

(d) is equal to 3

Q 79. If $\lim_{x\to 0}\frac{\log(3+x)-\log(3-x)}{x}=k$, the value of k is

[AIEEE]

(a) 0

(b) $-1/3$

(c) $2/3$

(d) $-2/3$

Q 80. Let f(a) = g(a) = k and their nth

derivatives $f^n(a), g^n(a)$ exist and are not equal for

some n. Further if $\lim_{x \to a} \frac{f(a)g(x) - f(a) - g(a)f(x) + g(a)}{g(x) - f(x)} = 4$, then

the value of k is equal to

[AIEEE]

(a) 4

(b) 2

(c) 1

(d) 0

Q 81. $\lim_{x\to 1}\frac{x^8-2x+1}{x^4-2x+1}$ equals

[DCE]

(a) 3

(b) 0

(c) −3

(d) 1

Q 82. The value of $\lim_{x\to\infty}\left(\frac{x+3}{x+1}\right)^{x+2}$ is

[UP SEE]

(a) 0

(b) 1

(c) e^2

(d) e^4

Q 83. The value of $\lim_{x \to 7} \frac{2-\sqrt{x-3}}{x^2-49}$ is

[MPPET]

(a) $2/9$

(b) $-2/49$

(c) $1/64$

(d) $-1/56$

Q 84. $\lim_{x\to\infty}\frac{(2x+1)^{40}(4x-1)^{5}}{(2x+3)^{45}}$ is equal to

[KPET]

(a) 16

(b) 24

(c) 32

(d) 8

Topic 2 Limits of Trigonometric Functions and Series

Q 1. The value of $\lim_{x \to 0} \frac{1-\cos(1-\cos x)}{x^4}$ is

[WBJEE]

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{6}$

(d) $\frac{1}{8}$

Q 2. $\lim_{x \to 0} \frac{\sin |x|}{x}$ is equal to

[WB JEE, BITSAT]

(a) 1

(b) 0

(c) positive infinity

(d) does not exist

Q 3. The value of $\lim_{x \to 0} \frac{\sin^2 x + \cos x - 1}{x^2}$ is

[WB JEE, BITSAT]

(a) 1

(b) $\frac{1}{2}$

(c) $-\frac{1}{2}$

(d) 0

Q 4. The value of $\lim_{x \to 0} (\cos x)^{\cot^2 x}$ is

[UP SEE]

(a) e^{-1}

(b) $e^{-1/2}$

(c) 1

(d) not existing

Q 5. For the function

$$f(x) = \lim_{n\to\infty} \frac{\log(2+x) - x^{2n}\sin x}{1+x^{2n}}$$

which of the following is true ?

[AMU]

(a) $\lim_{x \to 1^-} f(x)$ does not exist

(b) $\lim_{x \to 1^+} f(x)$ does not exist

(c) Both limits exist and $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x)$

(d) Both limits exist and $\lim_{x \to 1^-} f(x) \neq \lim_{x \to 1^+} f(x)$

Q 6. The value of $\lim_{x \to \frac{\pi}{2}} (\sin x)^{\tan x}$ is

[OJEE]

(a) 1

(b) 0

(c) e

(d) None of these

Q 7. The value of the limit $\lim_{x \to 1} \frac{\sin(e^{x-1} - 1)}{\log x}$ is

[WBJEE]

(a) 0

(b) e

(c) $\frac{1}{e}$

(d) 1

Q 8. $\lim_{x \to 0} \frac{1 - \cos^3 x}{x \sin x \cos x}$ is equal to

[BCECE]

(a) $\frac{2}{5}$

(b) $\frac{3}{5}$

(c) $\frac{3}{2}$

(d) $\frac{3}{4}$

Q 9. If $\lim_{x \to 0} \frac{(e^{kx} - 1) \sin kx}{x^2} = 4$, then k is equal to

[MHTCET]

(a) 2

(b) -2

(c) ± 2

(d) ± 4

Q 10. The value of $\lim_{x \to 0} \frac{(1-\cos 2x)}{x^2}$ is

[RPET]

(a) does not exist

(b) infinite

(c) 0

(d) 2

Q 11. $\lim_{x \to 0} \frac{e^{x^2} - \cos x}{x^2}$ is equal to

[J&KCET]

(a) 0

(b) $\frac{1}{2}$

(c) 1

(d) $\frac{3}{2}$

Q 12. $\lim_{x \to 1} \cos^{-1}\left(\frac{1-\sqrt{x}}{1-x}\right)$ is equal to

[Guj CET]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{4}$

Q 13. $\lim_{x \to 0}$ x $\log_e$ (sin x) is equal to

[DCE]

(a) -1

(b) $\log_e 1$

(c) 1

(d) None of these

Q 14. $\lim_{x \to 0} \left\{ \frac{1+\tan x}{1+\sin x} \right\}^{\operatorname{cosec} x}$ is equal to

[UP SEE]

(a) $\frac{1}{e}$

(b) 1

(c) e

(d) e^2

Q 15. $\lim_{x\to\infty} x \sin\left(\frac{2}{x}\right)$ is equal to

[KCET]

(a) ∞

(b) 0

(c) 2

(d) $\frac{1}{2}$

Q 16. $\lim_{x \to 0} \frac{(1 - e^x) \sin x}{x^2 + x^3}$ is equal to

[AMU]

(a) -1

(b) 0

(c) 1

(d) 2

Q 17. $\lim_{x \to \frac{\pi}{2}} \frac{a^{\cot x} - a^{\cos x}}{\cot x - \cos x}$, a > 0 is equal to

[WBJEE]

(a) $\log_e \frac{\pi}{2}$

(b) $\log_e 2$

(c) $\log_e a$

(d) a

Q 18. The value of $\lim_{x \to 2} \frac{e^{3x-6} - 1}{\sin(2-x)}$ is

[WBJEE]

(a) $\frac{3}{2}$

(b) 3

(c) -3

(d) -1

Q 19. If $f(x)=\sqrt{\frac{x-\sin x}{x+\cos^2 x}}$, then $\lim_{x\to\infty}$ f(x)is

[MHTCET]

(a) 0

(b) ∞

(c) 1

(d) None of these

Q 20. If $z_r = \cos\frac{r\alpha}{n^2} + i\sin\frac{r\alpha}{n^2}$, where r = 1,2,3,..., n,

then $\lim_{n\to\infty}$ z_1 z_2 ... z_n is equal to

[Manipal]

(a) cos α + i sin α

(b) $\cos\left(\frac{\alpha}{2}\right) - i\sin\left(\frac{\alpha}{2}\right)$

(c) $e^{i\alpha/2}$

(d) $\sqrt[3]{e^{i\alpha}}$

Q 21. Let $f(x)=\begin{cases}\sin x, & x \neq n\pi \\ 2, & x = n\pi\end{cases}$, where n $\in$ I and

$g(x)=\begin{cases}x^2+1, & x \neq 2 \\ 3, & x = 2\end{cases}$, then $\lim\limits_{x\to 0}$ g[f(x)] is

[Manipal]

(a) 1

(b) 0

(c) 3

(d) Does not exist

Q 22. The value of $\lim_{x \to 0} \frac{1 + \sin x - \cos x + \log(1 - x)}{x^3}$ is

[DCE]

(a) -1

(b) $\frac{1}{2}$

(c) $-\frac{1}{2}$

(d) 1

Q 23. $\lim_{x \to 1} \frac{\tan(x^2 - 1)}{x - 1}$ is equal to

[KCET]

(a) 2

(b) $\frac{1}{2}$

(c) -2

(d) $-\frac{1}{2}$

Q 24. The value of $\lim_{x\to\infty}\left(\frac{\pi}{2}-\tan^{-1}x\right)^{1/x}$ is

[AMU]

(a) 0

(b) 1

(c) -1

(d) e

Q 25. If $f(x)=\begin{cases}\frac{\sin(1+[x])}{[x]}, & \text{for } [x]\neq 0\\ 0, & \text{for } [x]=0\end{cases}$ where [x] denotes the greatest integer not exceeding x, then $\lim_{x\to 0^-} f(x)$ is equal to

[EAMCET]

(a) -1

(b) 0

(c) 1

(d) 2

Q 26. $\lim_{x\to 0}\frac{d}{dx}\int\frac{1-\cos x}{x^2}dx$ is equal to

[OJEE]

(a) 1

(b) 0

(c) 1/2

(d) None of these

Q 27. If $f(x) = \cot^{-1}[(3x - x^3)/(1 - 3x^2)]$ and

$g(x) = \cos^{-1}[(1 - x^2)/(1 + x^2)]$,then

$\lim_{x \to a} \frac{f(x) - f(a)}{g(x) - g(a)} \left(0 < a < \frac{1}{2}\right)$ is

[OJEE]

(a) $-\frac{3}{2}$

(b) $\frac{1}{2}$

(c) $\frac{3}{2}$

(d) None of these

Q 28. $\lim_{x\to 0}\frac{(1-\cos 2x)\sin 5x}{x^2\sin 3x}$ equals

[WB JEE, MHTCET]

(a) 10/3

(b) 3/10

(c) 6/5

(d) 5/6

Q 29. $\lim_{h\to 0}\frac{(a+h)^2\sin(a+h)-a^2\sin a}{h}$ is equal to

[BCECE]

(a) $2a \sin a$

(b) $a^2 \cos a$

(c) $a^2 \cos a + 2a \sin a$

(d) None of these

Q 30. Let α and β be the distinct roots of $ax^2 + bx + c = 0$, then $\lim_{x \to \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$ is equal to

[MPPET]

(a) $\frac{1}{2}(\alpha - \beta)^2$

(b) $-\frac{a^2}{2}(\alpha-\beta)^2$

(c) 0

(d) $\frac{a^2}{2}(\alpha-\beta)^2$

Q 31. The value of $\lim_{x\to 0}\left(\frac{\int_0^{x^2}\sec^2 t\ dt}{x\sin x}\right)$ is

[RPET]

(a) 3

(b) 2

(c) 1

(d) 0

Q 32. $\lim_{x \to 0} \frac{1}{x} \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ is equal to

[J&KCET]

(a) -2

(b) 0

(c) 2

(d) ∞

Q 33. If f(x) is the integral function of the function $\frac{2\sin x - \sin 2x}{x^3}$, $x \neq 0$, then $\lim_{x \to 0} f'(x)$ is equal to

[Guj. CET]

(a) 0

(b) 1

(c) −1

(d) None of these

Q 34. For $x > 0$, $\lim_{x\to 0}\left((\sin x)^{1/x} + \left(\frac{1}{x}\right)^{\sin x}\right)$ is

[IITJEE]

(a) 0

(b) −1

(c) 1

(d) 2

Q 35. The value of

$\lim_{\alpha\to 0}\frac{\left[\cos ec^{-1}(\sec\alpha) + \cot^{-1}(\tan\alpha) + \cot^{-1}\cos(\sin^{-1}\alpha)\right]}{\alpha}$ is

[DCE]

(a) 0

(b) -1

(c) -2

(d) 1

Q 36. $\lim_{x \to 0} \frac{\tan x - \sin x}{x^3}$ is equal to

[BITSAT, Guj. CET]

(a) $\frac{1}{2}$

(b) $-\frac{1}{2}$

(c) 0

(d) 1

Q 37. $\lim_{x \to \infty} \frac{\sin x}{x}$ is equal to

[UP SEE]

(a) ∞

(b) 1

(c) 0

(d) Does not exist

Q 38. $\lim_{n\to\infty}\left(\frac{1^2}{1-n^3}+\frac{2^2}{1-n^3}+.....+\frac{n^2}{1-n^3}\right)$ is equal to

[Kerala CEE]

(a) $\frac{1}{3}$

(b) $-\frac{1}{3}$

(c) $\frac{1}{6}$

(d) $-\frac{1}{6}$

(e) 0

Q 39. $\lim_{\theta \to \frac{\pi}{2}} \frac{\frac{\pi}{2} - \theta}{\cot \theta}$ is equal to

[AMU]

(a) 0

(b) −1

(c) 1

(d) ∞

Q 40. If $I_1 = \lim_{x \to 2^+} (x + [x])$, $I_2 = \lim_{x \to 2^-} (2x - [x])$ and $I_3 = \lim_{x \to \pi/2} \frac{\cos x}{(x - \pi/2)}$, then

[EAMCET]

(a) $I_1 < I_2 < I_3$

(b) $I_2 < I_3 < I_1$

(c) $I_3 < I_2 < I_1$

(d) $I_1 < I_3 < I_2$

Q 41. $\lim_{x \to 0} \frac{\sqrt{1-\cos 2x}}{\sqrt{2}x}$ is

[OJEE]

(a) λ

(b) -1

(c) 0

(d) Does not exist

Q 42. $\lim_{x \to 0} (\operatorname{cosec} x)^{1/\log x}$ is equal to

[WBJEE]

(a) 0

(b) 1

(c) $1/e$

(d) None of these

Q 43. $\lim_{x \to 0}$ x log sin x is equal to

[OJEE]

(a) 0

(b) ∞

(c) 1

(d) Cannot be determined

Q 44. $\lim_{x \to 0}\left[\frac{e^x - e^{\sin x}}{x - \sin x}\right]$ is equal to

[MP PET]

(a) -1

(b) 0

(c) 1

(d) None of these

Q 45. $\lim_{x \to 0} \frac{\sin^{-1} x - x}{x^3 \cos x}$ is equal to

[J&KCET]

(a) $1/2$

(b) $1/3$

(c) $1/6$

(d) $1/12$

Q 46. The value of $\lim_{n\to\infty} \frac{1}{1.3}+\frac{1}{3.5}+\frac{1}{5.7}+.....+\frac{1}{(2n-1)(2n+1)}$

is

[DCE]

(a) $1/2$

(b) $1/3$

(c) $1/4$

(d) None of these

Q 47. $\lim_{\theta \to 0} \frac{4\theta(\tan\theta - 2\theta\tan\theta)}{(1-\cos 2\theta)}$ is

[BITSAT, OJEE]

(a) $1/\sqrt{2}$

(b) $1/2$

(c) 1

(d) 2

Q 48. If α is a repeated root of ax + bx + c = 0,

then $\lim_{x \to \alpha} \frac{\sin(ax^2 + bx + c)}{(x - \alpha)^2}$ is

[AMU]

(a) 0

(b) a

(c) b

(d) c

Q 49. $\lim_{x \to 0} \frac{\log(1+x^3)}{\sin^3 x}$ is equal to

[AMU]

(a) 0

(b) 1

(c) 3

(d) None of these

Q 50. $\lim_{x \to 0} x^2 \sin \frac{\pi}{x}$ is equal to

[EAMCET]

(a) 1

(b) 0

(c) Does not exist

(d) ∞

Q 51. $\lim_{x \to 0} \left[\frac{8 \sin x + x \cos x}{3 \tan x + x^2} \right]$ is equal to

[MHT CET]

(a) 3

(b) 2

(c) -1

(d) 4

Q 52. The derivative of function f(x) is $\tan^4 x$. If f(x) = 0, then $\lim_{x \to 0} \frac{f(x)}{x}$ is equal to

[J&KCET]

(a) 1

(b) 0

(c) -1

(d) None of these

Q 53. Let $f(x)=\begin{cases}(1/2)\{g(x)+(x)\}\sin(x), x\geq 1\\ \sin x/x, x<1\end{cases}$ where

$g(x)=\begin{cases}1 & , \text{ if } x>0\\ -1 & , \quad x<0\\ 0 & , \text{ if } x=0\end{cases}$. Then, $\lim_{x\to 1} f(x)$ is equal to

[J&KCET]

(a) 0

(b) 2

(c) sin 1

(d) None of these

Q 54. $\lim_{n\to\infty}\left(1+\sin\frac{a}{n}\right)^n$ equals

[DCE]

(a) e^a

(b) e

(c) e^{2a}

(d) 0

Q 55. $\lim_{x\to 0}\frac{2\sin^2 3x}{x^2}$ is equal to

[BCECE]

(a) 0

(b) 1

(c) 18

(d) 36

Q 56. $\lim_{x \to 1}(1-x)\tan\left(\frac{\pi x}{2}\right)$ is equal to

[Jamia Millia Islamia]

(a) $\frac{\pi}{2}$

(b) π

(c) $\frac{2}{\pi}$

(d) 0

Q 57. The value of $\lim_{n\to\infty} \cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{4}\right)\cos\left(\frac{x}{8}\right).....\cos\left(\frac{x}{2^n}\right)$

is

[RPET]

(a) $\frac{x}{\sin x}$

(b) $\frac{x}{\cos x}$

(c) $\frac{(\sin x)}{x}$

(d) $\frac{(\cos x)}{x}$

Q 58. The value of $\lim_{x \to \pi/2} \frac{\cot x - \cos x}{(\pi - 2x)^3}$

[J&KCET]

(a) 1

(b) $\frac{1}{16}$

(c) 16

(d) None of these

Q 59. If $\lim_{x\to 0}\frac{[(a-n)nx-\tan x]\sin nx}{x^2}=0$, where n is non–zero real number, then a is equal to

[IIT JEE]

(a) 0

(b) $\frac{n+1}{n}$

(c) n

(d) $n+\frac{1}{n}$

Q 60. $\lim_{x\to\frac{\pi}{2}} \frac{\left[1-\tan\left(\frac{x}{2}\right)\right](1-\sin x)}{\left[1+\tan\left(\frac{x}{2}\right)\right](\pi-2x)^3}$ is

[AIEEE]

(a) 1/8

(b) 0

(c) 1/32

(d) ∞

Q 61. $\lim_{x\to\frac{\pi}{6}} \frac{\sin 2x}{\sin x}$ is equal to

[DCE]

(a) $\sqrt{3}$

(b) $\frac{1}{\sqrt{3}}$

(c) 2

(d) $\frac{1}{2}$

Q 62. $\lim_{x \to \frac{\pi}{6}} \frac{2\sin^2 x + \sin x - 1}{2\sin^2 x - 3\sin x + 1}$ is equal to

[Kerala CEE]

(a) 3

(b) -3

(c) 6

(d) 0

(e) 9

Q 63. If $f(x)=\begin{cases} x\sin\frac{1}{x} & x\neq 0 \\ 0 & x=0 \end{cases}$, then $\lim_{x\to 0} f(x)$ is equal to

[AMU]

(a) 1

(b) 0

(c) −1

(d) None of these

Q 64. $\lim_{x \to \frac{\pi}{6}} \left[\frac{3\sin x - \sqrt{3}\cos x}{6x - \pi} \right]$

[EAMCET]

(a) $\sqrt{3}$

(b) $\frac{1}{\sqrt{3}}$

(c) $-\frac{1}{\sqrt{3}}$

(d) $-\frac{1}{3}$

Q 65. If for some real number k $\lim_{x \to 0}$ kx cosec (x) = $\lim_{x \to 0}$ x cosec (kx), then the possible values of k are

[RPET]

(a) 1,−1

(b) 0, 1

(c) 1,2

(d) 0, π

Q 66. $\lim_{x \to \frac{\pi}{4}} \frac{\tan x - 1}{x - \frac{\pi}{4}}$ is equal to

[J&KCET]

(a) 1

(b) 1/2

(c) 2

(d) 0

Q 67. $\lim_{h\to 0}\frac{\sin\sqrt{x+h}-\sin\sqrt{x}}{h}$ is equal to

[J&K CET]

(a) $\cos\sqrt{x}$ (

b) $1/(2\sin\sqrt{x})$

(c) $(\cos\sqrt{x})/2\sqrt{x}$

(d) $\sin\sqrt{x}$

Q 68. $\lim_{x\to 1}\frac{\sqrt{1-\cos 2(x-1)}}{x-1}$

[J&KCET]

(a) exists and is equals $\sqrt{2}$

(b) exists and is equals $-\sqrt{2}$

(c) does not exist because x $-1 \to 0$

(d) does not exist because left hand limit is not

equal to right hand limit

Topic 3 Continuity

Q 1. Let f(x)= $\frac{(e^x-1)^2}{\sin\left(\frac{x}{a}\right)\log\left(1+\frac{x}{4}\right)}$ for $x \neq 0$ and $f(0)$ =

12.If f is continuous at x = 0, then the value of a is

equal to

[Kerala CEE]

(a) 1

(b) -1

(c) 2

(d) -2

(e) 3

Q 2. The value of $f(0)$, so that the function

$$f(x)=\frac{\sqrt{a^2-ax+x^2}-\sqrt{a^2+ax+x^2}}{\sqrt{a+x}-\sqrt{a-x}}$$

becomes continuous for all x, is given by

(a) $a^{3/2}$

(b) $a^{1/2}$

(c) $-a^{1/2}$

(d) $-a^{3/2}$

Q 3. The number of discontinuities of the greatest integer function f(x) = [x], $x \in \left(-\frac{7}{2}, 100\right)$ is equal to

[Kerala CEE]

(a) 104

(b) 100

(c) 102

(d) 101

(e) 103

Q 4. If $f : R \to R$ is defined by

$$f(x) = \begin{cases} \dfrac{2\sin x - \sin 2x}{2x\cos x}, & \text{if } x \neq 0 \\ a, & \text{if } x = 0 \end{cases}$$

then the value of a so that f is continuous at 0 is

[EAMCET]

(a) 2

(b) 1

(c) -1

(d) 0

Q 5. Let $f(x) = [x^3 - 3]$, where [x] is the

greatest integer function. Then the number of points in the interval $(1,2)$, where function is discontinuous is

[OJEE]

(a) 4

(b) 5

(c) 6

(d) 7

Q 6. Let a function f : R $\rightarrow$ R, where R is the

set of real numbers satisfying the equation f(x + y)

= f(x) + f(y), $\forall$ x, y if f(x) is continuous at x = 0,

then

[OJEE]

(a) f(x) is discontinuous, $\forall$ x $\in$ R

(b) f(x) is continuous, $\forall$ x $\in$ R

(c) f(x) is continuous for x $\in \{1,2,3, 4\}$

(d) None of the above

Q 7. $f(x)=x + |x|$ is continuous for

[WB JEE]

(a) $x \in (-\infty, \infty)$

(b) $x \in (-\infty, \infty) - \{0\}$

(c) only $x > 0$

(d) no value of x

Q 8. The function f(x) is defined as f(x) =

$\frac{2x - \sin^{-1} x}{2x + \tan^{-1} x}$, if $x \neq 0$. The value of/ to be assigned at x

= 0 so that the function is continuous there, is

[J&K CET]

(a) $-\frac{1}{3}$

(b) 1

(c) $\frac{2}{3}$

(d) $\frac{1}{3}$

Q 9. The function defined by

$$f(x)=\begin{cases}\left(x^2+e^{\frac{1}{2-x}}\right)^{-1}, & x\neq 2\\ k, & x=2\end{cases}$$ is continuous from

right at the point x = 2, then k is equal to

[DCE]

(a) 0

(b) $\frac{1}{4}$

(c) $-\frac{1}{2}$

(d) None of these

Q 10. If f : R $\rightarrow$ R is defined by

$$f(x)=\begin{cases}\dfrac{\cos 3x-\cos x}{x^2}, & \text{for } x\neq 0\\ \lambda, & \text{for } x=0\end{cases}$$

and if f is continuous at x = 0, then λ is equal to

[BITSAT, EAMCET]

(a) -2

(b) -4

(c) -6

(d) -8

Q 11. Let $f(x)=\begin{cases}5^{1/x}, & x<0\\ \lambda[x], & x\geq 0\end{cases}$ and $\lambda \in R$, then at x = 0

[UP SEE]

(a) f is discontinuous

(b) f is continuous only, if $\lambda = 0$

(c) f is continuous only, whatever λ may be

(d) None of the above

Q 12. If function $f(x)=\begin{cases}x, & \text{if } x \text{ is rational}\\ 1-x, & \text{if } x \text{ is irrational}\end{cases}$, then the

number of points at which f(x) is continuous, is

[UP SEE]

(a) ∞

(b) 1

(c) 0

(d) None of these

Q 13. If $f(x)=\begin{cases}\frac{1-\cos x}{x}, & x\neq 0\\ k, & x=0\end{cases}$ is continuous at x = 0,

then the value of k is

[Kerala CEE]

(a) 0

(b) $\frac{1}{2}$

(c) $\frac{1}{4}$

(d) $-\frac{1}{2}$

(e) None of these

Q 14. The function $f(x) = x - |x - x^2|$ is

[OJEE]

(a) continuous at x = 1

(b) discontinuous at x = 1

(c) not defined at x = 1

(d) None of the above

Q 15. $f(x)=\begin{cases} 2x, & x<0 \\ 2x+1, & x\geq 0 \end{cases}$. Then

[OJEE]

(a) f(x) is continuous at x = 0

(b) f (| x |) is continuous at x = 0

(c) f(x) is discontinuous at x = 0

(d) None of the above

Q 16. A function f(x) is defined as follows for real x,

$$f(x)=\begin{cases}1-x^2 & , \text{ for } x<1\\ 0 & , \text{ for } x=1\\ 1+x^2 & , \text{ for } x>1\end{cases}$$ Then,

[WB JEE]

(a) f(x) is not continuous at x = 1

(b) f(x) is continuous but not differentiable at x = 1

(c) f(x) is both continuous and differentiable at x =

1

(d) None of the above

Q 17. If $f(x)=\begin{cases}\frac{1-\sqrt{2}\sin x}{\pi-4x}, & \text{if } x\neq\frac{\pi}{4}\\ a, & \text{if } x=\frac{\pi}{4}\end{cases}$ is continuous at $\frac{\pi}{4}$,

then a is equal to

[BCECE]

(a) 4

(b) 2

(c) 1

(d) $1/4$

Q $18.$ The value of f(0) so that $\frac{(-e^x + 2^x)}{x}$ may be continuous at x = 0 is

[VITEEE]

(a) $\log\left(\frac{1}{2}\right)$

(b) 0

(c) 4

(d) $-1 + \log 2$

Q 19. Let [ ] denotes the greatest integer function and $f(x) = [\tan^2 x]$. Then,

[VITEEE]

(a) $\lim_{x \to 0} f(x)$ does not exist

(b) $f(x)$ is continuous at $x = 0$

(c) $f(x)$ is not differentiable at $x = 0$

(d) $f(x) = 1$

Q 20. If $f(x)=\begin{cases} x\sin\frac{1}{x}, & x\neq 0 \\ k, & x=0 \end{cases}$ is continuous at x = 0,

then the value of k is

[MHTCET]

(a) 1

(b) −1

(c) 0

(d) 2

Q 21. The points of discontinuity of tan x are

[Manipal]

(a) $n\pi$, $n \in I$

(b) $2n\pi$, $n \in I$

(c) $(2n+1)\frac{\pi}{2}$, $n \in I$

(d) None of these

Q 22. For the function $f(x) = \begin{cases} \frac{x^3 - a^3}{x - a}, & x \neq a \\ b, & x = a \end{cases}$, if f(x) is

continuous at x = a, then b is equal to

[MP PET]

(a) a^2

(b) $2a^2$

(c) $3a^2$

(d) $4a^2$

Q 23. If the function f : R $\rightarrow$ R given by

$f(x)=\begin{cases} x+a, & \text{if } x \leq 1 \\ 3-x^2, & \text{if } x>1 \end{cases}$ is continuous at x = 1, then a is equal to

[J&KCET]

(a) 4

(b) 3

(c) 2

(d) 1

Q 24. If $f(x) = \frac{\log_e(1+x^2\tan x)}{\sin x^3}$, $x \neq 0$, is to be continuous at $x = 0$, then $f(0)$ must be defined as

[Guj, CET]

(a) 1

(b) 0

(c) $\frac{1}{2}$

(d) -1

Q 25. The function f:R/{0} $\rightarrow$ R given by

$f(x)=\frac{1}{x}-\frac{2}{e^{2x}-1}$ can be made continuous at x = 0 by

defining f(0) as

[AIEEE]

(a) 2

(b) -1

(c) 0

(d) 1

Q 26. Function $f(x)=\begin{cases} x-1, & x<2 \\ 2x-3, & x\geq 2 \end{cases}$ is a continuous function

[UP SEE]

(a) for x = 2 only

(b) for all real values of x such that $x \neq 2$

(c) for all real values of x

(d) for all integral values of x only

Q 27. Let $f(x) = \begin{cases} \frac{\sin \pi x}{5x}, & x \neq 0 \\ k, & x = 0 \end{cases}$, if f(x) is continuous

at x = 0, then k is equal to

[UP SEE]

(a) $\frac{\pi}{5}$

(b) $\frac{5}{\pi}$

(c) 1

(d) 0

Q 28. If the function $f(x)=\begin{cases}\frac{1-\cos x}{x^2}, & \text{for } x \neq 0 \\ k, & \text{for } x=0\end{cases}$ is

continuous at x = 0, then the value of k is

[KCET]

(a) 1

(b) 0

(c) $\frac{1}{2}$

(d) -1

Q 29. If $f(x)=\begin{cases} mx+1, & x \le \frac{\pi}{2} \\ \sin x+n, & x > \frac{\pi}{2} \end{cases}$ is continuous at $x=\frac{\pi}{2}$,

then

[AMU]

(a) $m = 1, n = 0$

(b) $m=\frac{n\pi}{2}+1$

(c) $n=m\frac{\pi}{2}$

(d) $m=n=\frac{\pi}{2}$

Q 30. Let $f(x)=\begin{cases}\dfrac{\tan x-\cot x}{x-\dfrac{\pi}{4}}, & x\neq\dfrac{\pi}{2}\\ a, & x=\dfrac{\pi}{4}\end{cases}$ the value of a so that f(x) is continuous at $x=\dfrac{\pi}{4}$ is

[BCECE]

(a) 2

(b) 4

(c) 3

(d) 1

Q 31. Let $f(x)=\frac{1-\tan x}{4x-\pi}, x \neq \frac{\pi}{4}$, $x \in \left[0, \frac{\pi}{2}\right]$. If f(x) is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is

[Jamia Millia Islamia]

(a) 1

(b) ½

(c) $-1/2$

(d) -1

Q 32. At $x=\frac{3}{2}$ the function $f(x)=\frac{|2x-3|}{2x-3}$ is

[J&KCET]

(a) continuous

(b) discontinuous

(c) differentiable

(d) non−zero

Q 33. Let $f(x)=\begin{cases}\dfrac{x-4}{|x-4|}+a & , \quad x<4 \\ a+b & , \quad x=4 \\ \dfrac{x-4}{|x-4|} & , \quad x>4\end{cases}$

Then, f(x) is continuous at x = 4, when

[Guj. CET]

(a) a = 0, b = 0

(b) a = 1, b = 1

(c) a = -1,b = 1

(d) a = 1,b = -1

Q 34. If $f(x)=\begin{cases}\frac{\sin 5x}{x^2+2x}, & x\neq 0\\ k+\frac{1}{2}, & x=0\end{cases}$ is continuous at x = 0,

then the value of k is

[BITSAT, Guj. CET]

(a) 1

(b) -2

(c) 2

(d) $\frac{1}{2}$

Q 35. If the derivative of the function f(x) is everywhere continuous and is given by

$$f(x)=\begin{cases} bx^2+ax+4; & x\geq -1 \\ ax^2+b; & x<-1 \end{cases}$$, then

[Kerala CEE]

(a) a = 2, b = -3

(b) a =3, b = 2

(c) a = -2, b = -3

(d) a = -3,b = -2

(e) a =-1, b = -2

Q 36. For the function $f(x)=\frac{e^{1/x}-1}{e^{1/x}+1}$, x = 0, which of the following is correct ?

[WBJEE]

(a) $\lim_{x \to 0}$ f(x) does not exist

(b) $\lim_{x \to 0}$ f(x) = 1

(c) $\lim_{x \to 0}$ f(x) exists but f(x) is not continuous at x = 0

(d) f(x) is continuous at x = 0

Q 37. The value of f at x = 0 so that function

$f(x) = \frac{2^x - 2^{-x}}{x}$, $x \neq 0$ is continuous at x = 0, is

[MHTCET]

(a) 0

(b) $\log 2$

(c) 4

(d) $\log 4$

Q 38. If $f : R \to R$ given by

$$f(x) = \begin{cases} 2\cos x, & \text{if} \quad x \le -\frac{\pi}{2} \\ a\sin x + b, & \text{if} \quad -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 1 + \cos^2 x, & \text{if} \quad x \ge \frac{\pi}{2} \end{cases}$$ is a continuous

function on R, then (a, b) is equal to

[J&K CET]

(a) $(1/2, 1/2)$

(b) $(0, -1)$

(c) $(0, 2)$

(d) $(1, 0)$

Q 39. The value of k for which the function

$$f(x)=\begin{cases}\dfrac{1-\cos 4x}{8x^2}, & x\neq 0\\ k, & x=0\end{cases}$$ is continuous at x = 0, is

[AMU]

(a) k = 0

(b) k = 1

(c) k = −1

(d) None of these

Q 40. If f : R $\rightarrow$ R is defined by

$$f(x)=\begin{cases}\dfrac{x+2}{x^2+3x+2}, & \text{if} \quad x \in R-\{-1,-2\} \\ -1, & \text{if} \quad x=-2 \\ 0, & \text{if} \quad x=-1\end{cases}$$, then f is

continuous on the set

[EAMCET]

(a) R

(b) $R-\{-2\}$

(c) $R-\{-1\}$

(d) $R-\{-1,-2\}$

Q 41. $\lim_{x \to 0} [(1 + 3x)^{1/x}] = k$, then for continuity at x = 0, k is

[MHTCET]

(a) 3

(b) −3

(c) e^3

(d) e^{-3}

Q 42. If $f(x)=\begin{cases}\dfrac{2^x-1}{\sqrt{1+x}-1}, & -1\le x<\infty,\ x\neq 0\\ k, & x=0\end{cases}$ is continuous

everywhere, then k is equal to

[MP PET]

(a) $\frac{1}{2}\log_e 2$

(b) $\log_e 4$

(c) $\log_e 8$

(d) $\log_e 2$

Q 43. The function $f(x)=\frac{2x^2+7}{x^3+3x^2-x-3}$ is

discontinuous for

[J&K CET]

(a) x = 1 only

(b) x = 1 and x = −1 only

(c) x = 1, x = −1, x = − 3 only

(d) x = 1, x = −1, x = − 3 and other values of x

Q 44. The function f(x) = |x|+ $\frac{|x|}{x}$ is

[J&K CET]

(a) discontinuous at origin because | x | is

discontinuous there

(b) continuous at origin

(c) discontinuous at origin because both | x | and

$\frac{|x|}{x}$ are discontinuous there

(d) discontinuous at the origin because $\frac{|x|}{x}$ is

discontinuous there

Q 45. The number of points at which the function $f(x) = (|x - 1| + |x - 2| + \cos x)$ where $x \in [0, 4]$ is not continuous, is

[DCE]

(a) 1

(b) 2

(c) 3

(d) 0

Q 46. A function f on R into itself is continuous at a point a in R, iff for each $\in > 0$, there exists, such $\delta > 0$ that

[UP SEE]

(a) $|f(x) - f(a)| < \in \Rightarrow x - a| < \delta$

(b) $|f(x) - f(a)| > \in \Rightarrow x - a | > \delta$

(c) $|x - a > \delta \Rightarrow | f(x) - f(a)| > \in$

(d) $|x - a < \delta \Rightarrow |f(x) - f(a)| < \in$

Q 47. If $f(x)=\begin{cases}\dfrac{(4^x-1)^3}{\sin\left(\dfrac{x}{a}\right)\log\left(1+\dfrac{x^2}{3}\right)}, & x\neq 0\\ 9(\log 4)^3, & x=0\end{cases}$ is a continuous function at x = 0, then the value of a is equal to

[OJEE]

(a) 3

(b) 1

(c) 2

(d) 0

Q 48. If $f(x)=\begin{cases}\log_{(1-3x)}(1+3x), & \text{for } x\neq 0\\ k, & \text{for } x=0\end{cases}$ is continuous at $x = 0$, then k is equal to

[MHT CET]

(a) -2

(b) 2

(c) 1

(d) -1

Q 49. If $f(x) = (x + 1)^{\cot x}$ be continuous at $x = 0$, then $f(0)$ is equal to

[MP PET]

(a) 0

(b) $-e$

(c) e

(d) None of these

Q 50. $f(x)=\begin{cases}\frac{\sin 3x}{\sin x}, & x\neq 0\\ k, & x=0\end{cases}$ is continuous, if k is

[DCE]

(a) 3

(b) 0

(c) −3

(d) −1

Q 51. The function f(x) = [x] $\cos\left[\frac{2x-1}{2}\right]\pi$ where

[.] denotes the greatest integer function, is

discontinuous at

[AMU]

(a) all x

(b) no x

(c) all integer points

(d) x which is not an integer

Q 52. The function $f(x)=\frac{1-\sin x+\cos x}{1+\sin x+\cos x}$ is not defined

at x = π. The value of f(π), so that f(x) is

continuous at x = π, is

[OJEE]

(a) $-1/2$

(b) ½

(c) − 1

(d) 1

Q 53. For the function $f(x) = \frac{\log_e(1+x) + \log_e(1-x)}{x}$ to be

continuous at x = 0, the value of f(0) is

[RPET]

(a) -1

(b) 0

(c) -2

(d) 2

Topic 4 Differentiability

Q 1. Let f(x) = x | x | and g(x) = sin x

Statement I gof is differentiable at x = 0 and its

derivative is continuous at that point.

Statement II gof is twice differentiable at x = 0.

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement

I.

(b) Statement I is true, Statement II is true;

StatementII is not a correct explanation for

StatementI.

(c) Statement I is true, Statement II is false.

(d) Statement I is false, Statement II is true.

Q 2. If $f(x) = \begin{cases} x^p \cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ is differentiable at x

= 0, then

(a) $p < 0$

(b) $0 < p < 1$

(c) $p = 1$

(d) $p > 1$

Q 3. If $f(x) = ae^{|x|} + b|x|^2$; $a, b \in R$ and $f(x)$ is differentiable at $x = 0$. Then a and b are

[OJEE]

(a) $a = 0, b \in R$

(b) a = 1, b = 2

(c) b = 0, a $\in$ R

(d) a = 4, b = 5

Q 4. The set of points where the function f(x) = x|x| is differentiable is

[Jamia Millia Islamia]

(a) $(-\infty, \infty)$

(b) $(-\infty, 0) \cup (0, \infty)$

(c) $(0, \infty)$

(d) $[0, \infty)$

Q 5. If $f(x) = |\log_e |x||$, then $f'(x)$ equals

[Jamia Millia Islamia]

(a) $\frac{1}{|x|}, x \neq 0$

(b) $\frac{1}{x}$ for $|x| > 1$ and $\frac{-1}{x}$ for $|x| < 1$

(c) $\frac{-1}{x}$ for $|x| > 1$ and $\frac{1}{x}$ for $|x| < 1$

(d) $\frac{1}{x}$ for $x > 0$ and $-\frac{1}{x}$ for $x < 0$

Q 6. A function $f(x)=\begin{cases}1+x, & x \le 2\\ 5-x, & x>2\end{cases}$ is

[Jamia Millia Islamia]

(a) not continuous at x = 2

(b) differentiable at x = 2

(c) continuous but not differentiable at x = 2

(d) None of the above

Q 7. Function f(x) = |x − 1| + |x − 2|, x ∈ R is

[RPET]

(a) differentiable everywhere in R

(b) except x = 1 and x = 2 differentiable

everywhere in R

(c) not continuous at x = 1 and x = 2

(d) increasing in R

Q 8. The function $f(x)=\begin{cases} |x-3|, & \text{if } x \geq 1 \\ \frac{x^2}{4}-\frac{3x}{2}+\frac{13}{4}, & \text{if } x<1 \end{cases}$ is

[J&K CET]

(a) continuous and differentiable at x = 3

(b) continuous at x = 3, but not differentiable at x = 3

(c) continuous and differentiable everywhere

(d) continuous at x = 1, but not differentiable at x = 1

Q 9. f(x) = |x − 3| is .... at x = 3.

[Guj.CET]

(a) continuous and not differentiable

(b) continuous and differentiable

(c) discontinuous and not differentiable

(d) discontinuous and differentiable

Q 10. Let g(x) = $\frac{(x-1)^n}{\log\cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$, and let p be the left hand derivative of $|x-1|$ at $x = 1$. If $\lim_{x\to 1^+} g(x) = p$, then

[IITJEE]

(a) n = 1, m = 1

(b) n = 1,m = −1

(c) n = 2, m = 2

(d) n > 2, m = n

Q 11. Let $f(x)=\begin{cases}(x-1)\sin\frac{1}{x-1}, & \text{if } x\neq 1\\ 0, & \text{if } x=1\end{cases}$. Then, which one of the following is true ?

[AIEEE]

(a) f is differentiable at x = 1 but not at x = 0

(b) f is neither differentiable at x = 0 nor at x = 1

(c) f is differentiable at x = 0 and at x = 1

(d) f is differentiable at x = 0 but not at x = 1

Q 12. $f(x)=\begin{cases}2a-x & \text{in } -a<x<a\\ 3x-2a & \text{in } a\le x\end{cases}$. Then, which of the following is true ?

[KCET]

(a) f(x) is discontinuous at x = a

(b) f(x) is not differentiable at x = a

(c) f(x) is differentiable at $x \geq a$

(d) f(x) is continuous at all $x < a$

Q 13. If $f(x) = \begin{cases} \frac{x-1}{2x^2-7x+5}, & \text{for } x \neq 1 \\ -\frac{1}{3}, & \text{for } x = 1 \end{cases}$, then $f'(1)$ is equal to

[Kerala CEE]

(a) $-\frac{1}{9}$

(b) $-\frac{2}{9}$

(c) -13

(d) $1/3$

(e) None of these

Q 14. If $f(x)=\begin{cases} ax^2+b, & b\neq 0, x\leq 1 \\ x^2b+ax+c, & x>1 \end{cases}$, then, f(x) is

continuous and differentiable at x = 1, if

[Manipal]

(a) c = 0, a = 2b

(b) a = b, c $\in$ R

(c) a = b, c = 0

(d) a = b, $c \neq 0$

Q 15. The function $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 1. & x = 0 \end{cases}$, is

[MPPET]

(a) continuous but not differentiable at x = 0

(b) discontinuous at x = 0

(c) continuous and differentiable at x = 0

(d) not defined at x = 0

Q 16. Let f : R $\rightarrow$ R be a function defined by

f(x) = min {x + 1, | x| + l}. Then, which of the

following is true ?

[AEEEE]

(a) f(x) $\geq$ 1 for all x $\in$ R

(b) f(x) is not differentiable at x = 1

(c) f(x) is differentiable everywhere

(d) f(x) is not differentiable at x = 0

Q 17. At x = 0,thefunction f(x) = |x| is

[UPSEE]

(a) continuous but not differentiable

(b) discontinuous and differentiable

(c) discontinuous and not differentiable

(d) continuous and differentiable

Q 18. If $f(x)=\begin{cases} x-5, & \text{for} \quad x \le 1 \\ 4x^2-9, & \text{for} \quad 1<x<2 \\ 3x+4, & \text{for} \quad x \ge 2 \end{cases}$, then $f'(2^+)$ is

equal to

[EAMCET]

(a) 0

(b) 2

(c) 3

(d) 4

Q 19. If f(x) = | log | x | |, then

[WB JEE]

(a) f(x) is continuous and differentiable for all x in its domain

(b) f(x) is continuous for all x in its domain but not differentiable at $x = \pm 1$

(c) f(x) is neither continuous nor differentiable at $x = \pm 1$

(d) None of the above

Q 20. The set of points where the function $f(x) = |x - 1| e^x$ is differentiable, is

[MHT CET]

(a) R

(b) $R - \{1\}$

(c) $R - \{-1\}$

(d) $R - \{0\}$

Q 21. The function $f(x) = e^{-|x|}$ is

[MHT CET]

(a) continuous everywhere but not differentiable at

$x = 0$

(b) continuous and differentiable everywhere

(c) not continuous at $x = 0$

(d) None of the above

Q 22. Suppose $f(x)$ is differentiable at $x = l$ and

$\lim_{h \to 0} \frac{1}{h} f(1 + h) = 5$, then $f'(1)$ equals

[MP PET]

(a) 6

(b) 5

(c) 4

(d) 3

Q 23. If $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then $f(x)$ is differentiable on

[RPET]

(a) $[-1,1]$

(b) $R - \{-1, 1\}$

(c) R − $(-1,1)$

(d) None of these

Q 24. The set of points, where f(x) = $\frac{x}{1+|x|}$ is differentiable, is

[AIEEE]

(a) $(-\infty, -1) \cup (-1, \infty)$

(b) $(-\infty, \infty)$

(c) $(0, \infty)$

(d) $(-\infty, 0) \cup (0, \infty)$

Q 25. If $f(x) = x^{(\sqrt{x}+\sqrt{x+1})}$, then

[DCE]

(a) $f(x)$ is continuous but not differentiable at $x = 0$

(b) $f(x)$ is differentiable at $x = 0$

(c) $f(x)$ is not differentiable at $x = 0$

(d) None of the above

Q 26. Which one of the following is not true always?

[KCET]

(a) If f(x) is not continuous at x = a, then it is not differentiable at x = a

(b) If f(x) is continuous at x = a, then it is differentiable at x = a

(c) If f(x) and g(x) are differentiable at x = a, then f(x) + g(x) is also differentiable at x = a

(d) If a function f(x) is continuous at x = a , then $\lim_{x \to a}$ f(x) exists

Q 27. If f(x + y) = f(x) f(y) for all real x and y, f(6) = 3 and f' (0) = 10, then f' (6) is

[Kerala CEE]

(a) 30

(b) 13

(c) 10

(d) 0

(e) 6

Q 28. Let f(x)= ||x| − 1|, then points where f(x) is not differentiable, is / (are)

[IIT JEE]

(a) 0, ± 1

(b) ±1

(c) 0

(d) 1

Q 29. Let f be differentiable for all x. If $f(1) = -2$ and $f(x) \geq 2$ for $x \in [1,6]$, then

[AIEEE]

(a) $f(6) = 5$

(b) $f(6) < 5$

(c) $f(6) < 8$

(d) $f(6) > 8$

Q 30. Let f is a real-valued differentiable function satisfying $|f(x) - f(y)| \leq (x - y)^2$, $x, y \in R$ and $f(0) = 0$, then $f(1)$ equals

[AIEEE]

(a) 1

(b) 2

(c) 0

(d) −1

Q 31. Let $f(x)=\begin{cases} x^p \sin\frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then f(x) is

continuous but not differentiable at x = 0, if

[DCE]

(a) $0 < p \leq 1$

(b) $1 \leq p < \infty$

(c) $-\infty < p < 0$

(d) $p = 0$

Q 32. Let $f(x)=\begin{cases} 1, & \forall\ x<0 \\ 1+\sin x, & \forall\ 0\le x\le \pi/2 \end{cases}$, then what is the value of f' (x) at x = 0 ?

[BITSAT, OJEE]

(a) 1

(b) −1

(c) ∞

(d) does not exist

Q 33. The function f(x) = max [(1 − x), (1 + x,),2], x ∈ (−∞, ∞) is

[BCECE]

(a) continuous at all points

(b) differentiable at all points

(c) differentiable at all points except at x = 1 and x = −1

(d) None of the above

Q 34. If y = $\cos^{-1}$ cos (| x | − f(x)), where

$f(x)=\begin{cases}1, & \text{if} \quad x>0\\ -1, & \text{if} \quad x<0\\ 0, & \text{if} \quad x=0\end{cases}$. Then, $\left(\frac{dy}{dx}\right)_{x=\frac{5\pi}{4}}$ is equal to

[RPET]

(a) -1

(b) 1

(c) 0

(d) Cannot be determined

Q 35. If $f(x)=\begin{cases}x, & \text{for} \quad 0<x<1\\ 2-x, & \text{for} \quad 1\le x<2\\ x-(1/2)x^2, & \text{for} \quad x=2\end{cases}$. Then, f' (1) is

equal to

[J&K CET]

(a) -1

(b) 1

(c) 0

(d) None of these

Q 36. Let $f(x)=\begin{cases}\sin x, & \text{for } x \geq 0\\ 1-\cos x, & \text{for } x \leq 0\end{cases}$ and $g(x) = e^x$.

Then, $(gof)'(0)$ is

[UP SEE]

(a) 1

(b) −1

(c) 0

(d) None of these

Q 37. Define f on R into itself by

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$, then

[UP SEE]

(a) f is continuous at 0 but not differentiable at 0

(b) f is both continuous and differentiable at 0

(c) f is differentiable but not continuous at 0

(d) None of the above

Q 38. Given f(0) = 0 and f(x) $= \frac{1}{(1-e^{-1/x})}$ for x $\neq 0$.

Then only one of the following statements on f(x) is

true. That is f(x), is

[Kerala CEE]

(a) continuous at x = 0

(b) not continuous at x = 0

(c) both continuous and differentiable at x = 0

(d) not defined at x = 0

(e) continuous but not differentiable at x = 0

Q 39. The value of the derivative of |x − 1| + |x − 3| at x = 2 is

[AMU]

(a) 2

(b) 1

(c) 0

(d) -2

Q 40. If $f(x)=\begin{cases} xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & x\neq 0 \\ 0, & x=0 \end{cases}$, then f(x) is

[AIEEE]

(a) continuous as well as differentiable for all x

(b) continuous for all x but not differentiable at x = 0

(c) neither differentiable nor continuous at x = 0

(d) discontinuous everywhere

Q 41. f(x) = sin | x|. Then f(x) is not differentiable at

[DCE]

(a) x = 0 only

(b) all x

(c) multiples of π

(d) multiples of $\frac{\pi}{2}$

Q 42. If f (x) = | x |^3, then f' (0) equals

[Kerala CEE]

(a) 0

(b) ½

(c) -1

(d) $-1/2$

(e) None of these

Q 43. If $f(x)=\begin{cases}\dfrac{x-1}{2x^2-7x+5}, & \text{for } x\neq 1\\ -\dfrac{1}{3}, & \text{for } x=1\end{cases}$, then $f'(1)$ is equal to

[EAMCET]

(a) $-\frac{1}{9}$

(b) $-\frac{2}{9}$

(c) $-\frac{1}{3}$

(d) $\frac{1}{3}$

Answers

Topic 1 : Limits of Algebraic, Exponential and Logarithmic Functions

1. (b) 2. (a) 3. (d) 4. (a) 5. (d) 6. (b) 7. (a) 8. (c)

9. (c) 10. (b) 11. (b) 12. (a) 13. (b) 14. (c) 15. (c)

16. (c) 17. (b) 18. (c) 19. (a) 20. (a) 21. (b) 22.

(b) 23. (a) 24. (c) 25. (d) 26. (d) 27. (c) 28. (b)

29. (a) 30. (b) 31. (b) 32. (a) 33. (a) 34. (a) 35.

(a) 36. (c) 37. (c) 38. (d) 39. (b) 40. (a) 41. (d)

42. (a) 43. (b) 44. (b) 45. (a) 46. (a) 47. (a) 48.

(b) 49. (d) 50. (a) 51. (c) 52. (b) 53. (a) 54. (b)

55. (a) 56. (a) 57. (b) 58. (c) 59. (b) 60. (c)

61. (c) 62. (b) 63. (a) 64. (b) 65. (d) 66. (c) 67.

(d) 68. (a) 69. (c) 70. (a) 71. (c) 72. (c) 73. (a)

74. (d) 75. (d) 76. (a) 77. (b) 78. (d) 79. (c) 80.

(a) 81. (a) 82. (c) 83. (d) 84. (c)

Topic 2 : Limits of Trigonometric Functions and Series

1. (d) 2. (d) 3. (b) 4. (b) 5. (d) 6. (a) 7, (d) 8. (c)

9. (c) 10. (d) 11. (d) 12. (a) 13. (b) 14. (b) 15. (c)

16. (a) 17. (c) 18. (c) 19. (c) 20. (c) 21. (a) 22. (c) 23. (a) 24. (b) 25. (b) 26. (c) 27. (a) 28. (a) 29. (c) 30. (d) 31. (c) 32. (c) 33. (b) 34. (c) 35. (c) 36. (a) 37. (c) 38. (b) 39. (c) 40. (c) 41. (d) 42. (c) 43. (a) 44. (c) 45. (c) 46. (a) 47. (d) 48. (b) 49. (b) 50. (b) 51. (a) 52. (b) 53. (c) 54. (a) 55. (c) 56. (c) 57. (c) 58. (b) 59. (d) 60. (c) 61. (a) 62. (b) 63. (b) 64. (b) 65. (a) 66. (c) 67. (c) 68. (d)

Topic 3 : Continuity

1. (e) 2. (c) 3. (e) 4. (d) 5. (c) 6. (b) 7. (a) 8. (d)

9. (b) 10. (b) 11. (c) 12. (c) 13. (a) 14. (a) 15. (c)

16. (a) 17. (d) 18. (d) 19. (b) 20. (c) 21. (c) 22.

(c) 23. (d) 24. (a) 25. (d) 26. (c) 27. (a) 28. (c)

29. (c) 30. (b) 31. (c) 32. (b) 33. (d) 34. (c) 35.

(c) 36. (a) 37. (d) 38. (a) 39. (b) 40. (c) 41. (c)

42. (b) 43. (c) 44. (d) 45. (d) 46. (a) 47. (a) 48.

(d) 49. (c) 50. (a) 51. (c) 52. (c) 53. (b)

Topic 4 : Differentiability

1. (c) 2. (d) 3. (a) 4. (a) 5. (b) 6. (c) 7. (b) 8. (b)

9. (a) 10. (c) 11. (d) 12. (b) 13. (b) 14. (a) 15. (c)

16. (c) 17. (a) 18. (c) 19. (b) 20. (b) 21. (a) 22.

(b) 23. (b) 24. (b) 25. (c) 26. (b) 27. (a) 28. (a)

29. (d) 30. (c) 31. (a) 32. (d) 33. (c) 34. (b) 35.

(d) 36. (c) 37. (a) 38. (b) 39. (c) 40. (b) 41. (a)

42. (a) 43. (b)

Differentiation

Topic – 1

Differentiation of Algebraic and Transcendental Functions

Q 1. If f : $(-1, 1) \rightarrow$ R be a differentiable function with f(0) = -1 and f'(0) = 1. Let g(x) = [f$(2$f(x) = $2)]^2$.

Then, g'(0) is equal to

[AIEEE]

(a) 4

(b) − 4

(c) 0

(d) −2

Q 2. If $y = (1 + x)(1 + x^2)(1 + x^4) \ldots (1 + x^{2n})$, then the value of $\left(\frac{dy}{dx}\right)_{x=0}$ is

[WB JEE]

(a) 0

(b) -1

(c) 1

(d) 2

Q 3. If $y = \tan^{-1}x + \cot^{-1}x + \sec^{-1} x + \operatorname{cosec}^{-1} x$,

then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{x^2-1}{x^2+1}$

(b) π

(c) 0

(d) 1

Q 4. If $y = \sin^{-1}\sqrt{1-x}$, then $\frac{dy}{dx}$ is equal to

[BITSAT]

(a) $\frac{1}{\sqrt{1-x}}$

(b) $\frac{-1}{2\sqrt{1-x}}$

(c) $\frac{1}{\sqrt{x}}$

(d) $\frac{-1}{2\sqrt{x}\sqrt{1-x}}$

(e) $\frac{1}{\sqrt{x}\sqrt{1-x}}$

Q 5. If $f(x) = |x - 2| + |x + 1| = x$, then $f'(-10)$ is

equal to

[Kerala CEE]

(a) -3

(b) -2

(c) -1

(d) 0

(e) 1

Q 6. If $x^y = e^{2(x-y)}$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{2(1+\log x)}{(2+\log x)^2}$

(b) $\frac{1+\log x}{(2+\log x)^2}$

(c) $\frac{2}{2+\log x}$

(d) $\frac{2(1-\log x)}{(2+\log x)^2}$

(e) $\frac{2+\log x}{(2-\log x)^2}$

Q 7. If $y = \tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{2}$

(b) 2

(c) -2

(d) $-\frac{1}{2}$

(e) -1

Q 8. Let y be an implicit function of x defined by x^{2x} $- 2x^{x}$ cot y $- 1 = 0$. Then, y'(1) equals

[AIEEE]

(a) -1

(b) 1

(c) log 2

(d) $-$ log 2

Q 9. If $y = (\log_{\cos x} \sin x)(\log_{\sin x} \cos x) + \sin^{-1} \frac{2x}{1+x^2}$,

then $\frac{dy}{dx}$ at $x = \frac{\pi}{2}$ is equal to

[DCE]

(a) $\frac{8}{(4+\pi^2)}$

(b) 0

(c) $-\frac{8}{(4+\pi^2)}$

(d) 1

Q 10. Let $f(x) = 2^{2x-1}$ and $\phi(x) = -2^x + 2x \log 2$.

If $f'(x) > \phi'(x)$, then

[BITSAT]

(a) $0 < x < 1$

(b) $0 \leq x < 1$

(c) $x > 0$

(d) $x \geq 0$

Q 11. If $f(x) = \log_{x^2}(\log_e x)$, then $f'(x)$ at $x = e$ is

[KCET]

(a) 1

(b) $\frac{1}{e}$

(c) $\frac{1}{2e}$

(d) 0

Q 12. If $y = \sin^n x \cos nx$, then $\frac{dy}{dx}$ is

[KCET]

(a) $n \sin^{n-1} x \sin (n + 1)x$

(b) $n \sin^{n-1} x \cos (n-1)x$

(c) $n \sin^{n-1} x \cos nx$

(d) $n \sin^{n-1} x \cos (n+1)x$

Q 13. If $f(x) = \frac{g(x)+g(-x)}{2} + \frac{2}{[h(x)+h(-x)]^{-1}}$ where g and h are differentiable function, then $f'(0)$ is

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{3}{2}$

(d) 0

[KCET]

Q 14. If $y = \log_2 \log_2(x)$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{\log_2 e}{\log_e x}$

(b) $\frac{\log_2 e}{x \log_x 2}$

(c) $\frac{\log_2 x}{\log_e 2}$

(d) $\frac{\log_2 e}{\log_2 x}$

(e) $\frac{\log_2 e}{x \log_e x}$

Q 15. If $y = \sin[\cos^{-1}\{\sin(\cos^{-1}x)\}]$, then

$\frac{dy}{dx}$ at $x = \frac{1}{2}$ is equal to

[Kerala CEE]

(a) 0

(b) −1

(c) $\frac{2}{\sqrt{3}}$

(d) $\frac{1}{\sqrt{3}}$

Q 16. If $x^2 + y^2 = t - \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$,

then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{x^2y^3}$

(b) $\frac{1}{xy^3}$

(c) $\frac{1}{x^2y^2}$

(d) $\frac{1}{x^3y}$

(e) $\frac{-1}{x^3y}$

Q 17. If $y = \sec^{-1}[\text{cosec } x] + \text{cosec}^{-1}[\sec x] + \sin^{-1}[\cos x] + \cos^{-1}[\sin x]$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) 0

(b) 2

(c) −2

(d) −4

(e) 1

Q 18. If $y = e^{x}.\ e^{x}.e^{x^2}.e^{x^3}.....e^{x^n}$ ...., for $0 < x < 1$,

then $\frac{dy}{dx}$ at $x = \frac{1}{2}$ is

[Kerala CEE]

(a) e

(b) 4e

(c) 2e

(d) 3e

(e) 5e

Q 19. The derivative of $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ with respect to $\cos^{-1}\sqrt{1-x^2}$ is

[Kerala CEE]

(a) $\frac{\sqrt{1-x^2}}{1+x^2}$

(b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{2}{\sqrt{1-x^2}(1+x^2)}$

(d) $\frac{2}{1+x^2}$

(e) $\frac{2\sqrt{1-x^2}}{1+x^2}$

Q 20. Let $f(x) = (x - 7)^2 (x - 2)^7$, $x \in [2, 7]$ The value of $\theta \in (2, 7)$ such that $f'(\theta) = 0$ is equal to

[Kerala CEE]

(a) $\frac{49}{4}$

(b) $\frac{53}{4}$

(c) $\frac{53}{7}$

(d) $\frac{49}{9}$

(e) $\frac{45}{7}$

Q 21. If $y = \cos^{-1}(\cos x)$, then $\frac{dy}{dx}$ is

[AMU]

(a) 1 in the whole plane

(b) -1 in the whole plane

(c) 1 in the 2nd and 3rd quadrants of the plane

(d) -1 in the 3rd and 4th quadrants of the plane

Q 22. If $f'(x) = g(x)$ and $g'(x) = -f(x)$ for all x and $f(2) = 4 = f'(2)$, then $f^2(4) + g^2(4)$ is

[AMU]

(a) 8

(b) 16

(c) 32

(d) 64

Q 23. $x = \frac{1-\sqrt{y}}{1+\sqrt{y}} \Rightarrow \frac{dy}{dx}$ is equal to

[EAMCET]

(a) $\frac{4}{(x+1)^2}$

(b) $\frac{4(x-1)}{(1+x)^3}$

(c) $\frac{x-1}{(1+x)^3}$

(d) $\frac{4}{(x+1)^3}$

Q 24. $x=\cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right), y=\sin^{-1}\left(\frac{t}{\sqrt{1+t^2}}\right) \Rightarrow \frac{dy}{dx}$ is equal to

[EAMCET]

(a) 0

(b) tan t

(c) 1

(d) sin t cos t

Q 25. If 2f(x) = f'(x) and f(0) = 3, then f(2) is equal to

[OJEEE]

(a) $3e^4$

(b) $3e^2$

(c) e^4

(d) None of these

Q 26. If $y = \tan^{-1}\sqrt{\frac{1-\sin x}{1+\sin x}}$, then the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{6}$ is

[WB JEE]

(a) $-\frac{1}{2}$

(b) $\frac{1}{2}$

(c) 1

(d) -1

Q 27. If $8f(x) + 6f\left(\frac{1}{x}\right) = x + 5$ and $y = x^2 f(x)$,

then $\frac{dy}{dx}$ at $x = -1$ is equal to

[VITEEE]

(a) 0

(b) $\frac{1}{14}$

(c) $-\frac{1}{14}$

(d) 1

Q 28. If $x^2 + y^2 = t + \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$,

then $\frac{dy}{dx}$ is equal to

[VITEEE]

(a) $\frac{y}{x}$

(b) $-\frac{y}{x}$

(c) $\frac{x}{y}$

(d) $-\frac{x}{y}$

Q 29. If $y = x - x^2$, then the derivatives of y^2 w.r.t. x^2 is

[VITEEE]

(a) $2x^2 + 3x - 1$

(b) $2x^2 - 3x + 1$

(c) $2x^2 + 3x + 1$

(d) $2x^2 - 3x - 1$

Q 30. If $y = \sqrt{x \log_e x}$, then $\frac{dy}{dx}$ at x = e is

[VITEEE]

(a) $\frac{1}{e}$

(b) $\frac{1}{\sqrt{e}}$

(c) $\sqrt{e}$

(d) e^2

Q 31. Find $\frac{dy}{dx}$, if $x = 2\cos\theta - \cos 2\theta$ and $y = 2\sin\theta - \sin 2\theta$.

[MHTCET]

(a) $\tan\frac{3\theta}{2}$

(b) $-\tan\frac{3\theta}{2}$

(c) $\cot\frac{3\theta}{2}$

(d) $-\cot\frac{3\theta}{2}$

Q 32. Find the derivative of y with respect to x if

$e^x + e^y = e^{x+y}$

[MHTCET]

(a) $-e^{x-y}$

(b) e^{x-y}

(c) $-e^{y-x}$

(d) e^{y-x}

Q 33. If $xy = \tan^{-1}(xy) + \cot^{-1}(xy)$, then $\frac{dy}{dx}$ is equal to

[MHTCET]

(a) $\frac{y}{x}$

(b) $-\frac{y}{x}$

(c) $\frac{x}{y}$

(d) $-\frac{x}{y}$

Q 34. The derivative of $\cos^3 x$ w.r.t. $\sin^3 x$ is

[MHTCET]

(a) – cot x

(b) cot x

(c) tan x

(d) – tan x

Q 35. If $y = \log x^x$, then the value of $\frac{dy}{dx}$ is

[MPPET]

(a) $x^x(1 + \log x)$

(b) log (ex)

(c) $\log\left(\frac{e}{x}\right)$

(d) $\log\left(\frac{x}{e}\right)$

Q 36. The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$ is

[MPPET]

(a) $\sqrt{1-x^2}$

(b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{1}{2\sqrt{1-x^2}}$

(d) x

Q 37. If $x^y = e^{x-y}$, $x > 0$ then the value of $\frac{dy}{dx}$ at

$(1, 1)$ is

[RPET]

(a) 0

(b) $\frac{1}{2}$

(c) 1

(d) 2

Q 38. The value of differentiation of e^{x^2} with respect to e^{2x-1} at $x = 1$ is **[RPET]**

(a) e

(b) 0

(c) e^{-1}

(d) 1

Q 39. If $y = \log(\sin(x^2))$, $0 < x < \frac{\pi}{2}$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{\pi}}{2}$ is

[J&K CET]

(a) 0

(b) 1

(c) $\frac{\pi}{4}$

(d) $\sqrt{\pi}$

Q 40. $\frac{d}{dx}[\log_e e^{\sin(x^2)}]$ is equal to

[Guj. CET]

(a) $2\cos(x^2)$

(b) $2\cos x$

(c) $2x \cdot \cos x$

(d) $2x \cos(x^2)$

Q 41. $\frac{d}{dx}[\cos x°]$ is equal to

[Guj CET]

(a) $-\frac{\pi}{180}\sin x^{o}$

(b) – sin x°

(c) $\frac{\pi}{180}\sin x^{o}$

(d) $-\frac{\pi x}{180}\sin x$

Q 42. $\frac{d}{dx}[x^x + x^a + a^x + a^a] = \ldots.$, a is constant

[Guj CET]

(a) $x^x(1 + \log x) + a \cdot x^{a-1}$

(b) $x^x(1 + \log x) + a \cdot x^{a-1} + a^x \log a$

(c) $x^x(1 + \log x) + (a)\ (1 + \log a)$

(d) $x^x\ (1 + \log x) + a^a\ (1 + \log a) + ax^{a-1} + (a)\ (1 +$

$\log a)$

Q 43. If $y = \cot^{-1}\ (\cos 2x)^{1/2}$, then the value of

$\frac{dy}{dx}$ at $x = \frac{\pi}{6}$ will be

[UP SEE]

(a) $\left(\frac{2}{3}\right)^{1/2}$

(b) $\left(\frac{1}{3}\right)^{1/2}$

(c) $(3)^{1/2}$

(d) $(6)^{1/2}$

Q 44. The derivative of sin (x^3) w.r.t. cos (x^3) is

[KCET]

(a) – tan (x^3)

(b) tan (x^3)

(c) – cot (x^3)

(d) cot (x^3)

Q 45. If $y = \log_a x + \log_x a + \log_x x + \log_a a$,

then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{x} + x \log a$

(b) $\frac{\log a}{x} + \frac{x}{\log a}$

(c) $\frac{1}{x \log a} + x \log a$

(d) x log a

Q 46. If $f(x)=\frac{x-1}{4}+\frac{(x-1)^3}{12}+\frac{(x-1)^5}{20}+\frac{(x-1)^7}{28}+....,$ where 0 < x < 2, then f'(x) is equal to

[Kerala CEE]

(a) $\frac{1}{4x(2-x)}$

(b) $\frac{1}{4(x-2)^2}$

(c) $\frac{1}{2-x}$

(d) $\frac{1}{2+x}$

(e) $\frac{1}{(x-4)^2}$

Q 47. If $f(x + y) = 2f(x)\, f(y)$, $f'(5) = 1024(\log 2)$ and $f(2) = 8$, then the value of $f'(3)$ is

[Kerala CEE]

(a) $64\ (\log 2)$

(b) $128\ (\log 2)$

(c) 256

(d) $256\ (\log 2)$

(e) $1024\ (\log 2)$

Q 48. If $x = a\left\{\cos\theta + \log\tan\left(\frac{\theta}{2}\right)\right\}$ and y = a sin θ, then

$\frac{dy}{dx}$ is equal to

[AMU]

(a) cot θ

(b) tan θ

(c) sin θ

(d) cos θ

Q 49. The value of $\frac{dy}{dx}$ at x = $\frac{\pi}{2}$, where y is given by $y = x^{\sin x} + \sqrt{x}$, is

[WBJEE]

(a) $1+\frac{1}{\sqrt{2\pi}}$

(b) 1

(c) $\frac{1}{\sqrt{2\pi}}$

(d) $1-\frac{1}{\sqrt{2\pi}}$

Q 50. If $x^y = y^x$, then $x(x - y \log x)\ \frac{dy}{dx}$ is equal to

[BCECE, J&K CET]

(a) y(y – x log y)

(b) y(y + x log y)

(c) x(x + y log x)

(d) x(y – x log y)

Q 51. If $y = 1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + ...\infty$ with |x| > 1, then $\frac{dy}{dx}$ is

[Jamia Millia Islamia]

(a) $\frac{x^2}{y^2}$

(b) x^2y^2

(c) $\frac{y^2}{x^2}$

(d) $-\frac{y^2}{x^2}$

Q 52. The derivative of $a^{\sec x}$ w.r.t. $a^{\tan x}$ $(a > 0)$

is

[Jamia Millia Islamia]

(a) sec x $a^{\sec x - \tan x}$

(b) sin x $a^{\tan x - \sec x}$

(c) $\sin x\, a^{\sec x - \tan x}$

(d) $a^{\sec x - \tan x}$

Q 53. If $y = x^n \log x + x(\log x)^n$, then $\frac{dy}{dx}$ is equal to

[MHTCET]

(a) $x^{n-1}(1 + n \log x) + (\log x)^{n-1}\,[n + \log x]$

(b) $x^{n-2}(1 + n \log x) + (\log x)^{n-1}\,[n + \log x]$

(c) $x^{n-1}\,(1 + n \log x) + (\log x)^{n-1}\,[n - \log x]$

(d) None of the above

Q 54. $\frac{d}{dx}\sin^{-1}(x\sqrt{1-x}+\sqrt{x}\sqrt{1-x^2})$ is equal to

[Manipal]

(a) $-\frac{1}{2x\sqrt{1-x}}-\frac{1}{\sqrt{1-x^2}}$

(b) $\frac{1}{2x\sqrt{1-x}}-\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{1}{2\sqrt{x}\sqrt{1-x}}+\frac{1}{\sqrt{1-x^2}}$

(d) $-\frac{1}{2\sqrt{x}\sqrt{1-x}}+\frac{1}{\sqrt{1-x^2}}$

Q 55. The derivative of f(tan x) w.r.t g(sec x) at $x = \frac{\pi}{4}$, where f'(1) = 2 and g'($\sqrt{2}$) = 4, is

[Manipal]

(a) $\frac{1}{\sqrt{2}}$

(b) $\sqrt{2}$

(c) 1

(d) None of these

Q 56. If $y = \sqrt{\frac{1-x}{1+x}}$, then $(1 - x^2)\frac{dy}{dx} + y$ is equal to

[MPPET]

(a) 1

(b) -1

(c) 2

(d) 0

Q 57. A value of x in the interval $(1, 2)$ such that $f'(x) = 0$, where $f(x) = x^3 - 3x^2 + 2x + 10$ is

[RPET]

(a) $\frac{3+\sqrt{3}}{3}$

(b) $\frac{3+\sqrt{2}}{2}$

(c) $1+\sqrt{2}$

(d) $\sqrt{2}$

Q 58. If $y=\frac{3at^2}{1+t^3}, x=\frac{3at}{1+t^3}$, then $\frac{dy}{dx}$ is equal to

[J&K CET]

(a) $\frac{t(2-t^3)}{(1-2t^3)}$

(b) $\frac{t(2+t^3)}{(1-2t^3)}$

(c) $\frac{t(2-t^3)}{(1+2t^3)}$

(d) $\frac{t(2+t^3)}{(1+2t^3)}$

Q 59. If $\sec\left(\frac{x-y}{x+y}\right)=a$, then $\frac{dy}{dx}$ is

[Guj. CET]

(a) $\frac{y}{x}$

(b) $-\frac{y}{x}$

(c) $\frac{x}{y}$

(d) $-\frac{x}{y}$

Q 60. If $\phi(x)$ is the inverse of the function f(x) and $f'(x) = \frac{1}{1+x^5}$, then $\frac{d}{dx}\phi(x)$ is

[DCE]

(a) $\frac{1}{1+\{\phi(x)\}^5}$

(b) $\frac{1}{1+\{f(x)\}^5}$

(c) $1 + \{\phi(x)\}^5$

(d) $1 + f(x)$

Q 61. The derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is

[BITSAT]

(a) -1

(b) 1

(c) 2

(d) 4

Q 62. Differential coefficient of $\sqrt{\sec\sqrt{x}}$ is

[UP SEE]

(a) $\frac{1}{4\sqrt{x}}\sec\sqrt{x}\sin\sqrt{x}$

(b) $\frac{1}{4\sqrt{x}}(\sec\sqrt{x})^{3/2}.\sin\sqrt{x}$

(c) $\frac{1}{2}\sqrt{x}\sec\sqrt{x}\sin\sqrt{x}$

(d) $\frac{1}{2}\sqrt{x}(\sec\sqrt{x})^{3/2}.\sin\sqrt{x}$

Q 63. If $y = 2^{\log x}$, then $\frac{dy}{dx}$ is

[KCET]

(a) $\frac{2^{\log x}}{\log 2}$

(b) $2^{\log x} \cdot \log 2$

(c) $\frac{2^{\log x}}{x}$

(d) $\frac{2^{\log x} \cdot \log 2}{x}$

Q 64. If $\sec^{-1}\left(\frac{1+x}{1-y}\right) = a$, then $\frac{dy}{dx}$ is

[KCET]

(a) $\frac{y-1}{x+1}$

(b) $\frac{y+1}{x-1}$

(c) $\frac{x-1}{y-1}$

(d) $\frac{x-1}{y+1}$

Q 65. If $x=\frac{2at}{1+t^3}$ and $y=\frac{2at^2}{(1+t^3)^2}$, then $\frac{dy}{dx}$ is

[Kerala CEE]

(a) ax

(b) a^2x^2

(c) $\frac{x}{a}$

(d) $\frac{x}{2a}$

(e) 2a

Q 66. If $f(x) = (x - 2)(x - 4)(x - 6) \ldots (x - 2n)$,

then $f'(2)$ is

[Kerala CEE]

(a) $(-1)^n 2^{n-1} (n - 1)!$

(b) $(-2)^{n-1}(n-1)!$

(c) $(-2)^n n!$

(d) $(-1)^{n-1} 2^n (n - 1)!$

(e) $2^{n-1} (n - 1)!$

Q 67. If $2x^2 - 3xy + y^2 + x + 2y - 8 = 0$, then

$\frac{dy}{dx}$ is equal to

[EAMCET]

(a) $\frac{3y-4x-1}{2y-3x+2}$

(b) $\frac{3y+4x+1}{2y+3x+2}$

(c) $\frac{3y-4x+1}{2y-3x-2}$

(d) $\frac{3y-4x+1}{2y+3x+2}$

Q 68. If $y=\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}$, then $\frac{dy}{dx}$ is equal to

[OJEE]

(a) $\frac{1}{2}\cos ec^2\frac{x}{2}$

(b) $\frac{1}{2}\cos ec\frac{x}{2}$

(c) $\frac{1}{2}\cos ec^2 x$

(d) $\cos ec^2\frac{x}{2}$

Q 69. If $y = f\left(\frac{2x+3}{3-2x}\right)$ and f(x) = sin (log x), then $\frac{dy}{dx}$

is equal to

[OJEE]

(a) $\frac{12}{9-4x^2}\cos\left\{\log\left(\frac{2x+3}{3-2x}\right)\right\}$

(b) $\frac{12}{4x^2-9}\cos\left\{\log\left(\frac{2x+3}{3-2x}\right)\right\}$

(c) $\frac{12}{9-4x^2}\cos\left\{\log\left(\frac{3-2x}{2x+3}\right)\right\}$

(d) $\frac{12}{9-4x^2}\cos\left\{\log\left(\frac{2x+3}{2x-3}\right)\right\}$

Q 70. Let f(x) = e^x, g(x) = $\sin^{-1} x$ and h(x) = f[g(x)], then $\frac{h'(x)}{h(x)}$ is equal to

[WB JEE]

(a) $e^{\sin^{-1}x}$

(b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\sin^{-1} x$

(d) $\frac{1}{(1-x^2)}$

Q 71. If $y = \sin^{-1}\frac{x}{2} + \cos^{-1}\frac{x}{2}$, then the value of $\frac{dy}{dx}$ is

[BCECE]

(a) 1

(b) −1

(c) 0

(d) 2

Q 72. If y is a function of x and $\log(x + y) = 2xy$, then the value of $y'(0)$ is equal to

[BCECE]

(a) 1

(b) -1

(c) 2

(d) 0

Q 73. If $x = e^{y+e^{y+\ldots\ldots\infty}}$, $x > 0$ then $\frac{dy}{dx}$ is

[Jamia Millia Islamia]

(a) $\frac{x}{1+x}$

(b) $\frac{1}{x}$

(c) $\frac{1-x}{x}$

(d) $\frac{1+x}{x}$

Q 74. The derivative of log |x| is

[MHT CET]

(a) $\frac{1}{x}$, $x > 0$

(b) $\frac{1}{|x|}$, $x \neq 0$

(c) $\frac{1}{x}$, $x \neq 0$

(d) None of these

Q 75. If $y = \log_{\cos x} \sin x$, then $\frac{dy}{dx}$ is equal to

[MHTCET]

(a) $\frac{(\cot x \log \cos x + \tan x \log \sin x)}{(\log \cos x)^2}$

(b) $\frac{(\tan x \log \cos x + \cot x \log \sin x)}{(\log \cos x)^2}$

(c) $\frac{(\cot x \log \cos x + \tan x \log \sin x)}{(\log \sin x)^2}$

(d) None of the above

Q 76. Derivative of $\log_{10} x$ with respect to x^2 is

[J & K CET]

(a) $2x^2 \log_e 10$

(b) $\frac{\log_{10} e}{2x^2}$

(c) $\frac{\log_e 10}{2x^2}$

(d) $x^2 \log_e 10$

Q 77. The differential coefficient of f(log x), where f(x) = log x is

[Guj. CET]

(a) $\frac{x}{\log x}$

(b) $(x \log x)^{-1}$

(c) $\frac{\log x}{x}$

(d) None of these

Q 78. If $y = \cos^{-1}\left(\frac{1-\log x}{1+\log x}\right)$, then $\frac{dy}{dx}$ at x = e is

[DCE]

(a) $-\frac{1}{e}$

(b) $-\frac{1}{2e}$

(c) $\frac{1}{2e}$

(d) $\frac{1}{e}$

Q 79. For $0 < x < 2$, $\frac{d}{dx}\left(\tan^{-1}\sqrt{\frac{1+\cos\frac{x}{2}}{1-\cos\frac{x}{2}}}\right)$ is equal to

[UP SEE, WB JEE]

(a) $-1/4$

(b) $1/4$

(c) $-1/2$

(d) $1/2$

Q 80. If f(x) and g(x) are two functions with g(x) $= x - \frac{1}{x}$ and fog (x) $= x^3 - \frac{1}{x^3}$, then f'(x) is

[KCET]

(a) $3x^2 + 3$

(b) $x^2 - \frac{1}{x^2}$

(c) $1 + \frac{1}{x^2}$

(d) $3x^2 + \frac{3}{x^4}$

Q 81. Let f be twice differentiable function such that $f''(x) = -f(x)$, and $f'(x) = g(x)$, $h(x) = \{f(x)\}^2 + \{g(x)\}^2$. If $h(5) = 11$, then $h(10)$ is equal to

[Kerala CEE]

(a) 22

(b) 11

(c) 0

(d) 20

(e) None

Q 82. A differentiable function f(x) is defined for all $x > 0$ and satisfies $f(x^3) = 4x^4$ for all $x > 0$. The value of $f'(8)$ is

[Kerala CEE]

(a) $\frac{16}{3}$

(b) $\frac{32}{3}$

(c) $\frac{16\sqrt{2}}{3}$

(d) $\frac{32\sqrt{2}}{3}$

(e) $\frac{32(2)^{1/3}}{3}$

Q 83. If $y=\left(1+\frac{1}{x}\right)\left(1+\frac{2}{x}\right)\left(1+\frac{3}{x}\right)....\left(1+\frac{n}{x}\right)$ and $x \neq 0$, then

$\frac{dy}{dx}$ when $x = -1$ is

[Kerala CEE]

(a) $n!$

(b) $(n-1)!$

(c) $(-1)^n (n-1)!$

(d) $(-1)^n n!$

(e) $(n+1)!$

Q 84. If $\sin y = x \sin(a+y)$, then $\frac{dy}{dx}$ is

[OJEE]

(a) $\frac{\sin a}{\sin^2(a+y)}$

(b) $\frac{\sin^2(a+y)}{\sin a}$

(c) sin a $\sin^2$ (a + y)

(d) $\frac{\sin^2(a+y)}{\sin a}$

Q 85. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to

[Jamia Millia Islamia]

(a) $\frac{(2^x+2^y)}{(2^x-2^y)}$

(b) $\frac{(2^x+2^y)}{(1+2^{x+y})}$

(c) $2^{x-y}\left(\frac{2^y-1}{1-2^x}\right)$

(d) $\frac{2^{x+y}-2^{x}}{2^{y}}$

Q 86. If $y = 5^x x^5$, then $\frac{dy}{dx}$ is

[MHTCET]

(a) $5^x (x^5 \log 5 - 5x^4)$

(b) $x^5 \log 5 - 5x^4$

(c) $x^5 \log 5 + 5x^4$

(d) $5^x (x^5 \log 5 + 5x^4)$

Q 87. If $y = \tan^{-1}\left(\frac{a\cos x - b\sin x}{b\cos x + a\sin x}\right)$, then $\frac{dy}{dx}$ is equal to

[MHT CET]

(a) 2

(b) -1

(c) $\frac{a}{b}$

(d) 0

Q 88. Let $y = x^{x^{x^{\cdots\cdots}}}$, then $\frac{dy}{dx}$ is equal to

[MPPET]

(a) yx^{y-1}

(b) $\frac{y^2}{x(1-y\log x)}$

(c) $\frac{y}{x(1+y\log x)}$

(d) None of these

Q 89. If $x = y\sqrt{1-y^2}$, then $\frac{dy}{dx}$ is equal to

[RPET]

(a) yx^{y-1}

(b) $\frac{y^2}{x(1-y\log x)}$

(c) $\frac{y}{x(1+y\log x)}$

(d) None of these

Q 90. If $y = \frac{e^x + e^{-x}}{e^x - e^{-x}}$, then $\frac{dy}{dx}$ is equal to

[Guj. CET]

(a) sec h^2x

(b) cosec h^2x

(c) – sec h^2x

(d) – cosec h^2x

Q 91. If $r = [2\phi + \cos^2(2\phi + \pi/4)]^{1/2}$, then what is the value of the derivative of $dr/d\phi$ at $\phi = \pi/4$?

[BITSAT, OJEE]

(a) $2\left(\frac{1}{\pi+1}\right)^{1/2}$

(b) $2\left(\frac{2}{\pi+1}\right)^{2}$

(c) $\left(\frac{2}{\pi+1}\right)^{1/2}$

(d) $2\left(\frac{2}{\pi+1}\right)^{1/2}$

Q 92. If $y = \sec^{-1}\frac{x+1}{x-1} + \sin^{-1}\frac{x-1}{x+1}$, then $\frac{dy}{dx}$ is

[Kerala CEE]

(a) 1

(b) 0

(c) $\frac{x-1}{x+1}$

(d) $\frac{x+1}{x-1}$

(e) $\frac{x^2+1}{x^2-1}$

Q 93. The derivative of $y = (1 - x)(2 - x) \ldots (n - x)$ at $x = 1$ is equal to

[Kerala CEE]

(a) 0

(b) $(-1)(n-1)!$

(c) $n! - 1$

(d) $(-1)^{n-1}\,(n-1)!$

(e) $(-1)^{n}\,(n-1)!$

Q 94. Let $f(x)$ be a polynomial function of the second degree. If $f(1) = f(-1)$ and a_1, a_2, a_3 are in

AP, then $f'(a_1)$, $f'(a_2)$, $f'(a_3)$ are in

[AMU]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 95. If $f(x) = \cos x \cos 2x \cos 4x \cos 8x \cos 16x$, then $f'\left(\frac{\pi}{4}\right)$ is

[AMU]

(a) $\sqrt{2}$

(b) $\frac{1}{\sqrt{2}}$

(c) 0

(d) $\frac{\sqrt{3}}{2}$

Q 96. If $x\sqrt{1+y}+y\sqrt{1+x}=0$, then $\frac{dy}{dx}$ is equal to

[EAMCET]

(a) $\frac{1}{(1+x)^2}$

(b) $-\frac{1}{(1+x)^2}$

(c) $\frac{1}{1+x^2}$

(d) $\frac{1}{1-x^2}$

Q 97. $\frac{d}{dx}\sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$ is equal to

[BCECE]

(a) $\sec^2 x$

(b) $-\sec^2\left(\frac{\pi}{4}-x\right)$

(c) $\sec^2\left(\frac{\pi}{4}+x\right)$

(d) $\sec^2\left(\frac{\pi}{4}-x\right)$

Q 98. If $y = x^2 + \cfrac{1}{x^2 + \cfrac{1}{x^2 + \cfrac{1}{x^2 + ...\infty}}}$, then $\frac{dy}{dx}$ is equal to

[BCECE]

(a) $\frac{2xy}{2y-x^2}$

(b) $\frac{xy}{y+x^2}$

(c) $\frac{xy}{y-x^2}$

(d) $\frac{2x}{2+\frac{x^2}{y}}$

Q 99. If $f(x)=\frac{1}{1-x}$, then the derivative of the composite function f[f{f(x)}] is equal to

[Jamia Millia Islamia]

(a) 0

(b) $\frac{1}{2}$

(c) 1

(d) 2

Q 100. If $y = a \sin^3 \theta$ and $x = a \cos^3 \theta$, then at $\theta = \frac{\pi}{3}, \frac{dy}{dx}$ is equal to

[MHTCET]

(a) $\frac{1}{\sqrt{3}}$

(b) $-\sqrt{3}$

(c) $\frac{-1}{\sqrt{3}}$

(d) $\sqrt{3}$

Q 101. If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + ...\infty}}}$, then $(2y-1)\frac{dy}{dx}$ is equal to

[Kerala CEE, MHTCET]

(a) sin x

(b) – cos x

(c) cos x

(d) – sin x

Q 102. For $|x| < 1$, let $y = 1 + x + x^2 + \ldots$ to ∞,

then $\frac{dy}{dx} - y$ is

[MPPET]

(a) $\frac{x}{y}$

(b) $\frac{x^2}{y^2}$

(c) $\frac{x}{y^2}$

(d) xy^2

Q 103. If $y = \log^n x$, where $\log^n$ means log log log ..... (repeated n times), then

$x \log x \log^2 x \log^3 x \ldots\ldots \log^{n-1} x \log^n x \frac{dy}{dx}$ is equal to

[MP PET]

(a) log x

(b) x

(c) $\frac{1}{\log x}$

(d) $\log^n x$

Q 104. If $\sec\left(\frac{x^2-y^2}{x^2+y^2}\right) = e^a$, then $\frac{dy}{dx}$ is equal to

[RPET]

(a) $\frac{y^2}{x^2}$

(b) $\frac{y}{x}$

(c) $\frac{x}{y}$

(d) $\frac{x^2-y^2}{x^2+y^2}$

Q 105. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, then $\frac{dy}{dx}$ is equal to

[RPET]

(a) $\frac{x}{y}$

(b) $-\frac{x}{y}$

(c) $\frac{y}{x}$

(d) $-\frac{y}{x}$

Q 106. If $x^m y^n = (x + y)^{m+n}$, then $(dy/dx)_{x=1,\ y=2}$ is equal to

[J&KCET]

(a) $\cot\frac{\theta}{2}$

(b) $\tan\frac{\theta}{2}$

(c) $\frac{1}{2}\cos ec^2\frac{\theta}{2}$

(d) $-\frac{1}{2}\cos ec^2\frac{\theta}{2}$

Q 107. If $x = a(\theta - \sin\theta)$, $y = a(1 - \sin\theta)$, then

$\frac{dy}{dx}$ is equal to

[J&K CET]

(a) $\cos\frac{\theta}{2}$

(b) $\tan\frac{\theta}{2}$

(c) $\frac{1}{2}\cos ec^2\frac{\theta}{2}$

(d) $-\frac{1}{2}\cos ec^2\frac{\theta}{2}$

Q 108. If f(x) = x $\tan^{-1}$ x, then f'(1) is equal to

[DCE]

(a) $\frac{1}{2}+\frac{\pi}{4}$

(b) $-\frac{1}{2}+\frac{\pi}{4}$

(c) $-\frac{1}{2}-\frac{\pi}{4}$

(d) $\frac{1}{2}-\frac{\pi}{4}$

Q 109. If $y = \tan^{-1}(\sec x - \tan x)$, then $\frac{dy}{dx}$ is

[KCET]

(a) 2

(b) -2

(c) $\frac{1}{2}$

(d) $-\frac{1}{2}$

Q 110. If $y = e^{(1/2)\log(1+\tan^2 x)}$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{2}\sec^2 x$

(b) $\sec^2 x$

(c) sec x tan x

(d) $e^{1/2\log(1+\tan^2 x)}$

(e) $e^{1/2\log(1+\tan^2 x)} \cdot \frac{1}{2} \cdot \frac{1}{(1+\tan^2 x)}$

Q 111. If $y = x + x^2 + x^3 + \ldots \infty$, where $|x| < 1$, then for $|y| < 1$, $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $y + y^2 + y^3 + \ldots \infty$

(b) $1 - y + y^2 - y^3 + \ldots \infty$

(c) $1 - 2y + 3y^2 - \ldots \infty$

(d) $1 + 2y + 3y^2 + \ldots \infty$

(e) $y - y^2 + y^3 - \ldots \infty$

Q 112. If f'(x) = sin (log x) and $y = f\left(\frac{2x+3}{3-2x}\right)$, then

$\frac{dy}{dx}$ at x = 1 is equal to

[Kerala CEE]

(a) 6 sin log (5)

(b) 5 sin log (6)

(c) 12 sin log (5)

(d) 5 sin log (12)

(e) 12 sin log (6)

Q 113. If $f(x)=\sqrt{1+\cos^2(x^2)}$, then $f'\left(\frac{\sqrt{\pi}}{2}\right)$ is

[OJEE]

(a) $\frac{\sqrt{\pi}}{6}$

(b) $-\sqrt{\frac{\pi}{6}}$

(c) $\frac{1}{\sqrt{6}}$

(d) $\frac{\pi}{\sqrt{6}}$

Q 114. If $x = a(\cos\theta + \theta\sin\theta)$ and $y = a(\sin\theta - \theta\cos\theta)$, then $\frac{dy}{dx}$ is equal to

[BCECE]

(a) cos θ

(b) tan θ

(c) sec θ

(d) cosec θ

Q 115. Differential coefficient of $\sec^{-1}\frac{1}{2x^2-1}$ with respect to $\sqrt{1-x^2}$ at $x=\frac{1}{2}$ is equal to

[Jamia Millia Islamia]

(a) 2

(b) 4

(c) 6

(d) 1

Q 116. The derivative of $\left[\frac{e^x+1}{e^x}\right]$ is equal to

[MHTCET]

(a) 0

(b) $\frac{1}{e^x}$

(c) $-\frac{1}{e^x}$

(d) e^x

Q 117. If $x = \log(1 + t^2)$ and $y = t - \tan^{-1} t$.

Then, $\frac{dy}{dx}$ is equal to

[MHTCET]

(a) $e^x - 1$

(b) $t^2 - x$

(c) $\frac{\sqrt{e^x - 1}}{2}$

(d) $e^x - y$

Q 118. Derivative of sin x w.r.t. cos x is

[MHTCET]

(a) cos x

(b) cot x

(c) – cot x

(d) tan x

Q 119. $\frac{d}{dx}(\log x)^4$ is equal to

[MHTCET]

(a) $4 \log x^3$

(b) $4(\log x)^3$

(c) $\frac{4(\log x)^3}{x}$

(d) $\frac{(4 \log x)^3}{x}$

Q 120. If $y = (\cos x^2)^2$, then $\frac{dy}{dx}$ is equal to

[RPET]

(a) $-4x \sin 2x^2$

(b) $-x \sin x^2$

(c) $-2x \sin 2x^2$

(d) $-x \cos 2x^2$

Q 121. If $f(x) = \log_x (\log_e x)$, then $f'(x)$ at $x = e$ is equal to

[Kerala CEE]

(a) 1

(b) 2

(c) 0

(d) e

(e) $\frac{1}{e}$

Q 122. If y = cos (sin x^2), then at $x = \sqrt{\frac{\pi}{2}}, \frac{dy}{dx}$ is equal to

[AMU]

(a) -2

(b) 2

(c) $-2\sqrt{\frac{\pi}{2}}$

(d) 0

Q 123. If $y=\sqrt{\log x+\sqrt{\log x+\sqrt{\log x+\sqrt{\log x+.....\infty}}}}$, then $\frac{dy}{dx}$

is equal to

[AMU]

(a) $\frac{x}{2y-1}$

(b) $\frac{x}{2y+1}$

(c) $\frac{1}{x(2y-1)}$

(d) $\frac{1}{x(1-2y)}$

Differentiation by Substitution, Higher Order Derivatives

Q 1. If x = a $(1 + \cos\theta)$, y = a $(\theta + \sin\theta)$, then

$\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{2}$ is

[Kerala CEE]

(a) $-\frac{1}{a}$

(b) $\frac{1}{a}$

(c) -1

(d) -2

(e) $-\frac{2}{a}$

Q 2. The derivative of $\sin^{-1}(2x\sqrt{1-x^2})$ with respect to $\sin^{-1}(3x - 4x^3)$ is

[Kerala CEE]

(a) $\frac{2}{3}$

(b) $\frac{3}{2}$

(c) $\frac{1}{2}$

(d) 1

(e) 0

Q 3. If $y = \sin^{-1}\left(\frac{5x + 12\sqrt{1-x^2}}{13}\right)$, then $\frac{dy}{dx}$ is equal to

[DCE]

(a) $-\frac{1}{\sqrt{1-x^2}}$

(b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{3}{\sqrt{1-x^2}}$

(d) $\frac{x}{\sqrt{1-x^2}}$

Q 4. If $f(x) = 1 + nx + \frac{n(n-1)}{2}x^2 + \frac{n(n-1)(n-2)}{6}x^3 + \ldots +$ x^n, then $f''(1)$ is equal to

[KCET]

(a) $n(n-1)2^{n-1}$

(b) $(n-1)2^{n-1}$

(c) $n(n-1)2^{n-2}$

(d) $n(n-1)2^n$

Q 5. $y = e^{a\sin^{-1}x} \Rightarrow (1-x^2)\, y_{n+2} - (2n+1)xy_{n+1}$ is equal to

[EAMCET]

(a) $-(n^2 + a^2)y_n$

(b) $(n^2 - a^2)y_n$

(c) $(n^2 + a^2)\, y_n$

(d) $-(n^2 - a^2)y_n$

Q 6. The second order derivative of $a \sin^3 t$ with respect to $a \cos^3 t$ at $t = \frac{\pi}{4}$ is

[WBJEE]

(a) 2

(b) $\frac{1}{12a}$

(c) $\frac{4\sqrt{2}}{3a}$

(d) $\frac{3a}{4\sqrt{2}}$

Q 7. Let f be a twice differentiable function such that $f''(x) = -f(x)$ and $f'(x) = g(x)$. If $h'(x) = [f(x)^2 + g(x)^2]$ $h(1) = 8$ and $h(0) = 2$, then $h(2)$ is equal to

[Jamia Millia Islamia]

(a) 1

(b) 2

(c) 3

(d) None of these

Q 8. Let $x = \log_e t$, $t > 0$ and $y + 1 = t^2$. Then, $\frac{d^2x}{dy^2}$ is equal to

[J&KCET]

(a) $4e^{2x}$

(b) $-\frac{1}{2}e^{-4x}$

(c) $-\frac{3}{4}e^{5x}$

(d) $4e^{x}$

Q 9. Let g(x) = log f(x), where f(x) is a twice differentiable positive function on $(0, \infty)$ such that f(x + 1) = xf(x). Then, for N = $1, 2, 3, \ldots$,

$g''\left(N+\frac{1}{2}\right)-g''\left(\frac{1}{2}\right)$ is equal to

[IIT JEE]

(a) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N-1)^2}\right\}$

(b) $4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N-1)^2}\right\}$

(c) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N+1)^2}\right\}$

(d) $4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N+1)^2}\right\}$

Q 10. If f(x) = be^{ax} + ae^{bx}, then $f''(0)$ is equal to

e**[KCET]**

(a) 0

(b) 2ab

(c) ab(a + b)

(d) ab

Q 11. Let $f(x) = \sin x$, $g(x) = x^2$ and $h(x) = \log_e x$. If $F(x) = (hogof)(x)$, then $F''(x)$ is equal to

[Kerala CEE]

(a) a $csoec^3 x$

(b) $2 \cot x^2 - 4x^2 \operatorname{cosec}^2 x^2$

(c) $2x \cot x^2$

(d) $-2 \operatorname{cosec}^2 x$

(e) $4 \operatorname{cosec}^2 x$

Q 12. If $y = \sin(\log_e x)$, then $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx}$ is equal to

[EAMCET]

(a) $\sin(\log_e x)$

(b) $\cos(\log_e x)$

(c) y^2

(d) $-y$

Q 13. If $y = e^{ax} \sin bx$, then $\frac{d^2y}{dx^2} - 2a\frac{dy}{dx} + a^2y$ is equal to

[OJEE]

(a) 0

(b) 1

(c) $-b^2y$

(d) $-by$

Q 14. If $x = e^t \sin t$, $y = e^t \cot t$, then $\frac{d^2y}{dx^2}$ at $x = \pi$ is

[WB JEE]

(a) $2e^{\pi}$

(b) $\frac{1}{2}e^{\pi}$

(c) $\frac{1}{2e^{\pi}}$

(d) $\frac{2}{e^{\pi}}$

Q 15. $f(x) = e^x \sin x$, then $f''(x)$ is equal to

[BCECE]

(a) $e^{6x} \sin 6x$

(b) $2e^x \cos x$

(c) $8e^x \sin x$

(d) $8e^x \cos x$

Q 16. $\frac{d^2x}{dy^2}$ equals

[IIT JEE]

(a) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

(b) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-1}$

(c) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$

(d) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

Q 17. If $y = \cos^2\frac{3x}{2} - \sin^2\frac{3x}{2}$, then $\frac{d^2y}{dx^2}$ is

[KCET]

(a) $-3\sqrt{1-y^2}$

(b) $9y$

(c) $-9y$

(d) $3\sqrt{1-y^2}$

Q 18. $x = \cos\theta, y = \sin 5\theta \Rightarrow (1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx}$

is

[EAMCET]

(a) $-5y$

(b) $5y$

(c) $25y$

(d) $-25y$

Q 19. If $y = a^x.b^{2x-1}$, then $\frac{d^2y}{dx^2}$ is

[WB JEE]

(a) $y^2 \log ab^2$

(b) $y \log ab^2$

(c) y^2

(d) $y(\log ab^2)^2$

Q 20. $\frac{d^n}{dx^n}(\log x)$ is equal to

[WB JEE]

(a) $\frac{(n-1)!}{x^n}$

(b) $\frac{n!}{x^n}$

(c) $\frac{(n-2)!}{x^n}$

(d) $(-1)^{n-1}\frac{(n-1)!}{x^n}$

Q 21. If $y^2 = ax^2 + bx + c$, where a, b, c are constants, then $y^3 \frac{d^2y}{dx^2}$ is equal to

[MHTCET]

(a) a constant

(b) a function of x

(c) a function of y

(d) a function of x and y

Q 22. If $y^2 = ax^2 + bx + c$, where a, b, c are constants, then $y^3 \frac{d^2y}{dx^2}$ is equal to

[MHTCET]

(a) a constant

(b) a function of x

(c) a function of y

(d) a function of x and y both

Q 23. If $f''(x) = -f(x)$, where $f(x)$ is a continuous double differentiable function and $g(x) = f'(x)$.

If $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ and $f(5) = 5$, then $F(10)$ is

[IIT JEE]

(a) 0

(b) 5

(c) 10

(d) 25

Q 24. If $x = \sin t$, $y = \cos pt$, then

[BITSAT]

(a) $(1 - x^2)y_2 + xy_1 + p^2y = 0$

(b) $(1 - x^2)y_2 + xy_1 - p^2y = 0$

(c) $(1 + x^2)y_2 - xy_1 + p^2y = 0$

(d) $(1 - x^2)y_2 - xy_1 + p^2y = 0$

Q 25. If $y = \tan^{-1}\frac{\sqrt{1+x^2}-\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}}$, then $\frac{dy}{dx}$ is equal to

[BITSAT]

(a) $\frac{x^2}{\sqrt{1-x^4}}$

(b) $\frac{x^2}{\sqrt{1+x^4}}$

(c) $\frac{x}{\sqrt{1+x^4}}$

(d) $\frac{x}{\sqrt{1-x^4}}$

Q 26. If $y = x\log\left(\frac{x}{a+bx}\right)$, then $\frac{x^3 d^2y}{dx^2}$ is equal to

[UP SEE]

(a) $x\frac{dy}{dx} - y$

(b) $\left(x\frac{dy}{dx} - y\right)^2$

(c) $y\frac{dy}{dx} - x$

(d) None of these

Q 27. If sin (x + y) + cos (x + y) = log (x + y), then $\frac{d^2y}{dx^2}$ is

[KCET]

(a) $\frac{-y}{x}$

(b) 0

(c) −1

(d) 1

Q 28. Derivative of $\sec^{-1}\left(\frac{1}{1-2x^2}\right)$ w.r.t $\sin^{-1}(3x - 4x^3)$ is

[Kerala CEE]

(a) $\frac{1}{4}$

(b) $\frac{3}{2}$

(c) 1

(d) $\frac{2}{3}$

(e) $-2/3$

Q 29. If x = A cos 4t + B sin 4t, then $\frac{d^2x}{dt^2}$ is

equal to

[AMU]

(a) $-16x$

(b) $16\ x$

(c) x

(d) –x

Q 30. If $y = (x + \sqrt{1+x^2})^n$, then $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is

[OJEE]

(a) n^2y

(b) $-n^2y$

(c) $-y$

(d) $2x^2y$

Q 31. Let $y = t^{10} + 1$ and $x = t^8 + 1$, then $\frac{d^2y}{dx^2}$ is equal to

[MP PET]

(a) $\frac{5}{2}t$

(b) $20\ t^8$

(c) $\frac{5}{16t^6}$

(d) None of these

Q 32. $\frac{d}{dx}\left\{\tan^{-1}\left(\frac{2x}{1-x^2}\right)+\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)-\tan^{-1}\left(\frac{4x-4x^3}{1-6x^2+x^4}\right)\right\}$ is

equal to

[J&K CET]

(a) $\frac{1}{\sqrt{1-x^2}}$

(b) $-\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{1}{1+x^2}$

(d) $-\frac{1}{1+x^2}$

Q 33. The value of $\frac{d}{dx}\left[\tan^{-1}\left(\frac{\sqrt{x}(3-x)}{1-3x}\right)\right]$ is

[Kerala CEE]

(a) $\frac{1}{2(1+x)\sqrt{x}}$

(b) $\frac{3}{(1+x)\sqrt{x}}$

(c) $\frac{2}{(1+x)\sqrt{x}}$

(d) $\frac{3}{2(1-x)\sqrt{x}}$

(e) $\frac{3}{2(1+x)\sqrt{x}}$

Q 34. If $f : R \to R$ is an even function which is twice differentiable on R and $f''(\pi) = 1$, then $f''(-\pi)$ is equal to

(a) -1

(b) 0

(c) 1

(d) 2

[MHTCET]

Q 35. Observe the following statements

I. $f(x) = ax^{41} + bx^{-40} \Rightarrow \frac{f''(x)}{f(x)} = 1640x^{-2}$

II. $\frac{d}{dx}\tan^{-1}\left(\frac{2x}{1-x^2}\right) = \frac{1}{1+x^2}$

[EAMCET]

Which of the following is correct ?

(a) I is true, but II is false

(b) Both I and II are true

(c) Neither I nor II is true

(d) I is false, but II is true

Q 36. If $f(x) = 10 \cos x + (13 + 2x) \sin x$, then

$f''(x) + f(x)$ is equal to

[EAMCET]

(a) cos x

(b) 4 cos x

(c) sin x

(d) 4 sin x

Q $37.$ nth derivative of $(x + 1)^n$ is equal to

[MHTCET]

(a) $(n - 1)!$

(b) $(n + 1)!$

(c) $n!$

(d) $n[(n + 1)]^{n-1}$

Q 38. If $y = x^2 e^{mx}$, where m is a constant, then

$\frac{d^3y}{dx^3}$ is equal to

[MPPET]

(a) $me^{mx} (m^2x^2 + 6mx + 6)$

(b) $2m^3 xe^{mx}$

(c) $me^{mx} (m^2x^2 + 2mx + 2)$

(d) None of these

Q 39. If $y = 1 - x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} - \ldots.$, then $\frac{d^2y}{dx^2}$ is equal to

[RPET]

(a) –x

(b) x

(c) y

(d) –y

Q 40. If $y = 2^x \cdot 3^{2x-1}$, then $\frac{d^2y}{dx^2}$ is equal to

[Kerala CEE]

(a) (log 2) (log 3)

(b) (log 18)

(c) $(\log 18^2)y^2$

(d) (log 18)y

(e) $(\log 18)^2 y$

Q 41. If $f : R \to R$ is an even function having derivatives of all orders, then an odd function among the following is

[EAMCET]

(a) f''

(b) f'''

(c) $f' + f''$

(d) $f'' + f'''$

Q 42. $f(x)=\begin{vmatrix} x^3 & x^4 & 3x^2 \\ 1 & -6 & 4 \\ p & p^2 & p^3 \end{vmatrix}$, here p is a constant, then

$\frac{d^4 f(x)}{dx^4}$ is

[BCECE]

(a) proportional to x^2

(b) proportional to x

(c) proportional to x^3

(d) a constant

Q 43. f(x) is a polynomial of degree 2, $f(0) = 4$, $f'(0) = 3$ and $f''(0) = 4$, then $f(-1)$ is equal to

[MHTCET]

(a) 3

(b) -2

(c) 2

(d) -3

Q 44. If $f(x) = x^n$, then the value of

$$f(1) = \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + ... + \frac{(-1)^n f^n(1)}{n!}$$ is

[AIEEE]

(a) 2^n

(b) 2^{n-1}

(c) 0

(d) 1

Q 45. If $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{2}\right)$, then $y'(0)$ is

[DCE]

(a) $1/2$

(b) 0

(c) 1

(d) does not exist

Topic 3

Partial Derivative

Q 1. $z = \tan(y + ax) + \sqrt{y - ax} \Rightarrow z_{xx} - a^2 z_{yy}$ is equal to

[EAMCET]

(a) 0

(b) 2

(c) $z_x + z_y$

(d) $z_x z_y$

Q 2. If $z = \sec^{-1}\left(\frac{x^4 + y^4 - 8x^2y^2}{x^2 + y^2}\right)$, then $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}$ is equal to

[EAMCET]

(a) cot z

(b) 2 cot z

(c) 2 tan z

(d) 2 sec z

Q 3. If z = log (tan x + tan y), then (sin 2x)

$\frac{\partial z}{\partial x}+(\sin 2y)\frac{\partial z}{\partial y}$ is equal to

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 4. If $x^x\ y^y\ z^z = c$, then $\frac{\partial z}{\partial x}$ is equal to

[OJEE]

(a) $\left(\frac{1+\log x}{1+\log z}\right)$

(b) $-\left(\frac{1+\log x}{1+\log z}\right)$

(c) $\left(\frac{1+\log z}{1+\log x}\right)$

(d) None of these

Q 5. If $(x + y)\sin u = x^2y^2$, then $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}$ is equal to

[VITEEE]

(a) sin u

(b) cosec u

(c) 2 tan u

(d) 3 tan u

Q 6. If $u = \sin^{-1}\left(\frac{x^2 + y^2}{x + y}\right)$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to

[EAMCET]

(a) sin u

(b) tan u

(c) cos u

(d) cot u

Q 7. If $f(x, y) = \frac{\cos(x-4y)}{\cos(x+4y)}$, then $\left.\frac{\partial f}{\partial x}\right|_{y=\frac{\pi}{2}}$ is equal to

[EAMCET]

(a) -1

(b) 0

(c) 1

(d) 2

Q 8. If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, then the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 9. If $f(x, y) = 2(x - y)^2 - x^4 - y^4$, then $|(f_{xx}f_{yy} - f^2_{xy})|_{(0, 0)}$ is

[EAMCET]

(a) 32

(b) 16

(c) 0

(d) −1

Q 10. If $z = y + f(v)$, where $v = \left(\frac{x}{y}\right)$, then $v\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$ is

[OJEE]

(a) -1

(b) 1

(c) 0

(d) 2

Q 11. If $u(x, y) = y \log x + x \log y$, then $u_x u_y - u_x \log x - u_y \log y + \log x \log y$ is equal to

[EAMCET]

(a) 0

(b) -1

(c) 1

(d) 2

Q 12. If $u = \log\left(\frac{x^2 + y^2}{x + y}\right)$, then the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is

[OJEE]

(a) -1

(b) 0

(c) 1

(d) 2

Answers

Topic 1 : Differentiation of Algebraic and Transcendental Functions

1. (b) 2. (c) 3. (c) 4. (d) 5. (a) 6. (a) 7. (d) 8. (a)

9. (*) 10. (c) 11. (c) 12. (d) 13. (d) 14. (e) 15. (e)

16. (d) 17. (d) 18. (b) 19. (e) 20. (b) 21. (d) 22.

(c) 23. (b) 24. (c) 25. (a) 26. (a) 27. (c) 28. (b)

29. (b) 30. (b) 31. (a) 32. (c) 33. (b) 34. (a) 35.

(b) 36. (c) 37. (a) 38. (d) 39. (d) 40. (d), 41. (a)

42. (b) 43. (a) 44. (c) 45. (e) 46. (a) 47. (a) 48.

(b) 49. (a) 50. (a) 51. (d) 52. (c) 53. (a) 54. (c)

55. (a) 56. (d) 57. (a) 58. (a) 59. (a) 60. (c)

61. (b) 62. (b) 63. (d) 64. (a) 65. (c) 66. (b) 67.

(a) 68. (a) 69. (a) 70. (b) 71. (c) 72. (a) 73. (c)

74. (c) 75. (a) 76. (b) 77. (b) 78. (c) 79. (a) 80.

(a) 81. (b) 82. (b) 83. (c) 84. (b) 85. (c) 86. (d)

87. (b) 88. (b) 89. (b) 90. (d) 91. (d) 92. (b) 93.

(b) 94. (a) 95. (c) 96. (b) 97. (b) 98. (a) 99. (c)

100. (b) 101. (c) 102. (d) 103. (d) 104. (b) 105.

(b) 106. (b) 107. (a) 108. (a) 109. (d) 110. (c)

101. (c) 112. (c) 113. (b) 114. (b) 115. (b) 116.

(c) 117. (c) 118. (c) 119. (c) 120. (c) 121. (e)

122. (d) 123. (c)

Topic 2 : Differentiation by Substitution, Higher Order Derivatives

1. (a) 2. (a) 3. (b) 4. (c) 5. (c) 6. (c) 7. (d) 8. (b)

9. (a) 10. (c) 11. (d) 12. (d) 13. (c) 14. (d) 15.

(b> 16. (d) 17. (c) 18. (d) 19. (d) 20. (d)

21. (a) 22. (b) 23. (b) 24. (d) 25. (d) 26. (b) 27.

(b) 28. (d) 29. (a) 30. (a) 31. (c) 32. (c) 33. (e)

34. (c) 35. (a) 36. (b) 37. (c) 38. (a) 39. (c) 40.

(e) 41. (b) 42. (d) 43. (a) 44. (c) 45. (a)

Topic 3 : Partial Derivative

1. (a) 2. (b) 3. (b) 4. (b) 5. (d) 6. (b) 7. (b) 8. (a)

9. (c) 10. (b) 11. (c) 12. (c)

Application of Derivatives

Topic 1 Rate of Measure and Application in Mechanics

Q 1. An edge of a variable cube is increasing at the rate of 10 cm/s. How fast the volume of the cube will increase when the edge is 5 cm long?

[Kerala CEE]

(a) $750 cm^3/s$

(b) $75 cm^3/s$

(c) $300 cm^3/s$

(d) $150 cm^3/s$

(e) $25\ cm^3\ /\ s$

Q 2. If the error committed in measuring the radius of the circle is 0.05%, then the corresponding error in calculating the area is

[Kerala CEE]

(a) 0.05%

(b) 0.0025%

(c) 025%

(d) 0.1%

(e) 0.2%

Q 3. The smallest circle with centre on y−axis and passing through the point $(7, 3)$ has radius

[UP SEE]

(a) $\sqrt{58}$

(b) 7

(c) 3

(d) 4

Q 4. If sum of two numbers is 6, the minimum value of the sum of their reciprocals is

[UP SEE]

(a) $\frac{6}{5}$

(b) $\frac{3}{4}$

(c) $\frac{2}{3}$

(d) $\frac{1}{2}$

Q 5. A stone is thrown vertically upwards from the top of a tower 64 m high according to the law s $= 48t - 16t^2$. The greatest height attained by the stone above ground is

[KCET]

(a) 36 m

(b) 32 m

(c) 100 m

(d) 64 m

Q 6. The diagonal of a square is changing at the rate of 0. cm s^{-1}. Then, the rate of change of area, when the area. 400 cm^2, is equal to

[Kerala CEE]

(a) $20\sqrt{2}\text{cm}^2/\text{s}$

(b) $10\sqrt{2}\text{cm}^2/\text{s}$

(c) $\frac{1}{10\sqrt{2}}\text{cm}^2/\text{s}$

(d) $\frac{10}{\sqrt{2}}\text{cm}^2/\text{s}$

(e) $5\sqrt{2}\,\text{cm}^2/\text{s}$

Q 7. There is an error of $\pm\ 0.04$ cm in the measurement of the diameter of a sphere. When

the radius is 10 cm, the percentage error in the volume of the sphere is

[EAMCET]

(a) ± 1.2

(b) ± 1.0

(c) ± 0.8

(d) ± 0.6

Q 8. A particle is moving in a straight line. At time t, the distance between the particle from its

starting point is given by $x = t - 6t^2 + t^3$. Its acceleration will be zero at

[WB JEE]

(a) t = 1 unit time

(b) t = 2 units time

(c) t = 3 units time

(d) t = 4 units time

Q 9. The distance covered by a particle in t second is given by $x = 3 + 8t - 4t^2$. After 1 s its velocity will be

[WB JEE]

(a) 0 unit

(b) 3 units

(c) 4 units

(d) 7 units

Q 10. The equation of motion of a particle moving along a straight line is $s = 2t^3 - 9t^2 + 12t$, where the units of s and t are centimetre and second. The acceleration of the particle will be zero after

[MHT CET]

(a) $\frac{3}{2}$ s

(b) $\frac{2}{3}$ s

(c) $\frac{1}{2}$ s

(d) 1s

Q 11. If there is 2% error in measuring the radius of sphere, then ... will be the percentage error in the surface area.

[Guj.CET]

(a) 3%

(b) 1%

(c) 4%

(d) 2%

Q 12. A stone is thrown vertically upwards and the height x ft reached by the stone in t seconds is given by $x = 80t - 16t^2$. The stone reaches the maximum height in

[KCET]

(a) 2s

(b) 2.5s

(c) 3s

(d) 1.5s

Q 13. Gas is being pumped into a spherical balloon at the rate of 30 ft^3/min. Then, the rate at which the radius increases when it reaches the value 15 ft is

[Kerala CEE]

(a) $\frac{1}{15\pi}$ ft/min

(b) $\frac{1}{30\pi}$ ft/min

(c) $\frac{1}{20}$ ft/min

(d) $\frac{1}{25}$ ft/min

(e) None of these

Q 14. A spherical iron ball of radius 10 cm, coated with a layer of ice of uniform thickness, melts at a rate of $100\ \pi$ cm^3 /min. The rate at which the thickness of decreases when the thickness of ice is 5 cm, is

[Kerala CEE]

(a) 1 cm /min

(b) 2 cm/min

(c) $\frac{1}{376}$ cm/min

(d) 5 cm/min

(e) 3 cm/min

Q 15. Let y be the number of people in a village at time t. Assume that the rate of change of the population is proportional to the number of people in the village at any time and further assume that

the population never increases in time. Then, the population of the village at any fixed time t is given by

[VITEEE]

(a) $y = e^{kt} + c$, for some constants $c \leq 0$ and $k \geq 0$

(b) $y = ce^{kt}$, for some constants $c \geq 0$ and $k \leq 0$

(c) $y = e^{ct} + k$, for some constants $c \leq 0$ and $k \geq 0$

(d) $y = k\, e^{ct}$, for some constants $c \geq 0$ and $k \leq 0$

Q 16. A spherical balloon is expanding. If ths radius is increasing at the rate of 2 cm/min, the rate at which the volume increases (in cubic centimeters per minute) when the radius is 5 cm, is

[VITEEE]

(a) 10π

(b) 100π

(c) $200\ \pi$

(d) 50π

Q 17. A particle moves along a straight line with the law of motion given by $s^2 = at^2 + 2bt + c$. Then, the acceleration varies are

[J&KCET]

(a) $\frac{1}{s^3}$

(b) $\frac{1}{s}$

(c) $\frac{1}{s^4}$

(d) $\frac{1}{s^2}$

Q 18. A man of 2m height walks at a uniform speed of 6 km/h away from a lamp post of 6m height. The rate at which the length of his shadow increase is **[Guj. CET]**

(a) 2 km/h (b) 1 km/h (c) 3 km/h (d) 6 km/h

Q 19. The radius of a cylinder is increasing at the rate of 3 m/s and its altitude is decreasing at

the rate of 4 m/s. The rate of change of volume when radius is 4m and altitude is 6m, is

[BITSAT]

(a) 80π cu m/s

(b) 144π cu m/s

(c) 80 cu m/s

(d) 64 cu m/s

Q 20. If the radius of a circle be increasing at a uniform rate of 2 cm/s. The area of increasing of

area of circle, at the instant when the radius is 20 cm, is

[UP SEE]

(a) $70\ \pi$ cm^2/s

(b) 70 cm^2/s

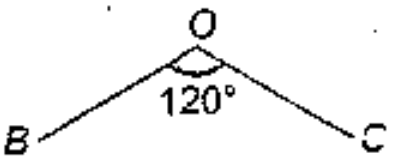
(c) $80\ \pi$ cm^2/s

(d) 80 cm^2/s

Q 21. OB and OC are two roads enclosing an angle of $120°$. X and Y start from 'O' at the same

time. X travels along OB with a speed of 4 km/h and Y travels along OC with a speed of 3 km/h. The rate at which the shortest distance between X and Y is increasing after 1 h is

[KCET]



(a) $\sqrt{37}$ km/h

(b) 37 km/h

(c) 13 km/h

(d) $\sqrt{13}$ km/h

Q 22. A missile is fired from the ground level rises x metres vertically upwards in t seconds where x $=100t - \frac{25}{2}t^2$. The maximum height reached is

[Kerala CEE]

(a) 200 m

(b) 125 m

(c) 160 m

(d) 190 m

(e) 300 m

Q 23. If the distance s covered by a particle in time t is proportional to the cube root of its velocity, then the acceleration is

[Kerala CEE]

(a) a constant

(b) $\propto s^3$

(c) $\propto \frac{1}{s^3}$

(d) $\propto s^5$

(e) $\propto \frac{1}{s^5}$

Q 24. The circumference of a circle is measured as 56 cm with an error 0.02 cm. The percentage error in its area is

[EAMCET]

(a) $1/7$

(b) $1/28$

(c) $1/14$

(d) $1/56$

Q 25. A particle moves along a straight line according to the law $s = 16 - 2t + 3t^3$, where smetres is the distance of the particle from a fixed point at the end of t seconds. The acceleration of the particle at the end of 2 s is

[WB JEE]

(a) 36 m/s^2

(b) 34 m/s^2

(c) 36 m

(d) None of these

Q 26. The distance travelled by a motor car in t, seconds after the brakes are applied is s feet, where s = 22t − 12t^2. The distance travelled by the car before it stops, is

[WB JEE]

(a) 10.08 ft

(b) 10ft

(c) 11 ft

(d) 11.5 ft

Q 27. A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of the abscissa, is

[Jamia Millia Islamia, MP PET]

(a) (2,4)

(b) $(2,-4)$

(c) $\left(-\frac{9}{8},\frac{9}{2}\right)$

(d) $\left(\frac{9}{8},\frac{9}{2}\right)$

Q 28. The radius of a circle is increasing at the rate of 0.1 cm/s. When the radius of the circle is 5 cm, the rate of change of its area, is

[Kerala CEE]

(a) $-\pi$ cm^2/s

(b) 10π cm^2/s

(c) $0.1\ \pi$ cm^2/s

(d) 5π cm^2/s

(e) π cm^2/s

Q 29. A particle moves in a straight line so that

$s = \sqrt{t}$, then its acceleration is proportional to

[WB JEE]

(a) $(\text{velocity})^3$

(b) velocity

(c) $(\text{velocity})^2$

(d) $(\text{velocity})^{3/2}$

Q 50. If a particle moves such that the displacement is proportional to the square of the velocity acquired, then its acceleration is

[MHT CET]

(a) proportional to s

(b) proportional to $\frac{1}{s^2}$

(c) proportional to $\frac{1}{s}$

(d) a constant

Q 31. A spherical balloon is being inflated at the rate of 35 cc/min. The rate of increase of the surface area of the balloon when its diameter is 14 cm, is

[Guj. CET]

(a) 7 sq cm/min

(b) 10 sq cm/min

(c) 17.5 sq cm/min

(d) 28 sq cm/min

Q 32. A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm^2/min. When the thickness of ice is 15 cm, then the rate at which the thickness of ice decreases, is

[AIEEE]

(a) $\frac{5}{6\pi}\text{cm}/\text{min}$

(b) $\frac{1}{54\pi}\text{cm}/\text{min}$

(c) $\frac{1}{18\pi}\text{cm}/\text{min}$

(d) $\frac{1}{36\pi}\text{cm}/\text{min}$

Q 33. A ladder 10 m long rests against a vertical wall with the lower end on the horizontal ground. The lower end of the ladder is pulled along the ground away from the wall at the rate of 3

cm/s. The height of the upper end while it is descending at the rate of 4 cm/s, is

[Kerala CEE]

(a) $4\sqrt{3}$ m

(b) $5\sqrt{3}$ m

(c) $5\sqrt{2}$ m

(d) 8 m

(e) 6 m

Q 34. A right circular cylinder which is open at the top and has a given surface area, will have the greatest volume if its height h and radius r are related by

[AMU]

(a) $2h = r$

(b) $h = 4r$

(c) $h = 2r$

(d) $h = r$

Q 35. The radius of a circular plate is increasing at the rate of 0.01 cm/s when the radius is 12 cm. Then, the rate at which the area increases, is

[EAMCET]

(a) $0.24\ \pi$ sq cm/s

(b) $60\ \pi$ sq cm/s

(c) $24\ \pi$ sq cm/s

(d) $1.2\ \pi$ sq cm/s

Q 36. A stone thrown upwards, has equation of motion $s = 490t - 4.9t^2$. Then, the maximum height reached by it, is

[EAMCET]

(a) 24500

(b) 12500

(c) 12250

(d) 25400

Q 37. The rate of change of the surface area of the sphere of radius r when the radius is increasing at the rate of 2 cm/s is proportional to

[Guj. CET]

(a) $\frac{1}{r^2}$

(b) $\frac{1}{r}$

(c) r^2

(d) r

Q 38. If the distance 's' metres traversed by a particle in t seconds is given by $s = t^3 - 3t^2$, then the velocity of the particle when the acceleration is zero, in metre/ second is

[KCET]

(a) 3

(b) −2

(c) −3

(d) 2

Q 39. If the volume of a sphere is increasing at a constant rate, then the rate at which its radius is increasing, is

[Kerala CEE]

(a) a constant

(b) proportional to the radius

(c) inversely proportional to the radius

(d) inversely proportional to the surface area

(e) proportional to its surface area

Q 40. A particle moves along the curve $y = x^2 + 2x$. Then, the point on the curve such that x and y coordinates of the particle change with the same rate is

[EAMCET]

(a) $(1, 3)$

(b) $\left(\frac{1}{2}, \frac{5}{2}\right)$

(c) $\left(-\frac{1}{2},-\frac{3}{4}\right)$

(d) $(-1, -1)$

Q 41. A point is moving on $y = 4 - 2x^2$. The x-coordinate of the point is decreasing at the rate of 5 unit per second. Then, the rate at which y-coordinate of the point is changing when the point is at $(1,2)$ is

[EAMCET]

(a) 5 units

(b) 10 units

(c) 15 units

(d) 20 units

Q 42. A particle is moving in a straight line such that the distance described 's' and the time taken 't' are given by $t = as^2 + hs + c$, $a > 0$. If v is the velocity of the particle at any time t, then acceleration is

[J&KCET]

(a) $-2av$

(b) $-2av^2$

(c) $-2av^3$

(d) None of these

Q 43. A particle is moving along the curve $x = at^2 + bt + c$. If $ac = b^2$, then the particle would be moving with uniform

[OJEE]

(a) rotation

(b) velocity

(c) acceleration

(d) retardation

Q 44. For a particle moving in a straight line, if time t be regarded as a function of velocity v, then the rate of change of the acceleration a is given by

[BCECE]

(a) $a^2 \frac{d^2t}{dv^2}$

(b) $a^3 \frac{d^2t}{dv^2}$

(c) $-a^3 \frac{d^2t}{dv^2}$

(d) None of these

Topic 2 Tangent and Normal, Maxima and Minima

2010

Q 1. Let f : R $\rightarrow$ R be a continuous function defined by

$$f(x) = \frac{1}{e^x + 2e^{-x}}$$

Statement I $f(c) = \frac{1}{3}$,, for some c $\in$ R.

Statement II $0 < f(x) \leq \frac{1}{2\sqrt{2}}$, for all $x \in R$.

[AIEEE]

(a) Statement I is true, Statement II is also true, Statement II is the correct explanation of Statement I.

(b) Statement I is true, Statement II is also true, Statement II is not the correct explanation of Statement I.

(c) Statement I is true, Statement II is false. (d)

Statement I is false, Statement II is true.

Q 2. The equation of the tangent to the curve $y = x + \frac{4}{x^2}$ that is parallel to the x–axis, is

[AIEEE]

(a) y = 0

(b) y = 1

(c) y = 2

(d) y = 3

Q 3. Let f : R $\rightarrow$ R be defined by ,

$f(x)=\begin{cases} k-2x, & \text{if } x \le -1 \\ 2x+3, & \text{if } x > -1 \end{cases}$. If f has a local minimum at x = −

1, then a possible value of k is

[WB JEE]

(a) 1

(b) 0

(c) $-\frac{1}{2}$

(d) −1

Q 4. The point in the interval $[0, 2\pi]$, where

$f(x) = e^x \sin x$ has maximum slope, is

[AIEEE]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) π

(d) $\frac{3\pi}{2}$

Q 5. The coordinates of the point on the curve $y = x^2 - 3x + 2$ where the tangent is perpendicular to the straight line $y = x$ are

[WB JEE, VITEEE]

(a) $(0, 2)$

(b) $(1,0)$

(c) $(-1,6)$

(d) $(2,-2)$

Q 6. If the normal to the curve y = f(x) at the point $(3, 4)$ make an angle $3\pi/4$ with the positive x−axis, then f'(3) is

[WB JEE, BITSAT]

(a) 1

(b) -1

(c) $-\frac{3}{4}$

(d) $\frac{3}{4}$

Q 7. If the line $ax + by + c = 0$ is a tangent to the curve $xy = 4$ then

[WB JEE]

(a) $a < 0, b > 0$

(b) $a \leq 0, b \geq 0$

(c) $a < 0, b < 0$

(d) $a \leq 0, b < 0$

Q 8. The minimum value of $f(x) = e^{(x^4 - x^3 + x^2)}$ is

[WB JEE]

(a) e

(b) −e

(c) 1

(d) -1

Q 9. The point on the curve $x^2 + y^2 = a^2$, $y \geq 0$

at which the tangent is parallel to x−axis is

[Kerala CEE]

(a) $(a, 0)$

(b) $(-a, 0)$

(c) $\left(\frac{a}{2}, \frac{\sqrt{3}}{2}a\right)$

(d) $(0, a)$

(e) $(0, a^2)$

Q 10. The angle between the curves, $y = x^2$ and $y^2 - x = 0$ at the point $(1,1)$, is

[Kerala CEE]

(a) $\frac{\pi}{2}$

(b) $\tan^{-1}\frac{4}{3}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{4}$

(e) $\tan^{-1}\frac{3}{4}$

Q 11. The distance between the origin and the normal to the ' curve $y = e^{2x} + x^2$ at $x = 0$ is

[Kerala CEE]

(a) 2

(b) $\frac{2}{\sqrt{3}}$

(c) $\frac{2}{\sqrt{5}}$

(d) $\frac{1}{2}$

(e) $\frac{1}{\sqrt{5}}$

Q 12. Given $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1, 1]$

[AIEEE]

(a) $P(-1)$ is the minimum, and $P(1)$ is the maximum of P

(b) $P(-1)$ is not minimum but $P(1)$ is the maximum of P

(c) P(-1) is the minimum and P(1) is not the

maximum of P

(d) Neither P(-1) is the minimum nor P(l) is the

maximum of P

Q 13. The shortest distance between the line y

− x = 1 and the curve x = y^2 is

[AIEEE]

(a) $\frac{3\sqrt{2}}{8}$

(b) $\frac{2\sqrt{3}}{8}$

(c) $\frac{3\sqrt{2}}{5}$

(d) $\frac{\sqrt{3}}{4}$

Q 14. The length of the normal to the curve x = $a(\theta + \sin\theta)$, y = $a(1 - \cos\theta)$ at $\theta = \frac{\pi}{2}$ is

[UP SEE]

(a) $2a$

(b) $\frac{a}{2}$

(c) $\frac{a}{\sqrt{2}}$

(d) $\sqrt{2}a$

Q 15. If $f(x) = x^2 + 4x + 1$, then

[J&KCET]

(a) $f(x) = f(-x)$, for all x

(b) $f(x) \neq 1$, for all $x = 0$

(c) $f''(x) > 0$, for all x

(d) $f(x) > 1$, for $x < 4$

(e) 2

Q 16. If the curves $\frac{x^2}{a^2}+\frac{y^2}{12}=1$ and $y^3 = 8x$ interesect at right angle, then the value of a^2 is equal to

[Kerala CEE]

(a) 16

(b) 12

(c) 8

(d) 4

(e) 2

Q 17. The equation of the tangent to the curve $y = 4e^{-\frac{x}{4}}$ at the point where the curve crosses y −axis is equal to

[Kerala CEE]

(a) $3x + 4y = 16$

(b) $4x+y = 4$

(c) $x + y = 4$

(d) $4x - 3y = -12$

(e) $x - y = -4$

Q 18. The equation of the tangent to the curve

$x^2 - 2xy + y^2 + 2x + y - 6 = 0$ at $(2, 2)$ is

[Kerala CEE]

(a) $2x + y - 6 = 0$

(b) $2y + x - 6 = 0$

(c) $x + 3y - 8 = 0$

(d) $3x + y - 8 = 0$

(e) $x + y - 4 = 0$

Q 19. The angle between the curves $y = a^x$ and $y = b^x$ is equal to

[Kerala CEE]

(a) $\tan^{-1}\left(\left|\frac{a-b}{1+ab}\right|\right)$

(b) $\tan^{-1}\left(\left|\frac{a+b}{1-ab}\right|\right)$

(c) $\tan^{-1}\left(\left|\frac{\log b+\log a}{1+\log a \log b}\right|\right)$

(d) $\tan^{-1}\left(\left|\frac{\log a+\log b}{1-\log a \log b}\right|\right)$

(e) $\tan^{-1}\left(\left|\frac{\log a-\log b}{1+\log a \log b}\right|\right)$

Q 20. If a and b are positive numbers such that $a > b$, then the minimum value of

$a\sec\theta - b\tan\theta \left(0 < \theta < \frac{\pi}{2}\right)$ is

[Kerala CEE]

(a) $\frac{1}{\sqrt{a^2 - b^2}}$

(b) $\frac{1}{\sqrt{a^2 + b^2}}$

(c) $\sqrt{a^2 + b^2}$

(d) $\sqrt{a^2 - b^2}$

(e) $a^2 - b^2$

Q 21. The function $f(x) = x^3 + ax^2 + bx + c$, $a^2 \leq 3b$ has

[EAMCET]

(a) one maximum value

(b) one minimum value

(c) no extreme value

(d) one maximum and one minimum value

Q 22. If $y = \frac{\sin(x+a)}{\sin(x+b)}$, $a \neq b$ then y is

[OJEE]

(a) minima at x = 0

(b) maxima at x = 0

(c) neither minima nor maxima at x = 0

(d) None of the above

Q 23. Angle between y = x and x = y at the

origin is

[WB JEE]

(a) $2\tan^{-1}\left(\frac{3}{4}\right)$

(b) $\tan^{-1}\left(\frac{4}{3}\right)$

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{4}$

Q 24. If θ is the semi vertical angle of a cone of maximum volume and given slant height, then $\tan\theta$ is given by

[BCECE]

(a) 2

(b) 1

(c) $\sqrt{2}$

(d) $\sqrt{3}$

Q 25. The difference between the greatest and least values of the function $f(x) = \cos x + \frac{1}{2}\cos 2x - \frac{1}{3}\cos 3x$ Is

[BCECE]

(a) $\frac{2}{3}$

(b) $\frac{8}{7}$

(c) $\frac{3}{8}$

(d) $\frac{9}{4}$

Q 26. Let $P(x) = a_0 + a_1x^2 + a_2x^4 + \ldots + a_nx^{2n}$ be a polynomial in a real variable x with $0 < a_0 < a_1 < a_2 < \ldots < a_n$. The function $P(x)$ has

[Jamia Millia Islamia]

(a) neither a maximum nor a minimum

(b) only one maximum

(c) only one minimum

(d) only one maximum and only one minimum

Q 27. The equation of the tangent to the curve y $= (2x - 1)\, e^{2(1 - x)}$ at the points its maximum, is

[Jamia Millia Islamia]

(a) $y - 1 = 0$

(b) $x - 1 = 0$

(c) $x + y - 1 = 0$

(d) $x - y + 1 = 0$

Q 28. The equation of the tangent to the curve $y = 4xe^x$ at $\left(-1, \frac{-4}{e}\right)$ is

[MHTCET]

(a) $y = -1$

(b) $y = -\frac{4}{e}$

(c) $x = -1$

(d) $x = \frac{-4}{e}$

Q 29. The maximum value of function $f(x) = \sin x (1 + \cos x)$, $x \in R$ is

[RPET]

(a) $\frac{3^{3/2}}{4}$

(b) $\frac{3^{5/3}}{4}$

(c) $\frac{3}{2}$

(d) $\frac{3^{7/5}}{4}$

Q 30. The normal at point $(1, 1)$ of the curve $y^2 = x^3$ is parallel to the line

[RPET]

(a) $3x - y - 2 = 0$

(b) $2x + 3y - 7 = 0$

(c) $2x - 3y + 1 = 0$

(d) $2y - 3x + 1 = 0$

Q 31. For the function $f(x) = xe^x$ the point

[J&KCET]

(a) $x = 0$ is a maximum

(b) $x = 0$ is a minimum

(c) $x = -1$ is a maximum

(d) x = −l is a minimum

Q 32. Suppose the cubic $x^3 - px + q$ has three distinct real roots where $p > 0$ and $q > 0$. Then, which one of the following holds ?

[AIEEE]

(a) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

(b) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$

(c) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$

(d) The cubic has minima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

Q 33. If $f(x) = 2x^3 - 21x^2 + 36x - 30$, then

which one of the following is correct

[DCE]

(a) f(x) has minimum at x = 1

(b) f(x) has maximum at x = 6

(c) f(x) has maximum at x = 1

(d) f(x) has no maxima or minima

Q 34. The minimum value of $x^2 + \frac{1}{1+x^2}$ is at

[UP SEE]

(a) x = 0

(b) x = I

(c) x = 4

(d) x = 3

Q 35. The maximum value of $\frac{\log x}{x} \text{in}(2,\infty)$ is

[KCET]

(a) 1

(b) $2/e$

(c) e

(d) $1/e$

Q 36. If $ax^2 + bx + 4$ attains its minimum value -1 at $x = 1$, then the values of a and b are respectively

[Kerala CEE]

(a) $5, -10$

(b) $5, -5$

(c) $5,5$

(d) $10,-5$

(e) $10,\ 10$

Q 37. Let $g(x)=\begin{cases} 2e & \text{if } x \le 1 \\ \log(x-1), & \text{if } x > 1 \end{cases}$. The equation of the normal to y = g(x) at the point $(3,\ \log\ 2)$, is

[Kerala CEE]

(a) $y - 2x = 6 + \log\ 2$

(b) $y + 2x = 6 + \log\ 2$

(c) $y + 2x = 6 - \log 2$

(d) $y + 2x = -6 + \log 2$

(e) $y - 2x = -6 + \log 2$

Q 38. If m and M respectively denote the minimum and maximum of $f(x) = (x-1)^2 + 3$ for $x \in [-3, 1]$, then the ordered pair (m, M) is equal to

[AMU]

(a) $(-3,19)$

(b) $(3, 19)$

(c) $(-19,3)$

(d) $(-19,-3)$

Q 39. The length of the subtangent at $(2, 2)$ to the curve $x^5 = 2y^4$ is

[EAMCET]

(a) $\frac{5}{2}$

(b) $\frac{8}{5}$

(c) $\frac{2}{5}$

(d) $\frac{5}{8}$

Q 40. The equation of the normal to the curve

$y^4 = ax^3$ at (a, a) is

[EAMCET]

(a) $x+2y = 3a$

(b) $3x - 4y+a = 0$

(c) $4x+3y = 7a$

(d) $4x - 3y = 0$

Q 41. The minimum value of $e^{(2x^2-2x+1)\sin^2 x}$ is

[OJEE]

(a) 0

(b) 1

(c) 2

(d) 3

Q 42. The greatest value of $f(x) = (x + 1)^{1/3} - (x - 1)^{1/3}$ on $[0,1]$ is

[OJEE]

(a) 0

(b) 1

(c) 2

(d) -1

Q 43. If θ is the angle between the curves xy = 2 and $x^2 + 4y = 0$, then tan θ is equal to

[BCECE]

(a) 1

(b) -1

(c) 2

(d) 3

Q 44. The function $x\sqrt{1-x^2}, (x > 0)$ has

[Jamia Millia Islamia]

(a) a local maxima

(b) a local minima

(c) neither a local maxima nor a local minima

(d) None of the above

Q 45. A circular sector of perimeter 60 m with maximum area is to be constructed. The radius the circular arc in metre must be

[Jamia Millia Islamia]

(a) 20

(b) 5

(c) 15

(d) 10

Q 46. The abscissae of the points, where the tangent to curve $y = x^3 - 3x^2 - 9x + 5$ is parallel to x−axis, are

[MHT CET]

(a) x = 0 and 0

(b) x = 1 and −1

(c) x = 1 and −3

(d) x = −1 and 3

Q 47. The point of the curve $y^2 = 2(x-3)$ at which the normal is parallel to the line $y - 2x + 1 = 0$ is

[MHT CET]

(a) $(5, 2)$

(b) $\left(-\frac{1}{2}, -2\right)$

(c) $(5, -2)$

(d) $\left(\frac{3}{2}, 2\right)$

Q 48. The function f(x)= $x+\frac{1}{x}$ has

[J&KCET]

(a) a local maxima at x = 1 and a local minima at x

= −1

(b) a local minima at x = 1 and a local maxima at x

= −1

(c) absolute maxima at x = 1 and absolute minima

at x = −l

(d) absolute minima at x = 1 and absolute maxima

at x = −l

Q 49. The slope of the tangent to the curves x =

$t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ at the point $(2, -3)$ is

[MPPET]

(a) 22/7

(b) 6/7

(c) −6

(d) −7

Q 50. Divide 12 into two parts such that the product of the square on one part and the fourth power of the second part is maximum, are

[MP PET]

(a) $6,6$

(b) $5,7$

(c) $4,8$

(d) $3,9$

Q 51. The abscissa of the point on the curve y = $a(e^{x/a} + e^{-x/a})$ where the tangent is parallel to the x−axis, is

[RPET]

(a) 0

(b) a

(c) 2a

(d) −2a

Q 52. The point on the curve $y = x^3$ at which the tangent to the curve is parallel to the x−axis, is

[J&K CET]

(a) $(2,2)$

(b) $(3,3)$

(c) $(4,4)$

(d) $(0,0)$

Q 53. The equation of normal to the curve $x^2 y = x^2 - 3x + 6$ at the point with abscissa $x = 3$ is

[J&K CET]

(a) $3x + 27y = 79$

(b) $27x - 3y = 79$

(c) $27x + 3y = 79$

(d) $3x - 27y = 79$

Q 54. If $y = 4x - 5$ is a tangent to the curve $y^2 = px^3 + q$ at $(2,3)$, then

[Guj.CET]

(a) $p = 2, q = -7$

(b) $p = -2, q = 7$

(c) $p = -2, q = -7$

(d) $p = 2, q = 7$

Q 55. The normal to a curve at P(x, y) meets the x−axis at G. If the distance of G from the origin is twice the abscissa of P, then the curve is a

[AIEEE]

(a) ellipse

(b) parabola

(c) circle

(d) hyperbola

Q 56. The equation of tangent to the curve

$\frac{x^2}{3}-\frac{y^2}{2}=1$, which is parallel to y = x, is

[DCE]

(a) $y = x \pm 1$

(b) $y = x - 1/2$

(c) $y = x + 1/2$

(d) $y = 1 - x$

Q 57. The minimum value of $2x + 3y$, when $xy = 6$, is

[BITSAT]

(a) 9

(b) 12

(c) 8

(d) 6

Q 58. The maximum value of f(x) = $\frac{x}{4+x+x^2}$ on [−1,1] is

[UP SEE]

(a) $-\frac{1}{3}$

(b) $-\frac{1}{4}$

(c) $\frac{1}{4}$

(d) $\frac{1}{6}$

Q 59. The point on the curve y =x, the tangent

at which makes an angle $45°$ with x−axis is **[KCET]**

(a) $\left(\frac{1}{4},\frac{1}{2}\right)$

(b) $\left(\frac{1}{2},\frac{1}{4}\right)$

(c) $\left(\frac{1}{2},-\frac{1}{2}\right)$

(d) $\left(\frac{1}{2},\frac{1}{2}\right)$

Q 60. The length of the subtangent to the curve $x^2 y^2 = a^4$ at $(-a, a)$ is

[KCET]

(a) $\frac{a}{2}$

(b) $2a$

(c) a

(d) $\frac{a}{3}$

Q 61. If the curves $x^2 = 9A(9 - y)$ and $x^2 = A(y + 1)$ intersect orthogonally, then the value of A is

[Kerala CEE]

(a) 3

(b) 4

(c) 5

(d) 7

(e) 9

Q 62. The length of tangent, subtangent, normal and subnormal for the curve $y = x^2 + x - 1$ at $(1, 1)$ are A, B, C and D respectively, then their increasing order is

[EAMCET]

(a) B,D,A,C

(b) B,A,C,D

(c) A,B,C,D

(d) B,A,D,C

Q 63. The condition $f(x) = x^3 + px^2 + qx + r (x \in R)$ to have no extreme value, is

[EAMCET]

(a) $p^2 < 3q$

(b) $2p^2 < q$

(c) $p^2 < \frac{1}{4}q$

(d) $p^2 > 3q$

Q 64. Observe the statements given below

Assertion (A) $f(x) = xe^{-x}$ has the maximum at $x = 1$

Reason (R) $f'(1) = 0$ and $f''(1) < 0$

which of the following is correct ?

[EAMCET]

(a) Both (A) and (R) are true and (R) is the correct reason for (A)

(b) Both (A) and (R) are true, but (R) is not the correct reason for (A)

(c) (A) is true, (R) is false

(d) (A) is false, (R) is true

Q 65. If $y = a \log x + bx^2 + x$ has its extremum at $x = -1$ and $x = 2$, then

[OJEE, BCECE]

(a) $a = 2, b = \frac{1}{2}$

(b) $a = 2, b = -\frac{1}{2}$

(c) $a = \frac{1}{2}, b = 2$

(d) $a = -\frac{1}{2}, b = 2$

Q 66. The line (x/a) + (y/b)=2, touches the curve $\frac{x^n}{a^n}+\frac{y^n}{b^n}=2$, at

[OJEE]

(a) (b, a)

(b) (−b,−a)

(c) (a, b)

(d) None of these

Q 67. The maximum slope of the curve $y = -x^3 + 3x^2 + 2x - 27$ is

[WB JEE]

(a) 5

(b) -5

(c) $1/5$

(d) None of these

Q 68. Let $f(x) = \begin{cases} |x|, & \text{for } 0 < |x| \leq 2 \\ 1, & \text{for } x = 0 \end{cases}$, then at x = 0, f has

[J&K GET]

(a) a local maximum

(b) a local minimum

(c) no local extremum

(d) no local maximum

Q 69. The function $f(x) = x^2e^{-2x}$, $x > 0$. Then, the maximum value of $f(x)$ is

[VITEEE]

(a) $\frac{1}{e}$

(b) $\frac{1}{2e}$

(c) $\frac{1}{e^2}$

(d) $\frac{4}{e^4}$

Q 70. The angle between the tangents at those points on the curve $x = t^2 + 1$ and $y = t^2 - t - 6$ where it meets x–axis is

[VITEEE]

(a) $\pm \tan^{-1}\left(\frac{4}{29}\right)$

(b) $\pm\tan^{-1}\left(\frac{5}{49}\right)$

(c) $\pm\tan^{-1}\left(\frac{10}{49}\right)$

(d) $\pm\tan^{-1}\left(\frac{8}{29}\right)$

Q 71. The maximum area of the rectangle that can be inscribed in a circle of radius r, is

[MHT CET]

(a) πr^2

(b) r^2

(c) $\pi r^2/4$

(d) $2r^2$

Q 72. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals

[MHT CET]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 73. On the interval $[0, 1]$ the function $x^{25}(1 - x)^{75}$ takes its maximum value at the point

[RPET]

(a) 0

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) $\frac{1}{3}$

Q 74. The greatest value of $\sin^3 x + \cos^3 x$ is

[J&K CET]

(a) 1

(b) 2

(c) $\sqrt{2}$

(d) $\sqrt{3}$

Q 75. The equation of the normal line to the curve $y = x \log_e x$ parallel to $2x - 2y + 3 = 0$ is

[J&K CET]

(a) $x + y = 3e^{-2}$

(b) $x - y = 6e^{-2}$

(c) $x - y = 3e^{-2}$

(d) $x - y = 6e^{2}$

Q 76. The function $f(x) = a \sin x + \frac{1}{3}\sin 3x$ has maximum value at $x = \frac{\pi}{3}$. The value of a is

[Guj.CET]

(a) 3

(b) $1/3$

(c) 2

(d) $1/2$

Q 77. The function $f(x)=\frac{x}{2}+\frac{2}{x}$ has a local minimum at

[AIEEE]

(a) x = -2

(b) x = 0

(c) x = l

(d) x=2

Q 78. Angle between the tangents to the curve

$y = x^2 - 5x + 6$ at the points $(2,0)$ and $(3,0)$ is

[AIEEE]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{3}$

Q 79. A population p(t) of 1000 bacteria introduced into nutrient medium grows according to the relation $p(t) = 1000 + \frac{1000t}{100 + t^2}$. The maximum size of this bacterial population is

[BITSAT, Guj, CET]

(a) 1100

(b) 1250

(c) 1050

(d) 5250

Q 80. The maximum value of $x^{1/x}$ is

[UP SEE]

(a) $1/e^{e}$

(b) e

(c) $e^{1/e}$

(d) $1/e$

Q 81. If the curve $y = 2x^3 + ax^2 + bx + c$ passes through the origin and the tangents drawn to it at $x = -1$ and $x = 2$ are parallel to the x−axis, then the values of a, b and c are respectively

[KCET]

(a) $12, -3$ and 0

(b) $-3, -12$ and 0

(c) $-3, 12$ and 0

(d) $3, -12$ and 0

Q 82. The tangent and the normal drawn to the curve $y = x^2 - x + 4$ at $P(1, 4)$ cut the x-axis at A and B respectively. If the length of the subtangent drawn to the curve at P is equal to the length of the subnormal, then the area of the triangle PAB in sq units is

[KCET]

(a) 4

(b) 32

(c) 8

(d) 16

Q 83. The equation of the tangent to the curve

$y=(1+x)^y + \sin^{-1}(\sin^2 x)$ at $x = 0$ is

[Kerala CEE]

(a) $x - y + 1 = 0$

(b) $x + y + l = 0$

(c) $2x - y + l = 0$

(d) $x+2y + 2 = 0$

(e) $2x + y - 1 = 0$

Q 84. The point on the curve $y = 2x^2 - 6x - 4$

at which the tangent is parallel to the x−axis, is

[Kerala CEE]

(a) $\left(\frac{3}{2}, \frac{13}{2}\right)$

(b) $\left(-\frac{5}{2},-\frac{17}{2}\right)$

(c) $\left(\frac{3}{2},\frac{17}{2}\right)$

(d) $(0, -4)$

(e) $\left(\frac{3}{2},-\frac{17}{2}\right)$

Q 85. If tangent to the curve $x = at^2$, $y = 2at$ is perpendicular to x−axis, then its point of contact is

[AMU]

(a) (a, a)

(b) $(0,a)$

(c) $(0,0)$

(d) $(a,0)$

Q 86. If θ is the angle between the curves $xy = 2$ and $x^2 + 4y = 0$, then $\tan\theta$ is equal to

[EAMCET]

(a) 1

(b) -1

(c) 2

(d) 3

Q 87. The perimeter of a sector is a constant. If its area is to be maximum, the sectorial angle is

[EAMCET]

(a) $\frac{\pi^c}{6}$

(b) $\frac{\pi^c}{4}$

(c) 4^c

(d) 2^c

Q 88. If $x = t^2$ and $y = 2t$, then equation of the normal at $t = 1$, is

[OJEE]

(a) $x + y - 3 = 0$

(b) $x + y - 1 = 0$

(c) $x + y + 1 = 0$

(d) $x + y + 3 = 0$

Q 89. If the line ax + by + c = 0 is a normal to the curve xy = 1, then

[WB JEE]

(a) a > 0, b > 0

(b) a > 0, b < 0

(c) a < 0, b < 0

(d) Data is insufficient

Q 90. If $f(x)=\frac{x^2-1}{x^2+1}$, for every real number x, then minimum value of f(x)

[BCECE]

(a) does not exist

(b) is equal to 1

(c) is equal to 0

(d) is equal to -1

Q 91. The length of the subtangent at any point

(x_1, y_1) on the curve $y = a^x$, $(a > 0)$ is

[J&K CET]

(a) $2 \log a$

(b) $\frac{1}{\log a}$

(c) $\log a$

(d) $a^{2x_1} \log a$

Q 92. The absolute maximum of $x^{40} - x^{20}$ on the interval $[0,1]$ is

[J&K CET]

(a) $-1/4$

(b) 0

(c) $1/4$

(d) $1/2$

Q 93. The set $\{x^3 - 12x : -3 \leq x \leq 3\}$ is equal to

[J&K CET]

(a) $\{x : -16 \leq x \leq 16\}$

(b) $\{x : -12 \leq x \leq 12\}$

(c) $\{x : -9 \leq x \leq 9\}$

(d) $\{x : 0 \leq x \leq 10\}$

Q 94. If ST and SN are the lengths of the subtangent and the subnormal at the point $\theta = \frac{\pi}{2}$ on the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, $a \neq 1$, then

[Guj, CET]

(a) ST = SN

(b) ST=2SN

(c) $ST^2 = aSN^3$

(d) $ST^3 = aSN$

Q 95. The normal to the curve $x = a(\cos\theta + \theta \sin\theta)$, $y = a(\sin\theta - \theta\cos\theta)$ at any point θ is such that

[AIEEE]

(a) it is at a constant distance from the origin

(b) it passes through $\left(\frac{a\pi}{2}, -a\right)$

(c) it makes angle $\frac{\pi}{2} - \theta$ with the x−axis

(d) it passes through the origin

Q 96. The number of values of x where f(x) = cos x + cos $\sqrt{2}$ x attains its maximum is

[DCE]

(a) 1

(b) 0

(c) 2

(d) infinite

Q 97. The largest value of $2x^3 - 3x - 12x + 5$

for $-2 \leq x \leq 4$ occurs at x is equal to

[BITSAT, OJEE]

(a) -4

(b) 0

(c) 1

(d) 4

Q 98. The maximum value of xy when x + 2y = 8 is

[Kerala CEE]

(a) 20

(b) 16

(c) 24

(d) 8

(e) 4

Q 99. Let $f(x) = 1 + 2x^2 + 2^2x^4 + \ldots\ldots + 2^{10}x^{20}$.

Then, f(x) has

[AMU]

(a) more than one minimum

(b) exactly one minimum

(c) at least one maximum

(d) None of the above

Q 100. The extreme values of

$4\cos(x^2)\cos\left(\frac{\pi}{3}+x^2\right)\cos\left(\frac{\pi}{3}-x^2\right)$ over R, are

[EAMCET]

(a) $-1,1$

(b) $-2,2$

(c) $-3,3$

(d) $-4,4$

Q 101. The two curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2 y - y^3 - 2 = 0$

[Jamia Millia Islamia]

(a) cut at right angle

(b) touch each other

(c) cut at an angle $\frac{\pi}{3}$

(d) cut at an angle $\frac{\pi}{4}$

Q 102. If $x + y = 8$, then maximum value of $x^2 y$ is

[MHT CET]

(a) $\frac{2048}{9}$

(b) $\frac{2048}{81}$

(c) $\frac{2048}{3}$

(d) $\frac{2048}{27}$

Q 103. The point on the curve $\sqrt{x}+\sqrt{y}=\sqrt{a}$, the normal at which is parallel to the x−axis, is

[MP PET]

(a) $(0,0)$

(b) $(0,a)$

(c) $(a, 0)$

(d) (a, a)

Q 104. The equation of the tangent to the curve $x = 2\cos^3\theta$ and $y = 3\sin^3\theta$ at the point, $\theta = \pi/4$ is

[RPET]

(a) $2x + 3y = 3\sqrt{2}$

(b) $2x - 3y = 3\sqrt{2}$

(c) $3x + 2y = 3\sqrt{2}$

(d) $3x - 2y = 3\sqrt{2}$

Q 105. The minimum value of $4e^{2x} + 9e^{-2x}$ is

[J&K CET]

(a) 11

(b) 12

(c) 10

(d) 14

Q $106.$ The point of parallel $2y = x^2$, which is nearest to the point $(0,3)$ is

[J&KCET]

(a) $(\pm 4,8)$

(b) $(\pm 1,1/2)$

(c) $(\pm 2, 2)$

(d) None of these

Q 107. The slope of the tangent to the curve $x = 3t^2 + 1$, $y = t^3 - 1$ at $x = 1$ is

[J&K CET]

(a) $\frac{1}{2}$

(b) 0

(c) -2

(d) ∞

Q 108. The normal to the curve x = a$(1 + \cos\theta)$, y = a sin θ at θ always passes through the fixed point

[AIEEE]

(a) $(a,0)$

(b) $(0,a)$

(c)$(0,0)$

(d) (a, a)

Q 109. For the curve $y^n = a^{n-1} x$ if the subnormal at any point is a constant, then n is equal to

[KCET]

(a) 1

(b) 2

(c) -2

(d) -1

Q 110. The length of the subtangent to the curve

$x^2 + xy + y^2 = 7$ at $(1, -3)$ is

[Kerala CEE]

(a) 3

(b) 5

(c) $\frac{3}{5}$

(d) 15

(e) 4

Q 111. The angle of intersection of the curves $y = x^2$, $6y = 7 - x^3$ at $(1, 1)$ is

[AMU]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$

(d) None of these

Q 112. Match the points on the curve $2y^2 = x + 1$ with the slopes of normals at those points and choose the correct answer.

[EAMCET]

Point	**Slope of the normal**
(A) (7, 2)	(1) $-4\sqrt{2}$
(B) $\left(0, \frac{1}{\sqrt{2}}\right)$	(2) -8
(C) $(1, -1)$	(3) 4

(D) $(3,\sqrt{2})$ (4) 0

(5) $-2\sqrt{2}$

	A	B	C	D		A	B	C	D
(a)	2	4	3	1	(b)	2	5	3	1
(c)	2	3	5	1	(d)	2	5	1	4

Q 113. Tangent is drawn to ellipse at $\frac{x^2}{27}+y^2=1$

$(3\sqrt{3}\cos\theta,\sin\theta)$ (where $\theta \in (0, \pi/2)$). Then the value of

θ such that sum of intercepts on axes made by this tangent is minimum, is

[IIT JEE]

(a) $\pi/3$

(b) $\pi/6$

(c) $\pi/8$

(d) $\pi/4$

Q 114. The real number x when added to its inverse gives the minimum value of the sum at x equals to

[AIEEE]

(a) 2

(b) 1

(c) -1

(d) -2

Q 115. The maximum value $x^3 - 3x$ in the interval $[0, 2]$ is

[DCE]

(a) -2

(b) 0

(c) 2

(d) 1

Q 116. If $x - 2y = 4$, the minimum value of xy is

[UP SEE]

(a) -2

(b) 0

(c) 0

(d) -3

Q 117. For the curve $xy = c^2$ the subnormal at any point varies as

[KCET]

(a) x^3

(b) x^2

(c) y^3

(d)∞

Q 118. If for a function f(x), f'(a)= 0, f"(a) = 0, f "'

(a) > 0, then at x = a, f (x) is

[Kerala CEE]

(a) minimum

(b) maximum

(c) not an extreme point

(d) extreme point

(e) None of these

Q 119. The sum of two numbers is 20. If the product of the square of one number and cube of the other is maximum, then the numbers are

[EAMCET]

(a) 12,8

(b) 3,4

(c) 9, 12

(d) 15,18

Q 120. A minimum value of $\int_0^x te^{t^2} dt$ is

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q $121.$ The minimum value of $2x^2 + x - 1$ is

[EAMCET]

(a) $-\frac{1}{4}$

(b) $\frac{3}{2}$

(c) $-\frac{9}{8}$

(d) $\frac{9}{8}$

Q 122. If $a^2x^4 + b^2y^4 = c^6$, then maximum value of xy is

[MP PET]

(a) $\frac{c^2}{\sqrt{ab}}$

(b) $\frac{c^3}{ab}$

(c) $\frac{c^3}{\sqrt{2ab}}$

(d) $\frac{c^3}{2ab}$

Q 123. The tangent to the curve $y = 2x - x + 1$ is parallel to the line $y = 3x + 9$ at the point

[RPET]

(a) $(3, 9)$

(b) $(2,-1)$

(c) $(2, 1)$

(d) $(1,2)$

Topic 3

Increasing and Decreasing Functions, Rolle's and Mean Value Theorems

2010

Q 1. Let f, g and h be real-valued functions defined on the interval $[0,1]$ by f(x) = $e^{x^2}+e^{-x^2}$, g(x) = $xe^{x^2}+e^{-x^2}$ and h (x) = $x^2e^{x^2}+e^{-x^2}$. If a, b and c denote respectively, the absolute maximum of f, g and h on $[0,1]$, then

[IITJEE]

(a) a = b and c ≠ b

(b) a = c and $a \neq b$

(c) $a \neq b$ and $c \neq b$

(d) a = b = c

Q 2. Let f be a real-valued function defined on the interval $(0, \infty)$ by $f(x) = \ln x + \int_0^x \sqrt{1+\sin t}\, dt$.

Then, which of the following statement (s) is (are) true?

[IIT JEE]

(a) $f''(x)$ exists for all $x \in (0, \infty)$

(b) f' (x) exists for all $x \in (0, \infty)$ and f' is

continuous on $(0, \infty)$, but not differentiable on $(0,$

$\infty)$

(c) there exists $\alpha > 1$ such that $|f'(x)| < |f(x)|$ for all

$x \in (\alpha, \infty)$

(d) there exists $\beta > 0$ such that $|f(x)| + |f'(x)| \leq \beta$

from all $x \in (0, \infty)$

Q 3. In which of the following functions, Rolle's

theorem is applicable?

[WB JEE]

(a) f(x) = |x| in $-2 \leq x \leq 2$

(b) f(x) = tan x in $0 \leq x \leq \pi$

(c) f(x) = $1 + (x - 2)^{2/3}$ in $1 \leq x \leq 3$

(d) f(x) = x $(x - 2)^2$ in $0 \leq x \leq 2$

Q 4. For what values of x, the function f (x) =

$x^4 - 4x^3 + 4x^2 + 40$ is monotonic decreasing?

[BITSAT, WB JEE]

(a) $0 < x < l$

(b) $1 < x < 2$

(c) $2 < x < 3$

(d) $4 < x < 5$

Q 5. The value of c in $(0,2)$ satisfying the

Mean value theorem for the function f(x) = x (x −

$1)^2$, $x \in [0, 2]$ is equal to

[Kerala CEE]

(a) $\frac{3}{4}$

(b) $\frac{4}{3}$

(c) $\frac{1}{3}$

(d) $\frac{2}{3}$

(e) $\frac{5}{3}$

Q 6. The value of a in order that f(x) = sin x − cos x−ax + b decreases for all real values of x is given by

[DCE]

(a) $a \geq \sqrt{2}$

(b) $a < \sqrt{2}$

(c) $a \geq 1$

(d) $a < 1$

Q 7. The Rolle's theorem is applicable in the interval $-1 \leq x \leq 1$ for the function

[WB JEE]

(a) f(x)= x

(b) f(x) = x^2

(c) f(x) = $2x^3+3$

(d) f(x) = |x|

Q 8. The interval of increase of the function F

(x) = x − e^x + tan $\left(\frac{2\pi}{7}\right)$ is

[Jamia Millia Islamia]

(a) $(0, \infty)$

(b) $(-\infty, 0)$

(c) $(1, \infty)$

(d) $(-\infty, -1)$

Q 9. Let $f(x) = x^3$ use Mean value theorem to write $\frac{f(x+h)-f(x)}{h} = f'(x+\theta h)$ with $0 < \theta < 1$. If $x \neq 0$, then $\lim_{h\to 0} \theta$ is equal to

[J&K CET]

(a) -1

(b) -0.5

(c) 0.5

(d) 1

Q 10. $f(x) = x^3 - 6x^2 - 36x + 2$ is decreasing function, then $x \in$

[Guj. CET]

(a) $(6, \infty)$

(b) $(-\infty, -2)$

(c) $(-2,6)$

(d) None of these

Q 11. Let the function g: $(-\infty, \infty) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2\tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is

[IITJEE]

(a) even and is strictly increasing in $(0, \infty)$

(b) odd and is strictly decreasing in $(-\infty, \infty)$

(c) odd and is strictly increasing in $(-\infty, \infty)$

(d) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

Q 12. How many real solutions does the equation $x^7 + 14x^5 + 16x^3 + 30x - 560 = 0$ have?

[AIEEE]

(a) 5

(b) 7

(c) 1

(d) 3

Q 13. The function $f(x) = x(x + 3)e^{-(1/2)x}$ satisfies the conditions of Rolle's theorem in $[-3,0]$. The value of c is

[DCE]

(a) 0

(b) -1

(c) -2

(d) -3

Q 14. $f(x)=\left(\frac{e^{2x}-1}{e^{2x}+1}\right)$ is

[B1TSAT]

(a) an increasing

(b) a decreasing

(c) an even

(d) None of these

Q 15. The function $f(x) = (9 - x^2)^2$ increases in

[Kerala CEE]

(a) $(-3, \infty) \cup (3, \infty)$

(b) $(-\infty, -3) \cup (3, \infty)$

(c) $(-\infty, -3) \cup (0, 3)$

(d) $(-3, 3)$ (e) $(3, \infty)$

Q 16. If a <0, the function $(e^{ax}+e^{-ax})$ is a decreasing function for all values of x, where

[OJEE]

(a) $x < 0$

(b) $x > 0$

(c) $x < 1$

(d) $x > 1$

Q 17. Select the correct statement from (a), (b), (c), (d). The function $f(x)=xe^{1-x}$

[WB JEE]

(a) strictly increases in the interval $\left(\frac{1}{2}, 2\right)$

(b) increases in the interval $(0, \infty)$

(c) decreases in the interval $(0, 2)$

(d) strictly decreases in the interval $(1, \infty)$

Q 18. Rolle's theorem is not applicable to the function f (x) = | x | for $-2 \leq x \leq 2$ because

[WB JEE]

(a) f is continuous for $-2 < x < 2$

(b) f is not derivable for x = 0

(c)f(-2)=f(2)

(d) f is not a constant function

Q 19. If the function f (x) = $ax^3 + bx^2 + 11x - 6$ satisfies the condition of Rolle's theorem in $[1, 3]$ and $f'\left(2+\frac{1}{\sqrt{3}}\right)=0$, then the values of a, b are respectively

[Manipal]

(a) $-1,6$

(b) $-2,1$

(c) $1,-6$

(d) $-1, \frac{1}{2}$

Q 20. If the Mean value theorem is f(b) − f(a) = (b − a)f'(c). Then, for the function $x^2 - 2x + 3$ in $\left[1, \frac{3}{2}\right]$, the value of c is

[MPPET]

(a) $6/5$

(b) $5/4$

(c) $4/3$

(d) $7/6$

Q 21. A function f is defined by $f(x) = e^x \sin x$ in $[0, \pi]$. Which of the following is not correct?

[MP PET]

(a) f is continuous in $[0, \pi]$

(b) f is differentiable in $[0, \pi]$

(c) $f(0) = f(\pi)$

(d) Rolle's theorem is not true in $[0, \pi]$

Q 22. The function f defined by $f(x) = x^3 - 6x^2 - 36x + 7$ is increasing, if

[RPET]

(a) $x > 2$ and also $x > 6$

(b) $x > 2$ and also $x < 6$.

(c) $x > -2$ and also $x < 6$

(d) $x < -2$ and also $x > 6$

Q 23. The function $f(x) = 2x^3 + 3x^2 - 12x + 1$ decreases in the interval

[J&K CET]

(a) $(2, 3)$

(b) $(1,2)$

(c) $(-2,1)$

(d) $(-3,-2)$

Q 24. The function $f(x) = x^2e^{-x}$ increases in the interval

[J&KCET]

(a) $(0,2)$

(b) $(2,3)$

(c) $(3,\ 4)$

(d) $(4,\ 5)$

Q 25. If $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$, then $f(x)$ is

[Guj CET]

(a) increasing in $(-\infty,\ -2)$ and in $(0,1)$

(b) increasing in $(-2,0)$ and in $(1,\ \infty)$

(c) decreasing in $(-2,0)$ and in $(0, 1)$

(d) decreasing in $(-\infty, -2)$ and in $(1, \infty)$

Q 26. A value of c for which the conclusion of Mean value theorem holds for the function f(x) = $\log_e x$ on the interval $[1, 3]$ is

[AIEEE]

(a) $2 \log_3 e$

(b) $\frac{1}{2}\log_e 3$

(c) $\log_3 e$

(d) $\log_e 3$

Q 27. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in

[AIEEE]

(a) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(b) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

(c) $\left(0, \frac{\pi}{2}\right)$

(d) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Q 28. For what value of a, $f(x) = -x^3 + 4ax^2 + 2x - 5$ is decreasing $\forall$ x ?

[DCE]

(a) $(1, 2)$

(b) $(3, 4)$

(c) R

(d) no value of a

Q 29. In the interval $[0,1]$, the function $x_2 - x + 1$ is

[Kerala CEE]

(a) increasing

(b) decreasing

(c) neither increasing nor decreasing

(d) do not say anything

(e) None of the above

Q 30. If $f(x) = kx - \sin x$ is monotonically increasing, then

[MHT CET]

(a) $k > 1$

(b) $k > -1$

(c) $k < 1$

(d) $k < -1$

Q 31. If $2a + 3b + 6c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval

[MP PET]

(a) $(0,1)$

(b) $(1,2)$

(c) $(2,3)$

(d) $(1,3)$

Q 32. The function $f(x) = \log (1 + x) - \frac{2x}{2+x}$ is increasing on

[RPET]

(a) $(0, \infty)$

(b) $(-\infty, 0)$

(c) $(-\infty, \infty)$

(d) None of these

Q 33. The function $f(x) = 1 - x^3$

[J&K CET]

(a) increases everywhere

(b) decreases in $(0, \infty)$

(c) increases in $(0, \infty)$

(d) None of these

Q 34. If f(x) = sin x/e^x in$[0, \pi]$ then f(x)

[J&K CET]

(a) satisfies Rolle's Theorem and, $c = \frac{\pi}{4}$ So that

$f'\left(\frac{\pi}{4}\right) = 4$

(b) does not satisfy Rolle's Theorem but $f'\left(\frac{\pi}{4}\right) > 0$

(c) satisfies Rolle's Theorem but $f'\left(\frac{\pi}{4}\right) = 0$

(d) satisfies Lagranges Mean Value Theorem but

$f'\left(\frac{\pi}{4}\right) \neq 0$

Q 35. The value of b for which the function

f(x)=sin x − bx + c is decreasing in the interval (−∞, ∞) is given by

[Guj. CET]

(a) b < 1

(b) b > 1

(c) b > 1

(d) $b < 1$

Q 36. A function f is defined by $f(x)=2 + (x - 1)^{2/3}$ in $[0,2]$. Which of the following is not correct ?

[UP SEE]

(a) f is not derivable in $(0, 2)$

(b) f is continuous in $[0, 2]$

(c) $f(0) = f(2)$

(d) Rolle's Theorem is true in $[0, 2]$

Q 37. Define f(x)= $\int_0^x \sin t\, dt, x \geq 0$. Then,

[Kerala CEE]

(a) f is increasing only in the interval $\left(0, \frac{\pi}{2}\right)$

(b) f is decreasing in the interval $[0, \pi]$

(c) f attains maximum at $x = \frac{\pi}{2}$

(d) f attains minimum at $x = \pi$

(e) f attains maximum at $x = \pi$

Q 38. In the interval $(-3,3)$ the function f(x) =

$\frac{x}{3}+\frac{3}{x}, x \neq 0$ is

[EAMCET]

(a) increasing

(b) decreasing

(c) neither increasing nor decreasing

(d) partly increasing and partly decreasing

Q 39. The function f(x) = $\cot^{-1}$ x+x increases in

the interval

[OJEE]

(a) $(1, \infty)$

(b) $(-1, \infty)$

(c) $(-\infty, \infty)$

(d) $(0, \infty)$

Q 40. The function f defined by $f(x) = 4x^4 - 2x +$ l in increasing for

[WB JEE]

(a) $x < 1$

(b) $x > 0$

(c) $x < \frac{1}{2}$

(d) $x > \frac{1}{2}$

Q 41. Let $f(x) = \sqrt{x-1} + \sqrt{x+24-10\sqrt{x-1}}$, $1 \leq x \leq 26$ be real valued function, then f' (x) for $1 < x < 26$ is

[MP PET]

(a) 0

(b) $\frac{1}{\sqrt{x-1}}$

(c) $2\sqrt{x-1}-5$

(d) None of these

Q 42. Let f be continuous on $[1,5]$ and differentiable in $(1,5)$. If $f(1) = -3$ and $f'(x) \geq 9$ for all $x \in (1, 5)$, then

[Kerala CEE]

(a) $f(5) \geq 33$

(b) $f(5) \geq 36$

(c) $f(5) \leq 36$

(d) $f(5) \geq 9$

(e) $f(5) \leq 9$

Q 43. If $(x) = x^2 - 2x + 4$ on $[1, 5]$, then the value of a constant c such that $\frac{f(5)-f(1)}{5-1} = f'(c)$, is

[AMU]

(a) 0

(b) 1

(c) 2

(d) 3

Q 44. Observe the following statements

Assertion (A) $f(x) = 2x^3 - 9x^2 + 12x - 3$ is increasing outside the interval $(1,2)$

Reason (R) $f'(x) < 0$ for $x \in (1, 2)$

Then which of the following is true ?

[EAMCET]

(a) Both A and R are true and R is the correct reason for A

(b) Both A and R are true and R is not the correct reason for A

(c) A is true but R is false

(d) A is false but R is true

Q 45. The function $f(x) = 2x^3 - 3x^2 + 90x + 174$

is increasing in the interval

[RPET]

(a) $\frac{1}{2} < x < 1$

(b) $\frac{1}{2} < x < 2$

(c) $3 < x < \frac{59}{4}$

(d) $-\infty < x < \infty$

Q 46. In the mean value theorem $f(b) - f(a) = (b - a)f'(c)$, if $a = 4$, $b = 9$ and $f(x) = \sqrt{x}$, then the value of c is

[J&KCET]

(a) 8.00

(b) 5.25

(c) 4.00

(d) 6.25

Q 47. If $f(x) = x^3 + bx^2 + cx + d$ and $0 < b^2 < c$,

then in $(-\infty, \infty)$

[IIT JEE]

(a) f(x) is strictly increasing function

(b) f (x) has a local maxima

(c) f(x) is strictly decreasing function

(d) f(x) is bounded

Q 48. If $f(x) = x^{\alpha} \log x$ and $f(0) = 0$, then the value of a for which Rolle's theorem can be applied in $[0, 1]$ is

[IIT JEE]

(a) -2

(b) -1

(c) 0

(d) $1/2$

Q 49. If f(x) satisfies the condition for Rolle's theorem is $[3,5]$, then $\int_3^5 f(x)dx$ equals

[DCE]

(a) 2

(b) -1

(c) 0

(d) $-4/3$

Q 50. The value of x for which the polynomial $2x^3 - 9x^2 + 12x + 4$ is a decreasing function of x, is

[Kerala CEE]

(a) $-1 < x < 1$

(b) $0 < x < 2$

(c) $x > 3$

(d) $1 < x < 2$

(e) $1<x<3$

Q 51. The function $f(x) = \sin^4 x + \cos^4 x$

increases, if

[Jamia Millia Islamia]

(a) $0 < x < \frac{\pi}{8}$

(b) $\frac{\pi}{4} < x < \frac{3\pi}{8}$

(c) $\frac{3\pi}{8} < x < \frac{5\pi}{8}$

(d) $\frac{5\pi}{8} < x < \frac{3\pi}{4}$

Q 52. The value of c, in the Lagrange's Mean value theorem $\frac{f(b)-f(a)}{b-a} = f'(c)$, for the function $f(x)=x(x-1)(x-2)$ in the interval $[0, 1/2]$, is

[MP PET]

(a) $\frac{1}{4}$

(b) $1-\frac{\sqrt{21}}{6}$

(c) $\frac{9}{8}$

(d) $1+\frac{\sqrt{21}}{6}$

Q 53. If $f(x)=\int_{x^2}^{x^2+1} e^{-t^2} dt$, then $f(x)$ increases in

[IIT JEE]

(a) $(2,2)$

(b) no value of x

(c) $(0, \infty)$

(d) $(-\infty, 0)$

Q 54. In $[0, 1]$ Lagranges Mean value theorem

is NOT applicable to

[IIT JEE]

(a) $f(x) = \begin{cases} \frac{1}{2} - x, & x < \frac{1}{2} \\ \left(\frac{1}{2} - x\right)^2, & x \geq \frac{1}{2} \end{cases}$

(b) $f(x)=\begin{cases}\frac{\sin x}{x}, & x \neq 0 \\ 1, & x=0\end{cases}$

(c) f(x)= x|x|

(d) f(x)=|x|

Q 55. For a given integer k, in the interval $\left[2\pi k+\frac{\pi}{2}, 2\pi k-\frac{\pi}{2}\right]$ the graph of sin x is

[DCE]

(a) increasing from -1 to 1

(b) decreasing from -1 to 0

(c) decreasing from 0 to 1

(d) None of the above

Q 56. If g(x) = min (x, x^2) where x is a real number, then

[Kerala CEE]

(a) g(x) is an increasing function

(b) g(x) is a decreasing function

(c) g(x) is a constant function

(d) g(x) is a continuous function except at x = 0

(e) g(x) is a continuous function except at x = 0

and x=1

Answers

Topic 1: Rate of Measure and Application in Mechanics

1. (a) 2. (d) 3. (b) 4. (c) 5. (c) 6. (b) 7. (d) 8. (b)

9. (a) 10. (a) 11. (c) 12. (b). 13. (b) 14. (a) 15.

(b) 16. (c) 17. (a) 18. (c) 19. (a) 20. (c) 21. (a)

22. (a) . 23. (d) 24. (c) 25. (a) 26. (a) 27. (d) 28.

(e) 29. (a) 30. (d) 31. (b) 32. (c) 33. (e) 34. (d)

35. (a) 36. (c) 37. (c) 38. (c) 39. (d) 40. (c)

41. (d) 42. (c) 43. (c) 44. (c)

Topic 2 : Tangent and Normal, Maxima and Minima

1. (c) 2. (d) 3. (d) 4. (b) 5. (b) 6. (a) 7. (c) 8. (c)

9. (d) 10. (e) 11. (c) 12. (b) 13. (a) 14. (d) 15. (c)

16. (d) 17. (c) 18. (a) 19. (e) 20. (d) 21. (c) 22.

(c) 23. (c) 24. (c) 25. (d) 26. (c) 27. (a) 28. (b)

29. (a) 30. (b) 31. (d) 32. (b) 33. (c) 34. (a) 35.

(d) 36. (a) 37. (b) 38. (b) 39. (b) 40. (c)

41. (b) 42. (c) 43. (d) 44. (a) 45. (c) 46. (d) 47.

(c) 48. (b) 49. (b) 50. (c) 51. (a) 52. (d) 53. (b)

54. (a) 55. (d) 56. (a) 57. (b) 58. (d) 59. (a) 60.

(c) 61. (b) 62. (d) 63. (a) 64. (a) 65. (b) 66. (c)

67. (a) 68. (a) 69. (c) 70. (c) 71. (d) 72. (c) 73.

(b) 74. (a) 75. (c) 76. (c) 77. (d) 78. (a) 79. (c)

80. (c) 81. (b) 82. (d) 83. (a) 84. (e) 85. (c) 86.

(d) 87. (d) 88. (a) 89. (b) 90. (d) 91. (b) 92. (b)

93. (a) 94. (a) 95. (a) 96. (a) 97. (d) 98. (d) 99.

(b) 100. (a) 101. (a) 102. (d) 103. (b) 104. (c)

105. (b) 106. (c) 107. (b) 108. (a) 109. (b) 110.

(d) 111. (c) 112. (b) 113. (b) 114. (b) 115. (c)

116. (a) 117. (c) 118. (c) 119. (a) 120. (a)

121. (c) 122. (c–) 123. (d)

Topic 3 : Increasing and Decreasing Functions, Rolle's and Mean Value Theorems

1. (d) 2. (b,c) 3. (d) 4. (b) 5. (b) 6. (a) 7. (b) 8.

(b) 9. (c) 10. (c) 11. (c) 12. (c) 13. (c) 14. (a) 15.

(a) 16. (a) 17. (d) 18. (b) 19. (c) 20. (b)

21. (d) 22. (d) 23. (c) 24. (a) 25. (b) 26. (a) 27.

(b) 28. (d) 29. (c) 30. (a) 31. (a) 32. (a) 33. (b)

34. (c) 35. (c) 36. (d) 37. (a) 38. (b) 39. (c) 40.

(d) 41. (a) 42. (a) 43. (b) 44. (b) 45. (d) 46. (d)

47. (a) 48. (d) 49. (d) 50. (d) 51. (b) 52. (b) 53.

(d) 54. (a) 55. (a) 56. (a)

Differential Equations

Useful Results & Formulae

1. Order of a differential equation is the order of the highest derivative.

Degree of a differential equation is the power of the . highest derivative.

In a transcendental differential equation, we cannot determine the degree ie, degree is undefined.

2. Standard form of differential equation for variable separable method is $f_1(x)dx = f_2(y)dy$, where $f_1(x)$ and $f_2(y)$ are the functions in x and y alone.

3. Standard form of homogeneous differential equations is

$$\frac{dy}{dx} = \frac{f_1(x, y)}{f_2(x, y)}$$

where f_1 (x, y) and f_2 (x, y) are homogeneous functions of x and y. For solution put y = vx and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ and then apply variable separable method.

4. Standard form of linear differential equation is

$$\frac{dy}{dx} + Py = Q$$

where, P and Q are functions of x only. Solution of this differential equation is

$$y.IF = \int Q.IF dx + c$$

where $IF = e^{\int P\,dx}$

5. Form of Bernoulli's equation is $\frac{dy}{dx} + Py = Qy^n$.

Firstly divide by y^n and then put $y^{1-n} = z$ and solve as it.

6. (i) d(x + y) = dx + dy

(ii) d(xy) = x dy + y dx

(iii) $d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2}$

(iv) $d\left(\frac{y}{x}\right) = \frac{xdy - ydx}{x^2}$

(v) $d[\log(xy)] = \frac{xdy + ydx}{xy}$

(vi) $d\left[\log\left(\frac{x}{y}\right)\right] = \frac{ydx - xdy}{xy}$

(vii) $d\left[\log\left(\frac{y}{x}\right)\right] = \frac{xdy - ydx}{xy}$

(viii) $d[\log(x + y)] = \frac{dx + dy}{x + y}$

(ix) $d\left[\tan^{-1}\left(\frac{y}{x}\right)\right] = \frac{xdy - ydx}{x^2 + y^2}$

(x) $d\left(-\frac{1}{xy}\right) = \frac{xdy + ydx}{x^2y^2}$

(xi) $d(x^m\ y^n) = x^{m-1}\ y^{n-1}\ (my\ dx + nx\ dy)$

Topic 1 Order and Degree, Formation of Differential Equation

Q 1. The degree of the differential equation

$$x = 1 + \left(\frac{dy}{dx}\right) + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + ...$$

[WBJEE]

(a) 3

(b) 2

(c) 1

(d) not defined

Q 2. The family of curves $y = e^{a \sin x}$, where a is an arbitrary constant, is represented by the differential equation

[Kerala CEE]

(a) $\log y = \tan x \frac{dy}{dx}$

(b) $y \log y = \tan x \frac{dy}{dx}$

(c) $y \log y = \sin x \frac{dy}{dx}$

(d) $\log y = \cos x \frac{dy}{dx}$

(e) $y \log y = \cos x \frac{dy}{dx}$

Q 3. The degree and order of the differential equation $y = px + \sqrt[3]{a^2p^2 + b^2}$, where $p = \frac{dy}{dx}$, are respectively

[BITSAT]

(a) 3,1

(b) 1,3

(c) 1, 1

(d) 3,3

(e) 3, 2

Q 4. The order and degree of the differential equation

$$\left(1+\left(\frac{dy}{dx}\right)^2\right)^{3/4}=\left(\frac{d^2y}{dx^2}\right)^{1/3}$$ is

[Kerala CEE]

(a) $(2,4)$

(b) $(2,3)$

(c) $(6,4)$

(d) $(6,9)$

(e) $(2, 12)$

Q 5. The differential equation representing the family of curves $y^2 = 2c(x + c^{2/3})$, where c is a

positive parameter, is of

[AMU]

(a) order 3, degree 3

(b) order 2, degree 4

(c) order 1, degree 5

(d) order 5, degree 1

Q 6. The differential equation of the family $y = ae^x + bx\,e^x + cx\,e^x$ of curves, where a, b, c are

arbitrary constants, is

[EAMCET]

(a) $y''' + 3y'' + 3y' + y = 0$

(b) $y''' + 3y'' - 3y' - y = 0$

(c) $y''' - 3y'' - 3y' + y = 0$

(d) $y'' - 3y'' + 3y' - y = 0$

Q 7. If $x^2 + y^2 = 1$, then

[WB JEE]

(a) $yy'' - (2y')^2 + 1 = 0$

(b) $yy''+ (y')^2 +1=0$

(c) $yy''- (y')^2 - 1 = 0$

(d) $yy'' + 2(y')^2 + 1 = 0$

Q 8. The order of the differential equation

$$\frac{d^2y}{dx^2} = \sqrt{1+\left(\frac{dy}{dx}\right)^2}$$ is

[WBJEE]

(a) 3

(b) 2

(c) 1

(d) 4

Q 9. The order of the differential equation of all tangent lines to the parabola $y = x^2$ is

[BCECE]

(a) 1

(b) 2

(c) 3

(d) 4

Q 10. The degree of the differential equation

satisfying $\sqrt{1-x^2}+\sqrt{1-y^2}$ = a (x − y) is

[Jamia Millia Manila]

(a) 1

(b) 2

(c) 3

(d) None of these

Q 11. The order and degree of the differential equation

$\frac{d^2y}{dx^2} = \sqrt[3]{1-\left(\frac{dy}{dx}\right)^4}$ are respectively

[MHT CET]

(a) 2, 3

(b) 3, 2

(c) 2, 4

(d) 2, 2

Q 12. Form of the differential equation of all family of lines y = mx + 4/m by eliminating the arbitrary constant m is

[MHT CET]

(a) $\frac{d^2y}{dx^2} = 0$

(b) $x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} + 4 = 0$

(c) $x\left(\frac{dy}{dx}\right)^2 + y\frac{dy}{dx} + 4 = 0$

(d) $\frac{dy}{dx} = 0$

Q 13. The order and degree of the differential equation $\left(1 + 4\frac{dy}{dx}\right)^{2/3} = 4\frac{d^2y}{dx^2}$ are respectively

[MP PET]

(a) $1, \frac{2}{3}$

(b) $3, 2$

(c) $2, 3$

(d) $2, \frac{2}{3}$

Q 14. $y = cx - c^2$, is the general solution of the differential equation

[RPET]

(a) $(y')^2 - xy' + y = 0$

(b) $y'' = 0$

(c) $y' = c$

(d) $(y')^2 + xy' + y = 0$

Q 15. The degree of the equation $e^x + \sin\left(\frac{dy}{dx}\right) = 3$ is

[Guj.CET]

(a) 2

(b) 0

(c) degree is not defined

(d) 1

Q 16. The differential equation of the family of circles with fixed radius 5 unit and centre on the line y = 2, is

[AIEEE]

(a) $(x-2)^2\, y'^2 = 25 - (y-2)^2$

(b) $(x-2)y'^2 = 25 - (y-2)^2$

(c) $(y-2)y'^2 = 25 - (y-2)^2$

(d) $(y-2)^2 y'^2 = 25 - (y-2)^2$

Q 17. The differential equation of all circles which passes through the origin and whose centre lies on y–axis, is

[UP SEE]

(a) $(x^2 - y^2)\frac{dy}{dx} - 2xy = 0$

(b) $(x^2 - y^2)\frac{dy}{dx} + 2xy = 0$

(c) $(x^2 - y^2)\frac{dy}{dx} - xy = 0$

(d) $(x^2 - y^2)\frac{dy}{dx} + xy = 0$

Q 18. The differential equation representing the family of curves $y = xe^{cx}$ (c is a constant) is

[Kerala CEE]

(a) $\frac{dy}{dx} = \frac{y}{x}\left(1 - \log\frac{y}{x}\right)$

(b) $\frac{dy}{dx} = \frac{y}{x}\log\left(\frac{y}{x}\right) + 1$

(c) $\frac{dy}{dx} = \frac{y}{x}\left(1 + \log\frac{y}{x}\right)$

(d) $\frac{dy}{dx} + 1 = \frac{y}{x}\log\left(\frac{y}{x}\right)$

(e) $\frac{dy}{dx} = \frac{x}{y}\left(1 + \log\frac{y}{x}\right)$

Q 19. The differential equation of the family of circles passing through the fixed points $(a, 0)$ and $(-a, 0)$ is

[WBJEE]

(a) $y_1(y^2 - x^2)+2xy + a^2 = 0$

(b) $y_1y^2 + xy + a^2x^2 = 0$

(c) $y_1(y^2 - x^2 +a^2)+2xy = 0$

(d) $y_1(y^2 + x^2) - 2xy + a^2 = 0$

Q 20. The differential equation of the family of curves $y = e^{2x}(a \cos x + b \sin x)$, where a and b are arbitrary constants, is given by

[WB JEE]

(a) $y_2 - 4y_1 + 5y = 0$

(b) $2y_2 - y_1 + 5y = 0$

(c) $y_2 + 4y_1 - 5y = 0$

(d) $y_2 - 2y_1 + 5y = 0$

Q 21. The order and degree of the following differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^{5/2}=\frac{d^3y}{dx^3}$ are respectively

[WB JEE]

(a) 3, 2

(b) 3, 10

(c) 2, 3

(d) 3, 5

Q 22. The differential equation of all straight lines touching the circle $x^2 + y^2 = a^2$ is

[VITEEE]

(a) $\left(y - \frac{dy}{dx}\right)^2 = a^2\left[1 + \left(\frac{dy}{dx}\right)^2\right]$

(b) $\left(y - x\frac{dy}{dx}\right)^2 = a^2\left[1 + \left(\frac{dy}{dx}\right)^2\right]$

(c) $\left(y - x\frac{dy}{dx}\right) = a^2\left[1 + \frac{dy}{dx}\right]$

(d) $\left(y - x\frac{dy}{dx}\right) = a^2\left[1 - \frac{dy}{dx}\right]$

Q 23. The order and degree of the differential equation $\sqrt{\frac{dy}{dx}} - 4\frac{dy}{dx} - 7x = 0$ are

[MHTCET]

(a) 1 and $1/2$

(b) 2 and 1

(c) 1 and 1

(d) 1 and 2

Q 24. The differential equation of all non−vertical lines in a plane is

[Manipal]

(a) $\frac{d^2y}{dx^2} = 0$

(b) $\frac{d^2x}{dy^2} = 0$

(c) $\frac{dy}{dx} = 0$

(d) $\frac{dx}{dy} = 0$

Q 25. The order of the differential equation

whose general solution is given by $y = (c_1 + c_2) \cos (x + c_3) - c_4 e^{x+c_5}$, where c_1, c_2, c_3, c_4, c_5 are arbitrary constants, is

[Manipal]

(a) 4

(b) 3

(c) 2

(d) 5

Q 26. The differential equation of all coaxial parabola $y^2 = 4a(x - b)$, where a and b are arbitrary constants, is

[MP PET]

(a) $y\frac{d^2y}{dx^2}+\frac{dy}{dx}=1$

(b) $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=1$

(c) $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$

(d) $y\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$

Q 27. The order and degree of the differential equation

$$5\left(\frac{d^2y}{dx^2}\right)^5 + 4\left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{dy}{dx}\right)^3 + 2y + x^3 = 0$$

are respectively

[RPET]

(a) (2, 5)

(b) (3, 2)

(c) (1, 3)

(d) $(2, 3)$

Q 28. $y = 2e^{2x} - e^{-x}$ is a solution of the differential equation

[Guj.CET]

(a) $y_2+y_1 +2y = 0$

(b) $y_2 - y_1 + 2y = 0$

(c) $y_2 + y_1 = 0$

(d) $y_2 - y_1 - 2y = 0$

Q 29. The differential equation of all circles passing through the origin and having their centres on the x−axis is

[AIEEE]

(a) $x^2 = y^2 + xy\frac{dy}{dx}$

(b) $x^2 = y^2 + 3xy\frac{dy}{dx}$

(c) $y^2 = x^2 + 2xy\frac{dy}{dx}$

(d) $y^2 = x^2 - 2xy\frac{dy}{dx}$

Q 30. The equation of family of a curve is $y^2 =$

$4a(x + a)$, then differential equation of the family is

[DCE]

(a) $y = y' + x$

(b) $y = y'' + x$

(c) $y = 2y'x + yy^2$

(d) $y'' + y' + y^2 = 0$

Q 31. The differential equation of the family of straight lines whose slope is equal to y−intercept, is

[KCET]

(a) $(x+1)\frac{dy}{dx}-y=0$

(b) $(x+1)\frac{dy}{dx}+y=0$

(c) $\frac{dy}{dx}=\frac{x-1}{y-1}$

(d) $\frac{dy}{dx}=\frac{x+1}{y+1}$

Q 32. If c_1, c_2, c_3, c_4, c_5 and c_6 are constants, then

the order of the differential equation whose general solution is given by $y = c_1 \cos (x + c_2) + c_3 \sin (x + c_4) + c_5 e^x + c_6$, is

[Kerala CEE]

(a) 6

(b) 5

(c) 4

(d) 3

(e) 2

Q 33. The differential equation obtained by eliminating the arbitrary constants a and b from xy = $ae^x + be^{-x}$ is

[EAMCET]

(a) $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy = 0$

(b) $\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} - xy = 0$

(c) $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + xy = 0$

(d) $\frac{d^2y}{dx^2}+\frac{dy}{dx}-xy=0$

Q 34. The degree of the differential equation

$2\left(\frac{d^2y}{dx^2}\right)+3\left(\frac{dy}{dx}\right)^2+4y^3=x$, is

[OJEE]

(a) 0

(b) 1

(c) 2

(d) 3

Q 35. The differential equation of all parabolas whose axes are parallel to y−axis, is

[WB JEE, MHT CET]

(a) $\frac{d^3y}{dx^3} = 0$

(b) $\frac{d^2x}{dy^2} = c$

(c) $\frac{d^3y}{dx^3} + \frac{d^2x}{dy^2} = 0$

(d) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = c$

Q 36. The differential equation of the family of the curves $x^2 + y^2 - 2ax = 0$ is

[BCECE]

(a) $x^2 - y^2 - 2xyy'' = 0$

(b) $y^2 - x^2 = 2xyy'$

(c) $x^2 + y^2 + 2y'' = 0$

(d) None of these

Q 37. The differential equation of the system of all circles of radius r in the xy −plane, is

[VITEEE]

(a) $\left[1+\left(\frac{dy}{dx}\right)^3\right]^2 = r^2\left(\frac{d^2y}{dx^2}\right)^2$

(b) $\left[1+\left(\frac{dy}{dx}\right)^3\right]^2 = r^2\left(\frac{d^2y}{dx^2}\right)^3$

(c) $\left[1+\left(\frac{dy}{dx}\right)^2\right]^3 = r^2\left(\frac{d^2y}{dx^2}\right)^2$

(d) $\left[1+\left(\frac{dy}{dx}\right)^2\right]^3 = r^2\left(\frac{d^2y}{dx^2}\right)^3$

Q 38. The order of the differential equation

whose solution is $y = a \cos x + b \sin x + ce^{-x}$, is

[MHTCET]

(a) 3

(b) 1

(c) 2

(d) 4

Q 39. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$ is a parameter is of order and degree as follows

[MP PET]

(a) order 2, degree 2

(b) order 1, degree 3

(c) order 1, degree 1

(d) order 1, degree 2

Q 40. The differential equation whose solution is

$Ax^2 + By^2 = 1$, where A and B are arbitrary

constants, is of

(a) first order and second degree

(b) first order and first degree

(c) second order and first degree

(d) second order and second degree

Q 41. The differential equation of the family of parabola with focus as the origin and the axis as x–axis, is

[BITSAT]

(a) $y\left(\frac{dy}{dx}\right)^2 + 4x\frac{dy}{dx} = 4y$

(b) $-y\left(\frac{dy}{dx}\right)^2 = 2x\frac{dy}{dx} - y$

(c) $y\left(\frac{dy}{dx}\right)^2 + y = 2xy\frac{dy}{dx}$

(d) $y\left(\frac{dy}{dx}\right)^2 + 2xy\frac{dy}{dx} + y = 0$

Q 42. The differential equation of system of concentric circles with centre $(1,2)$ is

[UP SEE]

(a) $(x-2) + (y-1)\frac{dy}{dx}=0$

(b) $(x-1) + (y-2)\frac{dy}{dx}= 0$

(c) $(x+1)\frac{dy}{dx}+(y-2)=0$

(d) $(x+2)\frac{dy}{dx}+(y-1) = 0$

Q 43. The order and degree of the differential equation $\sqrt{\sin x}(dx + dy) = \sqrt{\cos x}(dx - dy)$ are

[Kerala CEE]

(a) (1, 2)

(b) $(2, 2)$

(c) $(1, 1)$

(d) $(2, 1)$

(e) $(0, 1)$

Q 44. The order and degree of the differential equation $\left(1+3\frac{dy}{dx}\right)^{2/3} = 4\frac{d^3y}{dx^3}$ are

[OJEE]

(a) $\left(1, \frac{2}{3}\right)$

(b) $(3, 1)$

(c) $(3, 3)$

(d) $(l, 2)$

Q 45. The differential equation of all straight lines passing through origin is

[MP PET]

(a) $y = \sqrt{x}\frac{dy}{dx}$

(b) $\frac{dy}{dx} = y + x$

(c) $\frac{dy}{dx} = y - x$

(d) None of these

Q 46. If x = sin t,y = cos pt, then

[Guj. CET]

(a) $(1 - x^2)y_2 + xy_1 + p^2y = 0$

(b) $(1 - x^2)\, y_2 + xy_1 - p^2y = 0$

(c) $(1 + x^2)y_2 - xy_1 + p^2y = 0$

(d) $(1 - x^2)y_2 - xy_1 + p^2y = 0$

Q 47. The differential equation satisfied by the family of curves $y = ax \cos\left(\frac{1}{x} + b\right)$, where a, b are parameters, is

[BITSAT]

(a) $x^2y_2 + y - 0$

(b) $x^4y_2 + y = 0$

(c) $xy_2 - y = 0$

(d) $x^4y_2 - y = 0$

Q 48. The differential equation of the family of curves y = a cos (x + b) is

[BCECE]

(a) $\frac{d^2y}{dx^2} - y = 0$

(b) $\frac{d^2y}{dx^2} + y = 0$

(c) $\frac{d^2y}{dx^2} + 2y = 0$

(d) None of these

Q 49. If $y = (x+\sqrt{1+x^2})^n$, ,then$(1 + x^2) \frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is

[Jamia Millia Islamia]

(a) n^2y

(b) $-n^2y$

(c) $-y$

(d) $2x^2y$

Q 50. If y = a sin (5x + c), then

[MHT CET]

(a) $\frac{dy}{dx} = 5y$

(b) $\frac{dy}{dx} = -5y$

(c) $\frac{d^2y}{dx^2} = -25y$

(d) $\frac{d^2y}{dx^2} = 25y$

Q 51. A normal is drawn at a point P (x, y) of a curve. It meets the x-axis at Q. If PQ is of constant length k, then the differential equation describing such a curve is

[MP PET]

(a) $y\frac{dy}{dx} = \pm\sqrt{k^2 - y^2}$

(b) $x\frac{dy}{dx} = \pm\sqrt{k^2 - x^2}$

(c) $y\frac{dy}{dx} = \pm\sqrt{y^2 - k^2}$

(d) $x\frac{dy}{dx} = \pm\sqrt{x^2 - k^2}$

Q 52. The differential equation for which $\sin^{-1} x + \sin^{-1} y = c$ is given by

[Guj. CET]

(a) $\sqrt{1-x^2}dy + \sqrt{1-y^2}dx = 0$

(b) $\sqrt{1-x^2}dx+\sqrt{1-y^2}dy=0$

(c) $\sqrt{1-x^2}dx-\sqrt{1-y^2}dy=0$

(d) $\sqrt{1-x^2}dy-\sqrt{1-y^2}dx=0$

Q 53. The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant, is

[AIEEE]

(a) $2(x^2 - y^2)y' = xy$

(b) $2(x^2 + y^2)y' = xy$

(c) $(x^2 - y^2)y' = 2xy$

(d) $(x^2 + y^2)y' = 2xy$

Q 54. The degree and order of the differential equation whose solution is a parabola whose axis is x−axis, are

[DCE]

(a) 1,1

(b) 1,2

(c) 1,0

(d) 2,1

Q 55. If $x = A \cos 4t + B \sin 4t$, then $\frac{d^2x}{dt^2}$ is equal to

[KCET]

(a) $-16x$

(b) 16 x

(c) x

(d) −x

Q 56. The differential equation of all non−horizontal lines in a plane is

[Kerala CEE]

(a) $\frac{d^2y}{dx^2} = 0$

(b) $\frac{dx}{dy} = 0$

(c) $\frac{dy}{dx} = 0$

(d) $\frac{d^2x}{dy^2} = 0$

(e) $\frac{dy}{dx} + x = 0$

Q 57. The order and the degree of the differential equation

$$\sqrt{y + \frac{d^2y}{dx^2}} = x + \left(\frac{dy}{dx}\right)^{3/2}$$ are

[Kerala CEE]

(a) 2,2

(b) 2, 1

(c) 1,2

(d) 2, 3

(e) 3, 2

Q 58. If the function $y = \sin^{-1} x$, then $(1 - x^2)$

$\frac{d^2y}{dx^2}$ is equal to

[EAMCET]

(a) $-x\frac{dy}{dx}$

(b) 0

(c) $x\frac{dy}{dx}$

(d) $x\left(\frac{dy}{dx}\right)^2$

Q 59. $y = Ae^x + Be^{2x} + Ce^{3x}$ satisfies the differential equation

[EAMCET]

(a) $y''' - 6y'' + 11y' - 6y = 0$

(b) $y''' + 6y'' + 11y' + 6y = 0$

(c) $y''' + 6y'' - 11y' + 6y = 0$

(d) $y''' - 6y'' - 11y' + 6y = 0$

Q 60. The differential equation obtained by eliminating arbitrary constants from $y = ae^{bx}$ is

[MHT CET]

(a) $y\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

(b) $y\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

(c) $y\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 = 0$

(d) $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$

Q 61. The order and degree of the differential equation

$$y=\frac{dy}{dx}x+\sqrt{a^2\left(\frac{dy}{dx}\right)^2+b^2}$$ is

[RPET]

(a) 3, 1

(b) 1,2

(c) 2,1

(d) 1,3

Q 62. $y = ae^{mx} + be^{-mx}$ satisfies which of the following differential equations

[J&K CET]

(a) $\frac{dy}{dx} - my = 0$

(b) $\frac{dy}{dx} + my = 0$

(c) $\frac{d^2y}{dx^2} - m^2y = 0$

(d) None of these

Q 63. Let F denotes the family of ellipses whose centre is at the origin and major axis is the y–axis.

Then, equation of the family F is

[Kerala CEE]

(a) $\frac{d^2y}{dx^2}+\frac{dy}{dx}\left(x\frac{dy}{dx}-y\right)=0$

(b) $xy\frac{d^2y}{dx^2}-\frac{dy}{dx}\left(x\frac{dy}{dx}-y\right)=0$

(c) $xy\frac{d^2y}{dx^2}+\frac{dy}{dx}\left(x\frac{dy}{dx}-y\right)=0$

(d) $\frac{d^2y}{dx^2}-\frac{dy}{dx}\left(x\frac{dy}{dx}-y\right)=0$

(e) $xy\frac{d^2y}{dx^2}+\left(x\frac{dy}{dx}-y\right)=0$

Topic 2 Variable Separable Method

Q 1. Solution of the differential equation xdy − ydx = 0 represents a

[DCE]

(a) parabola

(b) circle

(c) hyperbola

(d) straight line

Q 2. The solution of the differential equation

$(x^2 - yx^2)\ \frac{dy}{dx} + y^2 + xy^2 = 0$ is

[UPSEE]

(a) $\log\left(\frac{x}{y}\right) = \frac{1}{x} + \frac{1}{y} + c$

(b) $\log\left(\frac{y}{x}\right) = \frac{1}{x} + \frac{1}{y} + c$

(c) $\log(xy) = \frac{1}{x} + \frac{1}{y} + c$

(d) $\log(xy) + \frac{1}{x} + \frac{1}{y} = c$

Q 3. The solution of $\frac{dy}{dx} = \frac{ax + g}{by + f}$ represents a circle when

[BITSAT]

(a) a = b

(b) a = − b

(c) a = -2b

(d) a = 2b

Q 4. The solution of differential equation $(1+x)$

$ydx + (1 - y)xdy = 0$ is

[UPSEE]

(a) $\log_e(xy) + x - y = c$

(b) $\log_e\left(\frac{x}{y}\right) + x + y = c$

(c) $\log_e\left(\frac{x}{y}\right) - x + y = c$

(d) $\log_e(xy) - x + y = c$

Q 5. The solution of $\cos y \frac{dy}{dx} = e^{x + \sin y} + x^2 e^{\sin y}$ is

[Kerala CEE]

(a) $e^x - e^{-\sin y} + \frac{x^3}{3} = c$

(b) $e^{-x} - e^{-\sin y} + \frac{x^3}{3} = c$

(c) $e^x + e^{-\sin y} + \frac{x^3}{3} = c$

(d) $e^x - e^{\sin y} - \frac{x^3}{3} = c$

(e) $e^x - e^{\sin y} + \frac{x^3}{3} = c$

Q 6. The solution of the differential equation $\frac{dy}{dx}$ = sin (x + y) tan (x + y) − 1 is

[EAMCET]

(a) cosec (x + y) + tan (x + y) = x + c

(b) x + cosec (x + y) = c

(c) x + tan (x + y) = c

(d) x + sec (x + y) = c

Q 7. A particular solution of $\log\left(\frac{dy}{dx}\right) = 3x + 4y$,

$y(0) = 0$ is

[BCECE]

(a) $e^{3x} + 3e^{-4y} = 4$

(b) $4e^{3x} - 3e^{-4y} = 3$

(c) $3e^{3x} + 4e^{-4y} = 7$

(d) $4e^{3x} + 3e^{-4y} = 7$

Q 8. General solution of the differential equation $\frac{dy}{dx} = \frac{x+y+1}{x+y-1}$ is given by

[MHT CET]

(a) $x + y = \log | x + y | + c$

(b) $x - y = \log | x + y | + c$

(c) $y = x + \log | x + y | + c$

(d) $y = x \log | x + y | + c$

Q 9. The solution of the differential equation $\frac{dy}{dx} = e^{x+y}$ is

[MP PET]

(a) $e^x + e^y = c$

(b) $e^x - e^y = c$

(c) $e^x + e^{-y} = c$

(d) $e^x - e^{-y} = c$

Q 10. The solution of the differential equation

$(x^2 - yx^2)\frac{dy}{dx} + y^2 + xy^2 = 0$ is

[DCE]

(a) $\log\left(\frac{x}{y}\right) = \frac{1}{x} + \frac{1}{y} + c$

(b) $\log\left(\frac{y}{x}\right) = \frac{1}{x} + \frac{1}{y} + c$

(c) $\log(xy) = \frac{1}{x} + \frac{1}{y} + c$

(d) $\log(xy) + \frac{1}{x} + \frac{1}{y} = c$

Q 11. The solution of the differential equation

$\frac{dy}{dx} = \frac{xy + y}{xy + x}$ is

[BITSAT]

(a) $x + y = \log\left(\frac{cy}{x}\right)$

(b) $x + y = \log(cxy)$

(c) $x - y = \log\left(\frac{cx}{y}\right)$

(d) $y - x = \log\left(\frac{cx}{y}\right)$

Q 12. The solution of the differential equation $(x + y)^2 \frac{dy}{dx} = a^2$ is

[UPSEE]

(a) $(x + y)^2 = \frac{a^2 x}{2} + c$

(b) $(x + y)^2 = a^2 x + c$

(c) $(x + y)^2 = 2a^2 x + c$

(d) None of these

Q 13. The differential equation $y\frac{dy}{dx} + x = c$ represents

[KCET]

(a) a family of hyperbolas

(b) a family of circles whose centres are on the y–axis

(c) a family of parabolas

(d) a family of circles whose centres are on the x–axis

Q 14. The solution of $\frac{dy}{dx} = 1 + y + y^2 + x + xy + xy^2$ is

[Kerala CEE]

(a) $\tan^{-1}\left(\frac{2y+1}{\sqrt{3}}\right) = x + x^2 + c$

(b) $4\tan^{-1}\left(\frac{4y+1}{\sqrt{3}}\right) = \sqrt{3}(2x + x^2) + c$

(c) $\sqrt{3}\tan^{-1}\left(\frac{3y+1}{3}\right) = 4(1 + x + x^2) + c$

(d) $\tan^{-1}\left(\frac{2y+1}{3}\right) = 4(2x + x^2) + c$

(e) $4\tan^{-1}\left(\frac{2y+1}{\sqrt{3}}\right) = \sqrt{3}(2x + x^2) + c$

Q 15. The solution of the differential equation

$\frac{dy}{dx} = \frac{x - 2y + 1}{2x - 4y}$ is

[EAMCET]

(a) $(x - 2y)^2 + 2x = c$

(b) $(x - 2y)^2 + x = c$

(c) $(x - 2y) + 2x^2 = c$

(d) $(x - 2y) + x^2 = c$

Q 16. The solution of the differential equation $9y\frac{dy}{dx} + 4x = 0$ is

[OJEE]

(a) $\frac{y^2}{9} + \frac{x^2}{4} = c$

(b) $\frac{y^2}{4} + \frac{x^2}{9} = c$

(c) $\frac{y^2}{9} - \frac{x^2}{4} = c$

(d) $y^2 - \frac{x^2}{9} = c$

Q 17. The solution of the differential equation

$\frac{d^2y}{dx^2} = e^{-2x}$ is $y = c_1e^{-2x} + c_2x + c_3$, where c_1 is

[OJEE]

(a) 1

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) 2

Q 18. The solution of the differential equation $\frac{dy}{dx}$ $= e^{y+x} + e^{y-x}$ is

[WBJEE]

(a) $e^{-y} = e^{x} - e^{-x} + c$

(b) $e^{-y} = e^{-x} - e^{x} + c$

(c) $e^{-y} = e^{x} + e^{-x} + c$

(d) $e^{-y} + e^{x} + e^{-x} = c$

Q 19. The solution of the differential equation e^{-}

^{x} (y + 1)dy + ($\cos^2$ x − sin 2x)y dx = 0 subjected to

the condition that y = 1 when x = 0 is

[Jamia Millia Islamia]

(a) y + log y + $e^x \cos^2$ x = 2

(b) log (y +1) + $e^x \cos^2$ x = 1

(c) y + log y = $e^x \cos^2$ x

(d) (y + l) + $e^x \cos^2$ x = 2

Q 20. The solution of the differential equation

$\frac{dy}{dx} = \frac{x - y + 3}{2(x - y) + 5}$ is

[MHTCET]

(a) 2 (x − y) + log (x − y) = x + c

(b) 2 (x − y) − log (x − y + 2) = x + c

(c) 2 (x − y) + log (x − y + 2) = x + c

(d) None of the above

Q 21. The solution of the equation $y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$ is

[Manipal]

(a) $y = c(x + a)(1 - ay)$

(b) $y = c(x + a)(1 + ay)$

(c) $y = c(x - a)(1 + ay)$

(d) None of these

Q 22. The solution of the differential equation $\frac{dy}{dx}$ $= e^{3x - 2y} + x^2 e^{-2y}$, is

[MP PET]

(a) $e^{2y} = e^{3x} + x^3 + c$

(b) $\frac{1}{2}e^{2y} = \frac{1}{3}(e^{3x} + x^3) + c$

(c) $\frac{1}{3}e^{2y} = \frac{1}{2}(e^{3x} + x^3) + c$

(d) $e^{2y} = e^{3x} + x^2 + c$

Q 23. The differential equation $(e^x + 1)y\, dy = (y + 1)e^x dx$, has the solution

[MPPET]

(a) $(y - 1)(e^x - 1) = ce^y$

(b) $(y - 1)(e^x + 1) = ce^y$

(c) $(y + 1)(e^x - 1) = ce^y$

(d) $(y + 1)(e^x + 1) = ce^y$

Q 24. The solution of the differential equation $\frac{dy}{dx} = (4x + y + 1)^2$, is

[RPET]

(a) $(4x + y + 1) = \tan(2x + c)$

(b) $(4x + y + 1)^2 = 2\tan(2x + c)$

(c) $(4x + y + 1)^3 = 3\tan(2x + c)$

(d) $(4x + y + 1) = 2\tan(2x + c)$

Q 25. The general solution of the differential equation $\frac{dy}{dx} = \frac{(1+y^2)}{xy(1+x^2)}$ is

[J&KCET]

(a) $(1 + x^2)(1 + y^2) = c$

(b) $(1 + x^2)(1 + y^2) = cx^2$

(c) $(1 - x^2)(1 - y^2) = c$

(d) $(1 + x^2)(1 + y^2) = cy^2$

Q 26. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$

determines a family of circles with

[IITJEE]

(a) variable radii and a fixed centre at $(0, 1)$(b)

variable radii and a fixed centre at $(0, -1)$

(c) fixed radius 1 and variable centres along the x−

axis

(d) fixed radius 1 and variable centres along the y−

axis

Q 27. The equation of curve through point $(1, 0)$

which satisfies the differential equation $(1 + y^2)$dx

− xy dy = 0, is

[DCE]

(a) $x^2 + y^2 = 4$

(b) $x^2 - y^2 = 1$

(c) $2x^2 + y^2 = 2$

(d) None of these

Q 28. The general solution of $y^2dx + (x^2 - xy + y^2)dy = 0$ is

[UP SEE]

(a) $\tan^{-1}\left(\frac{x}{y}\right) + \log y + c = 0$

(b) $2\tan^{-1}\left(\frac{x}{y}\right) + \log x + c = 0$

(c) $\log(y + \sqrt{x^2 + y^2}) + \log y + c = 0$ (d) $\sinh^{-1}$

$\left(\frac{x}{y}\right) + \log y + c = 0$

Q 29. The general solution of the differential

equation $(1 + y^2)dx + (1 + x^2)dy = 0$ is

[AMU]

(a) $x - y = c(1 - xy)$

(b) $x - y = c(1+xy)$

(c) $x + y = c\,(1 - xy)$

(d) $x + y = c\,(1 + xy)$

Q 30. The solution of $\frac{dy}{dx} + I = e^{x+y}$ is

[EAMCET]

(a) $e^{-(x+y)} + x + c = 0$

(b) $e^{-(x+y)} - x + c = 0$

(c) $e^{x+y} + x + c = 0$

(d) $e^{x+y} - x + c = 0$

Q 31. The differential equation $\frac{dy}{dx} = \frac{x(1+y^2)}{y(1+x^2)}$

represents a family of

[OJEE]

(a) parabola

(b) hyperbola

(c) circle

(d) ellipse

Q 32. If x sin $\left(\frac{y}{x}\right)dy = \left[y\sin\left(\frac{y}{x}\right) - x\right]dx$ and $y(1) = \frac{\pi}{2}$,

then the value of $\cos\left(\frac{y}{x}\right)$ is equal to

[VITEEE]

(a) x

(b) $\frac{1}{x}$

(c) log x

(d) e^x

Q 33. If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is

[MP PET]

(a) $\log\left(\frac{x}{y}\right) = cy$

(b) $\log\left(\frac{y}{x}\right) = cx$

(c) $x\log\left(\frac{y}{x}\right) = cy$

(d) $y\log\left(\frac{x}{y}\right) = cx$

Q 34. The solution of the differential equation $\frac{dy}{dx}$ $= e^{x-y} + x^2e^{-y}$ is

[UP SEE]

(a) $y = e^{x-y} - x^2e^{-y}+c$

(b) $e^y - e^x = \frac{1}{3}x^3 + c$

(c) $e^x+e^y=\frac{1}{3}x^3 + c$

(d) $e^x - e^y = \frac{1}{3}x^3 + c$

Q 35. The solution of the differential equation $(1 + y^2)\tan^{-1} x\, dx + y(1+x^2)dy = 0$ is

[MHTCET]

(a) $\log\left(\frac{\tan^{-1} x}{x}\right) + y(1 + x^2) = c$

(b) $\log(1 + y^2) + (\tan^{-1} x)^2 = c$

(c) $\log (1 + x^2) + \log (\tan^{-1} y) + c$

(d) $(\tan^{-1} x)(1 + y^2) + c = 0$

Q 36. The solution of the differential equation $\frac{dy}{dx}$

$\tan y = \sin(x + y) + \sin(x - y)$ is **[MHT CET]**

(a) $\sec y + 2 \cos x = c$

(b) $\sec y - 2 \cos x = c$

(c) $\cos y - 2 \sin x = c$

(d) $\tan y - 2 \sec y = c$

Q 37. The solution of the differential equation

$\sec^2 x \tan y \, dx + \sec^2 y \tan x \, dy = 0$ is

[MPPET]

(a) tan y tan x = c

(b) $\frac{\tan y}{\tan x} = c$

(c) $\frac{\tan^2 x}{\tan y} = c$

(d) None of these

Q 38. The solution of $e^{dy/dx} = (x + 1)$, $y(0) = 3$ is

[Kerala CEE]

(a) $y = x \log x - x + 2$

(b) $y = (x + 1) \log |x + 1| - x + 3$

(c) $y = (x + 1) \log |x+1| + x + 3$

(d) $y = x \log x + x + 3$

(e) $y = -(x + 1) \log |x + 1| + x + 3$

Q 39. Solution of $\frac{dy}{dx} = 3^{x+y}$ is

[MHTCET]

(a) $3^{x+y} = c$

(b) $3^x + 3^y = c$

(c) $3^{x-y} = c$

(d) $3^{x} + 3^{-y} = c$

Q 40. Solution of the equation $x\left(\frac{dy}{dx}\right)^2 + 2\sqrt{xy}\frac{dy}{dx} + y = 0$ is

[MP PET]

(a) $x + y = a$

(b) $\sqrt{x} - \sqrt{y} = \sqrt{a}$

(c) $x^2 + y^2 = a^2$

(d) $\sqrt{x} + \sqrt{y} = c$

Q 41. If c is an arbitrary constant, then the general solution of the differential equation y dx − x dy = xy dx is given by

[J&K CET]

(a) $y = cxe^{-x}$

(b) $y = cye^{-x}$

(c) $y + e^{x} = cx$

(d) $ye^x = cx$

Q 42. If y = y (x) and $\frac{2+\sin x}{y+1}\left(\frac{dy}{dx}\right) = -\cos x$, $y(0)$ $= 1$, then $y\left(\frac{\pi}{2}\right)$ equals

[IITJEE]

(a) $\frac{1}{3}$

(b) $\frac{2}{3}$

(c) $-\frac{1}{3}$

(d) 1

Q 43. The general solution of the differential equation $\frac{dy}{dx}+\frac{1+\cos 2y}{1-\cos 2x}=0$ is given by

[KCET]

(a) tan y + cot x = c

(b) tan y − cot x = c

(c) tan x − cot y − c

(d) tan x + cot y = c

Q 44. The solution of $\frac{dy}{dx} = \frac{ax + h}{by + k}$ represents a parabola, when

[Kerala CEE]

(a) a= 0, b = 0

(b) a = 1, b = 2

(c) a = 0, b $\neq 0$

(d) a =2,b = 1

(e) a = -2,b = −l

Q 45. The solution of $\frac{dy}{dx}+\sqrt{\left(\frac{1-y^2}{1-x^2}\right)}=0$ is

[BCECE]

(a) $\tan^{-1} x + \cot^{-1} x = c$

(b) $\sin^{-1} x + \sin^{-1} y = c$

(c) $\sec^{-1} x + \operatorname{cosec}^{-1} x = c$ (d) None of these

Q 46. Solution of differential equation sec x dy−

cosec y dx=0 is

[MHT CET]

(a) cosx + sin y = c

(b) sinx+cosy = c

(c) sin y − cos x = c

(d) cos y − sin x = c

Q 47. The solution of $\frac{dy}{dx}+1$ = cosec (x + y) is

[Kerala CEE]

(a) cos (x + y) + x = c

(b) $\cos (x + y) = c$

(c) $\sin (x + y) + x = c$

(d) $\sin (x + y) + \sin (x + y) = c$ (e) $x - \cos (x +$

$y)=c$

Q 48. The solution of differential equation $(\sin x$

$+ \cos x) dy + (\cos x - \sin x) dx = 0$ is

[AMU]

(a) $e^x (\sin x + \cos x) + c = 0$

(b) $e^y (\sin x + \cos x) = c$

(c) e^y (cos x − sin x) = c

(d) e^x (sin x − cos x) = c

Q 49. Solution of $\frac{dy}{dx} = \frac{x \log x^2 + x}{\sin y + y \cos y}$ is

[EAMCET]

(a) y sin y = x^2 log x + c

(b) y sin y = x^2 + c

(c) y sin y = x^2 +log x

(d) y sin y = x log x + c

Q 50. The solution of the equation $x^2 \frac{d^2y}{dx^2} = \log x$ when $x = 1$, $y = 0$ and $\frac{dy}{dx} = -1$ is

[OJEE]

(a) $\frac{1}{2}(\log x)^2 + \log x$

(b) $\frac{1}{2}(\log x)^2 - \log x$

(c) $-\frac{1}{2}(\log x)^2 + \log x$

(d) $-\frac{1}{2}(\log x)^2 - \log x$

Q 51. The solution of the differential equation $\frac{dy}{dx}$ = x log x is

[MP PET]

(a) $y = x^2 \log x - \frac{x^2}{2} + c$

(b) $y = \frac{x^2}{2} \log x - \frac{x^2}{4} + c$

(c) $y = \frac{x^2}{2} + \frac{x^2}{2} \log x + c$

(d) None of these

Q 52. If $\frac{dy}{dx} = 1 + x + y + xy$ and $y(-1) = 0$, then the function y is

[RPET]

(a) $e^{(1-x)^2/2}$

(b) $e^{(1+x)^2/2} - 1$

(c) $\log_e(1 + x) - 1$

(d) $(1 + x)$

Topic 3 Homogeneous Differential Equation

Q 1. The solution of the equation $\frac{dy}{dx} = \frac{x+y}{x-y}$ is

[DCE]

(a) $c(x^2+y^2)^{1/2} + e^{\tan^{-1}(y/x)} = 0$

(b) $c(x^2+y^2)^{1/2} + e^{\tan^{-1}(y/x)}$

(c) $c(x^2-y^2) = e^{\tan^{-1}(y/x)}$

(d) None of the above

Q 2. The solution of the differential equation

$xy^2\,dy - (x^3 + y^3)\,dx = 0$ is

[EAMCET]

(a) $y^3 = 3x^3 + c$

(b) $y^3 = 3x^3 \log(cx)$

(c) $y^3 = 3x^3 + \log(cx)$

(d) $y^3 + 3x^3 = \log(cx)$

Q 3. The solution of the differential equation

$xdy - ydx - \sqrt{x^2 + y^2}dx = 0$ is

[VITEEE]

(a) $y - \sqrt{x^2 + y^2} = cx^2$

(b) $y + \sqrt{x^2 + y^2} = cx^2$

(c) $y + \sqrt{x^2 + y^2} = cy^2$

(d) $x - \sqrt{x^2 + y^2} = cy^2$

Q 4. The solution of the differential equation

$(3xy + y^2)\, dx + (x^2 + xy)\, dy = 0$ is

[MHT CET]

(a) $x^2(2xy + y^2) = c^2$

(b) $x^2(2xy - y^2) = c^2$

(c) $x^2(y^2 - 2xy) = c^2$

(d) None of the above

Q 5. The solution of the differential equation

$x(x - y)\frac{dy}{dx} = y(x + y)$, is

[MPPET]

(a) $\frac{x}{y} + \log(xy) = c$

(b) $\frac{y}{x}$ + log (xy) = c

(c) $\frac{x}{y}$ + y logx = c

(d) $\frac{x}{y}$ + x log y=c

Q 6. The solution of $\frac{dy}{dx} = \frac{y^2}{xy - x^2}$ is

[EAMCET]

(a) $e^{y/x}$ = kx

(b) $e^{y/x}$ = ky

(c) $e^{x/y} = kx$

(d) $e^{-y/x} = ky$

Q 7. The solution of the differential equation $(x^2 + y^2)\,dx = 2xy\,dy$ is

[EAMCET]

(a) $x^2 + y^2 = cy$

(b) $c(x^2 - y^2) = x$

(c) $x^2 - y^2 = cy$

(d) $x^2 + y^2 = cx$

(here c is an arbitrary constant)

Q 8. If $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$, $y(1) = 1$, then one of the values of x_0 satisfying $y(x_0) = e$ is given by

[IIT JEE]

(a) $e\sqrt{2}$

(b) $e\sqrt{3}$

(c) $e\sqrt{5}$

(d) $e/\sqrt{2}$

Q 9. If dx + dy = (x + y) (dx − dy), then log (x + y) is equal to

[EAMCET]

(a) x + y + c

(b) x + 2y + c

(c) $x - y + c$

(d) $2x + y + c$

(In the above, c denotes a constant)

Q 10. If $\frac{dy}{dx} = \frac{y + x \tan \frac{y}{x}}{x}$, then $\sin \frac{y}{x}$ is equal to

[EAMCET]

(a) cx^2

(b) cx

(c) cx^3

(d) cx^4

Q 11. The general solution of $\frac{dy}{dx} = \frac{2x - y}{x + 2y}$ is

[J&KCET]

(a) $x^2 - xy + y^2 = c$

(b) $x^2 - xy - y^2 = c$

(c) $x^2 + xy - y^2 = c$

(d) $x^2 + xy^2 = c$

Q 12. The general solution of $y^2dx + (x^2 - xy + y^2)dy = 0$ is

[EAMCET]

(a) $\tan^{-1}\left(\frac{y}{x}\right) = \log y + c$

(b) $2\tan^{-1}\left(\frac{x}{y}\right) + \log x + c = 0$

(c) $\log\left(y+\sqrt{x^2+y^2}\right) + \log y + c = 0$

(d) $\sinh^{-1}\left(\frac{x}{y}\right) + \log y + c = 0$

Topic 4 Linear Differential Equation

Q 1. Solution of the differential equation cos x dy= x sin x − y)dx, 0 <x<$\frac{\pi}{2}$, is

[AIEEE]

(a) sec x = (tan x + c)y

(b) y sec x = tan x + c

(c) y tan x = sec x + c

(d) $\tan x = (\sec x + c)y$

Q 2. The equation of one of the curves whose slope at any point is equal to $y + 2x$ is

[WBJEE]

(a) $y = 2(e^x + x - 1)$

(b) $y = 2(e^x - x - 1)$

(c) $y = 2(e^x - x + 1)$

(d) $y = 2(e^x + x + 1)$

Q 3. The integrating factor of x dy/dx $(1 + x)y$ = x is

[VITEEE]

(a) x

(b) $2x$

(c) $e^{x \log x}$

(d) e^{x}

(e) xe^{x}

Q 4. The solution of the differential equation

$\left(e^{-2\sqrt{x}} - \frac{y}{\sqrt{x}}\right)\frac{dx}{dy} = 1$ is given by

[UP SEE]

(a) $ye^{2\sqrt{x}} = x + c$

(b) $ye^{-2\sqrt{x}} = \sqrt{x} + c$

(c) $y = \sqrt{x}$

(d) $y = 3\sqrt{x}$

Q 5. The integrating factor of the differential equation $(y \log y)\, dx = (\log y - x)\, dy$ is

[Kerala CEE]

(a) $\frac{1}{\log y}$

(b) log (log y)

(c) 1 + log y

(d) $\frac{1}{\log(\log y)}$

(e) log y

Q 6. The solution of the differential equation

$\frac{dy}{dx} = \frac{1}{x + y^2}$ is

[Kerala CEE]

(a) $y = -x^2 - 2x - 2 + ce^x$

(b) $y = x^2 + 2x + 2 - ce^x$

(c) $x = -y^2 - 2y + 2 - ce^y$

(d) $x = -y^2 - 2y - 2 + ce^y$

(e) $x = y^2 + 2y + 2 - ce^y$

Q 7. The integrating factor of the differential equation $x \log x \frac{dy}{dx} + y = 2 \log x$ is given by

[WBJEE]

(a) e^x

(b) log x

(c) log (logx)

(d) x

Q 8. Solution of the equation $x^2 y - x^3 \frac{dy}{dx} = y^4 \cos x$, when $y(0) = 1$ is

[VITEEE]

(a) $y^3 = 3x^3 \sin x$

(b) $x^3 = 3y^3 \sin x$

(c) $x^3 = y^3 \sin x$

(d) $x^3 = y^3 \cos x$

Q 9. The solution of the differential equation $(1 + y^2) + (x - e^{\tan^{-1} y})\frac{dy}{dx} = 0$ is

[J&KCET]

(a) $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + c$

(b) $xe^{\tan^{-1} y} = \tan^{-1} y + c$

(c) $xe^{2\tan^{-1} y} = e^{\tan^{-1} y} + c$

(d) $(x - 2) = ce^{-\tan^{-1} y}$

Q 10. The solution of the differential equation

$\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is

[AIEEE]

(a) $y = x \log x + x$

(b) $y = \log x + x$

(c) $y = x \log x + x^2$

(d) $y = xe^{(x-1)}$

Q 11. The integrating factor of the differential equation

$\frac{dy}{dx}+\frac{y}{(1-x)\sqrt{x}}=1-\sqrt{x}$ is

[Kerala CEE]

(a) $\frac{1-\sqrt{x}}{1+\sqrt{x}}$

(b) $\frac{1+\sqrt{x}}{1-\sqrt{x}}$

(c) $\frac{1-x}{1+x}$

(d) $\frac{\sqrt{x}}{1-\sqrt{x}}$

(e) $\frac{\sqrt{x}}{1+\sqrt{x}}$

Q 12. The solution of the differential equation $\frac{dy}{dx}$ − y tan x = e^x sec x is

[EAMCET]

(a) y = e^x cos x + c

(b) y cos x = e^x+c

(c) y = e^x sin x + c

(d) y sin x = e^x + c

Q 13. The solution of $(1 + x^2)\frac{dy}{dx} + 2xy - 4x^2 = 0$ is

[BCECE]

(a) $3x(1 + y^2) = 4y^3 + c$

(b) $3y(1 + x^2) = 4x^3 + c$

(c) $3x(1 - y^2) = 4y^3 + c$

(d) $3y(1 + y^2) = 4x^3 + c$

Q 14. The solution of the differential equation x

$\frac{dy}{dx} = 2y + x^3e^x$, where $y = 0$ when $x = 1$, is

[MPPET]

(a) $y = x^3(e^x - e)$

(b) $y = x^3(e - e^x)$

(c) $y = x^2(e^x - e)$

(d) $y = x^2(e - e^x)$

Q 15. The solution of the differential equation

$\frac{dy}{dx}+\frac{2x}{1+x^2}.y=\frac{1}{(1+x^2)^2}$ is

[DCE, AMU]

(a) $y(1-x^2) = \tan^{-1} x + c$

(b) $y(1+x^2) = \tan^{-1} x + c$

(c) $y(1+x^2)^2 = \tan^{-1} x + c$

(d) $y(1-x^2)^2 = \tan^{-1} x + c$

Q 16. The solution of the differential equation $\frac{dy}{dx}$

$= y \tan x - 2 \sin x$, is

[UP SEE]

(a) y sin x = c + sin $2x$

(b) y cos x = c+$\frac{1}{2}$sin$2x$

(c) ycosx = c − sin $2x$

(d) y cosx = c+$\frac{1}{2}$cos$2x$

Q 17. If the integrating factor of the differential equation $\frac{dy}{dx}$ + P (x)y = Q (x) is x, then P(x)is

[KeralaCEE]

(a) x

(b) $x^2/2$

(c) $1/x$

(d) $1/x^2$

(e) $1/2x$

Q 18. The solution of $(x + y + 1)\frac{dy}{dx} = 1$ is

[EAMCET]

(a) $y = (x + 2) + ce^x$

(b) $y = -(x+2) + ce^x$

(c) $x = -(y + 2) + ce^y$

(d) $x = (y + 2)^2 + ce^y$

Q 19. The solution of $\frac{dy}{dx} + y \tan x = \sec x$ is

[WB JEE]

(a) $y \sec x = \tan x + c$

(b) $y \tan x = \sec x + c$

(c) $\tan x - y \tan x + c$

(d) x sec x − y tan y + c

Q 20. The solution of $\frac{dy}{dx} + y = e^x$ is

[J&KCET]

(a) $2y = e^{2x} + c$

(b) $2ye^x = e^2 + c$

(c) $2ye^x = e^{2x} + c$

(d) $2ye^{2x} = 2e^x + c$

Q 21. The solution of the differential equation

$$\frac{dy}{dx} - \frac{\tan y}{x} = \frac{\tan y \sin y}{x^2}$$ is

[DCE]

(a) $\frac{x}{\sin y} + \log x = c$

(b) $\frac{y}{\sin x} + \log x = c$

(c) $\log x + x = c$

(d) $\log x + y = c$

Q 22. The solution of the differential equation x $\frac{dy}{dx} + 2y = x^2$ is

[Kerala CEE]

(a) $y = \frac{x^2 + c}{4x^2}$

(b) $y = \frac{x^2}{4} + c$

(c) $y = \frac{x^4 + c}{x^2}$

(d) $y = \frac{x^4 + c}{4x^2}$

(e) $y = \frac{x^3}{4} + \frac{c}{x^2}$

Q 23. An integrating factor of the differential equation $(1 + y + x^2y)dx + (x + x^3)dy = 0$ is

[Kerala CEE]

(a) logx

(b) x

(c) e^x

(d) $\frac{1}{x}$

(e) $-\frac{1}{x}$

Q 24. If y(t) is a solution of $(1 + t)\, dy/dx - ty = 1$ and $y(0) = -1$ then $y(1)$ is equal to

[IIT JEE]

(a) $-\frac{1}{2}$

(b) $e+\frac{1}{2}$

(c) $e-\frac{1}{2}$

(d) $\frac{1}{2}$

Q 25. An integrating factor of the differential equation $x\frac{dy}{dx}+y \log x = xe^{x}x^{-\frac{1}{2}\log x}$, $(x > 0)$, is

[MHTCET]

(a) $x^{\log x}$

(b) $(\sqrt{x})^{\log x}$

(c) $(\sqrt{e})^{(\log x)^2}$

(d) e^{x^2}

Q 26. To reduce the differential equation $\frac{dy}{dx}$ + $P(x) \cdot y = Q(x) \cdot y^n$ to the linear form, the substitution is

[MP PET]

(a) $v = \frac{1}{y^n}$

(b) $v = \frac{1}{y^{n-1}}$

(c) $v = y^n$

(d) $v = y^{n-1}$

Q 27. The solution of differential equation $(1 + y^2) + (x - e^{\tan^{-1} y})\frac{dy}{dx} = 0$ is

[DCE]

(a) $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + k$

(b) $2xe^{\tan^{-1} y} = e^{\tan^{-1} y} + k$

(c) $xe^{\tan^{-1} y} = e^{\tan^{-1} y} + k$

(d) $xe^{\tan^{-1} y} = e^{\tan^{-1} y} + k$

Q 28. The integrating factor of the differential equation $\frac{dy}{dx} + \frac{1}{x} \cdot y = 3x$ is

[UP SEE]

(a) x

(b) ln x

(c) 0

(d) ∞

Q 29. Solution of the differential equation $\frac{dy}{dx}+\frac{y}{x}$ = sin x is

[Kerala CEE]

(a) x (y + cos x) = sin x + c

(b) x (y − cosx) = sin x + c

(c) x (ycos x) = sinx + c

(d) x (y − cos x) = cos x + c

(e) x (y + cos x) = cos x + c

Q 30. The solution of the differential equation x

$\frac{dy}{dx}$ + y = x cos x + sin x, given that y = 1 when $x = \frac{\pi}{2}$

, is

[AMU]

(a) y = sin x − cos x

(b) y = cos x

(c) y = sin x

(d) y = sin x + cos x

Q 31. If $\frac{dy}{dx}$ +y = $2e^{2x}$, then y is equal to

[EAMCET]

(a) $ce^{x}+\frac{2}{3}e^{2x}$

(b) $(1-x)e^{-x}+\frac{2}{3}e^{2x}+c$

(c) $ce^{-x}+\frac{2}{3}e^{2x}$

(d) $e^{-x}+\frac{2}{3}e^{2x}+c$

Q 32. Observe the following statements

I. If dy + 2xydx= $2e^{-x^2}$ dx, then ye^{x^2} =2x + c

II. If . $ye^{x^2} - 2x = c$, then $dx = (2e^{-x^2} - 2xy)dy$

Which is/are correct statements ?

[EAMCET]

(a) Both I and II are true

(b) Neither I nor II is true

(c) I is true, II is false

(d) I is false, II is true

Q 33. The solution of dy = cos x $(2$ − y cosec x)dx, where y = $\sqrt{2}$, when x = $\pi/4$ is

[J&K CET]

(a) y = sin x + $\frac{1}{2}$cosec x

(b) y = tan $(x/2)$ + cot $(x/2)$

(c) y=$(1/\sqrt{2})$sec(x/2)+ $\sqrt{2}$cos(x/2)

(d) None of the above

Q 34. Solution of the differential equation

$\frac{dy}{dx}+\frac{y}{x}=x^2$ is

[DCE]

(a) $y=\frac{x^2}{4}+cx^{-2}$

(b) $y = x^{-1}+cx^{-3}$

(c) $y = \frac{x^3}{4} + cx^{-1}$

(d) $xy = x^2+c$

Q 35. Integrating factor of $(x + 2y^3)\ \frac{dy}{dx} = y^2$ is

[EAMCET]

(a) $e^{\left(\frac{1}{y}\right)}$

(b) $e^{-\left(\frac{1}{y}\right)}$

(c) y

(d) $\frac{-1}{y}$

Q 36. Observe the following statements

A : Integrating factor of $\frac{dy}{dx} + y = x^2$ is e^x.

R : Integrating factor of $\frac{dy}{dx} + P(x)\, y = Q(x)$ is $e^{\int P(x)dx}$

.

[EAMCET]

Then, the true statement among the following is

(a) A is true, R is false

(b) A is false, R is true

(c) A is true, R is true, R $\Rightarrow$ A

(d) A is false, R is false

Q 37. If integrating factor of $x(1 - x^2)dy + (2x^2y - y - ax^3)dx = 0$ is $e^{\int P(x)dx}$, then P is equal to

[RPET]

(a) $\frac{2x^2 - ax^3}{x(1-x^2)}$

(b) $2x^2 - 1$

(c) $\frac{2x^2 - 1}{ax^3}$

(d) $\frac{2x^2 - 1}{x(1-x^2)}$

Q 38. If y(t) is a solution of $(1 + t)\frac{dy}{dx} - ty = 1$ and $y(0) = -1$, then $y(1)$ is equal to

[IITJEE]

(a) $-\frac{1}{2}$

(b) $e + \frac{1}{2}$

(c) $e - \frac{1}{2}$

(d) $\frac{1}{2}$

Q 39. If $\frac{dy}{dx} + y = 2e^{2x}$, then y is equal to

[DCE]

(a) $ce^{x} + \frac{2}{3}e^{2x}$

(b) $(1+x)e^{-x} + \frac{2}{3}e^{2x} + c$

(c) $ce^{-x} + \frac{2}{3}e^{2x}$

(d) $e^{-x} + \frac{2}{3}e^{2x} + c$

Topic 5 Application of Differential Equations, Some Standard Results of Solutions

Q 1. The general solution of the differential equation $100\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + y = 0$ is

[WBJEE]

(a) $y = (c_1 + c_2x)e^x$

(b) $y = (c_1 + c_2x)e^{-x}$ (c) $y = (c_1 + c_2x)\, e^{\frac{x}{10}}$

(d) $y = c_1e^x + c_2e^{-x}$

Q 2. If $y'' - 3y' + 2y = 0$ where $y(0) = 1$, $y'(0) = 0$, then the value of y at x = log, 2 is

[WB JEE]

(a) 1

(b) −1

(c) 2

(d) 0

Q 3. The equation of the curve whose tangent at any point (x, y) makes an angle $\tan^{-1}(2x + 3y)$ with x−axis and which passes through $(1, 2)$ is

[DCE]

(a) $6x+9y + 2 = 26e^{3(x - 1)}$

(b) $6x-9y + 2 = 26e^{3(x - 1)}$

(c) $6x+9y-2 = 26e^{3(x -1)}$

(d) $6x - 9y - 2 = 26e^{3(x -1)}$

Q 4. The equation of the curve satisfying the equation $(xy - x^2)\frac{dy}{dx} = y^2$ and passing through the point

$(-1, 1)$ is

[BITSAT]

(a) $y = (\log y - 1)x$

(b) $y = (\log y + 1)x$

(c) $x = (\log x - 1)y$

(d) $x = (\log x + 1)y$

Q 5. A curve having the condition that the slope of tangent at some point is two times the slope of the straight line joining the same point to the origin of coordinates, is a/an

[UP SEE]

(a) circle

(b) ellipse

(c) parabola

(d) hyperbola

Q 6. The differential equation of all parabolas

whose axes are parallel to axis of x, is

[UP SEE]

(a) $\frac{d^3y}{dx^3} = 0$

(b) $\frac{d^3x}{dy^3} = 0$

(c) $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

(d) $\frac{d^2x}{dy^2} = 0$

Q 7. The equation of the curve satisfying the differential equation $y_2(x^2 + 1)=2xy_1$ passing through the point $(0,1)$ and having slope of tangent at x − 0 as 3 is

[AMU]

(a) $y = x^3 + 3x + 1$

(b) $y = x^3 - 3x + 1$

(c) $y = x^2+3x + 1$

(d) $y = x^2 - 3x + 1$

Q 8. The slope at any point of a curve $y = f(x)$ is given by $\frac{dy}{dx} = 3x^2$ and it passes through $(-1,1)$.

The equation of the curve is

[WB JEE]

(a) $y = x^3 + 2$

(b) $y = -x^3 - 2$

(c) $y = 3x^3 + 4$

(d) $y = -x^3 + 2$

Q 9. The equation of the curve whose subnormal is equal to a constant a is

[BCECE]

(a) $y = ax + b$

(b) $y^2 = 2ax + 2b$

(c) $ay^2 - x^3 = a$

(d) None of these

Q 10. If x dy = y (dx + y dy), y (1) = 1 and y (x) > 0, then y (− 3) is equal to

[Kerala CEE]

(a) 3

(b) 2

(c) 1

(d) 0

(e) -1

Q 11. The equation of curve passing through the point $\left(1, \frac{\pi}{4}\right)$ and having slope of tangent at any point (x, y) as $\frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$, is

[Manipal]

(a) $x = e^{1+\tan\left(\frac{y}{x}\right)}$

(b) $x = e^{1-\tan\left(\frac{y}{x}\right)}$

(c) $x = e^{1+\tan\left(\frac{x}{y}\right)}$

(d) $x = e^{1-\tan\left(\frac{x}{y}\right)}$

Q 12. The solution of the differential equation

$\frac{dy}{dx} = \frac{y}{x} + \frac{\phi\left(\frac{y}{x}\right)}{\phi'\left(\frac{y}{x}\right)}$ is

[Kerala CEE]

(a) $x\phi\left(\frac{y}{x}\right) = k$

(b) $\phi\left(\frac{y}{x}\right) = kx$

(c) $y\phi\left(\frac{y}{x}\right) = k$

(d) $\phi\left(\frac{y}{x}\right) = ky$

(e) $\phi\left(\frac{y}{x}\right) = ke^{\frac{y}{x}}$

Q 13. The equation of the curve through the point $(1,0)$ and whose slope is $\frac{y-1}{x^2+x}$, is

[OJEE]

(a) $2x + (y - 1)(x + 1) = 0$

(b) $2x - (y - 1)(x + l) = 0$

(c) $2x + (y - 1)(x - 1) = 0$

(d) None of these

Q 14. The general solution of the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 2e^{3x}$ is given by

[VITEEE]

(a) $y = (c_1 + c_2x)e^x + \frac{e^{3x}}{8}$

(b) $y = (c_1 + c_2x)e^{-x} + \frac{e^{-3x}}{8}$

(c) $y = (c_1 + c_2x)e^{-x} + \frac{e^{3x}}{8}$

(d) $y = (c_1 + c_2x)\, e^x + \frac{e^{-3x}}{8}$

Q 15. The solution of the differential equation $y\,dx + (x + x^2 y)\,dy = 0$ is

[Jamia Millia Islamia, MP PET]

(a) $-\frac{1}{xy} = c$

(b) $-\frac{1}{xy} + \log y = c$

(c) $\frac{1}{xy} + \log y = c$

(d) $\log y = cx$

Q 16. The differential equation of the curve for

which the initial ordinate of any tangent is equal to the corresponding subnormal, is

[Guj. CET]

(a) non−linear

(b) homogeneous

(c) in variable separable form

(d) None of the above

Q 17. The solution of the equation $\frac{d^2y}{dx^2} = e^{-2x}$ is

[OJEE]

(a) $\frac{e^{-2x}}{4}$

(b) $\frac{e^{-2x}}{4} + cx + d$

(c) $\frac{1}{4}e^{-2x} + cx^2 + d$

(d) $\frac{1}{4}e^{-2x} + c + d$

Q 18. The solution of $x\,dy - ydx + x^2e^x\,dx = 0$ is

[BCECE]

(a) $\frac{y}{x} + e^x = c$

(b) $\frac{x}{y} + e^x = c$

(c) $x + e^y = c$

(d) $y + e^x = c$

Q 19. A function $y = f(x)$ has a second order derivative $f'' = 6(x - 1)$. If its graph passes through the point $(2, 1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function is

[AIEEE, J&K CET]

(a) $(x - 1)^2$

(b) $(x - 1)^3$

(c) $(x + 1)^3$

(d) $(x + 1)^2$

Answers

Topic 1 : Order and Degree, Formation of Differential Equation

1. (c) 2. (b) 3. (a) 4. (a) 5. (c) 6. (d) 7. (b) 8. (b)

9. (a) 10. (a) 11. (a) 12. (b) 13. (c) 14. (a) 15. (c)

16. (d) 17. (a) 18. (c) 19. (c) 20. (a) 21. (a) 22.

(b) 23. (d) 24. (a) 25. (b) 26. (c) 27. (b) 28. (d)

29. (c) 30. (c) 31. (a) 32. (c) 33. (a) 34. (b) 35.

(a) 36. (b) 37. (c) 38. (a) 39. (b) 40. (c) 41. (b)

42. (b) 43. (c) 44. (c) 45. (d) 46. (d) 47. (b) 48.

(b) 49. (a) 50. (c) 51. (a) 52. (a) 53. (c) 54. (b)

55. (a) 56. (d) 57. (a) 58. (c) 59. (a) 60. (c) 61.

(b) 62. (c) 63. (c)

Topic 2 : Variable Separable Method

1. (d) 2. (a) 3. (b) 4. (a) 5. (c) 6. (b) 7. (c) 8. (c)

9. (c) 10. (a) 11. (d) 12. (d) 13. (d) 14. (e) 15. (a)

16. (b) 17. (b) 18. (b) 19. (a) 20. (c) 21. (a) 22.

(b) 23. (d) 24. (d) 25. (b) 26. (c) 27. (b) 28. (a)

29. (c) 30. (a) 31. (b) 32. (c) 33. (b) 34. (b) 35.

(b) 36. (a) 37. (a) 38. (b) 39. (d) 40. (d) 41. (d)

42. (a) 43. (b) 44. (c) 45. (b) 46. (b) 47. (a) 48.

(b) 49. (a) 50. (d) 51. (b) 52. (b)

Topic 3 : Homogeneous Differential Equation

1. (b) 2. (b) 3. (b) 4. (a) 5. (a) 6. (b) 7. (b) 8. (b)

9. (c) 10. (b) 11. (b) 12. (a)

Topic 4 : Linear Differential Equation

1. (a) 2. (b) 3. (e) 4. (a) 5. (e) 6. (d) 7. (b) 8. (b)

9. (a) 10. (a) 11. (b) 12. (b) 13. (b) 14. (c) 15. (b)

16. (d) 17. (c) 18. (c) 19. (a) 20. (c) 21. (a) 22.

(d) 23. (b) 24. (a) 25. (c) 26. (b) 27. (a) 28. (a)

29. (a) 30. (c) 31. (c) 32. (c) 33. (a) 34. (c) 35.

(a) 36. (c) 37. (d) 38. (a) 39. (c)

Topic 5 : Application of Differential Equations, Some Standard Results of Solutions

1. (c) 2. (d) 3. (a) 4. (a) 5. (c) 6. (b) 7. (a) 8. (a)

9. (b) 10. (a) 11. (b) 12. (b) 13. (a) 14. (c) 15. (b)

16. (b) 17. (b) 18. (a) 19. (b)

Definite Integrals

Useful Results & Formulae

1. (i) $\int_a^b f(x)dx = \int_a^b f(t)dt$

(ii) $\int_a^b f(x)dx = -\int_b^a f(x)dx$

(iii) $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx, (a < c < b)$

(iv) $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

(v) $\int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx, & \text{if } (2a-x) = f(x) \\ 0, & \text{if } (2a-x) = -f(x) \end{cases}$

(vi) $\int_0^{2a} f(x)dx = \int_0^a f(2a-x)dx + \int_0^a f(x)dx$

(vii) $\int_{-a}^{a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx, & \text{if } f(-x) = f(x) \\ 0, & \text{if } f(-x) = -f(x) \end{cases}$

(viii) $\int_a^{a+nT} f(x)dx = n\int_0^T f(x)dx$

(ix) $\left|\int_a^b f(x)dx\right| \le \int_a^b |f(x)|\,dx$

2. Summation of series by integration

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{r=1}^{n} f(a+rh)$$

where $h = \frac{b-a}{n}$

3. Gamma function, $\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$

(i) $\int_0^{\pi/2} \sin^m x \cos^n x \, dx = \frac{\lceil\left(\frac{m+1}{2}\right)\rceil\lceil\left(\frac{n+1}{2}\right)}{2\lceil\left(\frac{m+n+2}{2}\right)}$

(ii) $\int_0^{\pi/2} \sin^m x \cos^n x \, dx =$

$$\frac{(m-1)(m-3)....(1 \text{ or } 2)(n-1)(n-3)....(1 \text{ or } 2)}{(m+n)(m+n-2).....(1 \text{ or } 2)} \times \frac{\pi}{2}$$

when both m and n are even numbers.

or $\int_0^{\pi/2} \sin^m x \cos^n x \, dx = \frac{(m-1)(m-3)....(1 \text{ or } 2)(n-1)(n-3)....(1 \text{ or } 2)}{(m+n)(m+n-2).....(1 \text{ or } 2)}$

when either of m or n is odd number.

4. (i) $\lceil(1) = 1, \lceil(0) = \infty, \lceil\left(\frac{1}{2}\right) = \sqrt{\pi}$

(ii) $\lceil(n+1) = n\lceil n$

(iii) If $n \in N$, then $\lceil(n+1) = n!$

5. (i) Beta function = $\int_0^1 (x)^{m-1}(1-x)^{n-1}dx$, where m, n > 0

(ii) B(m, n) = $\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx$

(iii) B(m, n) = $\frac{\lceil m \lceil n}{\lceil m+n}; m$, n > 0

6. (i) $\int_0^{\pi/2} \log(\sin x)dx = \int_0^{\pi/2} \log(\cos x)dx = \frac{-\pi}{2}\log 2$

(ii) $\int_0^{\pi/2} \log(\tan x)dx = \int_0^{\pi/2} \log(\cot x)dx = 0$

(iii) $\int_0^{\pi} \sin mx \sin nx \; dx = \begin{cases} \pi/2, & \text{if } m = n \\ 0, & \text{if } m \neq n \end{cases}$

(iv)

$$\int_0^{\pi/2} f(\sin x)dx = \int_0^{\pi/2} f(\cos x)dx = \begin{cases} \frac{n-1}{n}.\frac{n-3}{n-2}.\frac{n-5}{n-4}.....\frac{2}{3}, & \text{when n is odd} \\ \frac{n-1}{n}.\frac{n-3}{n-2}.\frac{n-5}{n-4}.....\frac{3}{4}.\frac{1}{2}.\frac{\pi}{2}, & \text{when n is even} \end{cases}$$

7. Leibnitz's Rule

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)dt = f[v(x)]\frac{d}{dx}\{v(x)\} - f[u(x)]\frac{d}{dx}\{u(x)\}$$

Topic – 1 Fundamental of Definite Integration

Q 1. The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2}dx$ is (are)

[IIT JEE]

(a) $\frac{22}{7}-\pi$

(b) $\frac{2}{105}$

(c) 0

(d) $\frac{71}{15}-\frac{3\pi}{2}$

Q 2. If b > a, the $\int_a^b \frac{dx}{\sqrt{(x-a)(b-x)}}$ is equal to

[UP SEE]

(a) $\frac{\pi}{2}$

(b) π

(c) $\frac{\pi}{2}(b-a)$

(d) $\frac{\pi}{4}(b-a)$

Q 3. If $\int_0^p \frac{dx}{1+4x^2} = \frac{\pi}{8}$, then the value of p is

[OJEE]

(a) $\frac{1}{4}$

(b) $-\frac{1}{2}$

(c) $\frac{3}{2}$

(d) $\frac{1}{2}$

Q 4. If $I_1 = \int_0^{\pi/4} \sin^2 x dx$ and $I_2 = \int_0^{\pi/4} \cos^2 x\, dx$, then

[WB JEE]

(a) $I_1 = I_2$

(b) $I_1 < I_2$

(c) $I_1 > I_2$

(d) $I_2 = I_1 + \frac{\pi}{4}$

Q 5. The value of integral $\int_{-1}^{1} \frac{|x+2|}{x+2} dx$ is

[WB JEE]

(a) 1

(b) 2

(c) 0

(d) −1

Q 6. If $\int_{-1}^{4} f(x)dx = 4$ and $\int_{2}^{4} \{3 - f(x)\}dx = 7$, then the value of

$\int_{-1}^{2} f(x)dx$ is

[WB JEE]

(a) -2

(b) 3

(c) 4

(d) 5

Q 7. Which of the following is true ?

[MHT CET]

(a) $\int_0^1 e^x dx = e$

(b) $\int_0^1 2^x dx = \log 2$

(c) $\int_0^1 \sqrt{x}\, dx = \frac{2}{3}$

(d) $\int_0^1 x dx = \frac{1}{3}$

Q 8. If f(x) and g(x), x $\in$ R are continuous functions, then value of integral

$\int_{-\pi/2}^{\pi/2} [\{f(x) + f(-x)\}\{g(x) - g(-x)\}] dx$ is

[RPET]

(a) π

(b) $\frac{\pi}{2}$

(c) 1

(d) 0

Q 9. $\int_2^k (2x+1)dx = 6$, then k is equal to

[Guj. CET]

(a) 4

(b) -2

(c) -3

(d) 3

Q 10. $\int_{-1}^{1} \sin^3 x \cos^2 x \, dx$ is equal to

[Guj CET]

(a) -1

(b) 1

(c) 0

(d) None of these

Q 11. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$. Then, which one of the following is true ?

[AIEEE]

(a) $I > \frac{2}{3}$ and $J < 2$

(b) $I > \frac{2}{3}$ and $J > 2$

(c) $I < \frac{2}{3}$ and $J < 2$

(d) $I < \frac{2}{3}$ and $J > 2$

Q 12. The value of $\int_{-3}^{3} (ax^5 + bx^3 + cx + k) dx$, where a,

b, c, k are constant, depends only on

[WB JEE]

(a) a and k

(b) a and b

(c) a, b and c

(d) k

Q 13. $\int_0^{\lambda} \frac{ydy}{\sqrt{y+\lambda}}$ is equal to

[MP PET]

(a) $\frac{2}{3}(2-\sqrt{2})\lambda\sqrt{\lambda}$

(b) $\frac{2}{3}(2+\sqrt{2})\lambda\sqrt{\lambda}$

(c) $\frac{1}{3}(2-\sqrt{2})\lambda\sqrt{\lambda}$

(d) $\frac{1}{3}(2+\sqrt{2})\lambda\sqrt{\lambda}$

Q 14. $3a\int_0^1\left(\frac{ax-1}{a-1}\right)^2 dx$ is equal to

[RPET]

(a) $a - 1 + (a - 1)^{-2}$

(b) $a + a^{-2}$

(c) $a - a^2$

(d) $a^2 + \frac{1}{a^2}$

Q 15. $\int_0^3 |x^3 + x^2 + 3x| \, dx$ is equal to

[DCE]

(a) $\frac{171}{2}$

(b) $\frac{171}{4}$

(c) $\frac{170}{4}$

(d) $\frac{170}{3}$

Q 16. If $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$ and $I_4 = \int_1^2 2^{x^3} dx$, then

[BITSAT]

(a) $I_3 > I_4$

(b) $I_3 = I_4$

(c) $I_1 > I_2$

(d) $I_2 > I_1$

Q 17.

$\int_0^{\pi/4}(\cos x-\sin x)dx+\int_{\pi/4}^{5\pi/4}(\sin x-\cos x)dx+\int_{2\pi}^{\pi/4}(\cos x-\sin x)dx$ is equal to

[UP SEE]

(a) $\sqrt{2}-2$

(b) $2\sqrt{2}-2$

(c) $3\sqrt{2}-2$

(d) $4\sqrt{2}-2$

Q 18. The value of $\int_0^1 \frac{x^4+1}{x^2+1} dx$ is

[UP SEE]

(a) $\frac{1}{6}(3-4\pi)$

(b) $\frac{1}{6}(3\pi+4)$

(c) $\frac{1}{6}(3+4\pi)$

(d) $\frac{1}{6}(3\pi-4)$

Q 19. The value of the integral $\int_0^{\pi/2} (\sin^{100} x - \cos^{100} x) dx$

is

[KCET]

(a) $\frac{1}{100}$

(b) $\frac{100!}{(100)^{100}}$

(c) $\frac{\pi}{100}$

(d) 0

Q 20. The value of the integral $\int_{-\pi/4}^{\pi/4} \sin^{-4} x \, dx$ is

[WB JEE]

(a) $-\frac{8}{3}$

(b) $\frac{3}{2}$

(c) $\frac{8}{3}$

(d) None of these

Q 21. The value of $\int_3^5 \frac{x^2}{x^2-4} dx$ is

[WB JEE]

(a) $2 - \log_e\left(\frac{15}{7}\right)$

(b) $2 + \log_e\left(\frac{15}{7}\right)$

(c) $2 + 4 \log_e 3 - 4\log_e 7 + 4\log_e 5$

(d) $2-\tan^{-1}\left(\frac{15}{7}\right)$

Q 22. The value of $\int_0^{\infty}\frac{dx}{(a^2+x^2)}$ is equal to

[VITEEE]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{2a}$

(c) $\frac{\pi}{a}$

(d) $\frac{1}{2a}$

Q 23. The value of $\int_0^{\pi/2}\frac{\cos 3x+1}{2\cos x-1}dx$ is

[MHTCET]

(a) 2

(b) 1

(c) $\frac{1}{2}$

(d) 0

Q 24. The value of $\int_0^{\pi/2} \frac{(\sin x + \cos x)}{\sqrt{1+\sin 2x}} dx$ is

[RPET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 25. $\int_0^a \sqrt{a^2 - x^2}\,dx$ is equal to

[J&K CET]

(a) πa^2

(b) $\frac{1}{2}\pi a^2$

(c) $\frac{1}{3}\pi a^2$

(d) $\frac{1}{4}\pi a^2$

Q 26. $\int_{\pi/4}^{\pi/2} \operatorname{cosec}^2 x \, dx$ is equal to

[BITSAT, Guj. CET]

(a) -1

(b) 1

(c) 0

(d) $\frac{1}{2}$

Q 27. If $f(x) = \tan x - \tan^3 x + \tan^5 x - \ldots \infty$

with $0 < x < \frac{\pi}{4}$, then $\int_0^{\pi/4} f(x)dx$ is equal to

[Kerala CEE]

(a) 1

(b) 0

(c) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) $-\frac{1}{4}$

Q 28. $\int_{-1}^{1} \frac{\cosh x}{1+e^{2x}} dx$ is equal to

[EAMCET]

(a) 0

(b) 1

(c) $\frac{e^2-1}{2e}$

(d) $\frac{e^2+2}{2e}$

Q 29. If f is a continuous function, then

[MHTCET]

(a) $\int_{-2}^{2} f(x)dx = \int_{0}^{2} [f(x) - f(-x)]dx$

(b) $\int_{-3}^{5} 2f(x)dx = \int_{-6}^{10} f(x-1)dx$

(c) $\int_{-3}^{5} f(x)dx = \int_{-4}^{4} f(x-1)dx$

(d) $\int_{-3}^{5} f(x)dx = \int_{-2}^{6} f(x-1)dx$

Q 30. If $\int_{a}^{b} x^3 dx = 0$ and if $\int_{a}^{b} x^2 dx = \frac{2}{3}$, then the values of a and b are respectively

[AMU]

(a) $1, 1$

(b) $-1, -1$

(c) $1, -1$

(d) $-1, 1$

Q 31. If f(t) is an odd function, then $\int_0^x f(t)dt$ is

[MP PET]

(a) an odd function

(b) an even function

(c) neither even nor odd

(d) 0

Q $32.$ If $g(x) = \frac{f(x) - f(-x)}{2}$ defined over $[-3, 3]$ and $f(x) = 2x^2 - 4x + 1$, then $\int_{-3}^{3} g(x)dx$ is equal to

[RPET]

(a) 0

(b) 4

(c) -4

(d) 8

Q 33. If $2f(x) - 3f\left(\frac{1}{x}\right) = x$, then $\int_1^2 f(x)dx$ is equal to

[J&K CET]

(a) $(3/5)\log 2$

(b) $(-3/5)(1 + \log 2)$

(c) $(-3/5)\log 2$

(d) None of these

Q 34. $\int_0^{\pi/8} \cos^3 4\theta \, d\theta$ is equal to

[KCET]

(a) $\frac{5}{3}$

(b) $\frac{5}{4}$

(c) $\frac{1}{3}$

(d) $\frac{1}{6}$

Q 35. $\int_{-1}^{1} |1 - x| \, dx$ is equal to

[KCET]

(a) -2

(b) 0

(c) 2

(d) 4

Q 36. $\int_0^{\pi/2} \frac{\cos x}{1+\sin x} dx$ is equal to

[Kerala CEE]

(a) log 2

(b) 2 log 2

(c) $(\log 2)^2$

(d) $\frac{1}{2}\log 2$

(e) $2 \log 3$

Q 37. The integral $\int_{-1}^{3}\left[\tan^{-1}\frac{x}{x^2+1}+\tan^{-1}\frac{x^2+1}{x}\right]dx$ is equal to

[AMU]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) π

(d) 2π

Q 38. If $P = \int_0^{3\pi} f(\cos^2 x)dx$ and $Q = \int_0^{\pi} f(\cos^2 x)dx$, then

[OJEE]

(a) $P - Q = 0$

(b) $P - 2Q = 0$

(c) $P - 3Q = 0$

(d) $P - 5Q = 0$

Q 39. $\int_1^e \frac{1}{x} dx$ is equal to

[Jamia Millia Islamia]

(a) ∞

(b) 0

(c) 1

(d) $\log (1 + e)$

Q 40. $\int_0^1 \frac{x^2}{1+x^2} dx$ is equal to

[MHT CET]

(a) $\frac{\pi}{4}-1$

(b) $1-\frac{\pi}{2}$

(c) $\frac{\pi}{2}-1$

(d) $1-\frac{\pi}{4}$

Q 41. If f(x) is a function satisfying $f\left(\frac{1}{x}\right)+x^2 f(x)=0$ for all non−zero x, then $\int_{\sin\theta}^{\operatorname{cosec}\theta} f(x)dx$ is equal to

[J&K CET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 42. $\int_0^{\pi/3} \frac{\cos x + \sin x}{\sqrt{1+\sin 2x}} dx$ is equal to

[DCE]

(a) $\frac{4\pi}{3}$

(b) $\frac{2\pi}{3}$

(c) π

(d) $\frac{\pi}{3}$

Q 43. If $I_n = \int_0^{\pi/4} \tan^n \theta \, d\theta$, the $I_8 + I_6$ is equal to

[Kerala CEE]

(a) $\frac{1}{7}$

(b) $\frac{1}{4}$

(c) $\frac{1}{5}$

(d) $\frac{1}{6}$

(e) $\frac{1}{2}$

Q 44. The value of α which satisfying

$\int_{\pi/2}^{\alpha} \sin x \, dx = \sin 2\alpha$, $\alpha \in (0, 2\pi)$ is equal to

[AMU]

(a) $\frac{\pi}{2}$

(b) $\frac{3\pi}{2}$

(c) $\frac{7\pi}{6}$

(d) All of these

Q 45. $\int_0^3 \frac{3x+1}{x^2+9} dx$ is equal to

[EAMCET]

(a) $\log(2\sqrt{2}) + \frac{\pi}{12}$

(b) $\log(2\sqrt{2}) + \frac{\pi}{2}$

(c) $\log(2\sqrt{2}) + \frac{\pi}{6}$

(d) $\log(2\sqrt{2}) + \frac{\pi}{3}$

Topic – 2 Definite Integration by Substitution, Parts, Partial Fractions

Q 1. If a function f(x) satisfies f'(x) = g(x)

[WB JEE]

Then the value of $\int_a^b f(x)g(x)dx$ is

(a) $\frac{1}{2}[(f(b))^2 - (f(a))^2]$

(b) $\frac{1}{2}[(f(b))^2 + (f(a))^2]$

(c) $\frac{1}{2}[f(b) - f(a)]^2$

(d) None of these

Q 2. $\int_a^b \sqrt{(x-a)(b-x)}dx$, (b > a) is equal to

[DCE]

(a) $\frac{\pi(b-a)^2}{8}$

(b) $\frac{\pi(b+a)^2}{8}$

(c) $(b-a)^2$

(d) $(b+a)^2$

Q 3. The value of $\int_1^e 10^{\log_e x}dx$ is equal to

[Kerala CEE]

(a) $10 \log_e(10\ e)$

(b) $\frac{10e-1}{\log_e 10e}$

(c) $\frac{10e}{\log_e 10e}$

(d) $(10\ e)\log_e\ (10e)$

(e) $\frac{10e}{\log_e(10e)}$

Q 4. The value of $\int_{e^{-1}}^{e} \frac{dt}{t(1+t)}$ is equal to

[Kerala CEE]

(a) 0

(b) $\log\left(\frac{e}{1+e}\right)$

(c) $\log\left(\frac{1}{1+e}\right)$

(d) $\log(1 + e)$ (e) 1

Q 5. Limit of $\int_0^x \left[\frac{1}{\sqrt{1+t^2}} - \frac{1}{1+t}\right] dt$ as $x \to \infty$ is

[AMU]

(a) $\log_2 e$

(b) $\log_e 2$

(c) $\log_e\left(\frac{1}{e}\right)$

(d) $\log_{1/e} 2$

Q 6. $\int_0^{\pi} \frac{1}{1+\sin x} dx$ is equal to

[EAMCET]

(a) 1

(b) 2

(c) -1

(d) -2

Q 7. The value of $\int_0^{\infty} \frac{dx}{(x^2+4)(x^2+9)}$ is

[WB JEE]

(a) $\frac{\pi}{60}$

(b) $\frac{\pi}{20}$

(c) $\frac{\pi}{40}$

(d) $\frac{\pi}{80}$

Q 8. $\int_5^{10} \frac{1}{(x-1)(x-2)} dx$ is equal to

[MHTCET]

(a) $\log \frac{27}{32}$

(b) $\log \frac{32}{27}$

(c) $\log \frac{8}{9}$

(d) $\log \frac{3}{4}$

Q 9. $\int_0^1 \frac{1}{x+\sqrt{x}} dx$ is equal to

[Guj. CET]

(a) log 3

(b) log 1

(c) log 4

(d) log 2

Q 10. $\int_0^2 \frac{x^3 dx}{(x^2+1)^{3/2}}$ is equal to

[DCE]

(a) $(\sqrt{2}-1)^2$

(b) $\frac{(\sqrt{2}-1)^2}{2}$

(c) $\frac{\sqrt{2}-1}{2}$

(d) None of these

Q 11. $\int_0^1 x^{3/2}\sqrt{1-x}\,dx$ is equal to

[EAMCET]

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{9}$

(c) $\frac{\pi}{12}$

(d) $\frac{\pi}{16}$

Q 12. If $I_{10} = \int_0^{\pi/2} x^{10} \sin x \, dx$. Then, the value of $I_{10} + 90I_8$ is

[OJEE]

(a) $10\left(\frac{\pi}{2}\right)^3$

(b) $10\left(\frac{\pi}{2}\right)^9$

(c) $\frac{\pi}{2}$

(d) 0

Q 13. The value of $\int_0^a \sqrt{\frac{a-x}{x}} dx$ is

[VITEEE]

(a) $\frac{a}{2}$

(b) $\frac{a}{4}$

(c) $\frac{\pi a}{2}$

(d) $\frac{\pi a}{4}$

Q 14. The value of $\int_{\pi/4}^{\pi/2} e^x (\log \sin x + \cot x) dx$ is

[MHTCET]

(a) $e^{\pi/4} \log 2$

(b) $-e^{\pi/4} \log 2$

(c) $\frac{1}{2} e^{\pi/4} \log 2$

(d) $-\frac{1}{2} e^{\pi/4} \log 2$

Q 15. If $\int_2^e \left(\frac{1}{\log x} - \frac{1}{(\log x)^2} \right) dx = a + \frac{b}{\log 2}$, then

[Manipal]

(a) a = e, b = 2

(b) a = e, b = 2

(c) a = −e, b = 2

(d) None of these

Q 16. The value of $\int_1^2 \frac{dx}{(x+1)\sqrt{x^2-1}}$ is

[RPET]

(a) 1

(b) $\frac{1}{\sqrt{3}}$

(c) $\frac{2}{\sqrt{3}}$

(d) $\frac{-2}{\sqrt{3}}$

Q 17. The value of $\int_2^3 \frac{x+1}{x^2(x-1)}dx$ is

[UP SEE]

(a) $\log\frac{16}{9}+\frac{1}{6}$

(b) $\log\frac{16}{9}-\frac{1}{6}$

(c) $2\log 2-\frac{1}{6}$

(d) $\log\frac{4}{3}-\frac{1}{6}$

Q 18. The value of integral $\int_0^1 \sqrt{\frac{1-x}{1+x}}dx$ is

[AMU]

(a) $\frac{\pi}{2}+1$

(b) $\frac{\pi}{2}-1$

(c) -1

(d) 1

Q 19. The value of $\int_1^4 e^{\sqrt{x}} dx$ is

[OJEE]

(a) e^2

(b) $2e^2$

(c) $4e^2$

(d) $3e^2$

Q 20. $\int_0^{\infty} \frac{dx}{(x+\sqrt{x^2+1})^3}$ is equal to

[WB JEE]

(a) $\frac{3}{8}$

(b) $\frac{1}{8}$

(c) $\frac{-3}{8}$

(d) None of these

Q 21. The value of integral $\int_{-\pi/2}^{\pi/2} \sqrt{\cos x - \cos^3 x}\, dx$ is

[Guj. CET]

(a) 0

(b) $\frac{4}{3}$

(c) $\frac{2}{3}$

(d) $\frac{1}{5}$

Q 22. The value of $\int_1^2 \frac{dx}{x(1+x^4)}$ is

[UP SEE]

(a) $\frac{1}{4}\log\frac{17}{32}$

(b) $\frac{1}{4}\log\frac{32}{17}$

(c) $\log\frac{17}{2}$

(d) $\frac{1}{4}\log\frac{17}{2}$

Q 23. If $I_1 = \int_0^{\pi/2} x \sin x \, dx$ and $I_2 = \int_0^{\pi/2} x \cos x \, dx$, then

which one of the following is true **?**

[KCET]

(a) $I_1 + I_2 = \frac{\pi}{2}$

(b) $I_1 - I_2 = \frac{\pi}{2}$

(c) $I_1 + I_2 = 0$

(d) $I_1 = I_2$

Q 24. $I_n = \int_0^{\pi/4} \tan^n x\, dx$, then $\lim_{n\to\infty} n[I_n + I_{n+2}]$ us equal to

[OJEE]

(a) $\frac{1}{2}$

(b) 1

(c) ∞

(d) zero

Q 25. $\int_0^{\pi/2} x \sin x \, dx$ is equal to

[RPET]

(a) 0

(b) 1

(c) -1

(d) 2

Q 26. The value of $\int_{-2}^{0}[x^3+3x^2+3x+3+(x+1)\cos(x+1)]dx$

is

[IIT JEE]

(a) 0

(b) 3

(c) 4

(d) 1

Q 27. If $\int_{\log 2}^{x} \frac{du}{(e^u - 1)^{1/2}} = \frac{\pi}{6}$, then e^x is equal to

[BITSAT, OJEE]

(a) 1

(b) 2

(c) 4

(d) −1

Q 28. The value of $\int_1^2 [f\{g(x)\}]^{-1} f'\{g(x)\}\, g'(x)dx$,

where $g(1) = g(2)$ is equal to

[AMU]

(a) 1

(b) 2

(c) 0

(d) None of these

Q 29. $\int_1^2 e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)dx$ is equal to

[Jamia Millia Islamia, MHT CET]

(a) $e-\frac{e^2}{2}$

(b) $\frac{e^2}{2} - e$

(c) $\frac{e^2}{2} + e$

(d) $\frac{e^2}{2} - 2$

Q 30. $\int_1^3 \frac{\cos(\log x)}{x} dx$ is equal to

[MHTCET]

(a) 1

(b) cos (log 3)

(c) sin (log 3)

(d) $\pi/4$

Q 31. The value of $\int_1^{e^2} \frac{dx}{x(1+\log x)^2}$ is

[J&K CET]

(a) $\frac{2}{3}$

(b) $\frac{1}{3}$

(c) $\frac{3}{2}$

(d) $\ln 2$

Q 32. If the tangent to the graph function y = f(x) makes angles $\frac{\pi}{4}$ and $\frac{\pi}{3}$ with the x−axis is at the point x = 2 and x = 4 respectively, the value of $\int_2^4 f'(x)f''(x)dx$

[Kerala CEE]

(a) f(4) f(2)

(b) f(4)

(c) f(2)

(d) 0

(e) 1

Q 33. If $f(x)=\begin{vmatrix} \sin x+\sin 2x+\sin 3x & \sin 2x & \sin 3x \\ 3+4\sin x & 3 & 4\sin x \\ 1+\sin x & \sin x & 1 \end{vmatrix}$, then the value of $\int_0^{\pi/2} f(x)dx$ is

[AMU]

(a) 3

(b) $\frac{2}{3}$

(c) $\frac{1}{3}$

(d) 0

Q 34. $\int_0^2 \frac{2x-2}{2x-x^2}dx$ is equal to

[EAMCET]

(a) 0

(b) 2

(c) 3

(d) 4

Q 35. $\int_1^x \frac{\log(x^2)}{x}dx$ is equal to

[BCECE]

(a) $(\log x)^2$

(b) $\frac{1}{2}(\log x)^2$

(c) $\frac{\log x^2}{2}$

(d) None of these

Q 36. $\int_0^{\pi/6} \frac{\sin x}{\cos^3 x} dx$ is equal to

[RPET]

(a) $\frac{2}{3}$

(b) $\frac{1}{6}$

(c) 2

(d) $\frac{1}{3}$

Q 37. $\int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}} dx$ is equal to

[J & K CET]

(a) $\frac{\pi}{2}(\alpha-\beta)$

(b) $\frac{\pi}{2}(\beta-\alpha)$

(c) $\pi(\alpha-\beta)$

(d) $\pi(\beta-\alpha)$

Q 38. If $I(m, n) = \int_0^1 t^m (1+t)^n dt$, then the expression for $I(m, n)$ in terms of $I(m + 1, n - 1)$ is

[IIT JEE]

(a) $\frac{2^n}{m+1} - \frac{n}{m+1} I(m+1, n-1)$

(b) $\frac{n}{m+1} I(m+1, n-1)$

(c) $\frac{2^n}{m+1} + \frac{n}{m+1} I(m+1, n-1)$

(d) $\frac{m}{m+1} I(m+1, n-1)$

Q 39. If $f(y) = e^y$, $g(y) = y$; $y > 0$ and $F(t) = \int_0^t f(t-y)g(y)dy$, then

[AIEEE]

(a) $F(t) = 1 - e^{-t}(1 + t)$

(b) $F(t) = e^t - (1 + t)$

(c) $F(t) = te^t$

(d) $F(t) = te^{-t}$

Q 40. The value of the integral $I = \int_0^1 x(1-x)^n dx$ is

[AIEEE]

(a) $\frac{1}{n+1}$

(b) $\frac{1}{n+2}$

(c) $\frac{1}{n+1}-\frac{1}{n+2}$

(d) $\frac{1}{n+1}+\frac{1}{n+2}$

Q 41. Let $\frac{d}{dx}F(x)=\left(\frac{e^{\sin x}}{x}\right), x>0$.

If $\int_1^4 \frac{3}{x}e^{\sin x^3}dx = F(x) - F(1)$, then one of the possible value of k, is

[AIEEE]

(a) 15

(b) 16

(c) 63

(d) 64

Q 42. Let f(x) be a function satisfying f'(x) = f(x) with f(0) = 1 and g(x) be a function that satisfies f(x) + g(x) = x^2. Then, the value of the integral $\int_0^1 f(x)g(x)dx$, is

[AIEEE]

(a) $e-\frac{e^2}{2}-\frac{5}{2}$

(b) $e+\frac{e^2}{2}-\frac{3}{2}$

(c) $e-\frac{e^2}{2}-\frac{3}{2}$

(d) $e+\frac{e^2}{2}+\frac{5}{2}$

Q 43. $\int_8^{15}\frac{dx}{(x-3)\sqrt{x+1}}$ is equal to

[UP SEE]

(a) $\frac{1}{2}\log\frac{5}{3}$

(b) $\frac{1}{3}\log\frac{5}{3}$

(c) $\frac{1}{5}\log\frac{3}{5}$

(d) $\frac{1}{2}\log\frac{3}{5}$

Q 44. The value of $\int_0^{\pi}|\sin^3\theta|d\theta$ is

[UP SEE]

(a) 0

(b) π

(c) $\frac{4}{3}$

(d) $\frac{3}{8}$

Q 45. $\int_0^1 \cot^{-1}(1-x+x^2)dx$ is equal to

[OJEE, J & K CET]

(a) π − log 2

(b) π + log 2

(c) $\frac{\pi}{2}+\log 2$

(d) $\frac{\pi}{2}-\log 2$

Q 46. The value of $\int_0^{\pi/2} \frac{\cos\theta}{\sqrt{4-\sin^2\theta}}d\theta$ is

[BCECE]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{5}$

Q 47. $\int_0^{\pi/2} \frac{\cos x}{(1+\sin x)(2+\sin x)} dx$

[MP PET]

(a) $\log \frac{4}{3}$

(b) $\log\frac{1}{3}$

(c) $\log\frac{3}{4}$

(d) None of these

Q 48. $\int_0^2 \sqrt{\frac{2+x}{2-x}}dx$ is equal to

[MP PET]

(a) $\pi + 2$

(b) $\pi+\frac{3}{2}$

(c) $\pi + 1$

(d) π

Topic – 3 Properties of Summation of Series of Definite Integration

Q 1. Let p(x) be a function defined on R such that

$\lim_{x\to\infty}\frac{f(3x)}{f(x)}=1$, p'(x) = p'(1 – x), for all x $\in$ [0, 1], p(0) = 1 and p(1) = 41. Then, $\int_0^1 p(x)dx$ equals

[AIEEE]

(a) $\sqrt{41}$

(b) 21

(c) 41

(d) 42

Q 2. The value of $I = \int_{-\pi/2}^{\pi/2} |\sin x| \, dx$ is

[WB JEE]

(a) 0

(b) 2

(c) -2

(d) $-2 < I < 2$

Q 3. If $I_1 = \int_0^{3\pi} f(\cos^2 x)dx$ and $I_2 = \int_0^{\pi} f(\cos^2 x)dx$, then

[WB JEE, BITSAT]

(a) $I_1 = I_2$

(b) $3I_1 = I_2$

(c) $I_1 = 3I_2$

(d) $I_1 = 5I_2$

Q 4. If $I = \int_0^1 \frac{dx}{1+x^{\pi/2}}$, then

[WB JEE, VITEEE]

(a) $\log_e 2 < I < \pi/4$

(b) $\log_e 2 > I$

(c) $I = \pi/4$

(d) $I = \log_e 2$

Q 5. If $\int_0^a f(2a - x)dx = m$ and $\int_0^a f(x)dx = n$, then $\int_0^{2a} f(x)dx$ is equal to

[Kerala CEE]

(a) $2m + n$

(b) m + 2n

(c) m – n

(d) n – m

(e) m + n

Q 6. $\int_{-1}^{1}\left(e^{x^3}+e^{-x^3}\right)(e^x-e^{-x})dx$ is equal to

[BITSAT]

(a) $\frac{e^2}{2}-2e$

(b) $e^2 - 2e$

(c) $2(e^2 - e)$

(d) $2e^{-2} - 2e$

(e) 0

Q 7. The integral $\int_0^1 \frac{2\sin^{-1}\frac{x}{2}}{x} dx$ equals

[Kerala CEE]

(a) $\int_0^{\pi/6} \frac{x}{\tan x} dx$

(b) $\int_0^{\pi/6} \frac{2x}{\tan x} dx$

(c) $\int_0^{\pi/2} \frac{2x}{\tan x} dx$

(d) $\int_0^{\pi/6} \frac{x}{\sin x} dx$

(e) $\int_0^{\pi/6} \frac{2x}{\sin x} dx$

Q 8. $\int_{-100}^{100} f(x)dx$ is equal to

[Kerala CEE]

(a) $\int_{-100}^{100} f(x^2)dx$

(b) $\int_{-100}^{100} f(-x^2)dx$

(c) $\int_{-100}^{100} f\left(\frac{1}{x}\right)dx$

(d) $\int_{-100}^{100} f(-x)dx$

(e) $\int_{-100}^{100}[f(x)+f(-x)]dx$

Q 9. $\int_{0}^{\pi}[\cot x]dx$, [.] denotes the greatest integer function, is equal to

[AIEEE]

(a) $\frac{\pi}{2}$

(b) 1

(c) -1

(d) $-\frac{\pi}{2}$

Q 10. $\lim_{n\to\infty}\left(\frac{1}{n}+\frac{1}{n+1}+.....+\frac{1}{3n}\right)$ is equal to

[DCE]

(a) log 2

(b) log 3

(c) log 5

(d) 0

Q 11. $\int_{-1}^{2}|x|^3\,dx$ is equal to

[BITSAT]

(a) $\frac{5}{4}$

(b) $\frac{17}{4}$

(c) $\frac{15}{4}$

(d) $\frac{4}{5}$

Q 12. The value of $\int_2^4 \{|x-2|+|x-3|\}dx$ is

[UP SEE]

(a) 1

(b) 2

(c) 3

(d) 5

Q 13. If $f(x) = \int_{-1}^{x} |t| dt$, then for any $x \geq 0$, f(x) is equal to

[KCET]

(a) $1 - x^2$

(b) $\frac{1}{2}(1 + x^2)$

(c) $1 + x^2$

(d) $\frac{1}{2}(1 - x^2)$

Q 14. The value of $\int_{-2}^{4} |x+1| \, dx$ is equal to

[Kerala CEE]

(a) 12

(b) 14

(c) 13

(d) 16

(e) 15

Q 15. If $\left|\int_2^5 \frac{\sin x dx}{(1+x^2)}\right|$, then

[AMU]

(a) I $\geq \frac{1}{4}$

(b) I lies in interval $\left(\frac{1}{4},\frac{1}{5}\right)$

(c) I lies in the interval $\left(\frac{1}{5},\frac{1}{6}\right)$

(d) I $\leq \frac{3}{10}$

Q 16. The value of $\lim_{n\to\infty}\left[\frac{n}{n^2+1^2}+\frac{n}{n^2+2^2}+.....+\frac{n}{n^2+n^2}\right]$ is

[WB JEE]

(a) $\frac{\pi}{4}$

(b) log 2

(c) 0

(d) 1

Q 17. If f(x) = f(a – x), then $\int_0^a f(x)dx$ is equal to

[WB JEE]

(a) $\int_0^a f(x)dx$

(b) $\frac{a^2}{2}\int_0^a f(x)dx$

(c) $\frac{a}{2}\int_0^a f(x)dx$

(d) $-\frac{a}{2}\int_0^a f(x)dx$

Q 18. $\int_{-3}^{2}\{|x+1|+|x+2|+|x-1|\}dx$ is equal to

[BCECE]

(a) $\frac{31}{2}$

(b) $\frac{35}{2}$

(c) $\frac{37}{2}$

(d) $\frac{39}{2}$

Q 19. $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}$ equals

[Jamia Millia Islamia]

(a) $1+\sqrt{5}$

(b) $-1+\sqrt{5}$

(c) $-1+\sqrt{2}$

(d) $1+\sqrt{2}$

Q 20. Let f be a positive function. Let

$I_1 = \int_{1-k}^{k} xf\{x(1-x)\}dx,\ I_2 = \int_{1-k}^{k} f\{x(1-x)\}dx$ where $2k - 1 > 0$.

Then, $\frac{I_1}{I_2}$ is

[Jamia Millia Islamia]

(a) 2

(b) k

(c) $\frac{1}{2}$

(d) 1

Q 21. $\int_0^{\pi/2} \log \sin x \, dx$ is equal to

[MP PET]

(a) $-\pi \log 2$

(b) $\pi \log 2$

(c) $-\frac{\pi}{2}\log 2$

(d) $\frac{\pi}{2}\log 2$

Q 22. The value of integral $\int_0^4 |x-1| \, dx$ is

[RPET]

(a) 4

(b) 5

(c) 7

(d) 9

Q 23. $\int_{-\pi/2}^{\pi/2} \sin|x|\,dx$ is equal to

[BITSAT, EAMCET]

(a) 0

(b) 1

(c) 2

(d) π

Q 24. $\int_0^{1000} e^{x-[x]} dx$ is

[UP SEE]

(a) $e^{1000} - 1$

(b) $\frac{e^{1000}-1}{e-1}$

(c) $1000(e - 1)$

(d) $\frac{e-1}{1000}$

Q 25. The value of $\int_{-2}^{2}(ax^3+bx+c)dx$ depends on the

[KCET]

(a) value of b

(b) value of c

(c) value of a

(d) values of a and b

Q 26. If $\int_0^{\pi} xf(\sin x)dx = A\int_0^{\pi/2} f(\sin x)dx$, then A is equal to

[Kerala CEE]

(a) 0

(b) π

(c) $\frac{\pi}{4}$

(d) 2π

(e) 3π

Q 27. $\int_{-2}^{2} |[x]| dx$ is equal to

[Kerala CEE]

(a) 1

(b) 2

(c) 3

(d) 4

(e) 5

Q 28. $\int_{0}^{1} \frac{dx}{x+\sqrt{1-x^2}}$ is equal to

[RPET]

(a) $\pi/3$

(b) $\pi/2$

(c) $1/2$

(d) $\pi/4$

Q 29. The value of $\int_{-2}^{2}\left[p\log\left(\frac{1+x}{1-x}\right)+q\log\left(\frac{1-x}{1+x}\right)^{-2}+r\right]dx$

depends on

[OJEE]

(a) the value of p

(b) the value of q

(c) the value of r

(d) the values of p and q

Q 30. The value of $\int_0^1 \log[x]dx$ is

[OJEE]

(a) 0

(b) 1

(c) log 1

(d) None of these

Q 31. If $I = \int_0^{\pi/2} \cos(\sin x)dx$, $J = \int_0^{\pi/2} \sin(\cos x)dx$ and $K = \int_0^{\pi/2} \cos x\, dx$. Then,

[OJEE]

(a) K > I > J

(b) J > I > K

(c) I > J > K

(d) I > K > J

Q 32. If $I = \int_{-\pi}^{\pi} \frac{e^{\sin x}}{e^{\sin x} + e^{-\sin x}} dx$, then I equals

[WB JEE]

(a) $\frac{\pi}{2}$

(b) 2π

(c) π

(d) $\frac{\pi}{4}$

Q 33. The value of the integral $\int_0^2 |x^2 - 1| dx$ is

[WB JEE]

(a) 0

(b) 2

(c) $-\frac{1}{3}$

(d) -2

Q 34. The value of the integral $\int_{-a}^{a} \frac{xe^{x^2}}{1+x^2} dx$ is

[WB JEE]

(a) e^{a^2}

(b) 0

(c) c^{-a^2}

(d) a

Q 35. $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x}$ is equal to

[BCECE]

(a) π

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{4}$

(d) $\frac{3\pi}{2}$

Q 36. The value of $\int_{-\pi/2}^{\pi/2} \log\left(\frac{2-\sin\theta}{2+\sin\theta}\right) d\theta$ is

[BCECE]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 37. The value of $\int_0^{\pi} x \sin^3 x \, dx$ is

[MHTCET]

(a) $\frac{4\pi}{3}$

(b) $\frac{2\pi}{3}$

(c) 0

(d) None of these

Q 38. The value of $\int_0^{\pi/2} \frac{dx}{1+\cot x}$ is

[MP PET]

(a) π

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{4}$

Q 39. The value of $\int_0^{12a} \frac{f(x)}{f(x)+f(12a-x)}dx$ is

[MP PET]

(a) a

(b) 2a

(c) 3a

(d) 6a

Q 40. $\int_{-\pi/2}^{\pi/2} \frac{dx}{1+\cos x}$ is equal to

[J&K CET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 41. $\int_{\pi/6}^{\pi/3} \frac{1}{1+\tan^3 x} dx$ is

[Guj. CET]

(a) $\frac{\pi}{12}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{6}$

Q 42. Let F(x) = f(x) + $f\left(\frac{1}{x}\right)$, where $f(x)=\int_1^x \frac{\log t}{1+t}dt$.

Then, F(e) equals

[AIEEE]

(a) $1/2$

(b) 0

(c) 1

(d) 2

Q 43. $\int_0^{10} |x(x-1)(x-2)|\,dx$ is equal to

[DCE]

(a) 160.05

(b) 1600.5

(c) 16.005

(d) None of these

Q 44. $\int_0^8 |x-5|\,dx$ is equal to

[BITSAT]

(a) 17

(b) 9

(c) 12

(d) 18

Q 45. If $k\int_0^1 x.f(3x)dx = \int_0^3 t.f(t)dt$, then the value of k is

[KCET]

(a) 9

(b) 3

(c) $\frac{1}{9}$

(d) $\frac{1}{3}$

Q 46. $\int_{-1}^{1} \frac{17x^5 - x^4 + 29x^3 - 31x + 1}{x^2 + 1} dx$ is

[Kerala CEE]

(a) $\frac{4}{5}$

(b) $\frac{5}{4}$

(c) $\frac{4}{3}$

(d) $\frac{3}{4}$

(e) $\frac{2}{3}$

Q 47. If $g(x) = \int_0^x \cos^4 t \, dt$, then $g(x + \pi)$ is equal to

[Kerala CEE]

(a) $g(x) + g(\pi)$

(b) $g(x) - g(\pi)$

(c) $g(x) \cdot g(\pi)$

(d) $\frac{g(x)}{g(\pi)}$

(e) $\frac{g(\pi)}{g(x)}$

Q 48. The value of $I = \int_0^1 x \left| x - \frac{1}{2} \right| dx$ is

[AMU]

(a) $\frac{1}{3}$

(b) $\frac{1}{4}$

(c) $\frac{1}{8}$

(d) None of these

Q 49. If $f(t)=\int_{-t}^{t}\frac{e^{-|x|}}{2}dx$, then $\lim_{t\to\infty} f(t)$ is equal to

[EAMCET]

(a) 1

(b) $\frac{1}{2}$

(c) 0

(d) -1

Q 50. $\int_0^{2\pi} \sin^6 x \cos^5 x \, dx$ is equal to

[EAMCET]

(a) 2π

(b) $\pi/2$

(c) 0

(d) $-\pi$

Q 51. $\int_0^{\pi/2} \frac{2\sqrt{\cos\theta}}{3(\sqrt{\sin\theta}+\sqrt{\cos\theta})} d\theta$ is equal to

[OJEE]

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) None of these

Q 52. If for every integer n, $\int_{n}^{n+1} f(x)dx = n^2$, then the value of $\int_{-2}^{4} f(x)dx$ is

[WB JEE]

(a) 16

(b) 14

(c) 19

(d) None of these

Q 53. The value of integral $\int_0^{\pi} xf(\sin x)dx$ is

[WB JEE]

(a) $0\,\pi\int_0^{\pi/2} f(\sin x)dx$

(c) $\frac{\pi}{4}\int_0^{\pi} f(\sin x)dx$

(d) None of these

Q 54. The value of $\int_{-\pi/2}^{\pi/2}(x^3 + x\cos x + \tan^5 x + 1)dx$ is equal to

[WB JEE]

(a) 0

(b) 2

(c) π

(d) None of these

Q 55. If f(x) = f(a – x) and g(x) + g(a – x) = 2,

value of $\int_0^a f(x)g(x)dx$ is

[BCECE]

(a) $\int_0^a f(x)dx$

(b) $\int_0^a g(x)dx$

(c) $\int_0^a [g(x)-f(x)]dx$

(d) $\int_0^a [g(x)+f(x)]dx$

Q 56. The value of $\int_1^4 |x-3|dx$ is equal to

[VITEEE]

(a) 2

(b) $\frac{5}{2}$

(c) $\frac{1}{2}$

(d) $\frac{3}{2}$

Q 57. The value of $\int_{-2}^{3}|1-x^2|\,dx$ is

[Jamia Millia Islamia]

(a) $\frac{28}{3}$

(b) $\frac{14}{3}$

(c) $\frac{7}{3}$

(d) $\frac{1}{3}$

Q 58. The value of $\int_0^1 \tan^{-1}\left(\frac{2x-1}{1+x-x^2}\right)dx$ is

[MHTCET]

(a) 1

(b) 0

(c) −1

(d) None of these

Q 59. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx, a > 0$ is

[MP PET]

(a) 2π

(b) $\frac{\pi}{a}$

(c) $\frac{\pi}{2}$

(d) $a\pi$

Q 60. $\int_0^{\pi} \cos^3 x dx$ is equal to

[J&K CET]

(a) 0

(b) 1

(c) -1

(d) $\frac{1}{2\sqrt{2}}$

Q 61. To find the numerical value of

$\int_{-2}^{2}(px^3+qx+s)dx$, it is necessary to know the values of

the constants

[Guj. CET]

(a) p

(b) q

(c) s

(d) p and s

Q 62. The value of the integral $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}} dx$ is

[AIEEE]

(a) $\frac{3}{2}$

(b) 2

(c) 1

(d) $\frac{1}{2}$

Q 63. $\int_{-3\pi/2}^{-\pi/2}[(x+\pi)^3+\cos^2(x+3\pi)]dx$ is equal to

[AIEEE]

(a) $\left(\frac{\pi^4}{32}\right)+\left(\frac{\pi}{2}\right)$

(b) $\left(\frac{\pi}{2}\right)$

(c) $\left(\frac{\pi}{4}\right)-1$

(d) $\frac{\pi^4}{32}$

Q 64. The value of $\int_1^a [x]f'(x)dx$, a > 1, where [x] denotes the greatest integer not exceeding x, is

[AIEEE]

(a) [a]f(a) – {f(1) + f(2) + …. + f([a])}

(b) [a]f([a]) – {f(1) + f(2) +….+ f(a)}

(c) af ([a]) – {f(1) + f(2) +….+ f(a)}

(d) af(a) – {f(1) + f(2) + …. + f([a])}

Q 65. The value of $\int_0^{\pi/2} \frac{\sin^2 x - \cos^2 x}{\sin^3 x + \cos^3 x} dx$ is

[DCE]

(a) 0

(b) 1

(c) 2

(d) 3

Q 66. The value of $\int_{\pi}^{2\pi}[2\sin x]dx$ is

[DCE]

(a) $\frac{\pi}{3}$

(b) $-\frac{5\pi}{3}$

(c) $\frac{4\pi}{3}$

(d) $-\frac{\pi}{3}$

Q 67. $\int_{0}^{\pi/4}\log(1+\tan x)dx$ is equal to

[BITSAT, Guj. CET]

(a) $\frac{\pi}{8}\log_e 2$

(b) $\frac{\pi}{4}\log_2 e$

(c) $\frac{\pi}{4}\log_e 2$

(d) $\frac{\pi}{8}\log_e\left(\frac{1}{2}\right)$

Q 68. The value of the integral $\int_a^b \frac{\sqrt{x}dx}{\sqrt{x}+\sqrt{a+b-x}}$ is

[UP SEE, WB JEE]

(a) π

(b) $\frac{1}{2}(b-a)$

(c) $\frac{\pi}{2}$

(d) b – a

Q 69. If f(x) is defined $[-2, 2]$ by $f(x) = 4x^2 - 3x + 1$ and $g(x) = \frac{f(-x) - f(x)}{x^2 + 3}$, then $\int_{-2}^{2} g(x)dx$ is equal to

[KCET]

(a) $\int_0^{\pi} |\cos x| \, dx$ is equal to

[Kerala CEE]

(a) $\frac{1}{2}$

(b) -2

(c) 1

(d) -1

(e) 2

Q 71. $\int_0^2 [x^2]dx$ is

[Kerala CEE, OJEE]

(a) $2-\sqrt{2}$

(b) $2+\sqrt{2}$

(c) $\sqrt{2}-1$

(d) $-\sqrt{2}-\sqrt{3}+5$

(e) None of these

Q 72. $\int_{-10}^{10} \log\left(\frac{a+x}{a-x}\right) dx$ is equal to

[Kerala CEE]

(a) 0

(b) $-2 \log (a + 10)$

(c) $2\log\left(\frac{a+10}{a-10}\right)$

(d) $2 \log (a + 10)$

Q 73. Let $f(x) = x, -[x]$, for every real x, where

[x] is the greatest integer less than or equal to x.

Then, $\int_{-1}^{1} f(x)dx$ is

[Kerala CEE]

(a) 1

(b) 2

(c) 3

(d) 0

(e) $1/2$

Q 74. $\int_0^{\pi/2} \frac{\cos x - \sin x}{1+\cos x \sin x} dx$ is equal to

[AMU]

(a) 0

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{6}$

Q 75. $\int_{-1}^{1} |1-x| dx$ is equal to

[AMU]

(a) -2

(b) 0

(c) 2

(d) 4

Q 76. $\int_0^{10\pi} |\sin x| \, dx$ is

[OJEE]

(a) 20

(b) 8

(c) 10

(d) 18

Q 77. Evaluate $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx$ is

[OJEE]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) zero

(d) 1

Q 78. $\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x} dx$ is

[OJEE]

(a) $\frac{\pi^2}{4}$

(b) π^2

(c) zero

(d) $\frac{\pi}{2}$

Q 79. $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}} dx$ is equal to

[WB JEE]

(a) 1

(b) -1

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{4}$

Q 80. $\int_0^{\pi/2} x \sin^2 x \cos^2 x \, dx$ is equal to

[WB JEE]

(a) $\frac{\pi^2}{32}$

(b) $\frac{\pi^2}{16}$

(c) $\frac{\pi}{32}$

(d) None of these

Q 81. $\int_{-\pi/3}^{\pi/3} \frac{x \sin x}{\cos^2 x} dx$ is equal to

[Jamia Millia Islamia]

(a) $\frac{1}{3}(4\pi + 1)$

(b) $\frac{4\pi}{3} - 2 \log \tan \frac{5\pi}{12}$

(c) $\frac{4\pi}{3} + \log \tan \frac{5\pi}{12}$

(d) None of these

Q 82. $\int_a^b \frac{|x|}{x} dx$, a < 0 < b, is equal to

[BCECE]

(a) |b| − |a|

(b) |b| + |a|

(c) |a – b|

(d) None of these

Q 83. The value of $\int_{-\pi}^{\pi} \sin^3 x \cos^2 x dx$ is equal to

[MHTCET]

(a) 1

(b) 2

(c) 3

(d) 0

Q 84. The value of $\int_{-1}^{1} \log\left(\frac{x-1}{x+1}\right) dx$

[MHTCET]

(a) 1

(b) 2

(c) 0

(d) 4

Q 85. $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ is equal to

[MHTCET]

(a) $\frac{\pi}{12}$

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{6}$

(d) $\frac{\pi}{4}$

Q 86. $\int_{-2}^{2}(x-|x|)dx$ is equal to

[RPET]

(a) 0

(b) 2

(c) 4

(d) −4

Q 87. $\lim_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2}+\frac{2}{n^2}\sec^2\frac{4}{n^2}+....+\frac{n}{n^2}\sec^2 1\right]$ equals

[AIEEE]

(a) $\frac{1}{2}\tan 1$

(b) $\tan 1$

(c) $\frac{1}{2}\cos ec 1$

(d) $\frac{1}{2}\sec 1$

Q 88. The value of the integral $\int_0^{\pi/2}\log|\tan x|\,dx$ is

[DCE]

(a) $\pi \log 2$

(b) 0

(c) $-\pi \log 2$

(d) None of these

Q 89. $\int_0^{2n\pi} \left\{ |\sin x| - \left| \frac{1}{2} \sin x \right| \right\} dx$ equals

[BITSAT, OJEE]

(a) n

(b) $2n$

(c) $-2n$

(d) None of these

Q 90. $\int_{-\pi}^{\pi} \frac{\sin^4 x}{\sin^4 x + \cos^4 x} dx$ is equal to

[Kerala CEE]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{2}$

(d) 2π

(e) π

Q 91. The value of $\int_0^{\pi/2} \frac{2^{\sin x}}{2^{\sin x}+2^{\cos x}} dx$ is

[Kerala CEE]

(a) 2

(b) π

(c) $\frac{\pi}{4}$

(d) 2π

(e) $\frac{\pi}{2}$

Q 92. If f(x) = f(a + x) and $\int_0^a f(x)dx = k$, then

$\int_0^{na} f(x)dx$ is equal to

[AMU]

(a) nk

(b) (n – 1)k

(c) (n + 1)k

(d) 0

Q 93. $\int_0^{\pi/2} \frac{200\sin x + 100\cos x}{\sin x + \cos x} dx$ is equal to

[EAMCET]

(a) 50π

(b) 25π

(c) 75π

(d) 150π

Q 94. $\int_0^{\pi} \frac{\theta \sin\theta}{1+\cos^2\theta} d\theta$ is equal to

[EAMCET]

(a) $\frac{\pi^2}{2}$

(b) $\frac{\pi^3}{3}$

(c) π^2

(d) $\frac{\pi^2}{4}$

Q 95. $\int_0^{2\pi}(\sin x+|\sin x|)dx$ is equal to

[RPET]

(a) 4

(b) 0

(c) 1

(d) 8

Q 96. $\int_0^{\pi} \frac{x\ dx}{a^2 \cos^2 x + b^2 \sin^2 x}$ is equal to

[RPET]

(a) $\frac{\pi}{2ab}$

(b) $\frac{\pi}{ab}$

(c) $\frac{\pi^2}{2ab}$

(d) $\frac{\pi^2}{ab}$

Q 97. $\int_0^{\infty} \frac{x\,dx}{(1+x)(1+x^2)}$ is equal to

[Guj. CET]

(a) $\frac{\pi}{2}$

(b) 0

(c) 1

(d) $\frac{\pi}{4}$

Q 98. If $f(x)=\frac{e^x}{1+e^x}, I_1=\int_{f(-a)}^{f(a)} xg[x(1-x)]dx$ and

$I_2=\int_{f(-a)}^{f(a)} g[x(1-x)]dx$, then the value of $\frac{I_2}{I_1}$ is

[AIEEE]

(a) 2

(b) −3

(c) −1

(d) 1

Q 99. $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}e^{r/n}$ is

[AIEEE]

(a) e

(b) $e - 1$

(c) $1 - e$

(d) $e + 1$

Q 100. Value of the integral $\int_{-\pi/2}^{\pi/2} \cos x dx$ is

[DCE]

(a) 4

(b) 2

(c) 0

(d) 1

Q 101. $\int_0^{\pi} \frac{x\,dx}{1+\sin x}$ is equal to

[UP SEE]

(a) $-\pi$

(b) $\frac{\pi}{2}$

(c) π

(d) None of these

Q 102. $\int_0^1 \frac{x\ dx}{[x+\sqrt{1-x^2}]\sqrt{1-x^2}}$ is equal to

[Kerala CEE]

(a) 0

(b) 1

(c) $\frac{\pi}{4}$

(d) $\frac{\pi^2}{2}$

(e) $\frac{\pi}{2}$

Q 103. If $\int_{-1/2}^{1/2} \cos x \log\left(\frac{1+x}{1-x}\right) dx$ = k . log 2, then k equals

[Kerala CEE]

(a) 0

(b) -1

(c) -2

(d) $\frac{1}{2}$

(e) $-\frac{1}{2}$

Q 104. The value of $\int_a^b \frac{x}{|x|} dx$, a < b < 0 is

[OJEE]

(a) $-(|a| + |b|)$

(b) $|b| - |a|$

(c) $|a| - |b|$

(d) $|a| + |b|$

Q 105. The value of $\int_{-\pi/4}^{\pi/4} x^3 \sin^4 x \, dx$ is equal to

[Kerala CEE]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{8}$

(d) 0

(e) 1

Q 106. For any integer n, the integral

$\int_0^{\pi} e^{\cos^2 x} \cos^3(2n+1)x \, dx$ has the value

[AMU]

(a) π

(b) 1

(c) 0

(d) None of these

Q 107. If $f(x) = |x - 1|$, then $\int_0^2 f(x)dx$ is

[OJEE]

(a) 1

(b) 0

(c) 2

(d) -2

Q 108. $\int_0^{1.5}[x^2]dx$ is

[BCECE]

(a) $4+2\sqrt{2}$

(b) $2+\sqrt{2}$

(c) $2-\sqrt{2}$

(d) None of these

Q 109. If f(t) is an odd function, then $\int_0^x f(t)dt$, is

[Kerala CEE]

(a) an odd function

(b) an even function

(c) neither even nor odd

(d) 0

(e) 1

Q 110. $\int_0^{\pi/2} \frac{\sin^2 x}{\sin^2 x + \cos^2 x} dx$ is equal to

[J&K CET]

(a) $\frac{\pi}{4}$

(b) 2π

(c) π^2

(d) $\frac{1}{2}\pi^2$

Q 111. The value of $\int_0^{\pi}\left(\sum_{r=0}^{3} a_r \cos^{3-r} x \sin^r x\right)dx$ depends upon

[J&K CET]

(a) a_1 and a_2

(b) a_0 and a_3

(c) a_2 and a_3

(d) a_1 and a_3

Q 112. The value of $\int_{-1}^{1} x|x|dx$ is

[J&K CET]

(a) 2

(b) 1

(c) 0

(d) None of these

Q 113. If f(a + b – x) = f(x), then $\int_a^b xf(x)dx$ is equal to

[AIEEE]

(a) $\frac{a+b}{2}\int_a^b f(b-x)dx$

(b) $\frac{a+b}{2}\int_a^b f(x)dx$

(c) $\frac{b-a}{2}\int_a^b f(x)dx$

(d) $\frac{a+b}{2}\int_a^b f(a+b+x)dx$

Q 114. $\lim_{n\to\infty}\frac{1+2^4+3^4+....+n^4}{n^5}-\lim_{n\to\infty}\frac{1+2^3+3^3+....+n^3}{n^5}$ is

[AIEEE]

(a) $\frac{1}{30}$

(b) 0

(c) $\frac{1}{4}$

(d) $\frac{1}{5}$

Q 115. $\int_{-2}^{10}\sin(x-[x])dx$ equals, where [.] denotes

greatest integer function

[DCE]

(a) -12

(b) 10

(c) 8

(d) 12

Q 116. The value of $\int_{-\pi}^{\pi}(1-x^2)\sin x\cos^2 x\,dx$ is

[DCE]

(a) 0

(b) $\pi - \frac{\pi^3}{3}$

(c) $2\pi - \pi^3$

(d) $\frac{7}{2} - 2\pi^3$

Topic – 4 Definite Integration by Gamma Function Leibnitz Rule, Walli's Formula

Q 1. Let f be a real-valued function defined on the interval $(-1, 1)$ such that $e^{-x}f(x) = 2 + \int_0^x \sqrt{t^4+1}dt$, for all $x \in (-1, 1)$ and let f^{-1} be the inverse function of f. Then, $(f^{-1})'(2)$ is equal to

[IIT JEE]

(a) 1

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) $\frac{1}{e}$

Q 2. The value of integral $\int_0^{\pi/2} \sin^5 x \, dx$ is

[WB JEE]

(a) $\frac{4}{15}$

(b) $\frac{8}{5}$

(c) $\frac{8}{15}$

(d) $\frac{4}{5}$

Q 3. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1-(f'(t))^2}\,dt = \int_0^x f(t)dt, 0 \leq x \leq 1$ and $f(0) = 0$, then

[IIT JEE]

(a) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$

(b) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$

(c) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

(d) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

Q 4. $\int_0^{\pi} x \sin^4 x \, dx$ is equal to

[IIT JEE]

(a) $\frac{3\pi}{16}$

(b) $\frac{3\pi^2}{16}$

(c) $\frac{16\pi}{3}$

(d) $\frac{16\pi^2}{3}$

Q 5. If $\int_0^{\pi/2} \sin^6 x\, dx = \frac{5\pi}{12}$, then the value of

$\int_{-\pi}^{\pi} (\sin^6 x + \cos^6 x) dx$ is

[Kerala CEE]

(a) $5\pi/8$

(b) $5\pi/16$

(c) $5\pi/2$

(d) $5\pi/4$

(e) $5\pi/32$

Q 6. If $\int_0^{x^2} f(t)dt = x\cos\pi x$, then the value of $f(4)$ is

[Kerala CEE]

(a) 1

(b) $\frac{1}{4}$

(c) -1

(d) $\frac{-1}{4}$

(e) -4

Q 7. If $\int_{\sin x}^{1} t^2 f(t)dt = 1 - \sin x, \forall x \in [0, \pi/2]$, then

$f\left(\frac{1}{\sqrt{3}}\right)$ is

[IITJEE]

(a) 3

(b) $\sqrt{3}$

(c) $\frac{1}{3}$

(d) None of these

Q 8. The value $\left[\int_0^{\sin^2\theta} \sin^{-1}\sqrt{\phi}d\phi + \int_0^{\cos^2\theta} \cos^{-1}\sqrt{\phi}\, d\phi\right]$ is equal to

[BITSAT, OJEE]

(a) π

(b) $\pi/2$

(c) $\pi/3$

(d) $\pi/4$

Q 9. If f(x) is differentiable and $\int_0^{t^2} x f(x)dx = \frac{2}{5}t^5$, then $f\left(\frac{4}{25}\right)$ equals

[IITJEE]

(a) $\frac{2}{5}$

(b) $-\frac{5}{2}$

(c) 1

(d) $\frac{5}{2}$

Q 10. The value of $\int_0^{\pi/2} \sin^8 x dx$ is

[J&K CET]

(a) $\frac{105\pi}{32(4!)}$

(b) $\frac{105\pi}{14(4!)}$

(c) $\frac{105}{16\pi(4!)}$

(d) None of these

Q 11. $\int_0^1 \log\left\{\sin\left(\frac{\pi x}{2}\right)\right\} dx$ is equal to

[J&K CET]

(a) $-\frac{\pi}{2}\log 2$

(b) $-\log 2$

(c) $-\frac{2}{\pi}\log 2$

(c) $\frac{\pi}{2}\log 2$

Answers

Topic 1 : Fundamental of Definite Integration

1. (a) 2. (b) 3. (d) 4. (b) 5. (b) 6. (d) 7. (c) 8. (d)

9. (d) 10. (c) 11. (c) 12, (d) 13. (a) 14. (a) 15. (b)

16. (c) 17. (d) 18. (d) 19. (d) 20. (a) 21. (b) 22.

(b) 23. (b) 24. (c) 25. (d) 26. (b) 27. (c) 28. (c)

29. (b) 30. (d) 31. (b) 32. (a) 33. (b). 34. (d) 35.

(c) 36. (a) 37. (d) 38. (c) 39. (c) 40. (d) 41. (a)

42. (d) 43. (a) 44. (d) 45. (a)

Topic 2 : Definite Integration by Substitution, Parts, Partial Fractions

1. (a) 2. (a) 3. (b) 4. (e) 5. (b) 6. (b) 7. (a) 8. (b)

9. (c) 10. (d) 11. (d) 12. (b) 13. (c) 14. (c) 15. (a)

16. (b) 17. (b) 18. (b) 19. (b) 20. (a) 21. (b) 22.

(b) 23. (a) 24. (b) 25. (b) 26. (c) 27. (c) 28. (c)

29. (b) 30. (c) 31. (a) 32. (e) 33. (c) 34. (a) 35.

(a) 36. (b) 37. (b) 38. (a) 39. (b) 40. (c) 41. (d)

42. (c) 43. (a) 44. (c) 45. (d) 48. (b) 47. (a) 48.

(a)

Topic 3 : Properties and Summation of Series of Definite Integration

1. (b) 2. (b) 3. (c) 4. (a) 5. (e) 6. (e) 7. (b) 8. (d)

9. (d) 10. (b) 11. (b) 12. (c) 13. (b) 14. (c) 15. (d)

16. (a) 17. (c) 18. (*) 19. (b) 20. (c) 21. (c) 22.

(b) 23. (c) 24. (c) 25. (b) 26. (b) 27. (d) 28. (d)

29. (c) 30. (d) 31. (d) 32. (c) 33. (b) 34. (b) 35.

(c) 36. (a) 37. (b) 38. (d) 39. (d) 40. (c) 41. (a)

42. (a) 43. (b) 44. (a) 45. (a) 46. (c) 47. (a) 48.

(c) 49. (a) 50. (c) 51. (a) 52. (c) 53. (b) 54. (c)

55. (a) 56. (b) 57. (a) 58. (b) 59. (c) 60. (a)

61. (c) 62. (a) 63. (b) 64. (a) 65. (a) 66. (b) 67.

(a) 68. (b) 69. (c) 70. (e) 71. (d) 72. (a) 73. (a)

74. (a) 75. (c) 76. (a) 77. (a) 78. (b) 79. (d) 80.

(d) 81. (b) 82. (a) 83. (d) 84. (c) 85. (a) 86. (d)

87. (a) 88. (b) 89. (b) 90. (e) 91. (c) 92. (a) 93.

(c) 94. (d) 95. (a) 96. (c) 97. (d) 98. (a) 99. (b)

100. (b) 101. (c) 102. (c) 103. (a) 104. (b) 105.

(d) 106. (c) 107. (a) 108. (c) 109. (b) 110. (a)

111. (d) 112. (c) 113. (b) 114. (d) 115. (c) 116.

(a)

Topic 4 : Definite Integration by Gamma Function, Leibnitz Rule, Walli's Formula

1. (b) 2. (c) 3. (c) 4. (b) 5. (d) 6. (b) 7. (a) 8. (d)

9. (a) 10. (a) 11. (b)

Indefinite Integrals

Useful Results & Formulae

1. (i) $\int x^n dx = \frac{x^{n+1}}{n+1} + c, n \neq -1$

(ii) $\int \frac{1}{x} dx = \log |x| + c$

(iii) $\int e^x dx = e^x + c$

(iv) $\int a^x dx = \frac{a^x}{\log_e a} + c$

(v) $\int \sin x \, dx = -\cos x + c$

(vi) $\int \cos x \, dx = \sin x + c$

(vii) $\int \sec^2 x \, dx = \tan x + c$

(viii) $\int \operatorname{cosec}^2 x \, dx = -\cot x + c$

(ix) $\int \sec x \tan x \, dx = \sec x + c$

(x) $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$

(xi) $\int \cot x \, dx = \log |\sin x| + c$

(xii) $\int \tan x \, dx = -\log |\cos x| + c$

(xiii) $\int \sec x \, dx = \log |\sec x + \tan x| + c \quad = \log \tan\left(\frac{\pi}{4} + \frac{x}{2}\right) + c$

(xiv) $\int \operatorname{cosec} x \, dx = \log |\operatorname{cosec} x - \cot x| + c + c = \log \tan \frac{x}{2}$

(xv) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + c = -\cos^{-1}\left(\frac{x}{a}\right) + c$

(xvi) $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \sec^{-1} \frac{x}{a} + c$

(xviii) $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x - a}{x + a} \right| + c$

(xix) $\int \frac{1}{\sqrt{x^2 \pm a^2}} dx = \log \{| x + \sqrt{x^2 \pm a^2} |\} + c$

(xx) $\int \sqrt{a^2 - x^2} dx = \frac{1}{2} x \sqrt{a^2 - x^2} + \frac{1}{2} a^2 \sin^{-1}\left(\frac{x}{a}\right) + c$

(xxi) $\int \sqrt{x^2 \mp a^2} dx = \frac{1}{2} x \sqrt{x^2 \mp a^2} \mp \frac{1}{2} a^2 \log \{x + \sqrt{x^2 \mp a^2}\} + c$

2. (i) $\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2}$ (a sin bx – b cos bx) + c

(ii) $\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2}$ (a cos bx + b sin bx) + c

(iii) $\int e^x [f(x) + f'(x)]dx = e^x f(x) + c$

(iv) $\int [xf'(x) + f(x)]dx = xf(x) + c$

3. (i) $\int \sinh x \, dx = \cosh x + c$

(ii) $\int \cosh x \, dx = \sinh x + c$

(iii) $\int \sec h^2 x \, dx = \tanh x + c$

(iv) $\int \cos ec \, h^2 x \, dx = -\coth x + c$

(v) $\int \sec h \, x \tanh x \, dx = -\sec h \, x + c$

(vi) $\int \operatorname{cosec} h\, x \coth x\, dx = -\operatorname{cosec} h\, x + c$

4. When integral contains more than two functions, in those of them one is derivative of the other function i.e., $\int F(\phi(x)), \phi'(x)dx$, Put $\phi(x)$ = t and then integrate $\int f(t)dt$.

5. When integral contains more than one functions, in those of them neither derivative to the other function i.e. $\int \underset{I}{f(x)}\, \underset{II}{g(x)}\, dx$

$$= f(x)\int g(x)dx - \int\left[\frac{d}{dx}f(x)\int g(x)dx\right]dx$$

I and II function consider by the ILATE method.

6. Sometimes the integrals function of the form

$\int \frac{dx}{ax^2+bx+c}, \int \frac{px+q}{ax^2+bx+c}$ etc.

7. (i) Integrals of the form $I = \int \frac{a\sin x + b\cos x}{c\sin x + d\cos x} dx$

Numerator = M(differential of denominator) +

N(denominator) On comparing the coefficient of sin

x and cos x, we get M and N. Then,

$I = M\int \frac{c\cos x - d\sin x}{c\sin x + d\cos x} dx + N\int 1\, dx$

(ii) Integrals of the form $I = \int \frac{a \sin x + b \cos x + q}{c \sin x + d \cos x + r} dx$

Numerator = M(denominator) + N(differential of denominator) + P On comparing the coefficients of sin x, cos x and constant term, we get M, N and P.

Then,

$$I = M\int dx + N\int \frac{\text{differential of denominator}}{\text{denominator}} dx + \int \frac{dx}{c \sin x - d \cos x + r}$$

Topic – 1 Fundamental of Indefinite Integrals

2010

Q 1. $\int \sqrt{1+\cos x}\ dx$ is equal to

[WB JEE]

(a) $2\sqrt{2}\cos\frac{x}{2}+c$

(b) $2\sqrt{2}\sin\frac{x}{2}+c$

(c) $\sqrt{2}\cos\frac{x}{2}+c$

(d) $\sqrt{2}\sin\frac{x}{2}+c$

Q 2. $\int(x+1)(x+2)^7(x+3)dx$ is equal to

[Kerala CEE]

(a) $\frac{(x+2)^2}{2}-\frac{(x+2)^8}{8}+c$

(b) $\frac{(x+1)^2}{2}-\frac{(x+2)^8}{8}-\frac{(x+3)^2}{2}+c$

(c) $\frac{(x+2)^{10}}{10}+c$

(d) $\frac{(x+1)^2}{2}+\frac{(x+2)^8}{8}+\frac{(x+3)^2}{2}+c$

(e) $\frac{(x+2)^9}{9}-\frac{(x+2)^7}{7}+c$

Q 3. If $f\left(\frac{3x-4}{3x+4}\right)=x+2,$ then $\int f(x)\,dx$ is

[DCE]

(a) $e^{x+2}\log\left|\frac{3x-4}{3x+4}\right|+c$

(b) $-\frac{8}{3}\log|1-x|+\frac{2}{3}x+c$

(c) $\frac{8}{3}\log|1-x|+\frac{x}{3}+c$

(d) $e^{[(3x-4)/(3x+4)]}-\frac{x^2}{2}-2x+c$

Q 4. $\int\frac{d(\cos\theta)}{\sqrt{1-\cos^2\theta}}$ is equal to

[BITSAT]

(a) $\cos^{-1}\theta + c$

(b) $\theta + c$

(c) $\sin^{-1}\theta + c$

(d) $\sin^{-1}(\cos\theta) + c$

Q 5. $\int \operatorname{cosec}(x-a)\operatorname{cosec} x\,dx$ is equal to

[KCET]

(a) $\frac{-1}{\sin a}\log|\sin x \operatorname{cosec}(x-a)|+c$

(b) $\frac{-1}{\sin a}\log[\sin(x-a)\sin x]+c$

(c) $\frac{1}{\sin a}\log[\sin(x-a)\operatorname{cosec} x]+c$

(d) $\frac{1}{\sin a}\log[\sin(x-a)\sin x]+c$

Q 6. If $\frac{d}{dx}\{f(x)\}=\frac{1}{1+x^2}$, then $\frac{d}{dx}\{f(x^3)\}$ is

[Kerala CEE]

(a) $\frac{3x}{1+x^3}$

(b) $\frac{3x^2}{1+x^6}$

(c) $\frac{-6x^5}{(1+x^6)^2}$

(d) $\frac{-6x^5}{1+x^6}$

Q 7. If $u = -f''(\theta) \sin\theta + f'(\theta)\cos\theta$ and $v = f''(\theta)\cos\theta + f'(\theta)\sin\theta$, then

$\int\left[\left(\frac{du}{d\theta}\right)^2+\left(\frac{dv}{d\theta}\right)^2\right]^{\frac{1}{2}} d\theta$ is equal to

[Kerala CEE]

(a) $f(\theta) - f'(\theta) + c$

(b) $f(\theta) + f''(\theta) + c$

(c) $f'(\theta) + f''(\theta) + c$

(d) $f'(\theta) - f''(\theta) + c$

(e) $f(\theta) + f'(\theta) + c$

Q 8. $\int -\frac{x^4 - 1}{x^2(x^4 + x^2 + 1)^{1/2}} dx$ is equal to

[Kerala CEE]

(a) $\sqrt{\frac{x^4 + x^2 + 1}{x}} + c$

(b) $\frac{x^2}{\sqrt{x^4+x^2+1}}+c$

(c) $x(x^4 + x^2 + 1)^{3/2} + c$

(d) $\frac{\sqrt{x^4+x^2+1}}{x}+c$

(e) $\sqrt{x^4+x^2+1}+c$

Q 9. If $I_n = \int \sin^n x\,dx$, then $nI_n - (n-1)I_{n-2}$ equals

[EAMCET]

(a) $\sin^{n-1} x \cos x$

(b) $\cos^{n-1} x \sin x$

(c) $-\sin^{n-1} x \cos x$

(d) $-\cos^{n-1} x \sin x$

Q 10. If $\frac{d}{dx}\left[a\tan^{-1}x + b\log\left(\frac{x-1}{x+1}\right)\right] = \frac{1}{x^4-1}$, then $a - 2b$ is equal to

[EAMCET]

(a) 1

(b) -1

(c) 0

(d) 2

Q 11. $\int \frac{dx}{(1-x)(x-2)}$ is equal to

[BCECE]

(a) $\sin^{-1}(2x - 3) + c$

(b) $\sin^{-1}(2x + 5) + c$

(c) $\sin^{-1}(3 - 2x) + c$

(d) $\sin^{-1}(5 - 2x) + c$

Q 12. $\int \frac{(\sec^2 x - 7)}{\sin^7 x} dx$ is equal to

[BCECE]

(a) $\frac{\tan x}{\sin^7 x}+c$

(b) $\frac{\cos x}{\sin^7 x}+c$

(c) $\frac{\sin x}{\cos^7 x}+c$

(d) $\frac{\sin x}{\tan^7 x}+c$

Q 13. If $\int \frac{\sin^8 x-\cos^8 x}{1-2\sin^2 x\cos^2 x}dx = A\sin 2x + B$, then

[VITEEE]

(a) $A=-\frac{1}{2}$

(b) $A = \frac{1}{2}$

(c) A = -1

(d) A = 1

Q 14. $\int [\sin(\log x) + \cos(\log x)]dx$ is equal to

[MHTCET]

(a) x cos (log x) + c

(b) cos (log x) + c

(c) x sin (log x) + c

(d) sin (log x) + c

Q 15. $\int e^x \frac{(x-1)}{x^2} dx$ is equal to

[MHTCET]

(a) $\frac{e^x}{x^2} + c$

(b) $\frac{-e^x}{x^2} + c$

(c) $\frac{e^x}{x} + c$

(d) $\frac{-e^x}{x} + c$

Q 16. $\int \frac{x^4 + x^2 + 1}{x^2 - x + 1} dx$ is equal to

[MP PET]

(a) $\frac{x^3}{3}-\frac{x^2}{2}+x+c$

(b) $\frac{x^3}{3}+\frac{x^2}{2}+x+c$

(c) $\frac{x^3}{3}-\frac{x^2}{2}-x+c$

(d) $\frac{x^3}{3}+\frac{x^2}{2}-x+c$

Q 17. If f(x) = $\lim_{y\to x}\frac{\sin^2 y-\sin^2 x}{y^2-x^2}$, then $\int 4x\,f(x)dx$ is equal to

[J&K CET]

(a) cos 2x + e

(b) $2 \cos 2x + c$

(c) $-\cos 2x + c$

(d) $-2 \cos 2x + c$

Q 18. $\int [e^{a \log x} + e^{x \log a}]dx = ... + c;\ a, x < 1$

[Guj. CET]

(a) $\frac{x^{a+1}}{a+1} + \frac{a^x}{\log a}$

(b) $\frac{e^{a \log x}}{a \log x} + \frac{e^{x \log a}}{x \log a}$

(c) $\frac{x^{a-1}}{a-1} + a^x . \log a$

(d) $\frac{e^{a\log x}}{a/x}+\frac{e^{x\log a}}{x}$

Q 19. $\int\frac{dx}{\sin(x-a)\sin(x-b)}$ is

[DCE]

(a) $\frac{1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right|+c$

(b) $\frac{-1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right|+c$

(c) log sin(x – a) sin (x – b) + c

(d) $\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right|+c$

Q 20. $\int\frac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}}dx$ is equal to

[UP SEE]

(a) $\frac{1}{2}\sqrt{1+x}+c$

(b) $\frac{2}{3}(1+x)^{3/2}+c$

(c) $\sqrt{1+x}+c$

(d) $2(1+x)^{3/2}+c$

Q 21. $\int \text{cosec}^4 x\, dx$ is equal to

[UP SEE]

(a) $\cot x+\frac{\cot^3 x}{3}+c$

(b) $\tan x + \frac{\tan^3 x}{3} + c$

(c) $-\cot x - \frac{\cot^3 x}{3} + c$

(d) $-\tan x - \frac{\tan^3 x}{3} + c$

Q 22. $\int \frac{\sec x}{\sec x + \tan x} dx$ is equal to

[KCET]

(a) tan x – sec x + c

(b) log(1 + sec x) + c

(c) sec x + tan x + c

(d) log sin x + log cos x + c

Q 23. $\int \frac{\sec x \operatorname{cosec} x}{2\cot x - \sec x \operatorname{cosec} x} dx$ is equal to

[Kerala CEE]

(a) log |sec x + tan x| + c

(b) log |sec x + cosec x| + c

(c) $\frac{1}{2}$log|sec 2x + cosec 2x| + c

(d) log|sec 2x + cosec 2x| + c

(e) log |sec 2x cosec 2x| + c

Q 24. $\int \tan(\sin^{-1} x)dx$ is equal to

[Kerala CEE]

(a) $\frac{1}{\sqrt{1-x^2}}+c$

(b) $\sqrt{1-x^2}+c$

(c) $\frac{-x}{\sqrt{1-x^2}}+c$

(d) $\frac{x}{\sqrt{1-x^2}}+c$

(e) $-\sqrt{1-x^2}+c$

Q 25. $\int e^{3\log x}(x^4+1)^{-1}dx$ is equal to

[Kerala CEE]

(a) $e^{3\log x} + c$

(b) $\frac{1}{4}\log(x^4 + 1) + c$

(c) $\log(x^4 + 1) + c$

(d) $\frac{1}{2}\log(x^4 + 1) + c$

(e) $\frac{x^4}{x^4 + 1} + c$

Q 26. If $\int \frac{dx}{x^2 + 2x + 2} = f(x) + c$, then f(x) is equal to

[BCECE]

(a) $\tan^{-1}(x + 1)$

(b) $2\tan^{-1}(x + 1)$

(c) $-\tan^{-1}(x + 1)$

(d) $3\tan^{-1}(x + 1)$

Q 27. $\int \frac{(x^3 + 3x^2 + 3x + 1)}{(x+1)^5} dx$ is equal to

[Jamia Millia Islamia]

(a) $-\frac{1}{(x+1)} + c$

(b) $\frac{1}{5}\log(x+1) + c$

(c) log (x + 1) + c

(d) $\tan^{-1}$ x + c

Q 28. $\int \frac{\sqrt{x}}{x+1} dx$ is equal to

[J & K CET]

(a) $2(\sqrt{x} + \tan^{-1}\sqrt{x}) + c$

(b) $2(\sqrt{x} + \cot^{-1}\sqrt{x}) + c$

(c) $2(\sqrt{x} - \cot^{-1} - \sqrt{x}) + c$

(d) $2(\sqrt{x} - \tan^{-1} - \sqrt{x}) + c$

Q 29. Let $f(x)=\frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and

$g(x) = \underbrace{(fofo.....of)}_{n \text{ times}}(x)$. Then, $\int x^{n-2} g(x)\,dx$ equals

[IIT JEE]

(a) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}}+c$

(b) $\frac{1}{n-1}(1+nx^n)^{1-\frac{1}{n}}+c$

(c) $\frac{1}{n(n-1)}(1+nx^n)^{1+\frac{1}{n}}+c$

(d) $\frac{1}{n+1}(1+nx^n)^{1+\frac{1}{n}}+c$

Q 30. Let F(x) be an indefinite integral of $\sin^2 x$.

Statement I the function F(x) satisfies $F(x + \pi) =$

F(x) for all real x. Because

Statement II $\sin^2 (x + \pi) = \sin^2 x$ for all real x.

[IIT JEE]

(a) Statement I is True, Statement II is True;

Statement II is a correct explanation for Statement I

(b) Statement I is True, Statement II is

True;StatementII is not a correct explanation for

StatementI

(c) Statement I is True, Statement II is False

(d) Statement I is False, Statement II is True

Q 31. $\int \frac{dx}{\cos x + \sqrt{3}\sin x}$ equals

[AIEEE]

(a) $\frac{1}{2}\log \tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+c$

(b) $\frac{1}{2}\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+c$

(c) $\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+c$

(d) $\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+c$

Q 32. $\int\frac{dx}{x^2+4x+13}$ is equal to

[DCE]

(a) $\log(x^2 + 4x + 13) + c$

(b) $\frac{1}{3}\tan^{-1}\left(\frac{x+2}{3}\right)+c$

(c) $\log(2x + 4) + c$

(d) $\frac{2x+4}{(x^2+4x+13)^2}+c$

Q 33. The value of $\int \frac{x^2+1}{x^2-1}dx$ is

[KCET]

(a) $\log\left(\frac{x-1}{x+1}\right)+c$

(b) $\log\left(\frac{x+1}{x-1}\right)+c$

(c) $x+\log\left(\frac{x-1}{x+1}\right)+c$

(d) $\log(x^2-1)+c$

Q 34. $\int(\sin^6 x+\cos^6 x+3\sin^2 x\cos^2 x)dx$ is equal to

[Kerala CEE]

(a) x + c

(b) $\frac{3}{2}\sin 2x + c$

(c) $-\frac{3}{2}\sin 2x + c$

(d) $\frac{1}{3}\sin 3x - \cos 3x + c$

(e) $\frac{1}{3}\sin 3x + \cos 3x + c$

Q 35. $\int \frac{4^{x+1} - 7^{x-1}}{28^x} dx$ is equal to

[Kerala CEE]

(a) $\frac{1}{7\log_e 4}4^{-x} - \frac{4}{\log_e 7}7^{-x} + c$

(b) $\frac{1}{7\log_e 4}4^{-x} + \frac{4}{\log_e 7}7^{-x} + c$

(c) $\frac{4^{-x}}{\log_e 7} - \frac{7^{-x}}{\log_e 4} + c$

(d) $\frac{4^{-x}}{\log_e 4} - \frac{7^{-x}}{\log_e 7} + c$

(e) $\frac{1}{28}\log_e 4^{-x} + \frac{1}{7}\log_e 7^{-x} + c$

Q 36. The value of $\int \frac{5^{\log_e x} - e^{4\log_e x}}{e^{3\log_e x} - e^{2\log_e x}} dx$ is

[Kerala CEE]

(a) $x^2 + c$

(b) $\frac{x^2}{2}+c$

(c) $\frac{x^3}{3}+c$

(d) $\frac{x}{2}+c$

(e) e

Q 37. The value of $\int e^{\tan^{-1}x}.\frac{(1+x+x^2)}{1+x^2}dx$ is

[Kerala CEE]

(a) $\tan^{-1} x + c$

(b) $e^{\tan^{-1}x}+2x+c$

(c) $e^{\tan^{-1}x} + c$

(d) $e^{\tan^{-1}x} - x + c$

(e) $xe^{\tan^{-1}x} + c$

Q 38. $\int \frac{dx}{\sin x - \cos x + \sqrt{2}}$ equals

[AMU]

(a) $-\frac{1}{\sqrt{2}} \tan\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$

(b) $\frac{1}{2} \tan\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$

(c) $\frac{1}{\sqrt{2}} \cot\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$

(d) $-\frac{1}{\sqrt{2}}\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)+c$

Q 39. If $\int\frac{e^x-1}{e^x+1}dx = f(x) + c$, then $f(x)$ is equal to

[EAMCET]

(a) $2\log(e^x + 1)$

(b) $\log (e^{2x} - 1)$

(c) $2 \log (e^x + 1) - x$

(d) $\log(e^{2x} + 1)$

Q 40. $\int\left(\frac{x+2}{x+4}\right)^2 e^x dx$ is equal to

[WB JEE]

(a) $e^x\left(\frac{x}{x+4}\right)+c$

(b) $e^x\left(\frac{x+2}{x+4}\right)+c$

(c) $e^x\left(\frac{x-2}{x+4}\right)+c$

(d) $\left(\frac{2xe^x}{x+4}\right)+c$

Q 41. $\int\frac{dx}{\cos x-\sin x}$ is equal to

[Jamia Millia Islamia]

(a) $\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2}-\frac{\pi}{8}\right)\right|+c$

(b) $\frac{1}{\sqrt{2}}\log\left|\cot\left(\frac{x}{2}\right)\right|+c$

(c) $\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2}-\frac{3\pi}{8}\right)\right|+c$

(d) $\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2}+\frac{3\pi}{8}\right)\right|+c$

Q 42. $\int\frac{1}{1+\cos ax}dx$ is equal to

[J & K CET]

(a) $\cot\frac{ax}{2}+c$

(b) $\frac{1}{a}\tan\frac{ax}{2}+c$

(c) $\frac{1}{a}(\operatorname{cosec} ax-\cot ax)+c$

(d) $\frac{1}{a}(\cos ec\, ax + \cot ax) + c$

Q 43. Let f be a function such that $f(1) = 4$ and $f'(x) \geq 2$ for $1 \leq x < 4$. How small can $f(4)$ possibly be ?

[Kerala CEE]

(a) 8

(b) 12

(c) 16

(d) 2

(e) 10

Q 44. Let $f(x)=\frac{\sin^2 \pi x}{1+\pi^x}$. Then, $\int [f(x)+f(-x)]dx$ is equal to

[Kerala CEE]

(a) 0

(b) x + c

(c) $\frac{x}{2}-\frac{\cos \pi x}{2\pi}+c$

(d) $\frac{1}{1+\pi^x}.\frac{\cos^2 \pi x}{2\pi}+c$

(e) $\frac{x}{2}-\frac{\sin 2\pi x}{4\pi}+c$

Q 45. If $\int\sqrt{\frac{x}{a^3-x^3}}dx$ = g(x) + c, the g(x) is equal to

[AMU]

(a) $\frac{2}{3}\cos^{-1} x$

(b) $\frac{2}{3}\sin^{-1}\left(\frac{x^3}{a^3}\right)$

(c) $\frac{2}{3}\sin^{-1}\left(\sqrt{\frac{x^3}{a^3}}\right)$

(d) $\frac{2}{3}\cos^{-1}\left(\frac{x}{a}\right)$

Q 46. $\int \sqrt{\frac{x-1}{x+1}} dx$ is equal to

[MHT CET]

(a) $2\sqrt{x^2+1} + \sin^{-1} x + c$

(b) $\sqrt{x^2-1} - \sin^{-1} x + c$

(c) $2\sqrt{x^2-1} + \sin^{-1} x + c$

(d) $\sqrt{\frac{x^2-1}{2}} + \sin^{-1} x + c$

Q 47. If $I_1 = \int \sin^{-1} x \, dx$ and $I_2 = \int \sin^{-1} \sqrt{1-x^2} dx$, then

[MHTCET]

(a) $I_1 = I_2$

(b) $I_2 = \frac{\pi}{2} I_1$

(c) $I_1 + I_2 = \frac{\pi}{2} x$

(d) $I_1 + I_2 = \frac{\pi}{2}$

Q 48. If $f'(x) = x + \frac{1}{x}$, then the value of $f(x)$ is

[MHT CET]

(a) $x^2 + \log x + c$

(b) $\frac{x^2}{2} + \log x + c$

(c) $\frac{x}{2} + \log x + c$

(d) None of these

Q 49. If $\int \operatorname{cosec} x \, dx$ = f(x) + constant, then f(x) is equal to

[J & K CET]

(a) tan x/2

(b) log |tan (x /2) |

(c) log | sin x|

(d) log |cos x|

Q 50. $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx$ is equal to

[AIEEE]

(a) $\frac{x}{(\log x)^2 + 1} + c$

(b) $\frac{xe^x}{1 + x^2} + c$

(c) $\frac{x}{x^2 + 1} + c$

(d) $\frac{\log x}{(\log x)^2 + 1} + c$

Q 51. If $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx = k \cos 4x + c$, then

[DCE]

(a) $k = -1/2$

(b) $k = -1/8$

(c) $k = -1/4$

(d) None of these

Q 52. If $\int \frac{1}{x + x^5} dx = f(x) + c$, then the value of

$\int \frac{x^4}{x + x^5} dx$ equals

[DCE]

(a) log x – f(x) + c

(b) f(x) + log x + c

(c) f(x) – log x + c

(d) None of these

Q 53. $\int e^{x \log x} e^{x} dx$ is equal to

[Kerala CEE]

(a) $\frac{a^{x}}{\log ae} + c$

(b) $\frac{e^x}{1+\log_e a}+c$

(c) $(ae)^x + a$

(d) $\frac{(ae)^x}{\log_e ae}+c$

(e) $\frac{a^x e^x}{\log_x a}+c$

Q 54. $\int (1-\cos x)\cos ec^2 x\,dx$ is equal to

[MHT CET]

(a) $\tan\frac{x}{2}+c$

(b) $-\cot\frac{x}{2}+c$

(c) $2\tan\frac{x}{2}+c$

(d) $-2\cot\frac{x}{2}+c$

Q 55. $\int e^{-\log x}dx$ is equal to

[MP PET]

(a) $e^{-\log x} + c$

(b) $-xe^{-\log x} + c$

(c) $e^{\log x} + c$

(d) $\log|x| + c$

Q 56. $\int \sqrt{1+\sin\left(\frac{x}{4}\right)}dx$ is equal to

[RPET]

(a) $8\left(\sin\frac{x}{8}+\cos\frac{x}{8}\right)+c$

(b) $8\left(\sin\frac{x}{8}-\cos\frac{x}{8}\right)+c$

(c) $8\left(\cos\frac{x}{8}-\sin\frac{x}{8}\right)+c$

(d) $\frac{1}{8}\left(\sin\frac{x}{8}-\cos\frac{x}{8}\right)+c$

Q 57. $\int \cos\left[2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right]dx$ is equal to

[Kerala CEE]

(a) $\frac{1}{2}x^2 + c$

(b) $\frac{1}{2}\sin\left[2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right] + c$

(c) $-\frac{1}{2}x^2 + c$

(d) $\frac{1}{2}x + c$

(e) $-\frac{1}{2}x + c$

Q 58. If $f(x) = \cos x - \cos^2 x + \cos^3 x - \ldots \infty$,

the $\int f(x)dx$ equals

[Kerala CEE]

(a) $\tan\frac{x}{2}+c$

(b) $x+\tan\frac{x}{2}+c$

(c) $x-\frac{1}{2}\tan\frac{x}{2}+c$

(d) $\frac{x-\tan\frac{x}{2}}{2}+c$

(e) $x-\tan\frac{x}{2}+c$

Q 59. $\int\frac{dx}{\sin x\cos x}$ is equal to

[OJEE]

(a) log |sin x| + c

(b) log |tan x| + c

(c) log |cos x| + c

(d) None of these

Q 60. Primitive of $\cos^{-1}$ w.r.t. x is

[MHT CET]

(a) $x\cos^{-1}x-\frac{1}{2}\sqrt{1-x^2}+c$

(b) $x\cos^{-1}x-\sqrt{1-x^2}+c$

(c) $x\cos^{-1}x+\sqrt{1-x^2}+c$

(d) $x\cos^{-1}x+\frac{1}{2}\sqrt{1-x^2}+c$

Q 61. $\int\frac{1}{\sin^2 x.\cos^2 x}dx$ is equal to

[MHT CET]

(a) sin x – cos x + c

(b) tan x + cot x + c

(c) cos x + sin x + c

(d) tan x – cot x + c

Q 62. $\int (1+2\tan x(\tan x+\sec x)\}^{1/2} dx$ is equal to

[BCECE]

(a) log(sec x + tan x) + c

(b) $\log(\sec x + \tan x)^{1/2}$

(c) log sec x(sec x + tan x) + c

(d) None of the above

Q 63. The value of $\int \frac{(x+1)^2}{x(x^2+1)} dx$ is equal to

[MP PET]

(a) $\log_e x + c$

(b) $\log_e x + 2\tan^{-1} x + c$

(c) $\log_e\left(\frac{1}{x^2+1}\right)+c$

(d) None of these

Q 64. $\int\sqrt{1+\sin\frac{x}{2}}dx$ is equal to

[RPET]

(a) $\frac{1}{4}\left[\cos\frac{x}{4}-\sin\frac{x}{4}\right]+c$

(b) $4\left[\cos\frac{x}{4}-\sin\frac{x}{4}\right]+c$

(c) $4\left[\sin\frac{x}{4}-\cos\frac{x}{4}\right]+c$

(d) $4\left[\sin\frac{x}{4}+\cos\frac{x}{4}\right]+c$

Topic – 2 Integration by Substitution, Parts, Partial Fractions

2010

Q 1. $\int \frac{\log \sqrt{x}}{3x} dx$ is equal to

[WB JEE]

(a) $\frac{1}{3}(\log \sqrt{x})^2 + c$

(b) $\frac{2}{3}(\log \sqrt{x})^2 + c$

(c) $\frac{2}{3}(\log x)^2 + c$

(d) $\frac{1}{3}(\log x)^2 + c$

Q 2. $\int e^x \left(\frac{2}{x} - \frac{2}{x^2} \right) dx$ is equal to

[WB JEE, VITEEE]

(a) $\frac{e^x}{x} + c$

(b) $\frac{e^x}{2x^2} + c$

(c) $\frac{2e^x}{x} + c$

(d) $\frac{2e^x}{x^2} + c$

Q 3. The value of the integral $\int \frac{1}{(e^x + e^{-x})^2}$ is

[WB JEE, BITSAT]

(a) $\frac{1}{2}(e^{2x}+1)+c$

(b) $\frac{1}{2}(e^{-2x}+1)+c$

(c) $-\frac{1}{2}(e^{2x}+1)^{-1}+c$

(d) $\frac{1}{4}(e^{2x}-1)+c$

Q 4. $\int (x^2+1)\sqrt{x+1}dx$ is equal to

[Kerala CEE]

(a) $\frac{(x+1)^{7/2}}{7}-2\frac{(x+1)^{5/2}}{5}+2\frac{(x+1)^{3/2}}{3}+c$

(b) $2\left[\frac{(x+1)^{7/2}}{7}-2\frac{(x+1)^{5/2}}{5}+2\frac{(x+1)^{3/2}}{3}\right]+c$

(c) $\frac{(x+1)^{7/2}}{7}-2\frac{(x+1)^{5/2}}{5}+5$

(d) $\frac{(x+1)^{7/2}}{7}-3\frac{(x+1)^{5/2}}{5}+11(x+1)^{1/2}+c$

(e) $(x+1)^{7/2}+(x+1)^{5/2}+(x+1)^{3/2}+c$

Q 5. $\int\frac{1+x}{x+e^{-x}}dx$ is equal to

[Kerala CEE]

(a) $\log|(x-e^{-x})|+c$

(b) $\log|(x+e^{-x})|+c$

(c) $\log|(1+xe^{x})|+c$

(d) $(1 + xe^x)^2 + c$

(e) $\log|(1 - xe^x)| + c$

Q 6. $\int \frac{\cos x + x \sin x}{x^2 + x \cos x} dx$ is equal to

[Kerala CEE]

(a) $\log\left|\frac{\sin x}{1+\cos x}\right| + c$

(b) $\log\left|\frac{\sin x}{x+\cos x}\right| + c$

(c) $\log\left|\frac{2\sin x}{x+\cos x}\right| + c$

(d) $\log\left|\frac{x \sin x}{x+\cos x}\right| + c$

(e) $\log\left|\frac{x}{x+\cos x}\right|+c$

Q 7. $\int\frac{dx}{\sqrt{1-e^{2x}}}$ is equal to

[Kerala CEE]

(a) $\log|e^{-x}+\sqrt{e^{-2x}-1}|+c$

(b) $\log|e^{x}+\sqrt{e^{2x}-1}|+c$

(c) $-\log|e^{-x}+\sqrt{e^{-2x}-1}|+c$

(d) $-\log|e^{-2x}+\sqrt{e^{-2x}-1}|+c$

(e) $\log|e^{-2x}+\sqrt{e^{-2x}-1}|+c$

Q 8. If $\int \frac{x+2}{2x^2+6x+5}dx = P\int \frac{4x+6}{2x^2+6x+5}dx + \frac{1}{2}\int \frac{dx}{2x^2+6x+5}$, then the value of P is

[Kerala CEE]

(a) $\frac{1}{3}$

(b) $\frac{1}{2}$

(c) $\frac{1}{4}$

(d) 2

(e) 1

Q 9. $\int \frac{\log(x+\sqrt{1+x^2})}{\sqrt{1+x^2}} dx$ is equal to

[Kerala CEE]

(a) $[\log(x+\sqrt{1+x^2})]^2 + c$

(b) $x \log(x+\sqrt{1+x^2}) + c$

(c) $\frac{1}{2}\log(x+\sqrt{1+x^2}) + c$

(d) $\frac{1}{2}[\log(x+\sqrt{1+x^2})]^2 + c$

(e) $\frac{x}{2}\log(x+\sqrt{1+x^2}) + c$

Q 10. If $\int\left[\log(\log x)+\frac{1}{(\log x)^2}\right]dx = x[f(x) - g(x) + c,$

then

[DCE]

(a) f(x) = log(log x); g(x) = $\frac{1}{\log x}$

(b) f(x) = log x; g(x) = $\frac{1}{\log x}$

(c) f(x) = $\frac{1}{\log x}$; g(x) = log (log x)

(d) f(x) = $\frac{1}{x \log x}$; g(x) = $\frac{1}{\log x}$

Q 11. If $\int\frac{dx}{x^4+x^3}=\frac{A}{x^2}+\frac{B}{x}+\log\left|\frac{x}{x+1}\right|+C$, then

[BITSAT]

(a) $A = \frac{1}{2}$, B = 1

(b) A = 1, B = $-\frac{1}{2}$

(c) A = $-\frac{1}{2}$, B = 1

(d) A = 1, B = 1

Q 12. $\int e^x \frac{x^2+1}{(x+1)^2} dx$ is equal to

[UP SEE]

(a) $\frac{-e^x}{x+1} + c$

(b) $\frac{e^x}{x+1}+c$

(c) $e^x\frac{x-1}{x+1}+c$

(d) $\frac{xe^x}{x+1}+c$

Q 13. $\int(1+x-x^{-1})e^{x+x^{-1}}dx$ is equal to

[KCECE]

(a) $(1+x)e^{x+x^{-1}}+c$

(b) $(x-1)e^{x+x^{-1}}+c$

(c) $-xe^{x+x^{-1}}+c$

(d) $xe^{x+x^{-1}}+c$

Q 14. $\int (\sqrt[3]{x})\left(\sqrt[5]{1+\sqrt[3]{x^4}}\right)dx$ is equal to

[Kerala CEE]

(a) $\left(1+x^{\frac{3}{4}}\right)^{\frac{6}{5}}+c$

(b) $\left(1+x^{\frac{4}{3}}\right)^{\frac{6}{5}}+c$

(c) $\frac{5}{8}\left(1+x^{\frac{4}{3}}\right)^{\frac{6}{5}}+c$

(d) $\frac{1}{6}\left(1+x^{\frac{4}{3}}\right)^{6}+c$

Q 15. $\int \frac{\cos x - \sin x}{1 + 2\sin x \cos x} dx$ is equal to

[Kerala CEE]

(a) $-\frac{1}{\cos x - \sin x} + c$

(b) $\frac{\cos x + \sin x}{\cos x - \sin x} + c$

(c) $-\frac{1}{\sin x + \cos x} + c$

(d) $\frac{x}{\sin x + \cos x} + c$

(e) tan x sec x + c

Q 16. $\int \frac{1}{x}(\log_{ex} e) dx$ is equal to

[Kerala CEE]

(a) $\log_e(1 - \log_e x) + c$

(b) $\log_e(\log_e ex - 1) + c$

(c) $\log_e(\log_e x - 1) + c$

(d) $\log_e(\log_e x + x) + c$

(e) $\log_e(1 + \log_e x) + c$

Q 17. If $\int (\log x)^2 dx = x[f(x)]^2 + Ax[f(x) - 1] + c$,

then

[AMU]

(a) f(x) = log x, A = 2

(b) f(x) = log x, A = -2

(c) f(x) = −log x, A = 2

(d) f(x) = −log x, A = -2

Q 18. $\int \frac{dx}{(x+1)\sqrt{4x+3}}$ is equal to

[EAMCET]

(a) $\tan^{-1}\sqrt{4x+3}+c$

(b) $3\tan^{-1}\sqrt{4x+3}+c$

(c) $2\tan^{-1}\sqrt{4x+3}+c$

(d) $4\tan^{-1}\sqrt{4x+3}+c$

Q 19. Let $\int \frac{x^2}{\sqrt{1-x}}dx = p\sqrt{(1-x)}(3x^2+4x+8)$, then value of p is

[OJEE]

(a) $\frac{-2}{15}$

(b) $\frac{2}{15}$

(c) $\frac{4}{15}$

(d) None of these

Q 20. $\int \frac{(x+1)}{x(1+xe^x)}dx$ is

[OJEE]

(a) $\log\left(\frac{xe^x}{xe^x+1}\right)+\frac{1}{1+xe^x}+c$

(b) $\log\left(\frac{xe^x}{xe^x+1}\right)-\frac{1}{1+xe^x}+c$

(c) $\log\left(\frac{xe^x+1}{xe^x}\right)+\frac{1}{1+xe^x}+c$

(d) None of these

Q 21. If $f'(x) = \frac{dx}{(1+x^2)^{3/2}}$ and $f(0) = 0$, then $f(1)$ is equal to

[OJEE]

(a) $\sqrt{2}$

(b) $-\frac{1}{\sqrt{2}}$

(c) $\frac{1}{\sqrt{2}}$

(d) None of these

Q 22. $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$ is equal to

[WB JEE]

(a) $\log(\sin^{-1} x) + c$

(b) $\frac{1}{2}(\sin^{-1} x)^2 + c$

(c) $\log(\sqrt{1-x^2}) + c$

(d) $\sin(\cos^{-1} x) + c$

where c is an arbitrary constant

Q 23. $\int \frac{dx}{x(x+1)}$ equals

[WB JEE]

(a) $\log\left|\frac{x+1}{x}\right|+c$

(b) $\log\left|\frac{x}{x+1}\right|+c$

(c) $\log\left|\frac{x-1}{x}\right|+c$

(d) $\log\left|\frac{x-1}{x+1}\right|+c$

where c is an arbitrary constant

Q 24. $\int\frac{x^2-2}{x^3\sqrt{x^2-1}}dx$ is equal to

[VITEEE]

(a) $\frac{x^2}{\sqrt{x^2-1}}+c$

(b) $-\frac{x^2}{\sqrt{x^2-1}}+c$

(c) $\frac{\sqrt{x^2-1}}{x^2}+c$

(d) $-\frac{\sqrt{x^2-1}}{x^2}+c$

Q 25. If $\int uv''dx = uv' - vu' + a$, then a is equal to

[VITEEE]

(a) $\int u''v\,dx$

(b) $\int u'v\,dx$

(c) $\int uv'dx$

(d) $\int u''dx$

Q 26. $\int \frac{(1+x^4)}{(1-x^4)^{3/2}}dx$ is equal to

[VITEEE]

(a) $\frac{x}{\sqrt{1-x^4}}+c$

(b) $\frac{-x}{\sqrt{1-x^4}}+c$

(c) $\frac{2x}{\sqrt{1-x^4}}+c$

(d) $\frac{-2x}{\sqrt{1-x^4}}+c$

Q 27. If $\int \frac{1}{(\sin x+4)(\sin x-1)}dx$

$= A\dfrac{1}{\tan\dfrac{x}{2}-1} + B\tan^{-1}(f(x)) + C_1$. Then,

[Jamia Millia Islamia]

(a) $A = \dfrac{1}{5}, B = \dfrac{-2}{5\sqrt{15}}, f(x) = \dfrac{4\tan x + 3}{\sqrt{15}}$

(b) $A = -\dfrac{1}{5}, B = \dfrac{1}{\sqrt{15}}, f(x) = \dfrac{4\tan\left(\dfrac{x}{2}\right)+1}{\sqrt{15}}$

(c) $A = \dfrac{2}{5}, B = \dfrac{-2}{5}, f(x) = \dfrac{4\tan x + 1}{5}$

(d) $A = \dfrac{2}{5}, B = \dfrac{-2}{5\sqrt{15}}, f(x) = \dfrac{4\tan\left(\dfrac{x}{2}\right)+1}{\sqrt{15}}$

Q 28. $\int x \log x \, dx$ is equal to

[MHT CET]

(a) $\frac{x^2}{4}(2\log x - 1) + c$

(b) $\frac{x^2}{2}(2\log x - 1) + c$

(c) $\frac{x^2}{4}(2\log x + 1) + c$

(d) $\frac{x^2}{2}(2\log x + 1) + c$

Q 29. $\int \frac{1}{1+\cos x + \sin x} dx$ is equal to

[MP PET]

(a) $\log\left|1 + \tan\frac{x}{2}\right| + c$

(b) $\frac{1}{2}\log\left|1+\tan\frac{x}{2}\right|+c$

(c) $2\log\left|1+\tan\frac{x}{2}\right|+c$

(d) $\frac{1}{2}\log\left|1-\tan\frac{x}{2}\right|+c$

Q 30. $\int \sin^3 x.\cos^2 x\,dx$ is equal to

[MP PET]

(a) $\frac{\sin^5 x}{5}-\frac{\sin^3 x}{3}+c$

(b) $\frac{\sin^5 x}{5}+\frac{\sin^3 x}{3}+c$

(c) $\frac{\cos^5 x}{5}-\frac{\cos^3 x}{3}+c$

(d) $\frac{\cos^5 x}{5}+\frac{\cos^3 x}{3}+c$

Q 31. $\int \tan^{-1} x\,dx = \ldots\ldots + c$

[Guj. CET]

(a) $\frac{1}{1+x^2}$

(b) $x\tan^{-1} x+\frac{1}{2}\log|1+x^2|$

(c) $x\tan^{-1} x+\frac{1}{2}.\frac{\tan^{-1} x}{1+x^2}$

(d) $x\tan^{-1} x-\frac{1}{2}\log|1+x^2|$

Q 32. $\int \frac{f'(x)}{\sqrt{f(x)}}dx = \ldots\ldots + c;\ f(x) \neq 0$

[Guj. CET]

(a) $\frac{1}{2}\sqrt{f(x)}$

(b) $2\sqrt{f(x)}$

(c) $\frac{1}{2}f(x)$

(d) $2f(x)$

Q 33. $\int \frac{\sqrt{\tan x}}{\sin x \cos x} dx = ... + c;\ x \neq \frac{k\pi}{2}$ and tan x > 0

[Guj. CET]

(a) $\frac{1}{2\sqrt{\tan x}}$

(b) $\sqrt{2\tan x}$

(c) $2\sqrt{\tan x}$

(d) $\sqrt{\tan x}$

Q 34. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx, J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then,

for an arbitrary constant c, the value of J – I equals

[IIT JEE]

(a) $\frac{1}{2}\log\left|\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1}\right| + c$

(b) $\frac{1}{2}\log\left|\frac{e^{2x}+e^{x}+1}{e^{2x}-e^{2x}+1}\right|+c$

(c) $\frac{1}{2}\log\left|\frac{e^{2x}-e^{x}+1}{e^{2x}+e^{x}+1}\right|+c$

(d) $\frac{1}{2}\log\left|\frac{e^{4x}+e^{2x}+1}{e^{4x}-e^{2x}+1}\right|+c$

Q 35. The value of $\sqrt{2}\int\frac{\sin x\,dx}{\sin\left(x-\frac{\pi}{4}\right)}$ is

[AIEEE]

(a) $x-\log\left|\cos\left(x-\frac{\pi}{4}\right)\right|+c$

(b) $x+\log\left|\cos\left(x-\frac{\pi}{4}\right)\right|+c$

(c) $x - \log\left|\cos\left(x - \frac{\pi}{4}\right)\right| + c$

(d) $x + \log\left|\cos\left(x - \frac{\pi}{4}\right)\right| + c$

Q 36. If $\int e^x \left(\frac{1-\sin x}{1-\cos x}\right) dx$ = f(x) + constant, then f(x) is equal to

[BITSAT]

(a) $e^x \cot\left(\frac{x}{2}\right) + c$

(b) $e^{-x} \cot\left(\frac{x}{2}\right) + c$

(c) $-e^x \cot\left(\frac{x}{2}\right) + c$

(d) $-e^{-x}\cot\left(\frac{x}{2}\right)+c$

Q 37. If $\int e^{x}(1+x).\sec^{2}(xe^{x})dx$ = f(x) + constant, then

f(x) is equal to

[BITSAT]

(a) $\cos(xe^{x})$

(b) $\sin(xe^{x})$

(c) $2\tan^{-1}(x)$

(d) $\tan(x\,e^{x})$

Q 38. $\int \frac{\sin x \cos x}{\sqrt{1-\sin^4 x}} dx$ is equal to

[KCET]

(a) $\frac{1}{2}\sin^{-1}(\sin^2 x) + c$

(b) $\frac{1}{2}\cos^{-1}(\sin^2 x) + c$

(c) $\tan^{-1}(\sin^2 x) + c$

(d) $\tan^{-1}(2\sin^2 x) + c$

Q 39. If $\int f(x)dx = g(x)$, then $\int f(x)g(x)dx$ is equal to

[KCET]

(a) $\frac{1}{2}f^2(x)$

(b) $\frac{1}{2}g^2(x)$

(c) $\frac{1}{2}[g'(x)]^2$

(d) $f'(x)g(x)$

Q 40. $\int (\sin x - \cos x)^4 (\sin x + \cos x)dx$ is equal to

[Kerala CEE]

(a) $\frac{\sin x - \cos x}{5} + c$

(b) $\frac{(\sin x - \cos x)^3}{5} + c$

(c) $\frac{(\sin x - \cos x)^4}{4} + c$

(d) $\frac{(\sin x + \cos x)^5}{4} + c$

(e) None of these

Q 41. $\int e^{\sin\theta} [\log \sin \theta + \text{cosec}^2 \theta] \cos \theta \, d\theta$ is equal to

[Kerala CEE]

(a) $\int e^{\sin\theta} [\log \sin \theta + \text{cosec}^2 \theta] + c$

(b) $e^{\sin \theta} [\log \sin \theta + \text{cosec } \theta] + c$

(c) $e^{\sin \theta}$ [log sin θ − cosec θ] + c

(d) $e^{\sin \theta}$[log sin θ − cosec^2 θ] + c

(e) $e^{\sin \theta}$[log sin θ + $\cos^2$ θ] + c

Q 42. If $I_n = \int x^n . e^{cx} dx$ for $n \geq 1$, then c . I_n + n . I_{n-1} is equal to

[AMU]

(a) $x^n e^{cx}$

(b) x^n

(c) e^{cx}

(d) $x^n + e^{cx}$

Q 43. The value of $\int x(x^x)^x(2\log x + 1)dx$ is

[OJEE]

(a) $(x^x)^x + c$

(b) $x^x + c$

(c) $x^{\log x} + c$

(d) None of these

Q 44. $\int x^{51}(\tan^{-1} x + \cot^{-1} x)dx$

[BCECE]

(a) $\frac{x^{52}}{52}(\tan^{-1} x + \cot^{-1} x) + c$

(b) $\frac{x^{52}}{52}(\tan^{-1} - \cot^{-1} x) + c$

(c) $\frac{\pi x^{52}}{104} + \frac{\pi}{2} + c$

(d) $\frac{x^{52}}{52} + \frac{\pi}{2} + c$

Q 45. If $I = \int \frac{x^5}{\sqrt{1+x^3}} dx$, then I is equal to

[VITEEE]

(a) $\frac{2}{9}(1+x^3)^{\frac{5}{2}}+\frac{2}{3}(1+x^3)^{\frac{3}{2}}+c$

(b) $\log|\sqrt{x}+\sqrt{1+x^3}|+c$

(c) $\log|\sqrt{x}-\sqrt{1+x^3}|+c$

(d) $\frac{2}{9}(1+x^3)^{\frac{3}{2}}-\frac{2}{3}(1+x^3)^{\frac{1}{2}}+c$

Q 46. $\int\frac{\operatorname{cosec} x}{\cos^2\left(1+\log\tan\frac{x}{2}\right)}dx$ is equal to

[Jamia Millia Islamia]

(a) $\sin^2\left[1+\log\tan\frac{x}{2}\right]+c$

(b) $\tan\left[1+\log\tan\frac{x}{2}\right]+c$

(c) $\sec^2\left[1+\log\tan\frac{x}{2}\right]+c$

(d) $-\tan\left[1+\log\tan\frac{x}{2}\right]+c$

Q 47. $\int\frac{dx}{x\sqrt{x^6-16}}$ is equal to

[MHTCET]

(a) $\frac{1}{3}\sec^{-1}\left(\frac{x^3}{4}\right)+c$

(b) $\cos^{-1}\left(\frac{x^3}{4}\right)+c$

(c) $\frac{1}{12}\sec^{-1}\left(\frac{x^3}{4}\right)+c$

(d) $\sec^{-1}\left(\frac{x^3}{4}\right)+c$

Q 48. $\int \frac{x^{e-1}+e^{x-1}}{x^e+e^x}dx$ is equal to

[MHT CET]

(a) $\log(x^e+e^x)+c$

(b) $e\log(x^e+e^x)+c$

(c) $\frac{1}{e}\log(x^e+e^x)+c$

(d) None of these

Q 49. $\int \frac{x\tan^{-1}x}{\sqrt{(1+x^2)^{3/2}}}dx$ is equal to

[Manipal]

(a) $\frac{x-\tan^{-1}x}{1-x^2}+c$

(b) $\frac{x+\tan^{-1}x}{\sqrt{1-x^2}}+c$

(c) $\frac{x-\tan^{-1}x}{\sqrt{1+x^2}}+c$

(d) $\frac{x+\sqrt{1-x^2}}{\sqrt{1+x^2}}+c$

Q 50. $\int (e^x+e^{-x})^2.(e^x-e^{-x})dx$ is equal to

[MP PET]

(a) $e^x + c$

(b) $\frac{1}{2}(e^x-e^{-x})^2+c$

(c) $\frac{1}{2}(e^x + e^{-x})^2 + c$

(d) $\frac{1}{3}(e^x + e^{-x})^3 + c$

Q 51. $\int \frac{\sin x}{\sin(x-\alpha)} dx$ is

[MP PET]

(a) x sin α + cos α log sin (x + α) + c

(b) x sin α + cos α log sin (x − α) + c

(c) x cos α + sin α log cos (x + α) + c

(d) x cos α + sin α log sin (x − α) + c

Q 52. The value of $\int \frac{3x+2}{(x-2)^2(x-3)}dx$ is

[RPET]

(a) $11\log\frac{x-3}{x-2}-\frac{8}{x-2}+c$

(b) $11\log\frac{x+3}{x+2}-\frac{8}{x-2}+c$

(c) $11\log\frac{x-3}{x-2}+\frac{8}{x-2}+c$

(d) $11\log\frac{x+3}{x+2}+\frac{8}{x-2}+c$

Q 53. $\int \sin\sqrt{x}\,dx$ is equal to

[J & K CET]

(a) $\sin\sqrt{x}-\sqrt{x}\cos\sqrt{x}$

(b) $2(\sin\sqrt{x}-\sqrt{x}\cos\sqrt{x})+c$

(c) $\cos\sqrt{x}-\sqrt{x}\sin\sqrt{x}+c$

(d) $2(\cos\sqrt{x}-\sqrt{x}\sin\sqrt{x})+c$

Q 54. $\int\frac{dx}{x(x^7+1)}$ is equal to

[DCE]

(a) $\log\left(\frac{x^7}{x^7+1}\right)+c$

(b) $\frac{1}{7}\log\left(\frac{x^7}{x^7+1}\right)+c$

(c) $\log\left(\frac{x^7+1}{x^7}\right)+c$

(d) $\frac{1}{7}\log\left(\frac{x^7+1}{x^7}\right)+c$

Q 55. The value of $\int \frac{1}{[(x-1)^3(x+2)^5]^{1/4}}dx$ is

[DCE]

(a) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+c$

(b) $\frac{4}{3}\left(\frac{x+1}{x+2}\right)^{1/4}+c$

(c) $\frac{4}{3}\left(\frac{x+1}{x-2}\right)^{1/4}+c$

(d) $\frac{4}{3}\left(\frac{x-1}{x-2}\right)^{1/4}+c$

Q 56. $\int\frac{1+\tan^2 x}{1-\tan^2 x}dx$ is equal to

[BITSAT]

(a) $\log\left(\frac{1-\tan x}{1+\tan x}\right)+c$

(b) $\log\left(\frac{1+\tan x}{1-\tan x}\right)+c$

(c) $\frac{1}{2}\log\left(\frac{1-\tan x}{1+\tan x}\right)+c$

(d) $\frac{1}{2}\log\left(\frac{1+\tan x}{1-\tan x}\right)+c$

Q 57. $\int\frac{a^{x/2}}{a^{-x}-a^{x}}dx$ is equal to

[UPSEE]

(a) $\frac{1}{\log a}\sin^{-1}(a^x)+c$

(b) $\frac{1}{\log a}\tan^{-1}(a^x)+c$

(c) $2\sqrt{a^{-x}-a^x}+c$

(d) $\log(a^x - 1) + c$

Q 58. The value of $\int e^x(x^5+5x^4+1)dx$ is

[KCET]

(a) $e^x \cdot e^5 + c$

(b) $e^x \cdot x^5 + e^x + c$

(c) $e^{x+1} \cdot x^5 + c$

(d) $5x^4 \cdot e^x + c$

Q 59. If $f(x) = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$ and g(x) = $e^{\sin^{-1} x}$, then

$\int f(x)g(x)dx$ is equal to

[Kerala CEE]

(a) $e^{\sin^{-1} x}(\sin^{-1} x - 1) + c$

(b) $e^{\sin^{-1} x} + c$

(c) $e^{(\sin^{-1}x)^2}+c$

(d) $e^{2\sin^{-1}x}+c$

(e) $e^{\sin^{-1}x}\sin^{-1}x+c$

Q 60. The value of $\int\frac{x^2+1}{x^4-x^2+1}dx$ is

[Kerala CEE]

(a) $\tan^{-1}(2x^2-1)+c$

(b) $\tan^{-1}\frac{x^2+1}{x}+c$

(c) $\sin^{-1}\left(x-\frac{1}{x}\right)+c$

(d) $\tan^{-1} x^2 + c$

(e) $\tan^{-1}\left(\frac{x^2 - 1}{x}\right) + c$

Q 61. $\int \cos\left\{2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right\} dx$ is equal to

[Kerala CEE]

(a) $\frac{1}{8}(x^2 - 1) + c$

(b) $\frac{x^2}{4} + c$

(c) $\frac{x}{2} + c$

(d) $\frac{x}{4} + c$

(e) $\frac{x^2}{2}+c$

Q 62. If $\int \frac{2x^2+3}{(x^2-1)(x^2+4)}dx$

$= a\log\left(\frac{x-1}{x+1}\right)+b\tan^{-1}\left(\frac{x}{2}\right)+c$,then the value of a and b are

[AMU]

(a) $(1, -1)$

(b) $(-1, 1)$

(c) $\left(\frac{1}{2},-\frac{1}{2}\right)$

(d) $\left(\frac{1}{2},\frac{1}{2}\right)$

Q 63. $\int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x} dx$ is equal to

[EAMCET]

(a) 2 sec x + c

(b) 2 tan x + c

(c) tan x + c

(d) None of these

Q 64. $\int \frac{\sin x + 8\cos x}{4\sin x + 6\cos x} dx$ is equal to

[EAMCET]

(a) $x+\frac{1}{2}\log(4\sin x + 6\cos x) + c$

(b) $2x + \log(2\sin x + 3\cos x) + c$

(c) $x + 2\log(2\sin x + 3\cos x) + c$

(d) $\frac{1}{2}\log(4\sin x + 6\cos x) + c$

Q 65. $\int \frac{x}{(x^2-a^2)(x^2-b^2)}dx$ is equal to

[OJEE]

(a) $\frac{1}{a^2-b^2}\log\left|\frac{x^2-a^2}{x^2-b^2}\right| + c$

(b) $\frac{1}{a^2-b^2}\log\left|\frac{x^2-b^2}{x^2-a^2}\right| + c$

(c) $\frac{1}{2(a^2-b^2)}\log\left|\frac{x^2-a^2}{x^2-b^2}\right|+c$

(d) $\frac{1}{2(a^2-b^2)}\log\left|\frac{x^2-b^2}{x^2-a^2}\right|+c$

Q 66. The value of $\int\frac{dx}{x+\sqrt{x-1}}$ is

[BCECE]

(a) $\log(x+\sqrt{x-1})+\sin^{-1}\left(\sqrt{\frac{x-1}{x}}\right)+c$

(b) $\log(x+\sqrt{x-1})+c$

(c) $\log(x+\sqrt{x-1})-\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2\sqrt{x-1}+1}{\sqrt{3}}\right)+c$

(d) None of these

Q 67. The value of the integral $\int e^x \left(\frac{1-x}{1+x^2}\right)^2 dx$ is

[VITEEE]

(a) $e^x \left(\frac{1-x}{1+x^2}\right) + c$

(b) $e^x \left(\frac{1+x}{1+x^2}\right) + c$

(c) $\frac{e^x}{1+x^2} + c$

(d) $e^x(1 - x) + c$

Q 68. $\int \frac{x + \sin x}{1 + \cos x} dx$ is equal to

[MHTCET]

(a) $x \tan \frac{x}{2} + c$

(b) $\log(1 + \cos x) + c$

(c) $\cot \frac{x}{2} + c$

(d) $\log(x + \sin x) + c$

Q 69. $\int \cos^3 x . e^{\log \sin x} dx$ is equal to

[MHTCET]

(a) $-\frac{\sin^4 x}{4} + c$

(b) $-\frac{\cos^4 x}{4} + c$

(c) $\frac{e^{\sin x}}{4} + c$

(d) None of these

Q 70. $\int \frac{\log_e(\tan x)}{\sin x \cos x} dx$ is equal to

[J & K CET]

(a) $[\log_e (\tan x)]^2 + c$

(b) $\frac{1}{2}(\log_e \tan x)^2 + c$

(c) $\log_e(\log_e \tan x) + c$

(d) $\log_e \tan x + c$

Q 71. The value of the integral $\int \frac{\log(x+1)-\log x}{x(x+1)}dx$ is

[Guj. CET]

(a) $-\frac{1}{2}[\log(x+1)]^2-\frac{1}{2}(\log x)^2+\log(x+1)\log x+c$

(b) $-[\{\log(x+1)\}^2-(\log x)^2]+\log(x+1)\log x+c$

(c) $\left[\log\left(1+\frac{1}{x}\right)\right]^2+c$

(d) None of these

Q 72. If $\int f(x)\sin x\cos x dx=\frac{1}{2(b^2-a^2)}\log f(x)+c$, then f(x)

is equal to

[Guj. CET]

(a) $\frac{1}{a^2 \sin^2 x + b^2 \cos^2 x}$

(b) $\frac{1}{a^2 \sin^2 x - b^2 \cos^2 x}$

(c) $\frac{1}{a^2 \cos^2 x + b^2 \sin^2 x}$

(d) $\frac{1}{a^2 \cos^2 x - b^2 \sin^2 x}$

Q 73. The value of $\int \frac{(x^2 - 1)dx}{x^3 \sqrt{2x^4 - 2x^2 + 1}}$ is

[IIT JEE]

(a) $2\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}+c$

(b) $2\sqrt{2+\frac{2}{x^2}+\frac{1}{x^4}}+c$

(c) $\frac{1}{2}\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}+c$

(d) None of the above

Q 74. The value of $\int \frac{dx}{x^2(x^4+1)^{3/4}}$ is

[DCE]

(a) $-\frac{(x^4+1)^{1/4}}{x}+c$

(b) $\frac{(x^4+1)^{1/4}}{x}+c$

(c) zero

(d) None of these

Q 75. The value of the $\int \frac{\sin x + \cos x}{3\sin 2x} dx$ is

[DCE]

(a) $\frac{1}{4}\log\left(\frac{2-\sin x+\cos x}{2+\sin x-\cos x}\right)+c$

(b) $\frac{1}{2}\log\left(\frac{2+\sin x}{2-\sin x}\right)+c$

(c) $\frac{1}{4}\log\left(\frac{1+\sin x}{1-\sin x}\right)+c$

(d) None of these

Q 76. $\int \frac{\sin 2x}{1+\cos^2 x} dx$ is equal to

[BITSAT]

(a) $-\frac{1}{2}\log(1+\cos^2 x) + c$

(b) $2 \log (1 + \cos^2 x) + c$

(c) $\frac{1}{2}\log(1 + \cos 2x) + c$

(d) $c - \log(1 + \cos^2 x)$

Q 77. If $\int f(x)dx = g(x) + c$, then $\int f^{-1}(x)dx$ is equal to

[UP SEE, WB JEE]

(a) $xf^{-1}(x) + c$

(b) $f\{g^{-1}(x)\} + c$

(c) $xf^{-1}(x) - g\{f^{-1}(x)\} + c$

(d) $g^{-1}(x) + c$

Q 78. $\int \frac{x^2 + x - 6}{(x-2)(x-1)} dx$

[KCET]

(a) $x + 2\log(x - 1) + c$

(b) $2x + 2\log(x - 1) + c$

(c) $x + 4 \log (1 - x) + c$

(d) $x + 4 \log (x - 1) + c$

Q 79. If $\int x f(x) dx = \frac{f(x)}{2}$, then $f(x)$ is equal to

[Kerala CEE]

(a) e^x

(b) e^{-x}

(c) $\log x$

(d) $\frac{e^{x^2}}{2}$

(e) e^{x^2}

Q 80. $\int e^x (\log \sin x + \cot x) dx$ is equal to

[Kerala CEE]

(a) $e^x \cot x + c$

(b) $e^x \log \sin x + c$

(c) $e^x \log \sin x + \tan x + c$

(d) $e^x + \sin x + c$

(e) $\log(\sin x + \cos x) + e^x + c$

Q 81. If $\int \frac{\sqrt{x}}{x+1} dx = A\sqrt{x} + B\tan^{-1}\sqrt{x} + c$, then

[Kerala CEE]

(a) A = 1, B = 1

(b) A = 1, B = 2

(c) A = 2, B = 2

(d) A = 2, B = −2

(e) A = −2, B = −2

Q 82. $\int \frac{x^3 \sin[\tan^{-1}(x^4)]}{1+x^8} dx$ is equal to

[Kerala CEE]

(a) $\frac{1}{4}\cos[\tan^{-1}(x^4)]+c$

(b) $\frac{1}{4}\sin[\tan^{-1}(x^4)]+c$

(c) $-\frac{1}{4}\cos[\tan^{-1}(x^4)]+c$

(d) $-\frac{1}{4}\sec^{-1}[\tan^{-1}(x^4)]+c$

(e) $-\frac{1}{4}\cos^{-1}[\tan^{-1}(x^4)]+c$

Q 83. Observe the following statements

A. $\int\left(\frac{x^2-1}{x^2}\right)e^{\frac{x^2+1}{x}}dx=e^{\frac{x^2+1}{x}}+c$

R. $\int f'(x)e^{f(x)}dx = f(x) + c$

Then, which of the following is true ?

[EAMCET]

(a) Both A and R are true and R is the correct reason for A

(b) Both A and R are true and R is not the correct reason for A

(c) A is true, R is false

(d) A is false, R is true

Q 84. $\int \frac{dx}{x(x^n+1)}$ is equal to

[OJEE, MP PET]

(a) $\frac{1}{n}\log\left(\frac{x^n}{x^n+1}\right)+c$

(b) $\frac{1}{n}\log\left(\frac{x^n+1}{x^n}\right)+c$

(c) $\log\left(\frac{x^n}{x^n+1}\right)+c$

(d) None of the above

Q 85. $\int 32x^3(\log x)^2 dx$ is equal to

[WB JEE]

(a) $8x^4(\log x)^2 + c$

(b) $x^4\{8(\log x)^2 - 4 \log x + 1\} + c$

(c) $x^4\{8(\log x)^2 - 4 \log x\} + c$

(d) $x^3\{(\log x)^2 - 2 \log x\} + c$

Q 86. $\int \frac{\cos x - 1}{\sin x + 1} . e^x \, dx$ is equal to

[WB JEE]

(a) $\frac{e^x \cos x}{1 + \sin x} + c$

(b) $c - \frac{e^x \sin x}{1 + \sin x}$

(c) $c-\frac{e^x}{1+\sin x}$

(d) $c-\frac{e^x \cos x}{1+\sin x}$

Q 87. $\int \frac{(\sin\theta+\cos\theta)}{\sqrt{\sin 2\theta}} d\theta$ is equal to

[MHTCET]

(a) log $|\cos\theta - \sin\theta + \sqrt{\sin 2\theta}| + c$

(b) log $|\sin\theta - \cos\theta + \sqrt{\sin 2\theta}| + c$

(c) $\sin^{-1}(\sin\theta - \cos\theta) + c$

(d) $\sin^{-1}(\sin\theta + \cos\theta) + c$

Q 88. $\int \frac{(\tan^{-1} x)^3}{(1+x^2)} dx$ is equal to

[MP PET]

(a) $3(\tan^{-1} x)^2 + c$

(b) $\frac{(\tan^{-1} x)^4}{4} + c$

(c) $(\tan^{-1} x)^4 + c$

(d) None of these

Q 89. If $I_n = \int \tan^n x dx$ for $n \geq 2$, then $I_n + I_{n-2}$ is

equal to

[J & K CET]

(a) $\tan^n x + c$

(b) $\frac{(\tan^{n-1} x)}{n-1} + c$

(c) $\frac{\tan^n x}{n} + c$

(d) $n \tan^n x + c$

Q 90. If $\int \frac{dx}{x \log x} = f(x) +$ constant, then f(x) is equal to

[J & K CET]

(a) $1/\log x$

(b) $\log x$

(c) $\log \log x$

(d) $x/\log x$

Q 91. $\int \frac{e^x(1+\sin x)}{1+\cos x} dx$ is equal to

[Guj. CET]

(a) $e^x \tan\left(\frac{x}{2}\right) + c$

(b) $e^x \tan x + c$

(c) $e^x\left(\frac{1+\sin x}{1-\cos x}\right)+c$

(d) $c-e^x\cot\left(\frac{x}{2}\right)$

Q 92. $\int\frac{1+\tan x}{e^{-x}\cos x}dx$ is equal to

[KCET]

(a) $e^{-x}\tan x + c$

(b) $e^{-x}\sec x + c$

(c) $e^x\sec x + c$

(d) $e^x \tan x + c$

Q 93. If an anti-derivative of f(x) is e^x and that of g(x) is cos x, then $\int f(x)\cos x dx + \int g(x)e^x dx$ is equal to

[Kerala CEE]

(a) f(x) g(x) + c

(b) f(x) + g(x) + c

(c) $e^x \cos x + c$

(d) f(x) – g(x) + c

(e) $e^x \cos x + f(x)\, g(x) + c$

Q 94. $\int \sqrt{e^x - 1}\,dx$ is equal to

[Kerala CEE]

(a) $2[\sqrt{e^x - 1} - \tan^{-1}\sqrt{e^x - 1}] + c$

(b) $\sqrt{e^x - 1} - \tan^{-1}\sqrt{e^x - 1} + c$

(c) $\sqrt{e^x - 1} + \tan^{-1}\sqrt{e^x - 1} + c$

(d) $2[\sqrt{e^x - 1} + \tan^{-1}\sqrt{e^x - 1}] + c$

(e) $2[\sqrt{e^x - 1} - \tan^{-1}\sqrt{e^x + 1}] + c$

Q 95. $\int \cos^{-3/7} x \sin^{-11/7} x \, dx$ is equal to

[Kerala CEE]

(a) $\log |\sin^{4/7} x| + c$

(b) $\frac{4}{7} \tan^{4/7} x + c$

(c) $-\frac{7}{4} \tan^{-4/7} x + c$

(d) $\log |\cos^{3/7} x| + c$

(e) $\frac{7}{4} \tan^{-4/7} x + c$

Q 96. Let $f(x)=\int \frac{x^2 dx}{(1+x^2)(1+\sqrt{1+x^2})}$ and f(x) = 0. Then,

f(1) is

[AMU]

(a) $\log(1+\sqrt{2})$

(b) $\log(1+\sqrt{2})-\frac{\pi}{4}$

(c) $\log(1+\sqrt{2})+\frac{\pi}{4}$

(d) None of these

Q 97. If $\int \sin^{-1}\left(\frac{2x}{1+x^2}\right)dx = f(x) - \log(1 + x^2) + c,$

then f(x) is equal to

[EAMCET]

(a) $2x \tan^{-1} x$

(b) $-2x \tan^{-1} x$

(c) $x \tan^{-1} x$

(d) $-x \tan^{-1} x$

Q 98. If $\int \frac{\sin x}{\cos x(1+\cos x)} dx = f(x) + c$, the $f(x)$ is equal to

[EAMCET]

(a) $\log\left|\frac{1+\cos x}{\cos x}\right|$

(b) $\log\left|\frac{\cos x}{1+\cos x}\right|$

(c) $\log\left|\frac{\sin x}{1+\sin x}\right|$

(d) $\log\left|\frac{1+\sin x}{\sin x}\right|$

Q 99. $\int \frac{x^{49}\tan^{-1}(x^{50})}{(1+x^{100})}dx = k[\tan^{-1}(x^{50})]^2 + c$, then k is equal to

[EAMCET]

(a) $\frac{1}{50}$

(b) $-\frac{1}{50}$

(c) $\frac{1}{100}$

(d) $-\frac{1}{100}$

Q 100. $\int [f(x)g''(x) - f''(x)g(x)]dx$ is equal to

[BCECE]

(a) $\frac{f(x)}{g'(x)}$

(b) f'(x) g(x) – f(x) g'(x)

(c) f(x)g'(x) – f'(x)g(x)

(d) f(x)g'(x) + f'(x)g(x)

Q 101. $\int \frac{dx}{x+\sqrt{x}}$ is equal to

[MHTCET]

(a) $\frac{1}{2}\log(1+\sqrt{x})+c$

(b) $2\log(1+\sqrt{x})+c$

(c) $\frac{1}{4}\log(1+\sqrt{x})+c$

(d) $3\log(1+\sqrt{x})+c$

Q 102. The value of $\int\frac{x^2}{1+x^6}dx$ is

[MHTCET]

(a) $x^3 + c$

(b) $\frac{1}{3}\tan^{-1}(x^3)+c$

(c) $\log(1 + x^3)$

(d) None of these

Q 103. $\int \frac{f'(x)}{f(x)\log[f(x)]}dx$ is equal to

[MP PET]

(a) $\frac{f(x)}{\log f(x)}+c$

(b) f(x) . log f(x) + c

(c) log[log f(x)] + c

(d) $\frac{1}{\log[\log f(x)]}+c$

Q 104. $\int \frac{e^x}{(2+e^x)(e^x+1)}dx$ is equal to

[RPET]

(a) $\log\left(\frac{e^x+1}{e^x+2}\right)+c$

(b) $\log\left(\frac{e^x+2}{e^x+1}\right)+c$

(c) $\left(\frac{e^x+1}{e^x+2}\right)+c$

(d) $\left(\frac{e^x+2}{e^x+1}\right)+c$

Q 105. If x is any arbitrary constant, then

$\int 2^{2^{2^x}}.2^{2^x}.2^x dx$ is equal to

[J&K CEE]

(a) $\frac{\int 2^{2^x}\,dx}{(\log 2)^3}+c$

(b) $\frac{\int 2^{2^{2^x}}\,dx}{(\log 2)^3}+c$

(c) $\int 2^{2^{2^x}}(\log 2)^3\,dx+c$

(d) None of these

Q 106. $\int\frac{dx}{x(x^5+1)}$ is equal to

[UP SEE]

(a) $\frac{1}{5}\log x^5(x^5+1)+c$

(b) $\frac{1}{5}\log\left(\frac{x^5+1}{x^5}\right)+c$

(c) $\frac{1}{5}\log\left(\frac{x^5}{x^5+1}\right)+c$

(d) None of these

Q 107. $\int \sqrt{x}e^{\sqrt{x}}dx$ is equal to

[KCET]

(a) $2\sqrt{x}-e^{\sqrt{x}}-4\sqrt{xe^{\sqrt{x}}}+c$

(b) $(2x-4\sqrt{x}+4)e^{\sqrt{x}}+c$

(c) $(2x+4\sqrt{x}+4)e^{\sqrt{x}}+c$

(d) $(1-4\sqrt{x})e^{\sqrt{x}}+c$

Q 108. $\int \frac{(e^x - e^{-x})dx}{(e^x + e^{-x})\log(\cosh x)}$ is equal to

[Kerala CEE]

(a) log (tan h x) + c

(b) $2(\log e^x + e^{-x}) + c$

(c) $2 \log(e^x - e^{-x}) + c$

(d) $2\log[\log(e^x + e^{-x})] + c$

(e) log[log (cosh x)] + c

Q 109. If $\int \frac{dx}{(x+100)\sqrt{x+99}} = f(x) + c$, then f(x) is

[EAMCET]

(a) $2(x + 100)^{1/2}$

(b) $3(x + 100)^{1/2}$

(c) $2\tan^{-1}(\sqrt{x+99})$

(d) $2\tan^{-1}(\sqrt{x+100})$

Q 110. $\int \frac{3-x^2}{1-2x+x^2}.e^x dx = e^x f(x) + c \Rightarrow f(x)$

[EAMCET]

(a) $\frac{1+x}{1-x}$

(b) $\frac{1-x}{1+x}$

(c) $\frac{1+x}{x-1}$

(d) $\frac{x-1}{1+x}$

Q 111. $\int e^{2x}(2\sin 3x + 3\cos 3x)dx$ is equal to

[MHTCET]

(a) e^{2x} sin $3x$ + c

(b) e^{2x} cos $3x$ + c

(c) e^{2x+c}

(d) $e^{2x}(2 \sin 3x) + c$

Q 112. Primitive of $\frac{1}{4\sqrt{x}+x}$ is equal to

[MHTCET]

(a) $2\log|1+4\sqrt{x}|+c$

(b) $\frac{1}{2}\log|4-\sqrt{x}|+c$

(c) $2\log|4+\sqrt{x}|+c$

(d) $\frac{1}{2}\log|4+\sqrt{x}|+c$

Q 113. $\int e^x \left(\log x + \frac{1}{x} \right) dx$ is equal to

[MHTCET]

(a) $e^x \log x + c$

(b) $\frac{e^x}{\log x} + c$

(c) $\frac{\log x}{x} + c$

(d) $\frac{e^x}{x} + c$

Q 114. $\int \frac{x^2 - 1}{x^4 + x^2 + 1} dx$ is equal to

[RPET]

(a) $\log(x^4 + x^2 + 1) + c$

(b) $\log \frac{x^2 - x + 1}{x^2 + x + 1} + c$

(c) $\frac{1}{2} \log \frac{x^2 - x + 1}{x^2 + x + 1} + c$

(d) $\frac{1}{2} \log \frac{x^2 + x + 1}{x^2 - x + 1} + c$

Q 115. The value of the integral $\int \frac{dx}{x(1 + \log x)^2}$ is equal to

[RPET]

(a) $\frac{-1}{1 + x} + c$

(b) $\frac{-1}{1+\log x}+c$

(c) $\frac{1}{1+\log x}+c$

(d) $\frac{1}{1+x^2}+c$

Q 116. $\int \log 2x dx$ is equal to

[DCE]

(a) $x\log 2x-\frac{x^2}{2}+c$

(b) $x\log 2x-\frac{x}{2}+c$

(c) $x^2 \log 2x - \frac{x}{2} + c$

(d) $x \log 2x - x + c$

Q 117. If $I_n = \int (\log x)^n dx$, then $I_n + nI_{n-1}$ is equal to

[KCET]

(a) $(x \log x)^n$

(b) $x(\log x)^n$

(c) $n(\log x)^n$

(d) $(\log x)^{n-1}$

Q 118. $\int x^2(ax+b)^{-2}dx$ is equal to

[Kerala CEE]

(a) $\frac{2}{a^2}\left(x-\frac{b}{a}\log(ax+b)\right)+c$

(b) $\frac{2}{a^2}\left(x-\frac{b}{a}\log(ax+b)\right)-\frac{x^2}{a(ax+b)}+c$

(c) $\frac{2}{a^2}\left(x+\frac{b}{a}\log(ax+b)\right)+\frac{x^2}{a(ax+b)}+c$

(d) $\frac{2}{a^2}\left(x+\frac{b}{a}\log(ax+b)\right)-\frac{x^2}{a(ax+b)}+c$

(e) $\frac{2}{a^2}\left(x-\frac{b}{a}\log(ax+b)\right)+\frac{x^2}{a(ax+b)}+c$

Answers

Topic 1 : Fundamental of Indefinite Integrals

1. (b) 2. (a) 3. (b) 4. (d) 5. (a) 6. (b) 7. (b) 8. (d)

9. (c) 10. (b) 11. (a) 12. (a) 13. (a) 14. (c) 15. (c)

16. (b) 17. (c) 18. (a) 19. (a) 20. (b) 21 (c) 22.

(a) 23. (c) 24. (e) 25. (b) 26. (a) 27. (a) 28. (d)

29. (a) 30. (d) 31. (a) 32. (b) 33.(c) 34. (a) 35.

(a) 36. (c) 37. (e) 38. (d) 39. (c) 40. (a)

41 (d) 42. (b) 43. (e) 44. (e) 45. (c) 46. (b) 47.

(c) 48. (b) 49. (b) 50. (a) 51. (b) 52. (a) 53. (d)

54. (a) 55. (d) 56. (b) 57. (c) 58. (e) 59. (b) 60.

(b) 61. (d) 62. (c) 63. (b) 64. (c)

Topic 2 : Integration by Substitution, Parts, Partial Fractions

1. (a) 2. (c) 3. (c) 4. (b) 5. (c) 6. (e) 7. (c) 8. (c)

9. (d) 10. (a) 11. (c) 12 (c) 13. (d) 14. (c) 15. (c)

16. (e) 17. (b) 18. (c) 19. (a) 20. (a) 21. (c) 22.

(b) 23. (b) 24. (d) 25. (a) 26. (a) 27. (d) 28. (a)

29. (a) 30. (c) 31. d) 32. (b) 33. (c) 34. (c) 35.

(d) 36. (c) 37. (d) 38. (a) 39. (b) 40. (b)

41. (c) 42 (a) 43. (a) 44. (a) 45. (d) 46. (b) 47.

(c) 48. (c) 49. (c) 50. (d) 51. (d) 52. (c) 53. (b)

54. (b) 55. (a) 56. (d) 57. (a) 58. (b) 59. (a) 60.

(e) 61. (e) 62 (d) 63. (c) 64. (a) 65. (c) 66. (c)

67. (c) 68. (a) 69. (b) 70. (b) 71. (a) 72. (a) 73.

(c) 74. (a) 75. (a) 76. (d) 77. (c) 78. (d) 79. (d)

80. (b) 81. (d) 82. (c) 83. (c) 84. (a) 85. (b) 86.

(a) 87. (c) 88. (b) 89. (b) 90. (c) 91. (a) 92. (c)

93. (c) 94. (a) 95. (c) 96. (b) 97. (a) 98. (a) 99.

(c) 100. (c) 101. (b) 102. (b) 103. (c) 104. (a)

105. (d) 106. (c) 107. (b) 108. (e) 109. (c) 110.

(a) 111. (a) 112. (c) 113. (a) 114. (c) 115. (b)

116. (d) 117. (b) 118. (b)

Conic Section

Useful Results & Formulae

1. The second degree equation

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, represents

(i) a parabola if $\Delta \neq 0$, $ab - h^2 = 0$, $e = 1$

(ii) an ellipse if $\Delta \neq 0$, $ab - h^2 > 0$, $e < 1$

(iii) a hyperbola if $\Delta \neq 0$, $ab - h^2 < 0$, $e > 1$

(iv) a rectangular hyperbola, if $\Delta \neq 0$, ab – $h^2 < 0$, a + b = 0, e = $\sqrt{2}$

2.

Important Terms	**Parabola** $y^2 = 4ax$	**Ellipse** $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, **a > b**	**Hyperbola** $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$
Centre	$(0, 0)$	$(0, 0)$	$(0, 0)$

Focus/Foci	$(a, 0)$	$(\pm ae, 0)$	$(\pm ae, 0)$
Vertex/Vertices	$(0, 0)$	$(\pm a, 0)$	$(\pm a, 0)$
Equation of directrix/directrices	$x + a = 0$	$x = \pm \frac{a}{e}$	$x = \pm \frac{a}{e}$
Length of latusrectum	$4a$	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$

Ends of latusrectum	(a, ±2a)	$\left(\pm ae, \pm \frac{b^2}{a}\right)$	$\left(\pm ae, \pm \frac{b^2}{a}\right)$
Parametric equation	$x = at^2$, $y = 2at$	$x = a \cos \alpha$, $y = b \sin \alpha$	$x = a \sec \alpha$, $y = b \tan \alpha$
Distance between foci	−	$2ae$	$2ae$
Equation of tangent at	$yy_1 = 2a(x +$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

point (x_1, y_1)	$x_1)$		
Equation of tangent in slope form	$y = mx + \frac{a}{m}$	$y = mx \pm \sqrt{a^2m^2 + b^2}$	$y = mx \pm \sqrt{a^2m^2 - b^2}$
Equation of tangent in parametric form	$ty = x + at^2$	$\frac{x}{a}\cos\alpha + \frac{y}{b}\sin\alpha = 1$	$\frac{x}{a}\sec\alpha - \frac{y}{b}\tan\alpha = 1$

Line y = mx + c touches if	$c = \frac{a}{m}$	$c^2 = a^2m^2 + b^2$	$c^2 + a^2m^2 - b^2$
Point of contact of y = mx + c and curve	$\left(\frac{a}{m^2}, \frac{2a}{m}\right)$, $m \neq 0$	$\left(\frac{\pm a^2m}{\sqrt{a^2m^2 + b^2}}, \frac{\pm$	$\left(\frac{\pm a^2m}{\sqrt{a^2m^2 - b^2}}, \frac{\pm$
Equation of	$y - y_1 = \frac{-y_1}{2a}(x$	$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$	$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$

normal at point (x_1, y_1)			
Equation of normal in parametric form	$y + tx = 2at + at^3$	ax sec α − by cosec α = $a^2 - b^2$	ax cos α − by cot α = $a^2 + b^2$
Equation of normal in	$y = mx - 2am -$	$y = mx \pm \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2 m^2}}$	$y = mx \mp \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2 m}}$

slope form	am^3		
Condition for normally	c = – $2am$ – am^3	$c = \pm\frac{m(a^2-b^2)}{\sqrt{a^2+b^2m^2}}$	$c = \mp\frac{m(a^2-b^2)}{\sqrt{a^2-m^2b^2}}$
Equation of pair of tangents $SS_1 = T^2$	S = y^2 – $4ax$ S_1 = y_1^2 – $4ax_1$	$S = \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $S_1 = \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1$ $T = \frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1$	$S = \frac{x^2}{a^2}-\frac{y^2}{b^2}-1$ $S_1 = \frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1$ $T = \frac{xx_1}{a^2}-\frac{yy_1}{b^2}-1$

where	$T = yy_1 - 2a(x + x_1)$		
Equation of chord of contact	$yy_1 = 2a(x + x_1)$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$
Equation of chord with	$T = S_1$	$T = S_1$	$T = S_1$

mid point (x_1, y_1)			
Equation of diameter	$y = \frac{2a}{m}$	$y = -\frac{b^2}{a^2 m}x$	$y = \frac{b^2}{a^2 m}x$
Equation of polar	$yy_1 =$ $2a(x +$ $x_1)$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$
Coordinates	$\left(\frac{n}{l}, -\frac{2am}{l}\right)$	$\left(-\frac{a^2 l}{n}, -\frac{b^2 m}{n}\right)$	$\left(-\frac{a^2 l}{n}, \frac{b^2 m}{n}\right)$

of pole (line lx + my + n = 0)			
Auxiliary circle	–	$x^2 + y^2 = a^2$	$x^2 + y^2 = a^2$
Director circle	–	$x^2 + y^2 = a^2$ $+ b^2$	$x^2 + y^2 = a^2$ $- y^2$
Eccentricity	e = 1	$e = \sqrt{\left(\frac{a^2 - b^2}{a^2}\right)}$	$e = \sqrt{\left(\frac{a^2 + b^2}{a^2}\right)}$

3. (i) Point P(x_1, y_1) lies outside, on or inside the parabola $y^2 = 4ax$ according as $y_1^2 - 4ax_1 >, =, < 0$.

(ii) The point of intersection of tangent at two points P($at_1^2, 2at_1$) and Q($at_2^2, 2at_2$) on the parabola $y^2 = 4ax$ is $(at_1t_2, a(t_1 + t_2))$.

(iii) If the normal at P($at_1^2, 2at_1$), then $t_2 = -t_1 - \frac{2}{t_1}$

(iv) Equation of chord joining P($at_1^2, 2at_1$), Q($at_2^2, 2at_2$)

$$y - 2at_1 = \frac{2}{t_1 + t_2}(x - at_1^2)$$

If it passes through the focus, that $t_2 = -\frac{1}{t_1}$

$$\Rightarrow \quad t_1 t_2 = -1$$

4. Point P(x_1, y_1) lies outside, on or inside the ellipse according as $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$ >, =, < 0.

5. (i) Point (x_1, y_1) lies outside, on or inside the hyperbola according as <, =, > 0.

(ii) The equation of two asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm \frac{b}{a} x$ or $\frac{x}{a} \pm \frac{y}{b} = 0$.

6. (i) For rectangular hyperbola $xy = c^2$, the coordinates of foci are $(\pm c\sqrt{2}, \pm c\sqrt{2})$ and directrices are $x + y = \pm c\sqrt{2}$.

(ii) Equation of tangent to the hyperbola $xy = c^2$ at (x_1, y_1) is $xy_1 + yx_1 = 2c^2$ and at $\left(ct, \frac{c}{t}\right)$ is $\frac{x}{t} + yt = 2c$.

(iii) Equation of normal to the hyperbola $xy = c^2$ at (x_1, y_1) is $xx_1 - yy_1 = x_1^2 - y_1^2$ and at $\left(ct, \frac{c}{t}\right)$ is $xt^3 - ty - ct^4 + c = 0$.

Topic 1 Basic Terms of Conics

Q 1. For different values of α, the locus of the point of intersection of the two straight lines $\sqrt{3}x - y - 4\sqrt{3}\alpha = 0$ and $\sqrt{3}\alpha x + \alpha y - 4\sqrt{3} = 0$ is

[WBJEE]

(a) a hyperbola with eccentricity 2

(b) an ellipse with eccentricity $\sqrt{\frac{2}{3}}$

(c) an hyperbola with eccentricity $\sqrt{\frac{19}{16}}$

(d) an ellipse with eccentricity $\frac{3}{4}$

Q 2. S and T are the foci of an ellipse and B is end point of the minor axis. If STB is an equilateral triangle, the eccentricity of the ellipse is

[WB JEE]

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

Q 3. If t_1 and t_2 be the parameters of the end points of a focal chord for the parabola $y^2 = 4ax$, then which one is true?

[VITEEE]

(a) $t_1 t_2 = 1$

(b) $\frac{t_1}{t_2} = 1$

(c) $t_1 t_2 = -1$

(d) $t_1 + t_2 = -1$

Q 4. The distance between the foci of the conic $7x^2 - 9y^2 = 63$ is equal to

[Kerala CEE]

(a) 8

(b) 4

(c) 3

(d) 7

(e) 12

Q 5. One of the points on the parabola $y^2 = 12x$ with focal distance 12, is

[Kerala CEE]

(a) $(3, 6)$

(b) $(9, 6\sqrt{3})$

(c) $(7, 2\sqrt{21})$

(d) $(8, 4\sqrt{6})$ (e) $(1, \sqrt{12})$

Q 6. If a point P (x, y) moves along the ellipse

$\frac{x^2}{25}+\frac{y^2}{16}=1$ and if C is the centre of the ellipse, then

the sum of maximum and minimum values of CP is

[Kerala CEE]

(a) 25

(b) 9

(c) 4

(d) 5

(e) 16

Q 7. If the length of the major axis of an ellipse is $\frac{17}{8}$ times the length of the minor axis, then the eccentricity of the ellipse is

[Kerala CEE]

(a) $\frac{8}{17}$

(b) $\frac{15}{17}$

(c) $\frac{9}{17}$

(d) $\frac{2\sqrt{2}}{17}$

(e) $\frac{13}{17}$

Q 8. If 1 denotes the semi-latusrectum of the parabola $y^2 = 4ax$ and SP and SQ denote the segments of any focal chord PQ, S being the focus, then SP, l and SQ are in the relation.

[DCE]

(a) AP

(b) GP

(c) HP

(d) $l^2 = SP^2 + SQ^2$

Q 9. If distance between directrices of a rectangular hyperbola is 10, then distance between its foci will be

[UP SEE]

(a) $10\sqrt{2}$

(b) 5

(c) $5\sqrt{2}$

(d) 20

Q 10. The sum of the distances of a point $(2, -3)$ from the foci of an ellipse $16\,(x-2)^2 + 25(y+3)^2 = 400$ is

[Kerala CEE]

(a) 8

(b) 6

(c) 50

(d) 32

(e) 10

Q 11. If e_1 is the eccentricity of the ellipse $\frac{x^2}{16}+\frac{y^2}{7}=1$ and e_2 is the eccentricity of the hyperbola $\frac{x^2}{9}-\frac{y^2}{7}=1$, then $e_1 + e_2$ is equal to

[Kerala CEE]

(a) $\frac{16}{7}$

(b) $\frac{25}{4}$

(c) $\frac{25}{12}$

(d) $\frac{16}{9}$

(e) $\frac{23}{16}$

Q 12. In an ellipse, if the lines joining focus to

the extremities of the minor axis form an equilateral triangle with the minor axis, then the eccentricity of the ellipse is

[AMU]

(a) $\frac{\sqrt{3}}{2}$

(b) $\frac{\sqrt{3}}{4}$

(c) $\frac{1}{\sqrt{2}}$

(d) $\sqrt{\frac{2}{3}}$

Q 13. The coordinates of the focus of the parabola described parametrically by $x = 5t^2 + 2$, $y = -10t + 4$ are

[WB JEE]

(a) $(7, 4)$

(b) $(3, 4)$

(c) $(3, -4)$

(d) $(-7, 4)$

Q 14. The focal distance of a point P on the parabola $y^2 = 12x$, if the ordinate of P is 6, is

[MHT CET]

(a) 12

(b) 6

(c) 3

(d) 9

Q 15. The equation $y^2 - 8y - x + 19 = 0$ represents

[RPET]

(a) a parabola whose focus is $\left(\frac{1}{4},0\right)$ and directrix is $x=\frac{-1}{4}$

(b) a parabola whose vertex is $(3,\ 4)$ and directrix is $x=\frac{11}{4}$

(c) a parabola whose focus is $\left(\frac{13}{4},4\right)$ and vertex is $(0,0)$

(d) a curve which is not a parabola

Q 16. If in a hyperbola, the distance between the foci is 10 and the transverse axis has length 8, then the length of its latusrectum is

[J&K CET]

(a) 9

(b) $\frac{9}{2}$

(c) $\frac{32}{3}$

(d) $\frac{64}{3}$

Q 17. Equation of the directrix of parabola $2x^2$ $=14y$ is equal to

[Guj. CET]

(a) $y = -\frac{7}{4}$

(b) $x = -\frac{7}{4}$

(c) $y = \frac{7}{4}$

(d) $x = \frac{7}{4}$

Q 18. A parabola has the origin as its focus and the line x = 2 as the directrix. Then, the vertex of the parabola is at

[AIEEE]

(a) $(2,0)$

(b) $(0,2)$

(c) $(1, 0)$

(d) $(0, 1)$

Q 19. A focus of an ellipse is at the origin. The directrix is the line x = 4 and the eccentricity is $\frac{1}{2}$, then length of semi major axis is

[AIEEE]

(a) 5/3

(b) 8/3

(c) 2/3

(d) 4/3

Q 20. Eccentricity of the ellipse $x^2 + 2y^2 - 2x + 3y + 2 = 0$ is

[UP SEE]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{1}{2}$

(c) $\frac{1}{2\sqrt{2}}$

(d) $\frac{1}{\sqrt{3}}$

Q 21. If e_1 and e_2 are the eccentricities of a hyperbola $3x^2 - 3y^2 = 25$ and its conjugate, then

[KCET]

(a) $e_1^2 + e_2^2 = 2$

(b) $e_1^2 + e_2^2 = 4$

(c) $e_1 + e_2 = 4$

(d) $e_1 + e_2 = \sqrt{2}$

Q 22. A conic section is defined by the

equations x = -1 + sec t, y = 2 + 3 tan t. The coordinates of the foci are

[Kerala CEE]

(a) $(-1 - \sqrt{10}, 2)$ and $(-1 + \sqrt{10}, 2)$

(b) $(-1 - \sqrt{8}, 2)$ and $(-1 + \sqrt{8}, 2)$

(c) $(-1, 2 - \sqrt{8})$ and $(-1, 2 + \sqrt{8})$

(d) $(-1, 2 - \sqrt{10})$ and $(-1, 2 + \sqrt{10})$

(e) $(\sqrt{10}, 0)$ and $(-\sqrt{10}, 0)$

Q 23. If the lines joining the foci of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, where a > b and an extremity of its minor axis are inclined at an angle $60°$, then the eccentricity of the ellipse is

[Kerala CEE]

(a) $-\frac{\sqrt{3}}{2}$

(b) $\frac{1}{2}$

(c) $\frac{\sqrt{5}}{2}$

(d) $\frac{\sqrt{7}}{3}$

(e) $\sqrt{3}$

Q 24. The distance between the foci of the hyperbola $x^2 - 3y^2 - 4x - 6y - 11 = 0$ is

[EAMCET]

(a) 4

(b) 6

(c) 8

(d) 10

Q 25. The sum of the focal distances from any point on the ellipse $9x^2 + 16y^2 = 144$ is

[OJEE]

(a) 3

(b) 6

(c) 8

(d) 4

Q 26. The equation of the ellipse whose foci are at $(\pm 2,0)$ and eccentricity is $\frac{1}{2}$, is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Then,

[OJEE]

(a) $a^2 = 16, b^2 = 12$

(b) $a^2 = 12, b^2 = 16$

(c) $a^2 = 16, b^2 = 4$

(d) $a^2 = 4, b^2 = 16$

Q 27. The vertex of the parabola $x^2 + 2y = 8x - 7$ is

[WB JEE]

(a) $\left(\frac{9}{2}, 0\right)$

(b) $\left(4, \frac{9}{2}\right)$

(c) $\left(2, \frac{9}{2}\right)$

(d) $\left(4, \frac{7}{2}\right)$

Q 28. The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having its centre $(0, 3)$ is

[MP PET]

(a) 4

(b) $\frac{3}{7}$

(c) $\sqrt{12}$

(d) $\frac{7}{2}$

Q 29. Equation of the latusrectum of the ellipse

$9x^2 + 4y^2 - 18x - 8y - 23 = 0$ are

[BCECE]

(a) $y = \pm\sqrt{5}$

(b) $x = \pm\sqrt{5}$

(c) $y = 1 \pm \sqrt{5}$

(d) $x = -1 \pm \sqrt{5}$

Q 30. The directrix of the parabola $y^2 + 4x + 3 = 0$ is

[VITEEE]

(a) $x-\frac{4}{3}=0$

(b) $x+\frac{1}{4}=0$

(c) $x-\frac{3}{4}=0$

(d) $x-\frac{1}{4}=0$

Q 31. If the distance between the foci and the distance between the directrices of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ are in the ratio $3:2$, then a : b is

[Jamia Millia Islamia]

(a) $\sqrt{2} : 1$

(b) $\sqrt{3} : \sqrt{2}$

(c) $1 : 2$

(d) $2 : 1$

Q 32. The focal distance of a point on the parabola $y^2 = 16x$ whose ordinate is twice the abscissa, is

[MHT CET]

(a) 6

(b) 8

(c) 10

(d) 12

Q 33. If e and e' be the eccentricities of a hyperbola and its conjugate, then $\frac{1}{e^2}+\frac{1}{(e')^2}$ is equal to

[MP PET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 34. The eccentricity of the ellipse $9x^2 + 5y^2 - 30y = 0$ is

[RPET]

(a) $1/3$

(b) $2/3$

(c) $3/4$

(d) $4/5$

Q 35. If OAB is an equilateral triangle inscribed in the parabola $y^2 = 4ax$ with O as the vertex, then the length of the side of the ΔOAB is

[RPET]

(a) $8a\sqrt{3}$

(b) $4a\sqrt{3}$

(c) $2a\sqrt{3}$

(d) $a\sqrt{3}$

Q 36. One of the directrices of the ellipse $8x^2 + 6y^2 - 16x + 12y + 13 = 0$ is

[J&KCET]

(a) $3y - 3 = \sqrt{6}$

(b) $3y + 3 = \sqrt{6}$

(c) $y + 1 = \sqrt{3}$

(d) $y - 1 = -\sqrt{3}$

Q 37. Statement I The curve $y = -\frac{x^2}{2} + x + 1$ is symmetric with respect to the line $x = 1$. Because

Statement II A parabola is symmetric about its axis.

[IIT JEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

StatementI

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 38. For the hyperbola $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$, which of the following remains constant when α varies ?

[AIEEE]

(a) Eccentricity

(b) Directrix

(c) Abscissae of vertices

(d) Abscissae of foci

Q 39. The foci of the conic section $25x^2 + 16y^2 - 150x = 175$ are

[DCE]

(a) $(0, \pm 3)$

(b) $(0, \pm 2)$

(c) $(3, \pm 3)$

(d) $(0, \pm 1)$

Q 40. The focus of the parabola $y^2 - x - 2y + 2 = 0$ is

[UP SEE]

(a) $\left(\frac{1}{4}, 0\right)$

(b) $(1, 2)$

(c) $\left(\frac{5}{4}, 1\right)$

(d) $\left(\frac{3}{4},\frac{5}{2}\right)$

Q 41. If P is any point on the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$

and Sand S' are the foci, then PS + PS' is equal to

[KCET]

(a) 4

(b) 8

(c) 10

(d) 12

Q 42. The eccentricity of the conic $\frac{(x+2)^2}{7} + (y - 1)^2 = 14$ is

[Kerala CEE]

(a) $\sqrt{\frac{7}{8}}$

(b) $\sqrt{\frac{6}{17}}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\sqrt{\frac{6}{11}}$

(e) $\sqrt{\frac{6}{7}}$

Q 43. If for the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, y−axis is the minor axis and the length of the latusrectum is one half of the length of its minor axis, then its eccentricity is

[Kerala CEE]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{1}{2}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\frac{3}{4}$

(e) $\frac{3}{5}$

Q 44. For the parabola $y^2 + 6y - 2x + 5 = 0$

(I) The vertex is $(-2,-3)$

(II) The directrix is $y + 3$

$= 0$ Which of the following is correct ?

[EAMCET]

(a) Both I and II are true

(b) I is true, II is false

(c) I is false, II is true

(d) Both I and II are false

Q 45. Coordinates of foci of hyperbola are $(-5,3)$ and $(7,3)$ and eccentricity is $3/2$. Then, length of its latusrectum is

[OJEE]

(a) 20

(b) 10

(c) 40

(d) None of these

Q 46. Coordinates of the foci of the ellipse $5x^2 + 9y^2 + 10x - 36y - 4 = 0$, are

[OJEE]

(a) $(1, 2)$ and $(-3, 2)$

(b) $(2, 1)$ and $(-3, 2)$

(c) $(1, 2)$ and $(3, 2)$

(d) None of these

Q 47. The equation of a directrix of the ellipse

$\frac{x^2}{16}+\frac{y^2}{25}=1$ is

[VITEEE]

(a) $3y = \pm 5$

(b) $y = \pm 5$

(c) $3y = \pm 25$

(d) $y = \pm 3$

Q 48. The equation of the latusrectum of the parabola $x^2 + 4x + 2y = 0$, is equal to

[MHT CET]

(a) $2y + 3 = 0$

(b) $3y = 2$

(c) $2y = 3$

(d) $3y + 2 = 0$

Q 49. In an ellipse, the distances between its foci is 6 and minor axis is 8. Then, its eccentricity is

[AIEEE]

(a) $\frac{1}{2}$

(b) $\frac{4}{5}$

(c) $\frac{1}{\sqrt{5}}$

(d) $\frac{3}{5}$

Q 50. The ends of the latusrectum of the conic

$x^2 + 10x - 16y + 25 = 0$ are

[BITSAT, Guj. CET]

(a) $(3,-4), (13, 4)$

(b) $(-3,-4), (13,-4)$

(c) $(3, 4), (-13, 4)$

(d) $(5, -8), (-5, 8)$

Q 51. The ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and the hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ have in common

[KCET]

(a) centre only

(b) centre, foci and directrices

(c) centre, foci and vertices

(d) centre and vertices only

Q 52. The locus of the point which moves such

that the ratio of its distance from two fixed point in the plane is always a constant k (< 1) is

[KCET]

(a) hyperbola

(b) ellipse

(c) straight line

(d) circle

Q 53. The eccentricity of the hyperbola in the standard form $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, passing through $(3, 0)$ and $(3\sqrt{2}, 2)$ is

[Kerala CEE]

(a) $\frac{13}{3}$

(b) $\sqrt{13}$

(c) $\sqrt{3}$

(d) $\frac{\sqrt{13}}{3}$

(e) $\frac{5}{3}$

Q 54. If the eccentricities of the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ and the hyperbola $\frac{x^2}{64}-\frac{y^2}{b^2}=1$ are reciprocals of each other, then b^2 is equal to

[Kerala CEE]

(a) 192

(b) 64

(c) 16

(d) 32

(e) 128

Q 55. If the vertex of the parabola $y = x^2 - 16x +$ k lies on x−axis, then the value of k is

[Kerala CEE]

(a) 16

(b) 8

(c) 64

(d) -64

(e) -8

Q 56. For the ellipse $24x^2 + 9y^2 - 120x - 90y + 225 = 0$, the eccentricity is equal to

[AMU]

(a) $\frac{2}{5}$

(b) $\frac{3}{5}$

(c) $\sqrt{\frac{15}{24}}$

(d) $\frac{1}{5}$

Q 57. If the eccentricity of a hyperbola is $\sqrt{3}$, then the eccentricity of its conjugate hyperbola is

[EAMCET]

(a) $\sqrt{2}$

(b) $\sqrt{3}$

(c) $\sqrt{\frac{3}{2}}$

(d) $2\sqrt{3}$

Q 58. If b and c are the lengths of the segments of any focal chord of a parabola $y^2 = 4ax$, then the length of the semilatusrectum is

[EAMCET]

(a) $\frac{bc}{b+c}$

(b) $\sqrt{bc}$

(c) $\frac{b+c}{2}$

(d) $\frac{2bc}{b+c}$

Q 59. The equation of the directrix of parabola

$y^2 + 4y + 4x + 2 = 0$ is

[OJEE]

(a) $x = -1$

(b) $x = 1$

(c) $x = -\frac{3}{2}$

(d) $x = \frac{3}{2}$

Q 60. The eccentricity of the conic $4x^2 + 16y^2 - 24x - 32y = 1$ is

[WBJEE]

(a) $\frac{1}{2}$

(b) $\sqrt{3}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\frac{\sqrt{3}}{4}$

Q 61. The latusrectum of the parabola $y^2 = 4ax$,

whose focal chord is PSQ, such that SP = 3 and SQ = 2 is given by

[BCECE]

(a) $\frac{24}{5}$

(b) $\frac{12}{5}$

(c) $\frac{6}{5}$

(d) $\frac{1}{5}$

Q 62. The eccentricity of the hyperbola $9x^2 - 16y^2 - 18x - 64y - 199 = 0$ is

[Jamia Millia Islamia, MP PET]

(a) $\frac{16}{9}$

(b) $\frac{5}{4}$

(c) $\frac{25}{16}$

(d) zero

Q 63. Suppose S and S 'are foci of the ellips

$\frac{x^2}{25}+\frac{y^2}{16}=1$. If P is a variable point on the ellipse and if Δ is area of the triangles PSS', then the maximum value Δ is

[MHTCET]

(a) 8

(b) 12

(c) 16

(d) 20

Q 64. The equation of the directrix of the parabola $x^2 + 8y - 2x = 7$ is

[MPPET]

(a) $y = 3$

(b) $y = -3$

(c) $y = 2$

(d) $y = 0$

Q 65. The eccentricity of the hyperbola conjugate to $x^2 - 3y^2 = 2x + 8$ is

[MPPET]

(a) $\frac{2}{\sqrt{3}}$

(b) $\sqrt{3}$

(c) 2

(d) None of these

Q 66. If $(-1, -2\sqrt{2})$ is one of extremity of a focal chord of the parabola $y^2 = -8x$, then the other extremity is

[J&K CET]

(a) $(-1,-\sqrt{2})$

(b) $(2\sqrt{2},-1)$

(c) $(-4,4\sqrt{2})$

(d) $(4,4,\sqrt{2})$

Q 67. The sum of the focal distances of any point on the conic $\frac{x^2}{25}+\frac{y^2}{16}=1$ is

[Guj. CET]

(a) 10

(b) 9

(c) 41

(d) 18

Q 68. An ellipse has OB as semi minor axis, F and F' its foci and the angle FBF' is a right angle. Then, the eccentricity of the ellipse is

[AIEEE]

(a) $\frac{1}{\sqrt{3}}$

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) $\frac{1}{\sqrt{2}}$

Q 69. The eccentricity of the hyperbola $\frac{x^2}{16}-\frac{y^2}{25}=1$

is

[KCET]

(a) $\frac{3}{4}$

(b) $\frac{3}{5}$

(c) $\frac{\sqrt{41}}{4}$

(d) $\frac{\sqrt{41}}{5}$

Q 70. The eccentricity of the ellipse $25x^2 + 16y^2 - 150x - 175 = 0$ is

[Kerala CEE]

(a) $\frac{2}{5}$

(b) $\frac{2}{3}$

(c) $\frac{4}{5}$

(d) $\frac{3}{4}$

(e) $\frac{3}{5}$

Q 71. Distance between foci is 8 and distance between directrices is 6 of hyperbola, then length of latusrectum is

[MHT CET]

(a) $4\sqrt{3}$

(b) $\frac{4}{\sqrt{3}}$

(c) $\sqrt{\frac{3}{4}}$

(d) None of these

Q 72. In the standard form of an ellipse sum of the focal distances of a point is

[MHT CET]

(a) 1

(b) $-2a$

(c) $2a$

(d) None of these

Q 73. The distance between the directrices of the hyperbola $x = 8 \sec \theta$, $y = 8 \tan \theta$ is

[MP PET]

(a) $8\sqrt{2}$

(b) $16\sqrt{2}$

(c) $4\sqrt{2}$

(d) $6\sqrt{2}$

Q 74. The locus of a point which moves such that the difference of its distances from two fixed points is ways a constant, is

[RPET]

(a) a circle

(b) a straight line

(c) a hyperbola

(d) an ellipse

Q 75. If e and e_1 are the eccentricities of hyperbolas $xy = c^2$ and $x^2 - y^2 = c^2$, then $e^2 + e_1^2$ is

[DCE]

(a) 1

(b) 4

(c) 6

(d) 8

Q 76. The eccentricity of the hyperbola $5x^2 - 4y^2 + 20x + 8y = 4$ is

[UP SEE]

(a) $\sqrt{2}$

(b) $\frac{3}{2}$

(c) 2

(d) 3

Q 77. For the ellipse $25x^2 + 9y^2 - 150x - 90y + 225 = 0$, the eccentricity e is equal to

[KCET]

(a) $\frac{2}{5}$

(b) $\frac{3}{5}$

(c) $\frac{4}{5}$

(d) $\frac{1}{5}$

Q 78. The centre of the ellipse $9x^2 + 25y^2 - 18x - 100y - 166 = 0$ is

[Kerala CEE]

(a) $(1,1)$

(b) $(-1, 2)$

(c) $(-1, 1)$

(d) $(2, 2)$

(e) $(1, 2)$

Q 79. The latusrectum of the ellipse $9x^2 + 16y^2 = 144$ is

[Kerala CEE]

(a) 4

(b) $\frac{11}{4}$

(d) $\frac{9}{2}$

(e) $\frac{10}{3}$

Q 80. Any point on the hyperbola $\frac{(x+1)^2}{16} - \frac{(y-2)^2}{4} = 1$ is of the form

[Kerala CEE]

(a) $(4 \sec \theta, 2 \tan \theta)$

(b) $(4 \sec \theta + 1, 2 \tan \theta - 2)$

(c) $(4 \sec \theta - 1, 2 \tan \theta - 2)$

(d) $(\sec \theta - 4, \tan \theta = 2)$ (e) $(4 \sec \theta - 1, 2 \tan \theta + 2)$

Q 81. The eccentricity of the hyperbola $x^2 - y^2 =$ is

[Kerala CEE]

(a) $\sqrt{3}$

(b) 2

(c) $2\sqrt{2}$

(d) $\sqrt{2}$

(e) 1.5

Q 82. The equation of the directrix of $(x - 1)^2 = 2(y - 2)$ is

[Kerala CEE]

(a) $2y + 3 = 0$

(b) $2x + 1 = 0$

(c) $2x - 1 = 0$

(d) $2y - 1 = 0$

(e) $2y - 3 = 0$

Q 83. The eccentricity of the conic $36x^2 + 144y^2 - 36x - 96y - 119 = 0$ is

[EAMCET]

(a) $\frac{\sqrt{3}}{2}$

(b) $\frac{1}{2}$

(c) $\frac{\sqrt{3}}{4}$

(d) $\frac{1}{\sqrt{3}}$

Q 84. If $x^2 + 6x + 20y - 51 = 0$, then axis of parabola is

[OJEE]

(a) $x + 3 = 0$

(b) $x - 3 = 0$

(c) $x = 1$

(d) $x + 1 = 0$

Q 85. The difference in focal distances of any point on the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ is

[OJEE]

(a) 8

(b) 9

(c) 0

(d) 6

Q 86. P is any point on the ellipse $81\ x^2 + 144y^2 = 1944$, whose foci are S and S'. Then, SP + S'P equals

[BCECE]

(a) 3

(b) $4\sqrt{6}$

(c) 36

(d) 324

Q 87. The point $P(9/2, 6)$ lies on the parabola $y^2 = 4ax$, then parameter of the point P is

[MHTCET]

(a) $\frac{3a}{2}$

(b) $\frac{2}{3a}$

(c) $\frac{2}{3}$

(d) $\frac{3}{2}$

Q 88. The focus of the parabola $x^2 + 2y + 6x = 0$ is

[RPET]

(a) $(-3,4)$

(b) $(3, 4)$

(c) $(3,-4)$

(d) $(-3,-4)$

Q 89. The point on the curve $3x^2 - 4y^2 = 72$,

which is nearest to the line $3x + 2y - 1 = 0$, is

[J&K CET]

(a) $(6, 3)$

(b) $(6, -3)$

(c) $(6, 6)$

(d) $(6, 5)$

Q 90. The latusractum of the hyperbola $9x^2 - 16y^2 + 72x - 32y - 16 = 0$ is

[J&K CET]

(a) $\frac{9}{2}$

(b) $-\frac{9}{2}$

(c) $\frac{32}{3}$

(d) $-\frac{32}{3}$

Q 91. The foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide. Then, the value of b^2

is

[AIEEE]

(a) 1

(b) 5

(c) 7

(d) 9

Q 92. Length of major axis of ellipse $9x^2 + 7y^2 =$ 63 is **[DCE]**

(a) 3

(b) 9

(c) 6

(d) $2\sqrt{7}$

Q 93. The directrix of the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$ is

[UP SEE]

(a) $y=\frac{6}{\sqrt{13}}$

(b) $x=\frac{6}{\sqrt{13}}$

(c) $y = \frac{9}{\sqrt{13}}$

(d) $x = \frac{9}{\sqrt{13}}$

Q 94. The directrix of the parabola $x^2 - 4x - 8y + 12 = 0$ is

[KCET]

(a) $y = 0$

(b) $x = 1$

(c) $y = -1$

(d) $x = -1$

Q 95. If a focal chord of the parabola $y^2 = ax$ is $2x - y - 8 = 0$, then the equation of the directrix is

[Kerala CEE]

(a) $x + 4 = 0$

(b) $x - 4 = 0$

(c) $y - 4 = 0$

(d) $y + 4 = 0$

(e) None of these

Q 96. Vertex of the parabola $9x^2 - 6x + 36y + 9 = 0$ is

[AMU]

(a) $(1/3, -2/9)$

(b) $(-1/3, -1/2)$

(c) $(-1/3, 1/2)$

(d) $(1/3, 1/2)$

Q 97. If the foci and vertices of an ellipse be $(\pm 1,0)$ and $(\pm 2,0)$ then the minor axis of the ellipse is

[AMU]

(a) $2\sqrt{5}$

(b) 2

(c) 4

(d) $2\sqrt{3}$

Q 98. The eccentricity of the ellipse $9x^2 + 5y^2 - 18x - 207 - 16 = 0$ is

[EAMCET]

(a) $\frac{1}{2}$

(b) $\frac{2}{3}$

(c) $\frac{3}{2}$

(d) 2

Q 99. The point of the parabola $y^2 = 18x$, for which the ordinate is three times the abscissa is

[MP PET]

(a) $(6,2)$

(b) $(-2,-6)$

(c) $(3,18)$

(d) $(2,\ 6)$

Q 100. If transverse and conjugate axes of hyperbola are equal, then its eccentricity is

[RPET]

(a) $\sqrt{3}$

(b) $\sqrt{2}$

(c) $\frac{1}{\sqrt{2}}$

(d) 2

Topic 2 Equation of Conics and Intersection of Two Curves

Q 1. Passage The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B.

[IIT JEE]

I. Equation of a common tangent with positive slope to the circle as well as to the hyperbola is

(a) $2x - \sqrt{5}y - 20 = 0$

(b) $2x - \sqrt{5}y + 4 = 0$

(c) $3x - 4y + 8 = 0$

(d) $4x - 3y + 4 = 0$

II. Equation of the circle with AB as its diameter is

(a) $x^2 + y^2 - 12x + 24 = 0$

(b) $x^2 + y^2 + 12x + 24 = 0$

(c) $x^2 + y^2 + 24x - 12 = 0$

(d) $x^2 + y^2 - 24x - 12 = 0$

Q 2. The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the coordinate axes, which is turn in inscribed in another ellipse that passes through the point $(4, 0)$. Then, the equation of the ellipse is

[AIEEE]

(a) $x^2 + 12y^2 = 16$

(b) $4x^2 + 48y^2 = 48$

(c) $4x^2 + 64y^2 = 48$

(d) $x^2 + 16y^2 = 16$

Q 3. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points (x_i, y_i), for $i = 1, 2, 3$ and 4, then $y_1 + y_2 + y_3 + y_4$ equals

[EAMCET]

(a) 0

(b) c

(c) a

(d) c^4

Q 4. Focus of hyperbola is $(\pm 3, 0)$ and equation of tangent is $2x + y - 4 = 0$, find the equation of hyperbola

[MHT CET]

(a) $4x^2 - 5y^2 = 20$

(b) $4x^2 - 4y^2 = 20$

(c) $4x^2 - 5y^2 = 1$

(d) $5x^2 - 4y^2 = 1$

Q 5. Consider the two curves $C_1 : y^2 = 4x$

$C_2 : x^2 + y^2 - 6x + 1 = 0$, then

[IIT JEE]

(a) C_1 and C_2 touch each other only at one point

(b) C_1 and C_2 touch each other exactly at two points

(c) C_1 and C_2 intersect (but do not touch) at exactly two point

(d) C_1 and C_2 intersect nor touch each other

Q 6. The equation of ellipse whose distance between the foci is equal to 8 and distance between the directrix is 18, is

[DCE]

(a) $5x^2 - 9y^2 = 180$

(b) $9x^2 + 5y^2 = 180$

(c) $9x^2 - 5y^2 = 180$

(d) $5x^2 + 9y^2 = 180$

Q 7. The locus of the equation $x^2 - y^2 = 0$, is

[UP SEE]

(a) a circle

(b) a hyperbola

(c) a pair of lines

(d) a pair of lines at right angles

Q 8. The equation $\frac{x^2}{2-\lambda}-\frac{y^2}{\lambda-5}-1=0$, represent an ellipse, if

[KCET]

(a) $\lambda > 5$

(b) $\lambda < 2$

(c) $2 < \lambda < 5$

(d) $2 > \lambda > 5$

Q 9. The locus of the mid point of the line joining the focus and any point on the parabola $y^2 = 4ax$ is a parabola with the equation of directrix as

[KCET]

(a) $x + a = 0$

(b) $2x + a = 0$

(c) $x = 0$

(d) $x = \frac{a}{2}$

Q 10. For an ellipse with eccentricity $1/2$ the centre is at the origin. If one directrix is x = 4, then the equation of the ellipse is

[AMU]

(a) $3x^2 + 4y^2 = 1$

(b) $3x^2 + 4y^2 = 12$

(c) $4x^2 + 3y^2 = 1$

(d) $4x^2 + 3y^2 = 12$

Q 11. Length of the straight line $x - 3y = 1$ intercepted by the hyperbola $x^2 - 4y^2 = 1$ is

[OJEE]

(a) $\frac{3}{5}\sqrt{10}$

(b) $\frac{6}{5}\sqrt{10}$

(c) $\frac{5}{3}\sqrt{10}$

(d) $\frac{5}{6}\sqrt{10}$

Q 12. The two parabolas $x^2 = 4y$ and $y^2 = 4x$ meet in two distinct points. One of these is the origin and the other is

[WB JEE]

(a) $(2, 2)$

(b) $(4, -4)$

(c) $(4, 4)$

(d) $(-2, 2)$

Q 13. The equation of the ellipse having vertices at $(\pm 5,0)$ and foci $(\pm 4,0)$ is

[WBJEE]

(a) $\frac{x^2}{25}+\frac{y^2}{16}=1$

(b) $9x^2+25y^2=225$

(c) $\frac{x^2}{9}+\frac{y^2}{25}=1$

(d) $4x^2+5y^2=20$

Q 14. If t is a parameter, then $x = a\left(t + \frac{1}{t}\right), y = b\left(t - \frac{1}{t}\right)$ represents

[WB JEE]

(a) an ellipse

(b) a circle

(c) a pair of straight lines

(d) a hyperbola

Q 15. Let O be the origin and A be a point on the curve $y^2 = 4x$. Then, the locus of the mid point of OA, is

[BCECE]

(a) $x^2 = 4y$

(b) $x^2 = 2y$

(c) $x^2 = 16y$

(d) $y^2 = 2x$

Q 16. $x = 4(1 + \cos\theta)$ and $y = 3(1 + \sin\theta)$

are the parametric equations of

[Jamia Millia Islamia]

(a) $\frac{(x-3)^2}{9} + \frac{(y-4)^2}{16} = 1$

(b) $\frac{(x+4)^2}{16} + \frac{(y+3)^2}{9} = 1$

(c) $\frac{(x-4)^2}{16} - \frac{(y-3)^2}{9} = 1$

(d) $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$

Q 17. A parabola is drawn with its focus at $(3,4)$ and vertex at the focus of the parabola $y^2 - 12x - 4y + 4 = 0$. The equation of the parabola is

[Manipal]

(a) $y^2 - 8x - 6y + 25 = 0$

(b) $y^2 - 6x + 8y - 25 = 0$

(c) $x^2 - 6x - 8y + 25 = 0$

(d) $x^2 + 6x - 8y - 25 = 0$

Q 18. Locus of the point which divides double ordinate of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ in the ratio $1:2$ internally, is

[Manipal]

(a) $\frac{x^2}{a^2}-\frac{9y^2}{b^2}=\frac{1}{9}$

(b) $\frac{x^2}{a^2}+\frac{9y^2}{b^2}=1$

(c) $\frac{9x^2}{a^2}+\frac{9y^2}{b^2}=1$

(d) None of these

Q 19. The equation of the parabola whose focus is $(3, -4)$ and directrix $6x - 7y + 5 = 0$, is

[MP PET]

(a) $(7x + 6y)^2 - 570x + 750y + 2100 = 0$

(b) $(7x + 6y)^2 + 570x - 750y + 2100 = 0$

(c) $(7x - 6y)^2 - 570x + 750y + 2100 = 0$

(d) $(7x - 6y)^2 + 570x - 750y + 2100 = 0$

Q 20. A point P moves so that sum of its distances from $(-ae, 0)$ and $(ae, 0)$ is $2a$. Then, the locus of P is

[J&K CET]

(a) $\frac{x^2}{a^2}-\frac{x^2}{a^2(1-e^2)}=1$

(b) $\frac{x^2}{a^2}+\frac{y^2}{a^2(1-e^2)}=1$

(c) $\frac{x^2}{a^2}+\frac{y^2}{a^2(1+e^2)}=1$

(d) $\frac{x^2}{a^2}-\frac{y^2}{a^2(1+e^2)}=1$

Q 21. A hyperbola, having the transverse axis of length $2 \sin \theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then, its equation is

[IIT JEE]

(a) $x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$

(b) $x^2 \sec^2 \theta - y^2 \operatorname{cosec}^2 \theta = 1$

(c) $x^2 \sin^2 \theta - y^2 \cos^2 \theta = 1$

(d) $x^2 \cos^2 \theta - y^2 \sin^2 \theta = 1$

Q 22. Equation of the parabola with its vertex at $(1, 1)$ and focus $(3, 1)$ is

[BITSAT]

(a) $(x-1)^2 = 8(y-1)$

(b) $(y-1)^2 = 8(x-3)$

(c) $(y-1)^2 = 8(x-1)$

(d) $(x-3)^2 = 8(y-1)$

Q 23. If $\frac{x^2}{36} - \frac{y^2}{k^2} = 1$, is a hyperbola, then which of the following statements can by true ?

[KCET]

(a) $(-3,1)$ lies on the hyperbola

(b) $(3,1)$ lies on the hyperbola

(c) $(10, 4)$ lies on the hyperbola

(d) $(5, 2)$ lies on the hyperbola

Q 24. The parametric representation of a point of the ellipse whose foci are $(3, 0)$ and $(-1, 0)$ and eccentricity $2/3$ is

[Kerala CEE]

(a) $(1 + 3 \cos \theta, \sqrt{3} \sin \theta)$

(b) $(1 + 3 \cos \theta, 5 \sin \theta)$

(c) $(1 + 3 \cos \theta, 1 + \sqrt{5} \sin \theta)$

(d) $(1 + 3 \cos \theta, 1 + 5 \sin \theta)$

(e) $(1 + 3 \cos \theta, \sqrt{5} \sin \theta)$

Q 25. The line $x = at^2$ meets the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

in the real points, iff

[AMU]

(a) $|t| < 2$

(b) $|t| \leq 1$

(c) $|t| > 1$

(d) None of these

Q 26. The curve represented by $x = 3(\cos t + \sin t)$, $y = 4(\cos t - \sin t)$ is

[WB JEE]

(a) ellipse

(b) parabola

(c) hyperbola

(d) circle

Q 27. The curve described parametrically by x = $t^2 + 2t - 1$, y = $3t + 5$ represents

[VITEEE]

(a) an ellipse

(b) a hyperbola

(c) a parabola

(d) a circle

Q 28. The parametric representation of a point on the ellipse whose foci are $(-1,0)$ and $(7, 0)$ and eccentricity $1/2$ is

[MHT CET]

(a) $(3 + 8 \cos \theta, 4 \sqrt{3} \sin \theta)$

(b) $(8 \cos \theta, 4 \sqrt{3} \sin \theta)$

(c) $(3 + 4 \sqrt{3} \cos \theta, 8 \sin \theta)$

(d) None of these

Q 29. The equation of an ellipse whose eccentricity is $\frac{1}{2}$ and the vertices are $(4,0)$ and $(10,0)$ is

[MP PET]

(a) $3x^2 + 4y^2 - 42x + 120 = 0$

(b) $3x^2 + 4y^2 + 42x + 120 = 0$

(c) $3x^2 + 4y^2 + 42x - 120 = 0$

(d) $3x^2 + 4y^2 - 42x - 120 = 0$

Q 30. Let P be the point $(1, 0)$ and Q a point on the locus of $y^2 = 8x$. The locus of mid point of PQ is

[RPET]

(a) $x^2 - 4y + 2 = 0$

(b) $x^2 + 4y + 2 = 0$

(c) $y^2 + 4x + 2 = 0$

(d) $y^2 - 4x + 2 = 0$

Q 31. If e_1 is the eccentricity of the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ and e_2 is the eccentricity of the hyperbola passing through the foci of the ellipse and $e_1\ e_2$ = 1, then equation of the hyperbola is

[IIT JEE]

(a) $\frac{x^2}{9}-\frac{y^2}{16}=1$

(b) $\frac{x^2}{16}-\frac{y^2}{9}=-1$

(c) $\frac{x^2}{9}-\frac{y^2}{25}=1$

(d) None of these

Q 32. Axis of a parabola is y − x and vertex and focus are at a distance $\sqrt{2}$ and $2\sqrt{2}$ respectively from the origin. Then, equation of the parabola is

[IIT JEE]

(a) $(x - y)^2 = 8(x + y - 2)$

(b) $(x + y)^2 = 2(x + y - 2)$

(c) $(x - y)^2 = 4(x + y - 2)$

(d) $(x + y)^2 = 2(x - y + 2)$

Q 33. The locus of the vertices of the family of parabolas $y = \frac{a^3x^2}{3} + \frac{a^2x}{2} - 2a$ is

[AIEEE]

(a) $xy = \frac{3}{4}$

(b) $xy = \frac{35}{16}$

(c) $xy = \frac{64}{105}$

(d) $xy = \frac{105}{64}$

Q 34. The equation to the hyperbola having its eccentricity 2 and the distance between its foci is 8, is

[BITSAT]

(a) $\frac{x^2}{12}-\frac{y^2}{4}=1$

(b) $\frac{x^2}{4}-\frac{y^2}{12}=1$

(c) $\frac{x^2}{8}-\frac{y^2}{2}=1$

(d) $\frac{x^2}{16}-\frac{y^2}{9}=1$

Q 35. The equation of the parabola with vertex at $(-1,1)$ and focus $(2,1)$ is

[KCET]

(a) $y^2 - 2y - 12x - 11 = 0$

(b) $x^2 + 2x - 12y + 13 = 0$

(c) $y^2 - 2y + 12x + 11 = 0$

(d) $y^2 - 2y - 12x + 13 = 0$

Q 36. The equation of the hyperbola whose vertices are at $(5, 0)$ and $(-5, 0)$ and one of the directrices is $x = \frac{25}{7}$, is

[Kerala CEE]

(a) $\frac{x^2}{25} - \frac{y^2}{24} = 1$

(b) $\frac{x^2}{24} - \frac{y^2}{25} = 1$

(c) $\frac{x^2}{16} - \frac{y^2}{25} = 1$

(d) $\frac{x^2}{25} - \frac{y^2}{16} = 1$

(e) $\frac{x^2}{25} - \frac{y^2}{24} = -1$

Q 37. If $(0, 6)$ and $(0, 3)$ are respectively the vertex and focus of a parabola, then its equation is

[AMU]

(a) $x^2 + 12y = 72$

(b) $x^2 - 12y = 72$

(c) $y^2 - 12x = 72$

(d) $y^2 + 12x = 72$

Q 38. The equation of the ellipse whose foci are $(\pm 2, 0)$ and eccentricity $\frac{1}{2}$ is

[OJEE]

(a) $\frac{x^2}{12}+\frac{y^2}{16}=1$

(b) $\frac{x^2}{16}+\frac{y^2}{12}=1$

(c) $\frac{x^2}{16}+\frac{y^2}{8}=1$

(d) None of these

Q 39. Equation of the ellipse whose foci are $(2,$ $2)$ and $(4,\ 2)$ and the major axis is of length 10 is

[WB JEE]

(a) $\frac{(x+3)^2}{24}+\frac{(y+2)^2}{25}=1$

(b) $\frac{(x-3)^2}{24}+\frac{(y-2)^2}{25}=1$

(c) $\frac{(x+3)^2}{25}+\frac{(y+2)^2}{24}=1$

(d) $\frac{(x-3)^2}{25}+\frac{(y-2)^2}{24}=1$

Q 40. The equation of parabola with focus $(0, 0)$ and directrix $x + y = 4$, is [Jamia Millia Islamia]

(a) $x^2 + y^2 - 2xy + 8x + 8y - 16 = 0$

(b) $x^2 + y^2 - 2xy + 8x + 8y = 0$

(c) $x^2 + y^2 + 8x + 8y - 16 = 0$

(d) $x^2 - y^2 + 8x + 8y - 16 = 0$

Q 41. The equation of the hyperbola in the standard form (with transverse axis along the .x-

axis) having the length of the latusrectum = 9 unit and eccentricity = $\frac{5}{4}$, is

[MHT CET]

(a) $\frac{x^2}{16}-\frac{y^2}{18}=1$

(b) $\frac{x^2}{36}-\frac{y^2}{27}=1$

(c) $\frac{x^2}{64}-\frac{y^2}{36}=1$

(d) $\frac{x^2}{36}-\frac{y^2}{64}=1$

Q 42. The curve with parametric equation $x = e^t + e^{-t}$ $y = e^t - e^{-t}$ and is

[J&K CET]

(a) a circle

(b) an ellipse

(c) a hyperbola

(d) a parabola

Q 43. The equation of a parabola which passes through the intersection of a straight line $x + y = 0$ and the circle $x^2 + y^2 + 4y = 0$ is

[BITSAT, O JEE]

(a) $y^2 = 4x$

(b) $y^2 = x$

(c) $y^2 = 2x$

(d) None of these

Q 44. The point $(4,-3)$ with respect to the ellipse $4x^2 + 5y^2 = 1$ is

[BITSAT, O JEE]

(a) lies on the curve

(b) is inside the curve

(c) is outside the curve

(d) is focus of the curve

Q 45. The foci of an ellipse are $(0, \pm 4)$ and the equations for the directrices are $y = \pm 9$. The equation for the ellipse is

[UP SEE]

(a) $5x^2 + 9y^2 = 4$

(b) $2x^2 - 6y^2 = 28$

(c) $6x^2 + 3y^2 = 45$

(d) $9x^2 + 5y^2 = 180$

Q 46. The parabola with directrix $x+2y-1=0$ and focus $(1,0)$ is

[EAMCET]

(a) $4x^2 - 4xy + y^2 - 8x + 4y + 4 = 0$

(b) $4x^2 + 4xy + y^2 - 8x + 4y + 4 = 0$

(c) $4x^2 + 5xy + y^2 + 8x - 4y + 4 = 0$

(d) $4x^2 - 4xy + y^2 - 8x - 4y + 4 = 0$

Q 47. If a point P moves such that its distances from the point A $(1, 1)$ and the line $x + y + 2 = 0$ are equal, then the locus of P is

[EAMCET]

(a) a straight line

(b) a pair of straight lines

(c) a parabola

(d) an ellipse

Q 48. The length intercepted by the curve $y^2 = 4x$ on the line satisfying $dy/dx = 1$ and passing through point $(0, 1)$, is given by

[OJEE]

(a) 1

(b) 2

(c) 0

(d) None of these

Q 49. The equation of the parabola with vertex at the origin and directrix $y = 2$ is

[MPPET]

(a) $y^2 = 8x$

(b) $y^2 = -8x$

(c) $y^2 = \sqrt{8}x$

(d) $x^2 = -8y$

Q 50. If the foci of an ellipse are $(\pm\sqrt{5}, 0)$ and its eccentricity is $\sqrt{5}/3$, then the equation of the ellipse is

[J&K CET]

(a) $9x^2 + 4y^2 = 36$

(b) $4x^2 + 9y^2 = 36$

(c) $36x^2 + 9y^2 = 4$

(d) $9x^2 + 36y^2 = 4$

Q 51. If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$, passes through the points of intersection of the parabolas $y^2 = 4ax$ $x^2 = 4ay$, then

[AIEEE]

(a) $d^2 + (26 + 3c)^2 = 0$

(b) $d^2 + (36 + 2c)^2 = 0$

(c) $d^2 + (2b - 3c)^2 = 0$

(d) $d^2 + (3b - 2c)^2 = 0$

Q 52. The length of major and minor axis of an ellipse are 10 and 8 respectively and its major axis along the y−axis the equation of the ellipse referred to its centre as origin is

[RPET]

(a) $\frac{x^2}{25}+\frac{y^2}{16}=1$

(b) $\frac{x^2}{16}+\frac{y^2}{25}=1$

(c) $\frac{x^2}{100}+\frac{y^2}{64}=1$

(d) $\frac{x^2}{64}+\frac{y^2}{100}=1$

Q 53. The equation $|\sqrt{x^2+(y-1)^2}-\sqrt{x^2+(y+1)^2}|=k$ will represent a hyperbola for

[J&K CET]

(a) $k \in (0, 2)$

(b) $k \in (0, 1)$

(c) $k \in (1, \infty)$

(d) $k \in R^+$

Q 54. The curve represented by the equation

$4x^2 + 16y^2 - 24x - 32y - 12 = 0$ is

[Kerala CEE]

(a) a parabola

(b) a pair of straight lines

(c) an ellipse with eccentricity ½

(d) an ellipse with eccentricity $\sqrt{3}/2$

(e) a hyperbola with eccentricity $3/2$

Q 55. The equation $13[(x-1)^2 + (y-2)^2] = 3(2x + 3y - 2)^2$ represents

[AMU]

(a) parabola

(b) ellipse

(c) hyperbola

(d) None of these

Q 56. The one which does not represent a hyperbola is

[AMU]

(a) $xy = 1$

(b) $x^2 - y^2 = 5$

(c) $(X - 1)(Y - 3) = 0$

(d) $x^2 - Y^2 = 0$

Topic 3 Tangent and Normal of Conic

Q 1. Let A and B be two distinct points on the parabola $y^2 = 4x$. If the axis of the parabola touches a circle of radius r having AB as its diameter, then the slope of the line joining A and B can be

[IIT JEE]

(a) $-\frac{1}{r}$

(b) $\frac{1}{r}$

(c) $\frac{2}{r}$

(d) $-\frac{2}{r}$

Q 2. Passage Tangents are drawn from the point P $(3, 4)$ to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ touching the ellipse at points A and B.

[IIT JEE]

I. The coordinates of A and B are

(a) $(3, 0)$ and $(0,2)$

(b) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right) \text{and} \left(-\frac{9}{5}, \frac{8}{5}\right)$

(c) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right) \text{and} (0, 2)$

(d) $(3, 0) \text{and} \left(-\frac{9}{5}, \frac{8}{5}\right)$

II. The orthocentre of the ΔPAB is

(a) $\left(5, \frac{8}{7}\right)$

(b) $\left(\frac{7}{5}, \frac{25}{8}\right)$

(c) $\left(\frac{11}{5}, \frac{8}{5}\right)$

(d) $\left(\frac{8}{25}, \frac{7}{5}\right)$

III. The equation of the locus of the point whose distance from the point P and the line AB are equal, is

(a) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$

(b) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$

(c) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$

(d) $x^2 + Y^2 - 2XY + 27X + 31Y - 120 = 0$

Q 3. If two tangents drawn from a point P to the parabola $y = 4x$ are at right angles, then the locus of P is

[AIEEE]

(a) $x = 1$

(b) $2x + 1 = 0$

(c) $x = -1$

(d) $2x - 1 = 0$

Q 4. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x–axis at Q. If M is the mid point of the line segment PQ, then the locus of M intersects the latusrectum of the given ellipse at the points

[IIT JEE]

(a) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{2}{7}\right)$

(b) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{\sqrt{19}}{4}\right)$

(c) $\left(\pm 2\sqrt{3}, \pm \frac{1}{7}\right)$

(d) $\left(\pm 2\sqrt{3}, \pm \frac{4\sqrt{3}}{7}\right)$

Q 5. If tangents at extremities of a focal chord AB of the parabola $y^2 = 4ax$ intersect at a point C, then $\angle ACB$ is equal to

[UP SEE]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{6}$

Q 6. The number of values of c such that the line $y = 4x + c$ touches the curve $\frac{x^2}{4} + y^2 = 1$ is

[KCET]

(a) 1

(b) 2

(c) ∞

(d) 0

Q 7. The number of normals drawn to the parabola $y^2 = 4x$ from the point $(1, 0)$ is

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 8. The locus of the foot of the perpendicular from the centre of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ on any tangent is given by $(x^2 + y^2)^2 = lx^2 + my^2$ where

[OJEE]

(a) $l = a^2$, $m = b^2$

(b) $l = b^2$, $m = a^2$

(c) $l = m = a$

(d) $l = m = b$

Q 9. The equation of the tangent to the conic

$x^2 - y^2 - 8x + 2y + 11 = 0$ at $(2, 1)$ is

[WB JEE]

(a) $x + 2 = 0$

(b) $2x + 1 = 0$

(c) $x + y + 1 = 0$

(d) $x - 2 = 0$

Q 10. The total number of tangents through the points $(3,5)$ that can be drawn to the ellipses $3x^2 + 5y^2 = 32$ and $25x^2 + 9y^2 = 450$ is

[WBJEE]

(a) 0

(b) 2

(c) 3

(d) 4

Q 11. A line is drawn through the point $P(3,11)$ to cut the circle $x^2 + y^2 = 9$ at A and B. Then, PA . PB is equal to

[BCECE]

(a) 9

(b) 121

(c) 205

(d) 139

Q 12. The line $x + y = 6$ is a normal to the parabola $y^2 = 8x$ at the point

[BCECE]

(a) $(18, -12)$

(b) $(4, 2)$

(c) $(2, 4)$

(d) $(8, 8)$

Q 13. The tangent at $(1, 7)$ to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at

[Jamia Millia Islamia]

(a) $(6, 7)$

(b) $(-6, 7)$

(c) $(6, -7)$

(d) $(-6, -7)$

Q 14. If $4x - 3y + k = 0$ touches the ellipse $5x^2 + 9y^2 = 45$, then k is equal to

[MHT CET]

(a) $\pm 3\sqrt{21}$

(b) $3\sqrt{21}$

(c) $-3\sqrt{21}$

(d) $2\sqrt{21}$

Q 15. If m_1 and m_2 are the slopes of tangents to the circle $x^2 + y^2 = 4$ from the point $(3, 2)$, then $m_1 - m_2$ is equal to

[MHT CET]

(a) $\frac{5}{12}$

(b) $\frac{12}{5}$

(c) $\frac{3}{2}$

(d) 0

Q 16. Tangent to the ellipse $\frac{x^2}{32}+\frac{y^2}{18}=1$ having slope $-\frac{3}{4}$ meet the coordinate axes in A and B. Find the area of the Δ AOB, where O is the origin

[MHT CET]

(a) 12 sq unit

(b) 8 sq unit

(c) 24 sq unit

(d) 32 sq unit

Q 17. Equation of tangent to the parabola y^2 $=16x$ at $P(3,6)$ is

[MHT CET]

(a) $4x - 3Y + 12 = 0$

(b) $3Y - 4x - 12 = 0$

(c) $4x - 3y - 24 = 0$

(d) $3y - x - 24 = 0$

Q 18. If the line $y = 2x + \lambda$ be a tangent to the hyperbola $36x^2 - 25y^2 = 3600$, then λ is equal to

[DCE]

(a) 16

(b) -16

(c) ± 16

(d) None of these

Q 19. The equation of the normal to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at $(-4, 0)$ is

[KCET]

(a) $2x - 3y = 1$

(b) $x = 0$

(c) $x = 1$

(d) $y = 0$

Q 20. Three normals to the parabola $y = x$ through point $(a, 0)$. Then,

[OJEE]

(a) $a = \frac{1}{2}$

(b) $a = \frac{1}{4}$

(c) $a > \frac{1}{2}$

(d) $a < \frac{1}{2}$

Q 21. The tangent to the parabola $y^2 = 16x$, which is perpendicular to a line $y - 3x - 1 = 0$, is

[OJEE]

(a) $3y + x + 36 = 0$

(b) $3y - x - 36 = 0$

(c) $x + y - 36 = 0$

(d) $x - y + 36 = 0$

Q 22. The equation of the line which is tangent to both the circle $x^2 + y^2 = 5$ and the parabola $y^2 = 40x$ is

[Jamia Millia Islamia]

(a) $2x - y \pm 5 = 0$

(b) $2x - y + 5 = 0$

(c) $2x - y - 5 = 0$

(d) $2x + y + 5 = 0$

Q 23. If the line $x + y - 1 = 0$ is a tangent to the parabola $y^2 - y + x = 0$, then the point of contact is

[MHT CET]

(a) $(0, 1)$

(b) $(1,0)$

(c) $(0, -1)$

(d) $(-1,0)$

Q 24. If the line x cos α + y sin α = p be normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then

[MHTCET]

(a) $p^2(a^2\cos^2\alpha+b^2\sin^2\alpha) = a^2 - b^2$

(b) $p^2(a^2\cos^2\alpha+ b^2\sin^2\alpha) = (a^2- b^2)^2$

(c) $p^2(a^2\sec^2\alpha+b^2\,\mathrm{cosec}^2\alpha) = a^2 - b^2$

(d) $p^2(a^2\sec^2\alpha+b^2\mathrm{cosec}^2\alpha) = (a^2 - b^2)^2$

Q 25. The equation the tangent parallel to y − x + 5 = 0, drawn to $\frac{x^2}{3}-\frac{y^2}{2}=1$ is

[MHT CET]

(a) x − y − 1 = 0

(b) x − y + 2 = 0

(c) x+ y − 1 = 0

(d) x + y + 2 = 0

Q 26. The line $3x + 5y = 15\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$, at a point whose eccentric angle is

[MP PET]

(a) $\pi/6$

(b) $\pi/4$

(c) $\pi/3$

(d) $2\pi/3$

Q 27. The value of c, for which the line $y = 2x +$ c, is tangent to the parabola $y^2 = 4a(x + a)$, is

[MPPET]

(a) a

(b) $\frac{3a}{2}$

(c) $2a$

(d) $\frac{5a}{2}$

Q 28. The common tangent of the parabolas $y^2 = 4x$ and $x^2 = -8y$, is

[DCE]

(a) $y = x + 2$

(b) $y = x - 2$

(c) $y = 2x + 3$

(d) None of these

Q 29. The equation of normal at the point $(0,3)$ of the ellipse $9x^2 + 5y^2 = 45$, is

[UPSEE]

(a) x−axis

(b) y−axis

(c) $y + 3 = 0$

(d) $y - 3 = 0$

Q 30. Equation of tangents to the ellipse

$\frac{x^2}{9}+\frac{y^2}{4}=1$, which are perpendicular to the line $3x + 4y = 7$, are

[UP SEE]

(a) $4x-3y = \pm 6\sqrt{5}$

(b) $4x - 3y = \pm \sqrt{12}$

(c) $4x - 3y = \pm \sqrt{2}$

(d) $4x-3y = \pm 1$

Q 31. The slope of tangents drawn form a point

$(4, 10)$ to the parabola $y^2 = 9x$ are

[AMU]

(a) $\frac{1}{4}, \frac{3}{4}$

(b) $\frac{1}{4}, \frac{9}{4}$

(c) $\frac{1}{4}, \frac{1}{3}$

(d) None of these

Q 32. If the line lx + my = 1 is a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$ then $\frac{a^2}{l^2} - \frac{b^2}{m^2}$ is equal to

[EAMCET]

(a) $a^2 - b^2$

(b) $a^2 + b^2$

(c) $(a^2 + b^2)^2$

(d) $(a^2 - b^2)^2$

Q 33. A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$, is

[WBJEE, MHTCET]

(a) $y = \frac{3}{\sqrt{7}}x + \frac{15}{\sqrt{7}}$

(b) $y = 3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}}$

(c) $y = 2\sqrt{\frac{3}{7}}x + 15\sqrt{7}$

(d) none of these

Q 34. The equation of the tangents to the ellipse $4x^2 + 3y^2 = 5$, which are parallel to the line $y = 3x + 7$ are

[WB JEE]

(a) $y = 3x \pm \sqrt{\frac{155}{3}}$

(b) $y = 3x \pm \sqrt{\frac{155}{12}}$

(c) $y = 3x \pm \sqrt{\frac{95}{12}}$

(d) None of these

Q 35. The angle between the tangent drawn from the point $(1, 4)$ to the parabola $y^2 = 4x$ is

[BCECE]

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{2}$

Q 36. If the normal at $(ap , 2ap)$ on the parabola $y^2 = 4ax$, meets the parabola again at $(aq^2, 2aq)$, then

[VITEEE]

(a) $p^2 + pq + 2 = 0$

(b) $p^2 - pq + 2 = 0$

(c) $q^2 + pq + 2 = 0$

(d) $p^2 + pq + 1 = 0$

Q 37. The equation to the line touching both the parabolas $y^2 = 4x$ and $x^2 = -32y$, is

[MHT CET]

(a) $x + 2y + 4 = 0$

(b) $2x + y - 4 = 0$

(c) $x - 2y - 4 = 0$

(d) $x - 2y + 4 = 0$

Q 38. The locus of a point P(α, β) moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, is

[MP PET]

(a) a hyperbola

(b) a parabola

(c) a circle

(d) an ellipse

Q 39. The equation of a tangent parallel to y = x drawn to $\frac{x^2}{3}-\frac{y^2}{2}=1$, is

[Guj.CET]

(a) $x - y + 1 = 0$

(b) $x - y + 2 = 0$

(c) $x + y - 1 = 0$

(d) $x - y + 2 = 0$

Q 40. Two tangents are drawn from the point $(-2, -1)$ to the parabola $y^2 = 4x$. If α is the angle between these tangents, then $\tan \alpha$ is equal to

[Guj. CET]

(a) 3

(b) $1/3$

(c) 2

(d) $1/2$

Q 41. The value of m, for which the line y = mx +2 becomes a tangent to the conic $4x^2 - 9y^2 = 36$ are

[UP SEE]

(a) $\pm\frac{2}{3}$

(b) $\pm\frac{2\sqrt{2}}{3}$

(c) $\pm\frac{8}{9}$

(d) $\pm\frac{4\sqrt{2}}{3}$

Q 42. The value of m, for which the line y= mx + $\frac{25\sqrt{3}}{3}$ is a normal to the conic $\frac{x^2}{16}-\frac{y^2}{9}=1$, is

[UP SEE, WB JEE]

(a) $\pm\frac{2}{\sqrt{3}}$

(b) $\pm\sqrt{3}$

(c) $\pm\frac{\sqrt{3}}{2}$

(d) None of these

Q 43. The number of maximum normals which can be drawn from a point to ellipse is

[WBJEE]

(a) 4

(b) 2

(c) 1

(d) 3

Q 44. A line touches the circle $x^2 + y^2 - 2$ and the parabola $y^2 = 8x$, then equation of tangent is

[MHT CET]

(a) $y = x + 3$

(b) $y = x+2$

(c) $y = x + 4$

(d) $y = x + 1$

Q 45. If the parabola $y^2 = 4ax$ passes through the point $(1, -2)$, then tangent at this point is

[RPET]

(a) $x - y - 1 = 0$

(b) $x + y + 1 = 0$

(c) $x - y + 1 = 0$

(d) None of these

Q 46. Tangent at a point of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is drawn which cuts the coordinate axes at A and B. The minimum area of the ΔOAB is (O being the origin)

[IIT JEE]

(a) ab

(b) $\frac{a^3 + ab + b^3}{3}$

(c) $a^2 + b^2$

(d) $\frac{(a^2 + b^2)}{4}$

Q 47. The line among the following which touches the parabola $y^2 = 4ax$, is

[EAMCET]

(a) $x + my + am^3 = 0$

(b) $x - my + am^2 = 0$

(c) $x + my - am^2 = 0$

(d) $y + mx + am^2 = 0$

Q 48. The equation of the normal at the point $(a \sec\theta, b \tan\theta)$ of the curve $b^2x^2 - a^2y^2 = a^2b^2$ is

[RPET]

(a) $\frac{ax}{\cos\theta} + \frac{by}{\sin\theta} = a^2 + b^2$

(b) $\frac{ax}{\tan\theta}+\frac{by}{\sec\theta}=a^2+b^2$

(c) $\frac{ax}{\sec\theta}+\frac{by}{\tan\theta}=a^2+b^2$

(d) $\frac{ax}{\sec\theta}+\frac{by}{\tan\theta}=a^2-b^2$

Q 49. From the point $(-1, -6)$ two tangents are drawn to the parabola $y^2 = 4x$. Then, the angle between the two tangents is

[J&K CET]

(a) $30°$

(b) $45°$

(c) $60°$

(d) $90°$

Q 50. If the line $2x + \sqrt{6}y = 2$ touches the hyperbola $x^2 - 2y^2 = 4$, then the point of contact is

[IIT JEE]

(a) $(-2, \sqrt{6})$

(b) $(-5, 2\sqrt{6})$

(c) $\left(\frac{1}{2}, \frac{1}{\sqrt{6}}\right)$

(d) $(4, -\sqrt{6})$

Q 51. If tangents are drawn to the ellipse $x^2 + 2y^2 = 2$, then the locus of the mid point of the intercept made by the tangents between the coordinate axes is

[IIT JEE]

(a) $\frac{1}{2x^2}+\frac{1}{4y^2}=1$

(b) $\frac{1}{4x^2}+\frac{1}{2y^2}=1$

(c) $\frac{x^2}{2}+\frac{y^2}{4}=1$

(d) $\frac{x^2}{4}+\frac{y^2}{2}=1$

Q 52. How many real tangents can be drawn to the ellipse $5x^2+9y^2=32$ from the point $(2,3)$?

[DCE]

(a) 2

(b) 1

(c) 0

(d) 3

Q 53. The conditions that $ax + by + c = 0$ is tangent to the parabola $y^2 = 4ax$, is

[Kerala CEE]

(a) $a^2 = b^2 = c^2$

(b) $a = b$

(c) $b^2 = c$

(d) $b^2 = a$

(e) $a^2=b$

Q 54. On the ellipse $4x^2 + 9y^2 = 1$ the point at which the tangent are parallel to $8x = 9y$ are

[KeralaCEE]

(a) $\left(\frac{2}{5},\frac{1}{5}\right)\text{or}\left(-\frac{2}{5},\frac{1}{5}\right)$

(b) $\left(-\frac{2}{5},\frac{1}{5}\right)\text{or}\left(\frac{2}{5},-\frac{1}{5}\right)$

(c) $\left(-\frac{2}{5},-\frac{1}{5}\right)$

(d) $\left(-\frac{3}{5},-\frac{2}{5}\right)\text{or}\left(\frac{3}{5},\frac{2}{5}\right)$

(e) $\left(-\frac{3}{5},\frac{2}{5}\right)\text{or}\left(\frac{3}{5},-\frac{2}{5}\right)$

Q 55. The line y = mx +1 is a tangent to the parabola $y^2 = 4x$, if

[BCECE]

(a) m = 1

(b) m = 2

(c) m = 4

(d) m = 3

Q 56. The point of intersection of tangents at the ends of the latusrectum of the parabola $y^2 = 4x$, is equal to

[Jamia Millia Islamia]

(a) $(1,0)$

(b) $(-1, 0)$

(c). $(0, 1)$

(d) $(0, -1)$

Q 57. The equation of tangent to the ellipse $x^2 + 4y^2 = 5$ at $(-1,1)$, is

[MHT CET]

(a) $x + 4y + 5 = 0$

(b) $x - 4y - 5 = 0$

(c) $x + 4y - 5 = 0$

(d) $x - 4y + 5 = 0$

Q 58. The angle between the pair of tangents drawn to the ellipse $3x^2 + 2y^2 = 5$ from the point $(1, 2)$ is

[MP PET]

(a) $\tan^{-1}\left(\frac{12}{5}\right)$

(b) $\tan^{-1}(6\sqrt{5})$

(c) $\tan^{-1}\left(\frac{12}{\sqrt{5}}\right)$

(d) $\tan^{-1}(12\sqrt{5})$

Q 59. The value of λ, for which the line

$2x - \frac{8}{3}\lambda y = -3$ is a normal to the conic $x^2 + \frac{y^2}{4} = 1$ is

[MP PET]

(a) $-\frac{\sqrt{3}}{2}$

(b) $\frac{1}{2}$

(c) -3

(d) $\pm\frac{\sqrt{3}}{2}$

Q 60. The straight line x + y = c will be tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{16}=1$, c is equal to

[RPET]

(a) 8

(b) ±5

(c) +10

(d) ±6

Q 61. The area of the quadrilateral formed by the tangents at the end points of latus rectum to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$, is

[IITJEE]

(a) $27/4$ sq units

(b) 9 sq units

(c) $27/2$ sq units

(d) 27 sq units

Q 62. The normal at the point $(bt^2, 2bt_1)$ on a parabola $y^2 = 4bv$ meets the parabola again in the point $(bt_2^2, 2bt_2)$, then

[AIEEE]

(a) $t_2 = -t_1 - \frac{2}{t_1}$

(b) $t_2 = -t_1 + \frac{2}{t_1}$

(c) $t_2 = t_1 - \frac{2}{t_1}$

(d) $t_2 = t_1 + \frac{2}{t_1}$

Q 63. If the straight line $lx + my + n = 0$ touches the parabola, $y^2 = 4ax$, then

[BCECE]

(a) $nm = al^2$

(b) $nl = am^2$

(c) $nl = am$

(d) $ml = an$

Topic 4 Chord of Contact, Pole and Polar, Director Circle

Q 1. The middle point of the chord $x + 3y = 2$ of the conic $x^2 + xy - y^2 = 1$ is

[UPSEE]

(a) $(5,-1)$

(b) $(1,1)$

(c) $(2, 0)$

(d) $(-1,1)$

Q 2. The mid point of the chord $4x - 3y = 5$ of the hyperbola $2x^2 - 3y^2 = 12$ is

[EAMCET]

(a) $\left(0, -\frac{5}{3}\right)$

(b) $(2, 1)$

(c) $\left(\frac{5}{4}, 0\right)$

(d) $\left(\frac{11}{4}, 2\right)$

Q 3. The equation of the chord of the circle x^2 + $y^2 - 4x = 0$, whose mid point is $(1,0)$ is

[WB JEE]

(a) $y = 2$

(b) $y = 1$

(c) $x = 2$

(d) $x = 1$

Q 4. AB is a chord of the parabola $y^2 = 4$ ax with vertex at A. BC is drawn perpendicular to AB

meeting the axes at C. The projection of BC on the axis of the parabola is

[Jamia Millia Islamia]

(a) 2

(b) 2 a

(c) 4a

(d) 8a

Q 5. If $\tan \theta_1$, $\tan \theta_2 = -\frac{a^2}{b^2}$, then the chord joining two points θ_1 and θ_2 on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ will subtend a right angle at

[Jamia Millia Islamia]

(a) focus

(b) centre

(c) end of the major axis

(d) end of the minor axis

Q 6. Locus of mid point of any focal chord of $y^2 = 4ax$ is

[UP SEE]

(a) $y^2 = a(x - 2a)$

(b) $y^2 = 2a(x - 2a)$

(c) $y^2 = 2a(x - a)$

(d) None of these

Q 7. The locus of the middle points of the chords of the parabola $y^2 = 4ax$, which passes through the origin is

[OJEE]

(a) $y^2 = ax$

(b) $y^2 = 2ax$

(c) $y^2 = 4ax$

(d) $x^2 = 4ay$

Q 8. If $P(at^2, 2at)$ be one end of a focal chord of the parabola $y^2 = 4ax$, then the length of the chord is

[WB JEE]

(a) $a\left(t-\frac{1}{t}\right)^2$

(b) $a\left(t-\frac{1}{t}\right)$

(c) $a\left(t+\frac{1}{t}\right)$

(d) $a\left(t+\frac{1}{t}\right)^2$

Q 9. The length of the common chord of the parabolas $y^2 = x$ and $x^2 = y$ and is

[WB JEE]

(a) $2\sqrt{2}$

(b) 1

(c) $\sqrt{2}$

(d) $\frac{1}{\sqrt{2}}$

Q 10. The equation of the director circle of the hyperbola $9x^2 - 16y^2 = 144$ is

[MP PET]

(a) $x^2 + y^2 = 7$

(b) $x^2 + y^2 = 9$

(c) $x^2 + y^2 = 16$

(d) $x^2 + y^2 = 25$

Q 11. Let a focal chord of parabola $y^2 = 16x$ cuts it at points (f, g) and (h, k). Then, $f \cdot h$ is equal to

[OJEE]

(a) 12

(b) 16

(c) 14

(d) None of these

Q 12. The locus of middle points of chords of hyperbola $3x^2 - 2y^2 + 4x - 6y = 0$ parallel to $y = 2x$ is

[WB JEE]

(a) $3x - 4y = 4$

(b) $3y - 4x + 4 = 0$

(c) $4x - 3y = 3$

(d) $3x - 4y = 2$

Q 13. In the parabola $y^2 = 4ax$, the length of the chord passing through the vertex inclined to the axis at $\frac{\pi}{4}$ is

[J&K CET]

(a) $4a\sqrt{2}$

(b) $2a\sqrt{2}$

(c) $a\sqrt{2}$

(d) a

Q 14. The equation of the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy = c^2$ is

[OJEE]

(a) $\frac{x}{x_1 + x_2} + \frac{y}{y_1 + y_2} = 1$

(b) $\frac{x}{x_1 - x_2} + \frac{y}{y_1 - y_2} = 1$

(c) $\frac{x}{y_1 + y_2} + \frac{y}{x_1 + x_2} = 1$

(d) $\frac{x}{y_1 - y_2} + \frac{y}{x_1 - x_2} = 1$

Q 15. If the chords of contact of tangents from two points (x_1, y_1) and (x_2, y_2) to the hyperbola $4x^2 - 9y^2 - 36 = 0$ are at right angles, then $\frac{x_1x_2}{y_1y_2}$ is equal to

[BCECE]

(a) $\frac{9}{4}$

(b) $-\frac{9}{4}$

(c) $\frac{81}{16}$

(d) $-\frac{81}{16}$

Q 16. The length of the chord of the parabola y^2 = 4ax, which passes through the vertex and makes an angle α with the axis of the parabola is

[RPET]

(a) $4a \cos \alpha \operatorname{cosec}^2 \alpha$

(b) $4 \cos \alpha \operatorname{cosec}^2 \alpha$

(c) $a \cos \alpha \operatorname{cosec}^2 \alpha$

(d) a $\cos^2 \alpha$ cosec α

Q 17. If a focal chord of the parabola $y^2 = ax$ is $2x - y - 8 = 0$, then the equation of the directrix is

[MPPET]

(a) $x + 4 = 0$

(b) $x - 4 = 0$

(c) $y - 4 = 0$

(d) $y + 4 = 0$

Q 18. The equation of the director circle of the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$, is given by

[KCET]

(a) $x^2 + y^2 = 16$

(b) $x^2 + y^2 = 4$

(c) $x^2 + y^2 = 20$

(d) $x^2 + y^2 = 12$

Q 19. If (x_1, y_1) and (x_2, y_2) are the ends of a focal chord of $y^2 = 4ax$, then $x_1, x_2 + y_1 y_2$ is equal to

[Kerala CEE]

(a) $-3a^2$

(b) $3a^2$

(c) $-4a^2$

(d) $4a^2$

(e) $2a^2$

Q 20. The focal chord to $y^2 = 16x$ is tangent to $(x - 6)^2 + y^2 = 2$, then the possible values of the slope of this chord, are

[IIT JEE]

(a) $\{-1,1\}$

(b) $\{-2,2\}$

(c) $\{-2,1/2\}$

(d) $\{2,-1/2\}$

Q 21. Which of the following is a point on the common chord of the circles $x^2 + y^2 + 2x - 3y + 6 = 0$ and $x^2 + y^2 + x - 8y - 13 = 0$

[KCET]

(a) $(1,4)$

(b) $(1,-2)$

(c) $(1,-4)$

(d) $(1,2)$

Q 22. The locus of the point of intersection n of the perpendicular tangents to ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is

[KCET]

(a) $x^2 + y^2 = 4$

(b) $x^2 + y^2 = 9$

(c) $x^2 + y^2 = 5$

(d) $x^2 + y^2 = 13$

Topic 5 Diameter, Auxiliary Circle and Asymptotes

Q 1. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M. Then, the area of the triangle with vertices at A, M and the origin O is

[IIT JEE]

(a) $\frac{31}{10}$

(b) $\frac{29}{10}$

(c) $\frac{21}{10}$

(d) $\frac{27}{10}$

Q 2. Asymptotes of a hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ are

[Guj.CET]

(a) $x=\pm\frac{25}{16}y$

(b) $x=\pm\frac{4}{5}y$

(c) $y=\pm\frac{5}{4}x$

(d) $y = \pm \frac{4}{5} x$

Q 3. The product of perpendiculars drawn from any point of a hyperbola to its asymptotes is

[UP SEE]

(a) $\frac{a^2 b^2}{a^2 + b^2}$

(b) $\frac{a^2 + b^2}{a^2 b^2}$

(c) $\frac{ab}{\sqrt{a} + \sqrt{b}}$

(d) $\frac{ab}{a^2+b^2}$

Q 4. If 2 y = x and 3 y + 4x = 0 are the equations of a pair of conjugate diameters of an ellipse, then the eccentricity of the ellipse is

[WBJEE]

(a) $\sqrt{\frac{2}{3}}$

(b) $\sqrt{\frac{2}{5}}$

(c) $\sqrt{\frac{1}{3}}$

(d) $\sqrt{\frac{1}{2}}$

Q 5. If the area of the auxiliary circle of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b) is twice the area of the ellipse, then the eccentricity of the ellipse is

[KCET]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{\sqrt{3}}$

(d) $\frac{1}{2}$

Q 6. The value of k, if $(1,2)$, (k, -1) are conjugate points with respect to the ellipse $2x^2 + 3y^2 = 6$, is

[EAMCET]

(a) 2

(b) 4

(c) 6

(d) 8

Q 7. Equation of asymptotes of xy = 7x + 5y are

[DCE]

(a) x = 7,y = 5

(b) x = 5,y = 7

(c) $xy = 35$

(d) None of these

Q 8. The product of the lengths of perpendiculars drawn from any point on the hyperbola $x^2 - 2y^2 - 2 = 0$ to its asymptotes is

[EAMCET]

(a) $1/2$

(b) $2/3$

(c) 3/2

(d) 20

Answers

Topic 1 : Basic Terms of Conics

1. (a) 2. (c) 3. (c) 4. (a) 5. (b) 6. (b) 7. (b) 8. (c)

9. (d) 10. (b)11. (c) 12. (a) 13. (a) 14. (b) 15. (b)

16. (b) 17. (a) 18. (c) 19. (b) 20. (a) 21. (b) 22.

(a) 23. (b) 24. (c) 25. (c) 26. (a) 27. (b) 28. (a)

29. (c) 30. (d) 31. (a) 32. (b) 33. (b) 34. (b) 35.

(a) 36. (b) 37. (a) 38. (d) 39. (c) 40. (c) 41. (d)

42. (e) 43. (c) 44. (b) 45. (b) 46. (a) 47. (c) 48.

(c) 49. (d) 50. (c) 51. (d) 52. (b) 53. (d) 54. (a)

55. (c) 56. (c) 57. (c) 58. (d) 59. (d) 60. (c)

61. (a) 62. (b) 63. (b) 64. (a) 65. (c) 66. (c) 67.

(a) 68. (d) 69. (c) 70. (e) 71. (b) 72. (c) 73. (a)

74. (c) 75. (b) 76. (b) 77. (c) 78. (e) 79. (d) 80.

(e) 81. (d) 82. (e) 83. (a) 84. (a) 85. (a) 86. (b)

87. (d) 88. (a) 89. (b) 90. (a) 91. (c) 92. (c) 93.

(d) 94. (c) 95. (a) 96. (a) 97. (d) 98. (b) 99. (d)

100. (b)

Topic 2 : Equation of Conics and Intersection of Two Curves

1. (i) b (ii) a 2. (a) 3. (a) 4. (a) 5. (b) 6. (d) 7. (b)

8. (b) 9. (c) 10. (b) 11. (b) 12. (c) 13. (b) 14. (d)

15. (d) 16. (d) 17. (c) 18. (b) 19. (a) 20. (b)

21. (a) 22. (c) 23. (c) 24. (e) 25. (b) 26. (a) 27.

(c) 28. (a) 29. (a) 30. (d)

31. (b) 32. (a) 33. (d) 34. (b) 35. (a) 36. (a) 37.

(a) 38. (b) 39. (d) 40. (a) 41. (c) 42. (c) 43. (c)

44. (c) 45. (d) 46. (a) 47. (c) 48. (c) 49. (d) 50.

(b) 51. (a) 52. (b) 53. (a) 54. (d) 55. (c) 56. (d)

Topic 3 : Tangent and Normal of Conic

1. (c,d) 2. (i) d (ii) c (iii) a 3. (c) 4. (c) 5. (c) 6. (b)

7. (b) 8. (a) 9. (d) 10. (c) 11. (b) 12. (c) 13. (d)

14. (a) 15. (b) 16. (c) 17. (b) 18. (c) 19. (d) 20.

(c) 21. (a) 22. (a) 23. (a) 24. (d) 25. (a) 26. (b)

27. (d) 28. (d) 29. (b) 30. (a) 31. (b) 32. (c) 33.

(b) 34. (b) 35. (c) 36. (a) 37. (d) 38. (a) 39. (a)

40. (a) 41. (b) 42. (a) 43. (a) 44. (b) 45. (b) 46.

(a) 47. (b) 48. (c) 49. (d) 50. (d) 51. (a) 52. (a)

53. (c) 54. (b) 55. (a) 56. (b) 57. (d) 58. (c) 59.

(d) 60. (b) 61. (d) 62. (a) 63. (b)

Topic 4 : Chord of Contact, Pole and Polar, Director Circle

1. (d) 2. (b) 3. (d) 4. (c) 5. (b) 6. (c) 7. (b) 8. (d)

9. (c) 10. (a) 11. (b) 12. (a) 13. (a) 14. (a) 15. (d)

16. (b) 17. (a) 18. (d) 19. (a) 20. (a) 21. (c) 22.

(d)

Topic 5 : Diameter, Auxiliary Circle and Asymptotes

1. (d) 2. (d) 3. (a) 4. (c) 5. (b) 6. (c) 7. (b) 8. (b)

Probability

Useful Results & Formulae

1. $P(A) + P(\bar{A}) = 1,\ 0 \leq P(A) \leq 1$

2. If $(m + n + p)$ equally likely, mutually exclusive and exhaustive sample points are in favour of an event A. Then, m : n + p is called odds in favour of E and n + p : m is called odd against.

$\therefore \quad PE = \frac{m}{m+n+p}$

3. (i) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(ii) $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

(iii) If A and B are mutually exclusive events, then

$P(A \cap B) = 0$ $P(A \cup B) = P(A) + P(B)$

4. Let A and B be two events associated to a random experiment. Then

(i) $P(\bar{A} \cap B) = P(B) - P(A \cap B)$

(ii) $P(A \cap \bar{B}) = P(A) - P(A \cap B)$

(iii) If $B \subset A$, then $P(A \cap \bar{B}) = P(A) - P(B)$ and

$P(B) < P(A)$

(iv) If $A \subset B$, then $P(\bar{A} \cap B) = P(B) - P(A)$ and

$P(A) < P(B)$

(v) P (exactly one of A, B occurs)

$= P(A \cap \bar{B}) + P(\bar{A} \cap B)$

$= P(A) + P(B) - 2P(A \cap B)$

(vi) P(neither A nor B occurs) = $P_{(\bar{A} \cap \bar{B})} = 1 - P(A \cup B)$

5. (i) P (at least two of A, B, C occurs) = $P(A \cap B) + P(B \cap C) + P(C \cap A) - 2P(A \cap B \cap C)$

(ii) P (exactly two of A, B, C occurs) = $P(A \cap B) + P(B \cap C) + P(A \cap C) - 3P(A \cap B \cap C)$

(iii) P (exactly one of A, B, C occurs)

$= P(A) + P(B) + P(C) - 2\,P(A \cap B) - 2P(B \cap C) - 2P(A \cap C) + 3P(A \cap B \cap C)$

6. (i) If E_1, E_2 ,…, E_n are independent events,

then $P(E_1 \cap E_2 \cap E_3 \cap \ldots\ldots \cap E_n) = P(E_1)\, P(E_2) \ldots$

$P(E_n)$

(ii) If E_1, E_2, …,E_n are exhaustive events, then $P(E_1 \cup E_2 \cup \ldots\ldots \cup E_n) = 1$

7. (i) $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$, $P(B) \neq 0$

(ii) If A and B are independent events, then

$$P\left(\frac{A}{B}\right) = P(A)$$

(iii) If A and B are events such that $B \neq 0$, then

$$P\left(\frac{A}{B}\right)+P\left(\frac{A'}{B}\right)=1$$

8. If $E_1,E_1,\ldots,E_n$ and mutually exclusive and exhaustive events and A is any event

(i) $$P(A)=P(E_1)P\left(\frac{A}{E_1}\right)+P(E_2)P\left(\frac{A}{E_2}\right)+.....P(E_n)P\left(\frac{A}{E_n}\right)$$

(ii) $$P\left(\frac{E_i}{A}\right)=\frac{P(E_i)P\left(\frac{A}{E_i}\right)}{\sum_{k=1}^{n}P(E_k).P\left(\frac{A}{E_k}\right)}$$ (Baye's theorem)

9. If X is a discrete random variable which assumes values x_1, x_2, $x_3,\ldots$, x_n with respective probabilities P_1, P_2, $P_3,\ldots.P_n$. Then,

(i) Mean $\bar{X} = P_1x_1+P_2x_2 + P_3x_3+\ldots + P_nx_n = \sum_{i=1}^{n} P_i x_i$

(ii) Variance, var(x) = $P_1(x_1 - \bar{X})^2 + P_2(x_2 - \bar{X})^2+\ldots+$

$P_n(x_n - \bar{X})^2$

$$= \sum_{i=1}^{n} P_i (x_i - \bar{X})^2$$

(iii) Sum of total probability distribution is one.

10. In Binomial distribution

(i) $P(X = r) = {}^nC_r p^r q^{n-r}; r = 0,1,2,\ldots, n; p + q = 1$

(ii) Mean = np

(iii) Variance, σ^2 = npq

(iv) Standard deviation, $\sigma = \sqrt{(npq)}$

(v) mean > Variance

(vi) Let x be a mode, then np + p > x > np − q

11. In Poisson distribution

(i) P(X = x)= $\frac{\lambda^x e^{-\lambda}}{x!}$; x = 0, 1, 2,...

(ii) Mean = variance = np

12. Probability that exactly r letters are in right envelopes

$$=\frac{1}{r!}\left[\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-.....+(-1)^{n-r}\frac{1}{(n-r)!}\right]$$

Topic 1 Simple Problems on Probability

Q 1. Let ω be a complex cube root of unity with $\omega \neq 1$. A fair die is thrown three times. If r_1, r_2 and r_3 are the numbers obtained on the die, then the probability that $\omega^{r_1}+\omega^{r_2}+\omega^{r_3}=0$ is

[IITJEE]

(a) $\frac{1}{18}$

(b) $\frac{1}{9}$

(c) $\frac{2}{9}$

(d) $\frac{1}{36}$

Q 2. Four numbers are chosen at random (without replacement) from the set $(1, 2, 3, \ldots\ldots, 20)$.

Statement I The probability that the chosen numbers when arranged in some order will form an AP, is $\frac{1}{85}$.

Statement II If the four chosen numbers form an AP, then the set of all possible values of common difference is $(\pm 1, \pm 2, \pm 3, \pm 4, \pm 5)$.

[AIEEE]

(a) Statement I is True, Statement II is True; Statement II is a correct explanation for Statement I

(b) Statement I is True, Statement II is True;

Statement II is NOT a correct explanation for

Statement I

(c) Statement I is True, Statement II is False

(d) Statement I is False, Statement II is True

Q 3. If birth to male child and birth to female

child are equal–probable, then what is the

probability that at least one of the three children

born to a couple is male?

(a) $\frac{4}{5}$

(b) $\frac{7}{8}$

(c) $\frac{8}{7}$

(d) $\frac{1}{2}$

Q 4. Two dice are tossed once. The probability of getting even number at the first die or a total of 8 is

[WB JEE]

(a) $\frac{1}{36}$

(b) $\frac{3}{36}$

(c) $\frac{11}{36}$

(d) $\frac{5}{9}$

Q 5. The probability that atleast one of A and B occurs is 0.6. If A and B occur simultaneously with probability 0.3, then P (A') + P (B') is

[BITSAT]

(a) 0.9

(b) 0.15

(c) 1.1

(d) 1.2

Q 6. If n positive integers are taken at random and multiplied together, the probability that the last digit of the product is 2, 4, 6 or 8, is

[DCE]

(a) $\frac{4^n + 2^n}{5^n}$

(b) $\frac{4^n \times 2^n}{5^n}$

(c) $\frac{4^n - 2^n}{5^n}$

(d) None of these

Q 7. An unbiased die is tossed until a number greater than 4 appears. The probability that an even number of tosses is needed, is

[BITSAT]

(a) $\frac{1}{2}$

(b) $\frac{2}{5}$

(c) $\frac{1}{5}$

(d) $\frac{2}{3}$

Q 8. A bag contains a white and b black balls. Two players A and B alternately draw a ball from the bag replacing the ball each time after the draw till one of them draws a white ball and wins the game. A begins the game. If the probability of A winning the game is three times that of B, then the ratio a : b is

[BITSAT]

(a) $1 : 1$

(b) $1 : 2$

(c) $2 : 1$

(d) None of these

Q 9. In a college 25% boys and 10% girls offer Mathematics. There are 60% girls in the college. If a Mathematics student is chosen at random, then the probability that the student is a girl, will be

[UP SEE]

(a) $\frac{1}{6}$

(b) $\frac{3}{8}$

(c) $\frac{5}{8}$

(d) $\frac{5}{6}$

Q 10. Two dice are thrown n times in succession. The probability of obtaining a double six at least once is

[UP SEE]

(a) $\left(\frac{1}{36}\right)^n$

(b) $1-\left(\frac{35}{36}\right)^n$

(c) $\left(\frac{1}{12}\right)^n$

(d) None of these

Q 11. A and B toss a coin alternately on the understanding that the first to obtain head win the toss. The probability that A wins the toss

[UP SEE]

(a) $\frac{1}{3}$

(b) $\frac{2}{3}$

(c) $\frac{1}{4}$

(d) $\frac{3}{4}$

Q 12. Two dice are thrown together. Then, the probability that the sum of numbers appearing on them is a prime number, is

[UP SEE]

(a) $\frac{5}{12}$

(b) $\frac{7}{18}$

(c) $\frac{13}{36}$

(d) $\frac{11}{36}$

Q 13. A complete cycle of a traffic light takes 60 s. During each cycle the light is green for 25 s, yellow for 5 s and red for 30 s. At a randomly chosen time, the probability that the light will not be green, is

[Kerala CEE]

(a) $\frac{1}{3}$

(b) $\frac{1}{4}$

(c) $\frac{4}{12}$

(d) $\frac{7}{12}$

(e) $\frac{3}{4}$

Q 14. A determinant of second order is made with the elements $0, 1$. What is the probability that the determinant is positive ?

[AMU]

(a) $\frac{7}{12}$

(b) $\frac{11}{12}$

(c) $\frac{3}{16}$

(d) $\frac{15}{16}$

Q 15. A person draws a card from a pack of playing cards, replaces it and shuffles the pack. He continues doing this until he draws a spade. The chance that he will fail the first two times is

[AMU]

(a) $\frac{9}{64}$

(b) $\frac{1}{64}$

(c) $\frac{1}{16}$

(d) $\frac{9}{16}$

Q 16. The probability of choosing randomly a number c from the set $\{1, 2, 3, \ldots, 9\}$ such that the quadratic equation $x^2 + 4x + c = 0$ has real roots is

[EAMCET]

(a) $\frac{1}{9}$

(b) $\frac{2}{9}$

(c) $\frac{3}{9}$

(d) $\frac{4}{9}$

Q 17. Probability of all 3 digit numbers having all the digits same is

[OJEE]

(a) $\frac{1}{100}$

(b) $\frac{3}{100}$

(c) $\frac{7}{100}$

(d) None of these

Q 18. Three numbers are chosen at random from 1 to 20. The probability that they are consecutive, is

[WB JEE]

(a) $\frac{1}{190}$

(b) $\frac{1}{120}$

(c) $\frac{3}{190}$

(d) $\frac{5}{190}$

Q 19. A bag contains 6 red and 3 white balls.

Four balls are drawn one by one and not replaced.

The probability that they are alternatively of different colours, is

[VITEEE]

(a) $\frac{4}{42}$

(b) $\frac{5}{42}$

(c) $\frac{7}{42}$

(d) $\frac{8}{42}$

Q 20. If three students A, B, C can solve a problem with probabilities $\frac{1}{3}, \frac{1}{4}$ and $\frac{1}{5}$ respectively, then the probability that the problem will be solved is

[MP PET]

(a) $\frac{3}{5}$

(b) $\frac{4}{5}$

(c) $\frac{2}{5}$

(d) $\frac{47}{60}$

Q 21. Five persons A, B, C, D and E are in queue of a shop. The probability that A and E always together, is

[DCE]

(a) $\frac{1}{4}$

(b) $\frac{2}{3}$

(c) $\frac{2}{5}$

(d) $\frac{3}{5}$

Q 22. If A and B are independent events of a random experiment such that P $(A \cap B) = \frac{1}{6}$ and $P(\bar{A} \cap \bar{B}) = \frac{1}{3}$, then P(A) is equal to

[BITSAT AMU]

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{5}{7}$

(d) $\frac{2}{3}$

Q 23. Twelve tickets are numbered from 1 to 12. One ticket is drawn at random, then the probability of the number to be divisible by 2 or 3, is

[UP SEE]

(a) $\frac{2}{3}$

(b) $\frac{7}{12}$

(c) $\frac{5}{6}$

(d) $\frac{3}{4}$

Q 24. If A and B are any two events, then P(A $\cap$ B') is equal to

[Kerala CEE]

(a) P(A) + P(B')

(b) P(A)P(B)

(c)P(B)−P(A $\cap$ B)

(d) P(A) − P(A $\cap$ B)

(e) 1 − P(A $\cap$ B)

Q 25. Let S be the sample space of the random experiment of throwing simultaneously two unbiased dice with six faces (numbered 1 to 6) and let E_k = {(a, b) $\in$ S : ab = k} for k ≥ 1.

If p_k = P(E_k) for k ≥ 1, then the correct among the following, is

[EAMCET]

(a) $p_1 < p_{30} < p_4 < p_6$

(b) $p_{36} < p_6 < p_2 < p_4$

(c) $p_1 < p_{11} < p_4 < p_6$

(d) $p_{36} < p_{11} < p_6 < p_4$

Q 26. Let A = {1,3,5,7,9}, B = {2, 4,6,8}. If a cartesian product A x B, if chosen at random, the probability of a + b = 9 is

[OJEE]

(a) $\frac{1}{4}$

(b) $\frac{1}{5}$

(c) 1

(d) 0

Q 27. A person draws out two balls successively from a bag containing 6 red and 4 white balls. The probability that at least one of them will be red, is

[WB JEE]

(a) $\frac{78}{90}$

(b) $\frac{30}{90}$

(c) $\frac{48}{90}$

(d) $\frac{12}{90}$

Q 28. Probability of getting positive integral roots of the equation $x^2 - n = 0$ for the integer $n, 1 \leq n \leq 40$ is

[VITEEE]

(a) $\frac{1}{5}$

(b) $\frac{1}{10}$

(c) $\frac{3}{20}$

(d) $\frac{1}{20}$

Q 29. If two dice are thrown together, then the probability that the sum of numbers appearing on them is 9, is

[MP PET]

(a) $\frac{1}{9}$

(b) $\frac{1}{6}$

(c) $\frac{1}{4}$

(d) $\frac{1}{3}$

Q 30. The probability that number selected at random from the numbers $1, 2, 3, 4, 5, 6, 7, 8, \ldots, 100$ is a prime, is

[J&KCET]

(a) 0.4

(b) 0.25

(c) 0.45

(d) 0.43

Q 31. Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb only if the first misses the

target. The probability that the target is hit by the second plane, is

[AIEEE]

(a) 0.06

(b) 0.14

(c) 0.32

(d) 0.7

Q 32. The probability that the same number appear on throwing three dice simultaneously, is

[BITSAT]

(a) $\frac{1}{36}$

(b) $\frac{5}{36}$

(c) $\frac{1}{6}$

(d) $\frac{4}{13}$

Q 33. 5 boys and 5 girls are sitting in a row randomly. The probability that boys and girls sit alternatively, is

(a) $\frac{5}{126}$

(b) $\frac{1}{42}$

(c) $\frac{4}{126}$

(d) $\frac{1}{126}$

Q 34. An urn contains 3 red and 5 blue balls.

The probability that two balls are drawn in which

2nd ball drawn is blue without replacement, is

[OJEE]

(a) $\frac{5}{16}$

(b) $\frac{5}{56}$

(c) $\frac{5}{8}$

(d) $\frac{20}{56}$

Q 35. There are 12 white and 12 red balls in a bag. Balls are drawn one by one with replacement

from the bag. The probability that 7th drawn ball is 4th white, is

[BCECE]

(a) $\frac{1}{4}$

(b) $\frac{1}{8}$

(c) $\frac{1}{2}$

(d) $\frac{1}{3}$

Q 36. Three letters, to each of which corresponds an envelope, are placed in the envelopes at random. The probability that all the letters are not placed in the right envelopes, is

[MHT CET]

(a) $1/6$

(b) $5/6$

(c) $1/3$

(d) $2/3$

Q 37. Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house, is

[MP PET]

(a) $\frac{7}{9}$

(b) $\frac{8}{9}$

(c) $\frac{1}{9}$

(d) $\frac{2}{9}$

Q 38. When three dice are thrown the probability of getting 4 or 5 on each of the dice simultaneously, is

[RPET]

(a) $\frac{1}{72}$

(b) $\frac{1}{108}$

(c) $\frac{1}{24}$

(d) None of these

Q 39. The probability of choosing a number divisible by 6 or 8 from among 1 to 90 is

[J&K CET]

(a) $\frac{1}{6}$

(b) $\frac{1}{90}$

(c) $\frac{1}{30}$

(d) $\frac{23}{90}$

Q 40. A coin is tossed three times. The probability of getting a head once and a tail twice is

[J&K CET]

(a) $\frac{1}{8}$

(b) $\frac{1}{3}$

(c) $\frac{3}{8}$

(d) $\frac{1}{2}$

Q 41. The probability of forming a three digit numbers with the same digits when three digit numbers are formed out of the digits $0,2,4,6,8$, is

[J&K CET]

(a) $\frac{1}{16}$

(b) $\frac{1}{12}$

(c) $\frac{1}{645}$

(d) $\frac{1}{25}$

Q 42. Seven chits are numbered 1 to 7. Four chits are drawn one by one with replacement. The probability that the least number appearing on any selected chit is 5, is

[Guj. CET]

(a) $\left(\frac{3}{7}\right)^4$

(b) $\left(\frac{6}{7}\right)^3$

(c) $\frac{5.4.3}{7^3}$

(d) $\left(\frac{3}{4}\right)^3$

Q 43. For a party 8 guests are invited by a husband and his wife. They sit for a dinner around a round table. The probability that the husband and his wife sit together, is

[DCE]

(a) $\frac{2}{7}$

(b) $\frac{2}{9}$

(c) $\frac{1}{9}$

(d) $\frac{4}{9}$

Q 44. If M and N are any two events. The probability, that exactly one of them occurs, is

[UP SEE]

(a) $P(M) + P(N) - P(M \cap N)$

(b) $P(M) + P(N) + P(M \cap N)$

(c) P(M) + P(N)

(d) P(M) + P(N)-2P(M $\cap$ N)

Q 45. An anti-aircraft gun can take a maximum of four shots at any plane moving away from it. The probabilities of hitting the plane at the 1st, 2nd, 3rd and 4th shots are 0.4, 0.3, 0.2 and 0.1 respectively. What is the probability that at least one shot hits the plane ?

[Kerala CEE]

(a) 0.6976

(b) 0.3024

(c) 0.72

(d) 0.6431

(e) 0.7391

Q 46. A bag contains 3 black, 3 white and 2 red balls. One by one, three balls are drawn without replacement. The probability that the third ball is red, is equal to

[Kerala CEE]

(a) $\frac{2}{56}$

(b) $\frac{3}{56}$

(c) $\frac{1}{56}$

(d) $\frac{12}{56}$

(e) $\frac{14}{56}$

Q 47. In the random experiment of tossing two unbiased dice let E be the event of getting the sum

8 and F be the event of getting even numbers on

both the dice. Then,

I. $P(E) = \frac{7}{36}$ II. $P(F) = \frac{1}{3}$

Which of the following is a correct statement ?

[EAMCET]

(a) Both I and II are true

(b) Neither I nor II is true

(c) I is true, II is false

(d) I is false, II is true

Q 48. A number n is chosen at random from $\{1,2,3,4,\ldots,1000\}$. The probability that n is a number that leaves remainder 1 when divided by 7, is

[EAMCET]

(a) $\frac{71}{500}$

(b) $\frac{143}{1000}$

(c) $\frac{72}{500}$

(d) $\frac{71}{1000}$

Q 49. The probability of India winning a test match against West–Indies is $\frac{1}{2}$ assuming independence from match to match the probability that in a match series India's second win occurs at the third test, is

[OJEE]

(a) $\frac{1}{8}$

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

Q 50. Of a total of 600 bolts, 20% are too large and 10% are too small. The remainder are considered to be suitable. If a bolt is selected at random, the probability that it will be suitable is

[Jamia Millia Islamia]

(a) $\frac{1}{5}$

(b) $\frac{7}{10}$

(c) $\frac{1}{10}$

(d) $\frac{3}{10}$

Q 51. If $P(B)= \frac{3}{4}$, $P(A \cap B \cap \bar{C}) = \frac{1}{3}$ and $P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$, then $P(5 \cap C)$is

[Jamia Millia Islamia]

(a) $\frac{1}{12}$

(b) $\frac{1}{6}$

(c) $\frac{1}{15}$

(d) $\frac{1}{9}$

Q 52. 'X' speaks truth in 60% and 'Y' in 50% of the cases. The probability that they contradict each other narrating the same incident is

[MP PET]

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

Q 53. Probability of throwing 16 in one throw with three dice is

[MP PET]

(a) $\frac{1}{36}$

(b) $\frac{1}{18}$

(c) $\frac{1}{72}$

(d) $\frac{1}{9}$

Q 54. A number n is chosen at random from the set $\{11, 12, 13, \ldots, 30\}$. The probability that n is neither divisible by 3 nor divisible by 5 is

[J&K CET]

(a) $\frac{7}{20}$

(b) $\frac{9}{20}$

(c) $\frac{11}{20}$

(d) $\frac{13}{20}$

Q 55. The probability that a number n chosen at random from 1 to 30, to satisfy n + $(50/n)$ > 27 is

[J&K CET]

(a) $\frac{7}{30}$

(b) $\frac{3}{10}$

(c) $\frac{3}{5}$

(d) $\frac{1}{5}$

Q 56. Two unbiased dice are thrown simultaneously. The probability to get a sum more than 8 is

[J&K CET]

(a) $\frac{5}{36}$

(b) $\frac{5}{18}$

(c) $\frac{5}{12}$

(d) $\frac{2}{9}$

Q 57. A fair die is rolled. The probability that the first time 1 occurs at the even throw is

[IIT JEE]

(a) $\frac{1}{6}$

(b) $\frac{5}{11}$

(c) $\frac{6}{11}$

(d) $\frac{5}{36}$

Q 58. Two dice are thrown together. If the numbers appearing on the two dice are different, then what is the probability that the sum is 6 ?

[DCE]

(a) $\frac{5}{36}$

(b) $\frac{1}{6}$

(c) $\frac{2}{15}$

(d) None of these

Q 59. A number n is chosen at random from S = {1,2,3,..., 50}. Let A = $\left\{n \in S : n + \frac{50}{n} > 27\right\}$, B = {n $\in$ S : n is a prime} and C − {n $\in$ S : n is a square}. Then, correct order of their probabilities is

[EAMCET]

(a) P(A) < P(B) < P(C)

(b) P(A) > P(B) > P(C)

(c) P(B) < P(A) < P(C)

(d) P(A) > P(C) > P(B)

Q 60. The probability of getting at least one tail in 4 throws of a coin is

[BCECE]

(a) $\frac{15}{16}$

(b) $\frac{1}{16}$

(c) $\frac{1}{4}$

(d) None of these

Q 61. A and B play a game where each is asked to select a number from 1 to 25. If the two numbers match, both of them win a prize. The probability that they will not win a prize in a single trial, is

[Jamia Millia Islamia]

(a) $\frac{1}{25}$

(b) $\frac{24}{25}$

(c) $\frac{2}{25}$

(d) None of these

Q 62. If A and B are two mutually exclusive events, then

[Jamia Millia Islamia]

(a) $P(A) < P(\bar{B})$

(b) $P(A) > P(\bar{B})$

(c) $P(A) < P(B)$

(d) None of these

Q 63. A biased coin with probability p, $0 < p < 1$ of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is $\frac{2}{5}$, then p equals

[Jamia Millia Islamia]

(a) $\frac{1}{3}$

(b) $\frac{2}{3}$

(c) $\frac{2}{5}$

(d) $\frac{3}{5}$

Q 64. Three coins are tossed, then what is the probability that at least two heads appears on upper face ?

[MHT CET]

(a) $\frac{5}{8}$

(b) 40

(c) $\frac{8}{5}$

(d) None of these

Q 65. If events are independents and P(A) =$\frac{1}{3}$,

P(B) = $\frac{1}{3}$, P(C)= $\frac{1}{4}$ then P (A' $\cap$ B'$\cap$ C) is equal to

[MHT CET]

(a) $\frac{1}{4}$

(b) $\frac{1}{12}$

(c) $\frac{1}{3}$

(d) $\frac{5}{12}$

Q 66. A die is thrown. If it shows a six, we draw a ball from a bag consisting 2 black balls and 6 white balls. If it does not show a six, then we toss a coin. Then, the sample space of this experiment consists of

[J&K CET]

(a) 13 points

(b) 18 points

(c) 10 points

(d) None of these

Q 67. The probability of getting a total of at least 6 in the simultaneously throw of three dice is

[J&K CET]

(a) $\frac{6}{108}$

(b) $\frac{5}{27}$

(c) $\frac{1}{24}$

(d) $\frac{103}{108}$

Q 68. The probability that A speaks truth is $\frac{4}{5}$ while this probability for B is $\frac{3}{4}$. The probability that they contradict each other when asked to speak on a fact, is

[AIEEE]

(a) $\frac{3}{20}$

(b) $\frac{1}{5}$

(c) $\frac{7}{20}$

(d) $\frac{4}{5}$

Q 69. The probability of obtaining sum '8' in a single throw of two dice is

[Kerala CEE]

(a) $\frac{1}{36}$

(b) $\frac{5}{36}$

(c) $\frac{4}{36}$

(d) $\frac{6}{36}$

(e) None of these

Q 70. Six faces of an unbiased die are numbered with $2, 3, 5, 7, 11$ and 13. If two such dice are thrown, then the probability that the sum on the uppermost faces of the dice is an odd number, is

[EAMCET]

(a) $\frac{5}{18}$

(b) $\frac{5}{36}$

(c) $\frac{13}{18}$

(d) $\frac{25}{36}$

Q 71. A person puts three cards addressed to three different people in three envelopes with three different addresses without looking. What is the probability that the cards go into their respective envelopes ?

[OJEE]

(a) $\frac{2}{3}$

(b) $\frac{1}{6}$

(c) $\frac{1}{5}$

(d) $\frac{2}{5}$

Q 72. There are six vertices of a regular hexagon are chosen at random, then the possibility

that the triangle with three vertices is equilateral, is equal to

[BCECE]

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{1}{10}$

(d) $\frac{1}{20}$

Q 73. A box contains 10 good articles and 6 with defects, one article is chosen at random. What is the probability that it is either good or has a defect ?

[Jamia Millia Manila]

(a) $\frac{24}{64}$

(b) $\frac{40}{64}$

(c) $\frac{49}{64}$

(d) 1

Q 74. Unbiased die is thrown, probability that outcome is greater than 4, is

[MHT CET]

(a) $\frac{3}{4}$

(b) $\frac{4}{5}$

(c) $\frac{1}{3}$

(d) $\frac{5}{6}$

Q 75. If the probability that A and B will die with in a year are p and q respectively, then the probability that only one of them will be alive at the end of the year is

[KPET]

(a) $p + q$

(b) $p + q - 2pq$

(c) $p + q - pq$

(d) $p + q + pq$

Q 76. Two numbers are selected randomly from the set S = $\{1,2,3, 4, 5,6\}$ without replacement one by one. The probability that minimum of the two numbers is less than, 4 is

[IIT JEE]

(a) $1/15$

(b) $14/15$

(c) $1/5$

(d) $4/5$

Q 77. Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selected the winning horse, is

[AIEEE]

(a) $\frac{4}{5}$

(b) $\frac{3}{5}$

(c) $\frac{1}{5}$

(d) $\frac{2}{5}$

Q 78. A coin is tossed n times the probability of getting head at least once is greater than 0.8. Then the least value of such n is

[EAMCET]

(a) 2

(b) 3

(c) 4

(d) 5

Q 79. There are n letters and n addressed envelopes, the probability that all the letters are not kept in the right envelope, is

[AMU]

(a) $\frac{1}{n!}$

(b) $1-\frac{1}{n!}$

(c) $1-\frac{1}{n}$

(d) n!

Q 80. A five digits number is formed by writing the digits $1,2, 3, 4, 5$ in a random order without repetitions. Then the probability that the number is divisible by 4, is

[OJEE]

(a) $\frac{3}{5}$

(b) $\frac{18}{5}$

(c) $\frac{1}{5}$

(d) $\frac{6}{5}$

Q 81. Three dice are thrown simultaneously, then probability of throwing a total greater than 4 is

[J&K CET]

(a) $\frac{1}{54}$

(b) $\frac{53}{54}$

(c) $\frac{5}{108}$

(d) None of these

Q 82. To open a lock, a key is taken out from a collection of n keys at random. If the lock is not opened with this key, it is put back into the collection and another key is tried. The process is repeated again and again. If it is given that with only one key in the collection, the lock can be opened, then the probability that the lock will open in n trials, is

[J&KCET]

(a) $\left(\frac{1}{n}\right)^n$

(b) $\left(\frac{n-1}{n}\right)^n$

(c) $1-\left(\frac{n-1}{n}\right)^n$

(d) None of these

Q 83. The probability that the 6th day of a randomly chosen month of a year is a Sunday, is

[J&K CET]

(a) $\frac{1}{12}$

(b) $\frac{1}{17}$

(c) $\frac{1}{84}$

(d) None of these

Topic 2 Odds in Favour and Against, Use of Permutations and Combinations

Q 1. An urn contains nine balls of which three are red, four are blue and two are green. Three balls are drawn at random without replacement

from the urn. The probability that the three balls have different colours, is

[AIEEE]

(a) $\frac{1}{3}$

(b) $\frac{2}{7}$

(c) $\frac{1}{21}$

(d) $\frac{2}{23}$

Q 2. Out of $3n$ consecutive natural numbers, 3 natural numbers are chosen at random without replacement. The probability that the sum of the chosen numbers is divisible by 3, is

[DCE]

(a) $\frac{n(3n^2-3n+2)}{2}$

(b) $\frac{(3n^2-3n+2)}{2(3n-1)(3n-2)}$

(c) $\frac{(3n^2-3n+2)}{(3n-1)(3n-2)}$

(d) $\frac{n(3n-1)(3n-2)}{3(n-1)}$

Q 3. In an assembly of 4 persons the probability that at least 2 of them have the same birthday, is

[UP SEE]

(a) 0.293

(b) 0.24

(c) 0.0001

(d) 0.016

Q 4. A bag contains 4 brown and 5 white balls. A man pulls two balls at random without replacement. The probability that the man gets both the balls of the same colour is

[BCECE]

(a) $\frac{5}{108}$

(b) $\frac{1}{6}$

(c) $\frac{5}{18}$

(d) $\frac{4}{9}$

Q 5. From a group of 8 boys and 3 girls, a committee of 5 members to be formed. Find the probability that 2 particular girls are included in the committee.

[MHT CET]

(a) $\frac{4}{11}$

(b) $\frac{2}{11}$

(c) $\frac{6}{11}$

(d) $\frac{8}{11}$

Q 6. A pair of fair dice is thrown independently 4 times. The probability of getting a sum of exactly 7 twice is

[MPPET]

(a) $\frac{5}{81}$

(b) $\frac{25}{243}$

(c) $\frac{25}{216}$

(d) $\frac{125}{648}$

Q 7. The probability that the three cards drawn from a pack of 52 cards, are all black, is

[MP PET]

(a) $\frac{1}{17}$

(b) $\frac{2}{17}$

(c) $\frac{3}{17}$

(d) $\frac{2}{19}$

Q 8. If a fair coin is tossed 20 times and let we get head n times, then probability that n is odd, is

[RPET]

(a) $\frac{1}{2}$

(b) $\frac{1}{6}$

(c) $\frac{5}{8}$

(d) $\frac{7}{8}$

Q 9. Seven balls are drawn simultaneously from a bag containing 5 white and 6 green balls. The probability of drawing 3 white and 4 green balls is

[BCECE]

(a) $\frac{7}{{}^{11}C_7}$

(b) $\frac{{}^{5}C_{3} + {}^{6}C_{4}}{{}^{11}C_{7}}$

(c) $\frac{{}^{5}C_{2}\,{}^{6}C_{4}}{{}^{11}C_{7}}$

(d) $\frac{{}^{6}C_{3}\,{}^{5}C_{4}}{{}^{11}C_{7}}$

Q 10. A purse contains 4 copper and 3 silver coins. Another purse contains 6 copper and 2 silver coins. A coin is taken out from any purse, the probability that it is a silver coin, is

[MPPET]

(a) $\frac{37}{56}$

(b) $\frac{19}{56}$

(c) $\frac{4}{7}$

(d) $\frac{2}{3}$

Q 11. Out of 40 consecutive natural numbers, two are chosen at random. Probability that the sum of the numbers is odd, is

(a) $\frac{14}{29}$

(b) $\frac{20}{39}$

(c) $\frac{1}{2}$

(d) None of these

Q 12. Out of 15 persons 10 can speak Hindi and 8 can speak English. If two persons are chosen at random, then the probability that one

person speaks Hindi only and the other speaks both Hindi and English is

[Kerala CEE]

(a) $\frac{3}{5}$

(b) $\frac{7}{12}$

(c) $\frac{1}{5}$

(d) $\frac{2}{5}$

(e) $\frac{2}{3}$

Q 13. A committee of five is to be chosen from a group of 9 people. The probability that a certain married couple will either serve together or not at all is

[AMU]

(a) $\frac{1}{2}$

(b) $\frac{5}{9}$

(c) $\frac{4}{9}$

(d) $\frac{2}{3}$

Q 14. A letter is taken out at random from 'ASSISTANT' and another is taken out from 'STATISTICS'. The probability that they are the same letters, is **[AMU]**

(a) $\frac{1}{45}$

(b) $\frac{13}{90}$

(c) $\frac{19}{90}$

(d) None of these

Q 15. Four numbers are chosen at random from $\{1,2,3,\ldots, 40\}$. The probability that they are not consecutive, is

[EAMCET]

(a) $\frac{1}{2470}$

(b) $\frac{4}{7969}$

(c) $\frac{2469}{2470}$

(d) $\frac{7965}{7969}$

Q 16. A bag contains 6 white and 4 black balls. Two balls are drawn at random. The probability that they are of the same colours, is

[EAMCET]

(a) $\frac{1}{15}$

(b) $\frac{2}{5}$

(c) $\frac{4}{15}$

(d) $\frac{7}{15}$

Q 17. A box contains 9 tickets numbered 1 to 9 inclusive. If 3 tickets are drawn from the box without replacement, the probability that they are alternatively either {odd, even, odd} or {even, odd, even} is

[VITEEE]

(a) $\frac{5}{17}$

(b) $\frac{4}{17}$

(c) $\frac{5}{16}$

(d) $\frac{5}{18}$

Q 18. The odds against a certain event are $5:2$ and the odds in favour of another independent event are $6:5$. The probability that at least one of the events will happen, is

[MHT CET]

(a) $\frac{25}{77}$

(b) $\frac{52}{77}$

(c) $\frac{12}{77}$

(d) $\frac{65}{77}$

Q 19. A fair die is tossed eight times. The probability that a third six is observed on the eight throw, is

[OJEE]

(a) $\frac{^7C_2 \times 5^5}{6^7}$

(b) $\frac{^7C_2 \times 5^5}{6^8}$

(c) $\frac{^7C_2 \times 5^5}{6^6}$

(d) None of these

Q 20. Four persons are chosen at random from a group of 3 men, 2 women and 4 children. The chance that exactly 2 of them are children, is

[MHTCET]

(a) $\frac{10}{21}$

(b) $\frac{11}{13}$

(c) $\frac{13}{25}$

(d) $\frac{21}{32}$

Q 21. A bag contains 5 blue balls and unknown number of red balls, two balls are drawn at random. The probability of both of them are blue is $\frac{5}{14}$, then the number of red balls are

[MHT CET]

(a) 3

(b) 2

(c) 4

(d) 5

Q 22. If three natural numbers from 1 to 100 are selected randomly, then probability that all are divisible by both 2 and 3, is

[Kerala CEE]

(a) $\frac{4}{105}$

(b) $\frac{4}{33}$

(c) $\frac{4}{35}$

(d) $\frac{4}{1155}$

(e) $\frac{3}{1155}$

Q 23. A bag contains 5 white and 3 black balls and 4 balls are successively drawn out and not

replaced. The probability that they are alternately of different colours, is

[MP PET]

(a) $\frac{1}{196}$

(b) $\frac{2}{7}$

(c) $\frac{1}{7}$

(d) $\frac{13}{56}$

Q 24. A sample of a 4 items is drawn at a random without replacement from a lot of 10 items containing 3 defectives. If x denotes the number of defective items in the sample, then $P(0 < x < 3)$ is equal to

[DCE]

(a) $\frac{3}{10}$

(b) $\frac{4}{5}$

(c) $\frac{1}{2}$

(d) $\frac{1}{6}$

Q 25. A bag contains 8 red and 7 black balls.

Two balls are drawn at random. The probability that

both the balls are of the same colour, is

[UP SEE]

(a) $\frac{14}{15}$

(b) $\frac{11}{15}$

(c) $\frac{7}{15}$

(d) $\frac{4}{15}$

Q 26. A and B are two events. Odds against A are 2 to 1 and odds in favour of A $\cup$ B are 3 to 1. If $x \leq P(B) \leq y$, then ordered pair (x, y) is

[AMU]

(a) $\left(\frac{5}{12}, \frac{3}{4}\right)$

(b) $\left(\frac{2}{3}, \frac{3}{4}\right)$

(c) $\left(\frac{1}{3}, \frac{3}{4}\right)$

(d) None of these

Q 27. An unbiased coin is tossed to get 2 points for turning up a head and one point for the tail. If three unbiased coins are tossed simultaneously, then the probability of getting a total of odd number of points is

[EAMCET]

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{8}$

(d) $\frac{3}{8}$

Q 28. India plays two ODI matches each with Australia and Pakistan. The probability of India getting points $0,1,2$ are 0.45, 0.05, 0.50. The probability of India getting at least 7 points in the series is

[OJEE]

(a) 0.00875

(b) 0.875

(c) 0.0875

(d) None of these

Q 29. From a pack of cards two are accidently dropped. Probability that they are of opposite shade is

[UP SEE]

(a) $\frac{13}{51}$

(b) $\frac{1}{52 \times 51}$

(c) $\frac{26}{51}$

(d) None of these

Topic 3 Addition Theorem, Conditional Probability and Baye's Theorem

Q 1. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal

correctly is $\frac{3}{4}$. If the signal received at station B is green, then the probability that the original signal green is

[IIT JEE]

(a) $\frac{3}{5}$

(b) $\frac{6}{7}$

(c) $\frac{20}{23}$

(d) $\frac{9}{20}$

Q 2. One ticket is selected at random from 50 tickets numbered 00, 01, 02, ..., 49. Then, the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero equals

[AIEEE]

(a) $\frac{1}{14}$

(b) $\frac{1}{7}$

(c) $\frac{5}{14}$

(d) $\frac{1}{50}$

Q 3. Let A, B and C be the three events such that P(A) =0.3, P(B) =0.4, P(C) = $0.S$, P(A $\cap$ 5) = 0.08, P(A $\cap$ C) = 0.28, P(A $\cap$ B $\cap$ C) = 0.09. If P(A $\cup$ B $\cup$ C) $\geq$ 0.75, then P (B $\cap$ C) satisfies

[DCE]

(a) P (B $\cap$ C) $\leq$ 0.23

(b) P (B $\cap$ C)$\leq$ 0.48

(c) $0.23 \leq P(B \cap C) \leq 0.48$

(d) $0.23 \leq P(B \cap C) \geq 0.48$

Q 4. In tossing of a coin $(m + n)$ $(m > n)$

times, the probability of coming consecutive heads

at least m times is

[UP SEE]

(a) $\frac{n+2}{2^{m+1}}$

(b) $\frac{m-n}{2^{m+n}}$

(c) $\frac{m+n}{2^{m+n}}$

(d) $\frac{mn}{2^{m+n}}$

Q 5. If A and B are events of a random experiment such that $P(A \cup B) = \frac{4}{5}$, $P(\bar{A} \cup \bar{B}) = \frac{7}{10}$ and $P(B) = \frac{2}{5}$, then $P(A)$ equals

[EAMCET]

(a) $\frac{9}{10}$

(b) $\frac{8}{10}$

(c) $\frac{7}{10}$

(d) $\frac{3}{5}$

Q 6. Suppose that E_1 and E_2 are two events of a random experiment such that $P(E_1)= \frac{1}{4}$, $P\left(\frac{E_2}{E_1}\right)=\frac{1}{2}$ and $P\left(\frac{E_1}{E_2}\right)=\frac{1}{4}$, observe the lists given below

[EAMCET]

List I	List II
(a) $P(E_2)$	(i) 1 / 4

(b) $P(E_1 \cup E_2)$ (ii) $5/8$

(c) $P_{(\bar{E}_1 / \bar{E}_2)}$ (iii) $1/8$

(d) $P(E_1/\bar{E}_2)$ (iv) $1/2$

(v) $3/8$

(vi) $3/4$

The correct matching of the List I from the List II is

(A) (B) (C) (D)

(a) (ii) (iii) (vi) (i)

(b) (iv) (v) (vi) (i)

(c) (iv) (ii) (vi) (i)

(d) (i) (ii) (iii) (iv)

Q 7. A and B are two independent events such that P $(A \cup B')$ = 0.8 and P(A) = 0.3. Then, P(B) is

[WB JEE]

(a) $\frac{2}{7}$

(b) $\frac{2}{3}$

(c) $\frac{3}{8}$

(d) $\frac{1}{8}$

Q 8. Probability that in the toss of two dice we obtain an even sum or a sum less than 5, is

[VITEEE]

(a) $\frac{1}{2}$

(b) $\frac{1}{6}$

(c) $\frac{2}{3}$

(d) $\frac{5}{9}$

Q 9. Given P $(A \cup B) = 0.6, P(A \cap B) = 0.2,$

the probability of exactly one of the event occurs is

[MHT CET]

(a) 0.4

(b) 0.2

(c) 0.6

(d) 0.8

Q $10.$ Given P(A) = 0.5, P(B) = 0.4, P(A $\cap$ B) = 0.3, then $P\left(\frac{A'}{B'}\right)$ is equal to

[MHT CET]

(a) $\frac{1}{3}$

(b) $\frac{1}{2}$

(c) $\frac{2}{3}$

(d) $\frac{3}{4}$

Q 11. If A and B are two events such that P (A $\cup$ B) = $\frac{5}{6}$, P (A $\cap$ B)= $\frac{1}{3}$ and $P(\bar{B}) = \frac{1}{3}$, then the value of P(A) is

[MP PET]

(a) $\frac{1}{3}$

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

Q 12. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is

[IITJEE]

(a) 2, 4 or 8

(b) 3, 6 or 9

(c) 4 or 8

(d) 5 or 10

Q 13. It is given that the events A and B are such that P(A)= $\frac{1}{4}$, P(A | B)= $\frac{1}{2}$ and P(B | A)= $\frac{2}{3}$. Then, P (B) is

[AIEEE]

(a) $\frac{1}{2}$

(b) $\frac{1}{6}$

(c) $\frac{1}{3}$

(d) $\frac{2}{3}$

Q 14. A die is thrown. Let A be the event that the number obtained is greater than 3. Let B be the event that the number obtained is less than 5. Then $P(A \cup B)$ is

[AIEEE]

(a) $\frac{2}{5}$

(b) $\frac{3}{5}$

(c) 0

(d) 1

Q 15. A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is six. The probability that it is actually a six, is

[DCE]

(a) $\frac{3}{8}$

(b) $\frac{1}{5}$

(c) $\frac{3}{4}$

(d) None of these

Q 16. If A and B are two events such that P(A $\cup$ B) = $\frac{3}{4}$, P{A $\cap$ B)= $\frac{1}{4}$, P($\bar{A}$) = $\frac{2}{3}$, then P($\bar{A} \cap$ B) is equal to

[UP SEE]

(a) $\frac{5}{12}$

(b) $\frac{3}{8}$

(c) $\frac{5}{8}$

(d) $\frac{1}{2}$

Q 17. For k = $1,2,3$ the box B_k contains k red balls and $(k + 1)$ white balls. Let $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{3}$ and $P(B_3) = \frac{1}{6}$.

A box is selected at random and a ball is drawn from it. If a red ball is drawn, then the probability that it has come from box B_2, is

[EAMCET]

(a) $\frac{35}{78}$

(b) $\frac{14}{39}$

(c) $\frac{10}{13}$

(d) $\frac{12}{13}$

Q 18. If A and B are two independent events such that $P(B) = \frac{2}{7}$, $P(A \cup B^c) = 0.8$, then $P(A)$ is equal to

[BCECE]

(a) 0.1

(b) 0.2

(c) 0.3

(d) 0.4

Q 19. The chances to fail in Physics are 20% and the chances to fail in Mathematics are 10%. What are the chances to fail in at least one subject ?

[MHT CET]

(a) 28%

(b) 38%

(c) 72%

(d) 82%

Q 20. A parents has two children. If one of them is boy, then the probability that other is, also a boy, is

[MHT CET]

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{3}$

(d) None of these

Q 21. The probability that a leap year selected at random will contain either 53 Thursdays or 53 Fridays, is

[BITSAT]

(a) $\frac{3}{7}$

(b) $\frac{2}{7}$

(c) $\frac{5}{7}$

(d) $\frac{1}{7}$

Q 22. Let E_1, E_2 be two mutually exclusive events of an experiment with P (not E_2) = 0.6 = $P(E_1 \cup E_2)$. Then, $P(E_1)$ is equal to

[J&K CET]

(a) 0.1

(b) 0.3

(c) 0.4

(d) 0.2

Q 23. Let $H_1, H_2,\ldots,H_n$ be mutually exclusive events with $P(H_i) > 0$, $i = 1, 2,\ldots, n$. Let E be any other event with $0 < P(E) < 1$.

Statement I $P(H_i, / E) > P(E / H_i)P(H_i)$ for $i = 1,2,\ldots,n$.

Statement II $\sum_{i=1}^{n} P(H_i) = 1$.

[IIT JEE]

(a) Statement I is True, Statement II is True;

Statement II is a correct explanation for Statement I

(b) Statement I is True, Statement II is True, Statement II is NOT a correct explanation for Statement I

(c) Statement I is True, Statement II is False

(d) Statement I is False, Statement II is True

Q 24. One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to his wife given

that each American man is seated adjacent to his wife, is

[IIT JEE]

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{2}{5}$

(d) $\frac{1}{5}$

Q 25. For two events A and B, if P(A)= $P\left(\frac{A}{B}\right)=\frac{1}{4}$ and $P\left(\frac{B}{A}\right)=\frac{1}{2}$, then

[DCE]

(a) A and B are independent events

(b) $P\left(\frac{A'}{B}\right)=\frac{3}{4}$ (c) $P\left(\frac{B'}{A}\right)=\frac{1}{2}$ (d) All of the above

Q 26. If P(A) = P(B) = x and P(A $\cap$ B) = P(A' $\cap$ B') = $\frac{1}{3}$, then x is equal to

[UP SEE]

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{1}{4}$

(d) $\frac{1}{6}$

Q 27. The probability that at least one of the events A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, then $P(\bar{A}) + P(\bar{B})$ is

[AMU]

(a) 0.4

(b) 0.8

(c) 1.2

(d) 1.4

Q 28. If A and B are two events such that $P(A) > 0$ and $P(B) \neq 1$, then $P(\bar{A}|\bar{B})$ is equal to

[AMU]

(a) $1 - P(A|\bar{B})$

(b) $1-P(\bar{A}\,|\,B)$

(c) $\frac{1-P(A\cap B)}{P(\bar{B})}$

(d) $\frac{P(\bar{A})}{P(\bar{B})}$

Q 29. If A and B are mutually exclusive events with P{B) $\neq 1$, then P(A | $\bar{B}$) is equal to (Here, $\bar{B}$ is the complement of the event B)

[EAMCET]

(a) $\frac{1}{P(B)}$

(b) $\frac{1}{1-P(B)}$

(c) $\frac{P(A)}{P(B)}$

(d) $\frac{P(A)}{1-P(B)}$

Q 30. If P(A)= $\frac{1}{12}$, P(B)= $\frac{5}{12}$ and $P\left(\frac{B}{A}\right)=\frac{1}{15}$, then

$P(A \cup B)$ is equal to

[VITEEE]

(a) $\frac{89}{180}$

(b) $\frac{90}{180}$

(c) $\frac{91}{180}$

(d) $\frac{92}{180}$

Q 31. Let A and B be two events such that

$P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$, where $\bar{A}$

stands for complement of event A. Then, events A

and B are

[MP PET]

(a) mutually exclusive and independent

(b) independent but not equally likely

(c) equally likely but not independent

(d) equally likely and mutually exclusive

Q 32. If A and B are mutually exclusive events with $P(A) = \frac{1}{2} \times P(B)$ and $A \cup B = S$, (total sample space) then P (A) is equal to

[J&K CET]

(a) $\frac{2}{3}$

(b) $\frac{1}{3}$

(c) $\frac{1}{4}$

(d) $\frac{3}{4}$

Q 33. If two events A and B are such that $P(A^c) = 0.3$, $P(B) = 0.4$ and $P(A \cap B^c) = 0.5$, then $\left[\frac{B}{(A \cup B^c)}\right]$ is equal to

[DCE]

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{1}{4}$

(d) None of these

Q 34. An integer is chosen at random from first two hundred numbers. Then, the probability that the integer chosen is divisible by 6 or 8 is

[BCECE]

(a) $\frac{1}{4}$

(b) $\frac{2}{4}$

(c) $\frac{3}{4}$

(d) None of these

Q 35. The probability that in the toss of two dice, we obtain the sum 7 or 11, is

[Jamia Millia Islamia]

(a) $\frac{1}{6}$

(b) $\frac{1}{18}$

(c) $\frac{2}{9}$

(d) $\frac{23}{108}$

Q 36. If the probability of A to fail in an examination is 0.2 and that for B is 0.3, then probability that either A or B is fail, is

[Jamia Millia Islamia]

(a) 0.5

(b) 0.44

(c) 0.8

(d) 0.25

Q 37. A number is chosen at random among the first 120 natural numbers. The probability of the number chosen being a multiple of 5 or 15 is

[Jamia Millia Islamia]

(a) $\frac{1}{8}$

(b) $\frac{1}{5}$

(c) $\frac{1}{24}$

(d) $\frac{1}{6}$

Q 38. A bag A contains 4 green and 3 red balls and bag B contains 4 red and 3 green balls. One bag is taken at random and a ball is drawn and noted to be green. The probability that it comes from bag B, is

[DCE]

(a) $\frac{2}{7}$

(b) $\frac{2}{3}$

(c) $\frac{3}{7}$

(d) $\frac{1}{3}$

Q 39. Two dice are rolled one after the other.

The probability that the number on the first is

smaller than the number on the second is

(a) $\frac{1}{2}$

(b) $\frac{3}{4}$

(c) $\frac{7}{18}$

(d) $\frac{5}{12}$

Q 40. Box A contains 2 black and 3 red balls, while box B contains 3 black and 4 red balls. Out of these two boxes one is selected at random; and

the probability of choosing box A is double that of box B. If a red ball is drawn from the selected box, then the probability that it has come from box B, is

[EAMCET]

(a) $\frac{21}{41}$

(b) $\frac{10}{31}$

(c) $\frac{12}{31}$

(d) $\frac{13}{41}$

Q 41. A coin and six faced die, both unbiased, are thrown simultaneously. The probability of getting a head on the coin and an odd number on the die, is

[EAMCET]

(a) $\frac{1}{2}$

(b) $\frac{3}{4}$

(c) $\frac{1}{4}$

(d) $\frac{2}{3}$

Q 42. Let A and B are two events and P(A') = 0.3, P(B) = 0.4, P(A $\cap$ B')= 0.5, then P(A $\cup$ B')is

[OJEE]

(a) 0.5

(b) 0.8

(c) 1

(d) 0.1

Q 43. If E and F are two independent events such that $0 < P(E) < 1$ and $0 < P(F) < 1$, then

[BCECE]

(a) E and F^c are independent

(b) E^c and F^c are independent

(c) $P\left(\frac{E}{F}\right)+P\left(\frac{E^c}{F^c}\right)=1$

(d) None of these

Q 44. A problem in mathematics is given to three students E_1,E_2 and E_3 and their respective probability of solving the problem is $\frac{1}{2}, \frac{1}{3} \text{and} \frac{1}{4}$.

Probability that the problem is solved is

[Jamia Millia Islamia]

(a) $3/4$

(b) $1/2$

(c) $2/3$

(d) $1/3$

Q 45. A and B are two independent events such that P(A) $=\frac{1}{2}$ and P(B)= $\frac{1}{3}$, then P (neither A nor B) is equal to

[J&K CET]

(a) $\frac{2}{3}$

(b) $\frac{1}{6}$

(c) $\frac{5}{6}$

(d) $\frac{1}{3}$

Q 46. If $P(A \cap B) = \frac{1}{3}$, $P(A \cup B) = \frac{5}{6}$ and $P(A) = \frac{1}{2}$, then which one of the following is correct

[DCE]

(a) A and B are independent events

(b) A and B are mutually exclusive events

(c) $P(A) = P(B)$

(d) $P(A) < P(B)$

Q 47. In a certain population 10% of the people are rich, 5% are famous and 3% are rich and famous. The probability that a person picked at random from the population is either famous or rich but not both, is equal to

[UP SEE]

(a) 0.07

(b) 0.08

(c) 0.09

(d) 0.12

Q 48. In an entrance test there are multiple choice questions. There are four possible answers to each question, of which one is correct. The probability that a student knows the answer to a question is 90%. If he gets the correct answer to a question, then the probability that he was guessing, is

[Kerala CEE]

(a) $\frac{37}{40}$

(b) $\frac{1}{37}$

(c) $\frac{36}{37}$

(d) $\frac{1}{9}$

(e) None of these

Q 49. A and B are two independent witnesses

(ie, there is no collusion between them) in a case.

The probability that A will speak the truth is x and

the probability that B will speak the truth is y. A and B agree in a certain statement. The probability that the statement is true, is

[BITSAT]

(a) $\frac{x-y}{x+y}$

(b) $\frac{xy}{1+x+y+xy}$

(c) $\frac{x-y}{1-x-y+2xy}$

(d) $\frac{xy}{1-x-y+2xy}$

Q 50. Suppose E and F are two independent events of a random experiment. If the probability of occurrence of E is $\frac{1}{5}$ and the probability of occurrence of F given E is $\frac{1}{10}$, then the probability of non-occurrence of at least one of the events E and F is

[EAMCET]

(a) $\frac{1}{18}$

(b) $\frac{1}{2}$

(c) $\frac{49}{50}$

(d) $\frac{1}{50}$

Q 51. Probability $P(A) = \frac{4}{5}$, $P(B') = \frac{2}{5}$ and $P(A \cap 5) = \frac{1}{2}$, then $P(A \cap B')$ is equal to

[MHT CET]

(a) $\frac{3}{10}$

(b) $\frac{5}{2}$

(c) $\frac{2}{5}$

(d) $\frac{5}{7}$

Q 52. In a test, an examines either guesses or copies or knows the answer to a multiple choice questions with four choices. The probability that he makes a guess is $\frac{1}{3}$ and the probability that he copies the answer is $\frac{1}{6}$. The probability that his answer is correct given that he copied it is $\frac{1}{8}$. The probability that his answer is correct, given that he guessed it is $\frac{1}{4}$. The probability that they knew the

answer to the questions given that he correctly answered is

[J&K CET]

(a) $\frac{24}{31}$

(b) $\frac{31}{24}$

(c) $\frac{24}{29}$

(d) $\frac{29}{24}$

Q 53. A and B are the independent events. The probability that both occur simultaneously is $\frac{1}{6}$ and the probability that neither occur is $\frac{1}{3}$. The probability of occurrence of the events A and B is

[J&K CET]

(a) $\frac{1}{2}, \frac{3}{2}$

(b) $\frac{1}{2}, \frac{1}{3}$

(c) not possible

(d) None of these

Q 54. Events A, B, C are mutually exclusive events such that $P(A)=\frac{3x+1}{3}, P(B)=\frac{1-x}{4}$ and $P(C)=\frac{1-2x}{2}$.

The set of possible values of x are in the interval

[AIEEE]

(a) $\left[\frac{1}{3},\frac{1}{2}\right]$

(b) $\left[\frac{1}{3},\frac{2}{3}\right]$

(c) $\left[\frac{1}{3}, \frac{13}{3}\right]$

(d) [0, 1]

Q 55. If P(A) = P(B) = P(C) = $\frac{1}{4}$, P(AB) = P(CB) = 0 and P(AC)= $\frac{1}{8}$, then P(A + B) is equal to

[DCE]

(a) $\frac{5}{8}$

(b) $\frac{37}{64}$

(c) $\frac{3}{4}$

(d) $\frac{1}{2}$

Q 56. If $P(A) = P(B) = x$ and $P(A \cap B) = P(A' \cap B') = \frac{1}{3}$, then x is equal to

[UP SEE]

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{3}$

(d) $\frac{1}{6}$

Q 57. From a set of 100 cards numbered 1 to 100, one card is drawn at random. The probability that the number obtained on the card is divisible by 6 or 8 but not by 24, is

[Kerala CEE]

(a) $\frac{6}{25}$

(b) $\frac{1}{4}$

(c) $\frac{1}{6}$

(d) $\frac{1}{5}$

(e) $\frac{4}{5}$

Q 58. If A and B are two events such that P(A) $\neq 0$ and P (B) $\neq 1$, then $P\overline{\left(\frac{A}{B}\right)}$ is equal to

[AMU]

(a) $1-P\left(\frac{A}{B}\right)$

(b) $1-P\left(\frac{\bar{A}}{B}\right)$

(c) $\frac{1-P(A \cup B)}{P(\bar{B})}$

(d) $\frac{P(\bar{A})}{P(B)}$

Q 59. If A and B are two independent events such that $P(A \cap B') = \frac{3}{25}$ and $P(A' \cap B) = \frac{8}{25}$, then P(A) is equal to

[AMU]

(a) $\frac{1}{5}$

(b) $\frac{3}{8}$

(c) $\frac{2}{5}$

(d) $\frac{4}{5}$

Q 60. A bag X contains 2 white and 3 black balls and another bag Y contains 4 white and 2 black balls. One bag is selected at random and a ball is drawn from it. Then, the probability for the ball chosen be white, is

[EAMCET]

(a) $\frac{2}{15}$

(b) $\frac{7}{15}$

(c) $\frac{8}{15}$

(d) $\frac{14}{15}$

Q 61. Let $0 < P(A) < 1$, $0 < P(B) < 1$ and $P(A \cap B) = P(A) + P(B) - P(A)P(B)$, then

[BCECE]

(a) $P(B / A) = P(B) - P(A)$

(b) $P(A^c \cup B^c) = P(A^c) + P(B^c)$

(c) $P(A \cup B)^c = P(A^c)P(B^c)$

(d) $P(A / B) = P(A) + P(B^c)$

Q 62. If $4P(A) = 6P(B) = 10P(A \cap B) = 1$, then

$P\left(\frac{B}{A}\right)$ is equal to

[MP PET]

(a) $\frac{2}{5}$

(b) $\frac{3}{5}$

(c) $\frac{7}{10}$

(d) $\frac{19}{60}$

Topic 4 Binomial Poisson and Probability Distributions

Q 1. The probability distribution of a random variable X is given as

X	−5	−4	−3	−2	−1	0	1	2	3	4	5
P(X)	P	2p	3p	4p	5P	7p	8p	9p	10p	11p	12p

Then, the value of p is

[Kerala CEE]

(a) $\frac{1}{72}$

(b) $\frac{3}{73}$

(c) $\frac{5}{72}$

(d) $\frac{1}{74}$

(e) $\frac{1}{73}$

Q 2. In a binomial distribution $B\left(n, p=\frac{1}{4}\right)$, if the probability of at least one success is greater than or equal to $\frac{9}{10}$, then n is greater than

[AIEEE]

(a) $\frac{1}{\log_{10} 4 - \log_{10} 3}$

(b) $\frac{1}{\log_{10} 4 + \log_{10} 3}$

(c) $\frac{9}{\log_{10} 4 - \log_{10} 3}$

(d) $\frac{4}{\log_{10} 4 - \log_{10} 3}$

Q 3. If X and Y are independent binomial variates $B\left(5, \frac{1}{2}\right)$ and $B\left(7, \frac{1}{2}\right)$, then P (X + Y = 3) is

[DCE]

(a) $\frac{35}{47}$

(b) $\frac{55}{1024}$

(c) $\frac{220}{512}$

(d) $\frac{11}{204}$

Q 4. The mean and variance of a binomial variable X are 2 and 1 respectively. The probability that X takes values greater than 1, is

[BITSAT]

(a) $\frac{5}{16}$

(b) $\frac{8}{16}$

(c) $\frac{11}{16}$

(d) $\frac{1}{16}$

Q 5. If the random variable X takes the values $x_1, x_2, x_3, \ldots, x_{10}$ with probabilities $P(X = x_i) = ki$, then the value of k is equal to

[Kerala CEE]

(a) $\frac{1}{10}$

(b) $\frac{1}{4}$

(c) $\frac{1}{55}$

(d) $\frac{7}{12}$

(e) $\frac{3}{4}$

Q 6. If m and σ^2 are the mean and variance of the random variable X, whose distribution is given by

X	0	1	2	3
P(X)	$\frac{1}{3}$	$\frac{1}{2}$	0	$\frac{1}{6}$

Then,

[EAMCET]

(a) $m = \sigma^2 = 2$

(b) $m = 1, \sigma^2 = 2$

(c) $m = \sigma^2 = 1$

(d) $m = 2, \sigma^2 = 1$

Q 7. If X is a binomial variate with the range $\{0,1,2,3,4,5,6\}$ and P $(X = 2) = 4P\ (X = 4)$, then the parameter p of X is

[EAMCET]

(a) $\frac{1}{3}$

(b) $\frac{1}{2}$

(c) $\frac{2}{3}$

(d) $\frac{3}{4}$

Q 8. A random variable X has the probability distribution

X	1	2	3	4	5	6	7	8
P(X)	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

For the events E = {X is a prime number} and F = {X < 4}, then P $(E \cup F)$ is

[MP PET]

(a) 0.77

(b) 0.87

(c) 0.35

(d) 0.50

Q 9. Two cards are drawn successively with replacement from a well shuffled deck of 52 cards, then the mean of the number of aces is

[DCE]

(a) $\frac{1}{13}$

(b) $\frac{3}{13}$

(c) $\frac{2}{13}$

(d) None of these

Q 10. A die has four blank faces and two faces marked 3. The chance of getting a total of 12 in 5 throws is

[Kerala CEE]

(a) ${}^{5}C_{4}\left(\frac{1}{3}\right)^{4}\left(\frac{2}{3}\right)$

(b) ${}^5C_4\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^4$

(c) ${}^5C_4\left(\frac{1}{6}\right)^5$

(d) ${}^5C_4\left(\frac{1}{6}\right)^4\left(\frac{5}{6}\right)$

(e) ${}^5C_4\left(\frac{5}{6}\right)^4\left(\frac{1}{6}\right)$

Q 11. The distribution of a random variable X is given below

X	−2	−1	0	1	2	3

P(X)	$\frac{1}{10}$	k	$\frac{1}{5}$	2k	$\frac{3}{10}$	k

The value of k is

[AMU]

(a) $\frac{1}{10}$

(b) $\frac{2}{10}$

(c) $\frac{3}{10}$

(d) $\frac{7}{10}$

Q 12. If X is a poisson variate such that P(X = 1) = P(X = 2), then P(X = 4) is equal to

[EAMCET]

(a) $\frac{1}{2e^2}$

(b) $\frac{1}{3e^2}$

(c) $\frac{2}{3e^2}$

(d) $\frac{1}{e^2}$

Q 13. A manufacturer of cotter pins knows that 5% of his product is defective. He sells pins in boxes of 100 and guarantees that not more than one pin will be defective in a box. In order to find the probability that a box will fail to meet the guaranteed quality, the probability distribution one has to employ is

[VITEEE]

(a) binomial

(b) poisson

(c) normal

(d) exponential

Q 14. The probability that a certain kind of component will survive a given shock test is $\frac{3}{4}$. The probability that exactly 2 of the next 4 components tested survive is

[VITEEE]

(a) $\frac{9}{41}$

(b) $\frac{25}{128}$

(c) $\frac{1}{5}$

(d) $\frac{27}{128}$

Q 15. A random variable X follows binomial distribution with mean α and variance β. Then,

[VITEEE]

(a) $0 < \alpha < \beta$

(b) $0 < \beta < \alpha$

(c) $\alpha < 0 < \beta$

(d) $\beta < 0 < \alpha$

Q 16. The mode of the binomial distribution for which mean and standard deviation are 10 and $\sqrt{5}$ respectively, is

[MPPET]

(a) 7

(b) 8

(c) 9

(d) 10

Q 17. A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice is

[AIEEE]

(a) $\frac{1}{729}$

(b) $\frac{8}{9}$

(c) $\frac{8}{729}$

(d) $\frac{8}{243}$

Q 18. If in a trial the probability of success is twice the probability of failure. In six trials the probability of at least four successes is

[DCE]

(a) $\frac{496}{729}$

(b) $\frac{400}{729}$

(c) $\frac{500}{729}$

(d) $\frac{600}{729}$

Q 19. A dice is thrown 100 times, getting an even number is considered a success. The variance of the number of successes is

[DCE]

(a) 10

(b) 25

(c) 18

(d) 10

Q 20. A coin is tossed n times. The probability of getting head at least once is greater than 0.8, then the least value of n is

[BITSAT]

(a) 2

(b) 3

(c) 5

(d) 4

Q 21. The records of a hospital show that 10% of the cases of a certain disease are fatal. If 6 patients are suffering from the disease, then the probability that only three will die, is

[UP SEE]

(a) 8748×10^{-5}

(b) 1458×10^{-5}

(c) 1458×10^{-6}

(d) 41×10^{-6}

Q 22. A random variable X takes values $0, 1, 2,$ $3, \ldots$ with probability $P(X = x) = k(x + 1)\left(\frac{1}{5}\right)^x$, where k is constant, then $P(X = 0)$ is

[Kerala CEE]

(a) $\frac{7}{25}$

(b) $\frac{18}{25}$

(c) $\frac{13}{25}$

(d) $\frac{19}{25}$

(e) $\frac{16}{25}$

Q 23. The mean and standard deviation of a binomial variate X are 4 and $\sqrt{3}$ respectively.

Then, P(X >1) is equal to

[EAMCET]

(a) $1-\left(\frac{1}{4}\right)^{16}$

(b) $1-\left(\frac{3}{4}\right)^{16}$

(c) $1-\left(\frac{2}{3}\right)^{16}$

(d) $1-\left(\frac{1}{3}\right)^{16}$

Q 24. If the probability density function of a random variable X is f(x)= $\frac{x}{2}$ in $0 \leq x \leq 2$, then $P\left(\frac{X>1.5}{X>1}\right)$ is equal to

[VITEEE]

(a) $\frac{7}{16}$

(b) $\frac{3}{4}$

(c) $\frac{7}{12}$

(d) $\frac{21}{64}$

Q 25. If X follows a binomial distribution with parameters n = 100 and p = $\frac{1}{3}$, then P (X = r) is maximum when r is equal to

[VITEEE]

(a) 16

(b) 32

(c) 33

(d) None of these

Q 26. The mean and the variance of a binomial distribution are 4 and 2 respectively. Then, the probability of 2 successes is

[Jamia Millia Manila]

(a) $\frac{37}{256}$

(b) $\frac{219}{256}$

(c) $\frac{128}{256}$

(d) $\frac{28}{256}$

Q 27. In a binomial distribution the mean is 15 and variance is 10. Then, parameter n is

[MP PET]

(a) 28

(b) 16

(c) 45

(d) 25

Q 28. A random variable X has the following probability distribution

X	1	2	3	4
P(X)	k	2k	3k	4k

Then, the mean of X is

[J&K CET]

(a) 3

(b) 1

(c) 4

(d) 2

Q 29. The mean and variance of a random variable X having a binomial distribution are 4 and 2 respectively. Then, P $(X > 6)$ is equal to

[J&K CET]

(a) $\frac{1}{256}$

(b) $\frac{3}{256}$

(c) $\frac{9}{256}$

(d) $\frac{7}{256}$

Q 30. At a telephone enquiry system the number of phone calls regarding relevant enquiry follow poisson distribution with an average of 5 phone calls during 10 min time intervals. The

probability that there is at the most one phone call during a 10 min time period, is

[AIEEE]

(a) $\frac{5}{6}$

(b) $\frac{6}{55}$

(c) $\frac{6}{e^5}$

(d) $\frac{6}{5^e}$

Q 31. One hundred identical coins, each with probability p of showing heads are tossed once. If $0 < p < 1$ and the probability of head showing on 50 coins is equal to that of head showing on 51 coins, the value of p is

[DCE, RPET]

(a) $\frac{1}{2}$

(b) $\frac{51}{101}$

(c) $\frac{49}{101}$

(d) None of these

Q 32. In a book of 500 pages, it is found that there are 250 typing errors. Assume that poisson law holds for the number of errors per page. Then, the probability that a random sample of 2 pages will contain no error, is

[EAMCET]

(a) $e^{-0.3}$

(b) $e^{-0.5}$

(c) e^{-1}

(d) e^{-2}

Q 33. In a binomial distribution, the mean is 4 and variance is 3. Then, its mode is

[Jamia Millia Islamia, MP PET]

(a) 5

(b) 6

(c) 4

(d) None of these

Q 34. A random variable X can attain only the value $1,2,3,4,5$ with respective probabilities k, 2k, 3k, 2k, k. If m is the mean of the probability distribution, then (k, m) is equal to

[J&K CET]

(a) $\left(3,\frac{1}{9}\right)$

(b) $\left(\frac{1}{9},3\right)$

(c) $\left(\frac{1}{8}, 4\right)$

(d) $(1, 3)$

Q 35. If $f(x) = \lambda e^{-ax}$ $(a > 0)$ for $0 \leq x < \infty$ is a probability density, then λ is equal to

[J&K CET]

(a) a

(b) a^2

(c) $\frac{1}{a}$

(d) a^3

Q 36. A random variable X has poisson distribution with mean 2. Then, $P(X > 1.5)$ equals

[AIEEE]

(a) $\frac{3}{e^2}$

(b) $1-\frac{3}{e^2}$

(c) 0

(d) $\frac{2}{e^2}$

Q 37. If the range of a random variable X is $\{0, 1, 2, 3, 4, \ldots\}$ with $P(X = k) = \frac{(k+1)a}{3^k}$ for $k \geq 0$, then a is equal to

[EAMCET]

(a) $\frac{2}{3}$

(b) $\frac{4}{9}$

(c) $\frac{8}{27}$

(d) $\frac{16}{81}$

Q 38. For a binomial variate X with n = 6, if P(X = 2) = 9P(X = 4), then its variance is

[EAMCET]

(a) $\frac{8}{9}$

(b) $\frac{1}{4}$

(c) $\frac{9}{8}$

(d) 4

Q 39. An urn contains 4 white and 3 red balls. Three balls are drawn with replacement from this urn. Then, the standard deviation of the number of red balls drawn is

[J&K CET]

(a) $\frac{6}{7}$

(b) $\frac{36}{49}$

(c) $\frac{5}{7}$

(d) $\frac{25}{49}$

Q 40. If A and B each toss three coins. The probability that both get the same number of heads is

[J&K CET]

(a) $\frac{1}{9}$

(b) $\frac{3}{16}$

(c) $\frac{5}{16}$

(d) $\frac{3}{8}$

Q 41. A dice is thrown 5 times, then the probability that an even number will come up exactly 3 times, is

[DCE]

(a) $\frac{5}{16}$

(b) $\frac{1}{2}$

(c) $\frac{3}{16}$

(d) $\frac{3}{2}$

Q 42. A coin is tossed 10 times. The probability of getting exactly six heads, is

[UP SEE]

(a) $\frac{512}{513}$

(b) $\frac{105}{512}$

(c) $\frac{100}{153}$

(d) ${}^{10}C_6$

Q 43. A die is tossed thrice. If event of getting an even number is a success, then the probability of getting at least two successes is

[AMU]

(a) $\frac{7}{8}$

(b) $\frac{1}{4}$

(c) $\frac{2}{3}$

(d) $\frac{1}{2}$

Q 44. If X is a poisson variate with $P(X=0)$ $=0.8$, then the variance of X is

[EAMCET]

(a) $\log_e 20$

(b) $\log_{l0} 20$

(c) $\log_e \frac{5}{4}$

(d) 0

Q 45. In a binomial distribution, mean is 3 and standard deviation is $\frac{3}{2}$, then the probability distribution is

[Jamia Millia Islamia]

(a) $\left(\frac{3}{4}+\frac{1}{4}\right)^{12}$

(b) $\left(\frac{1}{4}+\frac{3}{4}\right)^{12}$

(c) $\left(\frac{1}{4}+\frac{3}{4}\right)^{9}$

(d) $\left(\frac{3}{4}+\frac{1}{4}\right)^{9}$

Q 46. If X has binomial distribution with mean np and variance npq, then $\frac{P(X=k)}{P(X=k-1)}$ is equal to

[RPET]

(a) $\frac{n-k}{k-1}\cdot\frac{p}{q}$

(b) $\frac{n-k+1}{k}\cdot\frac{p}{q}$

(c) $\frac{n+1}{k}\cdot\frac{q}{p}$

(d) $\frac{n-1}{k+1}\cdot\frac{q}{p}$

Q 47. The mean and variance of a random variable X having a binomial distribution are 4 and 2 respectively, then $P(X = 1)$ is

[AIEEE]

(a) $\frac{1}{32}$

(b) $\frac{1}{16}$

(c) $\frac{1}{8}$

(d) $\frac{1}{4}$

Q 48. The mean and variance of binomial distribution are 4 and 3 respectively. Then, the probability of getting exactly six success in this distribution is

[DCE]

(a) $^{16}C_6\left(\frac{1}{4}\right)^6\left(\frac{3}{4}\right)^{10}$

(b) $^{16}C_6\left(\frac{1}{4}\right)^{16}\left(\frac{3}{4}\right)^{20}$

(c) $^{16}C_6\left(\frac{1}{4}\right)^8\left(\frac{3}{4}\right)^{12}$

(d) ${}^{16}C_9\left(\frac{1}{4}\right)^{16}\left(\frac{3}{4}\right)^{20}$

Q 49. If a dice is thrown twice, the probability of occurrence of 4 at least once is

[UP SEE]

(a) $\frac{11}{36}$

(b) $\frac{35}{36}$

(c) $\frac{7}{12}$

(d) None of these

Q 50. The probability that in a family of 5 members, exactly 2 members have birthday on Sunday, is

[Kerala CEE]

(a) $\frac{12\times5^3}{7^5}$

(b) $\frac{10\times6^2}{7^5}$

(c) $\frac{2}{5}$

(d) $\frac{10\times6^3}{7^5}$

(e) 1

Q 51. For a poisson variate X, if $P(X = 2) = 3P(X = 3)$, then the mean of X is

[EAMCET]

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{1}{3}$

(d) $\frac{1}{4}$

Q 52. A random variate X takes the values $0, 1, 2, 3$ and its mean is 1.3. If $P(X = 3) = 2P(X = 1)$ and $P(X = 2) = 03$, then $P(X = 0)$ is equal to

[EAMCET]

(a) 0.1

(b) 0.2

(c) 0.3

(d) 0.4

Answers

Topic – 1 : Simple Problems on Probability

1. (c) 2. (c) 3. (b) 4. (d) 5. (c) 6. (c) 7. (b) 8. (c)

9. (b) 10. (b) 11. (b) 12. (a) 13. (d) 14. (c) 15. (d)

16. (d) 17. (a) 18. (c) 19. (b) 20. (a) 21. (c) 22.

(b) 23. (a) 24. (d) 25. (a) 26. (b) 27. (a) 28. (c)

29. (a) 30. (b) 31. (c) 32. (a) 33. (d) 34. (c) 35.

(c) 36. (b) 37. (c) 38. (d) 39. (d) 40. (c) 41. (d)

42. (a) 43. (b) 44. (d) 45. (a) 46. (e) 47. (b) 48.

(b) 49. (b) 50. (b) 51. (a) 52. (c) 53. (a) 54. (c)

55. (d) 56. (b) 57. (b) 58. (c) 59. (b) 60. (a)

61. (b) 62. (a) 63. (a) 64. (d) 65. (c) 66. (b) 67.

(d) 68. (c) 69. (b) 70. (a) 71. (b) 72. (c) 73. (d)

74. (c) 75. (b) 76. (d) 77. (d) 78. (b) 79. (b) 80.

(c) 81. (b) 82. (c) 83. (c)

Topic 2 : Odds in Favour and Against, Use of Permutations and Combinations

1. (b) 2. (c) 3. (d) 4. (d) 5. (b) 6. (c) 7. (b) 8. (a)

9. (c) 10. (b) 11. (b) 12. (c) 13. (c) 14. (c) 15. (c)

16. (d) 17. (d) 18. (b) 19. (b) 20. (a) 21. (a) 22.

(d) 23. (c) 24. (b) 25. (c) 26. (a) 27. (a) 28. (c)

29. (c)

Topic 3 : Addition Theorem, Conditional Probability and Baye's Theorems

1. (c) 2. (a) 3. (c) 4. (a) 5. (c) 6. (c) 7. (a) 8. (d)

9. (a) 10. (c) 11. (c) 12. (d) 13. (c) 14. (d) 15. (a)

16. (a) 17. (b) 18. (c) 19. (a) 20. (c) 21. (a) 22.

(d) 23. (d) 24. (c) 25. (d) 26. (a) 27. (c) 28. (a)

29. (d) 30. (a) 31. (b) 32. (b) 33. (c) 34. (a) 35.

(c) 36. (b) 37. (b) 38. (c) 39. (d) 40. (b) 41. (c)

42. (b) 43. (c) 44. (a) 45. (d) 46. (a) 47. (c) 48.

(b) 49. (d) 50. (c) 51. (a) 52. (c) 53. (b) 54. (a)

55. (d) 56. (a) 57. (d) 58. (c) 59. (a) 60. (c)

61. (c) 62. (a)

Topic 4 : Binomial, Poisson and Probability Distributions

1. (a) 2. (a) 3. (b) 4. (c) 5. (c) 6. (c) 7. (a) 8. (a)

9. (c) 10. (a) 11. (a) 12. (c) 13. (b) 14. (d) 15. (b)

16. (d) 17. (d) 18. (a) 19. (b) 20. (b) 21. (b) 22.

(e) 23. (b) 24. (c) 25. (c) 26. (d) 27. (c) 28. (a)

29. (c) 30. (c) 31. (b) 32. (c) 33. (c) 34. (b) 35.

(a) 36. (b) 37. (b) 38. (c) 39. (a) 40. (c) 41. (a)

42. (b) 43. (d) 44. (c) 45. (a) 46. (b) 47. (a) 48.

(a) 49. (a) 50. (d) 51. (a) 52. (d)

Miscellaneous

Useful Results & Formulae

1. (i) If $a * 0$, $b \in Z$, then 3 q, $r \in Z$ such that

$b = aq + r$,

$D < r < |a|$ (Division algorithm),

(ii) If $a \mid b$, then $a \mid -b$, $\forall$ $a, b \in Z$, $a \neq 0$

(iii) If $a \mid b$ and $b \mid c$, then $a \mid c$

(iv) If $a \mid b$ and $a \mid c$, then $a \mid bx + cy$ for some x, y

$\in Z$.

2. If $a = p_1^{\alpha_1}.p_2^{\alpha_2}.....p_n^{\alpha_n}$, where p_1, p_2, …,p_n are distinct primes and α_1, α_2,…..α_n are positive integers, then

(i) Number of positive divisors of 'a' is

$$T(a) = (\alpha_1 + 1)(\alpha_2 + 1).....(\alpha_n + 1)$$

(ii) Sum of all the positive divisors of 'a' is

$$S(a) = \frac{p_1^{\alpha_1+1} - 1}{p_1 - 1}.\frac{p_2^{\alpha_2+1} - 1}{p_2 - 1}.....\frac{p_n^{\alpha_n+1} - 1}{p_n - 1}$$

3. (i) The relation 'congruence modulo n' is an equivalence relation.

(ii) If $a \equiv b \pmod{m}$ and x is an integer, then $(a+x)$

$\equiv (b+x) \pmod{m}$ $(a-x) \equiv (b-x) \pmod{m}$

$ax = bx \pmod{m}$

(iii) $a \equiv b \pmod{m} \Rightarrow$ a and b leave the same

remainder when divided by m.

4. When n is a positive integer, then

(i) The last digit in 5^n is 5.

(ii) The last digit in 6^n is 6.

(iii) The last digit in 3^{4n} is 1.

(iv) The last digit in 9^n is 1 or 9 according as n is odd or even.

(v) The last digit in 7^{4n} is 1.

(vi) The last digit in $1! + 2! + \ldots + n!$, $n \geq 5$ is 0.

5. (i) To convert a decimal digit into its binary equivalent, divide the decimal number by 2 and note the remainder separately as the right most digit of binary equivalent. Again divide the quotient by 2 and note the remainder as next left digit of

binary number. Repeat this processes until quotient becomes 0.

(ii) To convert a fraction in decimal to binary form, multiply the fraction by 2 and note down its integral part. Repeat this processes unit, we get zeros in fractional part.

6. (i) To convert a binary number into its decimal equivalent, each digit of binary number is multiplied

by 2 having powers (0,1, 2, 3,...) and add all these numbers.

(ii) To convert a fraction in binary to decimal form, each digit of binary number is multiplied by 2 having negative powers (−l, + 2,−3...) and add all these numbers:

7. (i) Binary addition is similar to that of decimal numbers. 0 + 0=0, 0 + 1=1,1 + 0 = 1, 1 + 1=0 with a carry of 1.

(ii) Binary subtraction is similar to that in decimal numbers. $0-0=0$, $1-0 = 1$,$1-1=0$, $0-1=1$ with a borrow of 1 from next higher column.

8. (i) The octal number system has a base of 8. To convert an octal number to decimal number, we use the same processor as in the case of binary system, except that, we have a radix of 8 instead of 2.

(ii)

Three binary digits	000	001	010	011	100	101	110	111
Octal digits	0	1	2	3	4	5	6	7

9. The hexadecimal number system has a base of 16. In this case we need 16 symbols for representing various numbers.

Binary	00 00	00 01	0 0	0 0	01 00	0 1	0 1	01 11	1 0	1 0	10 10	1 0	11 00	11 01	1 1	11 11

			10	11		01	10		00	01		11			10	
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

10. Regula Falsi Method or False Position Method

Let

f(x) = 0 be a given algebraic equation whose one

real root lies in (a,b), then first approximation is

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

Second approximation $x_2 = \frac{x_1 f(b) - b f(x_1)}{f(b) - f(x_1)}$

11. In Newton–Raphson Method $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$

provided $f(x_{n-1}) \neq 0$.

12. (i) $\Delta y_{n-1} = y_n - y_{n-1}$ where Δ = forward

difference operator

(ii) In general $\Delta^n y_i = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i$

(iii) $\nabla f(x) = f(x) - f(x - h)$, where ∇ = Backward

difference operator and

$\nabla^2 f(x) = f(x) - 2f(x - h) + f(x - 2h)$.

13. (i) $E^n f(x) = f(x + nh)$, where E = shifting

operator (ii) $E^{-1} f(x) = f(x - h)$, where E^{-1} = Inverse

operator and $E^{-n} f(x) = f(x - nh)$

14. (i) $\Delta = E - 1$

(ii) $\nabla = 1 - E^{-1}$ and $E = (1 - \nabla)^{-1}$

15. (i) $\int_a^b f(x)dx = \int_{x_0}^{x_0+nh} ydx$

$= \frac{h}{2}[(y_o + y_n) + 2(y_1 + y_2 + \ldots + y_{n-1})]$ where $h = \frac{b-a}{n}$

[Trapezoidal Rule]

(ii) $\int_a^b f(x)dx = \int_{x_0}^{x_0+nh} ydx$

$= \frac{h}{3}[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + --^{+} y_{n-1}) + 2(y_2 + y_4 +$

$\ldots + y_{n-2})]$

where $h = \frac{b-a}{n}$ (Simpson's one third rule)

16. The total number of binary operation on a finite set containing n elements, is n^{n^2}.

17. A non-empty set G together with a binary operation '*' is called a group if

(i) $a, b \in G \Rightarrow a * b \in G$ (closure property)

(ii) $(a * b) * c = a*(b * c) \ \forall \ a, b, c \in G$

(Associative property)

(iii) For all $a \in G$, $\exists\ e \in G$ such that $a*e = a = e * a$, e is the identity element.

(iv) For all $a \in G \exists, b \in G$ such that $a * b = e = b * a$, is called inverse of a.

(v) If $a * b = b * a$ (commutative law), then group G is an abelian group.

18. (i) The identity element of a group (G, *) is unique,

(ii) The inverse of every element of (G, *) is unique.

(iii) $(a * b)^{-1} = b^{-1} * a^{-1}$

(iv) $a*a*a*\ldots*a = a^n$

19. (i) Addition modulo m is denoted by $+_m$ and defined as $a +_m b = r$, where r is the least non-negative remainder when a + b is divided by m, $0 \leq r < m$.

(ii) Multiplication modulo m is denoted by x_m and defined as $a x_m b = r$, where r is least non-negative remainder when ab is divided by m.

20. (i) A graph without loops and multiple edges is called a simple graph.

(ii) A graph with finite number of edges is called a finite graph.

(iii) The degree of a vertex is the number of edges incident on a vertex with self loops counted twice.

(iv) A vertex is called an even degree vertex if its degree is even and, an odd degree vertex if its degree is odd.

(v) If each edge of a graph G has a direction, then the graph is called a directed graph.

(vi) A graph G is connected if there exists a path between any two vertices in G.

Topic 1 Number Theory

Q 1. If ax + by = l, where a, b, x and y are integers, then which one of the following is not true ?.

[KCET]

(a) (a, y) = l

(b) (x, y) = 1

(c) (b, y) = 1

(d) (a, b) = l

Q 2. Thedigitintheunitplaceofthenumber! + 3^{7886}

is

[KCET]

(a) 7

(b) 3

(c) 1

(d) 9

Q 3. When 5^{99} is divided by 13, then the remainder is

[OJEE]

(a) 6

(b) 7

(c) 8

(d) 9

Q 4. The digit in the unit's place of 5^{834} is

[KCET]

(a) 0

(b) 1

(c) 3

(d) 5

Q 5. The remainder when $3^{100} \times 2^{50}$ is divided by 5 is

[KCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 6. If p and q are prime numbers satisfying the condition $p^2 - 2q^2 = 1$, then the value of $p^2 + 2q^2$ is

[KCET]

(a) 5

(b) 15

(c) 16

(d) 17

Q 7. The number of positive divisors of 252 is

[Manipal]

(a) 9

(b) 5

(c) 18

(d) 10

Q 8. The remainder when 5^{50} is divided by 7,

is

[Guj. CET]

(a) 1

(b) 2

(c) 4

(d) None of these

Q 9. If a and b are positive integers such that $a^2 - b^2$ is a prime number, then $a^2 - b^2$ is

[Guj. CET]

(a) a + b

(b) a – b

(c) ab

(d) 1

Q 10. The remainder obtained when 5^{124} is divided by 124 is

[KCET]

(a) 5

(b) 0

(c) 2

(d) 1

Q 11. Number of divisors of the form $(4n + 2)$, n ≥ 0 of the integer 240 is

[DCE]

(a) 4

(b) 8

(c) 10

(d) 3

Q 12. When 2^{301} is divided by 5, the least positive remainder is

[BITSAT]

(a) 4

(b) 8

(c) 2

(d) 6

Q 13. The GCD of 1080 and 675 is

[KCET]

(a) 145

(b) 135

(c) 225

(d) 125

Q 14. If $a \mid (b + c)$ and $a \mid (b - c)$ where a, b, c $\in$ N, then

[KCET]

(a) $b^2 \equiv c^2 \pmod{a^2}$

(b) $a^2+c^2 = b^2$

(c) $a^2 \equiv b^2 \pmod{c^2}$

(d) $c^2 \equiv a^2 \pmod{b^2}$

Q 15. If a, b and c $\in$ N, then which one of the following is not true ?

[KCET]

(a) a | b and a | c $\Rightarrow$ a | 3b + 2c

(b) a | b and b | c $\Rightarrow$ a | c

(c) a | (b + c) $\Rightarrow$ a | b and a | c

(d) a| b and a | c $\Rightarrow$ a | b + c

Q 16. The remainder obtained when

$(1!)^2 + (2!)^2 + (3!)^2 + \ldots + (100!)^2$

is divided by 10^2 is

[KCET]

(a) 27

(b) 28

(c) 17

(d) 14

Q 17. The last digit in 7^{300} is

[AMU]

(a) 7

(b) 9

(c) 1

(d) 3

Q 18. The unit's place digit in the number 13^{25} $+ 11^{25} - 3^{25}$ is

[BCECE]

(a) 0

(b) 1

(c) 2

(d) 3

Q 19. The number of divisors of 3×7^3, 7×11^2 and 2×61 are in

[BCECE]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 20. The digit in the units place in the number 7^{289} is

[KCET]

(a) 9

(b) 7

(c) 1

(d) 3

Q 21. The greatest integer which divides the number $(101^{100} - 1)$ is

[BCECE]

(a) 100

(b) 1000

(c) 10000

(d) None of these

Q 22. For any integer n > 1, the number of positive divisors of n is denoted by d(n). Then, for a prime P, $d(d(d(P^7)))$ =

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) P

Q 23. The least remainder when 17^{30} is divided by 5 is

[KCET]

(a) 2

(b) 1

(c) 4

(d) 3

Q 24. If x and y are odd integers, then $x^2 + y^2$ is

[J&K CET]

(a) an odd integer

(b) an even integer divisible by 4

(c) an even integer not divisible by 4

(d) None of the above

Topic 2 Binary Number

Q 1. The decimal equivalent of the binary number $(1111001101)_2$ is

[OJEE]

(a) $(970)_{10}$

(b) $(973)_{10}$

(c) $(975)_{10}$

(d) None of these

Q 2. The decimal equivalent to the binary number $(11001)_2$ is

[OJEE]

(a) $(28)_{10}$

(b) $(25)_{10}$

(c) $(20)_{10}$

(d) None of these

Q 3. The binary equivalent of 634.040625 is

[OJEE]

(a) 1001111010.101001

(b) 1110001001.101001

(c) 0111100101.110101

(d) None of the above

Q 4. The decimal equivalent to the binary number 10110110 is

[OJEE]

(a) 9

(b) 30

(c) 32

(d) 182

Q 5. What is the value of

[BITSAT]

$$\frac{(1001)_2^{(11)_2} - (101)_2^{(11)_2}}{(1001)_2^{(10)_2} + (1001)_2^{(01)_2}(101)_2^{(01)_2} + (101)_2^{(10)_2}}?$$

(a) $(1001)_2$

(b) $(101)_2$

(c) $(110)_2$

(d) $(100)_2$

Q 6. The difference of two numbers $(1100110011)_2$ and $(1101001011)_2$ in binary system is

[OJEE]

(a) $(100000)_2$

(b) $(101010)_2$

(c) $(11000)_2$

(d) $(10111)_2$

Q 7. $(10101101)_2 = (\ldots\ldots\ldots\ldots)_{10}$

[OJEE]

(a) 137

(b) 173

(c) 170

(d) None of these

Q 8. Give the binary equivalent of octal number $(5646)_8$

[OJEE]

(a) $(101111)2$

(b) $(1011101)_2$

(c) $(101110100110)_2$

(d) None of these

Q 9. The octal equivalent of the decimal number 0.3125 is

[OJEE]

(a) 0.24

(b) 0.42

(c) 0.39

(d) 0.98

Q 10. The hexadecimal equivalent of the binary number 111100010010001 is

[OJEE]

(a) 15C3

(b) C351

(c) 3C51

(d) C315

Topic 3 Numerical Method

Q 1. The velocity of a particle which starts from rest is given by the following table.

t (in second)	0	2	4	6	8	10
v (in m/s)	0	12	16	20	35	60

The total distance travelled (in metre) by the particles in 10 s, using Trapezoidal rule is given by

[EAMCET]

(a) 113

(b) 226

(c) 143

(d) 246

Q 2. For a certain function u_x, given that u_0 = 3, $u_1 = 12$, $u_2 = 81$, $u_3 = 200$, $u_4 = 100$, $u_5 = 8$, then $\Delta^5 u_x$ is equal to

[MHTCET]

(a) 750

(b) 778

(c) 765

(d) 755

Q 3. The value of $\left(\frac{\Delta^2}{E}\right)x^3$ at h = 1 is

[MHTCET]

(a) $8x$

(b) $6x$

(c) $5x^2$

(d) $6x^2$

Q 4. Consider four sub-intervals, the value of

$\int_0^1 \frac{1}{1+x} dx$ by

Trapezoidal rule, is

[MHT CET]

(a) 0.6870

(b) 0.6677

(c) 0.6977

(d) 0.5970

Q 5. By Simpson's rule, the value of $\int_1^2 \frac{dx}{x}$ dividing the interval $(1,2)$ into four equal parts, is

[MHT CET]

(a) 0.6932

(b) 0.6753

(c) 0.6692

(d) 7.1324

Q 6. The value of $\Delta \log f(x) + \Delta^2 (3^x)$ is

[MHTCET]

(a) $\log\left[1+\frac{\Delta f(x)}{f(x)}\right]+4.3^{x}$

(b) $\log\left[1+\frac{\Delta f(x)}{f(x)}\right]+3^{x}$

(c) $\log\left[\frac{\Delta f(x)}{1+f(x)}\right]+4.3^{x}$

(d) $\log\left[\frac{\Delta f(x)}{1+f(x)}\right]+3^{x}$

Q 7. By Simpson's rule the value of $\int_0^4 e^x\,dx$

where $e^0 = 1$, $e^1 = 2.72$, $e^2 = 739$, $e^3 = 20.09$ and

$e^4 = 54.60$, is

[MPPET]

(a) 53.79 sq unit

(b) 53.87 sq unit

(c) 53.98 sq unit

(d) 54.10 sq unit

Q 8. By using Newton–Raphsons method for finding the root of $x^4 - x - 10 = 0$ which is near to 2, the first iteration x_1 is equal to

[MP PET]

(a) 1.871

(b) 1.891

(c) 1.869

(d) 1.883

Q 9. In the usual notation the value of ΔV is equal to

[MHT CET]

(a) $\Delta - \nabla$

(b) $\Delta + \nabla$

(c) $\nabla - \Delta$

(d) None of these

Q 10. By the application of Simpson's one-third rule for numerical integration with two subintervals, the value of $\int_0^1 \frac{dx}{1+x}$ is

[MHT CET]

(a) $\frac{17}{36}$

(b) $\frac{17}{25}$

(c) $\frac{25}{36}$

(d) $\frac{17}{24}$

Q 11. The positive root of $x^2 - 78.8 = 0$ after first approximation by Newton–Raphson method assuming initial approximation to the root is 14, is

[MHT CET]

(a) 9.821

(b) 9.814

(c) 9.715

(d) 9.915

Q $12.$ For finding real root of the equation $x^2 - x = 2$ by Newton–Raphson method, choose $x_0 = 1$, then the value of x_2 is

[MPPET]

(a) $\frac{9}{15}$

(b) 3

(c) $\frac{11}{5}$

(d) $\frac{19}{5}$

Q 13. Dividing the interval $[0, 6]$ into 6 equal parts and by using trapezoidal rule the value of $\int_0^6 x^3 dx$ is approximately

[EAMCET]

(a) 330

(b) 331

(c) 332

(d) 333

Q 14. The value of $\left(\frac{\Delta^2}{E}\right)x^2$ at the interval h = 1 is

[MHTCET]

(a) 0

(b) 1

(c) 2

(d) 4

Q 15. Considering Four sub-intervals, the value of $\int_0^4 2^x dx$ by Simpson' s rule is

[MHT CET]

(a) $\frac{64}{8}$

(b) $\frac{65}{3}$

(c) $\frac{62}{12}$

(d) $\frac{61}{8}$

Q 16. By the False position method, the root of the equation $x^3 - 9x + 1 = 0$ lies in the interval $(2, 4)$ after first iteration. It is

[MP PET]

(a) 3

(b) 2.5

(c) 3.57

(d) 2.47

Q 17. Given $\log_e 2 = 0{,}69$. Approximate value of $\log_e 3$ by valuating $\int_2^3 \frac{dx}{x}$ by Simpson's rule with intervals, is

[DCE]

(a) 1.155

(b) 0.991

(c) 1.095

(d) 1.201

Q 18. If one root of the equation $x^3 + x^2 - 1 = 0$ is near to 1.0, then by Newton–Raphson method the first calculated approximate value of this root is

[BCECE]

(a) 0.9

(b) 0.6

(c) 1.2

(d) 0.8

Q 19. By Simpson's rule, the value of $\int_2^{10} \frac{dx}{1+x}$ by dividing the range into eight equal parts, is equal to

[BCECE]

(a) 1.2996

(b) 1.0996

(c) 1.1896

(d) 1.134

Q 20. If f(x) = a, a $\in$ R, then

[MHT CET]

(a) $\nabla f(x) = 0$

(b) $\nabla f(x) = a$

(c) $\nabla f(x) = 2a$

(d) $\nabla f(x) = a^2$

Q 21. $\Delta^2 (3e^x)$ is equal to

[MHT CET]

(a) $3e^x$

(b) $3(h-1)e^x$

(c) $3(e^h-1)^2 e^x$

(d) None of these

Q 22. $\Delta^2 y_0$ is equal to

[MHT CET]

(a) $2y_2 - 2y_1 - y_0$

(b) $y_2 - 2y_1 - y_0$

(c) $2y_2 - 2y_1 + y_0$

(d) $y_2 - 2y_1 + y_0$

Q 23. If $f(1) = 10$, $f(2) = 14$, then using

Newton's forward formula $f(1.3)$ is equal to

[MHT CET]

(a) 12.2

(b) 11.2

(c) 10.2

(d) 15.2

Q 33. For evaluating an integral $\int_a^b f(x)dx$.

Trapezoidal rule is employed as

[DCE]

(a) $[y_0 + y_1 + y_2 + \ldots\ldots + y_{n-1}]$

(b) $h[y_0 + y_1 + y_2 + \ldots\ldots + y_{n-1}]$

(c) $h/2\ [y_0 + y_n + 2(y_1 + y_2 + \ldots\ldots + y_{n-1})]$

(d) $h[y_0 + y_n + 2(y_1 + y_2 + \ldots\ldots + y_{n-1})]$

Q 34. Netwons–Raphson method for finding the root of the equation $f(x) = 0$ is

[DCE]

(a) $x_{n+1} = x_n - \frac{f'(x_n)}{f(x_n)}$

(b) $x_{n+1} = x_n + \frac{f'(x_n)}{f(x_n)}$

(c) $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

(d) $x_{n+1} = x_n + \frac{f(x_n)}{f'(x_n)}$

Q 35. By Simpson's rule taking n−A the value of the integral $\int_0^1 \frac{1}{1+x^2} dx$ is equal to

[DCE]

(a) 0.785

(b) 0.788

(c) 0.781

(d) None of these

Q 36. By Simpson's rule, the value of $\int_1^7 \frac{dx}{x}$ is

[AMU]

(a) 1.358

(b) 1.957

(c) 1.625

(d) 1.458

Q 37. The value of x_0 (the initial value of x) to get the solution in interval $(0.5, 0.75)$ of the equation $x^3 - 5x + 3 = 0$ by Newton–Raphson method is

[AMU]

(a) 0.5

(b) 0.75

(c) 0.625

(d) 0.60

Q 38. By bisection method, the real root of the equation $x^3 - 9x + 1 = 0$ lying between $x = 2$ and $x = 4$ is nearer to

[AMU]

(a) 2.2

(b) 2.75

(c) 5.5

(d) 4.0

Q 39. If $[2, 6]$ is divided into four intervals of equal length, then the approximate value of $\int_2^6 \frac{1}{x^2 - x} dx$ using Simpson's rule, is

[EAMCET]

(a) 0.3222

(b) 0.2333

(c) 0.5222

(d) 0.2555

Q 40. The root of the equation $x^3 + x - 3 = 0$, is in the interval $(1,2)$. After second iteration by False position method it will be in

[MPPET]

(a) $(1.178, 2.00)$

(b) $(1.25, 1.75)$

(c) (1.125, 1.375)

(d) (1.875, 2.00)

Q 41. An approximate evaluation of $\int_a^b f(x)dx$ using trapezoidal rule requires the division of [a, b] into

[J&K CET]

(a) an even number of sub−intervals

(b) an odd number of sub−intervals

(c) any number of sub−intervals

(d) None of the above

Topic 4 Group Theory

Q 1. For a, b $\in$ R, define a*b= $\frac{a}{a+b}$, where a + b $\neq 0$. If a * b = 5, then the value of b * a is

[Kerala CEE]

(a) 5

(b) −5

(c) 4

(d) -7

(e) -4

Q 2. In the group $17 = \{0,1,2,3,4,5\}$ under addition modulo 6, $(2 +_6 3^{-1} +_6 4)^{-1}$ is equal to

[KCET]

(a) 2

(b) 3

(c) 5

(d) 0

Q 3. Which one of the following is not true?

[KCET]

(a) Inverse of an element in a group is unique

(b) Fourth roots of unity form an additive abelian group

(c) Cancellation laws hold in a group

(d) Identity element in a group is unique

Q 4. The number of subgroups of the group $(Z_5, +_5)$ is

[KCET]

(a) 1

(b) 3

(c) 4

(d) 2

Q 5. On the set Z of all integers * is defined by $a*b = a + b - 5$. If $2*(x * 3) = 5$, then x is equal to

[KCET]

(a) 0

(b) 3

(c) 5

(d) 10

Q 6. Which one of the following is false?

[KCET]

(a) Addition is commutative in N

(b) Multiplication is associative in N

(c) If $a * b = a^b$ for all $a, b \in N$, then $*$ is commutative in N

(d) Addition is associative in N

Q 7. The set $\{-1,0,1\}$ is not a multiplicative group because of the failure of

[KCET]

(a) closure law

(b) associative law

(c) identity law

(d) inverse law

Q 8. Let S be a finite set containing n elements. Then, the total number of commutative binary operation on S is

[VITEEE]

(a) $n^{\left[\frac{n(n+1)}{2}\right]}$

(b) $n^{\left[\frac{n(n-1)}{2}\right]}$

(c) $n^{(n^2)}$

(d) $2^{(n^2)}$

Q 9. A group $(G,*)$ has 10 elements. The minimum number of elements of G, which are their own inverses is

[Manipal]

(a) 2

(b) 1

(c) 9

(d) 0

Q 10. In the group $(G, \otimes_{15})$, where G = $\{3,6,9,12\}$, $\otimes_{15}$ is multiplication modulo 15, the identity element is

[KCET]

(a) 3

(b) 6

(c) 12

(d) 9

Q 11. Which of the following is not a group with respect to the given operation ?

[KCET]

(a) The set of even integers under addition

(b) The set of odd integers under addition

(c) $\{0\}$ under addition

(d) $\{1,-1\}$ under multiplication

Q 12. In a group G = $\{1,3,7,9\}$ under

multiplication modulo 10, the inverse of 7 is

[VITEEE]

(a) 7

(b) 3

(c) 9

(d) 1

Q $13.$ In the group G – $\{1,2,3,4\}$ under multiplication modulo 5, $(3 \times_5 4)^{-1}$ is equal to

[VITEEE]

(a) 4

(b) 3

(c) 2

(d) 1

Q 14. On the set Q of all rational numbers the operation * which is both associative and commutative is given by a * b, is

[KCET]

(a) a + b + ab

(b) $a^2 + b^2$

(c) ab + 1

(d) $2a+3b$

Q 15. In the group G = $\{1,5,7,11\}$ under multiplication modulo 12, the solution of $7^{-1} \otimes_{12}$ (x $\otimes_{12} 11) = 5$ is equal to

[KCET]

(a) 5

(b) 1

(c) 7

(d) 11

Q 16. A subset of the additive group of real numbers which is not a subgroup is

[KCET]

(a) $(\{0\},+)$

(b) $(Z,+)$

(c) $(N, +)$

(d) $(Q, +)$

Q 17. The number of binary operations that can be defined on the set A = {a, b, c} is

[Kerala CEE]

(a) 3^3

(b) 3^4

(c) 3^9

(d) 9^3

(e) 3

Q 18. Let D = {1, 2, 3, 5, 6, 10, 15,30}. Define the operations '+','•' and '" on D as follows a + b =

LCM (a, b), a . b = GCD(a,b) and $a' = \frac{30}{a}$. Then, $(15'$ + 6) . 10 is equal to

[Kerala CEE]

(a) 1

(b) 2

(c) 3

(d) 5

(e) 10

Q 19. The identity element in the group

$$M = \left\{ \begin{bmatrix} x & x \\ x & x \end{bmatrix} \mid x \in R;\ x \neq 0 \right\}$$

with respect to matrix multiplication is

[KCET]

(a) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(b) $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Q 20. In the group $\{0, 1, 2, 4, 5\}$ under addition modulo 6 a subgroup is

[KCET]

(a) $\{0,2,5\}$

(b) $\{1,4,5\}$

(c) $\{0,1,3\}$

(d) $\{0,2,4\}$

Q 21. In the group $(Q^+, *)$ of positive rational numbers w.r.t. the binary operation $*$ defined by $a * b = \frac{ab}{3} \forall\ a, b \in Q^+$ the solution of the equation $5 * x = 4^{-1}$ in Q^+ is

[KCET]

(a) $\frac{27}{20}$

(b) $\frac{20}{27}$

(c) $\frac{1}{20}$

(d) 20

Q 22. Let $D_{70} = \{1, 2, 5, 7, 10, 14, 35, 70\}$.

Define '+', '.' and '' by $a + b = LCM(a, b)$, $a \cdot b = GCD(a,b)$ and $a' = \frac{70}{a}$ for all $a, b \in D_{70}$. The value of $(2 + 7)\ (14 \cdot 10)'$ is

[Kerala CEE]

(a) 7

(b) 4

(c) 35

(d) 5

(e) 1

Q 23. In Z, the set of all integers, the inverse of -7 w.r.t. * defined by $a * b = a + b + 7$ for all a, b $\in$ Z is

[AMU]

(a) -14

(b) 7

(c) 14

(d) -7

Q 24. Which of the following is a subgroup of

$G_1 = \{0, 1, 2, 3, 4, 5\}$, under addition modulo 6 ?

[AMU]

(a) $\{0,2\}$

(b) $\{0,1\}$

(c) $\{0,4\}$

(d) $\{0,3\}$

Q 25. Which of the following is true?

[KCET]

(a) The set of all fourth roots of unity is a multiplicative group

(b) The set of all cube roots of unity is an additive group

(c) $(ab)^{-1} = a^{-1} b^{-1}$ for all a, b in any group G

(d) If (ab) =a b for all a, b in any group G, then the group G is non−abelian

Q 26. The set of all integral multiples of 5 is a subgroup of

[KCET]

(a) The set of all rational numbers under multiplication

(b) The set of all integers under multiplication

(c) The set of all non−zero rational numbers under multiplication

(d) The set of all integers under addition

Q 27. If Q_1 is the set of all relations other than 1 with the binary operation * defined by a * b − a + b − ab for all a, b in Q_1, then the identity in Q_1 with respect to * is

[KCET]

(a) 1

(b) 0

(c) -1

(d) 2

Q 28. In the group $(1, 2, 3, 4, 5, 6)$ under multiplication modulo $7, 2^{-1} \times 4$ is equal to

[KCET]

(a) 1

(b) 4

(c) 2

(d) 3

Q 29. If (o) is a binary operation such that a o b = ab + 1 where a, b $\in$ Z, then the given function is

[DCE]

(a) commutative and associative

(b) commutative but not associative

(c) associative but not commutative

(d) neither associative nor commutative

Q 30. Which of the following is a group?

[KCET]

(a) $\{1, 2, 4, 8\}$ under multiplication

(b) $\{0, \pm 2, \pm 4, \pm 6, \ldots\}$ under addition

(c) $\{1, -1\}$ under addition

(d) $\{0, 1, 2, 3, 4\}$ under multiplication module 5

Q 31. z is a group of all positive rationals under the operation * defined by a * b = $\frac{ab}{2}$ the identity of z is

[AMU]

(a) 2

(b) 1

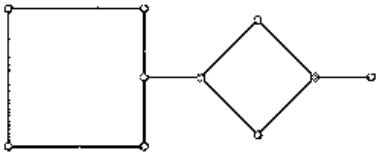
(c) 0

(d) 1/2

Topic 5 Graph Theory

Q 1. The adjoining graph is

[Manipal]



(a) Connected

(b) Disconnected

(c) Neither connected nor disconnected

(d) None of the above

Q 2. A graph G has 'm' vertices of odd degree and 'n' vertices of even degree. Then, which of the following statement is necessarily true?

[Manipal]

(a) m + n is an odd number

(b) m + n is an even number

(c) n + 1 is an even number

(d) m + 1 is an odd number

Answers

Topic 1 : Number Theory

1. (c) 2. (d) 3. (c) 4. (d) 5. (d) 6. (d) 7. (c) 8. (c)

9. (a) 10. (a) 11. (a) 12. (c) 13. (b) 14. (a) 15. (c)

16. (c) 17. (c) 18. (b) 19. (a) 20. (b) 21. (c) 22.

(c) 23. (c) 24. (c)

Topic 2 : Binary Number

1. (b) 2. (b) 3. (d) 4. (d) 5. (d) 6. (c) 7. (b) 8. (c)

9. (c) 10. (c)

Topic 3 : Numerical Method

1. (b) 2. (d) 3. (b) 4. (c) 5. (a) 6. (a) 7. (b) 8. (a)

9. (a) 10. (c) 11. (b) 12. (c) 13. (d) 14. (c) 15. (b)

16. (d) 17. (c) 18. (d) 19. (a) 20. (a) 21. (c) 22.

(d) 23. (b) 24. (a) 25. (a) 26. (c) 27. (b) 28. (d)

29. (b) 30. (c) 31. (d) 32. (d) 33. (c) 34. (c) 35.

(a) 36. (b) 37. (b) 38. (b) 39. (c) 40. (a) 41. (c)

Topic 4 : Group Theory

1. (e) 2. (b) 3. (b) 4. (d) 5. (d) 6. (c) 7. (d) 8. (a)

9. (b) 10. (b) 11. (b) 12. (b) 13. (b) 14. (a) 15. (b)

16. (c) 17. (c) 18. (b) 19. (b) 20. (d) 21. (a) 22.

(a) 23. (d) 24. (d) . 25. (a) 26. (d) 27. (b) 28. (c)

29. (b) 30. (b) 31. (a)

Topic 5 : Graph Theory

1. (a) 2. (b)

Sets, Relations and Functions

Type of Sets, Operation and Cartesian Product

Q 1. Let P = $\{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$ and Q

= $\{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$ be two sets. Then,

[IIT JEE]

(a) $P \subset Q$ and $Q - P \neq \varnothing$

(b) $Q \not\subset P$

(c) $P \not\subset Q$

(d) P = Q

Q 2. Let A, B, C be pair wise independent events with P(C) > 0 and P(A ∩ B ∩ C) = 0. Then, P(A^C ∩ B^C| C) is equal to

[AIEEE]

(a) P(A^C) − P(B)

(b) P(A) − P(B^C)

(c) P(A^C) + P(B^C)

(d) P(A^C) − P (B^C)

Q 3. The set A = {x : x $\in$ R, $x^2 = 16$ and $2x = 6$}

equals to

[BITSAT]

(a) ϕ

(b) {14, 3, 4}

(c) {3}

(d) {4}

Q 4. Let A and B be two sets, then (A $\cup$ B)' $\cap$

(A' $\cap$ B) is equal to

[GGSIPU]

(a) A'

(b) A

(c) B'

(d) None of these

Q 5. Let A = {l, 2}, B = {{1},{2}},C={{1, 2}}. Then,

which of the following relation is true?

[J&K CET]

(a) $A = B$

(b) $B \subseteq C$

(c) $A \in C$

(d) $A \subset C$

Relation, Equivalence Relation

Q 6. Let R be the set of real numbers.

Statement I $A = \{(x, y) \in R \times R : y - x$ is an integer$\}$ is an equivalence relation on R.

Statement II $B = \{(x, y) \in R \times R : x = \alpha y$ for some

rational number $\alpha\}$ is an equivalence relation on R.

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

Statement I.

(b) Statement I is true, Statement II is false.

(c) Statement I is false, Statement II is true.

(d) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement

I.

Q 7. If A and B are two equivalence relations

defined on set C, then

[UP SEE]

(a) $A \cap B$ is an equivalence relation

(b) $A \cap B$ is not an equivalence relation

(c) $A \cup B$ is an equivalence relation

(d) A $\cup$ B is not an equivalence relation

Type of Mapping

Q 8. The function f : $[0, \infty) \to [0, \infty)$ defined by f

(x) = $\frac{2x}{1+2x}$ is

[J&K CET]

(a) one–one and onto

(b) one–one but not onto

(c) not one–one but onto

(d) neither one–one nor onto

Domain–Range, Odd–Even and Periodic Function

Q 9. The even function of the following is

[WB JEE]

(a) $f(x)=\frac{a^x+a^{-x}}{a^x-a^{-x}}$

(b) $f(x)=\frac{a^x+1}{a^x-1}$

(c) $f(x)=x.\frac{a^x-1}{a^x+1}$

(d) $f(x)=\log_2(x+\sqrt{x^2+1})$

Q 10. The domain of $\cos^{-1}\frac{x-3}{2}-\log_{10}(4-x)$ is

[GGSIPU]

(a) $(1, 4)$

(b) $[1,4)$

(c) $(1,4]$

(d) $[1,4]$

Q 11. The range of the function $f(x) = \log_e(3x^2 + 4)$ is equal to

[Kerala CEE]

(a) $[\log_e 2, \infty)$

(b) $[\log_e 3, \infty)$

(c)$[2 \log_e . 3, \infty)$

(d) $[0, \infty)$ (e) $[2 \log_e 2, \infty)$

Q 12. The period of the function f(x) = | sin 2x |

+1 cos 8x | is

[Kerala CEE]

(a) 2π

(b) π

(c) $\frac{2\pi}{3}$

(d) $\frac{\pi}{2}$

(e) $\frac{\pi}{4}$

Q 13. The domain of $f(x) = \sin^{-1}\left[\log_2\left(\frac{x}{2}\right)\right]$ is

[KCET]

(a) $0 \leq x \leq 1$

(b) $0 \leq x \leq 4$

(c) $1 \leq x \leq 4$

(d) $4 \leq x \leq 6$

Q 14. If $f(x) = 3 - x, -4 \leq x \leq 4$, then the domain of $\log_e (f(x))$ is

[J&K CET]

(a) $[-4, 4]$

(b) $(-\infty, 3]$

(c) $(-\infty, 3)$

(d) $[-4, 3)$

Inverse, Composition and Different Types of Functions

Q 15. Let $f(x) = x^2$ and $g(x) = \sin x \ \forall \ x \in R$.

Then, the set of all x satisfying (fogogof) (x) =

(gogof) (x), where (fog)(x) = f(g(x)),is

[IITJEE]

(a) $\pm\sqrt{n\pi}$, $n \in \{0, 1, 2,.....\}$

(b) $\pm\sqrt{n\pi}$, $n \in \{1, 2,.....\}$

(c) $\frac{\pi}{2} + 2n\pi$, n $\in \{\ldots\ldots, -2, -1, 0, 1, 2, \ldots.\}$

(d) $2n\pi$, n $\in \{\ldots\ldots\ldots, -2, -1, 0, 1, 2, \ldots\ldots\}$

Q 16. Let f : $(0,1) \rightarrow$ R be defined by f(x) =

$\frac{b-x}{1-bx}$, where b is a constant such that $0 < b < 1$.

Then,

[IIT JEE]

(a) f is not invertible on $(0, 1)$

(b) $f \neq f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$

(c) $f = f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$

(d) f^{-1} is differentiable on $(0, 1)$

Q 17. If $f(x + 2y, x - 2y) = xy$, then $f(x, y)$ is equal to

[WB JEE]

(a) $\frac{1}{4}xy$

(b) $\frac{1}{4}(x^2 - y^2)$

(c) $\frac{1}{8}(x^2 - y^2)$

(d) $\frac{1}{2}(x^2 + y^2)$

Q 18. If f(x) is a polynomial function of the second degree such that $f(-3) = 6$, $f(0) = 6$ and $f(2) = 11$, then the graph of the function f(x) cuts the ordinate $x = 1$ at the point

[GGSIPU]

(a) $(1,8)$

(b) $(1,4)$

(c) $(1,-2)$

(d) None of these

Q 19. If $f(x) = 4x^3 + 3x^2 + 3x + 4$, then $x^3 f\left(\frac{1}{x}\right)$ is

[MP PET]

(a) $f(-x)$

(b) $\frac{1}{f(x)}$

(c) $\left[f\left(\frac{1}{x}\right)\right]^2$

(d) $f(x)$

Q 20. If the function f : $[1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$ then $f^{-1}(x)$ is

[Kerala CEE]

(a) $\left(\frac{1}{2}\right)^{x(x-1)}$

(b) $\frac{1}{2}(1-\sqrt{1+4\log_2 x})$

(c) $\frac{1}{2}\sqrt{1+4\log_2 x}$

(d) $\frac{1}{2}(1+\sqrt{1+4\log_2 x})$

(e) Not defined

Q 21. Let f = {(0, −1), (−1,−3), (2, 3,), (3, 5)} be a function from z to z defined by f(x) = ax + b.

Then,

[AMU]

(a) a = 1, b = − 2

(b) a = 2, b = 1

(c) a = 2, b = −1

(d) a = 1, b = 2

Q 22. If g (x) is the inverse of f(x) and f'(x) = cos x, then g' (x) is equal to

[J&K CET]

(a) sec x

(b) sec (g (x))

(c) cos(g (x))

(d) −sin(g (x))

2. Complex Numbers

Algebraic Operations and Equality of Complex Numbers

Q 23. If $x = \frac{-3+i\sqrt{3}}{2}$ is a complex number, then the value of $(x^2 + 3x)^2(x^2 + 3x + 1)$ is

[UP SEE]

(a) $-\frac{9}{8}$

(b) 6

(c) -18

(d) 36

Q 24. If $2x = 3 + 5i$, then the value of $2x^3 + 2x^2 - 7x + 72$ is

[MP PET]

(a) 4

(b) −4

(c) 8

(d) −8

Conjugate, Modulus and Argument

Q 25. If z is any complex number satisfying | z − 3 − 2i | ≤ 2, then the maximum value of |2z − 6 + 5i | is……

[IIT JEE]

Q 26. If $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$, then $\frac{z_1}{z_2}$ is

[MP PET]

(a) purely real

(b) purely imaginary

(c) zero of purely imaginary

(d) neither real nor imaginary

Q 27. If z_1 and z_2 are two non−zero complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$, then arg $\left(\frac{z_1}{z_2}\right)$ is

[Kerala CEE]

(a) 0

(b) $-\pi$

(c) $-\frac{\pi}{2}$

(d) $\frac{\pi}{2}$

(e) π

Q 28. The value of $\left|\frac{1+i\sqrt{3}}{\left(1+\frac{1}{i+1}\right)^2}\right|$ is

[KCET]

(a) 20

(b) 9

(c) $\frac{5}{4}$

(d) $\frac{4}{5}$

De–Moivre's Theorem, Square Root and Roots of Unity

Q 29. Let $\omega - e^{i\pi/3}$ and a, b, c, x, y, z be non–zero complex numbers such that a + b + c = x, a+ $b\omega + c\omega^2 = y$. $a + b\omega^2 + c\omega = z$. Then, the value of $\frac{|x|^2+|y|^2+|z|^2}{|a|^2+|b|^2+|c|^2}$ is.....

[IITJEE]

Q 30. $\left[\frac{1+\cos\left(\frac{\theta}{2}\right)-i\sin\left(\frac{\theta}{2}\right)}{1+\cos\left(\frac{\theta}{2}\right)+i\sin\left(\frac{\theta}{2}\right)}\right]^{4n}$ is equal to

[UP SEE]

(a) $\cos n\theta - i \sin n\theta$

(b) $\cos n\theta + i \sin n\theta$

(c) $\cos 2n\theta - i \sin 2n\theta$

(d) $\cos 2n\theta + i \sin 2n\theta$

Q 31. If $\omega \neq 1$ is a cube root of unity, then the sum of the series $S = 1 + 2\omega + 3\omega^2 + \ldots\ldots + 3n\omega^{3n-1}$ is

[WB JEE]

(a) $\frac{3n}{\omega-1}$

(b) $3n(\omega - 1)$

(c) $\frac{\omega-1}{3n}$

(d) 0

Applications in Geometry

Q 32. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n - \alpha^n - \beta^n$ for $n \leq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is

[IIT JEE]

(a) 1

(b) 2

(c) 3

(d) 4

Q 33. Let z_1, z_2 be two roots of the equation z^2 + az + b = 0, z being complex number. Further assume that the origin, z_1 and z_2 form an equilateral triangle. Then,

[GGSIPU]

(a) $a^2 = b$

(b) $a^2 = 2b$

(c) $a^2 = 3b$

(d) $a^2 = 4b$

Q 34. If P(x,y) denotes z=x + iy in Argand's plane and $\left|\frac{z-1}{z+2i}\right| = 1$, then the locus of P is a/an

[KCET]

(a) hyperbola

(b) ellipse

(c) circle

(d) straight line

3. Equations and Inequalities

Surds and Indices

Q 35. The value of $\frac{(\sqrt{\sqrt{3}+1}+\sqrt{\sqrt{3}-1})(\sqrt{3}-\sqrt{2})}{(\sqrt{\sqrt{3}+1}+\sqrt{\sqrt{3}-1})}$ is

[UP SEE]

(a) $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}}$

(b) 1

(c) $\sqrt{3}$

(d) $\frac{1}{\sqrt{3}}$

Q 36. Let a, b > 0 satisfy $a^3 + b^3 = a - b$, then

[AMU]

(a) $a^2 + b^2 > 1$

(b) $a^2 + b^2 < 0$

(c) $a^2 + b^2 = 1$

(d) $a^2 + ab + b^2 < 1$

Logarithm and Its Properties

Q 37. Let (x_0, y_0) be the solution of the following equations $(2x)^{\log 2} = (3y)^{\log 3}$, $3^{\log x} = 2^{\log y}$. Then, x_0 is

[IIT JEE]

(a) $\frac{1}{6}$

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) 6

Q 38. The sum of n terms of the series $\sum_{x=1}^{n} \log \frac{2^x}{3^{x-1}}$

is

[UP SEE]

(a) $\log\left(\frac{2^{n-1}}{3^{n+1}}\right)^{n/2}$

(b) $\log\left(\frac{2^{n+1}}{3^{n-1}}\right)^{n/2}$

(c) $\log\left(\frac{3^{n-1}}{2^{n+1}}\right)^{n/2}$

(d) $\log\left(\frac{3^{n+1}}{2^{n-1}}\right)^{n/2}$

Q 39. The sequence log a, $\log\frac{a^2}{b}, \log\frac{a^3}{b^2}$, ...is

[WBJEE]

(a) a GP

(b) an AP

(c) a HP

(d) both a GP and a HP

Q 40. $\log_4 2 - \log_8 2 + \log_{16} 2 - \ldots$ is equal to

[MP PET]

(a) e^2

(b) $\log_e 2 + 1$

(c) $\log_e 3 - 2$

(d) $1 - \log_e 2$

Q 41. If $\log_3 2$, $\log_3(2^x - 5)$ and $\log_3\left(2^x - \frac{7}{2}\right)$ are in AP, then x is equal to

[AMU]

(a) 8

(b) 3

(c) -8

(d) -3

Q 42. If $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$, then the value of x^{b+c}. y^{c+a} . z^{a+b} is

[KCET]

(a) 1

(b) 2

(c) 0

(d) -1

Relation Between Roots and Coefficients

Q 43. If α and β are the roots of the equation $x^2 - 2x + 4 = 0$, then the value of $\alpha^n + \beta^n$ will be

[BITSAT]

(a) $i2^{n+1} \sin(n\pi/3)$

(b) $2^{n+1}\cos(n\pi/3)$

(c) $i2^{n-1} \sin(n\pi/3)$

(d) $2^{n-1}\cos(n\pi/3)$

Q 44. If α, β and γ are the roots of $x^3 - 2x + 1 = 0$, then the value of $\sum\left(\frac{1}{\alpha+\beta-\gamma}\right)$ is

[KCET]

(a) $-\frac{1}{2}$

(b) -1

(c) 0

(d) $\frac{1}{2}$

Q 45. If $\sin\theta$ and $\cos\theta$ are the roots of the equation $ax^2 + bx + c = 0$, $a \neq 0$, then the relation between the coefficients of the equation is

[J&K CET]

(a) $a^2 - b^2 + 2ac = 0$

(b) $a^2 + b^2 + 2ac = 0$

(c) $a^2 - b^2 - 2ac = 0$

(d) $a^2 + b^2 - 2ac = 0$

Solution of Equations, Nature of Roots

Q 46. Let α, β be real and z be a complex number. If $z^2 + \alpha z + \beta = 0$ has two distinct roots on the line Re $(z) = 1$, then it is necessary that

[AIEEE]

(a) $\beta \in (-1, 0)$

(b) $|\beta| = 1$

(c) $\beta \in (1, \infty)$

(d) $\beta \in (0, 1)$

Q 47. If one of the root of $2x^2 - cx + 3 = 0$ is 3 and another equation $2x^2 - cx + d = 0$ has equal roots where c and d are real numbers, then d is

[UP SEE]

(a) 3

(b) $49/8$

(c) $8/49$

(d) -3

Q 48. Let a, b, c be three real numbers, such that $a + 2b + 4c = 0$. Then, the equation $ax^2 + bx + c = 0$

[WB JEE]

(a) has both the roots complex

(b) has its roots lying within $-1 < x < 0$

(c) has one of roots equal to $\frac{1}{2}$

(d) has its roots lying within $2 < x < 6$

Q 49. The number of integral values of b, for which the equation $x^2 + bx - 16 = 0$ has integral roots, is

[Kerala CEE]

(a) 2

(b) 3

(c) 4

(d) 5

(e) 6

Q 50. The number of real values of x which satisfy the equation $\left|\frac{x}{|x-1|}\right|+|x|=\frac{x}{|x-1|}$ is

[AMU]

(a) 2

(b) 1

(c) infinite

(d) zero

Condition for Common Roots, Quadratic Expressions and Position of Roots

Q 51. A value of b for which the equations $x^2 + bx - 1 = 0$, $x^2 + x + b = 0$ have one root in common, is

[IIT JEE]

(a) $-\sqrt{2}$

(b) $-i\sqrt{3}$

(c) $i\sqrt{5}$

(d) $\sqrt{2}$

Q 52. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $\frac{k}{\alpha}, \frac{k}{\beta}$ is

[MP PET]

(a) $cx^2 + kbx + k^2a = 0$

(b) $cx^2 + k^2bx + ka = 0$

(c) $kcx^2 + bx + k^2a = 0$

(d) $k^2cx^2 + bx + ka = 0$

Q 53. If both the roots of the equation $x^2 - 6ax + 2 - 2a + 9a^2 = 0$ exceed 3, then

[AMU]

(a) $a < \frac{1}{2}$

(b) $a > \frac{1}{2}$

(c) $a < 1$

(d) $a > \frac{11}{9}$

Solution of Inequalities

Q 54. The minimum value of the sum of real numbers a^{-5}, a^{-4}, $3a^{-3}$, 1, a^{8} and a^{10} with $a > 0$ is……

[IITJEE]

Q 55. If $|2x - 3| < |x + 5|$, then x lies in the interval

[Kerala CEE]

(a) $(-3, 5)$

(b) $(5, 9)$ (c) $\left(\frac{-2}{3}, 8\right)$

(d) $\left(-8, \frac{2}{3}\right)$

(e) $\left(-5, \frac{2}{3}\right)$

Q 56. The solution set of the inequality $4^{-x+\frac{1}{2}} - 7\,(2^{-x}) - 4 < 0, \forall\ x \in R$ is

[AMU]

(a) $(-\infty, 2)$

(b) $(-2, \infty)$

(c) $(-\infty, \infty)$

(d) $(2, \infty)$

4. Sequences and Series

Arithmetic Progression and Its Mean

Q 57. Let $a_1, a_2, a_3, \ldots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^{p} a_i$, $1 \leq p \leq 100$.

For any integer n with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n, then a_2 is……

[AIEEE]

Q 58. A man saves Rs 200 in each of the first three months of his service. In each of the subsequent months his saving increases by Rs 40 more than the saving of immediately previous month. His total saving from the start of service will be Rs 11040 after

[AIEEE]

(a) 19 months

(b) 20 months

(c) 21 months

(d) 18 months

Q 59. Let a_n be the nth term of an AP. If $\sum_{r=1}^{100} a_{2r} = \alpha$ and $\sum_{r=1}^{100} a_{2r-1} = \beta$, then the common difference of the AP is

[AIEEE]

(a) $\frac{\alpha - \beta}{200}$

(b) $\alpha - \beta$

(c) $\frac{\alpha - \beta}{100}$

(d) $\beta - \alpha$

Q 60. If the numbers a, b, c, d, e form an AP with a = 1, then a − 4b + 6c − 4d + e is equal to

[MP PET]

(a) 1

(b) 2

(c) 0

(d) 3

Q 61. The first four terms of an AP are a, 9, 3a − b, 3a + b. The 2011th term of the AP is

[Kerala CEE]

(a) 2015

(b) 4025

(c) 5030

(c) 8045

(e) 6035

Q 62. If S_1, S_2 and S_3 are the sum of n, $2n$ and $3n$ terms respectively of an arithmetic progression, then

[J&K CET]

(a) $S_3 = 2(S_1 + S_2)$

(b) $S_3 = S_1 + S_2$

(c) $S_3 = 3(S_2 - S_1)$

(d) $S_3 = 3(S_2 + S_1)$

Geometric Progression and Its Mean

Q 63. If x, y and z are in geometric progression, then $\log_x 10$, $\log_y 10$ and $\log_z 10$ are in

[UP SEE]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 64. The value of x which satisfies $8^{1+\cos x+\cos^2 x.....} = 64$ in $[-\pi, \pi]$ is

[GGSIPU]

(a) $\pm\frac{\pi}{2}, \pm\frac{\pi}{3}$

(b) $\pm\frac{\pi}{3}, \pm\frac{2\pi}{3}$

(c) $\pm\frac{\pi}{2}, \pm\frac{\pi}{6}$

(d) $\pm\frac{\pi}{6}, \pm\frac{\pi}{3}$

Q 65. If a, b and c are in GP and $a^{\frac{1}{x}} = b^{\frac{1}{y}} = c^{\frac{1}{3}}$,

then x, y, z are in

[GGSIPU]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 66. If the sequence $\{a_n\}$ is in GP, such that

$a_4/a_6 = 1/4$ and $a_2 + a_5 = 216$, then a_1 is equal to

[MP PET]

(a) 12 or 108/7

(b) 10

(c) 7 or 54/7

(d) None of these

Relation Between AP, GP and HP

Q 67. If AM and HM between two numbers are 27 and 12 respectively, then their GM is

[BITSAT]

(a) 9

(b) 18

(c) 24

(d) 36

Special Series, Sum to n Terms and Infinite Number of Terms

Q 68. If $H_n = 1 + \frac{1}{2} + ... + \frac{1}{n}$, then the value of

$S_n = 1 + \frac{3}{2} + \frac{5}{3} + ... + \frac{2n-1}{n}$ is

[AMU]

(a) $H_n + 2n$

(b) $n - 1 + H_n$

(c) $H_n - 2n$

(d) $2n - H_n$

Q 69. If n is an odd positive integer and $(1 + x + x^2 + x^3)^n = \sum_{r=0}^{3n} a_r x^r$, then $a_0 - a_1 + a_2 - a_3 + \ldots - a_{3n}$ is equal to

[KCET]

(a) 4^n

(b) 1

(c) -1

(d) 0

Logarithmic Series

Q 70. The value of $4 + 2(1 + 2) \log 2$

$+\frac{2(1+2^2)}{2!}(\log 2)^2 + \frac{2(1+2^3)}{3!}(\log 2)^3$ +... is

[UP SEE]

(a) 10

(b) 12

(c) $\log (3^2 \cdot 4^2)$

(d) $\log (2^2 \cdot 3^2)$

Q 71. The sum of the series $\frac{1}{1.2} - \frac{1}{2.3} + \frac{1}{3.4} - .....\infty$ is

[WB JEE]

(a) $21\text{og}_e 2 + 1$

(b) $2\ 1\text{og}_e 2$

(c) $2\ \log_e 2 - 1$

(d) $\log_e 2 - 1$

Q 72. $\frac{2}{1}.\frac{1}{3}+\frac{3}{2}.\frac{1}{9}+\frac{4}{3}.\frac{1}{27}+\frac{5}{4}.\frac{2}{81}+.....$ is equal to

[MP PET]

(a) $\frac{1}{2}+\log_e\left(\frac{3}{2}\right)$

(b) $\log_e 3 - \log_e 2$

(c) $\log 6$

(d) $\log_e 2 - \log_e 3$

Exponential Series

Q 73. The value of $\frac{2}{1!}+\frac{2+4}{2!}+\frac{2+4+6}{3!}+.....\infty$ is

[BITSAT]

(a) e

(b) 2e

(c) 3e

(d) None of these

Q 74. $\frac{1+\frac{1}{2!}+\frac{2}{3!}+\frac{2^2}{4!}+\frac{2^3}{5!}+.....}{1+\frac{1}{2!}+\frac{1}{4!}+\frac{1}{6!}+.....}$ is equal to

[MP PET]

(a) $e/4$

(b) $8e$

(c) $e/2$

(d) $\frac{e(e^2-1)}{2(e^2+1)}$

Q 75. The value of $\sum_{n=0}^{\infty}\frac{n^2+4}{n!}$ is equal to

[J&KCET]

(a) $6e$

(b) $5e$

(c) 4e

(d) None of these

5. Permutations and Combinations

Number of Permutations, Conditional Permutations

Q 76. In how many ways can 5 prizes be distributed among four students when every student can take one or more prizes?

[BITSAT]

(a) 1024

(b) 625

(c) 120

(d) 600

Q 77. The number of ways in which the digits 1, 2, 3, 4, 3, 2, 1 can be arranged so that the odd digits always occupy the odd places is

[MP PET]

(a) 6

(b) 12

(c) 18

(d) 24

Q 78. The number of integers greater than 6000 that can be formed with 3, 5, 6, 7 and 8, where no digit is repeated, is

[Kerala CEE]

(a) 120

(b) 192

(c) 216

(d) 72

(e) 202

Number of Combinations, Division into Groups and De arrangements

Q 79. A teacher takes 3 children from her class to the zoo at a time as often as she can, but she does not take the same three children to the zoo more than once. She finds that she goes to the zoo 84 times more that a particular child goes to the zoo. The number of children in her class is

[GGSIPU]

(a) 12

(b) 10

(c) 60

(d) None of these

Q 80. **Statement** I The number of ways distributing 10 identical balls in 4 distinct boxes such that no box is empty is ${}^{9}C_{3}$.

Statement II The number of ways of choosing any 3 places from 9 different places is ${}^{9}C_{3}$.

[AIEEE]

(a) Statement I is true, Statement **II is** true; Statement **II** is not a correct explanation for Statement I.

(b) Statement I is true, Statement **II** is false.

(c) Statement I is false, Statement **II** is true.

(d) Statement I is true, Statement **II** is true;

Statement **II** is a correct explanation for Statement

I.

Q 81. A student is allowed to select atmost n books from a collection of $(2n + 1)$ books. If the total number of ways in which he can select a book is 255, then the value of n equals to

[J&K CET]

(a) 6

(b) 5

(c) 4

(d) 3

Used in Geometrical Problem

Q 82. There are 10 points in a plane, out of these 6 are collinear. If N is the number of triangles formed by joining these points, then

[AIEEE]

(a) N > 190

(b) $N < 100$

(c) $100 < N \leq 140$

(d) $140 < N \leq 190$

6. Binomial Theorem and Mathematical Induction

Expansion of Binomial Theorem and General Term

Q 83. If A and B are coefficients of x^n in the expansions of $(1 + x)^{2n}$ and $(1 + x)^{2n-1}$ respectively, then A/B is equal to

[WB JEE, AMU]

(a) 4

(b) 2

(c) 9

(d) 6

Q 84. In the expansion of $\left(x^3 - \frac{1}{x^2}\right)^n$, $n \in N$, if the sum of the coefficient of x^5 and x^{10} is 0, then n is

[GGSIPU]

(a) 25

(b) 20

(c) 15

(d) None of these

Q 85. 5th term from the end in the expansion of $\left(\frac{x^3}{2}-\frac{2}{x^2}\right)^{12}$ is

[GGSIPU]

(a) $-7920\ x^{-4}$

(b) $7920\ x^4$

(c) $7920\ x^{-4}$

(d) $-7920\ x^4$

Q 86. In the expansion of $(1 - 3x + 3x^2 - x^3)^{2n}$,

the middle term is

[MP PET]

(a) $(n + l)$th term

(b) $(2n + 1)$th term

(c) $(3n + 1)$th term

(d) None of these

Q 87. If $(1 + ax)^n = 1 + 6x + \frac{27}{2}x^2 + \ldots + a^n x^n$,

then the values of a and n are respectively

[Kerala CEE]

(a) $2, 3$

(b) $3, 2$

(c) $\frac{3}{2}, 4$

(d) $1,6$

(e) $\frac{3}{2},6$

Q 88. If rth and $(r + l)$th terms in the expansion of $(p + q)^n$ are equal, then $\frac{(n+1)q}{r(p+q)}$ is

[KCET]

(a) 0

(b) 1

(c) $\frac{1}{4}$

(d) $\frac{1}{2}$

Properties of Binomial Coefficients

Q 89. Let $C_0, C_1, \ldots, C_n$ denotes the binomial coefficients in the expansion of $(1 + x)^n$. The value of $C_1 - 2C_2 + 3C_3 - 4C_4 + \ldots\ldots$(upton terms) is

[UP SEE]

(a) 2^n

(b) 2^{-n}

(c) 0

(d) 1

Q 90. If ${}^{n}C_4$, ${}^{n}C_5$ and ${}^{n}C_6$ are in AP, then n is

[WB JEE]

(a) 7 or 14

(b) 7

(c) 14

(d) 14 or 21

Q 91. ${}^{15}C_3 + {}^{15}C_5 + \ldots + {}^{15}C_{15}$ will be equal to

[WBJEE]

(a) 2^{14}

(b) $2^{14} - 15$

(c) $2^{14} + 15$

(d) $2^{14} - 1$

Q 92. The expression ${}^{n}C_0 + 2\ {}^{n}C_1 + 3\ {}^{3}C_2 + \ldots +$

$(n + 1) \cdot {}^{n}C_n$ is equal to

[AMU]

(a) $(n + l)2^n$

(b) $2^n(n + 2)$

(c) $(n + 2)2^{n-1}$

(d) $(n + 2)2^{n+1}$

Q 93. The coefficient of p^nq^n in the expansion of $t(l + p)(l + q)(p + q)]^n$ is

[J&K CET]

(a) $\sum_{k=0}^{n}[C(n, k)]^2$

(b) $\sum_{k=0}^{n}[C(n,k+2)]^2$

(c) $\sum_{k=0}^{n}[C(n,k+3)]^2$

(d) $\sum_{k=0}^{n}[C(n,k)]^3$

Mathematical Induction

Q 94. **Statement I** For each natural number n, $(n+1)^7 - n^7 - 1$ is divisible by 7.

Statement **II** For each natural number n, $n^7 - n$ is divisible by 7.

[AIEEE]

(a) Statement I is false, Statement **II** is true.

(b) Statement I is true, Statement **II** is true;

Statement II is correct explanation for Statement I.

(c) Statement I is true, Statement **II** is true;

Statement II is not a correct explanation for

Statement I.

(d) Statement I is true, Statement **II** is false.

Q 95. Which of the following result is valid ?

[AMU]

(a) $(1 + x)^n > (1 + nx)$ for all natural number n

(b) $(1 + x)^n \geq (1 + nx)$ for all natural number n,

where $x > -1$

(c) $(1 + x)^n \leq (1 + nx)$ for all natural number

(d) $(1 + x)^n < (1 + nx)$ for all natural number n

Q 96. If n is a natural number, then

[AMU]

(a) $1^2 + 2^2 + \ldots + n^2 < n^3/3$

(b) $1^2 + 2^2 + \ldots + n^2 = n^3/3$

(c) $1^2 + 2^2 + \ldots + n^2 > n^3$

(d) $1^2 + 2^2 + \ldots + n^2 > n^3/3$

7. Matrices and Determinants

Algebra of Matrices, Types of Matrices

Q 97. Let M be a 3×3 matrix satisfying

$$M\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}-1\\2\\3\end{bmatrix}, M\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\-1\end{bmatrix} \quad \text{and} \quad M\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}0\\0\\12\end{bmatrix}$$

Then, the sum of the diagonal entries of M is …

[IIT JEE]

Q 98. Let A and B be two symmetric matrices of order 3.

Statement I A (BA) and (AB) A are symmetric matrices.

Statement II AB is symmetric matrix, if matrix multiplication of A with B is commutative.

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

Statement I.

(b) Statement I is true, Statement II is false.

(c) Statement I is false, Statement II is true.

(d) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement

I.

Q 99. **Statement I** Determinant of a skew–symmetric matrix of order 3 is zero.

Statement II For any matrix A, $\det(A^T) = \det(A)$ and $\det(-A) = -\det(A)$.

[AIEEE]

Where det(B) denotes the determinant of matrix B.

Then,

(a) Statement I is true and Statement II is false.

(b) Both statements are true.

(c) Both statements are false.

(d) Statement I is false and Statement II is true.

Q 100. If X is any matrix of order nxp(n and p

are integers) and I is an identity matrix of order n x

n, then the matrix M = I – X(X' X)$^{-1}$ X' is

(I) idempotent matrix (II) MX = 0 Choose the

correct answer.

[UP SEE]

(a) (I) is correct

(b) (II) is correct

(c) (I) is incorrect

(d) (II) is incorrect

Q 101. If $A=\begin{bmatrix} 3 & x-1 \\ 2x+3 & x+2 \end{bmatrix}$ is a symmetric matrix,

then the value of x is

[WB JEE]

(a) 4

(b) 3

(c) -4

(d) -3

Q 102. If $A=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ a & b & -1\end{bmatrix}$ and I is the unit matrix of order 3, then $A^2 + 2A^4 + 4A^6$ is equal to

[Kerala CEE]

(a) $7A^8$

(b) $7A^7$

(c) $8I$

(d) $6I$

(e) I

Q 103. Let A and B both be 3×3 matrices.

Then, $(AB)^T = BA$, if

[J&K CET]

(a) A is skew–symmetric and B is symmetric

(b) B is skew–symmetric and A is symmetric

(c) A and B are skew–symmetric

(d) None of the above

Properties and Solution of Determinants

Q 104. If $z=\begin{vmatrix} 1 & 1+2i & -5i \\ 1-2i & -3 & 5+3i \\ 5i & 5-3i & 7 \end{vmatrix}$, then $(i=\sqrt{-1})$

[WB JEE]

(a) z is purely real

(b) 2 is purely imaginary

(c) $z+\overline{z}=0$

(d) $(z-\overline{z})i$ is purely imaginary

Q 105. If one of the cube roots of 1 be ω, then

$$\begin{vmatrix} 1 & 1+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -1+\omega & -1 \end{vmatrix}$$ is equal to

[WB JEE]

(a) ω

(b) i

(c) 1

(d) 0

Q 106. $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2-bc & b^2-ca & c^2-ab \end{vmatrix}$ is equal to

[GGSIPU]

(a) 0

(b) 1

(c) abc

(d) $(a-b)(b-c)(c-a)$

Q 107. If A and B are square matrices of order 3 such that $|A| = -1$, $|B| = 3$, then $|3AB|$ is equal to

[MP PET]

(a) − 9

(b) − 81

(c) − 27

(d) 81

Q 108. $\begin{vmatrix} \sin\alpha & \cos\alpha & \sin(\alpha+\delta) \\ \sin\beta & \cos\beta & \sin(\beta+\delta) \\ \sin\gamma & \cos\gamma & \sin(\gamma+\delta) \end{vmatrix}$ is equal to

[KCET]

(a) 0

(b) 1

(c) $1 + \sin\alpha \sin\beta \sin\gamma$

(d) $1 - (\sin\alpha - \sin\beta)(\sin\beta - \sin\gamma)(\sin\gamma - \sin\alpha)$

Q 109. If A is a square matrix of order 3 and α is a real number, then determinant $|\alpha A|$ is equal to

[IIT JEE]

(a) $\alpha^2|A|$

(b) $\alpha|A|$

(c) $\alpha^3|A|$

(d) None of these

Transpose, Adjoint and Inverse of Matrices

Q 110. Let M and N be two $3x3$ non-singular skew-symmetric matrices such that MN = NM. If P^T denotes the transpose of P, then $M^2N^2(M^TN)^{-1}(MN^{-1})^T$ is equal to

[IIT JEE]

(a) M^2

(b) $-N^2$

(c) $-M^2$

(d) MN

Q 111. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$, where each of a, b and c is either ω or ω^2. Then, the number of distinct matrices in the set S is

[IIT JEE]

(a) 2

(b) 6

(c) 4

(d) 8

Q 112. If $A_{3 \times 3}$ and det A = 6, then det $(2$ adj A$)$

is equal to

[GGSIPU]

(a) 48

(b) 8

(c) 288

(d) 12

Q 113. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying $AA^T = 9I_3$, then the values of a and b are respectively

[Kerala CEE]

(a) 1, 2

(b) $-1, 2$

(c) $-1, -2$

(d) $2, 1$

(e) $-2, -1$

Q 114. If A is a 2x 2 matrix and |A| = 2, then the matrix represented by A (adj (A)) is equal to

[J&K CET]

(a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

(c) $\begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$

(d) $\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$

System of Linear Equations, Rank of Matrices

Q 115. The number of values of k for which the linear equations $4x + ky + 2z = 0$, $kx + 4y + z = 0$ and $2x + 2y + z = 0$ posses a non−zero solution is

[AIEEE]

(a) 2

(b) 1

(c) zero

(d) 3

Q 116. If the system of linear equations x + 2y − 3z = 1, (p + 2)z = 3, (2p + l)y + z =2 has no solution, then

[AMU]

(a) p = 2

(b) $p = -2$

(c) $p = -\frac{1}{2}$

(d) $p = 3$

Q 117. Let m be a positive integer and $0 < r <$ m. The value of $\sum_{r=0}^{m} \begin{vmatrix} 2r-1 & {}^{m}C_{r} & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2 m & \cos^2 m & \tan^2 m \end{vmatrix}$ will be

[AMU]

(a) 2^m

(b) $m + 1$

(c) $m^2 - I$

(d) 0

Q 118. The system of equations $2x - y + z = 0$, $ax - y + 2z = 0$, $x - 2y + z = 0$ has non-zero solution, if a is equal to

[J&K CET]

(a) 1

(b) 2

(c) 4

(d) 5

8. Trigonometric Identities and Equations

Trigonometrical Ratios of Allied Angles and Basic Fundamental

Q 119. If A, B and C are the angles of a triangle such that sec(A − B), sec (A) and sec(A+B) are in arithmetic progression, then

[UP SEE]

(a) $\text{cosec}^2 A = 2\ \text{cosec}^2 \frac{B}{2}$

(b) $2\ \sec^2 A = \sec^2 \frac{B}{2}$

(c) $2\ \text{cosec}^2 A = \text{cosec}^2 \frac{B}{2}$

(d) $2\ \sec^2 B = \sec^2 \frac{A}{2}$

Q 120. If $\sin\theta = \frac{2t}{1+t^2}$ and θ lies in the second quadrant, then $\cos\theta$ is equal to

[WB JEE]

(a) $\frac{1-t^2}{1+t^2}$

(b) $\frac{t^2-1}{1+t^2}$

(c) $\frac{-|1-t^2|}{1+t^2}$

(d) $\frac{1+t^2}{|1-t^2|}$

Q 121. The sum $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \ldots +$ $\cos 179^\circ + \cos 180^\circ$ is equal to

[GGSIPU]

(a) 0

(b) 1

(c) -1

(d) 2

Trigonometrical Ratios of Sum and Difference of Two and Three Angles.

Q 122. If $\sin\theta = 3\sin(\theta + 2\alpha)$, then the value of $\tan(\theta + \alpha) + 2\tan\alpha$ is

[Kerala CEE]

(a) 3

(b) 2

(c) -1

(d) 0

(e) 1

Q 123. If $\alpha, \beta, \gamma \in [0, \pi]$ and if α, β, γ are in AP, then $\frac{\sin\alpha - \sin\gamma}{\cos\gamma - \cos\alpha}$ is equal to

[Kerala CEE]

(a) sin β

(b) cos β

(c) cot β

(d) cosec β

(e) $2 \cos \beta$

9. Inverse Trigonometric Functions

Solutions of Inverse Trigonometrical Functions

Q 134. The solution of equation

$\sin^{-1}\left(1-\frac{x}{2}\right)-2\sin^{-1}\frac{x}{2}=\frac{\pi}{2}$ is

[UP SEE]

(a) $x = 0$

(b) $x=\frac{1}{2}$

(c) x = 0 and $\frac{1}{2}$

(d) None of these

Q $135.$ The solutions set of inequation $\cos^{-1} x < \sin^{-1} x$ is

[WB JEE]

(a) $[-1,1]$

(b) $\left[\frac{1}{\sqrt{2}},1\right]$

(c) $[0, 1]$

(d) $\left(\frac{1}{\sqrt{2}},1\right]$

Q 136.

$$\tan^{-1}\left(\frac{1}{1+2}\right)+\tan^{-1}\left(\frac{1}{1+(2)(3)}\right)+\tan^{-1}\left(\frac{1}{1+(3)(4)}\right)+.....+\tan^{-1}\left(\frac{1}{1+n(n+1)}\right)=\tan^{-1}\theta$$

[GGSIPU]

(a) $\frac{n}{n+1}$

(b) $\frac{n+1}{n+2}$

(c) $\frac{n+2}{n+1}$

(d) $\frac{n}{n+2}$

Q 137. If $2 \sin^{-1} x - \cos^{-1} x = \frac{\pi}{2}$, then x is equal to

[Kerala CEE]

(a) $\frac{1}{\sqrt{2}}$

(b) $-\frac{1}{\sqrt{2}}$

(c) $\frac{-\sqrt{3}}{2}$

(d) $\frac{\sqrt{3}}{2}$

(e) $\frac{1}{2}$

Q 138. If $a \sin^{-1} x - b \cos^{-1} x = c$, then $a \sin^{-1} x + b \cos^{-1} x$ is equal to

[AMU]

(a) $\frac{\pi ab + c(a-b)}{a+b}$

(b) 0

(c) $\frac{\pi ab - c(a-b)}{a+b}$

(d) $\frac{\pi}{2}$

Q 139. If $\tan^{-1} x = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right)$, then x is

[KCET]

(a) $\frac{1}{3}$

(b) $\frac{1}{2}$

(c) $\frac{1}{4}$

(d) $\frac{1}{6}$

Q 140. The value of $\sin\left[\tan^{-1}\left(\frac{1-x^2}{2x}\right)+\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right]$ is

[J&K CET]

(a) 1

(b) 0

(c) -1

(d) $\frac{\pi}{2}$

10. Properties of Triangle

Relation Between Sides and Angles

Q 141. If angles A, B and C are in AP, then $\frac{a+c}{b}$

is equal to

[BITSAT]

(a) $2\sin\left(\frac{A-C}{2}\right)$

(b) $2\cos\left(\frac{A-C}{2}\right)$

(c) $\cos\left(\frac{A-C}{2}\right)$

(d) $\sin\left(\frac{A-C}{2}\right)$

Q 142. If $\tan^{-1} 2$, $\tan^{-1} 3$ are two angles of a triangle, then the third angle is

[GGSIPU]

(a) $30°$

(b) $45°$

(c) $60°$

(d) $75°$

Q 143. If a, b, c are the sides of a Δ ABC, then

$(a - b)^2 \cos^2 \frac{C}{2} + (a + b)^2 \sin^2 \frac{C}{2}$ is equal to

[MPPET]

(a) a^2

(b) c^2

(c) b^2

(d) $a^2 + b^2$

Q 144. In any Δ ABC, the simplified form of

$\frac{\cos 2A}{a^2} - \frac{\cos 2B}{b^2}$ is

[KCET]

(a) $a^2 - b^2$

(b) $\frac{1}{a^2 - b^2}$

(c) $\frac{1}{a^2} - \frac{1}{b^2}$

(d) $a^2 + b^2$

Q 145. Let C be right−angle of a ΔABC, then

$\frac{\sin^2 A}{\sin^2 B} - \frac{\cos^2 A}{\cos^2 B}$ is equal to

[J&K CET]

(a) $\frac{a^2 - b^2}{ab}$

(b) $\frac{a^4 - b^4}{a^2 b^2}$

(c) $\frac{a^4 + b^4}{a^2 b^2}$

(d) $\frac{a^2 + b^2}{ab}$

Q 146. In a ΔABC, a = 8 cm, b = 10 cm, c = 12 cm. The relation between angles of the triangle is

e**[J&K CET]**

(a) C = A + B

(b) C =2B

(c) C = 2A

(d) C = 3A

Circle Connected with Triangle

Q 147. If in a ΔABC, sin A, sin B, sin C are in AP, then

[WB JEE]

(a) the altitudes are in AP

(b) the altitudes are in HP

(c) the angles are in AP

(d) the angles are in HP

11. Heights and Distances

Q 148. A ladder rests against a wall so that its top touches the roof of the house. If the ladder makes an angle of $60°$ with the horizontal and height of the house be $6\sqrt{3}\,\text{m}$, then the length of the ladder is

[BITSAT]

(a) $12\sqrt{3}\ \text{m}$

(b) $12\ \text{m}$

(c) $\frac{12}{\sqrt{3}}\text{m}$

(d) None of these

Q 149. A man from the top of a 100 m high tower sees a car moving towards the tower at an angle of depression of $30°$. After some time, the angle of depression becomes $60°$. The distance travelled by the car during this time is

[MP PET]

(a) $100\sqrt{3}\,\text{m}$

(b) $\frac{200\sqrt{3}}{3}\text{m}$

(c) $\frac{100\sqrt{3}}{3}$ m

(d) $200\sqrt{3}$ m

Q 150. The angles of elevation of the top of a tower at two points, which are at distances a and b from the foot in the same horizontal line and on the same sides of the tower, are complementary. The height of the tower is

[MPPET]

(a) ab

(b) $\sqrt{ab}$

(c) $\sqrt{a/b}$

(d) $\sqrt{b/a}$

Q 151. Angles of elevation of the top of a tower from three points (collinear) A, B and C on a road leading to the foot of the tower are $30°$, $45°$ and $60°$ respectively. The ratio of AB to BC is

[KCET]

(a) $\sqrt{3}:1$

(b) $\sqrt{3}:2$

(c) $1:2$

(d) $2:\sqrt{3}$

12. Rectangular Cartesian Coordinates

System of Coordinates and Distance Between Points

Q 152. The line x + y = 4 divides the line joining the points $(-1, 1)$ and $(5, 7)$ in the ratio

[BITSAT]

(a) $2:1$

(b) $1 : 2$

(c) $1:2$ externally

(d) None of these

Q 153. The vertices of ΔABC are A $(2, 2)$ B$(-4,-4)$ and C$(5,-8)$. Find the length of a median of a triangle, which is passing through the point C

[Guj. CET]

(a) $\sqrt{65}$

(b) $\sqrt{117}$

(c) $\sqrt{85}$

(d) $\sqrt{116}$

Q 154. If algebraic sum of distances of a variable line from points $(2, 0)$, $(0, 2)$ and $(-2, -2)$ is zero, then the line passes through the fixed point

[AMU]

(a) $(-1,-1)$

(b) $(1,1)$

(c) $(2,2)$

(d) $(0,0)$

Q 155. The points A $(1, 2)$, B $(2, 4)$ and C $(4, 8)$ form a/an

[KCET]

(a) isosceles triangle

(b) equilateral triangle

(c) straight line

(d) right angled triangle

Q 156. The three distinct points $A(at_1^2, 2at_1), b(at_2^2, 2at_2)$ and C $(0, a)$ (where a is a real number) are collinear, if

[J&K CET]

(a) $t_1t_2 = -1$

(b) $t_1t_2 = 1$

(c) $2t_1t_2 = t_1 + t_2$

(d) $t_1 + t_2 = a$

Orthocentre, Circumcentre and Incentre

Q 157. The vertices of a triangle are A$(0, 0)$, B$(0, 2)$ and C$(2, 0)$, then find the distance between its orthocentre and circumcentre.

[Guj. CET]

(a) 0

(b) $\sqrt{2}$

(c) $\frac{1}{\sqrt{2}}$

(d) None of these

Q 158. Orthocentre of the triangle formed by the lines $x - y = 0$, $x + y = 0$, $x = 3$ is

[Guj. CET]

(a) $(0,0)$

(b) $(3,0)$

(c) $(0, 3)$

(d) can't be found

Area of Some Geometrical Figures

Q 159. If A = $(-3, 4)$, B = $(-1, -2)$, C = $(5, 6)$ and D = $(x, -4)$ are vertices of a quadrilateral such that $\Delta ABD = 2\Delta ACD$. Then, x is equal to

[GGSIPU]

(a) 6

(b) 9

(c) 69

(d) 96

Transformation of Axes and Locus

Q 160. The coordinates of a moving point P are $(2t^2 + 4, 4t + 6)$. Then, its locus will be a

[WB JEE]

(a) circle

(b) straight line

(c) parabola

(d) ellipse

Q 161. The equation of the locus of the point of intersection of the straight lines $x \sin\theta + (1 - \cos\theta)\, y = a \sin\theta$ and $x \sin\theta - (1 + \cos\theta)y + a \sin\theta = 0$ is

[WBJEE]

(a) $y = \pm ax$

(b) $x = \pm ay$

(c) $y^2 = 4x$

(d) $x^2 + y^2 = a^2$

Q 162. A variable line passes through a fixed point (a, b) and meets the coordinate axes in A and B. The locus of the point of intersection of lines through A, B parallel to coordinate axes is

[MP PET]

(a) $\frac{x}{a}+\frac{y}{b}=1$

(b) $\frac{a}{x}+\frac{b}{y}=1$

(c) $\frac{x}{a}+\frac{y}{b}=2$

(c) $\frac{x}{a}+\frac{y}{b}=3$

Q 163. Locus of a point which moves such that its distance from the x−axis is twice its distance from the line x − y = 0 is

[KCET]

(a) $x^2 + 4xy - y^2 = 0$

(b) $2x^2 - 4xy + y^2 = 0$

(c) $x^2 - 4xy + y^2 = 0$

(d) $x^2 - 4xy - y^2 = 0$

Q 164. A line segment of 8 units in length moves so that its end points are always on the coordinate axes. Then, the equation of locus of its mid-point is

[J&K CET]

(a) $x^2 + y^2 = 4$

(b) $x^2 + y^2 = 16$

(c) $x^2 + y^2 = 8$

(d) $|x| + |y| = 8$

13. Straight Line and Pair of Straight Lines

Equation of Line and Pair of Lines

Q 165. A straight line L through the point $(3, -2)$ is inclined at an angle $60°$ to the line $\sqrt{3}x + y = 1$. If L also intersects the x–axis, then the equation of L is

[IIT JEE]

(a) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

(b) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

(c) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

(d) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

Q 166. If a straight line passes through the points $\left(-\frac{1}{2}, 1\right)$ and $(1, 2)$, then its x−intercept is

[Kerala CEE]

(a) -2

(b) -1

(c) 2

(d) 1

(e) 0

Q 167. The equation of the line passing through $(0, 0)$ and intersection of $3x - 4y = 2$ and $x + 2y = -4$ is

[J&K CET]

(a) $7x = 6y$

(b) $6x=7y$

(c) $5x = 8y$

(d) $x = 0$

Angle Between Intersection Lines, Condition for Parallel and Perpendicular Lines

Q 168. The line parallel to the x−axis and passing through the point of intersection of the lines $ax + 2by + 3b = 0$ and $bx - 2ay - 3a = 0$, where $(a, b) \neq (0, 0)$ is

[Kerala CEE]

(a) above the x−axis at a distance of $\frac{3}{2}$

(b) above the x−axis at a distance of $\frac{2}{3}$

(c) below the x−axis at a distance of $\frac{2}{3}$

(d) below the x−axis at a distance of $\frac{3}{2}$

(e) below the x−axis at a distance of 3

Q 169. If lines represented by x + 3y − 6 = 0, 2x + y − 4 = 0 and kx − 3y + 1 = 0 are concurrent, then the value of k is

[KCET]

(a) $\frac{6}{19}$

(b) $\frac{19}{6}$

(c) $-\frac{19}{6}$

(d) $-\frac{6}{19}$

Q 170. The value of k for which the equation $x^2 - 4xy - y^2 + 6x + 2y + k = 0$ represents a pair of straight lines is

[J&K CET]

(a) $k = 4$

(b) $k = -1$

(c) $k = -\frac{4}{5}$

(d) $k = -\frac{22}{5}$

Position of a Point, Bisection of the Lines

Q 171. The lines L_1 : y − x = 0 and L_2 : 2x + y = 0 intersect the line L_3 : y + 2 = 0 at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

Statement I The ratio PR : RQ equals $2\sqrt{2} : \sqrt{5}$.

Statement II In any triangle, bisector of an angle divides the triangle into two similar triangles.

[AIEEE]

(a) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.

(b) Statement I is true, Statement II is false.

(c) Statement I is false, Statement II is true.

(d) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement

I.

Q 172. Equation of the bisector of the acute

angle between lines $3x + 4y + 5 = 0$ and $12x - 5y$

$- 7 = 0$ is

[BITSAT]

(a) $21x + 77y + 100 = 0$

(b) $99x - 27y + 30 = 0$

(c) $99x + 27y + 30 = 0$

(d) $21x - 77y - 100 = 0$

Q 173. The equation of the perpendicular bisector of the line segment joining $A(-2, 3)$ and $B(6, -5)$ is

[Kerala CEE]

(a) $x - y = -1$

(b) $x - y = 3$

(c) $x + y = 3$

(d) $x + y = 1$

(e) $x + y = -1$

14. Circle and System of Circles

Circle and Its Equations

Q 174. The circle passing through the point $(-1,0)$ and touching the y−axis at $(0, 2)$ also passes through the point

[IIT JEE]

(a) $\left(-\frac{3}{2},0\right)$

(b) $\left(-\frac{5}{2}, 2\right)$

(c) $\left(-\frac{3}{2}, \frac{5}{2}\right)$

(d) $(-4, 0)$

Q 175. The straight line $2x - 3y = 1$ divides the circular region $x^2 + y^2 \leq 6$ into two parts. If $S = \left\{\left(2, \frac{3}{4}\right), \left(\frac{5}{2}, \frac{3}{4}\right), \left(\frac{1}{4}, -\frac{1}{4}\right), \left(\frac{1}{8}, \frac{1}{4}\right)\right\}$, then the number of point(s) in S lying inside the smaller part is……

[IIT JEE]

Q 176. If the straight line y =mx lies outside the circle $x^2 + y^2 - 20y + 90 = 0$, then the value of m will satisfy

[WB JEE, Guj. CET]

(a) $m < 3$

(b) $| m | < 3$

(c) $m > 3$

(d) $| m | > 3$

Q 177. The intercept on the line y=x by the circle $x^2 + y^2 - 2x = 0$ is AB. Equation of the circle with AB as diameter is

[WB JEE]

(a) $x^2 + y^2 = 1$

(b) $x(x - 1) + y(y - 1) = 0$

(c) $x^2+y^2=2$

(d) $(x - 1)(x - 2) + (y - 1)(y - 2) = 0$

Q 178. The sum of the minimum distance and the maximum distance from the point $(4, -3)$ to the circle $x^2 + y^2 + 4x - 10y - 7 = 0$ is

[Kerala CEE]

(a) 20

(b) 12

(c) 10

(d) 16

(e) 22

Q 179. If the lines joining the origin to the intersection of the line $y = mx + 2$ and the circle $x^2 + y^2 = 1$ are at right angles, then

[AMU]

(a) $m = \sqrt{3}$

(b) $m = \pm\sqrt{7}$

(c) $m = 1$

(d) $m = \sqrt{5}$

Tangent and Normal to a Circle

Q 180. The length of tangent from point $(5, 1)$ to the circle $x^2 + y^2 + 6x - 4y - 3 = 0$ is

[BITSAT]

(a) 81

(b) 29

(c) 7

(d) 21

Q 181. Circle $ax^2 + ay^2 + 2gx + 2fy + c = 0$

touches x−axis, if

[MP PET]

(a) $f^2 > ac$

(b) $g^2 > ac$

(c) $f^2 = bc$

(d) $g^2 = ac$

Q 182. If the normal at the point P(8) to the ellipse $\frac{x^2}{14}+\frac{y^2}{5}=1$ intersects it again at the point Q(2θ), then cos θ is equal to

[AMU]

(a) $-\frac{2}{3}$

(b) $\frac{2}{3}$

(c) $\frac{3}{2}$

(d) $-\frac{3}{2}$

Chord of Contact of Tangents, Pole and Polar

Q 183. The locus of the point of intersection of the tangents at the extremeties of a chord of the circle $x^2 + y^2 = a^2$ which touches the circle $x^2 + y^2 - 2ax = 0$ passes through the point

[AMU]

(a) $(a/2,0)$

(b) $(0,a/2)$

(c) $(a, 0)$

(d) $(0, 0)$

System of Circles

Q 184. The circles $x^2 + y^2 + 2kx + 2y + 6 = 0$

and $x^2 + y^2 + 2kx + k = 0$ intersect orthogonally

when k equals to

[UP SEE]

(a) 2

(b) $\frac{1+\sqrt{193}}{16}$

(c) $\frac{1-\sqrt{193}}{16}$

(d) $-\frac{3}{2}$

Q 185. The total number of common tangents of

$x^2 + y^2 - 6x - 8y + 9 = 0$ and $x^2 + y^2 = 1$ is

[KCET]

(a) 4

(b) 2

(c) 3

(d) 1

Q $186.$ The centre of a circle which cuts $x^2 + y^2 + 6x - 1 = 0$, $x^2+y^2 - 3y + 2 = 0$ and $x^2 + y^2 + x + y - 3 = 0$ orthogonally is

[KCET]

(a) $\left(\frac{1}{7},\frac{9}{7}\right)$

(b) $\left(-\frac{1}{7},-\frac{9}{7}\right)$

(c) $\left(\frac{1}{7},-\frac{9}{7}\right)$

(d) $\left(-\frac{1}{7}, \frac{9}{7}\right)$

15. Conic Section

Basic Terms of Conics

Q 187. Let (x, y) be any point on the parabola y^2 = 4x. Let P be the point that divides the line segment from (0, 0) to (x, y) in the ratio 1 : 3. Then, the locus of P is

[IIT JEE]

(a) $x^2 = y$

(b) $y^2 = 2x$

(c) $y^2 = x$

(d) $x^2 = 2y$

Q 188. For the parabola $y^2 + 8x - 12y + 20 = 0$

[UP SEE]

(a) vertex is $(2, 6)$

(b) focus is $(0, 6)$

(c) latusrectum 4

(d) axis y = 6

Q 189. If the centre, one of the foci and semi–major axis of an ellipse be $(0, 0)$, $(0, 3)$ and 5, then its equation is

[BITSAT]

(a) $\frac{x^2}{16}+\frac{y^2}{25}=1$

(b) $\frac{x^2}{25}+\frac{y^2}{16}=1$

(c) $\frac{x^2}{9}+\frac{y^2}{25}=1$

(d) None of these

Q 190. The length of the latusrectum of the ellipse $16x^2 + 25y^2 = 400$ is

[WB JEE]

(a) $5/16$ unit

(b) $32/5$ unit

(c) $16/5$ unit

(d) $5/32$ unit

Q 191. The straight lines $y = \pm x$ intersect the parabola $y^2 = 8x$ in points P and Q, then length of PQ is

[MP PET]

(a) 4

(b) $4\sqrt{2}$

(c) 8

(d) 16

Q 192. The eccentricity of the hyperbola with latusrectum 12 and semiconjugate axis $2\sqrt{3}$, is

[AMU]

(a) 3

(b) $\sqrt{\frac{3}{2}}$

(c) $2\sqrt{3}$

(d) 2

Q 193. The sum of the reciprocals of focal distances of a focal chord PQ of $y^2 = 4ax$ is

[KCET]

(a) $\frac{1}{a}$

(b) a

(c) $2a$

(d) $\frac{1}{2a}$

Equation of Conics and Intersection of Two Curves

Q 194. Let the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be reciprocal to that of the ellipse $x^2 + 4y^2=4$. If the hyperbola passes through a focus of the ellipse, then

[IIT JEE]

(a) the equation of the hyperbola is $\frac{x^2}{3}-\frac{y^2}{2}=1$

(b) a focus of the hyperbola is $(2, 0)$

(c) the eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$

(d) the equation of the hyperbola is $x^2 - 3y^2 = 3$

Q 195. Equation of the ellipse whose axes are the axes of coordinates and which passes through the point $(-3,1)$ and has eccentricity $\sqrt{\frac{2}{5}}$ is

[AIEEE]

(a) $5x^2 + 3y^2 - 48 = 0$

(b) $3x^2 + 5y^2 - 15 = 0$

(c) $5x^2 + 3y^2 - 32 = 0$

(d) $3x^2 + 5y^2 - 32 = 0$

Q 196. The distance between the vertex of the parabola $y = x^2 - 4x + 3$ and the centre of the circle $x^2 = 9 - (y - 3)^2$ is

[Kerala CEE]

(a) $2\sqrt{3}$

(b) $3\sqrt{2}$

(c) $2\sqrt{2}$

(d) $\sqrt{2}$

(e) $2\sqrt{5}$

Q 197. The parabola $y^2 = 4x$ and the circle $x^2 + y^2 - 6x + 1 = 0$ will

[AMU]

(a) intersect at exactly one point

(b) touch each other at two distinct points

(c) touch each other at exactly one point

(d) intersect at two distinct points

Tangent and Normal of Conic

Q 198. Consider the parabola $y^2 = 8x$. Let Δ_1 be the area of the triangle formed by the end points of its latusrectum and the point $P\left(\frac{1}{2}, 2\right)$ on the parabola and Δ_2 be the area of the triangle formed by drawing tangents at P and at the end points of the latusrectum. Then, $\frac{\Delta_1}{\Delta_2}$ is.....

[IIT JEE]

Q 199. Let P$(6, 3)$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal at the point P intersects the x–axis at $(9, 0)$, then the eccentricity of the hyperbola is

[IIT JEE]

(a) $\sqrt{\frac{5}{2}}$

(b) $\sqrt{\frac{3}{2}}$

(c) $\sqrt{2}$

(d) $\sqrt{3}$

Q 200. Let L be a normal to the parabola $y^2 = 4x$.

If L passes through the point $(9, 6)$, then L is given

by

[IIT JEE]

(a) $y - x + 3 = 0$

(b) $y + 3x - 33 = 0$

(c) $y + x - 15 = 0$

(d) $y - 2x + 12 = 0$

Q 201. The equation to the common tangents to the two hyperbolas $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and $\frac{y^2}{a^2}-\frac{x^2}{b^2}=1$ are

[MP PET]

(a) $y=\pm x\pm\sqrt{(b^2-a^2)}$

(b) $y=\pm x\pm\sqrt{(a^2-b^2)}$

(c) $y=\pm x\pm(a^2-b^2)$

(d) $y=\pm x\pm\sqrt{(a^2+b^2)}$

Q 202. The equation of the common tangent to the parabola $y^2 = 8x$ and rectangular hyperbola xy $= -1$ is

[AMU]

(a) $x - y + 2 = 0$

(b) $9x - 3y + 2 = 0$

(c) $2x - y + 1 = 0$

(d) $x + 2y - 1 = 0$

Q 203. Area of a triangle formed by tangent and normal to the curve $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at $P\left(\frac{a}{\sqrt{2}},\frac{b}{\sqrt{2}}\right)$ with the x−axis is

[KCET]

(a) 4ab

(b) $\frac{ab\sqrt{a^2+b^2}}{4}$

(c) $\frac{ab\sqrt{a^2-b^2}}{4}$

(d) $\frac{b(a^2+b^2)}{4a}$

Q 204. The equation of tangent of $y^2 = 12x$ and making an angle $\frac{\pi}{3}$ with x−axis is

[Guj. CET]

(a) $\pm y - \sqrt{3}x + \sqrt{3} = 0$

(b) $\pm y + \sqrt{3}x + 3 = 0$

(c) $\pm y - \sqrt{3}x - \sqrt{3} = 0$

(d) $\pm y + \sqrt{3}x - 3 = 0$

Diameter, Auxiliary Circle and Asymptotes

Q 205. Equation of auxiliary circle of $\frac{x^2}{16}-\frac{y^2}{25}=-1$ is

[Guj. CET]

(a) $x^2 + y^2 = 16$

(b) $x^2 + y^2 = 25$

(c) $x^2 + y^2 = 9$

(d) $x^2 + y^2 = 41$

Q 206. Find the measure of angle between the asymptotes of $x^2 - y^2 = 16$.

[Guj. CET]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{6}$

(d) $\frac{\pi}{2}$

Q 207. The equation of a hyperbola whose asymptotes are $3x \pm 5y = 0$ and vertices are $(\pm 5, 0)$ is

[KCET]

(a) $3x^2 - 5y^2 = 25$

(b) $5x^2 - 3y^2 = 225$

(c) $25x^2 - 9y^2 = 225$

(d) $9x^2 - 25y^2 = 225$

16. Limits, Continuity and Differentiability

Limits of Algebra, Exponential and Logarithmic Functions

Q 208. If $\lim_{x\to 0}[1+x\log(1+b^2)]^{\frac{1}{x}} = 2b\sin^2\theta$, b > 0 and $\theta \in$ $(-\pi, \pi]$, then the value of θ is

[IIT JEE]

(a) $\pm\frac{\pi}{4}$

(b) $\pm\frac{\pi}{3}$

(c) $\pm\frac{\pi}{6}$

(d) $\pm\frac{\pi}{2}$

Q 209. Let $f : R \to [0, \infty)$ be such that $\lim_{x \to 5} f(x)$ exists and $\lim_{x \to 5} \frac{[f(x)]^2 - 9}{\sqrt{|x-5|}} = 0$. Then, $\lim_{x \to 6} f(x)$ equals to

[AIEEE]

(a) 3

(b) 0

(c) 1

(d) 2

Q 210. If function f(x) is differentiable at x = a,

then $\lim_{x \to a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$

[AIEEE]

(a) $2a\, f(a) + a^2\, f'(a)$

(b) $-a^2\, f'(a)$

(c) $af(a) - a^2\, f'(a)$

(d) $2af(a) - a^2\, f'(a)$

Q 211. $\lim_{x \to 0} \left(\frac{16^x + 9^x}{2} \right)^{1/x}$ is equal to

[GGSIPU]

(a) 25/2

(b) 12

(c) 1

(d) 1/4

Q 212. $\lim_{x\to\infty}\left(1+\frac{4}{x-1}\right)^{x+3}$

[Guj. CET]

(a) e^4

(b) e^2

(c) e^3

(d) e

Q 213. If f'(x) = f(x), f(0) = 1, then $\lim_{x \to 0} \frac{f(x)-1}{x}$ is

equal to

[Guj. CET]

(a) 0

(b) 1

(c) -1

(d) 2

Q 214. $\lim_{k\to\infty}\left(\frac{1^3+2^3+3^3+.....+k^3}{k^4}\right)$ is equal to

[Kerala CEE]

(a) 0

(b) 2

(c) $\frac{1}{3}$

(d) ∞

(e) $\frac{1}{4}$

Q 215. $\lim_{x \to a}\left[\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}\right]$ is equal to

[KCET]

(a) $\frac{2}{3}$

(b) $\frac{2}{\sqrt{3}}$

(c) $\frac{3\sqrt{3}}{2}$

(d) $\frac{2}{3\sqrt{3}}$

Limits of Trigonometric Functions and Series

Q 216. $\lim_{x \to 2} \left(\frac{\sqrt{1-\{\cos 2(x-2)\}}}{x-2} \right)$

[AIEEE]

(a) equals $\sqrt{2}$

(b) equals $-\sqrt{2}$

(c) equals $\frac{1}{\sqrt{2}}$

(d) does not exist

Q 217. The value of $\lim_{x \to \frac{\pi}{2}} \frac{\sin(\cos x)\cos x}{\sin x - \cos ecx}$ is

[UP SEE]

(a) ∞

(b) 1

(c) 0

(d) -1

Q 218. $\lim_{x \to 0} \frac{\sin(\pi \sin^2 x)}{x^2}$ is equal to

[WB JEE]

(a) π^2

(b) 3π

(c) 2π

(d) π

Q 219. $\lim_{x\to 0}\left(\frac{1+\tan x}{1+\sin x}\right)^{\operatorname{cosec} x}$ is equal to

[MP PET]

(a) e

(b) $1/e$

(c) 1

(d) 0

Q 220. If $\lim_{x \to 0} \frac{ae^x - b\cos x + ce^{-x}}{x \sin x} = 2$; then

[AMU]

(a) a = 1, b = 2, c = 1

(b) a = 1, b = 1, c = 2

(c) a = 2, b = 1, c = 1

(d) a = b = c = 1

Q 221. Define F(x) as the product of two real functions $f_1(x) = x$, $x \in R$,

and $f(x) = \begin{cases} \sin\frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ as follows

$$F(x) = \begin{cases} f_1(x).f_2(x), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Statement I F(x) is continuous on R.

Statement II $f_1(x)$ and $f_2(x)$ are continuous on R.

[AIEEE]

(a) Statement I is false, Statement II is true.

(b) Statement I is true, Statement II is true;

Statement II is correct explanation for statement I.

(c) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

statement I.

(d) Statement I is true, Statement II is false.

Q 222. The values of p and q for which the

function

$$f(x)=\begin{cases}\dfrac{\sin(p+1)x+\sin x}{x} & , \quad x<0\\ q & , \quad x=0\\ \dfrac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}} & , \quad x>0\end{cases}$$

is continuous for all x in R, are

[AIEEE]

(a) $p=\frac{5}{2}, q=\frac{1}{2}$

(b) $p=-\frac{3}{2}, q=\frac{1}{2}$

(c) $p=\frac{1}{2}, q=\frac{3}{2}$

(d) $p=\frac{1}{2}, q=-\frac{3}{2}$

223. Let $f(x)=\begin{cases} -2\sin x & , \quad -\pi \le x \le -\frac{\pi}{2} \\ a\sin x + b & , \quad -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \cos x & , \quad \frac{\pi}{2} \le x \le \pi \end{cases}$

If f(x) is continuous on $[-\pi, \pi]$, then

[UP SEE]

(a) a = 1, b = 1

(b) a = −1, b = −1

(c) a = −1, b = 1

(d) a = 1, b = −1

Q 224. $f(x)=\begin{cases}[x]+[-x], & \text{where } x \neq 2 \\ \lambda, & \text{when } x=2\end{cases}$

If f(x) is continuous at x = 2, then the value of λ will be

[WB JEE]

(a) -1

(b) 1

(c) 0

(d) 2

Q 225. If $f(x)=\begin{cases} cx+1; & x \le 3 \\ cx^2-1; & x>3 \end{cases}$ is continuous at x = 3,

then c is equal

[Guj. CET]

(a) $1/3$

(b) $2/3$

(c) $3/2$

(d) 3

Q 226. If $f(x)=\begin{cases}\frac{\log x}{x-1}, & \text{if } x\neq 1\\ k, & \text{if } x=1\end{cases}$ is continuous at x = 1,

then the value of k is

[KCET]

(a) 0

(b) −1

(c) 1

(d) e

Differentiability

Q 227. Let f : R$\to$R be a function such that f(x + y) = f(x) + f(y), $\forall$x, y $\in$ R. If f(x) is differentiable at x = 0, then

[IIT JEE]

(a) f(x) is differentiable only in a finite interval containing zero

(b) f(x) is continuous $\forall$ x $\in$ R

(c) f' (x) is constant $\forall$ x $\in$ R

(d) f(x) is differentiable except at finitely many points

Q 228. If $f(x)=\begin{cases} -x-\frac{\pi}{2} & , \quad x \le -\frac{\pi}{2} \\ -\cos x & , \quad -\frac{\pi}{2} < x \le 0 \\ x-1 & , \quad 0 < x \le 1 \end{cases}$, then

[IIT JEE]

(a) f(x) is continuous at $x=-\frac{\pi}{2}$

(b) f(x) is not differentiable at x = 0

(c) f(x) is differentiable at x = 1

(d) f(x) is differentiable at $x = -\frac{3}{2}$

Q 229. f(x) = x sin $\frac{1}{x}$ is

[GGSIPU]

(a) continuous but not differentiable at x = 0

(b) discontinuous but differentiable at x = 0

(c) differentiable at x = 0

(d) cannot be determined

Q 230. If $f(x) = p|\sin x| + qe^{|x|} = r|x|^3$ and if $f(x)$ is differentiable at $x = 0$, then

[AMU]

(a) $p = 0$, $q = 0$ and $r = 0$

(b) $p + q = 0$ and r is any real number

(c) $p + q + r = 0$

(d) $-p + q - r = 0$

17. Differentiation

Differentiation of Algebraic and Transcendental Functions

Q 231. Let y'(x) + y(x) g'(x) = g(x)g'(x), y(0) = 0, $x \in R$, where f'(x) denotes $\frac{df(x)}{dx}$ and g(x) is a given non-constant differentiable function on R with g(0) = g(2) = 0. Then, the value of y(2) is.....

[IITJEE]

Q 232. If f (x) = $\log_e$ ($\log_e$ x), then f'(e) is equal to

[MP PET]

(a) e^{-1}

(b) e

(c) 1

(d) 0

Q 233. If F (x) = $\int_{x^2}^{x^3} \log t$. If [x > 0]. then F' (x) is

equal to

[MPPET]

(a) $(9x^2 - 4x) \log x$

(b) $(4x - 9x^2) \log x$

(c) $(9x^2 + 4x) \log x$

(d) $(3x^2 - 2x) \log x$

Q 234. $\frac{d}{dx}\left[\sin^{-1}\frac{x}{a}\right] = ...;$ a < 0 and $\left|\frac{x}{a}\right| < 1$

[Guj. CET]

(a) $\frac{1}{\sqrt{a^2 - x^2}}$

(b) $\frac{1}{\sqrt{x^2 - a^2}}$

(c) $\frac{-1}{\sqrt{a^2 - x^2}}$

(d) $\frac{-1}{\sqrt{x^2 - a^2}}$

Q 235. If $\pi < x < 2\pi$, then $\frac{d}{dx}\left[\tan^{-1}\left(\frac{1-\cos x}{1+\cos x}\right)^{1/2}\right]$ is

[Guj. CET]

(a) 0

(b) 1

(c) $1/2$

(d) $-1/2$

Q 236. If $f(x)=\frac{\sin^2 x}{1+\cot x}+\frac{\cos^2 x}{1+\tan x}$, then $f'\left(\frac{\pi}{4}\right)$ is

[KCET]

(a) $\sqrt{3}$

(b) $\frac{1}{\sqrt{3}}$

(c) 0

(d) $-\sqrt{3}$

Q 237. If y = sin x . sin 2x − sin 3x… sin nx,

then y' is

[J&K CET]

(a) $\sum_{k=1}^{n} k \tan kx$

(b) $y.\sum_{k=1}^{n} k \cot kx$

(c) $y.\sum_{k=1}^{n} k \tan kx$

(d) $\sum_{k=1}^{n} \cot kx$

Q 238. If $x \neq 0$ and $y = \log_e |2x|$, then $\frac{dy}{dx}$ is equal to

[J&K CET]

(a) $\frac{1}{x}$

(b) $-\frac{1}{x}$

(c) $\pm\frac{1}{2x}$

(d) None of these

Differentiation by Substitution, Higher Order Derivatives

Q 239. Let $f(\theta)=\sin\left(\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)\right)$, where $-\frac{\pi}{4}<\theta<\frac{\pi}{4}$.

Then, the value of $\frac{d}{d(\tan\theta)}(f(\theta))$ is.....

[IITJEE]

Q 240. Let $f(x) = \tan^{-1} x$. Then, $f'(x) + f''(x) = 0$, when x is equal to

[WB JEE]

(a) 0

(b) 1

(c) i

(d) −i

Q 241. If $y = \cos(3 \cos^{-1} x)$, then $\frac{d^3y}{dx^3}$ is equal to

[MP PET]

(a) 24

(b) 27

(c) $3-12x^2$

(d) -24

Q 242. If $x = \sin^{-1}(3t - 4t^3)$ and $y = \cos^{-1}(\sqrt{1-t^2})$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{2}$

(b) $\frac{2}{3}$

(c) $\frac{1}{3}$

(d) $\frac{2}{5}$

(e) $\frac{1}{5}$

18. Applications of Derivatives

Rate of Measures and Application in Mechanics

Q 243. The distance (in metre) travelled by a vehicle in time t (in seconds) is given by the equation $s = 3t^3 + 2t^2 + t + 1$. The difference in the acceleration between $t = 2$ and $t = 4$ is

[UP SEE]

(a) $36\ m/s^2$

(b) $38\ m/s^2$

(c) 45 m/s^2

(d) 46 m/s^2

Q 244. If $y = 2x^3 - 2x^2 + 3x - 5$, then for $x = 2$ and $\Delta x = 0.1$ value of Δy is

[WB JEE]

(a) 2.002

(b) 1.9

(c) 0

(d) 0.9

Q 245. The approximate value of $\sqrt[5]{33}$ correct to 4 decimal places is

[WB JEE]

(a) 2.0000

(b) 2.1001

(c) 2.0125

(d) 2.0500

Q 246. The total revenue in rupees received from the sale of x units of a product is given by $R(x) = 13x^2 + 26x + 15$. Then, the marginal revenue in rupees, when $x = 15$ is

[Kerala CEE]

(a) 116

(b) 126

(c) 136

(d) 416

(e) 146

Q 247. A sphere increases its volume at the rate of π cc/s. The rate at which its surface area increases when the radius is 1 cm is

[KCET]

(a) 2π sq cm/s

(b) π sq cm/s

(c) $\frac{3\pi}{2}$ sq cm/s

(d) $\frac{\pi}{2}$sq cm/s

Q 248. The distance travelled by a bus in t− seconds after the brakes are applied is $1 + 2t - 2t^2$ metres. The distance travelled by the bus before it stops is equal to

[J&K CET]

(a) 0.5 m

(b) 1 m

(c) 1.5 m

(d) 2.5 m

Tangent and Normal, Maxima and Minima

Q 249. For x $\in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t} \sin t \, dt$. Then,

f has

[AIEEE]

(a) local minimum at π and 2π

(b) local minimum at π and local maximum at 2π

(c) local maximum at π and local minimum at 2π

(d) local maximum at π and 2π

Q 250. Let f be a function defined by

$$f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

[AIEEE]

Statement I x = 0 is point of minima of f.

Statement II f' $(0) = 0$

(a) Statement I is false, Statement **II** is true.

(b) Statement I is true, Statement **II** is true;

Statement **II** is correct explanation for Statement I.

(c) Statement I is true, Statement **II** is true;

Statement **II** is not a correct explanation for

Statement I.

(d) Statement I is true, Statement **II** is false.

Q 251. The maximum value of f(x) = 2 sin x +

cos 2x, $0 \leq x \leq \frac{\pi}{2}$ occurs at x is equal to

[UP SEE]

(a) 0

(b) $\pi/6$

(c) $\pi/2$

(d) None of these

Q 252. The point $(0, 5)$ is closer to the curve $x^2 = 2y$ at

[BITSAT]

(a) $(2\sqrt{2}, 0)$

(b) $(0, 0)$

(c) $(2, 2)$

(d) None of these

Q 253. Let $f(x) = x^3e^{-3x}$, $x > 0$. Then, the maximum value of $f(x)$is

[WBJEE]

(a) e^{-3}

(b) $3e^{-3}$

(c) $27e^{-9}$

(d) ∞

Q 254. The length of the normal to the curve y = a cosh $\left(\frac{x}{a}\right)$ at any point varies as

[GGSIPU]

(a) ordinate

(b) abscissa

(c) square of the abscissa

(d) square of the ordinate

Q 255. The point on the curve $y = x^3$ at which tangent is parallel to the point $(1, 1)$ is **[MP PET]**

(a) $(0,0)$

(b) $(1,-1)$

(c) $(-1,1)$

(d) $(-1,-1)$

Q 256. If $f(x) = \frac{x^2-1}{x^2+1}$ for every real number x, then the minimum value of f

[AMU]

(a) -1

(b) does not exist

(c) 0

(d) 1

Q 257. If P is a point on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with focii S and S', then the maximum value of $\Delta SPS'$ is

[KCET]

(a) ab

(b) abe^2

(c) abe

(d) ab/e

QIncreasing and Decreasing Functions, Rolle's and Mean Value Theorems

Q 258. $f(x)=\begin{cases} 0, & x=0 \\ x-3, & x>0 \end{cases}$ the function f(x) is

[WB JEE]

(a) increasing when $x \geq 0$

(b) strictly increasing when $x > 0$

(c) strictly increasing at $x = 0$

(d) not continuous at $x = 0$ and so it is not

increasing when $x > 0$

Q 259. The function $f(x) = ax+b$ is strictly

increasing for all real x, if

[WB JEE]

(a) $a > 0$

(b) $a < 0$

(c) $a = 0$

(d) $a \leq 0$

Q 260. The value of c in mean value theorem for the function $f(x) = 2x^2 + 3x + 4$ in the interval $[1, 2]$ is

[MP PET]

(a) $1/2$

(b) $1/3$

(c) $3/2$

(d) $2/3$

Q 261. The function $f(x) = (x(x-2))^2$ is

increasing in the set

[Kerala CEE]

(a) $(-\infty, 0) \cup (2, \infty)$

(b) $(-\infty, 1)$

(c) $(0, 1) \cup (2, \infty)$

(d) $(1, 2)$ (e) $(0,2)$

Q 262. If $f(x) = xe^{x(1-x)}$ then $f(x)$ is

[AMU]

(a) increasing on R

(b) decreasing on $\left[-\frac{1}{2},1\right]$

(c) increasing on $\left[-\frac{1}{2},1\right]$

(d) decreasing on R

Q 263. If $f(x) = kx - \cos x$ is monotonically increasing for all $x \in R$, then

[J&KCET]

(a) $k > -1$

(b) $k < 1$

(c) $k > 1$

(d) None of these

Q 264. If $f(x) = 2x^2 - |x| + 4$, $x \in [-1,2]$. Then, for some $c \in (-1, 2)$, $f'(c)$ is equal to

[J&K CET]

(a) $\frac{f(2)-f(0)}{2-0}$

(b) $\frac{f(2)-f(-1)}{2-(-1)}$

(c) $\frac{f(1)-f(-1)}{1-(-1)}$

(d) None of the above

19. Indefinite Integrals

Fundamental of Indefinite Integrals

Q 256. The value of integral $I=\int \frac{\sin x+\cos x}{\sqrt{1+\sin 2x}}dx$ is

[UP SEE]

(a) $\sqrt{1+\cos 2x}$

(b) $\sqrt{x}$

(c) x

(d) $\sqrt{1+2x}$

Q 257. $\int \frac{\sin^8 x - \cos^8 x}{1-2\sin^2 x \cos^2 x} dx$ is equal to

[WB JEE]

(a) $-\frac{1}{2}\sin 2x + C$

(b) $\frac{1}{2}\sin 2x + C$

(c) $\frac{1}{2}\sin x + C$

(d) $-\frac{1}{2}\sin x + C$

Q 267. $\int \frac{1}{\sin(x-a)\sin(x-b)} dx$ is equal to

[GGSIPU]

(a) $\frac{1}{\sin(b-a)} \log \left| \frac{\sin(x+b)}{\sin(x+a)} \right| + C$

(b) $\frac{1}{\sin(b+a)} \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$

(c) $\frac{1}{\sin(b-a)} \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$

(d) None of these

Q 268. $\int e^{e^{e^x}} . e^{e^x} . e^x \, dx = ... + C$

[Guj. CET]

(a) e^{e^x}

(b) $\frac{1}{2} e^2 . e^x$

(c) $e^{e^{e^x}}$

(d) $\frac{1}{2} e^{e^x}$

Q 269. $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx$ is equal to

[Kerala CEE]

(a) $\frac{3}{2}x+\frac{35}{35}\log|9e^{2x}-4|+C$

(b) $\frac{3}{2}x-\frac{36}{36}\log|9e^{2x}-4|+C$

(c) $-\frac{3}{2}x-\frac{35}{36}\log|9e^{2x}-4|+C$

(d) $-\frac{5}{2}x+\frac{35}{36}\log|9e^{2x}-4|+C$

(e) $\frac{5}{2}x+\frac{35}{36}\log|9e^{2x}-4|+C$

Integration by Substitution, Parts, Partial Fractions

Q 270. $\int\frac{\sin 2x}{\sin^4 x+\cos^4 x}dx$ is equal to

[BITSAT]

(a) $2\ \tan^{-1}(\tan^2 x)+C$

(b) $\tan^{-1}(x\ \tan^2 x) + C$

(c) $\tan^{-1}\ (\tan^2 x) + C$

(d) None of these

Q 271. $\int \frac{dx}{x^2\sqrt{4-x^2}}$ is equal to

[GGSIPU]

(a) $\frac{1}{4}\left(\frac{\sqrt{4-x^2}}{x}\right)+C$

(b) $\frac{1}{2}\left(\frac{\sqrt{4-x^2}}{x}\right)+C$

(c) $-\frac{1}{4}\left(\frac{\sqrt{4-x^2}}{x}\right)+C$

(d) $-\frac{1}{2}\left(\frac{\sqrt{4-x^2}}{x}\right)+C$

Q 272. $\int\frac{1+\cot^2 x}{1+\cot x}dx$ is equal to

[MP PET]

(a) – log (1 + cot x) + C

(b) log (1 + tan x) + C

(c) log (1 + cot x) + C

(d) log cot x + C

Q 273. $\int \sqrt{\frac{e^x + a}{e^x - a}} dx$ is equal to

[MP PET]

(a) $\cosh^{-1}\left(\frac{e^x}{a}\right) + \sec^{-1}\left(\frac{e^x}{a}\right) + C$

(b) $\cosh^{-1}\left(\frac{e^x}{a}\right) - \sec^{-1}\left(\frac{e^x}{a}\right) + C$

(c) $\frac{1}{a}\left[\cosh^{-1}\left(\frac{e^x}{a}\right) + \sec^{-1}\left(\frac{e^x}{a}\right)\right] + C$

(d) $\frac{1}{a}\left[\cosh^{-1}\left(\frac{e^x}{a}\right) - \sec^{-1}\left(\frac{e^x}{a}\right)\right] + C$

Q 274. $\int e^x \left[\frac{\sin x + \cos x}{1 - \sin^2 x}\right] dx$ is

[KCET]

(a) $(e^x \cdot \text{cosec } x) + C$

(b) $e^x \cot x + C$

(c) $(e^x \cdot \sec x) + C$

(d) $e^x \tan x + C$

20. Definite Integrals

Fundamental of Definite Integration

Q 275. Let [.] denote the greatest integer function, then the value of $\int_0^{1.5} x\,|x^2|\,dx$ is

[AIEEE]

(a) $\frac{5}{4}$

(b) 0

(c) $\frac{3}{2}$

(d) $\frac{3}{4}$

Q 276. If $\frac{d}{dx}(\phi(x)) = f(x)$, then $\int_1^2 f(x)dx$ is equal to

[BITSAT]

(a) $f(l) - f(2)$

(b) $\phi(1) - \phi(2)$

(c) f(2)− f(l)

(d) $\phi(2) - \phi(1)$

Q 277. The value of $\int_{-2}^{2}(x\cos x + \sin x + 1)dx$ is

[WB JEE]

(a) 2

(b) 0

(c) −2

(d) 4

Q 278. The value of $\int_0^{\pi} \sin^{50} x \cos^{49} x \, dx$ is

[WB JEE]

(a) 0

(b) $\pi/4$

(c) $\pi/2$

(d) 1

Q 279. If $I_1 = \int_0^{\pi/2} f(\sin 2x) \sin x \, dx$ and $I_2 = \int_0^{\pi/4} f(\cos 2x) \cos x \, dx$,

then I_1 / I_2 is equal to

(a) 1

(b) $\sqrt{2}$

(c) $\frac{1}{\sqrt{2}}$

(d) 2

Q 280. The value of $\int_0^4 |x-1| dx$ is

[KCET]

(a) $\frac{5}{2}$

(b) 5

(c) 4

(d) 1

Definite Integration by Substitution, Parts, Partial Fractions

Q 281. The value of $\int_{\sqrt{\log 2}}^{\sqrt{\log 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\log 6 - x^2)} dx$ is

[IIT JEE]

(a) $\frac{1}{4}\log\frac{3}{2}$

(b) $\frac{1}{2}\log\frac{3}{2}$

(c) $\log\frac{3}{2}$

(d) $\frac{1}{6}\log\frac{3}{2}$

Q 282. The value of $\int_0^1 \frac{8\log(1+x)}{1+x^2}dx$ is

[AIEEE]

(a) $\frac{\pi}{8}\log 2$

(b) $\frac{\pi}{2}\log 2$

(c) $\log 2$

(d) $\pi \log 2$

Q 283. $\int_0^2 |1-x|\,dx$ is equal to

[BITSAT]

(a) 0

(b) 1

(c) $\frac{3}{2}$

(d) $\frac{1}{2}$

Q 284. $\int_{\pi}^{16\pi} |\sin x| \, dx$ is equal to

[WB JEE]

(a) 0

(b) 32

(c) 30

(d) 28

Q 285. $\int_0^{\pi/2} \frac{x \sin x \cos x}{\cos^4 x + \sin^4 x} dx$ is equal to

[MP PET]

(a) $\frac{\pi^2}{4}$

(b) $\frac{\pi^2}{8}$

(c) $\frac{\pi^2}{16}$

(d) $\frac{\pi^2}{32}$

Q 286. If [x] denotes the greatest integer less than or equal to x, then the value of $\int_0^2 (|x-2|+[x])dx$ is equal to

[Kerala CEE]

(a) 2

(b) 3

(c) 1

(d) 4

(e) $\frac{3}{2}$

Q 287. If $I_n = \int_0^{\pi/4} \tan^n x \, dx$, where n is a positive integer, then $I_{10} + I_8$ is

[KCET]

(a) $\frac{1}{9}$

(b) $\frac{1}{8}$

(c) $\frac{1}{7}$

(d) 9

Properties and Summation of Series of Definite Integration.

Q 288. The value of $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^3}{r^4+n^4}$ is

[WB JEE]

(a) $\frac{1}{2}\log_e(1/2)$

(b) $\frac{1}{4}\log_e(1/2)$

(c) $\frac{1}{4}\log_e 2$

(d) $\frac{1}{2}\log_e 2$

Definite Integration by Gamma Function, Leibnitz Rule, Walli's Formula.

Q 289. Let f : $[1, \infty) \to [2, \infty)$ be a differentiable function such that $f(1) = 2$. If $6\int_1^x f(t)dt = 3x\, f(x) - x^3$ for all $x > 1$, then the value of $f(2)$ is.....

[IIT JEE]

21. Applications of Integrals

Area of Bounded Region

Q 290. Let the straight line x = b divide the area enclosed by $y = (1 - x)^2$, $y = 0$ and $x = 0$ into two

parts $R_1(0 \leq x \leq b)$ and $R_2(b \leq x \leq 1)$ such that R_1 – R_2 = Then, $\frac{1}{4}$. b equals

[IITJEE]

(a) $\frac{3}{4}$

(b) $\frac{1}{2}$

(c) $\frac{1}{3}$

(d) $\frac{1}{4}$

Q 291. The area of the region enclosed by the curves $y = x$, $x = e$, $y = \frac{1}{x}$ and the positive x−axis is

[AIEEE]

(a) 1 sq unit

(b) $\frac{3}{2}$ sq unit

(c) $\frac{5}{2}$ sq unit

(d) $\frac{1}{2}$ sq unit

Q 292. Let f : $[-1, 2] \to [0, \infty)$ be a continuous function such that f(x) = f(1 − x) for all x $\in$ $[-1, 2]$. Let $R_1 = \int_{-1}^{2} xf(x)dx$ and R_2 be the area of the region bounded by y = f(x), x = −1, x = 2 and the x−axis. Then,

[IITJEE]

(a) $R_1 = 2R_2$

(b) $R_1 = 3R_2$

(c) $2R_1 = R_2$

(d) $3R_1 = R_2$

Q 293. The area enclosed between $y^2 = x$ and $y = x$ is

[WBJEE]

(a) $\frac{2}{3}$ sq unit

(b) $\frac{1}{2}$ sq unit

(c) $\frac{1}{3}$ sq unit

(d) $\frac{1}{6}$ sq unit

Q 294. The area in the positive quadrant enclosed by the circle $x^2 + y^2 = 4$, the line $x = y\sqrt{3}$ and x−axis is

[MP PET]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) π

Q 295. Find the area, if a curve xy = 4, bounded by the lines x = 1 and x = 3 and X−axis

[Guj. CET]

(a) log 12

(b) log 64

(c) log 81

(d) log 27

Q 296. If the area between $y = mx^2$ and $x = my^2 (m>0)$ is $1/4$ sq units, then the value of m is

[KCET]

(a) $\pm 3\sqrt{2}$

(b) $\pm \frac{2}{\sqrt{3}}$

(c) $\sqrt{2}$

(d) $\sqrt{3}$

Q 297. The area bounded by the curve y = 1 + $\log_e$ x, the x−axis and the straight line x = e is equal to (in square units)

[J&K CET]

(a) $3e-2$

(b) e

(c) $e-\frac{1}{e}$

(d) $e+\frac{1}{e}$

22. Differential Equations

Order and Degree, Formation of Differential Equation

Q 298. $\frac{d^2x}{dy^2}$ equals to

[A1EEE]

(a) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$

(b) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$

(c) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

(d) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

Q 299. If $y = \frac{A}{x} + Bx^2$, then $x^2 \frac{d^2y}{dx^2}$ is equal to

[WBJEE]

(a) 2y

(b) y^2

(c) y^3

(d) y^4

Q 300. The order of differential equation

$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + \int ydx = x^3$ is

[MP PET]

(a) 2

(b) 3

(c) 1

(d) 4

Q 301. If $y = (\sin^{-1} x)^2$, then $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx}$ is

equal to

[Kerala CEE]

(a) 0

(b) -1

(c) -2

(d) 1

(e) 2

Q 302. $\sqrt{r} = ac^{\theta \cot \alpha}$ where a and α are real numbers, then $\frac{d^2r}{d\theta^2} - 4r\cot^2\alpha$ is

[KCET]

(a) r

(b) $\frac{1}{r}$

(c) 1

(d) 0

Q 303. The degree of the differential equation

$\frac{d^2y}{dx^2} = \frac{5y + \frac{dy}{dx}}{\sqrt{\frac{d^2y}{dx^2}}}$ is equal to

[J&K CET]

(a) 2

(b) 3

(c) 4

(d) $\frac{5}{2}$

Variable Separable Method

Q 304. Let f be the purchase value of an equipment and V(t) be the value after it has been used for t years. The value V(t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T - t)$, where k > 0 is a constant and T is the total life in

years of the equipment. Then, the scrap value V (T) of the equipment is

[AIEEE]

(a) $I-\frac{kT^2}{2}$

(b) $I-\frac{k(T-t)^2}{2}$

(c) e^{-kT}

(d) $T^2-\frac{1}{k}$

Q 305. The solution of differential equation

$4xy\frac{dy}{dx}=\frac{3(1+x)^2(1+y^2)}{(1+x^2)}$ is

[UP SEE]

(a) log(l + y) = log x +2tan x + constant

(b) log(1 + y^2) = 3 log $\left(\frac{1}{x}\right)$+ 6 $\tan^{-1}$ x + constant

(c) 2 log(l + y^2) = 3 log x + 6 $\tan^{-1}$ x + constant

(d) None of the above

Q 306. The general solution of the differential equation $\frac{dy}{dx} = e^y (e^x + e^{-x} + 2x)$ is

[Kerala CEE]

(a) $e^{-y} = e^x - e^{-x} + x^2 + C$

(b) $e^{-y} = e^{-x} - e^x - x^2 + C$

(c) $e^{-y} = -e^{-x} - e^x - x^2 + C$

(d) $e^y = e^{-x} + e^x + x^2 + C$

(e) $e^y = e^{-x} + e^x + C$

Q 307. The general solution of $\left(\frac{dy}{dx}\right)^2 = 1 - x^2 - y^2 + x^2y^2$ is

[KCET]

(a) $2 \sin^{-1} y = x\sqrt{1-x^2} + \sin^{-1} x + C$

(b) $\cos^{-1} y = x \cos^{-1} x + C$

(c) $\sin^{-1} y = \frac{1}{2}\sin^{-1} x + C$

(d) $2 \sin^{-1} y = x\sqrt{1-y^2} + C$

Homogeneous Differential Equation

Q 308. The solution of $\frac{dy}{dx} = \frac{y}{x} + \tan\frac{y}{x}$ is

[WB JEE]

(a) x = C sin(y/x)

(b) x = C sin(xy)

(c) y = C sin(y /x) (d) xy = C sin (x /y)

Linear Differential Equation

Q 309. Consider the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If $y(1) = 1$, then x is given by

[AIEEE]

(a) $1 - \frac{1}{y} + \frac{e^{\frac{1}{y}}}{e}$

(b) $4 - \frac{2}{y} - \frac{e^{\frac{1}{y}}}{e}$

(c) $3 - \frac{1}{y} + \frac{e^{\frac{1}{y}}}{e}$

(d) $1 + \frac{1}{y} - \frac{e^{\frac{1}{y}}}{e}$

Q 310. Integrating Factor (IF) of the differential equation $\frac{dy}{dx} - \frac{3x^2y}{1+x^3} = \frac{\sin^2(x)}{1+x}$ is

[WBJEE]

(a) e^{1+x^3}

(b) $\log(1+x^3)$

(c) $1 + x^3$

(d) $\frac{1}{1+x^3}$

Q 311. The solution of $y\,dx - x\,dy + \log x\,dx = 0$ is

[GGSIPU]

(a) $y - \log x - 1 = cx$

(b) $x + \log y + 1 = cx$

(c) $y + \log x + 1 + cx$

(d) $y + \log x - 1 = cx$

Q 312. An integrating factor of the differential equation $(1 + x^2)\frac{dy}{dx} + xy = x$ is

[Kerala CEE]

(a) $\frac{x}{1+x^2}$

(b) $\frac{1}{2}\log(1+x^2)$

(c) $\sqrt{1+x^2}$

(d) x

(e) $\frac{1}{1+x^2}$

Application of Differential Equations, Some Standard Results of Solutions

Q 313. The general solution of the differential equation $\frac{d^2y}{dx^2}+8\frac{dy}{dx}+16y=0$ is

[WB JEE]

(a) $(A + Bx)\, e^{5x}$

(b) $(A + Bx)\, e^{-4x}$

(c) $(A + Bx^2)e^{4x}$

(d) $(A + Bx^4)e^{4x}$

23. Vector Algebra

Algebra and Modulus of Vectors

Q 314. If S is circumcentre, G the centroid, O the orthocentre of ΔABC, then $\overline{SA}+\overline{SB}+\overline{SC}$ isequal to

[GGSIPU]

(a) $\overline{SG}$

(b) $\overline{OS}$

(c) $\overline{SO}$

(d) $\overline{OG}$

Q 315. In R^2, find the unit vector orthogonal to unit vector $\bar{x} = (\cos \alpha, \sin \alpha)$

[Guj. CET]

(a) $(\cos \alpha/2, \sin \alpha/2)$

(b) $(-\cos \alpha, -\sin \alpha)$

(c) $(-\sin \alpha, \cos \alpha)$

(d) $(\cos \alpha, \sin \alpha)$

Q 316. A unit vector in the XOY−plane that makes an angle $30°$ with the vector **i** + j and makes an angle $60°$ with i − j is

[Kerala CEE]

(a) $\frac{1}{4}[(\sqrt{6}+\sqrt{2})\mathrm{i}-(\sqrt{6}-\sqrt{2})\mathrm{j}]$

(b) $\frac{1}{2}[(\sqrt{6}-\sqrt{2})\mathrm{i}+(\sqrt{6}+\sqrt{2})\mathrm{j}]$

(c) $\frac{1}{4}[(\sqrt{6}-\sqrt{2})\mathrm{i}+(\sqrt{6}+\sqrt{2})\mathrm{j}]$

(d) $\frac{1}{3}[(\sqrt{6}+\sqrt{2})\mathrm{i}+(\sqrt{6}-\sqrt{2})\mathrm{j}]$

(e) $\frac{1}{4}[(\sqrt{6}+\sqrt{2})\mathrm{i}+(\sqrt{6}-\sqrt{2})\mathrm{j}]$

Scalar or Dot Product of Two Vectors and Its Applications

Q 317. Let a = i + j + k, b = i − j + k and c = i − j − k be three vectors. A vector v in the plane of a and b, whose projection on **c** is $\frac{1}{\sqrt{3}}$, is given by

[IITJEE]

(a) **i** − 3j + 3k

(b) $-3\mathrm{i}-3\mathrm{j}-\mathrm{k}$

(c) $3i-j+3k$

(d) $\mathbf{i} + 3j - 3k$

Q 318. A unit vector perpendicular to both $\mathbf{i} + \mathbf{j} + k$ and $2i + j + 3k$ is

[KCET]

(a) $(2i - j - k)\sqrt{6}$

(b) $\frac{(2i - j - k)}{\sqrt{6}}$

(c) $2i + j + k$

(d) $\frac{3i + j - 2k}{\sqrt{6}}$

Vector or Cross Product of Two Vectors and Its Applications

Q 319. Let a = − i − k, b = − i + j and c = i + 2j + 3 k be three given vectors. If r is a vector such that r x b =c x b and r • a = 0, then the value of r • b is……

[IIT JEE]

Q 320. The vectors a and b are not perpendicular and c and d are two vectors satisfying $b \times c = b \times d$ and $a \cdot d = 0$. Then, the vectors d is equal to

[AIEEE]

(a) $c + \left(\frac{a.c}{a.b}\right) b$

(b) $b + \left(\frac{b.c}{a.b}\right) c$

(c) $c - \left(\frac{a.c}{a.b}\right) b$

(d) $b - \left(\frac{b.c}{a.b}\right) c$

Q 321. If a and b represent the adjacent sides of a parallelogram whose area is 15 unit, then the area of the parallelogram whose adjacent sides are 3a + 2b and a + 3b is

[MP PET]

(a) 45 units

(b) 75 units

(c) 105 units

(d) 165 units

Scalar Triple Product and Its Applications

Q 322. If $a = \frac{1}{\sqrt{10}}(3i + k)$ and $b = \frac{1}{7}(2i + 3j - 6k)$,

then the value of $(2a - b) \cdot [(a \times b) \times (a + 2b)]$ is

[AIEEE]

(a) -3

(b) 5

(c) 3

(d) -5

Q 323. If a, b and c are non−coplanar, then the value of $a.\left\{\frac{b\times c}{3b.(c\times a)}\right\}-b.\left\{\frac{c\times a}{2c.(a\times b)}\right\}$ is

[KCET]

(a) $-\frac{1}{2}$

(b) $-\frac{1}{3}$

(c) $-\frac{1}{6}$

(d) $\frac{1}{6}$

Q 324. For the non-zero vectors, a, b and c, the relation $a \cdot (b \times c) = 0$ is true, if

[J&K CET]

(a) $b \perp c$

(b) $a \perp b$

(c) $a \parallel c$

(d) $a \perp c$

Vector Triple Product and Its Applications

Q 325. The vector(s) which is/are coplanar with vectors **i** + j + 2k and **i** + 2j + k, are perpendicular to the vector **i** + j + k is/are

[IIT JEE]

(a) j − k

(b) −i + j

(c) i − j

(d) $-j + k$

Q 326. If a is perpendicular to b, then the vector

$\mathbf{a} \times \{a \times \{a \times (a \times b)\}\}$ is equal to

[Kerala CEE]

(a) $|a|^2 b$

(b) $|a| b$

(c) $|a|^3 b$

(d) $|a|^4 b$

(e) 0

Applications of Vectors in Geometry

Q 327. Find the distance between the planes $\vec{r} \cdot (2\hat{i} - \hat{j} + 3\hat{k}) = 4$ and $\vec{r} \cdot (6\hat{i} - 3\hat{j} + 9\hat{k}) + 13 = 0$

[Guj. CET]

(a) $\frac{5}{3(\sqrt{14})}$

(b) $\frac{10}{3(\sqrt{14})}$

(c) $\frac{25}{3(\sqrt{14})}$

(d) None of these

Q 328. The angle between the line $\mathbf{r} = (\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) + \lambda(2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k})$ and the plane $\mathbf{r} \cdot (\mathbf{i} + \mathbf{j} - 2\mathbf{k}) = 0$ is

[Kerala CEE]

(a) $0°$

(b) $60°$

(c) $30°$

(d) $90°$

(e) $45°$

24. Three Dimensional Geometry

System of Coordinates, Direction Cosines, Direction Ratios and Projection

Q 329. If projection of any line on coordinate axes 3, 4 and 5, then its length is

[BITSAT]

(a) 12

(b) 50

(c) $5\sqrt{2}$

(d) $3\sqrt{2}$

Q 330. If $(1, -2, -2)$ and $(0, 2, 1)$ are direction ratios of two lines, then the direction cosines of a perpendicular to both the lines are

[GGSIPU]

(a) $\left(\frac{1}{3}, -\frac{1}{3}, \frac{2}{3}\right)$

(b) $\left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right)$

(c) $\left(-\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right)$

(d) $\left(\frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$

Q 331. The shortest distances of the point $(1, 2, 3)$ from the axes x, y and z respectively are

[MP PET]

(a) $1, 2, 3$

(b) $\sqrt{5}, \sqrt{13}, \sqrt{10}$

(c) $\sqrt{10}, \sqrt{13}, \sqrt{5}$

(d) $\sqrt{13}, \sqrt{10}, \sqrt{5}$

Q 332. The projections of a directed line segment on the coordinate axes are $12, 4, 3$. The direction cosines of the line are

[AMU]

(a) $\frac{12}{13}, \frac{4}{13}, \frac{3}{13}$

(b) $\frac{12}{13}, \frac{4}{13}, -\frac{3}{13}$

(c) $-\frac{12}{13}, \frac{4}{13}, \frac{3}{13}$

(d) $\frac{12}{13}, -\frac{4}{13}, -\frac{3}{13}$

Q 333. The ratio in which ZX−plane divides the line segment AB joining the points A $(4, 2, 3)$ and B $(-2, 4, 5)$ is equal to

[J&K CET]

(a) $1 : 2$ internally

(b) $1 : 2$ externally

(c) $-2 : 1$

(d) None of these

Line

Q 334. Statement I The point A $(1, 0, 7)$ is the mirror image of the point B $(1, 6, 3)$ in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$.

Statement II The line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ bisects the line segment joining A $(1,0, 7)$ and B $(1, 6, 3)$.

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

Statement I.

(b) Statement I is true, Statement II is false.

(c) Statement I is false, Statement II is true.

(d) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement

I.

Q 335. If lines x = l + s, y=−3− λs, z = 1 + λs and $x = \frac{t}{2}$, y =1 + t, z = 2 − t, where s and t are parameters, are coplanar, then λ is equal to

[MP PET]

(a) − 1/2

(b) − 1

(c) − 2

(d) 0

Q 336. Find the shortest distance between the lines $\frac{x-3}{2}=\frac{y+15}{-7}=\frac{z-9}{5}$ and $\frac{x+1}{2}=\frac{y-1}{1}=\frac{z-9}{-3}$

[Guj. CET]

(a) $2\sqrt{3}$

(b) $3\sqrt{3}$

(c) $4\sqrt{3}$

(d) None of these

Plane

Q 337. The plane ax + by + cz =1 meets the coordinate axes in A, B, C. The centroid of the ΔABC is

[MP PET]

(a) (3a, 3b, 3c)

(b) $\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)$

(c) $\left(\frac{1}{3a},\frac{1}{3b},\frac{1}{3c}\right)$

(d) (a, b, c)

Q 338. The image of the point $(1, 3, 4)$ with respect to the plane $2x-y+z+3=0$ is

[MP PET]

(a) $(-1,4,3)$

(b) $(-3,5,2)$

(c) $(1,3,4)$

(d) $(-1,-3,-4)$

Q 339. An equation of the plane through the points $(1, 0, 0)$ and $(0, 2, 0)$ and at a distance $\frac{6}{7}$ units from the origin is

[Kerala CEE]

(a) $6x + 3y + z - 6 = 0$

(b) $6x + 3y + 2z - 6 = 0$

(c) $6x + 3y + z + 6 = 0$

(d) $6x + 3y + 2z + 6 = 0$

(e) $6x + 2y + 3z + 6 = 0$

Q 340. The equation of the plane that contains the point $(1, -1, 2)$ and is perpendicular to each of the planes $2x + 3y - 2z = 5$ and $x + 2y - 3z = 8$ is

[AMU]

(a) $5x + 4y - z = 7$

(b) $5x - 4y + z = 7$

(c) $-5x + 4y - z = 7$

(d) $5x - 4y - z = 7$

Q 341. The equation of the plane perpendicular to the Z−axis and passing through $(2, -3, 5)$ is

[J&K CET]

(a) $x - 2 = 0$

(b) $y + 3 = 0$

(c) $z - 5 = 0$

(d) $2x - 3y + 5z + 4 = 0$

Sphere

Q 342. The centre and radius of the sphere $r^2 - 2r(3i + 4j - 5k) + 1 = 0$ are

[GGSIPU]

(a) $3i + 4j - 5k, 1$

(b) $-3i - 4j + 5k, 7$

(c) $-3i-4j + 5k,7$

(d) $3i + 4j - 5k, 7$

Q 343. The equation of a sphere having centre $(1, 2, 3)$ and radius 3 units is

[Guj. CET]

(a) $x^2 + y^2 + z^2 - 2x - 4y - 6z = 0$

(b) $x^2 + y^2 + z^2 - 2x - 4y - 6z + 5 = 0$

(c) $x^2 + y^2 + z^2 - 2x - 4y - 6z - 5 = 0$

(d) None of the above

Q 344. The plane $2x - 2y + z + 5 = 0$ is a tangent to the sphere $(x - 2)^2 + (y - 2)^2 + (z - 1)^2 = r^2$, if r equals

[J&K CET]

(a) 1

(b) 2

(c) 4

(d) None of these

25. Statistics

Mean, Median and Mode

Q 345. A scientist is weighing each of 30 fishes. Their mean weight worked out is 30 g and a standard deviation of 2 g. Later, it was found that the measuring scale was misaligned and always under reported every fish weight by 2 g. The correct mean and standard deviation (in gram) of fishes are respectively

[AIEEE]

(a) 28, 4

(b) 32, 2

(c) 32, 4

(d) 28, 2

Q 346. The AM of 9 terms is 15. If one more term is added to this, series, then the AM becomes 16. The value of the added term is

[Kerala CEE]

(a) 30

(b) 27

(c) 25

(d) 23

(e) 20

Q 347. If the median of $\frac{x}{5}, x, \frac{x}{4}, \frac{x}{2}, \frac{x}{3} (x > 0)$ is 8,

then the value of x is

[Kerala CEE]

(a) 24

(b) 32

(c) 8

(d) 16

(e) 40

Measure of Dispersion

Q 348. Mean deviation of 6, 8, 12, 15, 10, 9

through mean is

[GGSIPU]

(a) 10

(b) 2.33

(c) $2, 5$

(d) None of these

Q 349. Coefficient of variation of two distribution are 50% and 60% and their arithmetic means are 30 and 25 respectively. Difference of their standard deviation is

[GGSIPU]

(a) 1

(b) 1.5

(c) 2.5

(d) 0

Q 350. If $r = 0.5$, $\Sigma xy = 120$, $\Sigma x^2 = 90$, $\sigma_y = 8$,

then n is equal to

[MP PET]

(a) 100

(b) 10

(c) 15

(d) 50

Q 351. If the standard deviation of 3, 8, 6,10,12, 9,11,10,12, 7 is 2.71, then the standard deviation of 30, 80, 60, 100, 120, 90, 110, 100, 120, 7 is

[Kerala CEE]

(a) 2.71

(b) 27.1

(c) $(2.71)\sqrt{10}$

(d) $(2.71)\sqrt{2}$

(e) 0.271

Q 352. If the values observed are $1, 2, 3, \ldots,$ n each with frequency 1 and n is even, then the mean deviation from mean equals to

[J&K CET]

(a) n

(b) $\frac{n}{2}$

(c) $\frac{n}{4}$

(d) None of these

26. Probability

Simple Problem on Probability

Q 353. A card is drawn from a well shuffled pack of playing cards. The probability that it is a heart of a king is

[UP SEE]

(a) $\frac{1}{52}$

(b) $\frac{15}{52}$

(c) $\frac{18}{52}$

(d) $\frac{19}{52}$

Q 354. 4 boys and 2 girls occupy seats in a row at random. Then, the probability that the two girls occupy seats side by side is

[WB JEE]

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{3}$

(d) $\frac{1}{6}$

Q 355. The probability that a leap year will have only 52 Sundays is

[GGSIPU]

(a) $\frac{4}{7}$

(b) $\frac{5}{7}$

(c) $\frac{6}{7}$

(d) $\frac{1}{7}$

Q 356. A die is rolled three times. The probability of getting a larger number than the previous number each line is

[MP PET]

(a) $\frac{5}{72}$

(b) $\frac{5}{54}$

(c) $\frac{13}{216}$

(d) $\frac{1}{18}$

Q 357. A fair coin is tossed 100 times the probability of getting tails an odd number of time is

[AMU]

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) 0

(d) 1

Q 358. A bag has four pair of balls of four distinct colours. If four balls are picked at random (without replacement), the probability that there is atleast one pair among them have the same colour is

[J&K CET]

(a) $\frac{1}{7!}$

(b) $\frac{8}{35}$

(c) $\frac{19}{35}$

(d) $\frac{27}{35}$

Odds in Favour and Against, Use of Permutations and Combinations

Q 359. A bag contains 3 white and 5 black balls. One ball is drawn at random. Then, the probability that it is white, is

[BITSAT]

(a) $\frac{1}{8}$

(b) $\frac{3}{8}$

(c) $\frac{5}{8}$

(d) $\frac{3}{5}$

Addition Theorem, Conditional Probability and Baye's Theorem

Paragraph for Questions Nos. 360 to 361

Let U_1 and U_2 be two urns such that U_1 contains 3 white and 2 red balls and U_2 contains only 1 white ball. A fair coin is tossed. If head appears, then 1 ball is drawn at random from U_1 and put into U_2. However, if tail appears, then 2 balls are drawn at random from U_1 and put into U_2. Now, 1 ball is drawn at random from U_2.

Q 360. The probability of the drawn ball from U_2 being white is

[IIT JEE]

(a) $\frac{13}{30}$

(b) $\frac{23}{30}$

(c) $\frac{19}{30}$

(d) $\frac{11}{30}$

Q 361. Given that the drawn ball from U_2 is white, the probability that head appeared on the coin is

[IIT JEE]

(a) $\frac{17}{23}$

(b) $\frac{11}{23}$

(c) $\frac{15}{23}$

(d) $\frac{12}{23}$

Q 362. Let E and F be two independent events.

The probability that exactly one of them occurs is

$\frac{11}{25}$ and the probability of none of them occurring is

$\frac{2}{25}$. If P(T) denotes the probability of occurrence of

the event T, then

[IITJEE]

(a) $P(E)=\frac{4}{5}, P(F)=\frac{3}{5}$

(b) $P(E)=\frac{1}{5}, P(F)=\frac{2}{5}$

(c) $P(E) = \frac{2}{5}, P(F) = \frac{1}{5}$

(d) $P(E) = \frac{3}{5}, P(F) = \frac{4}{5}$

Q 363. If A and B are two non-empty sets with B^c denoting the complement of set B such that $B \subset A$, then which of the following probability statements hold true?

[UP SEE]

(a) $P(A \cap B^c) = P(B) - P(A)$

(b) $P(A \cap B^C) = P(A) - P(B)$

(c) $P(B) \leq P(A)$

(d) $P(A) \leq P(B)$

Binomial, Poisson and Probability Distributions

Q 364. Consider 5 independent Bernoulli's trials each with probability of success p. If the probability of at least one failure is greater than or equal to $\frac{31}{32}$, then p lies in the interval

[AIEEE]

(a) $\left(\frac{3}{4}, \frac{11}{12}\right]$

(b) $\left[0, \frac{1}{2}\right]$

(c) $\left(\frac{11}{12}, 1\right]$

(d) $\left(\frac{1}{2}, \frac{3}{4}\right]$

Q 365. Suppose f(x) = $\frac{k}{2^x}$ is a probability distribution of a random variable X that can take on the values x = $0, 1, 2, 3, 4$. Then, k is equal to

[J & K CET]

(a) $\frac{16}{15}$

(b) $\frac{15}{16}$

(c) $\frac{31}{16}$

(d) None of these

27. Mathematical Logic and Boolean Algebra

Mathematical Logic

Q 366. Consider the following statements p : Suman is brilliant. q : Suman is rich. r : Suman is honest.

The negative of the statement. "Suman is brilliant and dishonest if and only if Suman is rich" can be expressed as

[AIEEE]

(a) $\sim q \leftrightarrow (p \wedge \sim r)$

(b) $\sim q \leftrightarrow p \wedge r$

(c) $\sim(p \wedge \sim r) \leftrightarrow q$

(d) $\sim p \wedge (q \leftrightarrow \sim r)$

Q 367. The only statement among the followings that is a tautology is

[AIEEE]

(a) $B \rightarrow [A \wedge (A \rightarrow B)]$

(b) $A \wedge (A \vee B)$

(c) $A \vee (A \wedge B)$

(d) $[A \wedge (A \rightarrow B)] \rightarrow B$

Q 368. Which of the following is not a logical statement?

[GGSIPU]

(a) 8 is less than 6

(b) Every set is a finite set

(c) Kashmir is far from here

(d) The sun is a star

Q 369. Let p : roses are red and q : the sun is a star.Then, the verbal translation of (~ p)/q is

[Kerala CEE]

(a) roses are not red and the sun is not a star

(b) it is not true that roses are red or the sun is not a star

(c) it is not true that roses are red and the sun is not a star

(d) roses are not red or the sun is a star

(e) it is not true that roses are red and the sun is a star

Q 370. The negation of $p \rightarrow (\sim p \vee q)$ is

[KCET]

(a) $p \vee (p \vee \sim q)$ (b) $p \rightarrow \sim (p \vee q)$ (c) $p \rightarrow q$

(d) $p \wedge \sim q$

Q 371. The contrapositive statement of the proposition $p \rightarrow \sim q$ is

[J&K CET]

(a) $\sim p \rightarrow q$

(b) $\sim q \rightarrow p$

(c) $q \rightarrow \sim p$

(d) None of these

28. Linear Programming

Q 372. The point which provides the solution of the linear programming problem. Max $(45x+55y)$ subject to constraintsx, y $\geq 0.6x + 4y \leq 120$, $3x + 10y \leq 180$ is

[MPPET]

(a) $(15,10)$

(b) $(10, 15)$

(c) $(0, 18)$

(d) $(20,0)$

Q 373. Variables of the objective function of the linear programming problem are

[MP PET]

(a) zero

(b) zero or positive

(c) negative

(d) zero or negative

29. Statics and Dynamics

Velocity and Acceleration, Relative Velocities

Q 374. The maximum and minimum magnitude of resultants of two forces are P_1 and P_2 respectively. The magnitude of the resultant when two forces are at a right angles is equal to

[J&K CET]

(a) $2\sqrt{P_1P_2}$

(b) $\sqrt{P_1 + P_2}$

(c) $\frac{\sqrt{P_1^2 + p_2^2}}{2}$

(d) $\frac{\sqrt{P_1^2 + P_2^2}}{2}$

Rectilinear Motion with Acceleration, Motion under Gravity and Laws of Motion

Q 375. Two stones A and B are thrown at an angle of $50°$ from the horizontal. The velocity of B is twice of the velocity of A. The ratio in the greatest heights attained by A and B is

[UP SEE]

(a) $1/2$

(b) $1/4$

(c) $1/8$

(d) None of these

Q 376. The acceleration of a particle starting from rest moving in a straight line with uniform acceleration is 8 m/s^2. The time taken by the particle to move the second metre is

[WB JEE]

(a) $\frac{\sqrt{2}-1}{2}$s

(b) $\frac{\sqrt{2}+1}{2}$s

(c) $(1+\sqrt{2})$s

(d) $(\sqrt{2}-1)$s

Q 377. A particle is thrown vertically upwards with velocity 24.5 cm per minute. It will return to the original position after

[J&K CET]

(a) 1 s

(b) 3 s

(c) 1.5 s

(d) None of these

Projectiles, Momentum, Energy and Direct Impact of Smooth Bodies

Q 378. Let two like parallel forces are acting, at points A and B such that the force at A is double

than the force at B. If C is the point of action of the resultant, then AC/BC is

[UP SEE]

(a) 2

(b) ½

(c) 1/3

(d) 1/4

Q 379. A force of 25 N is applied to two objects A and B. This force produces the accelerations in

the ratio $4 : 3$. If the force is increased to 30 N, then the difference in the ratios of new and earlier accelerations is

[UP SEE]

(a) zero

(b) $4/3$

(c) $8/3$

(d) $3/4$

Q 380. A body projected in vertical direction attains maximum height 50 m. Find its velocity at 25 m height

[Guj. CET]

(a) $2\sqrt{10}$ m/s

(b) $3\sqrt{10}$ m/s

(c) $5\sqrt{10}$ m/s

(d) $7\sqrt{10}$ /s

Q 381. Two balls are projected from the same point in directions inclined at $45°$ and $60°$ to the horizontal respectively. If they attain the same height, the ratio of their velocities of projection is equal to

[J&K CET]

(a) $\sqrt{3}:1$

(b) $3:1$

(c) $3:2$

(d) $\sqrt{3}:\sqrt{2}$

Composition of Coplanar, Concurrent and Parallel Forces, Lami's Theorem

Q 382. Three forces A, B and C acting at a point are in equilibrium. The ratio of angles between A and B; B and C; and A and C is $1:2:3$. Then, A:B : C is

[UP SEE]

(a) $1:\frac{\sqrt{3}}{2}:\frac{1}{2}$

(b) $\frac{1}{2}:\frac{\sqrt{3}}{2}:1$

(c) $1:\frac{1}{2}:\frac{1}{3}$

(d) $1:\frac{1}{\sqrt{2}}:\frac{1}{\sqrt{3}}$

Q 383. Three forces P, Q and R acting at a point O in the plane. The measure of angles $\angle$POQ and $\angle$QOR are $120°$ and $90°$ respectively. Then, the equilibrium forces P, Q and R are in the ratio

[J&K CET]

(a) $3:1:2$

(b) $2:1:3$

(c) $\sqrt{3}:1:2$

(d) $2:1:\sqrt{3}$

Moment, Couple and Friction

Q 384. A force 2 units acts along the line $x-4=y-5$. The moment of the force about the point $(1,1)$ along Z−axis is equal to

[J&K CET]

(a) 0

(b) $\frac{1}{\sqrt{2}}$

(c) $\sqrt{2}$

(d) $2\sqrt{2}$

30. Miscellaneous

Number Theory

Q 385. If n > 1 and n divides $(n-1)! + 1$, then

[AMU]

(a) n is always even

(b) n has to be a composite number

(c) n is divisible by exactly two primes

(d) n has to be a prime

Q 386. The digit in the unit's place of 7^{171} + $(177)!$ is

[KCET]

(a) 3

(b) 2

(c) 1

(d) 0

Q 387. The sum of all positive divisors of 242 except 1 and itself is

[KCET]

(a) 156

(b) 242

(c) 342

(d) 399

Numerical Method

Q 388. If $e^0 = 1$, $e^1 = 2.72$, $e^2 = 7.39$, $e^3 = 20.09$, $e^4 = 54.60$, then the value of $\int_0^4 e^x dx$ by using Simpson's $\frac{1}{3}$ rule is

[MP PET]

(a) 53.8733

(b) 53.78333

(c) 53.60

(d) 53.98333

Q 389. By using Newton–Raphson method, the root of $x^4 - x - 10 = 0$ which is nearer to $x = 2$, correct to three places of decimal, is

[MP PET]

(a) 2.021

(b) 1.856

(c) 1.956

(d) 1.586

Group Theory

Q 390. On the set of all non−zero reals, an operation * is defined as a * b = $\frac{3ab}{2}$. In this group, a solution of $(2 * x) * 3^{-1} = 4^{-1}$ is

[KCET]

(a) 6

(b) 1

(c) 1/6

(d) 3/2

Q 391. $G = \left\{ \begin{bmatrix} x & x \\ x & x \end{bmatrix}, x \text{ is a non} - \text{zero real number} \right\}$ is a group

with respect to matrix multiplication. In this group,

the inverse of $\begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix}$ is

[KCET]

(a) $\begin{bmatrix} 4/3 & 4/3 \\ 4/3 & 4/3 \end{bmatrix}$

(b) $\begin{bmatrix} 3/4 & 3/4 \\ 3/4 & 3/4 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Answers

1. (d)

2. (a)

3. (a)

4. (d)

5. (c)

6. (a)

7. (a)

8. (b)

9. (c)

10. (b)

11. (c)

12. (d)

13. (c)

14. (c)

15. (b)

16. (a)

17. (c)

18. (a)

19. (d)

20. (d)

21. (c)

22. (b)

23. (c)

24. (a)

25. (5)

26. (b)

27. (a)

28. (d)

29. (3)

30. (c)

31. (a)

32. (c)

33. (c)

34. (d)

35. (b)

36. (d)

37. (c)

38. (b)

39. (b)

40. (d)

41. (c)

42. (a)

43. (b)

44. (b)

45. (a)

46. (c)

47. (b)

48. (c)

49. (a)

50. (b)

51. (b)

52. (a)

53. (d)

54. (8)

55. (c)

56. (b)

57. (3,9)

58. (c)

59. (c)

60. (c)

61. (e)

62. (c)

63. (b)

64. (b)

65. (a)

66. (a)

67. (b)

68. (d)

69. (d)

70. (b)

71. (c)

72. (a)

73. (c)

74. (c)

75. (a)

76. (a)

77. (c)

78. (b)

79. (d)

80. (a)

81. (c)

82. (b)

83. (b)

84. (c)

85. (c)

86. (c)

87. (c)

88. (b)

89. (c)

90. (a)

91. (b)

92. (c)

93. (d)

94. (b)

95. (b)

96. (d)

97. (9)

98. (d)

99. (a)

100. (a)

101. (c)

102. (a)

103. (c)

104. (a)

105. (d)

106. (a)

107. (b)

108. (a)

109. (c)

110. (c)

111. (a)

112. (c)

113. (e)

114. (b)

115. (a)

116. (c)

117. (d)

118. (d)

119. (b)

120. (c)

121. (c)

122. (d)

123. (c)

124. (d)

125. (7)

126. (a)

127. (*)

128. (a)

129. (d)

130. (d)

131. (b)

132. (a)

133. (b)

134. (a)

135. (d)

136. (d)

137. (d)

138. (a)

139. (b)

140. (a)

141. (b)

142. (b)

143. (b)

144. (c)

145. (b)

146. (c)

147. (b)

148. (b)

149. (b)

150. (b)

151. (a)

152. (c)

153. (c)

154. (d)

155. (c)

156. (c)

157. (b)

158. (a)

159. (*)

160. (c)

161. (d)

162. (b)

163. (b)

164. (b)

165. (b)

166. (a)

167. (a)

168. (d)

169. (b)

170. (c)

171. (b)

172. (c)

173. (b)

174. (d)

175. (2)

176. (b)

177. (b)

178. (a)

179. (b)

180. (c)

181. (d)

182. (a)

183. (a)

184. (a)

185. (c)

186. (d)

187. (c)

188. (a)

189. (a)

190. (b)

191. (d)

192. (d)

193. (a)

194. (b, d)

195. (d)

196. (e)

197. (d)

198. (2)

199. (b)

200. (abd)

201. (b)

202. (a)

203. (d)

204. (c)

205. (b)

206. (d)

207. (d)

208. (d)

209. (a)

210. (d)

211. (b)

212. (a)

213. (b)

214. (e)

215. (d)

216. (d)

217. (d)

218. (d)

219. (c)

220. (a)

221. (d)

222. (b)

223. (c)

224. (a)

225. (a)

226. (c)

227. (b,c)

228. (abcd)

229. (a)

230. (b)

231. (0)

232. (a)

233. (a)

234. (c)

235. (d)

236. (c)

237. (b)

238. (a)

239. (1)

240. (b)

241. (d)

242. (c)

243. (a)

244. (b

245. (c)

246. (d)

247. (a)

248. (c)

249. (c)

250. (c)

251. (b)

252. (d)

253. (a)

254. (d)

255. (d)

256. (a)

257. (c)

258. (b)

259. (a)

260. (c)

261. (c)

262. (c)

263. (a)

264. (a)

265. (c)

266. (a)

267. (c)

268. (c)

269. (c)

270. (d)

271. (c)

272. (a)

273. (a)

274. (c)

275. (d)

276. (d)

277. (d)

278. (a)

279. (b)

280. (b)

281. (a)

282. (b)

283. (b)

284. (c)

285. (c)

286. (b)

287. (a)

288. (c)

289. (8/3)

290. (b)

291. (b)

292. (c)

293. (d)

294. (c)

295. (c)

296. (b)

297. (b)

298. (c)

299. (a)

300. (b)

301. (e)

302. (d)

303. (b)

304. (a)

305. (c)

306. (b)

307. (a)

308. (a)

309. (d)

310. (d)

311. (c)

312. (c)

313. (b)

314. (c)

315. (c)

316. (e)

317. (c)

318. (b)

319. (9)

320. (c)

321. (c)

322. (d)

323. (c)

324. (c)

325. (a,d)

326. (d)

327. (c)

328. (a)

329. (c)

330. (b)

331. (d)

332. (a)

333. (b)

334. (d)

335. (b)

336. (d)

337. (c)

338. (b)

339. (b)

340. (d)

341. (c)

342. (d)

343. (b)

344. (b)

345. (b)

346. (c)

347. (a)

348. (b)

349. (d)

350. (b)

351. (b)

352. (b)

353. (a)

354. (c)

355. (b)

356. (b)

357. (a)

358. (d)

359. (b)

360. (b)

361. (d)

362. (a,d)

363. (b,c)

364. (b)

365. (d)

366. (a)

367. (d)

368. (c)

369. (d)

370. (d)

371. (c)

372. (b)

373. (b)

374. (d)

375. (b)

376. (a)

377. (d)

378. (b)

379. (a)

380. (d)

381. (d)

382. (*)

383. (d)

384. (c)

385. (d)

386. (c)

387. (a)

388. (a)

389. (b)

390. (c)

391. (b)

Sets, Relations and Functions

Types of Sets, Operation and Cartesian Product

Q 1. If A = $\{4^n - 3n - 1 : n \in N\}$ and B = $\{9(n-1) : n \in N\}$, then

[AMU]

(a) $B \subset A$

(b) $A \cup B = N$

(c) $A \subset B$

(d) None of these

Q 2. Out of 64 students, the number of students taking Mathematics is 45 and number of students taking both Mathematics and Biology is 10. Then, the number of students taking only Biology is

[OJEE]

(a) 18

(b) 19

(c) 20

(d) None of these

Q 3. There are 100 students in a class. In an examination, 50 of them failed in Mathematics, 45 failed in Physics, 40 failed in Biology and 32 failed in exactly two of the three subjects. Only one student passed in all the subject. Then, the number of students failing in all the three subjects

[WB JEE]

(a) is 12

(b) is 4

(c) is 2

(d) cannot be determined from the given

information

Relation, Equivalence Relation

Q 4. Let R be a relation on the set N, defined

by $\{(x, y) : 2x - y = 10\}$, then R is

[AMU]

(a) reflexive

(b) symmetric

(c) transitive

(d) None of the above

Q 5. The function f : $[0, 3] \to [1,29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is

[IIT JEE]

(a) one–one and onto

(b) onto but not one–one

(c) one–one but not onto

(d) neither one–one nor onto

Q 6. The total number of injections (one–one into mappings) from $\{a_1, a_2, a_3, a_4\}$ to $\{b_1, b_2, b_3, b_4, b_5, b_6, b_7\}$ is

[WB JEE]

(a) 400

(b) 420

(c) 800

(d) 840

Domain–Range, Odd–Even and Periodic Function

Q 7. If $f(x) = \sin\sqrt{x}$, then period of $f(x)$ is

[OJEE]

(a) π

(b) $\frac{\pi}{2}$

(c) 2π

(d) None of these

Inverse, Composition and Different Types of Functions

Q 8. Let f : $(-1,1) \to$ R be such that f(cos 4θ) = $\frac{2}{2-\sec^2\theta}$ for $\theta \in \left(0,\frac{\pi}{4}\right) \cup \left(\frac{\pi}{4},\frac{\pi}{2}\right)$. Then, the value (s) of $f\left(\frac{1}{3}\right)$ is/are

[IIT JEE]

(a) $1-\sqrt{\frac{3}{2}}$

(b) $1+\sqrt{\frac{3}{2}}$

(c) $1-\sqrt{\frac{2}{3}}$

(d) $1+\sqrt{\frac{2}{3}}$

Q 9. If f : R $\rightarrow$ R is defined by f(x) = 2x + 3,

then $f^{-1}(x)$

[KCET]

(a) is given by $\frac{x-3}{2}$

(b) is given by $\frac{1}{2x+3}$

(c) does not exist because 'f' is not injective

(d) does not exist because 'f' is not surjective

Q 10. Let R be the set of real numbers and the functions $f : R \to R$ and $g : R \to R$ be defined by $f(x) = x^2 + 2x - 3$ and $g(x) = x + 1$. Then, the value of x for which $f(g(x)) = g(f(x))$ is

[WB JEE]

(a) -1

(b) 0

(c) 1

(d) 2

Q 11. If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ are such that min $f(x) >$ max $g(x)$, then relation between b and c is

[Manipal]

(a) no relation

(b) $0 < c < b/2$

(c) $|c| < \sqrt{2}|b|$

(d) $|c| > \sqrt{2}|b|$

2. Complex Numbers

Algebraic Operations and Equality of Complex Numbers

Q 12. Let z be a complex number such that the imaginary part of z is non-zero and $a = z^2 + z + 1$ is real. Then, a cannot take the value

[IIT JEE]

(a) -1

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) $\frac{3}{4}$

Q 13. If $\left(\frac{3}{2}+i\frac{\sqrt{3}}{2}\right)^{50} = 3^{25}(x + iy)$, where x and y are real, then the ordered pair (x, y) is

[WB JEE, MP PET]

(a) $(-3, 0)$

(b) $(0, 3)$

(c) $(0, -3)$

(d) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

Conjugate, Modulus and Argument

Q 14. If $|z - \bar{z}| + |z + \bar{z}| = 2$, then z lies on

[AMU]

(a) a circle

(b) a square

(c) an ellipse

(d) a line

Q 15. The number of solutions of the equation

$z^2 + \bar{z} = 0$, where $z \in C$ are

[KCET]

(a) 1

(b) 4

(c) 5

(d) 6

Q 16. If the conjugate of $(x + iy)(1 - 2i)$ is $1 +$ i, then

[KCET]

(a) $x - iy = \frac{1+i}{1-2i}$

(b) $x + iy = \frac{1-i}{1-2i}$

(c) $x = \frac{1}{5}$

(d) $x = -\frac{1}{5}$

Q 17. The points representing the complex number z for which arg $\left(\frac{z-2}{z+2}\right) = \frac{\pi}{3}$ lie on

[WB JEE]

(a) a circle

(b) a straight line

(c) an ellipse

(d) a parabola

Q 18. If z_1 and z_2 are two complex numbers such that $|z_1| = |z_2| + |z_1 - z_2|$, then

[Manipal]

(a) $\text{Im}\left(\frac{z_1}{z_2}\right) = 0$

(b) $\text{Re}\left(\frac{z_1}{z_2}\right) = 0$

(c) $\text{Re}\left(\frac{z_1}{z_2}\right) = \text{Im}\left(\frac{z_1}{z_2}\right)$

(d) None of these

Applications in Geometry

Q 19. If $z \neq 1$ and $\frac{z^2}{z-1}$ is real, then the point represented by the complex number z lies

[AIEEE]

(a) either on the real axis or on a circle passing through the origin

(b) on a circle with centre at the origin

(c) either on the real axis or on a circle not passing through the origin

(d) on the imaginary axis

Q 20. Area of the triangle in the arg and diagram formed by the complex numbers z, iz, z +iz, where z = x + iy is

[BITSAT]

(a) | z |

(b) $| z |^2$

(c) $2|z|^2$

(d) $\frac{1}{2}|z|^2$

3. Equations and Inequalities Logarithm and its Properties

Q 21. The value of

$$6+\log_{3/2}\left(\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}.....}}}\right)$$ is

[IIT JEE]

Q 22. If $\log_2(9^{x-1}+7)-\log_2(3^{x-1}+1)=2$,

then x values are

[KCET]

(a) $0, 2$

(b) $0, 1$

(c) $1, 4$

(d) $1, 2$

Relation Between Roots and Coefficients

Q $23.$ If the product of the roots of the equation

$x^2 - 2\sqrt{2}\,kx + 2e^{2\log k} - 1 = 0$ is 31, then the roots

of the equation are real for k is equal to

[AMU]

(a) -4

(b) 1

(c) 4

(d) 0

Q 24. If α, β, γ are the roots of the equation $x^3 + 4x + 2 = 0$, then $\alpha^3 + \beta^3 + \gamma^3$ is equal to

[KCET]

(a) 2

(b) 6

(c) -2

(d) -6

Q 25. If $(\alpha + \sqrt{\beta})$ and $(\alpha - \sqrt{\beta})$ are the roots of the equation $x^2 + px + q = 0$, where α, β, p and q are real, then the roots of the equation $(p^2 - 4q)(p^2x^2 + 4px) - 16q = 0$ are

[WB JEE]

(a) $\left(\frac{1}{\alpha}+\frac{1}{\sqrt{\beta}}\right)\text{and}\left(\frac{1}{\alpha}-\frac{1}{\sqrt{\beta}}\right)$

(b) $\left(\frac{1}{\sqrt{\alpha}}+\frac{1}{\beta}\right)\text{and}\left(\frac{1}{\sqrt{\alpha}}-\frac{1}{\beta}\right)$

(c) $\left(\frac{1}{\sqrt{\alpha}}+\frac{1}{\sqrt{\beta}}\right)\text{and}\left(\frac{1}{\sqrt{\alpha}}-\frac{1}{\sqrt{\beta}}\right)$

(d) $(\sqrt{\alpha}+\sqrt{\beta})$ and $(\sqrt{\alpha}-\sqrt{\beta})$

Q 26. The harmonic mean of the root of the equation $(5+\sqrt{2})x^2-(4+\sqrt{5})x+8+2\sqrt{5}=0$ is

[Manipal]

(a) 2

(b) 4

(c) 6

(d) 8

Solution of Equations, Nature of Roots

Q 27. If a, b, c axe in arithmetic progression, then the roots of the equation $ax^2 - 2bx + c = 0$ are

[WB JEE]

(a) 1 and $\frac{c}{a}$

(b) $-\frac{1}{a}$ and – c

(c) -1 and $-\frac{c}{a}$

(d) -2 and $-\frac{c}{2a}$

4. Sequences and Series

Arithmetic Progression and its Mean

Q 28. If 100 times the 100th term of an AP with non-zero common difference equals the 50 times its 50th term, then the 150th term of this AP is

[AIEEE]

(a) −150

(b) 150 times its 50th term

(c) 150

(d) Zero

Q 29. If $\log_{10} 2$, $\log_{10} (2^x - 1)$ and $\log_{10} (2^x + 3)$ are in AP then x is equal to

[AMU]

(a) $\log_2 5$

(b) $\log_2 (-1)$

(c) $\log_2 (1/5)$

(d) $\log_5 2$

Q 30. Six numbers are in AP such that their sum is 3. The first term is 4 times the third term. Then, the fifth term is

[WB JEE]

(a) -15

(b) -3

(c) 9

(d) -4

Geometric Progression and its Mean

Q 31. Let S_k be the sum of an infinite GP series whose first term is k and common ratio is $\frac{k}{k+1}$ (k > 0). Then, the value of $\sum_{k=1}^{\infty} \frac{(-1)^k}{S_k}$ is equal to

[WB JEE]

(a) $\log_e 4$

(b) $\log_e 2 - 1$

(c) $1 - \log_e 2$

(d) $1 - \log_e 4$

Q 32. $\sum_{r=1}^{\infty} \frac{1+a+a^2+.....+a^{r-1}}{r!}$ is equal to

[MP PET]

(a) $\frac{e^a - e}{a-1}$

(b) $e^a - e$

(c) $\frac{e^a}{a-1}$

(d) $\frac{e^a - e}{e-1}$

Q 33. Suppose a, b, c are in AP and a^2, b^2, c^2 are in GP. If a<b< c and a + b + c = $\frac{3}{2}$, then the value of a is

[Manipal]

(a) $\frac{1}{2\sqrt{2}}$

(b) $\frac{1}{2\sqrt{3}}$

(c) $\frac{1}{2}-\frac{1}{\sqrt{3}}$

(d) $\frac{1}{2}-\frac{1}{\sqrt{2}}$

Q 34. $0.2 + 0.22 + 0.222 + \ldots$ to n terms is equal to

[BITSAT]

(a) $\left(\frac{2}{9}\right)-\left(\frac{2}{81}\right)(1-10^{-n})$

(b) $n-\left(\frac{1}{9}\right)(1-10^{-n})$

(c) $\left(\frac{2}{9}\right)\left[n-\left(\frac{1}{9}\right)(1-10^{-n})\right]$

(d) $\left(\frac{2}{9}\right)$

Harmonic Progression and its Mean

Q 35. Let $a_1, a_2, a_3, \ldots$ be in a harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is

[IIT JEE]

(a) 22

(b) 23

(c) 24

(d) 25

Special Series, Sum to n terms and Infinite Number

Q 36. **Statement 1**. The sum of the series 1 + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) + ... + (361 + 380 + 400) is 800. Statement 2.

$\sum_{k=1}^{n}(k^3-(k-1)^3) = n^3$, for any natural number n.

[AIEEE]

(a) Statement 1 is false, statement 2 is true

(b) Statement 1 is true, statement 2 is true;

statement 2 is a correct explanation for statement 1

(c) Statement 1 is true, statement 2 is true;

statement 2 is not a correct explanation for

statement 1

(d) Statement 1 is true, statement 2 is false

Q 37. For a natural number n, which one is the

correct statement?

[AMU]

(a) $1^3 + 2^3 + \ldots + n^3 = (1 + 2 + \ldots + n)^2$

(b) $1^3 + 2^3 + \ldots + n^3 > (1 + 2 + \ldots + n)^2$

(c) $1^3 + 2^3 + \ldots + n^3 < (1 + 2 + \ldots + n)$

(d) $1^3 + 2^3 + \ldots + n^3 \neq (\text{l} + 2 + \ldots + n)$

Q 38. The sum of the infinite series

$1+\frac{1}{3}+\frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}+\frac{1.3.5.7}{3.6.9.12}+\ldots\ldots$ is equal to

[WB JEE]

(a) $\sqrt{2}$

(b) $\sqrt{3}$

(c) $\sqrt{\frac{3}{2}}$

(d) $\sqrt{\frac{1}{3}}$

Q 39. The sum of n terms of the series

$\frac{3}{1^2}+\frac{5}{1^2+2^2}+\frac{7}{1^2+2^2+3^2}+....$ is

[MP PET]

(a) $\frac{n+1}{n}$

(b) $\frac{n}{n+1}$

(c) $\frac{6n}{n+1}$

(d) $\frac{6(n-1)}{n}$

5. Permutations and Combinations

Number of Permutations, Conditional Permutations

Q 40. Sum of digits in the units place formed by the digits $1, 2, 3, 4$ taken all at a time is

[OJEE]

(a) 30

(b) 60

(c) 59

(d) 61

Q 41. Four speakers will address a meeting where speaker Q will always speak after speaker P. Then, the number of ways in which the order of speakers can be prepared is

[WB JEE]

(a) 256

(b) 128

(c) 24

(d) 12

Number of Combinations, Division into Groups and Dearrangements

Q 42. The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets atleast one ball is

[IIT JEE]

(a) 75

(b) 150

(c) 210

(d) 243

Paragraph

Let a_n denote the number of all n−digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = The number of such n−digit integers ending with digit 1

and c_n = The number of such n–digit integers

ending with digit 0.

Q 43. Which of the following is correct?

[IIT JEE]

(a) $a_{17} = a_{16} + a_{15}$

(b) $c_{17} \neq c_{16} + c_{15}$

(c) $b_{17} \neq b_{16} + c_{16}$

(d) $a_{17} = c_{17} + b_{16}$

Q 44. The value of b_6 is

[IIT JEE]

(a) 7

(b) 8

(c) 9

(d) 11

Q 45. Let X = {1, 2, 3, 4, 5}. The number of different ordered pairs (Y, Z) that can formed such that $Y \subseteq X$, $Z \subseteq X$ and $Y \cap Z$ is empty, is

[AIEEE]

(a) 5^2

(b) 3^5

(c) 2^5

(d) 5^3

6. Binomial Theorem and Mathematical Induction

Expansion of Binomial Theorem and General Term

Q 46. If n is a positive integer, then $(\sqrt{3}+1)^{2n} - (\sqrt{3} - 1)^{2n}$ is

[AIEEE]

(a) an irrational number

(b) an odd positive integer

(c) an even positive integer

(d) a rational number other than positive integers

Q 47. The coefficient of x^{10} in the expansion of

$1 + (1 + x) + .., + (1 + x)^{20}$ is

[WB JEE]

(a) $^{19}C_9$

(b) $^{20}C_{10}$

(c) $^{21}C_{11}$

(d) $^{22}C_{12}$

Q 48. If $\frac{e^x}{1-x} = B_0 + B_1x + B_2x^2 + \ldots + B_nx^n + \ldots,$

then $B_n - B_{n-1}$ equals

[Manipal]

(a) $\frac{1}{n!}$

(b) $\frac{1}{(n-1)!}$

(c) $\frac{1}{n!}-\frac{1}{(n-1)!}$

(d) 1

Independent Term, Middle Term and Greatest Term

Q 49. If the $(3r)$th and $(r+2)$th terms in the binomial expansion of $(1+x)^{2n}$ are equal, then

[AMU]

(a) n = r

(b) $n = r + 1$

(c) $n = 2r$

(d) $n = 2r - 1$

Q 50. The 13th term in the expansion of $\left(x^2 + \frac{2}{x}\right)^n$ is independent of x, then the sum of the divisors of n is

[KCET]

(a) 36

(b) 37

(c) 38

(d) 39

Q 51. The middle term in the expansion

$\left(x^2+\frac{1}{x^2}+2\right)^n$ is

[Manipal]

(a) $\frac{n}{[(n/2)!]^2}$

(b) $\frac{(2n!)}{[(n/2)!]^2}$

(c) $\frac{1.3.5.....(2n+1)}{n!}2^n$

(d) $\frac{(2n)!}{(n!)^2}$

Properties of Binomial Coefficients

Q 52. Let $(1 + x)^{10} = \sum_{r=0}^{10} C_r x^r$ and $(1 + x)^7 = \sum_{r=0}^{7} d_r x^r$.

If $P = \sum_{r=0}^{5} C_{2r}$ and $Q = \sum_{r=0}^{3} d_{2r+1}$, then $\frac{P}{Q}$ is equal to

[WB JEE]

(a) 4

(b) 8

(c) 16

(d) 32

7. Matrices and Determinant

Algebra of Matrices, Types of Matrices

Q 53. If P is a 3 x 3 matrix such that $P^T = 2P +$ I, where P^T is the transpose of P and I is the 3x3 identity matrix, then there exists a column matrix, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that

[IIT JEE]

(a) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(b) PX = X

(c) PX = 2X

(d) PX = − X

Q 54. Let, $A=\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 3 & 2 & 1\end{bmatrix}$. If u_1 and u_2 are column matrices such that $Au_1=\begin{bmatrix}1\\0\\0\end{bmatrix}$ and $Au_2=\begin{bmatrix}0\\1\\0\end{bmatrix}$, then u_x + u_2 is equal to

[AIEEE]

(a) $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

(b) $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$

(d) $\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

Properties and Solution of Determinants

Q 55. Let P = $[a_{ij}]$ be a 3×3 matrix and let Q = $[b_{ij}]$, where $b_{ij} = 2^{i+j} a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is

[IIT JEE]

(a) 2^{10}

(b) 2^{11}

(c) 2^{12}

(d) 2^{13}

Q 56. Let P and Q be $3x3$ matrices $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$; then determinant of $(P^2 + Q^2)$ is equal to

[AIEEE]

(a) -2

(b) 1

(c) 0

(d) -1

Q 57. If $f(x)=\begin{vmatrix} x+1 & x & 1 \\ x(x+1) & x(x-1) & 2x \\ x(x+1)(x-1) & x(x-1)(x-2) & 3x(x-1) \end{vmatrix}$, then

f(1000) is equal to

[AMU]

(a) 1

(b) 1000

(c) −1000

(d) 0

Q 58. If $ax^4 + bx^3 + cx^2 + dx +$

$e = \begin{vmatrix} x^3+3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}$, then e is equal to

[KCET]

(a) 1

(b) 0

(c) 2

(d) −1

Q 59. If P, Q and R are angles of ΔPQR, then the value of $\begin{vmatrix} -1 & \cos R & \cos Q \\ \cos R & -1 & \cos P \\ \cos Q & \cos P & -1 \end{vmatrix}$ is equal to

[WE JEE]

(a) -1

(b) 0

(c) $\frac{1}{2}$

(d) 1

Q 60. If $P = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 1 \end{bmatrix}$, $Q = PP^T$, then the value of the determinant of Q is equal to

[WB JEE]

(a) 2

(b) -2

(c) 1

(d) 0

Q 61. The value of the determinant

$$\begin{vmatrix} a^2+1 & ab & ac \\ ba & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix}$$ is

[MPPET]

(a) $(a + b + c)^2$

(b) $a^2 + b^2 + c^2$

(c) $a^2 + b^2 + c^2 + 1$

(d) 1

Transpose, Adjoint and Inverse of Matrices

Q 62. If the adjoint of a 3×3 matrix P is

$\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value (s) of the

determinant of P is/are

[IIT JEE]

(a) -2

(b) -1

(c) 1

(d) 2

System of Linear Equations, Rank of Matrices

Q 63. The system of linear equations

$x - y - 2z = 6$

$-x + y + z = \mu$

$\lambda x + y + z = 3$ has

[WB JEE]

(a) infinite number of solutions for $\lambda \neq -1$ and all μ

(b) infinite number of solutions for $\lambda = -1$ and $\mu = 3$

(c) no solution for $\lambda \neq -1$

(d) unique solution for $\lambda = -1$ and $\mu = 3$

Q 64. The system of equations $-2x + y + z = a$, $x - 2y + z = b$, $x + y - 2z = c$ is consistent

[MP PET]

(a) $a+ b - c = 0$

(b) $a - b + c = 0$

(c) a + b + c $\neq 0$

(d) a + b + c = 0

8. Trigonometric Identities and Equations

Trigonometrical Ratios of Allied Angles and Basic Fundamental

Q 65. In a ΔPQR, if 3 sin P + 4 cos Q = 6 and

4 sin Q + 3 cos P = 1, then the angle R is equal to

[IIT JEE]

(a) $\frac{5\pi}{6}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{4}$

(d) $\frac{3\pi}{4}$

Q 66. If $\cos \alpha + 2 \cos \beta + 3 \cos \gamma = 0$, $\sin \alpha + 2 \sin \beta + 3 \sin \gamma = 0$ and $\alpha + \beta + \gamma = \pi$, then $\sin 3\alpha + 8 \sin 3\beta + 27 \sin 3\gamma$ is equal to

[KCET]

(a) -18

(b) 0

(c) 3

(d) 9

Q 67. If $\sin A + \sin B + \sin C = 3$, then $\cos A + \cos B + \cos C$ is equal to

[MP PET]

(a) 3

(b) 2

(c) 1

(d) 0

Q 68. If $\alpha + \beta + \gamma = \pi$, then the value of $\sin^2 \alpha + \sin^2 \beta - \sin^2 \gamma$ is equal to

[Manipal]

(a) $2 \sin \alpha$

(b) $2 \sin \alpha \cos \beta \sin \gamma$

(c) $2 \sin \alpha \sin \beta \cos \gamma$

(d) $2 \sin \alpha \sin \beta \sin \gamma$

Q 69. If cos A = m cos B, and $\cot\frac{A+B}{2}=\lambda\tan\frac{B-A}{2}$,

then λ is

[Manipal]

(a) $\frac{m}{m-1}$

(b) $\frac{m+1}{m}$

(c) $\frac{m+1}{m-1}$

(d) None of these

Trigonometrical Ratios of Sum and Difference of Two and Three Angles

Q 70. If A + B = $\frac{\pi}{4}$, then $(1 + \tan A)(1 + \tan B)$

is equal to

[MP PET]

(a) 2

(b) 1

(c) 3

(d) $\sqrt{3}$

Maximum–Minimum Values and Conditional Trigonometrical Identities

Q 71. Least value of $3 \sin^2 \theta + 4 \cos^2 \theta$ is

[OJEE]

(a) 2

(b) 3

(c) 0

(d) 1

Hyperbolic Functions

Q 72. The curve represented by

$x = a(\sinh \theta + \cosh \theta)$

$y = b(-\sinh \theta + \cosh \theta)$ is

[AMU]

(a) a hyperbola

(b) a parabola

(c) an ellipse

(d) a circle

Solution of Trigonometrical Equations

Q 73. Let $\theta, \phi \in [0, 2\pi]$ be such that $2 \cos\theta(1 - \sin\phi) = \sin^2\theta$, $\left(\tan\frac{\theta}{2} + \cot\frac{\theta}{2}\right)\cos\phi - 1$, $\tan(2\pi - \theta) > 0$ and $-1 < \sin\theta < -\frac{\sqrt{3}}{2}$. Then, ϕ cannot satisfy

[IITJEE]

(a) $0 < \phi < \frac{\pi}{2}$

(b) $\frac{\pi}{2} < \phi < \frac{4\pi}{3}$

(c) $\frac{4\pi}{3} < \phi < \frac{3\pi}{2}$

(d) $\frac{3\pi}{2} < \phi < 2\pi$

Q 74. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has

[AIEEE]

(a) infinite number of real roots

(b) no real roots

(c) exactly one real root

(d) exactly four real roots

Q 75. Value of θ satisfying $\cos\theta + \sqrt{3}\sin\theta = 2$ is

[OJEE]

(a) $30°$

(b) $60°$

(c) $45°$

(d) $90°$

Q 76. If $\sin 2x = 4\cos x$, then x is equal to

[KCET]

(a) $n\frac{\pi}{2} \pm \frac{\pi}{4}$, $n \in Z$

(b) no value

(c) $n\pi + (-1)^n \frac{\pi}{4}$, $n \in Z$

(d) $2n\pi \pm \frac{\pi}{2}$, $n \in Z$

Q 77. Number of solutions of the equation tan x + sec x = 2 cos x, x $\in [0, \pi]$ is

[WB JEE]

(a) 0

(b) 1

(c) 2

(d) 3

9. Inverse Trigonometrical Functions

Solutions of Inverse Trigonometrical Functions

Q 78. Value of $\sin(\cot^{-1} x)$ is

[OJEE]

(a) $-\frac{1}{1+x^2}$

(b) $\frac{1}{\sqrt{1+x^2}}$

(c) $-\frac{1}{\sqrt{1+x^2}}$

(d) $\frac{1}{1+x^2}$

Q 79. The number of real solutions of the equation $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{2}$ is

[KCET]

(a) one

(b) four

(c) two

(d) infinitely many

Q 80. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, then the value of $x^9 + y^9 + z^9 - \frac{1}{x^9 y^9 z^9}$ is equal to

[WBJEE]

(a) 0

(b) 1

(c) 2

(d) 3

Q 81. If $x + y + z = xyz$, then $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z$ is equal to

[Manipal]

(a) π

(b) $\frac{\pi}{2}$

(c) 0

(d) None of these

Q 82. If $4 \sin^{-1} x + \cos^{-1} x = \pi$, then x is equal to

[BITSAT]

(a) $\frac{1}{2}$

(b) 2

(c) 1

(d) $\frac{1}{3}$

10. Properties of Triangle

Relation Between Sides and Angles

Q 83. Let PQR be a triangle of area A with a = 2, b = $\frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R, respectively. Then, $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals to

[IIT JEE]

(a) $\frac{3}{4\Delta}$

(b) $\frac{45}{4\Delta}$

(c) $\left(\frac{3}{4\Delta}\right)^2$

(d) $\left(\frac{45}{4\Delta}\right)^5$

Q 84. Let p, q and r be the sides opposite to the angles P, Q, R, respectively in a ΔPQR. If r^2 sin P sin Q = pq, then the triangle is

[WB JEE]

(a) equilateral

(b) acute angled but not equilateral

(c) obtuse angled

(d) right angled

Q 85. The base BC of a ΔABC is 6 cm and $\angle$B $= 112.5°$, $\angle$C $= 22.5°$, then its altitude is

[MP PET]

(a) 12 cm

(b) 6 cm

(c) 1.5 cm

(d) 3 cm

Q 86. If A + B + C = $180°$, then $\frac{\cot A + \cot B + \cot C}{\cot A \cot B \cot C}$ is

equal to

[BITSAT]

(a) 1

(b) cot A cos B cot C

(c) -1

(d) 0

12. Rectangular Cartesian Coordinates System of Coordinates and Distance Between Points

Q 87. If the line $2x + y = k$ passes through the point which divides the line segment joining the points $(1, 1)$ and $(2, 4)$ in the ratio $3 : 2$, then k equals

[AIEEE]

(a) $\frac{29}{5}$

(b) 5

(d) 6

(d) $\frac{11}{5}$

Q 88. The line joining A(b cos α, b sin α) and B(a cos β, a sin β), where a $\neq$ b, is produced to the point M(x,y) so that AM : MB = b : a. Then, $x\cos\frac{\alpha+\beta}{2}+y\sin\frac{\alpha+\beta}{2}$ is equal to

[WB JEE]

(a) 0

(b) 1

(c) -1

(d) $a^2 + b^2$

Q 89. The equation of the base BC of an equilateral ΔABC is $x + y = 2$ and A is $(2, -1)$. The length of the side of the triangle is

[BITSAT]

(a) $\sqrt{2}$

(b) $\left(\frac{3}{2}\right)^{1/2}$

(c) $\left(\frac{1}{2}\right)^{1/2}$

(d) $\left(\frac{2}{3}\right)^{1/2}$

Orthocentre, Circumcentre and Incentre

Q 90. The equation of the circle circumscribing the triangle formed by the lines $x + y = 6$, $2x + y = 4$ and $x + 2y = 5$ is

[BITSAT]

(a) $x^2 + y^2 + 17x + 19y - 50 = 0$

(b) $x^2+y^2-17x-19y-50=0$

(c) $x^2+y^2+17x-19y-50=0$

(d) $x^2+y^2-17x-19y+50=0$

Area of Some Geometrical Figures

Q 91. A line is drawn through the point $(1, 2)$ to meet the coordinate axes at P and Q such that it forms a ΔOPQ, where O is the origin, if the area of the ΔOPQ is least, then the slope of the line PQ is

[AIEEE]

(a) $-\frac{1}{4}$

(b) -4

(c) -2

(d) $-\frac{1}{2}$

Q 92. The tangent to the curve xy = 25 at any point on it cuts the coordinate axes at A and B, then the area of the ΔOAB is

[KCET]

(a) 50 sq units

(b) 25 sq units

(c) 75 sq units

(d) 100 sq units

13. Straight Line and Pair of Straight Lines

Equation of Line and Pair of Lines

Q 93. The point of intersection of lines

represented by the equation $3x^2 + 8xy - 3y^2 + 29x$

$- 3y + 18 = 0$ is

[MP PET]

(a) $\left(\frac{3}{2}, \frac{5}{2}\right)$

(b) $\left(\frac{-3}{2}, \frac{-5}{2}\right)$

(c) $(-3, -5)$

(d) $(3, 5)$

Angle Between Intersection Lines, Condition for Parallel and Perpendicular Lines

Q 94. If lines $(\tan^2 \theta + \cos^2 \theta) x^2 - 2 \tan \theta \cdot xy + \sin^2 \theta \cdot y^2 = 0$ make with x–axis angles α, β, then $\tan \alpha - \tan \beta$ is equal to

[MP PET]

(a) 2

(b) 4

(c) $\tan\theta$

(d) $2\tan\theta$

Position of a Point, Bisector of the Lines

Q 95. If the two pair of lines $x^2 - 2mxy - y^2 = 0$ and $x^2 - 2nxy - y^2 = 0$ are such that one of them

represents the bisector of the angles between the

other, then

[AMU]

(a) $mn = 1$

(b) $m + n = mn$

(c) $mn = -1$

(d) $m - n = mn$

14. Circle and System of Circles

Circle and its Equations

Q 96. The length of the diameter of the circle which touches the x−axis at the point $(1,0)$ and passes through the point $(2, 3)$ is

[AIEEE]

(a) $\frac{10}{3}$

(b) $\frac{3}{5}$

(c) $\frac{6}{5}$

(d) $\frac{5}{3}$

Q 97. Equation of unit circle concentric with

circle $x^2 + y^2 + 8x + 4y - 8 = 0$ is

[OJEE]

(a) $x^2 + y^2 + 8x + 4y + 19 = 0$

(b) $x^2 + y^2 - 8x + 4y + 19 = 0$

(c) $x^2 + y^2 - 8x - 4y + 19 = 0$

(d) None of the above

Q 98. The least and the greatest distances of the point $(10, 7)$ from the circle $x^2 + y^2 - 4x - 2y - 20 = 0$ are

[KCET]

(a) $10, 5$

(b) $15, 20$

(c) $12, 16$

(d) $5, 15$

Tangent and Normal to a Circle

Paragraph

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3},1)$. A straight line l, perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$.

Q 99. A common tangent of the two circles is

[IIT JEE]

(a) $x = 4$

(b) $y = 2$

(c) $x + \sqrt{3}y = 4$

(d) $x + 2\sqrt{2}y = 6$

Q 100. A possible equation of L is

(a) $x - \sqrt{3}y = 1$

(b) $x + \sqrt{3}y = 1$

(c) $x - \sqrt{3}y = -1$

(d) $x + \sqrt{3}y = 5$

Q 101. The equations of the two tangents from $(-5, -4)$ to the circle $x^2 + y^2 + 4x + 6y + 8 = 0$ are

[KCET]

(a) $x + 2y + 13 = 0, 2x - y + 6 = 0$

(b) $2x + y + 13 = 0, x - 2y - 6$

(c) $3x + 2y + 23 = 0, 2x - 3y + 4 = 0$

(d) $x - 7y = 23, 6x + 13y = 4$

Q 102. If a > 2b > 0, then positive value of m for which $y = mx - b\sqrt{1+m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x - a)^2 + y^2 = b^2$ is

[Manipal]

(a) $\frac{2b}{\sqrt{a^2 - 4b^2}}$

(b) $\frac{\sqrt{a^2 - 4b^2}}{2b}$

(c) $\frac{2b}{a - 2b}$

(d) $\frac{b}{a - 2b}$

Chord of Contact of Tangent, Pole and Polar

Q 103. The locus of the mid-point of the chord of contact of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ is

[IIT JEE]

(a) $20(x^2 + y^2) - 36x + 45y = 0$

(b) $20(x^2 + y^2) + 36x - 45y = 0$

(c) $36(x^2 + y^2) - 20x + 45y = 0$

(d) $36(x^2 + y^2) + 20x - 45y = 0$

Q 104. The locus of the mid-points of the chord of the circle $x^2 + y^2 = 4$ which subtends a right angle at the origin, is

[Manipal]

(a) $x + y = 2$

(b) $x^2 + y^2 = 1$

(c) $x^2 + y^2 = 2$

(d) $x + y = 1$

System of Circles

Q 105. If the circles $x^2 + y^2 + 2x + 2ky + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally, then k is equal to

[WB JEE]

(a) 2 or $-\frac{3}{2}$

(b) -2 or $-\frac{3}{2}$

(c) 2 or $\frac{3}{2}$

(d) -2 or $\frac{3}{2}$

Q 106. The circles $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 6x + 8 = 0$ are

[MP PET]

(a) intersecting in two points

(b) non–intersecting

(c) touching externally

(d) touching internally

15. Conic Section

Basic Terms of Conics

Q 107. The equation $y^2 + 4x + 4y + k = 0$

represents a parabola whose latusrectum is

[WB JEE]

(a) 1

(b) 2

(c) 3

(d) 4

Q 108. A hyperbola passes through $(3, 3)$ and the length of its conjugate axis is 8. The length of latusrectum is

[MP PET]

(a) $\frac{20}{3}$

(b) $\frac{40}{3}$

(c) $\frac{50}{3}$

(d) None of these

Q 109. If the length of the major axis of the ellipse $\left(\frac{x^2}{a^2}\right)+\left(\frac{y^2}{b^2}\right)=1$ is three times the length of minor axis, its eccentricity is

[BITSAT]

(a) $\frac{1}{3}$

(b) $\frac{1}{\sqrt{3}}$

(c) $\sqrt{\frac{2}{3}}$

(d) $\frac{2\sqrt{2}}{3}$

Equation of Conics and Intersection of Two Curves

Q 110. The ellipse $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ is inscribed in a rectangle R whose sides are parallel to the coordinate axes. Another ellipse E_2 passing through the point $(0, 4)$ circumscribes the rectangle R. The eccentricity of the ellipse E_2 is

[IIT JEE]

(a) $\frac{\sqrt{2}}{2}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$

(d) $\frac{3}{4}$

Q 111. Let S be the focus of the parabola $y^2 = 8x$ and let PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the ΔOPS is

[IIT JEE]

Q 112. An ellipse is drawn by taking a diameter of the circle $(x - 1)^2 + y^2 = 1$ as its semi-minor axis and a diameter of the circle $x^2 + (y - 2)^2 = 4$ is semi-major axis. If the centre of the ellipse is at the origin and its axis are the coordinate axes, then the equation of the ellipse is

[AIEEE]

(a) $4x^2+y^2 = 4$

(b) $x^2 + 4y^2 = 8$

(c) $4x^2 + y^2 = 8$

(d) $x^2 + 4y^2 = 16$

Q 113. The line $x = 2y$ intersects the ellipse $\frac{x^2}{4} + y^2 = 1$ at the points P and Q. The equation of the circle with PQ as diameter is

[WB JEE]

(a) $x^2 + y^2 = \frac{1}{2}$

(b) $x^2 + y^2 = 1$

(c) $x^2 + y^2 = 2$

(d) $x^2 + y^2 = \frac{5}{2}$

Q 114. Let the foci of the ellipse $\frac{x^2}{9} + y^2 = 1$ subtend a right angle at a point P. Then, the locus of P is

[WB JEE]

(a) $x^2 + y^2 = 1$

(b) $x^2 + y^2 = 2$

(c) $x^2 + y^2 = 4$

(d) $x^2 + y^2 = 8$

Q 115. Let P and Q be the points on the parabola $y^2 = 4x$ so that the line segment PQ subtends right angle at the vertex. If PQ intersects the axis of the parabola at R, then the distance of the vertex from R is

[WB JEE]

(a) 1

(b) 2

(c) 4

(d) 6

Q 116. S and T are the foci of the ellipse $\left(\frac{x^2}{a^2}\right)+\left(\frac{y^2}{b^2}\right)=1$ and B is an end of the minor axis. If STB is an equilateral triangle, then eccentricity of the ellipse is

[BITSAT]

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) $\sqrt{\frac{3}{2}}$

Tangents and Normal of Conic

Q 117. Tangents are drawn to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$, parallel to the straight line $2x-y = l$. The points of contacts of the tangents on the hyperbola are

[IIT JEE]

(a) $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

(b) $\left(-\frac{9}{2\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

(c) $(3\sqrt{3}, -2\sqrt{2})$

(d) $(-3\sqrt{3}, 2\sqrt{2})$

Q 118. Statement **1** An equation of a common tangent to the parabola $y^2 = 16\sqrt{3}x$ and the ellipse $2x^2 + y^2 = 4$ is $y = 2x + 2\sqrt{3}$.

Statement 2 If the line y = mx + $\frac{4\sqrt{3}}{m}$, $(m \neq 0)$ is a common tangent to the parabola $y^2 = 16\sqrt{3}x$ and the ellipse $2x^2 + y^2 = 4$, then m satisfies $m^4 + 2m^2 =$ 24.

[AIEEE]

(a) Statement 1 is false, statement 2 is true

(b) Statement 1 is true, statement 2 is true; statement −2 is a correct explanation for statement 1

(c) Statement 1 is true, statement 2 is true;

statement 2 is not a correct explanation for

statement 1

(d) Statement 1 is true, statement 2 is false

Q 119. The values of m, for which the line y =

mx + 2 is a tangent to the hyperbola $4x^2 - 9y^2 =$

36, are

[MP PET]

(a) $\pm\frac{4\sqrt{2}}{3}$

(b) $\pm\frac{2}{3}$

(c) $\pm\frac{8}{9}$

(d) $\pm\frac{2\sqrt{2}}{3}$

16. Limits, Continuity and Differentiability

Limits of Algebraic, Exponential and Logarithmic Functions

Q 120. If $\lim_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4$, then

[IIT JEE]

(a) a = l, b = 4

(b) a = 1, b = − 4

(c) a = 2, b = − 3

(d) a = 2, b = 3

Q 121. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a}-1)x^2-(\sqrt{1+a}-1)x+(\sqrt[6]{1+a}-1)=0$, where a > − 1.

Then, $\lim_{a\to 0^+}\alpha(a)$ and $\lim_{a\to 0^+}\beta(a)$ are

[IIT JEE]

(a) $-\frac{5}{2}$ and 1

(b) $-\frac{1}{2}$ and -1

(c) $-\frac{7}{2}$ and 2

(d) $-\frac{9}{2}$ and 3

Q 122. $\lim_{x \to 0} \frac{x 2^x - x}{1 - \cos x}$ is equal to

[KCET]

(a) $2 \log 2$

(b) $\log 2$

(c) $\frac{1}{2} \log 2$

(d) $\frac{1}{2}$

Q 123. $\lim_{x \to 0} \frac{\pi^x - 1}{\sqrt{1+x} - 1}$

[WB JEE]

(a) does not exist

(b) equals $\log_e (\pi^2)$

(c) equals 1

(d) lies between 10 and 11

Q 124. The value of $\lim_{n \to \infty} \frac{(n!)^{1/n}}{n}$ is

[WBJEE]

(a) 1

(b) $\frac{1}{e^2}$

(c) $\frac{1}{2e}$

(d) $\frac{1}{e}$

Q $125.$ Let f : R $\rightarrow$ R be such that f(1) = 3 and f'

(1) = $6.$

Then, $\lim_{x \to 0} \left[\frac{f(1+x)}{f(1)} \right]^{1/x}$ equals

[Manipal]

(a) 1

(b) $e^{1/2}$

(c) e^{2}

(d) e^{3}

Limits of Trigonometric Functions and Series

Q 126. If $Z_n = \cos \frac{\pi}{2^r} + i \sin \frac{\pi}{2^r}$, then $Z_0 Z_1 Z_2 \ldots$ is equal to

[AMU]

(a) 1

(b) −1

(c) i

(d) −i

Q 127. Let $f(x)=\int_1^x \frac{\sin t}{t}dt$, $\lim_{x\to\infty} f'(x)$ is equal to

[OJEE]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 128. If $f(x)=\begin{vmatrix} \sin x & \cos x & \tan x \\ x^3 & x^2 & x \\ 2x & 1 & x \end{vmatrix}$, then $\lim_{x\to 0}\frac{f(x)}{x^2}$ is equal to

[KCET]

(a) 0

(b) 3

(c) 2

(d) 1

Q 129. The value of $\lim_{x\to 0}\frac{x\cos x-\log_e(1+x)}{x^2}$ is

[MP PET]

(a) 1/5

(b) 1/4

(c) 1/3

(d) 1/2

Q 130. If $f(x)=\begin{cases}\frac{\sin[x]}{[x]}, & [x]\neq 0\\ 0, & [x]=0\end{cases}$ where [x] denotes the greatest integer less than or equal to x, then $\lim_{x\to 0} f(x)$ is equal to

[Manipal]

(a) 1

(b) 0

(c) -1

(d) None of these

Q 131. $\lim_{x\to 0}\left(\frac{(2+x)\sin(2+x)-2\sin 2}{x}\right)$ is equal to

[BITSAT]

(a) sin 2

(b) cos 2

(c) 1

(d) 2 cos 2 + sin 2

Continuity

Q 132. For every integer n, let a_n and b_n be real numbers. Let function f :R $\rightarrow$ R be given by

$$f(x)=\begin{cases} a_n+\sin \pi x, & \text{for } x\in[2n,2n+1] \\ b_n+\cos \pi x, & \text{for } x\in(2n-1,2n) \end{cases}$$, for all integers n.

If f is continuous, then which of the following hold(s) for all n?

[IIT JEE]

(a) $a_{n-1} - b_{n-1} = 0$

(b) $a_n - b_n = 1$

(c) $a_n - b_{n+1} = l$

(d) $a_{n-1} - b_n = -l$

Q 133. If f:R $\rightarrow$ R is a function defined by f(x) = [x] $\cos\left(\frac{2x-1}{2}\right)\pi$, where [x] denotes the greatest integer function, then f is

[AIEEE]

(a) continuous for every real x

(b) discontinuous only at x = 0

(c) discontinuous only at non−zero integral values

of x

(d) continuous only at x = 0

Q 134. If f(x) = $x(\sqrt{x}-\sqrt{x+1})$, then f(x) is continuous

in

[OJEE]

(a) [0, ∞)

(b) (0, ∞)

(c) (−∞, ∞)

(d) $(3, \infty)$

Q 135. If $f(x) = \begin{cases} \frac{x^2 - (a+2)x + a}{x-2}, & x \neq 2 \\ 2, & x = 2 \end{cases}$ is continuous at

$x = 2$, then the value of a is

[KCET]

(a) -6

(b) 0

(c) 1

(d) -1

Q 136. If $f(x)=\frac{\log(1+ax)-\log(1-bx)}{x}$ for $x \neq x$ and $f(0)$ = k and f(x) is continuous at x = 0, then k is equal to

[BITSAT]

(a) a + b

(b) a – b

(c) a

(d) b

Differentiability

Q 137. Let $f(x)=\begin{cases} x^2\left|\cos\frac{\pi}{x}\right|, & x\neq 0, x\in R \\ 0, & x=0 \end{cases}$, then f is

[IIT JEE]

(a) differentiable both at x = 0 and at x = 2

(b) differentiable at x = 0 but not differentiable at x = 2

(c) not differentiable at x = 0 but differentiable at x = 2

(d) differentiable neither at x = 0 nor at x = 2

Q 138. Consider the function, f(x) = |x − 2| + |x − 5|, x ∈ R,

Statement 1 f' (4) = 0

Statement 2 f is continuous in [2, 5], differentiable in (2, 5) and f(2) = f(5).

[AIEEE]

(a) Statement 1 is false, statement 2 is true

(b) Statement 1 is true, statement 2 is true;

statement 2 is a correct explanation for statement 1

(c) Statement 1 is true, statement 2 is true;

statement 2 is not a correct explanation for

statement 1

(d) Statement 1 is true, statement 2 is false

Q 139. The number of points of f(x) = | x − 1| +

|x − 3| + sin x, x ∈ [0, 4) where f(x) is not

differentiable, is

[OJEE]

(a) 0

(b) 1

(c) 2

(d) 3

Q 140. A function f is defined by $f(x) = 2 + (x - 1)^{2/3}$ in $[0, 2]$. Which of the following is not correct?

[MP PET]

(a) f is continuous in $[0, 2]$

(b) $f(0) = f(2)$

(c) f is not derivable in $[0, 2]$

(d) Rolle's theorem is true in $[0, 2]$

Q 141. The function $f(x) = |x - 1|$ is

[MP PET]

(a) continuous everywhere

(b) continuous everywhere except at $x = 1$

(c) differentiable everywhere

(d) differentiable nowhere

17. Differentiation

Differentiation of Algebraic and Transcendental Functions

Q 142. If $f(x) = \log_x \{\ln (x)\}$, then $f'(e)$ is equal to

[AMU]

(a) $1/e$

(b) e

(c) $-e$

(d) e^2

Q 143. If $e^x + e^y = e^{x+y}$, then $\frac{dy}{dx}$ at $(2, 2)$ is

[OJEE]

(a) 1

(b) -1

(c) 2

(d) None of these

Q 144. If $y = e^{\log_e[1+x+x^2+.....1]}$, then $\frac{dy}{dx}$ is equal to

[KCET]

(a) $\frac{1}{(1+x)^2}$

(b) $\frac{1}{(1-x)^2}$

(c) $\frac{-1}{(1+x)^2}$

(d) $\frac{-1}{(1-x)^2}$

Q 145. Let $y=\left(\frac{3^x-1}{3^x+1}\right)\sin x + \log_e (1+x)$, $x>-1$.

Then, at $x = 0$, $\frac{dy}{dx}$ equals

[WB JEE]

(a) 1

(b) 0

(c) -1

(d) -2

Q 146. For $-\frac{\pi}{2} < x < \frac{3\pi}{2}$, the value of $\frac{d}{dx}\left\{\tan^{-1}\frac{\cos x}{1+\sin x}\right\}$ is equal to

[WB JEE]

(a) $\frac{1}{2}$

(b) $-\frac{1}{2}$

(c) 1

(d) $\frac{\sin x}{(1+\sin x)^2}$

Q 147. $\frac{d}{dx}\left\{\tan^{-1}\sqrt{\frac{1+\sin\frac{x}{2}}{1-\sin\frac{x}{2}}}\right\}$ is equal to

[MP PET]

(a) $-1/2$

(b) $1/2$

(c) $1/4$

(d) $-1/4$

Q 148. If x is measured in degrees, then $\frac{d}{dx}(\cos x)$ is equal to

[BITSAT]

(a) – sin x

(b) $-\frac{180}{\pi}\sin x$

(c) $-\frac{\pi}{180}\sin x$

(d) sin x

Q 149. If $\Delta_1 = \begin{vmatrix} x & a & b \\ b & x & a \\ a & b & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$ are the given determinants, then

[BITSAT]

(a) $\Delta_1 = 3(\Delta_2)^2$

(b) $\left(\frac{d}{dx}\right)(\Delta_1) = 3\Delta_2$

(c) $\left(\frac{d}{dx}\right)(\Delta_1) = 3(\Delta_2)^2$

(d) $\Delta_1 = 3(\Delta_2)^{3/2}$

Differentiation by Substitution, Higher Order Derivatives

Q 150. $\frac{d}{dx}\left(\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}\right)$ is equal to

[MPPET]

(a) $\frac{x^2}{2\sqrt{1+x^2}(\sqrt{1+x^2}-1)}$

(b) $\frac{1}{1+x^2}$

(c) $\frac{1}{2(1+x^2)}$

(d) $\frac{2}{1+x^2}$

Q 151. Let f be twice differentiable function satisfying $f(1) = 1$, $f(2) = 4$, $f(3) = 9$, then

[Manipal]

(a) $f''(x) = 2, \forall\ x \in R$

(b) $f'(x) = 5 = f''(x)$. For some $x \in (1, 3)$

(c) there exists atleast one $x \in (1, 3)$ such that $f''(x) = 2$ (d) None of the above

Partial Derivative

Q 152. If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, then $x\frac{\delta u}{\delta x}+y\frac{\delta u}{\delta y}$ is equal to

[OJEE]

(a) $-\sin 2u$

(b) $\sin 2u$

(c) $\cos 2u$

(d) $-\cos 2u$

18. Applications of Derivatives

Rate of Measure and Application in Mechanics

Q 153. A spherical balloon is filled with 4500π cu m of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cu m/min, then the rate (in m/min) at which the radius of the balloon decreases 49 min after the leakage began is

[AIEEE]

(a) $\frac{9}{7}$

(b) $\frac{7}{9}$

(c) $\frac{2}{9}$

(d) $\frac{9}{2}$

Q 154. The population $p(t)$ at time t of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is

[AIEEE]

(a) $2 \log 18$

(b) $\log 9$

(c) $\frac{1}{2} \log 18$

(d) $\log 18$

Q 155. A particle moves in a straight line with a velocity given by $\frac{dx}{dt} = x + 1$, where x is the distance travelled in time t. The time taken by the particle to traverse a distance of 999 m is

[AMU]

(a) $\log_{1000} e$

(b) $\log_e 100$

(c) $3\ 1\log_e 10$

(d) $4\ \log_e 10$

Q 156. If displacement x of a moving particle in a plane at time t is given by $x = 2\cos(t\sqrt{3}) + 3\sin(t\sqrt{3})$, then its acceleration is proportional to

[MP PET]

(a) $1/x$

(b) x

(c) x^2

(d) $1/x^2$

Tangent and Normal, Maxima and Minima

Q 157. Let p(x) be a real polynomial of least degree which has a local maximum at x = 1 and a local minimum at x = 3. If p(1) = 6 and p(3) = 2, then p' (0) is

[IIT JEE]

Q 158. Let f : R $\rightarrow$ R be defined as f(x) = | x | + | $x^2 - 1$|. The total number of points at which f attains either a local maximum or a local minimum is

[IIT JEE]

Q 159. If f(x)= $\int_0^x e^{t^2} (r - 2)(r - 3)\, dt$, $\forall$ x $\in$ $(0, \infty)$, then

[IIT JEE]

(a) f has a local maximum at x = 2

(b) f is decreasing on $(2, 3)$

(c) there exists some c $\in$ $(0, \infty)$ such that f" (c) = 0

(d) f has a local minimum at x = 3

Q 160. Let a,b $\in$ R be such that the function f

given by f(x) = log | x | + bx^2 + ax, x $\neq 0$ has

extreme values at x = -1 and x = 2.

Statement 1. f has local maximum at x = -1 and

at x = 2.

Statement 2.

$a = \frac{1}{2}$ and $b = \frac{-1}{4}$.

[AIEEE]

(a) Statement 1 is false, statement 2 is true

(b) Statement 1 is true, statement 2 is true;

statement 2 is a correct explanation for statement 1

(c) Statement 1 is true, statement 2 is true;

statement 2 is not a correct explanation for

statement 1

(d) Statement 1 is true, statement 2 is false

Q 161. The least value of the function f(x) = ax + b/x, a > 0, b > 0,x > 0 is

[AMU]

(a) $\sqrt{ab}$

(b) $2\sqrt{\frac{a}{b}}$

(c) $2\sqrt{\frac{b}{a}}$

(d) $2\sqrt{ab}$

Q 162. The maximum value of xe^{-x} is

[KCET]

(a) e

(b) $\frac{1}{e}$

(c) –e

(d) $-\frac{1}{e}$

Q 163. Maximum value of the function $f(x)=\frac{x}{8}+\frac{2}{x}$

on the interval $[1, 6]$ is

[WBJEE]

(a) 1

(b) $\frac{9}{8}$

(c) $\frac{13}{12}$

(d) $\frac{17}{8}$

Increasing and Decreasing Function, Rolle's Mean Value Theorems

Paragraph

Let $f(x) = (1 - x)^2 \sin^2 x + x^2$ for all $x \in R$ and let

$g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t \right) f(t)\, dt$ for all $x \in (1, \infty)$.

Q 164. Which of the following is true?

[IIT JEE]

(a) g is increasing on $(1, \infty)$

(b) g is decreasing on $(1, \infty)$

(c) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$

(d) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

Q 165. Consider the statements

P : There exists some x $\in$R such that

f(x) + $2x = 2(1 + x^2)$ Q : There exists some x $\in$ R such that

$2f(x) + 1 = 2x(1 + x)$ Then,

[IIT JEE]

(a) Both P and Q are true

(b) P is true and Q is false

(c) P is false and Q is true

(d) Both P and Q are false

Q 166. Which of the following function is decreasing on $(0, \pi/2)$?

[AMU]

(a) sin $2x$

(b) cos $3x$

(c) tan x

(d) cos $2x$

Q $167.$ If f is a real−valued differentiable function such that $f(x)f'(x) < 0$ for all real x, then

[WB JEE]

(a) f(x) must be an increasing function

(b) f(x) must be a decreasing function

(c) | f(x) | must be an increasing function

(d) | f(x) | must be a decreasing function

Q 168. Rolle's theorem is applicable in the interval $[-2, 2]$ for the function

[WB JEE]

(a) $f(x) = x^3$

(b) $f(x) = 4x^4$

(c) $f(x) = 2x^3 + 3$

(d) $f(x) = \pi |x|$

Q 169. The function $x - \cot x$

[MP PET]

(a) always increases

(b) always decreases

(c) never decreases

(d) sometimes increases and sometimes decreases

19. Indefinite Integrals

Fundamental of Indefinite Integrals

Q 170. If f(x) and g(x) are twice differentiable functions on $(0, 3)$ satisfying $f''(x) = g''(x)$, $f'(1) = 4$, $g'(1) = 6$, $f(2) = 3$, $g(2) = 9$, then $f(1) - g(1)$ is

[WB JEE]

(a) 4

(b) -4

(c) 0

(d) -2

Q 171. $\int e^x . \frac{x^2+1}{(x+1)^2} dx$ is equal to

[MP PET]

(a) $-\frac{e^x}{x+1}+C$

(b) $\frac{e^x}{x+1}+C$

(c) $\frac{xe^x}{x+1}+C$

(d) $e^x \frac{x-1}{x+1}+C$

Integration by Substitution, Parts, Partial Fractions

Q 172. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals to (for some arbitrary constant C)

[IIT JEE]

(a) $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$

(b) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$

(c) $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$

(d) $\frac{1}{(\sec x + \tan x)^{11/2}}\left\{\frac{1}{11} + \frac{1}{7}(\sec x + \tan x)^2\right\} + C$

Q 173. If the integrals $\int \frac{5\tan x}{\tan x - 2} dx = x + a \ln|\sin x - 2\cos x| + k$, then a is equal to

[AIEEE]

(a) -1

(b) -2

(c) 1

(d) 2

Q 174. $\int \frac{2^x}{\sqrt{1-4^x}} dx$ is equal to

[AMU]

(a) $(\log 2) \sin^{-1} 2^x + C$

(b) $\frac{1}{2} \sin^{-1} 2^x + C$

(c) $\frac{1}{\log 2} \sin^{-1} 2^x + C$

(d) $2 \log 2 \sin^{-1} 2^x + C$

Q 175. If $f(x) = \int \frac{dx}{(1+x^2)^{3/2}}$ and $f(0) = 0$, then value of

$f(1)$ is equal to

[OJEE]

(a) $\frac{1}{\sqrt{2}}$

(b) $-\frac{1}{\sqrt{2}}$

(c) $\sqrt{2}$

(d) $-\sqrt{2}$

Q 176. If $\int \frac{\sqrt{\cot x}}{\sin x \cos x} dx = P\sqrt{\cot x} + Q$, then the value of P is

[OJEE]

(a) -2

(b) 2

(c) 3

(d) -3

Q 177. If $n \in N$ and $I_n = \int (\log x)^n dx$, then $I_n + nI_{n-1}$

is equal to

[KCET]

(a) $\frac{(\log x)^{n+1}}{n+1}$

(b) $x(\log x)^n + C$

(c) $(\log x)^{n-1}$

(d) $\frac{(\log x)^n}{n}$

Q 178. $\int \frac{2dx}{(e^x + e^{-x})^2}$ is equal to

[BITSAT]

(a) $-\frac{e^{-x}}{(e^x + e^{-x})} + C$

(b) $-\frac{1}{(e^x + e^{-x})} + C$

(c) $\frac{1}{(e^x + 1)^2} + C$

(d) $\frac{1}{(e^x - e^{-x})^2} + C$

20. Definite Integrals

Fundamental of Definite Integration

Q 179. If [x] is the greatest integer function not greater than x, then $\int_0^{11} [x] dx$ is equal to

[KCET]

(a) 45

(b) 66

(c) 35

(d) 55

Q 180. The value of the integral

$\int_{\pi/6}^{\pi/2}\left(\frac{1+\sin 2x+\cos 2x}{\sin x+\cos x}\right)dx$ is equal to

[WB JEE]

(a) 16

(b) 8

(c) 4

(d) 1

Q 181. The value of the integral $\int_1^5 (|x-3|+|1-x|)dx$ is equal to

[WB JEE]

(a) 4

(b) 8

(c) 12

(d) 16

Q 182. $\int_0^{2\pi} |\sin x|\, dx$ is equal to

[MP PET]

(a) 2

(b) $\sqrt{3}$

(c) 4

(d) 0

Definite Integration by Substitution, Parts, Partial Fractions

Q 183. The value of the integral $\int_{-2}^{2}(1+2\sin x)e^{|x|}dx$ is

[WB JEE]

(a) 0

(b) $e^2 - 1$

(c) $2(e^2 - 1)$

(d) 1

Q 184. The value of the integral $\int_0^{\pi/2} \frac{1}{1+(\tan x)^{101}} dx$ is equal to

[WBJEE]

(a) 1

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{8}$

(d) $\frac{\pi}{4}$

Q 185. Let [x] denote the greatest integer less than or equal to x, then the value of the integral $\int_{-1}^{1}(|x|-2[x])dx$ is equal to

[WB JEE]

(a) 3

(b) 2

(c) −2

(d) −3

Properties and Summation of Series of Definite Integration

Q 186. The value of the integral

$\int_{-\pi/2}^{\pi/2}\left(x^2+\log\frac{\pi-x}{\pi+x}\right)\cos x\,dx$ is

[IIT JEE]

(a) 0

(b) $\frac{\pi^2}{2}-4$

(c) $\frac{\pi^2}{2}+4$

(d) $\frac{\pi^2}{2}$

Q 187. If g (x) = $\int_0^x \cos 4t\, dt$, then g(x + π) equals

[AIEEE]

(a) $\frac{g(x)}{g(\pi)}$

(b) g(x) + g(π)

(c) g(x) − g(π)

(d) $g(x) - g(\pi)$

Q 188. The value of $\int_0^{2\pi} \frac{dx}{e^{\sin x}+1}$ is

[MP PET]

(a) π

(b) 0

(c) 2π

(d) $\frac{\pi}{2}$

21. Applications of Integrals

Area of Bounded Region

Q 189. If S be the area of the region enclosed by

$y = e^{-x^2}$, y = 0, x = 0 and x = 1. Then,

[IIT JEE]

(a) $S \geq \frac{1}{e}$

(b) $S \geq 1 - \frac{1}{e}$

(c) $S \leq \frac{1}{4}\left(1 + \frac{1}{\sqrt{e}}\right)$

(d) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}}\left(1 - \frac{1}{\sqrt{2}}\right)$

Q 190. The area bounded between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$ and the straight line $y = 2$ is

[AIEEE, OJEE]

(a) $20\sqrt{2}$

(b) $\frac{10\sqrt{2}}{3}$

(c) $\frac{20\sqrt{2}}{3}$

(d) $10\sqrt{2}$

Q 191. The area of the region bounded by the curves $y = x^3$, $y = \frac{1}{x}$, $x = 2$ is

[WB JEE]

(a) $4 - \log_e 2$

(b) $\frac{1}{4} + \log_e 2$

(c) $3 - \log_e 2$

(d) $\frac{15}{4} - \log_e 2$

22. Differential Equations

Order and Degree, Formation of Differential Equation

Q 192. The degree of the differential equation

satisfying $\sqrt{1-x^2}+\sqrt{1-y^2}$ = a(x − y) is

[AMU]

(a) 1

(b) 3

(c) 2

(d) None of these

Q 193. Order of the equation $\left(1+5\frac{dy}{dx}\right)^{3/2}=10\frac{d^3y}{dx^3}$ is

[OJEE]

(a) 2

(b) 3

(c) 1

(d) 0

Q 194. The order of degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^{4/5}-2\frac{dy}{dx}\left(\frac{d^2y}{dx^2}\right)^2=0$ are respectively,

[MPPET]

(a) 2, 10

(b) 3, 10

(c) 3, 5

(d) 3, 4

Q 195. The differential equation of system of circles touching the y−axis at origin is

[MP PET]

(a) $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$

(b) $\frac{dy}{dx} = \frac{y^2 - x^2}{xy}$

(c) $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$

(d) $\frac{dy}{dx} = \frac{x^2 - y^2}{2xy}$

Variable Separable Method

Q 196. Solution of $e^{dy/dx} = x$ when $x = 1$ and $y = 0$ is

[KCET]

(a) $y = x(\log x - 1) + 4$

(b) $y = x(\log x - 1) + 3$

(c) $y = x(\log x + 1) + 1$

(d) $y = x(\log x - 1) + 1$

Q 197. The area of the region, bounded by the curves $y = \sin^{-1} x + x(1 - x)$ and $y = \sin^{-1} x - x(1 - x)$ in the first quadrant is

[WB JEE]

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{1}{3}$

(d) $\frac{1}{4}$

Homogeneous Differential Equation

Q 198. The general solution of the differential equation $\frac{dy}{dx} = \frac{x+y+1}{2x+2y+1}$ is

[WB JEE]

(a) $\log_e |3x + 3y + 2| + 3x + 6y = C$

(b) $\log_e |3x + 3y + 2| - 3x + 6y = C$

(c) $\log_e |3x + 3y + 2| - 3x - 6y = C$

(d) $\log, |3x + 3y + 2| + 3x - 6y = C$

Linear Differential Equation

Q 199. If y(x) satisfies the differential equation y' − y tan x = 2 x sec x and y (0), then

[IIT JEE]

(a) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$

(b) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$

(c) $y\left(\frac{\pi}{3}\right) = \frac{\pi}{9}$

(d) $y'\left(\frac{\pi}{3}\right)=\frac{4\pi}{3}+\frac{2\pi^2}{3\sqrt{3}}$

Q 200. Solution of $(x + 2y^3)\ \frac{dy}{dx}=y$ is

[OJEE]

(a) $x = y^3 + cy$

(b) $x = y^3 + cy$

(c) $x^3 = y^3 + cy$

(d) None of these

Q 201. The solution of the differential equation $(x + a)\frac{dy}{dx} - 3y = (x + a)^5$, when $y = 16a^5$ at $x = a$, is

[MPPET]

(a) $y = (x + a)^5$

(b) $2y = (x + a)^5$.

(c) $2y = (x + a)^3 + (x + a)^5$

(d) $y = (x + a)^3 - (x + a)^5$

Q 202. If y(t) is a solution of $(1 + t)\frac{dy}{dt} - ty = 1$ and $y(0) = -1$, then $y(1)$ is equal to

[Manipal]

(a) $-\frac{1}{2}$

(b) $e+\frac{1}{2}$

(c) $e-\frac{1}{2}$

(d) $\frac{1}{2}$

23. Vector Algebra

Algebra and Modulus of Vectors

Q 203. The figure formed by four points i + j + k, $2i+3j$, $3i+5j-2k$, k− j is a

[MPPET]

(a) parallelogram

(b) rectangle

(c) trapezium

(d) square

Scalar or Dot Product of Two Vectors and its Applications

Q 204. If a, b and c are unit vectors satisfying $|a - b|^2 + |b - c|^2 + |c - a|^2 = 9$, then $| 2a + 5b + 5c |$, is

[IITJEE]

Q 205. Let **a** and b be two vectors. If the vectors **c** = **a** + 2b and **d** = 5a − 4b are perpendicular to each other, then the angle between **a** and b is

[AIEEE]

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{4}$

Q 206. Let ABCD be a parallelogram such that AB= q, AB= **p** and $\angle$BAD be an acute angle. If **r** is the vector that coincides with the altitude directed from the vertex B to the side AD, then **r** is given by

[AIEEE]

(a) $r = 3q - \frac{3(p.q)}{(p.p)}p$

(b) $r = -q + \left(\frac{p.q}{p.p}\right)p$

(c) $r = q - \left(\frac{p.q}{p.p}\right)p$

(d) $r = -3q + \frac{3(p.q)}{(p.p)}p$

Q 207. If a + b + **c** = **0** and | **a** | = 3, | b | = 5, |

c | = 7, then angle between **a** and b is

[OJEE]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{4}$

(d) π

Vector or Cross Product of Two Vectors and its Applications

Q 208. If **a** and b are vectors such that | **a** + **b** | $= \sqrt{29}$ and **a** $\times (2i + 3j + 4k) = (2i + 3j + 4k) \times b$, then a possible value of $(a + b) \cdot (-7i + 2j + 3k)$ is

[IIT JEE]

(a) 0

(b) 3

(c) 4

(d) 8

Q 209. If $(a \times b) + (\mathbf{a} \cdot \mathbf{b})^2 = 144$ and $|\mathbf{a}| = 4$,

then | b | is equal to

[KCET]

(a) 16

(b) 8

(c) 3

(d) 12

Scalar Triple Product and its Applications

Q 210. If a, b and **c** are three non−coplanar vectors and p, q and **r** are vectors defined by $p = \frac{b \times c}{[a\ b\ c]}, q = \frac{c \times a}{[a\ b\ c]}$ and $r = \frac{a \times b}{[a\ b\ c]}$, then the value of (a + b) • **p** + (b + c) . q + (c + a) • **r** is equal to

[KCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 211. If the volume of the parallelopiped formed by three non−coplanar vectors a, band **c** is 4 cu units, then [a × b b × **c** **c** × a] is equal to

[KCET]

(a) 64

(b) 16

(c) 4

(d) 8

Vector Triple Product and its Application

Q 212. If **a**, **b**, **c** are unit vectors such that a × (b × c) = $\frac{1}{2}$b, then angle between **a** and **c** is

[MP PET]

(a) $\pi/6$

(b) $\pi/4$

(c) $\pi/2$

(d) $\pi/3$

Application of Vector in Geometry

Q 213. The vector equation **r** = i − 2j − k + t(6j − k) represents a straight line passing through the points

[BITSAT]

(a) $(0, 6, -1)$ and $(1, -2, -1)$

(b) $(0, 6, -1)$ and $(-1, -4, -2)$

(c) $(1, -2, -1)$ and $(1, 4, -2)$

(d) $(1, -2, -1)$ and $(0, -6, 1)$

24. Three Dimensional Geometry

System of Coordinates, Directional Cosines, Direction Ratios and Projection

Q 214. The distance of the point $(1, 2, 3)$ from the line through $(-1, 2, 5)$ and $(2, 3, 4)$ is

[MP PET]

(a) $\sqrt{\frac{6}{11}}$

(b) $2\sqrt{6}$

(c) $2\sqrt{\frac{6}{11}}$

(d) $\sqrt{6}$

Q 215. The distance of the point $(-1, -5, -10)$ from the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 5$ is

[MP PET]

(a) 11

(b) 17

(c) 13

(d) 19

Line

Q 216. Equation of the line passing through $(2, -1, 1)$ and parallel to the line $\frac{x-5}{4}=\frac{y+2}{-3}=\frac{z}{5}$ is

[OJEE]

(a) $\frac{x-2}{4}=\frac{y+1}{-3}=\frac{z-1}{5}$

(b) $\frac{x-2}{4}=\frac{y+1}{3}=\frac{z-1}{5}$

(c) $\frac{x-2}{-4}=\frac{y+1}{-3}=\frac{z-1}{5}$

(d) None of the above

Q 217. The point of intersection of line

$\frac{x-5}{3}=\frac{y-7}{-1}=\frac{z+2}{1}$ and $\frac{x+3}{-36}=\frac{y-3}{2}=\frac{z-6}{4}$ is

[MP PET]

(a) $(5,7,-2)$

(b) $(-3,3,6)$

(c) $(2,\ 10,\ 4)$

(d) $\left(21,\frac{5}{3},\frac{10}{3}\right)$

Plane

Q 218. The equation of a plane passing through the line of intersection of the planes x + 2y + 3z = 2 and x − y + z = 3 and at a distance $2/\sqrt{3}$ from the point $(3,\ 1,\ -1)$ is

[IIT JEE]

(a) $5x - 11y + z = 17$

(b) $\sqrt{2}x + y = 3\sqrt{2} - 1$ (c) $x + y + z = \sqrt{3}$

(d) $x - \sqrt{2}y = 1 - \sqrt{2}$

Q 219. A equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is

[AIEEE]

(a) $x - 2y + 2z - 3 = 0$

(b) $x - 2y + 2z + 1 = 0$

(c) $x - 2y + 2z - 1 = 0$

(d) $x - 2y + 2z + 5 = 0$

Line and Plane

Q 220. The point P is the intersection of the straight line joining the points $Q(2, 3, 5)$ and $R(1, -1, 4)$ with the plane $5x - 4y - z = 1$. If S is the foot of the perpendicular drawn from the point $T(2, 1, 4)$ to QR, then the length of the line segment PS is

[IIT JEE]

(a) $\frac{1}{\sqrt{2}}$

(b) $\sqrt{2}$

(c) 2

(d) $2\sqrt{2}$

Q 221. If the straight line $\frac{x-1}{2}=\frac{y+1}{k}=\frac{z}{2}$ and $\frac{x+1}{5}=\frac{y+1}{2}=\frac{z}{k}$ are coplanar, then the plane (s) containing these two lines is/are

[IIT JEE]

(a) $y + 2z = -1$

(b) $y + z = -1$

(c) $y - z = -1$

(d) $y - 2z = -1$

Q 222. If the line $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-k}{2}=\frac{z}{1}$ intersect, then k is equal to

[AIEEE]

(a) -1

(b) $\frac{2}{9}$

(c) $\frac{9}{2}$

(d) 0

Q 223. The equation of the plane containing the line $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$ is $a(x-x_1) + b(y-y_1) + c(z-z_1) = 0$ where

[AMU]

(a) $ax_1 + by_1 + cz_1 = 1$

(b) $a/l = b/m = c/n$

(c) $lx_1 + my_1 + nz_1 = 0$

(d) $al + bm + cn = 0$

25. Statistics

Measure of Dispersion

Q 224. Let $x_1, x_2, \ldots, x_n$ be n observations and let $\bar{x}$ be their arithmetic mean and σ^2 be the variance.

Statement 1 Variance of $2x_1, 2x_2, \ldots, 2x_n$ is $4\sigma^2$.

Statement 2 Arithmetic mean $2x_1, 2x_2, \ldots, 2x_n$ is $4\bar{x}$.

[AIEEE]

(a) Statement 1 is false, statement 2 is true

(b) Statement 1 is true, statement 2 is true;

statement 2 is a correct explanation for Statement 1

(c) Statement 1 is true, statement 2 is true;

statement 2 is not a correct explanation for

statement 1

(d) Statement 1 is true, statement 2 is false

26. Probability

Simple problems on Probability

Q 225. Probability of product of a perfect square when 2 dice are thrown together is

[OJEE]

(a) $\frac{2}{9}$

(b) $\frac{1}{9}$

(c) $\frac{5}{18}$

(d) None of these

Q 226. An urn contains 8 red and 5 white balls. Three balls are drawn at random. Then, the probability that balls of both colours are drawn is

[WB JEE]

(a) $\frac{40}{143}$

(b) $\frac{70}{143}$

(c) $\frac{3}{13}$

(d) $\frac{10}{13}$

Q 227. A and B throw a die alternatively till one of them gets a six and wins the game. If starts the, game first, the probability of A winning the game is

[MP PET]

(a) $3/11$

(b) $5/11$

(c) $6/11$

(d) $7/11$

Odds in Favour and Against, Use of Permutations and Combinations

Q 228. Two decks of playing cards are well shuffled and 26 cards are randomly distributed to a player. Then, the probability that the player gets all distinct cards is

[WB JEE]

(a) $^{52}C_{26}/^{104}C_{26}$

(b) $2 \times {}^{52}C_{26}/^{104}C_{26}$

(c) $2^{13} \times {}^{52}C_{26} / {}^{104}C_{26}$

(d) $2^{26} \times {}^{52}C_{26} / {}^{104}C_{26}$

Addition Theorem, Conditional Probability and Baye's Theorem

Q 229. A ship is fitted with three engines E_1, E_2 and E_3. The engines function independently of each other with respective probabilities $1/2, 1/4$ and $1/4$. For the ship to be operational atleast two of its engines must function. Let X denote the event that the ship is operational and let X_1, X_2 and X_3

denotes, respectively the events that the engines

E_1, E_2 and E_3 are functioning. Which of the

following is/are true?

[IIT JEE]

(a) $P[X_1^C \mid X] = \frac{3}{16}$

(b) P [Exactly two engines of the ship are

functioning x] = $\frac{7}{8}$

(c) P [X | X_2] = $\frac{5}{16}$

(d) $P[X \mid X_1] = \frac{7}{16}$

Q 230. Four fair dice D_1, D_2, D_3 and D_4 each having six faces numbered $1, 2, 3, 4, 5$ and 6 are rolled simultaneously. The probability that D_4 shows a number appearing on one of D_1, D_2 and D_3 is

[IIT JEE]

(a) $\frac{91}{216}$

(b) $\frac{108}{216}$

(c) $\frac{125}{216}$

(d) $\frac{127}{216}$

Q 231. Let X and Y be two events such that P(X | Y) = $\frac{1}{2}$, P(Y | X) = $\frac{1}{3}$ and P(X $\cap$ Y) = $\frac{1}{6}$. Which of the following is/are correct?

[IIT JEE]

(a) P(X $\cup$ Y) = $\frac{2}{3}$

(b) X and Y are independent

(c) X and Y are not independent

(d) $P(X^c \cap Y) = \frac{1}{3}$

Q 232. Three numbers are chosen at random without replacement from $\{1, 2, 3, \ldots, 8\}$. The probability that their minimum is 3, given that their maximum is 6, is

[AIEEE]

(a) $\frac{3}{8}$

(b) $\frac{1}{5}$

(c) $\frac{1}{4}$

(d) $\frac{2}{5}$

Q 233. Let A and B be two events with $P(A^C)$ = 0.3, $P(B) = 0.4$ and $P(A \cap B^c) = 0.5$. Then, $P[B \mid (A \cup B^c)]$ is equal to

[WB JEE]

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

27. Linear Programming

Q 234. Which of the following sets are not convex?

[AMU]

(a) $\{(x, y) : 8x^2 + 6y^2 \leq 24\}$

(b) $\{(x, y) : 6 \leq x^2 + y^2 \leq 36\}$

(c) $\{(x, y) : y \geq 3, y \leq 30\}$

(d) $\{(x, y) : x^2 \leq y\}$

Q 235. Consider the linear programming problem

[AMU]

Max $z = 4x + y$

Subject to

$x + y \leq 50$

$x + y \geq 100$

$x, y \geq 0$

Then, max value of z is

(a) 0

(b) 50

(c) 100

(d) Does not exist

28. Miscellaneous

Number Theory

Q 236. Which of the following is incorrect? If $a \equiv b \pmod{m}$ and x is an integer, then

[KCET]

(a) $(a + x) \equiv (b + x) \pmod{m}$

(b) $(a - x) \equiv (b - x \pmod{m}$

(c) $ax \equiv bx \pmod{m}$

(d) $(a \div x) \equiv (b \div x) \pmod{m}$

Q 237. The last digit of number 7^{886} is

[KCET]

(a) 9

(b) 7

(c) 3

(d) 1

Q 238. The sum $1 \times 1! + 2 \times 2! + \ldots + 50 \times 50!$

Equals

[WB JEE]

(a) $51!$

(b) $51! - 1$

(c) $51! + 1$

(d) $2 \times 51!$

Q 239. The least positive remainder, when $123 \times 125 \times 127$ is divided by 124 is

[BITSAT]

(a) 4

(b) 120

(c) 121

(d) 130

Binary Number

Q 240. $(100101)_2$ is equal to

[OJEE]

(a) 37

(b) 36

(c) 34

(d) 32

Numerical Method

Q 241. On dividing the interval into four equal parts the value of the integral $\int_1^5 x^2 dx$, by trapezoidal rule, is

[MPPET]

(a) 42

(b) 41.3

(c) 41

(d) 40

Q 242. For finding the root of $x^2 - 5x + 2 = 0$ near 0 by Newton–Raphson method, the second iteration x_2 is equal to

[MP PET]

(a) 0.400000

(b) 0.438095

(c) 0.438000

(d) 0.430000

Group Theory

Q 243. Which of the following is a subgroup of the group G = $\{2^n \mid n \in Z\}$ under multiplication?

[KCET]

(a) $\{4^n \mid n \in N\}$

(b) $\{3^n \mid n \in Z\}$

(c) $\{6^n \mid n \in N\}$

(d) $\{4^n \mid n \in Z\}$

Q 244. In the group $G = \{1, 2, 3, 4, 5, 6\}$ under $\otimes_7$, the solution of $4 \otimes_7 x = 5$ is

[KCET]

(a) 3

(b) 2

(c) 4

(d) 5

Q 245. Which of the following is false?

[KCET]

(a) (N, .) is a group

(b) (N, +) is a semi−group

(c) (Z,+) is a group

(d) Set of even integers is a group under usual addition

Graph Theory

Q 246. The vertex connectivity of any tree is

[Manipal]

(a) one

(b) two

(c) three

(d) None of these

Answers

1. (c) 2. (b) 3. (c) 4. (a) 5. (b) 6. (d) 7. (d) 8. (b)

9. (a) 10. (a) 11. (d) 12. (d) 13. (d) 14. (d) 15. (b)

16. (b) 17. (a) 18. (a) 19. (a) 20. (d) 21. (4) 22.

(d) 23. (c) 24. (d) 25. (a) 26. (b) 27. (a) 28. (d)

29. (a) 30. (d) 31. (d) 32. (a) 33. (d) 34. (c) 35.

(d) 36. (b) 37. (a) 38. (b) 39. (c) 40. (b) 41. (d)

42. (b) 43. (a) 44. (b) 45. (b) 46. (a) 47. (c) 48.

(a) 49. (c) 50. (d) 51. (d) 52. (b) 53. (d) 54. (d)

55. (d) 56. (c) 57. (d) 58. (b) 59. (b) 60. (a) 61.

(c) 62. (a) 63. (b) 64. (d) 65. (b) 66. (b) 67. (d)

68. (c) 69. (c) 70. (a) 71. (b) 72. (a) 73. (a,c,d)

74. (b) 75. (b) 76. (b) 77. (c) 78. (b) 79. (c) 80.

(c) 81. (a) 82. (a) 83. (c) 84. (d) 85. (d) 86. (a)

87. (c) 88. (a) 89. (d) 90. (d) 91. (c) 92. (a) 93.

(b) 94. (a) 95. (c) 96. (a) 97. (a) 98. (d) 99. (d)

100. (a) 101. (a) 102. (a) 103. (a) 104. (c) 105.

(a) 106. (d) 107. (d) 108. (b) 109. (d) 110. (c)

111. (4) 112. (d) 113. (d) 114. (d) 115. (c) 116.

(c) 117. (a,b) 118. (c) 119. (d) 120. (b) 121. (b)

122. (b) 123. (b) 124. (d) 125. (c) 126. (b) 127.

(a) 128. (c) 129. (d) 130. (d) 131. (d) 132. (b,d)

133. (a) 134. (b) 135. (b) 136. (a) 137. (a) 138. (c) 139. (c) 140. (c) 141. (a) 142. (a) 143. (b) 144. (b) 145. (a) 146. (b) 147. (c) 148. (c) 149. (b) 150. (c) 151. (c) 152. (b) 153. (c) 154. (a) 155. (c) 156. (b) 157. (9) 158. (5) 159. (a,b,c,d) 160. (c) 161. (d) 162. (b) 163. (d) 164. (b) 165. (c) 166. (d) 167. (d) 168. (b) 169. (a) 170. (b) 171. (d) 172. (c) 173. (d) 174. (c) 175. (a) 176. (a) 177. (b) 178. (a) 179. (d) 180. (d) 181. (c)

182. (c) 183. (c) 184. (d) 185. (a) 186. (b) 187. (b,c) 188. (a) 189. (a,b,d) 190. (c) 191. (d) 192. (a) 193. (b) 194. (d) 195. (c) 196. (d) 197. (c) 198. (d) 199. (d) 200. (a) 201. (b) 202. (a) 203. (c) 204. (3) 205. (c) 206. (b) 207. (a) 208. (c) 209. (c) 210. (d) 211. (b) 212. (d) 213. (c) 214. (c) 215. (c) 216. (a) 217. (d) 218. (a) 219. (a) 220. (a) 221. (b,c) 222. (c) 223. (d) 224. (d) 225. (a) 226. (d) 227. (c) 228. (d) 229. (b,d) 230. (a)

231. (a,b) 232. (b) 233. (a) 234. (d) 235. (d) 236. (d) 237. (a) 238. (b) 239. (c) 240. (a) 241. (a) 242. (b) 243. (d) 244. (a) 245. (a) 246. (a)

Sets, Relations and Functions

Q 1. Given the function $f(x)=\frac{a^x+a^{-x}}{2}$, $(a > 2)$.

Then, $f(x + y) + f(x-y)$ is equal to

[Manipal]

(a) $f(x)- f(y)$

(b) $f(y)$

(c) $2f(x)\ f(y)$

(d) None of these

Q 2. Let f(x) = cos ax + sin x be periodic.

Then, a must be

[OJEE]

(a) irrational

(b) rational

(c) positive real number

(d) None of these

Q 3. Let A={(x, y):y = e^{-x}} and B = {(x, y) :y =

− x}. Then,

[BITSAT]

(a) $A \cap B = \phi$

(b) $A \subset B$

(c) $B \subset A$

(d) $A \cap B = \{(0, 1), (0, 0)\}$

Q 4. Let $f(x) = (x + 2)^2 - 2, x \geq -2$. Then, $f^{-1}(x)$ is equal to

[AMU]

(a) $-\sqrt{2+x}-2$

(b) $\sqrt{2+x}+2$

(c) $\sqrt{2+x}-2$

(d) $-\sqrt{2+x}+2$

Q 5. Let z be the set of integers. Then, the relation R = {(a, b) : 1 + ab > 0}on z is

[AMU]

(a) reflexive and transitive but not symmetric

(b) symmetric and transitive but not reflexive

(c) reflexive and symmetric but not transitive

(d) an equivalence relation

2. Complex Numbers

Q 6. If z is a complex number of unit modulus and argument θ, then arg $\left(\frac{1+z}{1+\bar{z}}\right)$ is equal to

[JEE Main]

(a) $-\theta$

(b) $\frac{\pi}{2}-\theta$

(c) θ

(d) $\pi - \theta$

Q 7. Let complex numbers α and $\frac{1}{\alpha}$ lie on circles $(x - x_0)^2 + (y - y_0)^2 = r^2$ and $(x - x_0)^2 + (y - y_0)^2 = 4r^2$, respectively. If $z_0 = x_0 + iy_0$ satisfies the equation $2|z_0|^2 = r^2 + 2$, then $|\alpha|$ is equal to

[JEE Advanced]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{1}{2}$

(c) $\frac{1}{\sqrt{7}}$

(d) $\frac{1}{3}$

Q 8. Let $w = \frac{\sqrt{3}+i}{2}$ and P = $\{w^n : n = 1, 2, 3,\ldots\}$.

Further $H_1 = \left\{z \in C : \text{Re}\, z > \frac{1}{2}\right\}$ and $H_2 \left\{z \in C : \text{Re}\, z < -\frac{1}{2}\right\}$,

where C is the set of all complex numbers, if $z_1 \in$

$P \cap H_1$, $z_2 \in P \cap H_2$ and O represents the origin,

then $\angle z_1$ O z_2 is equal to

[JEE Advanced]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{6}$

(c) $\frac{2\pi}{3}$

(d) $\frac{5\pi}{6}$

Paragraph

Directions (Q. Nos. 9 and 10) Let S = $S_1 \cap S_2$

$\cap S_3$, where

$S_1 = \{z \in C : |z| < 4\}$, $S_2 = \left\{z \in C : \text{Im}\left[\frac{z-1+\sqrt{3}i}{1-\sqrt{3}i}\right] > 0\right\}$ and

$S_3 = \{z \in C : \text{Re}(z) > 0\}$

[JEE Advanced]

Q 9. Area of S is equal to

(a) $\frac{10\pi}{3}$

(b) $\frac{20\pi}{3}$

(c) $\frac{16\pi}{3}$

(d) $\frac{32\pi}{3}$

Q 10. $\min_{z \in S} |1-3i-z|$ is equal to

(a) $\frac{2-\sqrt{3}}{2}$

(b) $\frac{2+\sqrt{3}}{2}$

(c) $\frac{3-\sqrt{3}}{2}$

(d) $\frac{3+\sqrt{3}}{2}$

Q 11. If $z(2-i2\sqrt{3})^2 = i(\sqrt{3}+i)^4$, then amplitude of z is

[UPSEE]

(a) $\frac{-\pi}{6}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{6}$

(d) None of these

Q 12. If the fourth roots of unity are z_1, z_2, z_3 and z_4, then $z_1 + z_2 + z_3 + z_4$ is equal to

[KCET]

(a) 0

(b) C

(c) 3

(d) None of these

Q 13. If z_1, z_2 and z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right| = 1$, then $|z_1 + z_2 + z_3|$ is

[AMU]

(a) 3

(b) 1

(c) greater than 3

(d) less than 1

Q 14. Among the complex number z satisfying condition $|z + 1 - i| < 1$, the number having the least positive argument is

[OJEE]

(a) $1 - i$

(b) $1 + i$

(c) − i

(d) None of these

Q 15. Let $z_1 \neq z_2$ and $|z_1| = |z_2|$. If z_1 has positive real part and z_2 has negative imaginary part. Then, $\frac{z_2 + z_1}{z_1 - z_2}$ may be

[Manipal]

(a) 0

(b) real and positive

(c) real and negative

(d) None of these

3. Equations and Inequalities

Q 16. The real number k for which the equation, $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$.

[JEE Main]

(a) lies between 1 and 2

(b) lies between 2 and 3

(c) lies between -1 and 0

(d) does not exist

Q 17. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$; a, b, c $\in$ R, have a common root, then a : b : c is

[JEE Main]

(a) $1 : 2 : 3$

(b) $3 : 2 : 1$

(c) $1 : 3 : 2$

(d) $3 : 1 : 2$

Q 18. If $3^x = 4^{x-1}$, then x is equal to

[JEE Advanced]

(a) $\frac{2\log_3 2}{2\log_3 2 - 1}$

(b) $\frac{2}{2 - \log_2 3}$

(c) $\frac{1}{1 - \log_4 3}$

(d) $\frac{2\log_2 3}{2\log_2 3 - 1}$

Q 19. Let a, b and c be real numbers, $a \neq 0$. If α is a root of $a^2x^2 + bx + c = 0$, β is a root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$. Then, the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies

[UPSEE]

(a) $\gamma = \alpha$

(b) $\alpha < \beta < \gamma$

(c) $\alpha < \gamma < \beta$

(d) $\gamma = \frac{\alpha + \beta}{2}$

Q 20. The number of real solutions of the equation $\sin e^x = 5^x + 5^{-x}$ is

[KCET]

(a) 1

(b) 2

(c) 0

(d) None of these

Q 21. If the arithmetic mean of the root of a quadratic equation is 8 and the geometric mean is 5, then the equation is

[BITSAT]

(a) $x^2 - 16x - 25 = 0$

(b) $x^2 + 16x - 25 = 0$

(c) $x^2 - 16x + 25 = 0$

(d) $x^2 - 8x + 5 = 0$

Q 22. If x, y and z are three positive real numbers, then minimum values of $\frac{y+z}{x}+\frac{z+x}{y}+\frac{x+y}{z}$ is

[BITSAT]

(a) 1

(b) 2

(c) 3

(d) 6

Q 23. If the roots of the equation $ax^2 + bx + c = 0$ are of the form $\frac{k+1}{k}$ and $\frac{k+2}{k+1}$, then $(a + b + c)^2$ is equal to

[Manipal]

(a) $b^2 - 4ac$

(b) $b^2 - 2ac$

(c) $2b^2 - ac$

(d) Σa^2

Q 24. If the equation $2ax^2 - 3bx + 4c = 0$ and $3x^2 - 4x + 5 = 0$ have a common root, then $(a + b)/(b + c)$ is equal to $(a, b, c \in R)$

[Kerala]

(a) $1/2$

(b) $3/35$

(c) $34/31$

(d) $29/23$

(e) None of these

4. Sequences and Series

Q 25. The sum of first 20 terms of the sequence

$0.7, 0.77, 0.777, \ldots$, is

[JEE Main]

(a) $\frac{7}{81}(179 - 10^{-20})$

(b) $\frac{7}{9}(99 - 10^{-20})$

(c) $\frac{7}{81}(179 + 10^{-20})$

(d) $\frac{7}{9}(99 + 10^{-20})$

Q 26. Let $S_n = \sum_{2}^{4n} (1-)^{\frac{k(k+1)}{2}} k^2$. Then, S_n can take value (s)

[JEE Advanced]

(a) 1056

(b) 1088

(c) 1120

(d) 1332

Q 27. If $\log_x ax$, $\log_x bx$ and $\log_x cx$ are in HP, where a, b, c, x belong to $(1, +\infty)$, then a, b and c are in

[UPSEE]

(a) AP

(b) HP

(c) GP

(d) None of these

Q 28. Consider an infinite geometric series with

first term c and common ratio r. If its sum is 4 and the second term is $\frac{3}{4}$, then

[AMU]

(a) a = 2, $r = \frac{1}{2}$

(b) a = 2, $r = \frac{3}{8}$

(c) a = 1, $r = \frac{3}{4}$

(d) None of these

Q 29. In an arithmetical progression a_1, a_2,a_3,...., sum (S) = $a_1^2 - a_2^2 + a_3^2 - a_4^2 + ... - a_{2k}^2$ is equal to

[BITSAT]

(a) $\frac{k}{2k-1}(a_1^2 - a_{2k}^2)$

(b) $\frac{2k}{k-1}(a_{2k}^2 - a_1^2)$

(c) $\frac{k}{k+1}(a_1^2 - a_{2k}^2)$

(d) none of these

Q 30. Sum of the series $\frac{1}{1.2.3}+\frac{5}{3.4.5}+\frac{9}{5.6.7}+....$ is equal to

[Manipal]

(a) $\frac{3}{2}-3\log_e 2$

(b) $\frac{5}{2}-3\log_e 2$

(c) $1-4\log_e 2$

(d) None of these

Q 31. If $0 < \phi < \frac{\pi}{2}$, $x = \sum_{n=0}^{\infty} \cos^{2n} \phi$, $y = \sum_{n=0}^{\infty} \sin^{2n} \phi$ and $z = \sum_{n=0}^{n} \cos^{2n} n\phi$, then

[OJEE]

(a) xyz = xz + y

(b) xyz = xy + z

(c) xyz = x + y + z

(d) xyz = yz + x

5. Permutations and Combinations

Q 32. Let A and B two sets containing 2 elements and 4 elements, respectively. The number of subsets of A x B having 3 or more elements is

[JEE Main]

(a) 256

(b) 220

(c) 219

(d) 211

Q 33. Let T_n be the number of all possible triangles formed by joining vertices of an n-sided regular polygon. If $T_{n+1} - T_n = 10$, then the value of n is

[JEE Main]

(a) 7

(b) 5

(c) 10

(d) 8

Q 34. Consider the set of eight vectors V = [ai + bj + ck : a, b, c $\in\{-1,1\}$]. Three non-coplanar vectors can be chosen from V in 2^p ways. Then, p is

(Answer lies between 0 to 9)

[JEE Advanced]

Q 35. A pack contains n card numbered from 1

to n. Two consecutive numbered card are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller of the numbers on the removed cards is k, then k – 20 is equal to

[JEE Advanced]

(Answer lies between 0 to 9)

Q 36. Total number of ways in which five '+' and

three '–' signs can be arranged in a line such that

no two '–' sign occur together is

[AMU]

(a) 10

(b) 20

(c) 15

(d) None of these

Q 37. A box contains 2 white balls, 3 black balls

and 4 red balls. In how many ways can 3 balls be drawn from the box, if atleast one black ball is to be included in the draw

[AMU]

(a) 64

(b) 24

(c) 3

(d) 12

Q 38. The number of positive integers which can be formed by using, any number of digits from $0, 1, 2, 3, 4, 5$ but using each digit not more than once in each number is

[KCET]

(a) 1200

(b) 1500

(c) 1600

(d) 1630

Q 39. A student is to answer 10 out of 13 questions in an examination such that he must choose atleast 4 from the first five questions. The number of choices available to him is

[Manipal]

(a) 140

(b) 196

(c) 280

(d) 346

6. Binomial Theorem and Mathematical Induction

Q 40. The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$ is

[JEE Main]

(a) 4

(b) 120

(c) 210

(d) 310

Q 41. The coefficients of three consecutive terms of $(1 + x)^{B+5}$ are in the ratio $5 : 10 : 14$. Then, n is equal to (Answer lies between 0 to 9)

[JEE Advanced]

(a) 1

(b) 3

(c) 4

(d) 6

Q 42. If $49^n + 16n + p$ is divisible by 64 for all n $\in$N, then the least negative integral value of p is

[AMU]

Q 43. If the 4th term in the expansion of $\left(ax + \frac{1}{x}\right)^n$ is $\frac{5}{2}$, for all x $\in$ R, then the values of a and n are

[AMU]

(a) $\frac{1}{2}, 6$

(b) $6, \frac{1}{2}$

(c) $2, 6$

(d) None of these

Q 44. If the 6th term in the expansion of the binomial $[\sqrt{2^{\log(10-3x)}} + \sqrt[5]{2^{(x-2)\log 3}}]^m$ is equal to 21 and it is known that the binomial coefficient of 2nd, 3rd and 4th terms in the expansion represent respectively

the first, third and fifth terms of an AP, then x is

equal to

[OJEE]

(a) 0

(b) 1

(c) -2

(d) 3

Q 45. $\sum_{0\le r<s\le n} \sum ({}^{n}C_{r} + {}^{n}C_{s})$ is equal to

[Manipal]

(a) 2^{n+1}

(b) $2^{n+1} - 1$

(c) $(n - 1)2^n$

(d) $(n + 1)\ 2^n$

Q 46. $\left(1+\frac{C_1}{C_0}\right)\left(1+\frac{C_2}{C_1}\right)\left(1+\frac{C_3}{C_2}\right).....\left(1+\frac{C_n}{C_{n-1}}\right)$ is equal to

[Kerala CEE]

(a) $\frac{n+1}{n!}$

(b) $\frac{(n+1)^n}{(n-1)!}$

(c) $\frac{(n-1)^n}{n!}$

(d) $\frac{(n+1)^n}{n!}$

(e) None of these

7. Matrices and Determinants

Q 47. The number of values of k, for which the system of equations

[JEE Main]

$(k + l)x + 8y = 4k$

$kx + (k+3)y = 3k - 1$ has no solution, is

(a) infinite

(b) 1

(c) 2

(d) 3

Q 48. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3

matrix A and $|A| = 4$, then α is equal to

[JEE Main]

(a) 4

(b) 11

(c) 5

(d) 0

Q 49. For 3×3 matrices M and N, which of the following statement(s) is (are) not correct?

[JEE Advanced]

(a) $N^T M N$ is symmetric or skew–symmetric, according as M is symmetric or skew–symmetric

(b) MN − NM is symmetric for all symmetric

matrices M and N

(c) MN is symmetric for all symmetric matrices M

and N

(d) (adj M) (adj N) = adj(MN) for all invertible

matrices M and N

Q 50. Let ω be a complex cube root of unity

.with $\omega \neq 1$ and P = $[p_{ij}]$ be a n x n matrix with p_{ij} =

ω^{i+j}. Then, $P^2 \neq 0$, when n is equal to

[JEE Advanced]

(a) 57

(b) 55

(c) 58

(d) 56

Q 51. The value of $\begin{bmatrix} \sin\alpha & \cos\alpha & \sin(\alpha+\gamma) \\ \sin\beta & \cos\beta & \sin(\beta+\gamma) \\ \sin\delta & \cos\delta & \sin(\gamma+\delta) \end{bmatrix}$ is

[AMU]

(a) sin α sin β sin δ

(b) $\cos \alpha \cos \beta \cos \delta$

(c) 1

(d) 0

Q 52. If ω is the cube root of unity, then a root of the following equation is $\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix} = 0$

[OJEE]

(a) x = 1

(b) x = ω

(c) $x = \omega^2$

(d) $x = 0$

Q 53. The value of $\theta \in [0, \pi/2]$ and satisfying the equation $\begin{vmatrix} 1+\cos^2\theta & \sin^2\theta & 4\sin 4\theta \\ \cos^2\theta & 1+\sin^2\theta & 4\sin 4\theta \\ \cos^2\theta & \sin^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$ is

[BITSAT]

(a) $\frac{11\pi}{24}$

(b) $\frac{17\pi}{24}$

(c) $\frac{5\pi}{24}$

(d) $\frac{\pi}{24}$

Q 54. If $f(x)=\begin{vmatrix} x-3 & 2x^2-18 & 3x^3-81 \\ x-5 & 2x^2-50 & 4x^3-500 \\ 1 & 2 & 3 \end{vmatrix}$, then $f(1) \cdot f(3)$ + $f(3) \cdot f(5)$ + $f(5) \cdot f(1)$ is equal to

[Manipal]

(a) $f(1)$

(b) $f(3)$

(c) $f(1) + f(3)$

(d) $f(1) + f(5)$

Q 55. If $A_i = \begin{bmatrix} a^i & b^i \\ b^i & a^i \end{bmatrix}$ and if | a | < 1, | b | < 1, then

$\sum_{i=1}^{\infty} \det(A_i)$ is equal to

[Kerala CEE]

(a) $\frac{a^2}{(1-a)^2} - \frac{b^2}{(1-b)^2}$

(b) $\frac{a^2 - b^2}{(1-a^2)(1-b^2)}$

(c) $\frac{a^2}{(1-a^2)(1-b^2)}$

(d) $\frac{a^2}{(1+a)^2} - \frac{b^2}{(1+b)^2}$

(e) None of these

8. Trigonometric Identities and Equations

Q 56. If x, y and z are in AP and $\tan^{-1} x$, $\tan^{-1} y$ and $\tan^{-1} z$ are also in AP, then

[JEE Main]

(a) x = y = z

(b) 2x = 3y = 6z

(c) 6x = 3y = 2z

(d) 6x = 4y = 3z

Q 57. ABCD is a trapezium such that AB and CD are parallel and BC $\perp$ CD. If $\angle$ADB = θ, BC = p and CD = q, then AB is equal to

[JEE Main]

(a) $\frac{(p^2+q^2)\sin\theta}{p\cos\theta+q\sin\theta}$

(b) $\frac{p^2+q^2\cos\theta}{p\cos\theta+q\sin\theta}$

(c) $\frac{p^2+q^2}{p^2\cos\theta+q^2\sin\theta}$

(d) $\frac{(p^2+q^2)\sin\theta}{(p\cos\theta+q\sin\theta)^2}$

Q 58. The expression $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$ can be written as

[JEE Main]

(a) sin A cos A + 1

(b) sec A cosec A + 1

(c) tan A + cot A

(d) sec A + cosec A

Q 59. The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$, is

[JEE Advanced]

(a) 6

(b) 4

(c) 2

(d) 0

Q 60. Let $f(x) = x \sin \pi x$, $x > 0$. Then, for all natural numbers n, $f'(x)$ vanishes at

[JEE Advanced]

(a) a unique point in the interval $\left(n, n+\frac{1}{2}\right)$

(b) a unique point in the interval $\left(n+\frac{1}{2}, n+1\right)$

(c) a unique point in the interval $(n, n+1)$

(d) two points in the interval $(n, n+1)$

Q 61. If $\log_{\cos x} \tan x + \log_{\sin x} \cot x = 0$, then the most general solution of x is

[UPSEE]

(a) $2n\pi - \frac{3\pi}{4}$, $n \in Z$

(b) $2n\pi + \frac{\pi}{4}$, $n \in Z$

(c) $n\pi + \frac{\pi}{4}$, $n \in Z$

(d) None of these

Q 62. If $0 < \beta < \alpha < \frac{\pi}{4}$, $\cos(\alpha + \beta) = \frac{3}{5}$ and $\cos(\alpha - \beta) = \frac{4}{3}$, then $\sin 2\alpha$ is equal to

[KCET]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 63. The most general solution of the equation

$\sec^2 x = \sqrt{2}(1 - \tan^2 x)$ are give by

[BITSAT]

(a) $n\pi \pm \frac{\pi}{4}$

(b) $2n\pi + \frac{\pi}{4}$

(c) $n\pi \pm \frac{\pi}{8}$

(d) None of these

Q 64. The number of real solutions of the equations $x^3 + x^2 + 4x + 2\sin x = 0$ in $0 \leq x \leq 2\pi$ is

[OJEE]

(a) four

(b) two

(c) one

(d) 0

Q 65. Which one of the following is not true?

[AMU]

(a) $|\sin x| \leq 1$

(b) $-1 \leq \cos x \leq 1$

(c) $|\sec x| < 1$

(d) $\operatorname{cosec} x \geq 1$ or $\operatorname{cosec} x \leq -1$

Q 66. The general solution of $\sin 3x + \sin x - 3$

$\sin 2x = \cos 3x + \cos x - 3 \cos 2x$ is

[AMU]

(a) $\frac{n\pi}{2}+\frac{\pi}{8}$ for n integer

(b) $\frac{n\pi}{2}-\frac{\pi}{8}$ for n integer

(c) $n\pi + \frac{\pi}{8}$ for n integer

(d) $n\pi - \frac{\pi}{8}$ for n integer

Q 67. The number of solutions of the equation

$(1 - \cos 2x)(\cos 2x + \cot^2 x) = 0,\ 0 \leq x \leq 2\pi$ is

[Manipal]

(a) 3

(b) 2

(c) 1

(d) 0

9. Inverse Trigonometric Functions

Q 68. The value of $\cot\left\{\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right\}$ is

[JEE Advanced]

(a) $\frac{23}{25}$

(b) $\frac{25}{23}$

(c) $\frac{23}{24}$

(d) $\frac{24}{23}$

Q 69. Let $\tan^{-1}\left(\tan\frac{5\pi}{4}\right)=\alpha, \tan^{-1}\left(-\tan\frac{2\pi}{3}\right)=\beta$, then

[UPSEE]

(a) $4\alpha - 4\beta = 0$

(b) $4\alpha - 3\beta = 0$

(c) $\alpha > \beta$

(d) None of these

Q 70. The value of $4\tan^{-1}\frac{1}{5}-\tan^{-1}\frac{1}{70}+\tan^{-1}\frac{1}{99}$ is

[KCET]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) π

(d) ∞

Q 71. $\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{a}{b}\right)+\tan\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\frac{a}{b}\right)$ is equal to

[BITSAT]

(a) $\frac{a}{2b}$

(b) $\frac{b}{2a}$

(c) $\frac{2b}{a}$

(d) $\frac{2a}{b}$

10. Properties of Triangle

Q 72. In a ΔPQR, P is the largest angle and $\cos P = \frac{1}{3}$. Further in circle of the triangle touches the sides PQ, QR and RP at N,L and M respectively,

such that the lengths of PN, QL and RM are consecutive even integers. Then, possible length(s) of the side(s) of the triangle is (are)

[JEE Advanced]

(a) 16

(b) 18

(c) 24

(d) 22

Q 73. The area of a ΔABC is $a^2 - (b - c)^2$.

Then, tan A is equal to

[UPSEE]

(a) $\frac{3}{4}$

(b) $\frac{8}{15}$

(c) $\frac{4}{3}$

(d) None of these

Q 74. If in a ΔABC, $\frac{a}{\cos A} = \frac{b}{\cos B}$, then

[KCET]

(a) $\sin^2 A + \sin^2 B = \sin^2 C$

(b) $2 \sin A \cos B = \sin C$

(c) $2 \sin A \sin B \sin C = 1$

(d) None of the above

Q 75. If in ΔABC,

[BITSAT]

$a \tan A + b \tan B = (a + b) \tan \frac{(A+B)}{2}$, then

(a) A = B = C

(b) C = A

(c) A = B

(d) B = C

Q 76. The area of a circle and the area of a regular polygon of n sides and of perimeter equal to that of the circle are in the ratio

[OJEE

(a) tan $\pi/n : \pi/n$

(b) cos $\pi/n : \pi/n$

(c) sin $\pi/n : \pi/n$

(d) cot $\pi/n : \pi/n$

Q 77. $a^3\cos(B - C) + b^3\cos(C - A) + c^3 \cos(A -$

B) is equal to

[Manipal]

(a) $3abc$

(b) $3(a + b + c)$

(c) $abc(a + b + c)$

(d) 0

Q 78. If the sides of the triangle are p, q, $\sqrt{p^2 + q^2 + pq}$, then the greatest angle is

[Kerala CEE]

(a) $\frac{\pi}{2}$

(b) $\frac{5\pi}{4}$

(c) $\frac{2\pi}{3}$

(d) $\frac{7\pi}{4}$

(e) None of these

11. Heights and Distances

Q 79. A verticle pole PO is standing at the centre O of a square ABCD. If AC subtends an $\angle 90°$ at the top P of the pole, then the angle subtended by a side of the square at P is

[UPSEE]

(a) $30°$

(b) $45°$

(c) $60°$

(d) None of these

Q 80. ABCD is a square plot. The angle of elevation of the top of a pole standing at D from A or C is $30°$ and that from B is θ, then tan θ is equal to

[KCET]

(a) $\sqrt{6}$

(b) $1/\sqrt{6}$

(c) $\sqrt{3}/2$

(d) $\sqrt{2}/3$

12. Rectangular cartesian coordinates

Q 81. The x−coordinate of the incentre of the triangle that has the coordinates of mid−points of its sides as $(0, 1)$, $(1,1)$ and $(1, 0)$ is

[JEE Main]

(a) $2+\sqrt{2}$

(b) $2-\sqrt{2}$

(c) $1+\sqrt{2}$

(d) $1-\sqrt{2}$

Q 82. The triangle joining the point P$(2, 7)$, Q$(4, -1)$ and R$(-2, 6)$ is

[UPSEE]

(a) isosceles triangle

(b) equilateral triangle

(c) right angled triangle

(d) None of these

Q 83. ABC is a triangle with vertices A$(-1, 4)$, B$(6, -2)$ and C$(-2, 4)$.D,E and F are the points which divide each AB, BC and CA respectively in the ratio $3:1$ internally. Then, the centroid of the

triangle DEF is

[Kerala CEE]

(a) $(3, 6)$

(b) $(1, 2)$

(c)$(4,8)$

(d) $(-3,6)$

(e) None of these

13. Straight Line and Pair of Straight Lines

Q 84. A ray of light along $x+\sqrt{3}y=\sqrt{3}$ gets reflected upon reaching x−axis, the equation of the reflected ray is

[JEE Main]

(a) $y=x+\sqrt{3}$

(b) $\sqrt{3}y=x-\sqrt{3}$

(c) $y=\sqrt{3}x-\sqrt{3}$

(d) $\sqrt{3}y=x-1$

Q 85. For a > b > c > 0, the distance between

$(1,1)$ and the point of intersection of the lines ax +

by + c = 0 and bx + ay + c = 0 is less than $2\sqrt{2}$.

Then,

[JEE Advanced]

(a) a + b – c > 0

(b) a – b + c < 0

(c) a – b + c > 0

(d) a + b – c < 0

Q 86. The equation of second degree

$x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ represents a pair of straight lines. The distance between is

[UPSEE]

(a) $2\sqrt{3}$

(b) $2\sqrt{5}$

(c) 2

(d) 0

Q 87. Let P = $(-1, 0)$, O = $(0, 0)$ and Q = $(3, 3\sqrt{3})$ be three points. Then, the equation of the bisector of the $\angle$POQ is

[BITSAT]

(a) $y = \sqrt{3}x$

(b) $\sqrt{3}y = x$

(c) $y = -\sqrt{3}x$

(d) $\sqrt{3}y = -x$

14. Circle and System of Circles

Q 88. The circle passing through $(1, -2)$ and touching the axis of x at $(3, 0)$ also passes through the point

[JEE Main]

(a) $(-5, 2)$

(b) $(2, -5)$

(c) $(5, -2)$

(d) $(-2, 5)$

Q 89. Circle (s) touching x−axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ on y−axis is (are)

[JEE Advanced]

(a) $x^2 + y^2 - 6x + 8y + 9 = 0$

(b) $x^2 + y^2 - 6x + 7y + 9 = 0$

(c) $x^2+y^2 - 6x - 8y + 9 = 0$

(d) $x^2+y^2-6x - 7y + 9 = 0$

Q 90. The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle of area 154 sq unit. Then, the equation of the circle is

[AMU]

(a) $x^2 + y^2 - 2x + 2y + 47 = 0$

(b) $x^2+y^2 + 2x-2y-47 = 0$

(c) $x^2 + y^2 - 2x + 2y - 47 = 0$

(d) $x^2+y^2-2x-2y-47 = 0$

Q 91. Circles are drawn through the point $(2, 0)$ to cut intercept of length 5 units on the x−axis. If their centres lie in the first quadrant, then their equation is

[UPSEE]

(a) $x^2 + y^2 - 9x + 2fy + 14 = 0$

(b) $3x^2 + 3y^2 + 27x - 2fy + 42 = 0$

(c) $x^2 + y^2 - 9x - 2fy + 14 = 0$

(d) $x^2 + y^2 - 2fx - 9y + 14 = 0$

Q 92. Two vertices of an equilateral triangle are $(-1, 0)$ and $(1, 0)$ and its circumcircle is

[OJEE]

(a) $x^2 + \left(y - \frac{1}{\sqrt{3}}\right) = \frac{4}{3}$

(b) $x^2 - \left(y + \frac{1}{\sqrt{3}}\right)^2 = \frac{4}{3}$

(c) $x^2 + \left(y - \frac{1}{\sqrt{3}}\right)^2 = -\frac{4}{3}$

(d) None of these

Q 93. If the equation of the tangent to the circle $x^2 + y^2 - 2x + 6y - 6 = 0$ parallel to $3x - 4y + 7 = 0$ is $3x - 4y + k = 0$, then the values of k are

[Kerala CEE]

(a) $5, -35$

(b) $-5, 35$

(c) $7, -32$

(d) $-7, 32$

(e) None of these

15. Conic Section

Q 94. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having centre at $(0, 3)$ is

[JEE Main]

(a) $x^2 + y^2 - by - 7 = 0$

(b) $x^2 + y^2 - 6y + 7 = 0$

(c) $x^2 + y^2 - 6y - 5 = 0$

(d) $x^2 + y^2 - 6y + 5 = 0$

Q 95. A circle, $2x^2 + 2y^2 = 5$ and a parabola, $y^2 = 4\sqrt{5}x$.

[JEE Main]

Statement I An equation of a common tangent to these curves is $y = x + \sqrt{5}$.

Statement n If the line, $y = mx + \frac{\sqrt{5}}{m}(m \neq 0)$ is the common tangent, then m satisfies $m^4 - 3m^2 + 2 = 0$.

(a) Statement I is true; Statement II is true;

Statement II is a correct explanation for statement I

(b) Statement I is true; Statement II is true;

Statement II is not a correct explanation for

statement I

(c) Statement I is true; Statement II is false

(d) Statement I is false; Statement II is true

Paragraph

Directions (Q. Nos. 96 and 97)

Let PQ be a focal chord of the parabola $y^2 = 4ax$.

The tangents to the parabola at P and Q meet at a point lying on the line $y = 2x + a$, $a > 0$.

Q 96. Length of chord PQ is

[JEE Advanced]

(a) $7a$

(b) $5a$

(c) $2a$

(d) $3a$

Q 97. If chord PQ subtends an angle θ at the vertex of $y^2 = 4ax$, then $\tan\theta$ is equal to

(a) $\frac{2}{3}\sqrt{7}$

(b) $\frac{-2}{3}\sqrt{7}$

(c) $\frac{2}{3}\sqrt{5}$

(d) $\frac{-2}{3}\sqrt{5}$

Q 98. The line $2x + y + k = 0$ is a normal to the parabola $y^2 = -8x$, if k is equal to

[AMU]

(a) -24

(b) 12

(c) 24

(d) -12

Q 99. The angle between the tangents drawn from the point $(1, 4)$ to the parabola $y^2 = 4ax$ and $x^2 = 4y$

[AMU]

(a) 0

(b) $\pi/6$

(c) $\pi/4$

(d) $\pi/3$

Q 100. The image of the parabola $y^2 = 4x$ about the line $x - y + l = 0$ is

[KCET]

(a) $(x + 1)^2 = 4(y - 1)$

(b) $(x - 1)^2 = 4(y + 1)$

(c) $(x + 1)^2 = -4(y + 1)$

(d) $(x - 1)^2 = -4(y - 1)$

Q 101. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then, the value of b^2, is

[OJEE]

(a) 1

(b) 5

(c) 7

(d) 9

Q 102. Suppose S and S' are foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$. If P is a variable point on the ellipse and if Δ is area of the ΔPSS', then the maximum value of Δ is

[Kerala CEE]

(a) 8

(b) 12

(c) 16

(d) 20

(e) None of these

16. Limits, Continuity and Differentiability

Q 103. $\lim\limits_{x\to 0}\dfrac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$ is equal to

[JEE Main]

(a) $-\dfrac{1}{4}$

(b) $\dfrac{1}{2}$

(c) 1

(d) 2

Q 104. For a $\in$ R (the set of all real numbers), a

$\neq -1$, $\lim_{n\to\infty} \frac{(1^a + 2^a + ... + n^a)}{(n+1)^{a-1}[(na+1)+(na+2)+.....+(na+n)]} = \frac{1}{60}$.

Then, a is equal to

[JEE Advanced]

(a) 5

(b) 7

(c) $\frac{-15}{2}$

(d) $\frac{-17}{2}$

Q 105. The graph of the function y = f(x) has a unique tangent at the point (a, θ) through which the graph passes. Then, $\lim_{x\to 0}\left[\frac{\log_e\{(1+6y(x)\}}{3f(x)}\right]$ is

[UPSEE]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 106. The value of $\lim_{n\to\infty}\frac{1+2+.....+n}{3n^2+5}$ is equal to

[OJEEE]

(a) $\frac{1}{3}$

(b) $\frac{1}{5}$

(c) $\frac{1}{6}$

(d) 6

Q 107. The $f(x)=\frac{\log(1+ax)-\log(1-bx)}{x}$ is not defined at $x = 0$. The value which should be assigned to f at $x = 0$, so that it is continuous at $x = 0$, is

[AMU]

(a) a − b

(b) a + b

(c) b − a

(d) None of these

Q 108. Let f : R $\rightarrow$ R be such that f(1) = 3 and f'

(1) = 6. Then, $\lim_{x\to 0}\left[\frac{f(1+x)}{f(1)}\right]^{1/x} = 1$

[AMU]

(a) 1

(b) $e^{1/2}$

(c) e^{2}

(d) e^{3}

Q 109. $\lim_{h\to 0}\frac{(a+h)^2\sin(a+h) - a^2\sin a}{h}$ is equal to

[Manipal]

(a) $a^2 \cos a + a \sin a$

(b) $a^2 \cos a + 2a \sin a$

(c) $2a^2 \cos a + a \sin a$

(d) None of these

Q 110. The value of f at $x=0$ so that function $f(x)$ $= \frac{2^x - 2^{-x}}{x}$, $x \neq 0$, is continuous at $x = 0$, is

[Manipal]

(a) 0

(b) $\log 4$

(c) 4

(d) e^4

Q 111. $\lim_{x \to 0} \left(\frac{1+\tan x}{1+\sin x} \right)^{\operatorname{cosec} x}$ is equal to

[Kerala CEE]

(a) $\frac{1}{e}$

(b) 1

(c) e

(d) e^2

(e) None of these

Q 112. Let $f(x + y) = f(x)f(y)$ and $f(x) = 1 + \sin(3x)g(x)$, where $g(x)$ is continuous, then $f'(x)$ is

[Kerala CEE]

(a) $f(x)g(0)$

(b) $3g(0)$

(c) $f(x)\cos 3x$

(d) $3\ f(x)g(0)$

(e) None of these

17. Differentiation

Q 113. If $y = \sec(\tan^{-1} x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to

[JEE Main]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{1}{2}$

(c) 1

(d) $\sqrt{2}$

Q 114. If $e^{y+e^{y+e^{y+\ldots+e^{\infty}}}}$, then $\frac{dy}{dx}$ is

[UPSEE]

(a) $\frac{1}{x}$

(b) $\frac{1-x}{x}$

(c) $\frac{x}{1+x}$

(d) None of these

Q 115. If $\sin(x + y) = \log(x + y)$, then dy/dx is equal to

[BITSAT]

(a) 2

(b) -2

(c) 1

(d) -1

Q 116. The derivative of $y = (1 - x)(2 - x)\ldots(n - x)$ at $x = 1$ is equal to

[OJEE]

(a) 0

(b) $(-1)(n-1)!$

(c) $n! - 1$

(d) $(-1)^{n-1}\,(n-1)!$

Q 117. If $y = \tan^{-1}\left(\frac{a\cos x - b\sin x}{b\cos x + a\sin x}\right)$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) 2

(b) -1

(c) $\frac{a}{b}$

(d) 0(e) None of these

18. Applications of Derivatives

Q 118. The intercepts on x−axis made by tangents to the curve, $y = \int_0^x |t|\,dt$, $x \in R$, which are parallel to the line y = 2x, are equal to

[JEE Main]

(a) ± 1

(b) ± 2

(c) ± 3

(d) ± 4

Q 119. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio $8:15$ is converted into an open rectangular box by folding after removing squares of equal area from all four comers. If the total area of removed squares is

100, the resulting box has maximum volume. The lengths of the sides of the rectangular sheet are

[JEE Advanced]

(a) 24

(b) 32

(c) 45

(d) 60

Q 120. The function f(x) = 2| x | + | x + 2| − | | x+ 2| − 2| x | | has a local minimum or a local maximum at x is equal to

[JEE Advanced]

(a) -2

(b) $\frac{-2}{3}$

(c) 2

(d) $\frac{2}{3}$

Paragraph

Directions (Q. Nos. 121 and 122)

Let f : $[0,1] \to$ R (the set of all real numbers) be a function. Suppose the function f is twice differentiable, f(0) = f(1) = 0 and satisfies f'' (x) − 2f' (x) + f(x) $\geq e^x$, x $\in$ [0, 1]

[JEE Advanced]

Q 121. Which of the following is true for 0 < x < 1 ?

(a) $0 < f(x) < \infty$

(b) $-\frac{1}{2} < f(x) < \frac{1}{2}$

(c) $-\frac{1}{4} < f(x) < 1$

(d) $-\infty < f(x) < 0$

Q 122. If the function $e^{-x} f(x)$ assumes its minimum in the interval $[0,1]$ a $x = \frac{1}{4}$, which of the following is true?

(a) $f'(x) < f(x), \frac{1}{4} < x < \frac{3}{4}$

(b) f' (x) > f(x), $0 < x < \frac{1}{4}$

(c) f' (x) < f(x), $0 < x < \frac{1}{4}$

(d) f' (x) < f(x), $\frac{3}{4} < x < 1$

Q 123. A line L : y = mx + 3 meets y−axis at E$(0, 3)$ and the arc of the parabola $y^2 = 16x$, $0 \leq y \leq 6$ at the point F(x_0, y_0). The tangent to the parabola at F(x_0, y_0) intersects the y−axis at G$(0, y_1)$. The slope m of the line L is chosen such that the area of the ΔEFG has a local maximum Match

List I with List II and select the correct answer

using the code given below the lists

	List I		**List II**
P.	m =	1.	$1/2$
Q.	Maximum area of ΔEFG is	2.	4
R.	y_0 =	3.	2
S.	y_1 =	4.	1

Codes

[JEE Advanced]

	P	Q	R	S		P	Q	R	S
(a)	4	1	2	3	(b)	3	4	1	2
(c)	1	3	2	4	(d)	1	3	4	2

Q 124. If at each point of the curve $y = x^3 - ax^2 + x + 1$ the tangent is inclined at an acute angle with the positive direction of the x−axis, then

[UPSEE]

(a) $a \leq \sqrt{3}$

(b) $a > 0$

(c) $-\sqrt{3} \leq a \leq \sqrt{3}$

(d) None of these

Q 125. The equation of the tangent to the curve $y = e^{-|x|}$ at the point, where the curve cuts the line $x = 1$, is

[KCET]

(a) $e(x + y) = 1$

(b) $y + ex = 1$

(c) $x + y = e$

(d) None of these

Q 126. If $f(x)$ is differentiable and strictly increasing function, then the value of $\lim_{x\to 0}\frac{f(x^2)-f(x)}{f(x)-f(0)}$ is equal to

[AMU]

(a) -1

(b) 0

(c) 1

(d) 2

Q 127. The difference between the greatest and the least values of the function f(x) = sin 2x − x on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is

[AMU]

(a) π

(b) $\sqrt{3} - \pi/3$

(c) $-\sqrt{3}+\pi/3$

(d) None of these

Q 128. Let $P(x) = a_0 + a_1x^2 + a_2x^4 + \ldots + a_nx^{2n}$

be a polynomial in a real variable x with $0 < a_0 < a_1 < a_2 < \ldots < a_n$. The function P(x) has

[BITSAT]

(a) neither maximum nor minimum

(b) only one maximum

(c) only one minimum

(d) None of the above

19. Indefinite Integrals

Q 129. If $\int f(x)dx = \psi(x)$, then $\int x^5 f(x^3)dx$ is equal to

[JEE Main]

(a) $\frac{1}{3}[x^3\psi(x^3) - \int x^2\psi(x^3)dx] + C$

(b) $\frac{1}{3}x^3\psi(x^3) - 3\int x^3\psi(x^3)dx + C$

(c) $\frac{1}{3}x^3\psi(x^3) - \int x^2\psi(x^3)dx + C$

(d) $\frac{1}{3}[x^3\psi(x^3) - \int x^3\psi(x^3)dx] + C$

Q 130. $\int \frac{(x^2-1)}{x^3\sqrt{2x^4-2x^2+1}}dx$ is equal to

[AMU]

(a) $-\frac{1}{2}\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}+C$

(b) $\frac{1}{2}\log(2x^4 - 2x^2 + 1) + C$

(c) $\frac{1}{2}\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}+C$

(d) None of these

Q 131. If $I_1 = \int \sin^{-1} x\,dx$ and $I_2 = \int \sin^{-1}\sqrt{1-x^2}\,dx$, then

[Kerala CEE]

(a) $I_1 = I_2$

(b) $I_2 = \frac{\pi}{2} I_1$

(c) $I_1 + I_2 = \frac{\pi}{2} x$

(d) $I_1 + I_2 = \frac{\pi}{2}$

(e) None of these

20. Definite Integrals

Q 132. **Statement I** The value of the integral

$\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ is equal to $\pi/6$.

Statement II $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

[JEE Main]

(a) Statement I is true; Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true; Statement II is true;

Statement II is not a correct explanation for

Statement I

(c) Statement I is true; Statement II is false

(d) Statement I is false; Statement II is true

Q 133. Let $f:\left[\frac{1}{2},1\right]\rightarrow$ R (the set of all real numbers) be a positive, non-constant and differentiable function such that f' (x) < 2f(x) and $f\left(\frac{1}{2}\right)=1$. Then, the value of $\int_{1/2}^{1} f(x)dx$ lies in the interval

[JEE Advanced]

(a) (2e − 1, 2e)

(b) (e−1, 2e−l)

(c) $\left(\frac{e-1}{2}, e-1\right)$

(d) $\left(0, \frac{e-1}{2}\right)$

Q 134. The value of $\int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$ is equal to

[UPSEE]

(a) $\frac{\pi^2}{32}$

(b) $\frac{\pi^2}{16}$

(c) $\frac{\pi^2}{12}$

(d) None of these

Q 135. $\int_{-\pi}^{\pi} \sin x[f(\cos x)]dx$ is equal to

[AMU]

(a) $2f(\pi)$

(b) $2/(2)$

(c) $2/(1)$

(d) None of these

Q 136. $\int_{\pi/6}^{\pi/3} \frac{dx}{\sqrt{\tan x}}$ is equal to

[Manipal]

(a) $\frac{\pi}{12}$

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{6}$

(d) $\frac{\pi}{4}$

21. Applications of Integrals

Q 137. The area (in sq units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x−axis and lying in the first quadrant is

[JEE Main]

(a) 9

(b) 6

(c) 18

(d) $\frac{27}{4}$

Q 138. The area enclosed by the curves y = sin x + cos x and y = | cos x − sin x| over the interval $\left[0, \frac{\pi}{2}\right]$ is

[JEE Advanced]

(a) $4(\sqrt{2}-1)$

(b) $2\sqrt{2}(\sqrt{2}-1)$

(c) $2(\sqrt{2}+1)$

(d) $2\sqrt{2}(\sqrt{2}+1)$

Q 139. A vertical line passing through the point $(h, 0)$ intersects the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ at the points P and Q.

Let the tangents to the ellipse at P and Q meet at the point R. If $\Delta(h)$ = area of the ΔPQR, $\Delta_1 = \max_{1/2 \le h \le 1} \Delta(h)$ and $\Delta_2 = \min_{1/2 \le h \le 1}$ then $\frac{8}{\sqrt{5}}\Delta_1 - 8\Delta_2$ is equal to

(Answer lies between 0 to 9)

[JEE Advanced]

Q 140. The area bounded by the curves y = x and $y = x^3$ is equal to **[AMU]**

(a) 0

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) 1

Q 141. The area bounded by the curves $y = e^x$, $y = e^{-x}$, the ordinates $x = 0$ and $x = 1$ is given by

[AMU]

(a) $e + e^{-1} - 2$

(b) $e - e^{-1}$

(c) $e + e^{-1}$

(d) $e + e^{-1} + 2$

Q 142. Area included between the parabola $y = \frac{x^2}{4a}$

and the curve $y = \frac{8a^3}{x^2 + 4a^2}$ is

[UPSEE]

(a) $\frac{a^2}{3}(6\pi - 4)$

(b) $\frac{a^2}{3}(4\pi + 3)$

(c) $\frac{a^2}{3}(8\pi + 3)$

(d) None of these

Q 143. The area of the region bounded by $y^2 = 4ax$ and $x^2 = 4ay$, $a > 0$ (in sq unit) is

[Kerala CEE]

(a) $16\frac{a^2}{3}$

(b) $14\frac{a^2}{3}$

(c) $13\frac{a^2}{3}$

(d) $16a^2$

(e) None of these

22. Differential Equations

Q 144. At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P with respect to additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employees 25 more workers, then the new level of production of items is

[JEE Main]

(a) 2500

(b) 3000

(c) 3500

(d) 4500

Q 145. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$.

Let the slope of the curve at each point (x, y) be

$\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$. Then, the equation of the curve is

[JEE Advanced]

(a) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$

(b) $\cos ec\left(\frac{y}{x}\right) = \log x + 2$

(c) $\sec\left(\frac{2y}{x}\right) = \log x + 2$

(d) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$

Q 146. A solution of the differential equation

$\left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + y = 0$ is

[AMU]

(a) y = 2x

(b) $y = -2x$

(c) $y = 2x - 4$

(d) $y = 2x + 4$

Q 147. The solution of the equation $(x^2 + xy)dy = (x^2+y^2)dx$ is

[KCET]

(a) $\log x = \log(x - y) + \frac{y}{x} + c$

(b) $\log x = 2\log(x - y) + \frac{y}{x} + c$

(c) $\log x = \log(x - y) + \frac{x}{y} + c$

(d) None of the above

Q 148. An integrating factor of the differential equation $x\frac{dy}{dx} + y \log x = xe^x x^{-\frac{1}{2}\log x} (x > 0)$ is

[Manipal]

(a) $x^{\log x}$

(b) $(\sqrt{x})^{\log x}$

(c) $(\sqrt{e})^{(\log x)^2}$

(d) e^{x^2}

23. Vector Algebra

Q 149. If the vectors AB = 3i + 4k and AC = 5i− 2j + 4k are the sides of a ΔABC, then the length of the median through A is

[JEE Main]

(a) $\sqrt{18}$

(b) $\sqrt{72}$

(c) $\sqrt{33}$

(d) $\sqrt{45}$

Q 150. Let PR = 3 i + j − 2k and SQ = i − 3 j − 4k determine diagonals of a parallelogram PQRS and PT = i + 2j + 3k be another vector. Then, the volume of the parallelopiped determined by the vectors PT, PQ and PS is

[JEE Advanced]

(a) 5

(b) 20

(c) 10

(d) 30

Q 151. A line l passing through the origin is perpendicular to the lines $l_1 : (3 + 1)\,i + (-1 + 2t)\,j + (4 + 2t)k$, $-\infty < t < \infty$ $l_2 : (3 + 2s)\,i + (3 + 2s)\,j + (2 + s)\,k$, $-\infty < s < \infty$

Then, the coordinate (s) of the point(s) on l_2 at a distance of $\sqrt{17}$ from the point of intersection of l and l_1 is (are)

[JEE Advanced]

(a) $\left(\frac{7}{3},\frac{7}{3},\frac{5}{3}\right)$

(b) $(-1, -1, 0)$

(c) $(1,1,1)$

(d) $\left(\frac{7}{9},\frac{7}{9},\frac{8}{9}\right)$

Q 152. Match List I with List II and select the correct answer using the code given below the lists

	List I		List II

P.	Volume of parallelopiped determined by vectors a,b and c is 2. Then, the volume of the parallelopiped determined by vectors $2(a \times b)$, $3(b \times c)$ and $(c \times a)$ is	1.	100
Q.	Volume of parallelopiped determined by vectors a, b and c is 5. Then, the volume of the parallelopiped determined by vectors $3(a + b)$, $(b + c)$ and $2(c + a)$	2.	30

	is		
R.	Area of a triangle with adjacent sides determined by vector a and b is 20. Then, the area of the triangle with adjacent sides determined by vectors $(2a + 3b)$ and $(a - b)$ is	3.	24
S.	Area of a parallelogram with adjacent sides determined by vectors a and b is	4.	60

	30. Then, the area of the parallelogram with adjacent sides determined by vectors (a + b) and a is		

Codes

[JEE Advanced]

	P	Q	R	S
(a)	4	2	3	1
(b)	2	3	1	4
(c)	3	4	1	2

(d) 1 4 3 2

Q 153. Let a = i + j + k, b = i − j + k and c = i − j − k be three vectors. A vector v in the plane of a and b, whose projection on c is $\frac{1}{\sqrt{3}}$, is given by

[AMU]

(a) 3i + j − 3k

(b) 3i− j + 3k

(c) 3i + j + 3k

(d) $-3i + j + 3k$

Q 154. If $r \cdot a = r \cdot b = r \cdot c = 1$ where a, b, c

are any three non–coplanar vectors, then r is

[UPSEE]

(a) coplanar with a, b, c

(b) parallel to $a + b + c$

(c) parallel to $b \times c + c \times a + a \times b$

(d) parallel to $(a \times b) \times c$

Q 155. Let a = 2i + j−2k and b = i + j. If c is a vector such that a . c = | c|, | c . a| = $2\sqrt{2}$ and the angle between a × b and c is $30°$, then | (a × b) × c| is equal to

[KCET]

(a) $2/3$

(b) $3/2$

(c) 2

(d) 3

Q 156. The position vector of the point, where the line $\mathbf{r}=\mathbf{i}-\mathbf{j}+k+t(i+j-k)$ meets the plane $r \cdot (i+j+k)=5$ is

[BITSAT]

(a) $5i+j-k$

(b) $5i+3j-3k$

(c) $2i+j+2k$

(d) $5i+j+k$

24. Three Dimensional Geometry

Q 157. Distance between two parallel planes $2x + y + 2z = 8$ and $4x+2y + 4z + 5 = 0$ is

[JEE Main]

(a) $\frac{3}{2}$

(b) $\frac{5}{2}$

(c) $\frac{7}{2}$

(d) $\frac{9}{2}$

Q 158. If the lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ and

$\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar, then k can have

[JEE Main]

(a) any value

(b) exactly one value

(c) exactly two values

(d) exactly three values

Q 159. Perpendicular are drawn from points on the line $\frac{x+2}{2}=\frac{y+1}{-1}=\frac{z}{3}$ to the plane x + y + z = 3. The feet of perpendiculars lie on the line

[JEE Advanced]

(a) $\frac{x}{5}=\frac{y-1}{8}=\frac{z-2}{-13}$

(b) $\frac{x}{2}=\frac{y-1}{3}=\frac{z-2}{-5}$

(c) $\frac{x}{4}=\frac{y-1}{3}=\frac{z-2}{-7}$

(d) $\frac{x}{2}=\frac{y-1}{-7}=\frac{z-2}{5}$

Q 160. Two lines L_1 : x = 5, $\frac{y}{3-\alpha}=\frac{z}{-2}$ and L_2 : x = α, $\frac{y}{-1}=\frac{z}{2-\alpha}$ are coplanar. Then, α can take value (s)

[JEE Advanced]

(a) 1

(b) 2

(c) 3

(d) 4

Q 161. Consider the lines $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}$,

$L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$ and the planes P_1 : $7x + y + 2z = 3$, P_2 : $3x + 5y - 6z = 4$. Let ax+by + cz = d the equation of the plane passing through the point of intersection of lines L_1 and L_2 and perpendicular to planes P_1 and P_2.

Match List I with List II and select the correct answer using the code given below the lists.

[JEE Advanced]

	List I		List II
P.	a =	1.	13
Q.	6 =	2.	− 13
R.	c =	3.	1
S.	d =	4.	−2

Codes

	P	Q	R	S
(a)	3	2	4	1

(b) 1 3 4 2

(c) 3 2 1 4

(d) 2 4 1 3

Q 162. If a line makes angles α, β, γ with the coordinate axes, then $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is

[OJEE]

(a) -1

(b) -2

(c) 2

(d) -3

25. Statistics

Q 163. All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given?

[JEE Main]

(a) Mean

(b) Median

(c) Mode

(d) Variance

Q 164. The mean of five observations is 4 and their variance is 5.2. If three of these observations are 1, 2 and 6, then the other two are

[UPSEE]

(a) 2 and 9

(b) 3 and 8

(c) 4 and 7

(d) 5 and 6

26. Probability

Q 165. A multiple choice examination has 5 questions. Each question has three alternative answers of which exactly one is correct. The probability that a student will get 4 or more correct answers just by guessing is

[JEE Main]

(a) $\frac{17}{3^5}$

(b) $\frac{13}{3^5}$

(c) $\frac{11}{3^5}$

(d) $\frac{10}{3^5}$

Q 166. Four persons independently solve a certain problem correctly with probabilities $\frac{1}{2}, \frac{3}{4}, \frac{1}{4}, \frac{1}{8}$.

Then, the probability that the problem is solved correctly by atleast one of them is

[JEE Advanced]

(a) $\frac{235}{256}$

(b) $\frac{21}{256}$

(c) $\frac{3}{256}$

(d) $\frac{253}{256}$

Paragraph

Directions (Q.Nos. 167 and 168)

A box B_1 contains 1 white ball, 3 red balls and 2 black balls. Another box B_2 contains 2 white balls, 3 red balls and 4 black balls. A third box B_3 contains 3 white balls, 4 red balls and 5 black balls.

[JEE Advanced]

Q 167. If 1 ball is drawn from each of the boxes B_1, B_2 and B_3, the probability that all 3 drawn balls are of the same colour is

(a) $\frac{82}{648}$

(b) $\frac{90}{648}$

(c) $\frac{558}{648}$

(d) $\frac{566}{648}$

Q 168. If 2 balls are drawn (without replacement) from a randomly selected box and one of the balls is white and the other ball is red, the probability that these 2 balls are drawn from box B_2 is

(a) $\frac{116}{181}$

(b) $\frac{126}{181}$

(c) $\frac{65}{181}$

(d) $\frac{55}{181}$

Q 169. Of the three independent events E_1, E_2 and E_3, the probability that only E_1 occurs is α, only E_2 occurs is β and only E_3 occurs is y. Let the probability p that none of events E_1, E_2 or E_3 occurs satisfy the equations $(\alpha - 2\beta)$, $p = \alpha\beta$ and $(\beta - 3\gamma)$

$p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval $(0,1)$.

Then, $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3}$ is equal to

[JEE Advanced]

(Answer lies between 0 to 9)

Q 170. If n integers taken at random are multiplied together, then the probability that the last digit of the product is $1, 3, 7$ or 9 is

[UPSEE]

(a) $\frac{2^n}{5^n}$

(b) $\frac{4^n - 2^n}{5^n}$

(c) $\frac{4^n}{5^n}$

(d) None of these

Q 171. A fair dice is thrown twenty times. The probability that on the tenth throw the fourth six appears is

[OJEE]

(a) $\frac{^{20}C_{10} \times 5^6}{6^{20}}$

(b) $\frac{120 \times 5^7}{6^{10}}$

(c) $\frac{84 \times 5^6}{6^{10}}$

(d) None of these

27. Mathematical Logic and Boolean Algebra

Q 172. Statement I $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is a fallacy.

Statement II $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ is a tautology.

[JEE Main]

(a) Statement I is true; Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true; Statement II is true;

Statement II is not a correct explanation for

Statement I

(c) Statement I is true; Statement II is false

(d) Statement I is false; Statement II is true

Q 173. The negative of the compound proposition

$p \vee (\sim p \vee q)$ is

[Kerala CEE]

(a) $(p \wedge \sim q) \wedge \sim p$

(b) $(p \wedge \sim q) \vee \sim p$

(c) $(p \wedge \sim q) \vee \sim p$

(d) $(p \wedge q) \wedge q$

(e) None of these

Q 174. Match List I with List II and select the correct answer using the code given below the lists

	List I	List II	
P.	$\left[\frac{1}{y^2}\left\{\frac{\cos(\tan^{-1}y)+y\sin(\tan^{-1}y)}{\cot(\sin^{-1}y)+\tan(\sin^{-1}y)}\right\}+y^4\right]^{1/2}$ takes value	1.	$\frac{1}{2}\sqrt{\frac{5}{3}}$
Q.	If cos x + cos y + cos z = 0 = sin x + sin y + sin z, then possible value of $\cos\frac{x-y}{2}$ is	2.	$\sqrt{2}$
R	If cos $\left(\frac{\pi}{4}-x\right)$ cos 2x + sin x sin 2x secx =	3	$\frac{1}{2}$

.	cos x sin $2x$ sec x + cos $\left(\frac{\pi}{4}+x\right)$ cos $2x$, then possible value of sec x is		
S .	If $\cot\left(\sin^{-1}\sqrt{1-x^2}\right)$ = sin [$\tan^{-1}(x\sqrt{6})$], x = 0. Then, possible value of x is	4 .	1

Codes

[JEE Advanced]

	P	Q	R	S
(a)	4	3	1	2

(b) 4 3 2 1

(c) 3 4 2 1

(d) 3 4 1 2

Q 175. The sum of all the proper divisors of 9900 is

[KCET]

(a) 29351

(b) 23951

(c) 33851

(d) None of the above

Q 176. The last three digits in 3^{32} is

[BITSAT]

(a) 841

(b) 541

(c) 941

(d) None of the above

Answers

1.(c) 2.(b) 3.(a) 4.(c) 5.(c) 6.(c) 7.(c) 8.(c,d) 9. (b)

10.(c) 11. (a) 12. (a) 13. (b) 1,4. (d) 15. (a) 16.

(d) 17. (a) 18. (a,b,c) 19. (c) 20. (c) 21. (c) 22.

(d) 23. (a) 24. (c) 25. (c) 26. (a) 27. (c) 28. (c)

29. (a) 30. (b) 31. (b) 32. (c) 33. (b) 34.(5) 35.(5)

36. (b) 37. (a) 38. (d) 39. (b) 40. (c) 41.(6) 42.

(d) 43. (a) 44. (a) 45. (c) 46. (d) 47. (b) 48. (b)

49. (c,d) 50. (b,c,d) 51. (d) 52. (d) 53. (a) 54. (b)

55. (b) 56. (a) 57. (a) 58. (b) 59. (c) 60. (b,c)

61. (b) 62. (b) 63. (c) 64. (c) 65. (c) 66. (a) 67.

(d) 68. (b) 69. (b) 70. (a) 71. (c) 72. (b,d) 73. (b)

74. (b) 75. (c) 76. (a) 77. (a) 78. (c) 79. (c) 80.

(b) 81. (b) 82. (c) 83. (b) 84. (b) 85. (a) 86. (c)

87. (d) 88. (c) 89. (a,c) 90. (c) 91. (c) 92. (a) 93.

(a) 94. (a) 95. (b) 96. (b) 97. (c) 98. (c) 99. (a)

100. (a) 101. (c) 102. (b) 103. (d) 104. (b,d) 105.

(c) 106. (c) 107. (b) 108. (c) 109. (b) 110. (b)

111. (b) 112. (d) 113. (a) 114. (b) 115. (d) 116.

(b) 117. (b) 118. (a) 119. (a,c) 120. (a,b) 121. (b)

122. (c) 123. (a) 124. (c) 125. (d) 126. (a) 127.

(b) 128. (c) 129. (c) 130. (c) 131. (c) 132. (d)

133. (d) 134. (b) 135. (d) 136. (a) 137. (a) 138.

(b) 139.(9) 140. (c) 141. (a) 142. (a) 143. (a)

144. (c) 145. (a) 146. (c) 147. (b) 148. (c) 149.

(c) 150. (c) 151. (b,d). 152. (c) 153. (b) 154. (c)

155. (b) 156. (b) 157. (c) 158. (c) 159. (d) 160.

(a,d) 161. (a) 162. (a) 163. (d) 164. (c) 165. (c)

166. (a) 167. (a) 168. (d) 169.(6) 170. (a) 171.

(c) 172. (b) 173. (a) 174. (b) 175. (b) 176. (a)

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